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Inventor(s)	Liu; Jing et al.

Tropomyosin receptor kinase (TRK) degradation compounds and methods of use

Abstract

This disclosure relates to bivalent compounds (e.g., bi-functional small molecule compounds), compositions comprising one or more of the bivalent compounds, and to methods of use the bivalent compounds for the treatment of certain disease in a subject in need thereof. The disclosure also relates to methods for identifying such bivalent compounds.

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Background/Summary

CROSS-REFERENCE

(1) This application is a § 371 U.S. National Stage Entry of International Application No. PCT/CN2021/078240, filed Feb. 26, 2021, which claims the benefit of International Patent Application No. PCT/CN2020/076748, filed Feb. 26, 2020, which are incorporated herein by reference in their entirety.

BACKGROUND OF THE INVENTION

(2) This disclosure relates to bivalent compounds (e.g., bi-functional small molecule compounds), compositions comprising one or more of the bivalent compounds, and to methods of use of the bivalent compounds for the treatment of certain diseases in a subject in need thereof. The disclosure also relates to methods for identifying such bivalent compounds.

SUMMARY OF THE INVENTION

(3) In one aspect, provided herein is a compound of Formula I:

(4) ##STR00001## or a pharmaceutically acceptable salt thereof, wherein X^{sup.1} and X^{sup.2} are independently selected from CH and N; X^{sup.3} and X^{sup.4} are independently selected from C(O) and CR^{sup.4}R^{sup.5}; R^{sup.1} is selected from H, —NR^{sup.2}R^{sup.3}, halogen, optionally substituted C_{sub.1-6}alkyl, optionally substituted C_{sub.1-6}heteroalkyl, optionally substituted C_{sub.3-10}carbocyclyl, optionally substituted 3- to 10-membered heterocyclyl, optionally substituted 5- to 10-membered heteroaryl, optionally substituted C_{sub.1-6}heteroalkyl, optionally substituted C_{sub.1-6}haloalkyl, and optionally substituted C_{sub.1-6}alkoxy; R^{sup.2}, R^{sup.3}, R^{sup.4}, and R^{sup.5} are independently selected from H, optionally substituted C_{sub.1-6}alkyl, optionally substituted C_{sub.1-6}heteroalkyl, optionally substituted C_{sub.3-10}carbocyclyl, optionally substituted 3- to 10-membered heterocyclyl, optionally substituted 5- to 10-membered heteroaryl, optionally substituted C_{sub.1-6}heteroalkyl, optionally substituted C_{sub.1-6}haloalkyl, and optionally substituted C_{sub.1-6}alkoxy; and

(5) L is selected from a bond,

(6) ##STR00002## ##STR00003##

or one of X^{sup.3} and X^{sup.4} is C(O) and the other is CR^{sup.4}R^{sup.5}; and L is

(7) ##STR00004##

(8) In some embodiments, X^{sup.1} and X^{sup.2} are each N.

(9) In some embodiments, X^{sup.3} is C(O) and X^{sup.4} is CR^{sup.4}R^{sup.5}. In some embodiments, X^{sup.3} is C(O) and X^{sup.4} is CR^{sup.4}R^{sup.5}.

(10) In some embodiments, X^{sup.3} and X^{sup.4} are both C(O). In some embodiments, X^{sup.3} and X^{sup.4} are both CR^{sup.4}R^{sup.5}.

(11) In some embodiments, R^{sup.1} is —NR^{sup.2}R^{sup.3}. In some embodiments, R^{sup.1} is

(12) ##STR00005##

(13) In one aspect, provided herein is a compound of Formula Ia:

(14) ##STR00006## or a pharmaceutically acceptable salt thereof, wherein X^{sup.1} and X^{sup.2} are independently selected from CH and N; X^{sup.3} and X^{sup.4} are independently selected from C(O), CR^{sup.4}R^{sup.5}, and NR^{sup.6}; R^{sup.1} is selected from H, —NR^{sup.2}R^{sup.3}, halogen, optionally substituted C_{sub.1-6}alkyl, optionally substituted C_{sub.1-6}heteroalkyl, optionally substituted C_{sub.3-10}carbocyclyl, optionally substituted 3- to 10-membered heterocyclyl, optionally substituted 5- to 10-membered heteroaryl, optionally substituted aryl, optionally substituted C_{sub.1-6}heteroalkyl, optionally substituted C_{sub.1-6}haloalkyl, and optionally substituted C_{sub.1-6}alkoxy; R^{sup.2}, R^{sup.3}, R^{sup.4}, R^{sup.5}, and R^{sup.6} are independently

selected from H, halogen, optionally substituted C.sub.1-6alkyl, optionally substituted C.sub.1-C.sub.6 heteroalkyl, optionally substituted C.sub.3-10carbocyclyl, optionally substituted 3-to 10-membered heterocyclyl, optionally substituted 5- to 10-membered heteroaryl, optionally substituted C.sub.1-6heteroalkyl, optionally substituted C.sub.1-6haloalkyl, optionally substituted C.sub.1-6alkoxy, and optionally substituted 2,6-dioxopiperidin-3-yl; L is selected from a bond, R^{sup.r}—R'', R'COR'', R'CO.sub.2R'', R'C(O)N(R^{sup.7})R'', R'C(S)N(R^{sup.7})R'', R'OR'', R'SR'', R'SOR'', R'SO.sub.2R'', R'SON(R^{sup.7})R'', R'N(R^{sup.7})R'', R''N(R^{sup.7})COR'', R'N(R^{sup.7})CON(R^{sup.8})R'', R'N(R^{sup.7})C(S)R'', optionally substituted C.sub.1-C.sub.8alkylene, optionally substituted C.sub.1-C.sub.8 heteroalkylene, optionally substituted C.sub.2-C.sub.8 alkenylene, optionally substituted C.sub.2-C.sub.8 alkynylene, optionally substituted C.sub.1-C.sub.8heteroalkylene, optionally substituted C.sub.1-C.sub.8alkoxyC.sub.1-C.sub.8alkylene, optionally substituted C.sub.1-C.sub.8alkylaminoC.sub.1-C.sub.8alkylene, optionally substituted C.sub.1-C.sub.5 haloalkylene, optionally substituted C.sub.1-C.sub.8 hydroxyalkylene, optionally substituted C.sub.1-C.sub.5 aminoalkylene, optionally substituted C.sub.4-C.sub.13 fused carbocyclyl, optionally substituted 5-13 membered fused heterocyclyl, optionally substituted C.sub.5-C.sub.13 bridged carbocyclyl, optionally substituted 5-13 membered bridged heterocyclyl, optionally substituted C.sub.5-C.sub.13 spiro carbocyclyl, optionally substituted 5-13 membered spiro heterocyclyl, optionally substituted 3-10 membered carbocyclyl, optionally substituted 3-10 membered heterocyclyl, optionally substituted aryl, and optionally substituted heteroaryl; wherein L is optionally attached to X^{sup.3} or X^{sup.4}; R' and R'' are independently selected from null, optionally substituted C.sub.1-C.sub.8 alkylene, optionally substituted C.sub.1-C.sub.5 heteroalkylene, optionally substituted C.sub.2-C.sub.8 alkenylene, optionally substituted C.sub.2-C.sub.8 alkynylene, optionally substituted C.sub.1-C.sub.8heteroalkylene, optionally substituted C.sub.1-C.sub.5 hydroxyalkylene, optionally substituted C.sub.1-C.sub.5 aminoalkylene, optionally substituted C.sub.1-C.sub.8alkoxyC.sub.1-C.sub.8alkylene, optionally substituted C.sub.1-C.sub.5alkylaminoC.sub.1-C.sub.5alkylene, optionally substituted C.sub.1-C.sub.5 haloalkylene, optionally substituted 3-10 membered carbocyclyl, optionally substituted 3-10 membered heterocyclyl, optionally substituted C.sub.4-C.sub.13 fused carbocyclyl, optionally substituted 5-13 membered fused heterocyclyl, optionally substituted C.sub.5-C.sub.13 bridged carbocyclyl, optionally substituted 5-13 membered bridged heterocyclyl, optionally substituted C.sub.5-C.sub.13 spiro carbocyclyl, optionally substituted 5-13 membered spiro heterocyclyl, optionally substituted aryl, and optionally substituted heteroaryl; and R^{sup.7} and R^{sup.8} are independently selected from hydrogen, optionally substituted C.sub.1-C.sub.8 alkyl, optionally substituted C.sub.1-C.sub.5 heteroalkyl, optionally substituted C.sub.1-C.sub.5 heteroalkyl, optionally substituted C.sub.1-C.sub.8 alkoxy, optionally substituted C.sub.2-C.sub.8 alkenyl, optionally substituted C.sub.2-C.sub.8 alkynyl, optionally substituted C.sub.1-C.sub.8 alkoxy-C.sub.1-C.sub.5alkyl, optionally substituted C.sub.1-C.sub.5 haloalkyl, optionally substituted C.sub.1-C.sub.8 hydroxyalkyl, optionally substituted C.sub.1-C.sub.5 aminoalkyl, optionally substituted C.sub.1-C.sub.5alkylaminoC.sub.1-C.sub.8alkyl, optionally substituted 3-10 membered carbocyclyl, optionally substituted 3-10 membered heterocyclyl, optionally substituted aryl, and optionally substituted heteroaryl; or R' and R'', R^{sup.7} and R^{sup.8}, R' and R^{sup.7}, R' and R^{sup.8}, R'' and R^{sup.7}, R'' and R^{sup.8} together with the atom to which they are connected optionally form a 3-20 membered carbocyclyl or 3-20 membered heterocyclyl ring;

(15) In some embodiments, L is selected from

(16) ##STR00007##

(17) In some embodiments, X' and X^{sup.2} are each N.

(18) In some embodiments, at least one of X^{sup.3} and X^{sup.4} is NR^{sup.6}. In some embodiments, X^{sup.3} and X^{sup.4} are both NR^{sup.6}.

(19) In some embodiments, either X^{sup.3} or X^{sup.4} is —N-(2,6-dioxopiperidin-3-yl).

(20) In some embodiments, R^{sup.1} is —NR^{sup.2}R^{sup.3}. In some embodiments, R^{sup.1} is

(21) ##STR00008##

(22) In some embodiments, L is connected to X^{sup.3}. In some embodiments, L is connected to X^{sup.4}.

(23) In one aspect, provided herein is a compound of Formula II:

(24) ##STR00009## or a pharmaceutically acceptable salt thereof, wherein X^{sup.1} and X^{sup.2} are independently selected from CH and N; one of X^{sup.3} and X^{sup.4} is C(O) and the other is CR^{sup.1}R^{sup.2}; and L is selected from

(25) ##STR00010##

or L is selected from X^{sup.3} and X^{sup.4} are each C(O);

(26) ##STR00011##

and R^{sup.1} and R^{sup.2} are independently selected from H, halogen, optionally substituted C_{sub.1-6}alkyl, optionally substituted C_{sub.1-6}heteroalkyl, optionally substituted C_{sub.3-10}carbocyclyl, optionally substituted 3- to 10-membered heterocyclyl, optionally substituted 5- to 10-membered heteroaryl, optionally substituted C_{sub.1-6}heteroalkyl, optionally substituted C_{sub.1-6}haloalkyl, and optionally substituted C_{sub.1-6}alkoxy.

(27) In some embodiments, X^{sup.1} and X^{sup.2} are each N.

(28) In one aspect, provided herein is a compound of Formula III:

(29) ##STR00012## or a pharmaceutically acceptable salt thereof, wherein X^{sup.1} and X^{sup.3} are independently selected from CR^{sup.1}, CR^{sup.1}R^{sup.2}, O, N, and NR^{sup.1}; X^{sup.2} is selected from N, CO, and CH; Y is selected from O, NR^{sup.8} and CR^{sup.8}R^{sup.9}; Ar is selected from C_{sub.6-10}aryl and 5- to 10-membered heteroaryl, each of which is optionally substituted with one or more substituents independently selected from hydrogen, halogen, CN, NO_{sub.2}, OR^{sup.17}, SR^{sup.17}, NR^{sup.18}R^{sup.19}, COR^{sup.17}, CO_{sub.2}R^{sup.17}, CONR^{sup.18}R^{sup.19}, SOR^{sup.17}, SO_{sub.2}R^{sup.17}, SO_{sub.2}NR^{sup.18}R^{sup.19}, NR^{sup.17}COR^{sup.19}, NR^{sup.17}C(O)NR^{sup.18}R^{sup.19}, NR^{sup.18}SOR^{sup.17}, NR^{sup.18}SO_{sub.2}R^{sup.17}, optionally substituted C_{sub.1-8}alkyl, optionally substituted C_{sub.1-8}heteroalkyl, optionally substituted C_{sub.2-8}alkenyl, optionally substituted C_{sub.2-8}alkynyl, optionally substituted C_{sub.1-8}heteroalkyl, optionally substituted C_{sub.1-8}alkoxy, optionally substituted C_{sub.1-8}alkyl amino, optionally substituted C_{sub.3-10}carbocyclyl, —O-(optionally substituted C_{sub.3-10}carbocyclyl), —NH-(optionally substituted C_{sub.3-10}carbocyclyl), optionally substituted 3- to 10-membered heterocyclyl, —O-(optionally substituted 3- to 10-membered heterocyclyl), —NH-(optionally substituted 3- to 10-membered heterocyclyl), optionally substituted C_{sub.6-10}aryl, and optionally substituted 5- to 10-membered heteroaryl; L is selected from a bond, R^{sup.r}—R'', R'COR'', R'CO_{sub.2}R'', R'C(O)N(R^{sup.21})R'', R'C(S)N(R^{sup.21})R'', R'OR'', R'SR'', R'SOR'', R'SO_{sub.2}R'', R'SON(R^{sup.21})R'', R'N(R^{sup.21})R'', R'N(R^{sup.21})COR'', R'N(R^{sup.21})CON(R^{sup.22})R'', R'N(R^{sup.21})C(S)R'', optionally substituted C_{sub.1-8}alkylene, optionally substituted C_{sub.1-8}heteroalkylene, optionally substituted C_{sub.2-8}alkenylene, optionally substituted C_{sub.2-8}alkynylene, optionally substituted C_{sub.1-8}heteroalkylene, optionally substituted C_{sub.1-8}alkoxyC_{sub.1-8}alkylene, optionally substituted C_{sub.1-8}alkylaminoC_{sub.1-8}alkylene, optionally substituted C_{sub.1-5}haloalkylene, optionally substituted C_{sub.1-8}hydroxyalkylene, optionally substituted C_{sub.1-5}aminoalkylene, optionally substituted C_{sub.4-13}fused carbocyclyl, optionally substituted 5-13 membered fused heterocyclyl, optionally substituted C_{sub.5-13}bridged carbocyclyl, optionally substituted 5-13 membered bridged heterocyclyl, optionally substituted C_{sub.5-13}spiro carbocyclyl, optionally substituted 5-13 membered spiro heterocyclyl, optionally substituted 3-10 membered carbocyclyl, optionally substituted 3-10 membered heterocyclyl, optionally substituted aryl, and optionally substituted heteroaryl; wherein L is optionally attached to X^{sup.1} or X^{sup.3}; R' and R'' are independently selected from null, optionally substituted C_{sub.1-6}alkylene, optionally substituted C_{sub.1-5}heteroalkylene, optionally

substituted C.sub.2-C.sub.8 alkenylene, optionally substituted C.sub.2-C.sub.8 alkynylene, optionally substituted C.sub.1-C.sub.8 heteroalkylene, optionally substituted C.sub.1-C.sub.5 hydroxyalkylene, optionally substituted C.sub.1-C.sub.8 aminoalkylene, optionally substituted C.sub.1-C.sub.8alkoxyC.sub.1-C.sub.8alkylene, optionally substituted C.sub.1-C.sub.8alkylaminoC.sub.1-C.sub.5alkylene, optionally substituted C.sub.1-C.sub.8 haloalkylene, optionally substituted 3-10 membered carbocyclyl, optionally substituted 3-10 membered heterocyclyl, optionally substituted C.sub.4-C.sub.13 fused carbocyclyl, optionally substituted 5-13 membered fused heterocyclyl, optionally substituted C.sub.5-C.sub.13 bridged carbocyclyl, optionally substituted 5-13 membered bridged heterocyclyl, optionally substituted C.sub.5-C.sub.13 spiro carbocyclyl, optionally substituted 5-13 membered spiro heterocyclyl, optionally substituted aryl, and optionally substituted heteroaryl; R.sup.1 and R.sup.2 are independently selected at each occurrence from H, halogen, optionally substituted C.sub.1-8 alkyl, optionally substituted C.sub.1-8heteroalkyl, optionally substituted C.sub.3-10carbocyclyl, optionally substituted 3- to 10-membered heterocyclyl, optionally substituted 5- to 10-membered heteroaryl, optionally substituted C.sub.1-6heteroalkyl, optionally substituted C.sub.1-6haloalkyl, optionally substituted C.sub.1-6alkoxy, and optionally substituted 2,6-dioxopiperidin-3-yl; R.sup.3 is selected from a bond, —OR.sup.14—, —SR.sup.14—, —N(R.sup.15)R.sup.14—, —COR.sup.14—, —CO.sub.2R.sup.14—, —CON(R.sup.15)R.sup.14—, —SOR.sup.14—, —SO.sub.2R.sup.14—, —SON(R.sup.15)R.sup.14—, —N(R.sup.16)COR.sup.14—, —N(R.sup.16)CON(R.sup.15)R.sup.14—, N(R.sup.16)SOR.sup.14—, —N(R.sup.16)SO.sub.2R.sup.14—optionally substituted C.sub.1-8alkylene, optionally substituted C.sub.1-C.sub.5 heteroalkylene, optionally substituted C.sub.2-C.sub.8alkenylene, optionally substituted C.sub.2-C.sub.8alkynylene, optionally substituted C.sub.1-8heteroalkylene, optionally substituted C.sub.3-10carbocyclyl, optionally substituted 3- to 10-membered heterocyclyl, optionally substituted C.sub.6-10aryl, and optionally substituted 5- to 10-membered heteroaryl; R.sup.4, R.sup.3, and R.sup.6 are independently selected from hydrogen, halogen, CN, NO.sub.2, OR.sup.10, SR.sup.11, NR.sup.12R.sup.13, COR.sup.10, CO.sub.2R.sup.10, C(O)NR.sup.12R.sup.13, SOR.sup.10, SO.sub.2R.sup.10, SO.sub.2NR.sup.12R.sup.13, NR.sup.10C(O)R.sup.13, NR.sup.10C(O)NR.sup.12R.sup.13, NR.sup.10SOR.sup.13, NR.sup.10SO.sub.2R.sup.13, optionally substituted C.sub.1-8alkyl, optionally substituted C.sub.1-C.sub.5 heteroalkyl, optionally substituted C.sub.2-C.sub.8alkenyl, optionally substituted C.sub.2-C.sub.8alkynyl, optionally substituted C.sub.1-8 heteroalkyl, optionally substituted C.sub.1-8alkoxy, optionally substituted C.sub.3-10carbocyclyl, and optionally substituted 3- to 10-membered heterocyclyl; R.sup.7 is selected from optionally substituted C.sub.1-8alkyl, optionally substituted C.sub.1-8heteroalkyl, optionally substituted C.sub.3-10carbocyclyl, optionally substituted 3- to 10-membered heterocyclyl, optionally substituted C.sub.6-10aryl, and optionally substituted 5- to 10-membered heteroaryl; R.sup.8 and R.sup.9 are independently selected from hydrogen, halogen, OH, optionally substituted C.sub.1-8alkyl, optionally substituted C.sub.1-8heteroalkyl, optionally substituted C.sub.1-8alkoxy, optionally substituted C.sub.3-10 carbocyclyl, —O-(optionally substituted C.sub.3-10carbocyclyl), optionally substituted C.sub.1-8alkylamino, —NH-(optionally substituted C.sub.3-10carbocyclyl), and optionally substituted 3- to 10-membered heterocyclyl; or R.sup.8 and R.sup.9 are taken together with the atom to which they are connected to form an optionally substituted C.sub.3-10carbocyclyl or an optionally substituted 3- to 10-membered heterocyclyl; R.sup.10, R.sup.11, R.sup.12, and R.sup.13 are independently selected from hydrogen, optionally substituted C.sub.1-8alkyl, optionally substituted C.sub.1-C.sub.5 heteroalkyl, optionally substituted C.sub.2-C.sub.8 alkenyl, optionally substituted C.sub.2-C.sub.8 alkynyl, optionally substituted C.sub.1-8 heteroalkyl, optionally substituted C.sub.3-10carbocyclyl, optionally substituted 3- to 10-membered heterocyclyl, optionally substituted C.sub.6-10aryl, and optionally substituted 5-to 10-membered heteroaryl; or R.sup.12 and R.sup.13 are taken together with the atom to which they are connected to form an optionally substituted 3- to 10-membered heterocyclyl; R.sup.14 is selected

from null, optionally substituted C.sub.1-8alkylene, optionally substituted C.sub.1-C.sub.5 heteroalkylene, optionally substituted C.sub.2-C.sub.8 alkenylene, optionally substituted C.sub.2-C.sub.8 alkynylene, optionally substituted C.sub.1-8heteroalkylene, optionally substituted C.sub.1-8alkoxy, optionally substituted C.sub.3-10 carbocyclyl, —O-(optionally substituted C.sub.3-10 carbocyclyl), optionally substituted C.sub.1-8alkylamino, —NH-(optionally substituted C.sub.3-10 carbocyclyl), optionally substituted C.sub.3-10carbocyclyl, optionally substituted 3- to 10-membered heterocyclyl, optionally substituted C.sub.6-10aryl, and optionally substituted 5- to 10-membered heteroaryl; R.sup.15 and R.sup.16 are independently selected from hydrogen, optionally substituted C.sub.1-8alkyl, optionally substituted C.sub.1-C.sub.5 heteroalkyl, optionally substituted C.sub.2-C.sub.8 alkenyl, optionally substituted C.sub.2-C.sub.8 alkynyl, optionally substituted C.sub.1-8heteroalkyl, optionally substituted C.sub.1-8alkoxy, optionally substituted C.sub.3-10 carbocyclyl, optionally substituted 3- to 10-membered heterocyclyl, optionally substituted C.sub.6-10aryl, and optionally substituted 5- to 10-membered heteroaryl; or R.sup.14 and R.sup.15, together with the atom to which they are connected, optionally form an optionally substituted C.sub.3-10carbocyclyl or an optionally substituted 3- to 10-membered heterocyclyl; R.sup.17, R.sup.18, and R.sup.19 are independently selected from hydrogen, optionally substituted C.sub.1-8alkyl, optionally substituted C.sub.1-C.sub.5 heteroalkyl, optionally substituted C.sub.2-C.sub.8 alkenyl, optionally substituted C.sub.2-C.sub.8 alkynyl, optionally substituted C.sub.1-8heteroalkyl, optionally substituted C.sub.1-8alkoxy, optionally substituted C.sub.3-10carbocyclyl, —O-(optionally substituted C.sub.3-10carbocyclyl), optionally substituted 3- to 10-membered heterocyclyl, optionally substituted C.sub.6-10aryl, and optionally substituted 5- to 10-membered heteroaryl; or R.sup.18 and R.sup.19 are together with the atom to which they are connected to form an optionally substituted C.sub.3-10carbocyclyl or an optionally substituted 3- to 10-membered heterocyclyl, and

(30) R.sup.21 and R.sup.22 are independently selected from hydrogen, optionally substituted C.sub.1-C.sub.8 alkyl, optionally substituted C.sub.1-C.sub.5 heteroalkyl, optionally substituted C.sub.1-C.sub.8 alkoxy, optionally substituted C.sub.2-C.sub.8 alkenyl, optionally substituted C.sub.2-C.sub.8 alkynyl, optionally substituted C.sub.1-C.sub.8 alkoxy-C.sub.1-C.sub.8alkyl, optionally substituted C.sub.1-C.sub.5 haloalkyl, optionally substituted C.sub.1-C.sub.5 hydroxyalkyl, optionally substituted C.sub.1-C.sub.5 aminoalkyl, optionally substituted C.sub.1-C.sub.5alkylaminoC.sub.1-C.sub.5alkyl, optionally substituted C.sub.3-10carbocyclyl, optionally substituted 3-10 membered heterocyclyl, optionally substituted aryl, and optionally substituted heteroaryl; or R' and R'', R.sup.21 and R.sup.22, R' and R.sup.21, R' and R.sup.22, R'' and R.sup.21, or R'' and R.sup.22 together with the atom to which they are connected optionally form a 3-20 membered carbocyclyl or 3-20 membered heterocyclyl ring.

(31) In some embodiments, L is selected from

(32) ##STR00013##

(33) In some embodiments, R.sup.4, R.sup.5, and R.sup.6 are each hydrogen.

(34) In some embodiments, Y is CR.sup.8R.sup.9. In some embodiments, Y is CH.sub.2.

(35) In some embodiments, R.sup.7 is optionally substituted C.sub.6-10aryl. In some embodiments, R.sup.7 is

(36) ##STR00014##

(37) In some embodiments, Ar is C.sub.6-10aryl substituted with NR.sup.18R.sup.19. In some embodiments, Ar is

(38) ##STR00015##

(39) In some embodiments, R.sup.3 is optionally substituted 3- to 10-membered heterocyclyl. In some embodiments, R.sup.3 is

(40) ##STR00016##

(41) In some embodiments, X.sup.1 is CR.sup.1, X.sup.2 is CH, and X.sup.3 is NR.sup.1. In some embodiments, X' is NR.sup.1, X.sup.2 is CH, and X.sup.3 is CR.sup.1. In some embodiments,

X.sup.1 is CR.sup.1, X.sup.2 is N, and X.sup.3 is NR.sup.1. In some embodiments, XL is NR.sup.1, X.sup.2 is N, and X.sup.3 is CR.sup.1. In some embodiments, X.sup.1 is NR.sup.1, X.sup.2 is CH, and X.sup.3 is N. In some embodiments, X.sup.1 is N, X.sup.2 is CH, and X.sup.3 is NR.sup.1. In some embodiments, XL is CR.sup.1R.sup.2, X.sup.2 is CO, and X.sup.3 is NR.sup.1. In some embodiments, X.sup.1 is NR.sup.1, X.sup.2 is CO, and X.sup.3 is NR.sup.1. In some embodiments, X.sup.1 is O, X.sup.2 is CO, and X.sup.3 is NR.sup.1. In some embodiments, X.sup.1 is CR.sup.1, X.sup.2 is CO, and X.sup.3 is NR.sup.1. In some embodiments, X.sup.1 is N, X.sup.2 is CO, and X.sup.3 is NR.sup.1.

(42) In some embodiments, R.sup.1 is

(43) ##STR00017##

(44) In one aspect, provided herein is a compound of Formula IV:

(45) ##STR00018## or a pharmaceutically acceptable salt thereof, wherein X.sup.1 and X.sup.3 are independently selected from CR.sup.1, N, and NR.sup.1; X.sup.2 is selected from N and CH; Y.sup.1 is selected from N and CR.sup.6; Y.sup.2, Y.sup.3, and Y.sup.4 are independently selected from N and C, with the proviso that only one of Y.sup.2, Y.sup.3, and Y.sup.4 is N; Z is selected from null, a bond, C(R.sup.5).sub.2, C(R.sup.5).sub.2C(R.sup.5).sub.2, CO, C(R.sup.5).sub.2CO, CONR.sup.3, C(R.sup.5), O, C(R.sup.5).sub.2NR.sup.5 and CH.sub.2NR.sup.5; Ar.sup.1 and Ar.sup.2 are independently selected from C.sub.6-10aryl and 5- to 10-membered heteroaryl, each of which is optionally substituted with one or more substituents independently selected from halogen, CN, NO.sub.2, OR.sup.10, SR.sup.10, NR.sup.11R.sup.12, COR.sup.11, CO.sub.2R.sup.10, CONR.sup.11R.sup.12, SOR.sup.10, SO.sub.2R.sup.10, SO.sub.2NR.sup.11R.sup.12, NR.sup.10COR.sup.12, NR.sup.10C(O)NR.sup.11R.sup.12, NR.sup.10SOR.sup.12, NR.sup.10SO.sub.2R.sup.12, optionally substituted C.sub.1-8alkyl, optionally substituted C.sub.1-C.sub.8 heteroalkyl, optionally substituted C.sub.2-8alkenyl, optionally substituted C.sub.2-8alkynyl, optionally substituted C.sub.1-8 heteroalkyl, optionally substituted C.sub.1-8haloalkyl, optionally substituted C.sub.3-10carbocyclyl, optionally substituted 3- to 10-membered heterocyclyl, optionally substituted C.sub.6-10aryl, and optionally substituted 5- to 10-membered heteroaryl; L is selected from a bond, R.sup.r—R", R'COR", R'CO.sub.2R", R'C(O)N(R.sup.13)R", R'C(S)N(R.sup.13)R", R'OR", R'SR", R'SOR", R'SOR", R'SO.sub.2N(R.sup.13)R", R'N(R.sup.13)R", R"N(R.sup.13)COR", R'N(R.sup.13)CON(R.sup.14)R", R'N(R.sup.13)C(S)R", optionally substituted C.sub.1-C.sub.8 alkylene, optionally substituted C.sub.1-C.sub.5 heteroalkylene, optionally substituted C.sub.2-C.sub.8 alkenylene, optionally substituted C.sub.2-C.sub.8 alkynylene, optionally substituted C.sub.1-C.sub.8heteroalkylene, optionally substituted C.sub.1-C.sub.8alkoxyC.sub.1-C.sub.8alkylene, optionally substituted C.sub.1-C.sub.8alkylaminoC.sub.1-C.sub.8alkylene, optionally substituted C.sub.1-C.sub.5 haloalkylene, optionally substituted C.sub.1-C.sub.8 hydroxyalkylene, optionally substituted C.sub.1-C.sub.5 aminoalkylene, optionally substituted C.sub.4-C.sub.13 fused carbocyclyl, optionally substituted 5-13 membered fused heterocyclyl, optionally substituted C.sub.5-C.sub.13bridged carbocyclyl, optionally substituted 5-13 membered bridged heterocyclyl, optionally substituted C.sub.5-C.sub.13 spiro carbocyclyl, optionally substituted 5-13 membered spiro heterocyclyl, optionally substituted 3-10 membered carbocyclyl, optionally substituted 3-10 membered heterocyclyl, optionally substituted aryl, and optionally substituted heteroaryl; wherein L is optionally attached to X.sup.1 or X.sup.3; R' and R" are independently selected from null, optionally substituted C.sub.1-C.sub.6 alkylene, optionally substituted C.sub.1-C.sub.5 heteroalkylene, optionally substituted C.sub.2-C.sub.8 alkenylene, optionally substituted C.sub.2-C.sub.8 alkynylene, optionally substituted C.sub.1-C.sub.8heteroalkylene, optionally substituted C.sub.1-C.sub.5 hydroxyalkylene, optionally substituted C.sub.1-C.sub.5 aminoalkylene, optionally substituted C.sub.1-C.sub.8alkoxyC.sub.1-C.sub.8alkylene, optionally substituted C.sub.1-C.sub.5alkylaminoC.sub.1-C.sub.5alkylene, optionally substituted C.sub.1-C.sub.8 haloalkylene, optionally substituted 3-10 membered

carbocyclyl, optionally substituted 3-10 membered heterocyclyl, optionally substituted C.sub.4-C.sub.15 fused carbocyclyl, optionally substituted 5-13 membered fused heterocyclyl, optionally substituted C.sub.5-C.sub.13 bridged carbocyclyl, optionally substituted 5-13 membered bridged heterocyclyl, optionally substituted C.sub.5-C.sub.13 spiro carbocyclyl, optionally substituted 5-13 membered spiro heterocyclyl, optionally substituted aryl, and optionally substituted heteroaryl; R.sup.1 is selected at each occurrence from H, halogen, optionally substituted C.sub.1-6alkyl, optionally substituted C.sub.1-heteroalkyl, optionally substituted C.sub.3-10carbocyclyl, optionally substituted 3- to 10-membered heterocyclyl, optionally substituted 5- to 10-membered heteroaryl, optionally substituted C.sub.1-6 heteroalkyl, optionally substituted C.sub.1-6haloalkyl, optionally substituted C.sub.1-6alkoxy, and optionally substituted 2,6-dioxopiperidin-3-yl; R.sup.3 is selected from a bond, —OR.sup.7—, —SR.sup.7—, —N(R.sup.8)R.sup.7—, —COR.sup.7—, —CO.sub.2R.sup.7—, —CON(R.sup.8)R.sup.7—, —SOR.sup.7—, —SO.sub.2R.sup.7—, —SO.sub.2N(R.sup.8)R.sup.7—, —N(R.sup.9)COR.sup.7—, —N(R.sup.9)CON(R.sup.8)R.sup.7—, N(R.sup.9)SOR.sup.7—, —N(R.sup.9)SO.sub.2R.sup.7—, optionally substituted C.sub.1-8alkylene, optionally substituted C.sub.1-C.sub.5 heteroalkylene, optionally substituted C.sub.2-C.sub.8alkenylene, optionally substituted C.sub.2-C.sub.8alkynylene, optionally substituted C.sub.1-8heteroalkylene, optionally substituted C.sub.3-10carbocyclyl, optionally substituted 3- to 10-membered heterocyclyl, optionally substituted C.sub.6-10aryl, and optionally substituted 5- to 10-membered heteroaryl; R.sup.4 and R.sup.5 are independently selected at each occurrence from hydrogen, halogen, OH, NH.sub.2, CN, NO.sub.2, optionally substituted C.sub.1-4alkyl, optionally substituted C.sub.1-C.sub.6 heteroalkyl, optionally substituted C.sub.1-4alkoxy, optionally substituted C.sub.1-4heteroalkyl, optionally substituted C.sub.1-4haloalkyl, optionally substituted C.sub.3-10carbocyclyl, —O-(optionally substituted C.sub.3-10carbocyclyl), —NH-(optionally substituted C.sub.3-10carbocyclyl), and optionally substituted 3- to 10-membered heterocyclyl; R.sup.6 is selected from hydrogen, halogen, CN, NO.sub.2, optionally substituted C.sub.1-6alkyl, optionally substituted C.sub.1-C.sub.6 heteroalkyl, optionally substituted C.sub.3-10carbocyclyl, and optionally substituted 3- to 10-membered heterocyclyl; R.sup.7 is selected from null, optionally substituted C.sub.1-8alkylene, optionally substituted C.sub.1-C.sub.8 heteroalkylene, optionally substituted C.sub.2-C.sub.8 alkenylene, optionally substituted C.sub.2-C.sub.8 alkynylene, optionally substituted C.sub.1-8heteroalkylene, optionally substituted C.sub.1-8alkoxy, optionally substituted C.sub.3-10 carbocyclyl, —O-(optionally substituted C.sub.3-10 carbocyclyl), optionally substituted C.sub.1-8alkylamino, —NH-(optionally substituted C.sub.3-10 carbocyclyl), optionally substituted C.sub.3-10carbocyclyl, optionally substituted 3- to 10-membered heterocyclyl, optionally substituted C.sub.6-10aryl, and optionally substituted 5- to 10-membered heteroaryl; R.sup.8 and R.sup.9 are independently selected from hydrogen, optionally substituted C.sub.1-8alkyl, optionally substituted C.sub.1-C.sub.5 heteroalkyl, optionally substituted C.sub.2-C.sub.8 alkenyl, optionally substituted C.sub.2-C.sub.8 alkynyl, optionally substituted C.sub.1-8heteroalkyl, optionally substituted C.sub.1-8alkoxy, optionally substituted C.sub.3-10 carbocyclyl, optionally substituted 3- to 10-membered heterocyclyl, optionally substituted C.sub.6-10aryl, and optionally substituted 5- to 10-membered heteroaryl; or R.sup.7 and R.sup.8, together with the atom to which they are connected, optionally form an optionally substituted C.sub.3-10carbocyclyl or an optionally substituted 3- to 10-membered heterocyclyl; R.sup.10, R.sup.11, and R.sup.12 are independently selected from hydrogen, optionally substituted C.sub.1-8alkyl, optionally substituted C.sub.1-C.sub.5 heteroalkyl, optionally substituted C.sub.2-8alkenyl, optionally substituted C.sub.2-8-alkynyl, optionally substituted C.sub.3-10carbocyclyl, optionally substituted 3- to 10-membered heterocyclyl, optionally substituted C.sub.6-10aryl, and optionally substituted 5- to 10-membered heteroaryl; or R.sup.11 and R.sup.12 are together with the atom to which they are connected to form an optionally substituted C.sub.3-C.sub.10carbocyclyl or an optionally substituted 3- to 10-membered heterocyclyl; R.sup.13 and R.sup.14 are independently selected from hydrogen, optionally substituted C.sub.1-C.sub.8alkyl, optionally substituted C.sub.1-C.sub.5

heteroalkyl, optionally substituted C.sub.1-C.sub.6 alkoxy, optionally substituted C.sub.2-C.sub.8 alkenyl, optionally substituted C.sub.2-C.sub.8 alkynyl, optionally substituted C.sub.1-C.sub.6 alkoxy-C.sub.1-C.sub.5alkyl, optionally substituted C.sub.1-C.sub.8 haloalkyl, optionally substituted C.sub.1-C.sub.5 hydroxyalkyl, optionally substituted C.sub.1-C.sub.8 aminoalkyl, optionally substituted C.sub.1-C.sub.5alkylaminoC.sub.1-C.sub.5alkyl, optionally substituted 3-10 membered carbocyclyl, optionally substituted 3-10 membered heterocyclyl, optionally substituted aryl, and optionally substituted heteroaryl; or R' and R'', R.sup.13 and R.sup.14, R' and R.sup.13, R' and R.sup.14, R'' and R.sup.13, R'' and R.sup.14 together with the atom to which they are connected optionally form a 3-20 membered carbocyclyl or 3-20 membered heterocyclyl ring; and n is 0, 1, 2, 3, or 4.

(46) In some embodiments, L is selected from

(47) ##STR00019##

(48) In some embodiments, Y.sup.1 is N, Y.sup.2 is N, Y.sup.3 is C, and Y.sup.4 is C.

(49) In some embodiments, Ar.sup.1 is C.sub.6-10aryl optionally substituted with halogen. In some embodiments, Ar.sup.1 is

(50) ##STR00020##

(51) In some embodiments, Ar.sup.2 is C.sub.6-10aryl optionally substituted with NR.sup.11R.sup.12. In some embodiments, Ar.sup.2 is

(52) ##STR00021##

(53) In some embodiments, R.sup.3 is optionally substituted 3- to 10-membered heterocyclyl. In some embodiments, R.sup.3 is

(54) ##STR00022##

(55) In some embodiments, R.sup.4 is hydrogen.

(56) In some embodiments, Z is C(R.sup.3).sub.2. In some embodiments, Z is CH.sub.2.

(57) In some embodiments, n is 0.

(58) In some embodiments, X.sup.1 is CR.sup.1, X.sup.2 is CH, and X.sup.3 is NR.sup.1. In some embodiments, X.sup.1 is NR.sup.1, X.sup.2 is CH, and X.sup.3 is CR.sup.1. In some embodiments, X.sup.1 is CR.sup.1, X.sup.2 is N, and X.sup.3 is NR.sup.1. In some embodiments, X.sup.1 is NR.sup.1, X.sup.2 is N, and X.sup.3 is CR.sup.1. In some embodiments, X.sup.1 is NR.sup.1, X.sup.2 is CH, and X.sup.3 is N. In some embodiments, X.sup.1 is N, X.sup.2 is CH, and X.sup.3 is NR.sup.1

(59) ##STR00023##

(60) In some embodiments, R.sup.1 is methyl. In some embodiments, R.sup.1 is According to one aspect of the present disclosure, a bivalent compound disclosed herein comprises a tropomyosin receptor kinase (TRK) ligand conjugated to a degradation tag, or a pharmaceutically acceptable salt or analog thereof.

(61) In one embodiment, the TRK ligand is capable of binding to a TRK protein comprising a TRK, a TRK mutant, a TRK deletion, a TRK splicing or a TRK fusion protein.

(62) In another embodiment, the TRK ligand is a TRK kinase inhibitor or a portion of TRK kinase inhibitor.

(63) In another embodiment, the TRK ligand is selected from the group consisting of entrectinib (RXDX-101), GNF-8625, larotrectinib (LOXO-101; ARRY-470), altiratinib (DCC.sub.2701, DCC-270, DP-5164), sitravatinib (MGCD516), cabozantinib (XL-184, BMS-907351), dovitinib (TKI-258, CHIR-258), milciclib (PHA-848125AC), belizatinib (TSR-011), GZ389988, pegcantratinib, AZD7451, TPX-0005, LOXO-195, regorafenib, DS-6051b, F17752, PLX7486, AZD-6918, ASP7962, ONO-4474, PF-06273340, and analogs thereof.

(64) In another embodiment, the degradation tag binds to an ubiquitin ligase, or is a hydrophobic group or a tag that leads to misfolding of the TRK protein.

(65) In another embodiment, the ubiquitin ligase is an E3 ligase.

(66) In another embodiment, the E3 ligase is selected from the group consisting of a cereblon E3

ligase, a VHL E3 ligase, an IAP ligase, a MDM2 ligase, a TRIM24 ligase, a TRIM21 ligase, a KEAP1 ligase, DCAF16 ligase, RNF4 ligase, RNF114 ligase, and AhR ligase.

(67) In another embodiment, the degradation tag is selected from the group consisting of pomalidomide, thalidomide, lenalidomide, VHL-1, adamantane, 1-((4,4,5,5,5-pentafluoropentyl)sulfinyl)nonane, nutlin-3a, RG7112, RG7338, AMG232, AA-115, bestatin, MV-1, LCL161, CPD36, GDC-0152, CRBN-1, CRBN-2, CRBN-3, CRBN-4, CRBN-5, CRBN-6, CRBN-7, CRBN-8, CRBN-9, CRBN-10, CRBN-11, and analogs thereof.

(68) In another embodiment, the TRK ligand is conjugated to the degradation tag via a linker moiety.

(69) In another embodiment, the TRK ligand comprises a moiety of Formula 1

(70) ##STR00024## wherein X is selected from CR'R'', CO, O, S, SO, SO.sub.2, and NR', wherein R' and R'' are independently selected from hydrogen, halogen, OH, optionally substituted C.sub.1-C.sub.5 alkyl, optionally substituted C.sub.1-C.sub.5 heteroalkyl, optionally substituted C.sub.1-8 alkoxy, optionally substituted C.sub.1-C.sub.8alkoxyC.sub.1-C.sub.8alkyl, optionally substituted C.sub.1-C.sub.6 alkylamino, optionally substituted C.sub.1-C.sub.8alkylaminoC.sub.1-C.sub.8alkyl, optionally substituted C.sub.3-C.sub.10 carbocyclyl, optionally substituted C.sub.3-C.sub.10 cycloalkoxy, and optionally substituted 3-10 membered heterocyclyl; or R' and R'' together with the atom to which they are connected optionally form an optionally substituted 3-8 membered carbocyclyl or heterocyclyl ring; R is selected from optionally substituted C.sub.1-C.sub.8 alkyl, optionally substituted C.sub.1-C.sub.6 heteroalkyl, optionally substituted C.sub.3-C.sub.10 carbocyclyl, optionally substituted 3-10 membered heterocyclyl, optionally substituted aryl, and optionally substituted heteroaryl; R.sup.1, R.sup.2, and R.sup.3 are independently selected from hydrogen, halogen, CN, NO.sub.2, OR.sup.5, SR.sup.6, NR.sup.7R.sup.8, COR.sup.5, CO.sub.2R.sup.5, C(O)NR.sup.7R.sup.8, SOR.sup.8, SO.sub.2R.sup.5, SO.sub.2NR.sup.7R.sup.8, NR.sup.7C(O)R.sup.8, NR.sup.5C(O)NR.sup.1R.sup.8, NR.sup.10SOR.sup.8, NR.sup.7SO.sub.2R.sup.8, optionally substituted C.sub.1-C.sub.8 alkyl, optionally substituted C.sub.1-C.sub.5 heteroalkyl, optionally substituted C.sub.1-C.sub.8 alkoxy, optionally substituted C.sub.1-C.sub.8alkoxyC.sub.1-C.sub.5alkyl, optionally substituted C.sub.1-C.sub.8alkylaminoC.sub.1-C.sub.8alkyl, optionally substituted C.sub.3-C.sub.10 carbocyclyl, optionally substituted C.sub.3-C.sub.10 cycloalkoxy, optionally substituted 3-10 membered heterocyclyl, optionally substituted C.sub.2-C.sub.8 alkenyl, and optionally substituted C.sub.2-C.sub.8 alkynyl, wherein

(71) R.sup.5, R.sup.6, R.sup.7, and R.sup.8 are independently selected from hydrogen, optionally substituted C.sub.1-C.sub.8alkyl, optionally substituted C.sub.1-C.sub.5 heteroalkyl, optionally substituted C.sub.1-C.sub.8alkoxyC.sub.1-C.sub.5alkyl, optionally substituted C.sub.1-C.sub.5alkylaminoC.sub.1-C.sub.8alkyl, optionally substituted C.sub.3-C.sub.10 carbocyclyl, optionally substituted 3-10 membered heterocyclyl, optionally substituted C.sub.2-C.sub.8 alkenyl, optionally substituted C.sub.2-C.sub.8 alkynyl, optionally substituted aryl, or optionally substituted heteroaryl, or

(72) R.sup.7 and R.sup.8 together with the atom to which they are connected optionally form an optionally substituted 3-8 membered heterocyclyl ring; R.sup.4 is connected to the linker moiety of the bivalent compound, and is selected from a bond, —OR.sup.9—, —SR.sup.9—, —NR.sup.10R.sup.11—, —COR.sup.9—, —CO.sub.2R.sup.9—, —CONR.sup.10R.sup.11—, —SOR.sup.9—, —SO.sub.2R.sup.9—, —SO.sub.2NR.sup.10R.sup.11—, —NR.sup.10COR.sup.11—, —NR.sup.9 CONR.sup.10R.sup.11—, —NR.sup.10SOR.sup.11—, —NR.sup.10SO.sub.2R.sup.11—, optionally substituted C.sub.1-C.sub.8 alkylene, optionally substituted C.sub.1-C.sub.5 heteroalkylene, optionally substituted C.sub.1-C.sub.8 alkylene-O—, optionally substituted C.sub.1-C.sub.8alkylene-O—C.sub.1-C.sub.5alkylene, optionally substituted C.sub.1-C.sub.5alkylene-N(C.sub.1-C.sub.5alkyl)-C.sub.1-C.sub.5alkylene, optionally substituted C.sub.3-C.sub.8 carbocyclylene, optionally substituted C.sub.3-C.sub.8 carbocyclylene-O—,

optionally substituted 3-8 membered carbocyclyl, optionally substituted 3-8 membered heterocyclyl, optionally substituted C.sub.2-C.sub.8 alkenylene, optionally substituted C.sub.2-C.sub.8 alkynylene, aryl, and optionally substituted heteroaryl, wherein (73) R.sup.9, R.sup.10, and R.sup.11 are independently selected from null, a bond, hydrogen, optionally substituted C.sub.1-C.sub.6 alkyl, optionally substituted C.sub.1-C.sub.5 heteroalkyl, optionally substituted C.sub.1-C.sub.8 alkoxy, optionally substituted C.sub.1-C.sub.8alkoxyC.sub.1-C.sub.5alkyl, optionally substituted C.sub.1-C.sub.5alkylaminoC.sub.1-C.sub.5alkyl, optionally substituted C.sub.3-C.sub.8 carbocyclyl, optionally substituted C.sub.3-C.sub.8 cycloalkoxy, optionally substituted 3-8 membered carbocyclyl, optionally substituted 3-8 membered heterocyclyl, optionally substituted C.sub.2-C.sub.8 alkenyl, optionally substituted C.sub.2-C.sub.8 alkynyl, optionally substituted C.sub.1-C.sub.8 alkylene, optionally substituted C.sub.1-C.sub.5 heteroalkylene, optionally substituted C.sub.1-C.sub.8 alkylene-O—, optionally substituted C.sub.1-C.sub.8 alkylene-N(C.sub.1-C.sub.8alkyl)-, optionally substituted C.sub.1-C.sub.8alkylene-O-alkylene, optionally substituted C.sub.1-C.sub.8alkylene-N(C.sub.1-C.sub.8alkyl)-C.sub.1-C.sub.5alkylene, optionally substituted C.sub.3-C.sub.8 carbocyclylene, optionally substituted C.sub.3-C.sub.8 carbocyclylene-O—, optionally substituted aryl, and optionally substituted heteroaryl, or R.sup.10 and R.sup.11 together with the atom to which they are connected optionally form a 3-8 membered carbocyclyl or heterocyclyl ring; and Ar is selected from aryl and heteroaryl group, each of which is optionally substituted with one or more substituents independently selected from hydrogen, halogen, CN, NO.sub.2, OR.sup.12, SR.sup.12, NR.sup.13R.sup.14, COR.sup.12, CO.sub.2R.sup.12, CONR.sup.13R.sup.14, SOR.sup.12, SO.sub.2R.sup.12, SO.sub.2NR.sup.13R.sup.14, NR.sup.13COR.sup.14, NR.sup.15C(O)NR.sup.13R.sup.14, NR.sup.13SOR.sup.14, NR.sup.13SO.sub.2R.sup.14, optionally substituted C.sub.1-C.sub.8 alkyl, optionally substituted C.sub.1-C.sub.5 heteroalkyl, optionally substituted C.sub.1-C.sub.8 alkoxy, optionally substituted C.sub.1-C.sub.8alkoxyC.sub.1-C.sub.5alkyl, optionally substituted C.sub.1-C.sub.8alkylaminoC.sub.1-C.sub.5alkyl, optionally substituted C.sub.3-C.sub.8 carbocyclyl, optionally substituted C.sub.3-C.sub.8 cycloalkoxy, optionally substituted 3-8 membered heterocyclyl, optionally substituted C.sub.2-C.sub.8 alkenyl, optionally substituted C.sub.2-C.sub.8 alkynyl, optionally substituted aryl, and optionally substituted heteroaryl, wherein

(74) R.sup.12, R.sup.13, R.sup.14, and R.sup.15 are independently selected from hydrogen, optionally substituted C.sub.1-C.sub.6 alkyl, optionally substituted C.sub.1-C.sub.8 heteroalkyl, optionally substituted C.sub.1-C.sub.6 alkoxy, optionally substituted C.sub.1-C.sub.8alkoxyC.sub.1-C.sub.8alkyl, optionally substituted C.sub.1-C.sub.5alkylaminoC.sub.1-C.sub.8alkyl, optionally substituted C.sub.3-C.sub.8 carbocyclyl, optionally substituted C.sub.3-C.sub.8 cycloalkoxy, optionally substituted 3-8 membered heterocyclyl, optionally substituted C.sub.2-C.sub.8 alkenyl, optionally substituted C.sub.2-C.sub.8 alkynyl, optionally substituted aryl, and optionally substituted heteroaryl, or

(75) R.sup.13 and R.sup.14 together with the atom to which they are connected optionally form a 3-8 membered carbocyclyl or heterocyclyl ring.

(76) In one embodiment, X is selected from CR'R'', O, and NR'; wherein R' and R'' are independently selected from hydrogen, F, OH, optionally substituted C.sub.1-C.sub.3 alkyl, and optionally substituted C.sub.1-C.sub.3 alkoxy; or R' and R'' together with the atom to which they are connected optionally form an optionally substituted 3-6 membered carbocyclyl or heterocyclyl ring.

(77) In another embodiment, X is selected from CH.sub.2, cyclopropylene, CHF, CF2, O, NH, NCH3, NCH2CH.sub.3, and N-isopropyl.

(78) In another embodiment, R is selected from optionally substituted C.sub.3-C.sub.8 carbocyclyl, optionally substituted 3-8 membered heterocyclyl, optionally substituted aryl, and optionally substituted heteroaryl.

(79) In another embodiment, R is selected from optionally substituted phenyl and optionally substituted heteroaryl.

(80) In another embodiment, X is CH₂; and R is 3,5-difluorophenyl.

(81) In another embodiment, R¹, R², and R³ are independently selected from hydrogen, F, C₁, and OH. In another embodiment, R⁴—Ar is selected from a moiety of formulae A1, A2, A3, and A4;

(82) ##STR00025## wherein * indicates the connection to the linker moiety of the bivalent compound; and R^a is selected from hydrogen, halogen, CN, NO₂, OR¹², SR¹², NRBR¹⁴, COR¹², CO₂R¹², CONR¹³R¹⁴, SOR¹², SO₂R¹², SO₂NR¹³R¹⁴, NR¹⁵COR¹⁴, NR¹⁵C(O)NR¹³R¹⁴, NR¹⁵SOR¹⁴, NR¹⁰SSO₂R¹⁴, optionally substituted C₁-C₈alkyl, optionally substituted C₁-C₅ heteroalkyl, optionally substituted C₁-C₈alkoxy, optionally substituted C₁-C₈alkoxyC₁-C₈alkyl, optionally substituted C₁-C₈alkylaminoC₁-C₅alkyl, optionally substituted C₃-C₈ carbocyclyl, optionally substituted C₃-C₈ cycloalkoxy, optionally substituted 3-8 membered heterocyclyl, optionally substituted C₂-C₈ alkenyl, optionally substituted C₂-C₈ alkynyl, optionally substituted aryl, and optionally substituted heteroaryl, wherein

(83) R¹², R¹³, R¹⁴, and R¹⁵ are independently selected from hydrogen, optionally substituted C₁-C₅ alkyl, optionally substituted C₁-C₈ heteroalkyl, optionally substituted C₁-C₆ alkoxy, optionally substituted C₁-C₈alkoxyC₁-C₈alkyl, optionally substituted C₁-C₅alkylaminoC₁-C₈alkyl, optionally substituted C₃-C₈ carbocyclyl, optionally substituted C₃-C₈ cycloalkoxy, optionally substituted 3-8 membered heterocyclyl, optionally substituted C₂-C₈ alkenyl, optionally substituted C₂-C₈ alkynyl, aryl, and optionally substituted heteroaryl, or R¹³ and R¹⁴ together with the atom to which they are connected optionally form a 3-8 membered carbocyclyl or heterocyclyl ring.

(84) In another embodiment, R⁴—Ar is selected from a moiety of formulae A1, A3, A3, and A4;

(85) ##STR00026## wherein * indicates the connection to the linker moiety of the bivalent compound; and R^a is selected from hydrogen, halogen, NR¹³R¹⁴, and NR¹³COR¹⁴, wherein R¹³ and R¹⁴ are independently selected from hydrogen, optionally substituted C₁-C₈alkyl, optionally substituted C₁-C₅ heteroalkyl, optionally substituted C₁-C₈alkoxyC₁-C₅alkyl, optionally substituted C₁-C₈alkylaminoC₁-C₅alkyl, optionally substituted arylalkyl, optionally substituted heteroarylalkyl, optionally substituted C₃-C₈ carbocyclyl, optionally substituted C₃-C₈ cycloalkoxy, optionally substituted 3-8 membered heterocyclyl, optionally substituted C₂-C₈ alkenyl, optionally substituted C₂-C₈ alkynyl, phenyl, and optionally substituted C₅-C₆ heteroaryl, or R¹³ and R¹⁴ together with the atom to which they are connected optionally form a 3-8 membered carbocyclyl or heterocyclyl ring.

(86) In another embodiment, R^a is selected from H, (tetrahydro-2H-pyran-4-yl)amino, and 2-fluoroethylamino.

(87) In another embodiment, R⁴ is selected from optionally substituted

(88) ##STR00027##

(89) In another embodiment, the TRK ligand comprises a moiety of Formula 2;

(90) ##STR00028## wherein

(91) X¹, X², X³, and X⁴ are independently selected from C, CR', and N (preferly, X¹ is selected from CR' and N; and X², X³, and X⁴ are independently selected from C and N), wherein

(92) R' is selected from hydrogen, halogen, CN, NO₂, optionally substituted C₁-C₆

alkyl, optionally substituted C.sub.1-C.sub.8 heteroalkyl, optionally substituted C.sub.3-C.sub.6 carbocyclyl, and optionally substituted 3-6 membered heterocyclyl; X is selected from null, a bond, C(R.sup.2).sub.2, C(R.sup.2).sub.2C(R.sup.2).sub.2, CO, C(R.sup.2).sub.2CO, CONR.sup.2, C(R.sup.2).sub.2O, C(R.sup.2).sub.2NR.sup.2 and CH.sub.2NR.sup.2; R.sup.1 and R.sup.2, at each occurrence, are independently selected from hydrogen, halogen, OH, NH.sub.2, CN, NO.sub.2, optionally substituted C.sub.1-C.sub.4 alkyl, optionally substituted C.sub.1-C.sub.4 heteroalkyl, optionally substituted C.sub.1-C.sub.4 alkoxy, optionally substituted C.sub.1-C.sub.4 alkylamino, optionally substituted C.sub.1-C.sub.4 alkoxyalkyl, optionally substituted C.sub.1-C.sub.4 haloalkyl, optionally substituted C.sub.1-C.sub.4 hydroxyalkyl, optionally substituted C.sub.1-C.sub.4alkylaminoC.sub.1-C.sub.4alkyl, optionally substituted C.sub.3-C.sub.6 carbocyclyl, optionally substituted C.sub.3-C.sub.6 cycloalkoxy, and optionally substituted 3-6 membered heterocyclyl; n is 1 to 4; R.sup.3 is connected to the linker moiety of the bivalent compound either directly or through R.sup.4; R.sup.3 and R.sup.4 are independently selected from null, a bond, —OR.sup.5—, —SR.sup.5—, —NR.sup.6R.sup.7—, —COR.sup.5—, —CO.sub.2R.sup.5—, —CONR.sup.6R.sup.7—, —SOR.sup.5—, —SOR.sup.5—, —SO.sub.2NR.sup.6R.sup.7—, —NR.sup.9 COR.sup.7—, —NR.sup.5C(O)NR.sup.6R.sup.7—, —NR.sup.6SOR.sup.7—, —NR.sup.6SO.sub.2R.sup.7—, optionally substituted C.sub.1-C.sub.8 alkylene, optionally substituted C.sub.1-C.sub.5alkylene-O—C.sub.1-C.sub.8alkylene, optionally substituted C.sub.1-C.sub.6 heteroalkylene, optionally substituted C.sub.1-C.sub.5 haloalkylene, optionally substituted C.sub.1-C.sub.5 hydroxyalkylene, optionally substituted C.sub.1-C.sub.5alkylene-N(C.sub.1-C.sub.8alkyl)-C.sub.1-C.sub.5alkylene, optionally substituted C.sub.3-C.sub.8 carbocyclylene, optionally substituted C.sub.3-C.sub.8 carbocyclylene-O—, optionally substituted 3-8 membered carbocyclyl, optionally substituted 3-8 membered heterocyclyl, optionally substituted C.sub.2-C.sub.8 alkenylene, optionally substituted C.sub.2-C.sub.8 alkynylene, optionally substituted aryl, and optionally substituted heteroaryl, wherein

(93) R.sup.5, R.sup.6 and R.sup.7 are independently selected from null, a bond, hydrogen, optionally substituted C.sub.1-C.sub.8 alkyl, optionally substituted C.sub.1-C.sub.8 heteroalkyl, optionally substituted C.sub.2-C.sub.8 alkenyl, optionally substituted C.sub.2-C.sub.8 alkynyl, optionally substituted C.sub.3-C.sub.8 carbocyclyl, optionally substituted 3-8 membered heterocyclyl, optionally substituted heterocarbocyclyl, optionally substituted arylalkyl, optionally substituted heteroarylalkyl, optionally substituted C.sub.1-C.sub.8 alkylene, optionally substituted C.sub.1-C.sub.8 heteroalkylene, optionally substituted C.sub.1-C.sub.8alkylene-O—, optionally substituted C.sub.1-C.sub.8 alkylene-N(C.sub.1-C.sub.8 alkyl)-, optionally substituted C.sub.1-C.sub.8 alkylene-O-alkylene, optionally substituted C.sub.1-C.sub.5alkylene-N(C.sub.1-C.sub.8 alkyl)-C.sub.1-C.sub.8alkylene, optionally substituted C.sub.3-C.sub.8 carbocyclylene, optionally substituted C.sub.3-C.sub.8 carbocyclylene-O—, optionally substituted 3-8 membered carbocyclyl, optionally substituted 3-8 membered heterocyclyl, optionally substituted aryl, and optionally substituted heteroaryl, or

(94) R.sup.6 and R.sup.7 together with the atom to which they are connected optionally form a 3-8 membered carbocyclyl or 3-8 membered heterocyclyl ring; and Ar.sup.1 and Ar.sup.2 are independently selected from aryl and heteroaryl, each of which is optionally substituted with one or more substituents independently selected from halogen, CN, NO.sub.2, OR.sup.10, SR.sup.10, NR.sup.11R.sup.12 COR.sup.10, CO.sub.2R.sup.10, CONR.sup.11R.sup.12, SORT, SO.sub.2R.sup.10SO.sub.2NR.sup.11R.sup.12, NR.sup.11COR.sup.12, NR.sup.10C(O)NR.sup.11R.sup.12, NR.sup.10SOR.sup.12, NR.sup.10SO.sub.2R.sup.12, optionally substituted C.sub.1-C.sub.6 alkyl, optionally substituted C.sub.1-C.sub.5 heteroalkyl, optionally substituted C.sub.1-C.sub.8alkoxyC.sub.1-C.sub.8alkyl, optionally substituted C.sub.1-C.sub.5 haloalkyl, optionally substituted C.sub.1-C.sub.6 hydroxyalkyl, optionally substituted C.sub.1-C.sub.5alkylaminoC.sub.1-C.sub.5alkyl, optionally substituted C.sub.3-C.sub.7 carbocyclyl, optionally substituted 3-7 membered heterocyclyl, optionally substituted C.sub.2-

C.sub.8 alkenyl, optionally substituted C.sub.2-C.sub.8 alkynyl, optionally substituted aryl, and optionally substituted heteroaryl, wherein

(95) R.sup.10, R.sup.11, and R.sup.12, at each occurrence, are independently selected from null, hydrogen, optionally substituted C.sub.1-C.sub.6 alkyl, optionally substituted C.sub.1-C.sub.8 heteroalkyl, optionally substituted C.sub.2-C.sub.8 alkenyl, optionally substituted C.sub.2-C.sub.8 alkynyl, optionally substituted C.sub.3-C.sub.7 carbocyclyl, optionally substituted 3-7 membered heterocyclyl, optionally substituted aryl, and optionally substituted heteroaryl, or

(96) R.sup.11 and R.sup.12 together with the atom to which they are connected optionally form a 3-8 membered carbocyclyl or heterocyclyl ring.

(97) In one embodiment, X.sup.1 is selected from CR' and N, wherein R' is selected from hydrogen, F, Cl, CH.sub.3, CF.sub.3, and cyclopropyl.

(98) In another embodiment, X.sup.2, X.sup.3, and X.sup.4 are independently selected from C and N.

(99) In another embodiment, X is selected from a bond, CH.sub.2, CH.sub.2CH.sub.2, CO, CH.sub.2CO, CONH, CONCH₃, CH.sub.2O, CH.sub.2NH, and CH.sub.2NCH₃.

(100) In another embodiment, R.sup.1 and R.sup.2, at each occurrence, are independently selected from hydrogen, F, Cl, OH, optionally substituted C.sub.1-C.sub.4 alkyl, optionally substituted C.sub.1-C.sub.4 heteroalkyl, optionally substituted C.sub.1-C.sub.4 alkoxy, optionally substituted C.sub.1-C.sub.4 alkylamino, optionally substituted C.sub.1-C.sub.4 haloalkyl, optionally substituted C.sub.3-C.sub.6 carbocyclyl, optionally substituted C.sub.3-C.sub.6 cycloalkoxy, and optionally substituted 3-6 membered heterocyclyl.

(101) In another embodiment, X is CH.sub.2; and Ar.sup.1 is 3-fluorophenyl.

(102) In another embodiment, R.sup.3 is connected to the linker moiety of the bivalent compound directly, and R.sup.3 is selected from null, a bond, —OR.sup.5—, —SR.sup.5—, —NR.sup.6R.sup.7—, —COR.sup.5—, —CO.sub.2R.sup.5—, —CONR.sup.6R.sup.7—, —SOR.sup.5—, —SO.sub.2R.sup.5—SO.sub.2NR.sup.6R.sup.7—, —NR.sup.5COR.sup.7—, —NR.sup.5COR.sup.7—, —NR.sup.5C(O)NR.sup.6R.sup.7—, —NR.sup.5SOR.sup.7—, —NR.sup.5SO.sub.2R.sup.7—, optionally substituted C.sub.1-C.sub.8 alkylene, optionally substituted C.sub.1-C.sub.5 heteroalkylene, optionally substituted C.sub.1-C.sub.8alkylene-O—C.sub.1-C.sub.5alkylene, optionally substituted C.sub.1-C.sub.8 haloalkylene, optionally substituted C.sub.1-C.sub.8 hydroxyalkylene, optionally substituted C.sub.1-C.sub.5alkylene-N(C.sub.1-C.sub.8 alkyl)-C.sub.1-C.sub.5alkylene, optionally substituted C.sub.3-C.sub.8 carbocyclylene, optionally substituted 3-8 membered carbocyclyl, optionally substituted 3-8 membered heterocyclyl, optionally substituted C.sub.2-C.sub.8 alkenylene, optionally substituted C.sub.2-C.sub.8 alkynylene, optionally substituted aryl, and optionally substituted heteroaryl, wherein

(103) R.sup.5, R.sup.6 and R.sup.7 are independently selected from null, a bond, hydrogen, optionally substituted C.sub.1-C.sub.8 alkyl, optionally substituted C.sub.1-C.sub.8 heteroalkyl, optionally substituted C.sub.2-C.sub.8 alkenyl, optionally substituted C.sub.2-C.sub.8 alkynyl, optionally substituted C.sub.3-C.sub.8 carbocyclyl, optionally substituted 3-8 membered heterocyclyl, optionally substituted heterocarbocyclyl, optionally substituted arylalkyl, optionally substituted heteroarylalkyl, optionally substituted C.sub.1-C.sub.8 alkylene, optionally substituted C.sub.1-C.sub.5 heteroalkylene, optionally substituted C.sub.1-C.sub.8alkylene-O—, optionally substituted C.sub.1-C.sub.8 alkylene-N(C.sub.1-C.sub.6 alkyl)-, optionally substituted C.sub.1-C.sub.6 alkylene-O-alkylene, optionally substituted C.sub.1-C.sub.5alkylene-N(C.sub.1-C.sub.6 alkyl)-C.sub.1-C.sub.8alkylene, optionally substituted C.sub.3-C.sub.8 carbocyclyl, optionally substituted C.sub.3-C.sub.8 carbocyclyl-O—, optionally substituted 3-8 membered carbocyclyl, optionally substituted 3-8 membered heterocyclyl, optionally substituted aryl, and optionally substituted heteroaryl, or

(104) R.sup.6 and R.sup.7 together with the atom to which they are connected optionally form a 3-

8 membered carbocyclyl or heterocyclyl ring.

(105) In another embodiment, R.sup.3 is connected to the linker moiety of the bivalent compound through R.sup.4, and

(106) R.sup.3 and R.sup.4 are independently selected from null, a bond, —OR.sup.5—, —SR.sup.5—, —NR.sup.6R.sup.7—, —COR.sup.5—, —CO.sup.3R.sup.5—, —CONR.sup.6R.sup.7—, —SOR.sup.5—, —SO.sub.2R.sup.5—, —SO.sub.2NR.sup.6R.sup.7—, —NR.sup.5COR.sup.7—, —NR.sup.5C(O)NR.sup.6R.sup.7—, —NR.sup.5SOR.sup.7—, —NR.sup.5SO.sub.2R.sup.7—, optionally substituted C.sub.1-C.sub.8alkylene, optionally substituted C.sub.1-C.sub.5 heteroalkylene, optionally substituted C.sub.1-C.sub.8alkylene-O—C.sub.1-C.sub.8alkylene, optionally substituted C.sub.1-C.sub.5 haloalkylene, optionally substituted C.sub.1-C.sub.8 hydroxyalkylene, optionally substituted C.sub.1-C.sub.5alkylene-N(C.sub.1-C.sub.5alkyl)-C.sub.1-C.sub.5alkylene, optionally substituted C.sub.5-C.sub.8carbocyclylene, optionally substituted 3-8 membered carbocyclyl, optionally substituted 3-8 membered heterocyclyl, optionally substituted C.sub.2-C.sub.8 alkenylene, optionally substituted C.sub.2-C.sub.8 alkynylene, optionally substituted aryl, and optionally substituted heteroaryl, wherein

(107) R.sup.5, R.sup.6 and R.sup.7 are independently selected from null, a bond, hydrogen, optionally substituted C.sub.1-C.sub.8 alkyl, optionally substituted C.sub.1-C.sub.8 heteroalkyl, optionally substituted C.sub.2-C.sub.8 alkenyl, optionally substituted C.sub.2-C.sub.8 alkynyl, optionally substituted C.sub.3-C.sub.8 carbocyclyl, optionally substituted 3-8 membered heterocyclyl, optionally substituted heterocarbocyclyl, optionally substituted arylalkyl, optionally substituted heteroarylalkyl, optionally substituted C.sub.1-C.sub.8 alkylene, optionally substituted C.sub.1-C.sub.8 heteroalkylene, optionally substituted C.sub.1-C.sub.8 alkylene-O—, optionally substituted C.sub.1-C.sub.8 alkylene-N(C.sub.1-C.sub.6 alkyl)-, optionally substituted C.sub.1-C.sub.8 alkylene-O-alkylene, optionally substituted C.sub.1-C.sub.8alkylene-N(C.sub.1-C.sub.5 alkyl)-C.sub.1-C.sub.5alkylene, optionally substituted C.sub.3-C.sub.8 carbocyclylene, optionally substituted C.sub.3-C.sub.8 carbocyclylene-O—, optionally substituted 3-8 membered carbocyclyl, optionally substituted 3-8 membered heterocyclyl, optionally substituted aryl, and optionally substituted heteroaryl, or R.sup.6 and R.sup.7 together with the atom to which they are connected optionally form a 3-8 membered carbocyclyl or heterocyclyl ring.

(108) In another embodiment, Ar.sup.1 is selected from C.sub.1-C.sub.10 aryl and C.sub.5-C.sub.10 heteroaryl, each of which is optionally substituted with one or more substituents independently selected from F, Cl, CN, NO.sub.2, OR.sup.10, NR.sup.11R.sup.12, COR.sup.10, CO.sub.2R.sup.10, CONR.sup.11R.sup.12, SOR.sup.10, SO.sub.2R.sup.10, SO.sub.2NR.sup.11R.sup.12, NR.sup.10OR.sup.12, NR.sup.10C(O)NR.sup.11R.sup.12, NR.sup.10SOR.sup.12, NR.sup.10SO.sub.2R.sup.12, optionally substituted C.sub.1-C.sub.6 alkyl, optionally substituted C.sub.1-C.sub.6 heteroalkyl, optionally substituted C.sub.1-C.sub.6 alkoxyalkyl, optionally substituted C.sub.1-C.sub.8 haloalkyl, optionally substituted C.sub.1-C.sub.8 hydroxyalkyl, optionally substituted C.sub.1-C.sub.8alkylaminoC.sub.1-C.sub.8alkyl, optionally substituted C.sub.3-C.sub.7 carbocyclyl, optionally substituted 3-7 membered heterocyclyl, optionally substituted C.sub.2-C.sub.6 alkenyl, optionally substituted C.sub.2-C.sub.6 alkynyl, optionally substituted aryl, and optionally substituted C.sub.4-C.sub.5 heteroaryl, wherein R.sup.10, R.sup.11, and R.sup.12, at each occurrence, are independently selected from null, hydrogen, optionally substituted C.sub.1-C.sub.6 alkyl, optionally substituted C.sub.1-C.sub.6 heteroalkyl, optionally substituted C.sub.2-C.sub.6 alkenyl, optionally substituted C.sub.2-C.sub.6 alkynyl, optionally substituted C.sub.3-C.sub.7 carbocyclyl, optionally substituted 3-7 membered heterocyclyl, optionally substituted aryl, and optionally substituted heteroaryl, or R.sup.11 and R.sup.12 together with the atom to which they are connected optionally form a 3-8 membered carbocyclyl or heterocyclyl ring.

(109) In another embodiment, Ar.sup.2 is selected from C.sub.6-C.sub.10 aryl and C.sub.5-C.sub.10 heteroaryl, each of which is optionally substituted with one or more substituents

independently selected from F, Cl, CN, NO.sub.2, OR.sup.13, NR.sup.14R.sup.15, COR.sup.13, CO.sub.2R.sup.13, CONR.sup.14R.sup.15, SOR.sup.13, SO.sub.2R.sup.13, SO.sub.2NR.sup.14R.sup.15, NR.sup.13COR.sup.14, NR.sup.13C(O)NR.sup.14R.sup.15, NR.sup.13SOR.sup.14, NR.sup.13SO.sub.2R.sup.14, optionally substituted C.sub.1-C.sub.6 alkyl, optionally substituted C.sub.1-C.sub.6 heteroalkyl, optionally substituted C.sub.1-C.sub.6 alkoxyalkyl, optionally substituted C.sub.1-C.sub.6 haloalkyl, optionally substituted C.sub.1-C.sub.6 hydroxyalkyl, optionally substituted C.sub.1-C.sub.5alkylaminoC.sub.1-C.sub.5alkyl, optionally substituted C.sub.3-C.sub.7 carbocyclyl, optionally substituted 3-7 membered heterocyclyl, optionally substituted C.sub.2-C.sub.6 alkenyl, optionally substituted C.sub.2-C.sub.6 alkynyl, optionally substituted aryl, and optionally substituted C.sub.4-C.sub.5 heteroaryl, wherein R.sup.13, R.sup.14, and R.sup.15, at each occurrence, are independently selected from null, hydrogen, optionally substituted C.sub.1-C.sub.6 alkyl, optionally substituted C.sub.1-C.sub.6 heteroalkyl, optionally substituted C.sub.2-C.sub.6 alkenyl, optionally substituted C.sub.2-C.sub.6 alkynyl, optionally substituted C.sub.3-C.sub.7 carbocyclyl, optionally substituted 3-7 membered heterocyclyl, optionally substituted aryl, and optionally substituted heteroaryl, or R.sup.14 and R.sup.15 together with the atom to which they are connected optionally form a 3-8 membered carbocyclyl or heterocyclyl ring.

(110) In another embodiment, R.sup.3—Ar.sup.2 is selected from a moiety of formulae B1 and B2;

(111) ##STR00029## wherein * indicates the connection to the linker moiety of the bivalent compound; Y.sup.1, Y.sup.2, Y.sup.3, and Y.sup.4 are independently selected from CH and N, with the proviso that up to 3 of Y.sup.1, Y.sup.2, Y.sup.3, and Y.sup.4 are N; each R.sup.a is independently selected from hydrogen, halogen, CN, NO.sub.2, OR.sup.13, NR.sup.14R.sup.15, COR.sup.13, CO.sub.2R.sup.13, CONR.sup.14R.sup.15, SOR.sup.13, SO.sub.2R.sup.13, SO.sub.2NR.sup.14R.sup.15, NR.sup.13COR.sup.14, NR.sup.13C(O)NR.sup.14R.sup.15, NR.sup.13SOR.sup.14, NR.sup.13SO.sub.2R.sup.14, optionally substituted C.sub.1-C.sub.6 alkyl, optionally substituted C.sub.1-C.sub.5 heteroalkyl, optionally substituted C.sub.1-C.sub.8alkoxy, optionally substituted C.sub.1-C.sub.8alkoxyC.sub.1-C.sub.8alkyl, optionally substituted C.sub.1-C.sub.8alkylaminoC.sub.1-C.sub.8alkyl, optionally substituted C.sub.3-C.sub.8 carbocyclyl, optionally substituted C.sub.3-C.sub.8 cycloalkoxy, optionally substituted 3-8 membered heterocyclyl, optionally substituted C.sub.2-C.sub.8 alkenyl, optionally substituted C.sub.2-C.sub.8 alkynyl, optionally substituted aryl, and optionally substituted heteroaryl, wherein R.sup.13, R.sup.14, and R.sup.15, at each occurrence, are independently selected from hydrogen, optionally substituted C.sub.1-C.sub.8alkyl, optionally substituted C.sub.1-C.sub.5 heteroalkyl, optionally substituted C.sub.1-C.sub.8 alkoxy, optionally substituted C.sub.1-C.sub.8alkoxyC.sub.1-C.sub.6alkyl, optionally substituted C.sub.1-C.sub.5alkylaminoC.sub.1-C.sub.8alkyl, optionally substituted C.sub.3-C.sub.8 carbocyclyl, optionally substituted C.sub.3-C.sub.8 cycloalkoxy, optionally substituted 3-8 membered heterocyclyl, optionally substituted C.sub.2-C.sub.8 alkenyl, optionally substituted C.sub.2-C.sub.8 alkynyl, optionally substituted aryl, and optionally substituted heteroaryl, or R.sup.14 and R.sup.15 together with the atom to which they are connected optionally form a 3-8 membered carbocyclyl or heterocyclyl ring; m is 0 to 4; and R.sup.3 is the same as defined in Formula 2.

(112) In another embodiment, R.sup.3—Ar.sup.2 is selected from a moiety of formula B3;

(113) ##STR00030## wherein * indicates the connection to the linker moiety of the bivalent compound; Y.sup.1, Y.sup.2, Y.sup.3, and Y.sup.4 are independently selected from CR.sup.a, N, O, and S, with the proviso that up to 3 of Y.sup.1, Y.sup.2, Y.sup.3, and Y.sup.4 are N; each R.sup.a is independently selected from hydrogen, halogen, CN, NO.sub.2, OR.sup.13, NR.sup.14R.sup.15, COR.sup.13, CO.sub.2R.sup.13, CONR.sup.14R.sup.15, SOR.sup.13, SO.sub.2R.sup.13, SO.sub.2NR.sup.14R.sup.15, NR.sup.13COR.sup.14, NR.sup.13C(O)NR.sup.14R.sup.15, NR.sup.10SOR.sup.14, NR.sup.13SO.sub.2R.sup.14, optionally substituted C.sub.1-C.sub.6 alkyl, optionally substituted C.sub.1-C.sub.5 heteroalkyl, optionally substituted C.sub.1-C.sub.8 alkoxy,

optionally substituted C.sub.1-C.sub.8alkoxyC.sub.1-C.sub.8alkyl, optionally substituted C.sub.1-C.sub.8alkylaminoC.sub.1-C.sub.8alkyl, optionally substituted C.sub.3-C.sub.8 carbocyclyl, optionally substituted C.sub.3-C.sub.8 cycloalkoxy, optionally substituted 3-8 membered heterocyclyl, optionally substituted C.sub.2-C.sub.8 alkenyl, optionally substituted C.sub.2-C.sub.8 alkynyl, optionally substituted aryl, and optionally substituted heteroaryl, wherein R.sup.13, R.sup.14, and R.sup.15 are independently selected from hydrogen, optionally substituted C.sub.1-C.sub.8 alkyl, optionally substituted C.sub.1-C.sub.5 heteroalkyl, optionally substituted C.sub.1-C.sub.8 alkoxy, optionally substituted C.sub.1-C.sub.8alkoxyC.sub.1-C.sub.5alkyl, optionally substituted C.sub.1-C.sub.5alkylaminoC.sub.1-C.sub.5alkyl, optionally substituted C.sub.3-C.sub.8 carbocyclyl, optionally substituted C.sub.3-C.sub.8 cycloalkoxy, optionally substituted 3-8 membered heterocyclyl, optionally substituted C.sub.2-C.sub.8 alkenyl, optionally substituted C.sub.2-C.sub.8 alkynyl, optionally substituted aryl, and optionally substituted heteroaryl, or R.sup.14 and R.sup.15 together with the atom to which they are connected optionally form a 3-8 membered carbocyclyl or heterocyclyl ring; m is 0 to 4; and R.sup.3 is the same as defined in Formula 2.

(114) In another embodiment, X.sup.1 is N; X.sup.2 is N; X.sup.3 is C; X.sup.4 is C; and X is CH.sub.2.

(115) In another embodiment, Ar.sup.1 is 3-fluorophenyl.

(116) In another embodiment, Ar.sup.2 is 2-pyridyl

(117) ##STR00031##

(118) In another embodiment, R.sup.3 is selected from optionally substituted

(119) In another embodiment, the TRK ligand comprises a moiety of FORMULA 3;

(120) ##STR00032##

wherein X.sup.1, X.sup.2, X.sup.3, and X.sup.4 are independently selected from C, CR', and N (preferably, X.sup.1 and X.sup.4 are independently selected from CR' and N; X.sup.2 and X.sup.3 are independently selected from C and N), wherein R' is selected from hydrogen, halogen, CN, NO.sub.2, and optionally substituted C.sub.1-C.sub.6 alkyl, C.sub.3-C.sub.6 carbocyclyl, or 3-6 membered heterocyclyl; X is selected from null, a bond, C(R.sup.2).sub.2,

C(R.sup.2).sub.2C(R.sup.2).sub.2, CO, C(R.sup.2), CO, NR.sup.2CO, OC(R.sup.2).sub.2, and NR.sup.2C(R.sup.2).sub.2; R.sup.1 and each R.sup.2 are independently selected from hydrogen, halogen, OH, NH.sub.2, CN, NO.sub.2, optionally substituted C.sub.1-C.sub.4 alkyl, optionally substituted C.sub.1-C.sub.4 heteroalkyl, optionally substituted C.sub.1-C.sub.4 alkoxy, optionally substituted C.sub.1-C.sub.4 alkylamino, optionally substituted C.sub.1-C.sub.6 alkoxyalkyl, optionally substituted C.sub.1-C.sub.4 haloalkyl, optionally substituted C.sub.1-C.sub.4 hydroxyalkyl, optionally substituted C.sub.1-C.sub.8alkylaminoC.sub.1-C.sub.8alkyl, optionally substituted C.sub.3-C.sub.6 carbocyclyl, optionally substituted C.sub.3-C.sub.8 cycloalkoxy, and optionally substituted 3-6 membered heterocyclyl; n is 1 to 4; R.sup.3 is selected from hydrogen, optionally substituted C.sub.1-C.sub.6 alkyl, optionally substituted C.sub.1-C.sub.6 heteroalkyl, optionally substituted C.sub.3-C.sub.6 carbocyclyl, optionally substituted 3-6 membered heterocyclyl, optionally substituted C.sub.1-C.sub.6 alkoxyalkyl, optionally substituted C.sub.1-C.sub.8 haloalkyl, optionally substituted C.sub.1-C.sub.6 hydroxyalkyl, and optionally substituted C.sub.1-C.sub.8alkylaminoC.sub.1-C.sub.8alkyl; R.sup.4 is connected to the linker moiety of the bivalent compound either directly or through R.sup.5,

wherein

R.sup.4 and R.sup.5 are independently selected from null, —OR.sup.6—, —SR.sup.6—, —N(R.sup.7)R.sup.6—, —COR.sup.6—, —CO.sub.2R.sup.6—, —CON(R.sup.7)R.sup.6—, —SOR.sup.6—, —SO.sub.2R.sup.6—, —SO.sub.2N(R.sup.7)R.sup.6—, —NR.sup.5COR.sup.6—, —N(R.sup.8)C(O)N(R.sup.7)R.sup.6—, —NR.sup.5SOR.sup.6—, —NR.sup.5SO.sub.2R.sup.6—, optionally substituted C.sub.1-C.sub.8alkylene, optionally substituted C.sub.1-C.sub.5alkylene-O—C.sub.1-C.sub.5alkylene, optionally substituted C.sub.1-C.sub.5 haloalkylene, optionally

substituted C.sub.1-C.sub.8 hydroxyalkylene, optionally substituted C.sub.1-C.sub.8alkylene-N(C.sub.1-C.sub.5alkyl)-C.sub.1-C.sub.5alkylene, optionally substituted C.sub.3-C.sub.8 carbocyclylene, optionally substituted 3-8 membered carbocyclyl, optionally substituted 3-8 membered heterocyclyl, optionally substituted C.sub.2-C.sub.8 alkenylene, optionally substituted C.sub.2-C.sub.8 alkynylene, optionally substituted aryl, and optionally substituted heteroaryl; R.sup.6 is selected from null, optionally substituted C.sub.1-C.sub.8 alkylene, optionally substituted C.sub.1-C.sub.8 heteroalkylene, optionally substituted C.sub.1-C.sub.8 alkylene-O—, optionally substituted C.sub.1-C.sub.8 alkylene-N(C.sub.1-C.sub.8 alkyl)-, optionally substituted C.sub.1-C.sub.8alkylene-O—C.sub.1-C.sub.8 alkylene, optionally substituted C.sub.1-C.sub.5alkylene-N(C.sub.1—C.sub.8 alkyl)-C.sub.1-C.sub.8alkylene, optionally substituted C.sub.3-C.sub.8 carbocyclyl, optionally substituted C.sub.3-C.sub.8 carbocyclyl-O—, optionally substituted 3-8 membered heterocyclyl, optionally substituted aryl, and optionally substituted heteroaryl; R.sup.7 and R.sup.8 are independently selected from null, hydrogen, optionally substituted C.sub.1-C.sub.8 alkyl, optionally substituted C.sub.1-C.sub.5 heteroalkyl, optionally substituted C.sub.2-C.sub.8 alkenyl, optionally substituted C.sub.2-C.sub.8 alkynyl, optionally substituted C.sub.3-C.sub.8 carbocyclyl, optionally substituted heterocyclyl, optionally substituted arylalkyl, optionally substituted heteroarylalkyl, optionally substituted 3-8 membered carbocyclyl, optionally substituted 3-8 membered heterocyclyl, optionally substituted aryl, and optionally substituted heteroaryl; or R.sup.6 and R.sup.7 together with the atom to which they are connected optionally form a 3-8 membered carbocyclyl or heterocyclyl ring; Ar is selected from aryl and heteroaryl, each of which is optionally substituted with one or more substituents independently selected from halogen, CN, NO.sub.2, OR.sup.10, SR.sup.10, NR.sup.11R.sup.12, COR.sup.10, CO.sub.2R.sup.10, CONR.sup.11R.sup.12, SOR.sup.10, SO.sub.2R.sup.10, SO.sub.2NR.sup.11R.sup.12, NR.sup.10COR.sup.12, NR.sup.10C(O)NR.sup.11R.sup.12, NR.sup.10SOR.sup.12, NR.sup.10SO.sub.2R.sup.12, optionally substituted C.sub.1-C.sub.6 alkyl, optionally substituted C.sub.1-C.sub.5 heteroalkyl, optionally substituted C.sub.1-C.sub.8alkoxyC.sub.1-C.sub.5alkyl, optionally substituted C.sub.1-C.sub.5 haloalkyl, optionally substituted C.sub.1-C.sub.5 hydroxyalkyl, optionally substituted C.sub.1-C.sub.5alkylaminoC.sub.1-C.sub.8alkyl, optionally substituted C.sub.3-C.sub.7 carbocyclyl, optionally substituted 3-7 membered heterocyclyl, optionally substituted C.sub.2-C.sub.8 alkenyl, optionally substituted C.sub.2-C.sub.8 alkynyl, optionally substituted aryl, and optionally substituted heteroaryl, wherein R.sup.10, R.sup.11, and R.sup.12 are independently selected from null, hydrogen, optionally substituted C.sub.1-C.sub.5 alkyl, optionally substituted C.sub.1-C.sub.8 heteroalkyl, optionally substituted C.sub.2-C.sub.8 alkenyl, optionally substituted C.sub.2-C.sub.8 alkynyl, optionally substituted C.sub.3-C.sub.7 carbocyclyl, optionally substituted 3-7 membered heterocyclyl, optionally substituted aryl, and optionally substituted heteroaryl, or R.sup.11 and R.sup.12 together with the atom to which they are connected optionally form a 3-8 membered carbocyclyl or heterocyclyl rings.

(121) In one embodiment, X.sup.1 and X.sup.4 is selected from CR' and N, and R' is selected from hydrogen, F, C.sub.1, CH.sub.3, CF.sub.3, and cyclopropyl.

(122) In one embodiment, X.sup.1 is N.

(123) In one embodiment, X.sup.4 is CH.

(124) In another embodiment, X.sup.2 and X.sup.3 are independently selected from C and N.

(125) In one embodiment, X.sup.2 is C and X.sup.3 is N.

(126) In one embodiment, X.sup.3 is C and X.sup.2 is N.

(127) In another embodiment, X is selected from a bond, CH.sub.2, CH.sub.2CH.sub.2, CO, CH.sub.2CO, CONH, CONCH.sub.3, CH.sub.2O, CH.sub.2NH, and CH.sub.2NCH.sub.3.

(128) In another embodiment, X is CH.sub.2.

(129) In another embodiment, R.sup.1 and each R.sup.2 are independently selected from hydrogen, F, Cl, OH, optionally substituted C.sub.1-C.sub.4 alkyl, optionally substituted C.sub.1-C.sub.4

heteroalkyl, optionally substituted C.sub.1-C.sub.4 heteroalkyl, optionally substituted C.sub.1-C.sub.4 alkoxy, optionally substituted C.sub.1-C.sub.4 alkylamino, optionally substituted C.sub.1-C.sub.4 haloalkyl, optionally substituted C.sub.3-C.sub.6 carbocyclyl, optionally substituted C.sub.3-C.sub.6 cycloalkoxy, and optionally substituted 3-6 membered heterocyclyl.

(130) In another embodiment, R.sup.1 and R.sup.2 are hydrogen.

(131) In another embodiment, R.sup.3 is selected from hydrogen, CH.sub.3, CH.sub.2CH.sub.3, propyl, isopropyl, cyclopropyl, CH.sub.2F, CHF.sub.2, and CF.sub.3.

(132) In another embodiment, R.sup.3 is selected from hydrogen.

(133) In another embodiment, R.sup.4 is connected to the linker moiety of the bivalent compound directly, and R.sup.4 is selected from null, —OR.sup.6—, —SR.sup.6—, —N(R.sup.7)R.sup.6—, —COR.sup.6—, —CO.sub.2R.sup.6—, —CON(R.sup.7)R.sup.6—, —SOR.sup.6—, —

SO.sub.2RE—, —SO.sub.2N(R.sup.7)R.sup.6—, —NR.sup.8COR.sup.6—, —

N(R.sup.8)C(O)N(R.sup.7)R.sup.6—, —NR.sup.8SOR.sup.6—, —NR.sup.8SO.sub.2R.sup.6—,

optionally substituted C.sub.1-C.sub.8 alkylene, optionally substituted C.sub.1-C.sub.5

heteroalkylene, optionally substituted C.sub.1-C.sub.5alkylene-O—C.sub.1-C.sub.8alkylene,

optionally substituted C.sub.1-C.sub.5 haloalkylene, optionally substituted C.sub.1-C.sub.8

hydroxyalkylene, optionally substituted C.sub.1-C.sub.5alkylene-N(C.sub.1-C.sub.5alkyl)-C.sub.1-

C.sub.8alkylene, optionally substituted C.sub.3-C.sub.8 carbocyclylene, optionally substituted 3-8

membered carbocyclyl, optionally substituted 3-8 membered heterocyclyl, optionally substituted

C.sub.2-C.sub.8 alkenylene, optionally substituted C.sub.2-C.sub.8 alkynylene, optionally

substituted aryl, and optionally substituted heteroaryl; R.sup.6 is selected from null, optionally

substituted C.sub.1-C.sub.8alkylene, optionally substituted C.sub.1-C.sub.5 heteroalkylene,

optionally substituted C.sub.1-C.sub.8 alkylene-O—, optionally substituted C.sub.1-C.sub.6

alkylene-N(C.sub.1-C.sub.8 alkyl)-, optionally substituted C.sub.1-C.sub.8 alkylene-O—C.sub.1-

C.sub.8 alkylene, optionally substituted C.sub.1-C.sub.5alkylene-N(C.sub.1-C.sub.8 alkyl)-

C.sub.1-C.sub.8alkylene, optionally substituted C.sub.3-C.sub.8 carbocyclyl, optionally substituted

C.sub.3-C.sub.8 carbocyclyl-O—, optionally substituted 3-8 membered heterocyclyl, optionally

substituted aryl, and optionally substituted heteroaryl; R.sup.7 and R.sup.8 are independently

selected from null, hydrogen, optionally substituted C.sub.1-C.sub.8 alkyl, optionally substituted

C.sub.1-C.sub.5 heteroalkyl, optionally substituted C.sub.2-C.sub.8 alkenyl, optionally substituted

C.sub.2-C.sub.8 alkynyl, optionally substituted C.sub.3-C.sub.8 carbocyclyl, optionally substituted

heterocyclyl, optionally substituted arylalkyl, optionally substituted heteroarylalkyl, optionally

substituted 3-8 membered carbocyclyl, optionally substituted 3-8 membered heterocyclyl,

optionally substituted aryl, and optionally substituted heteroaryl; or R.sup.6 and R.sup.7 together

with the atom to which they are connected optionally form a 3-8 membered carbocyclyl or

heterocyclyl ring; In another embodiment, R.sup.4 is connected to the linker moiety of the bivalent

compound through R.sup.5, and R.sup.4 and R.sup.5 are independently selected from null, —

OR.sup.6—, —SR.sup.6—, —N(R.sup.7)R.sup.6—, —COR.sup.6—, —CO.sub.2R.sup.6—, —

CON(R.sup.7)R.sup.6—, —SOR.sup.6—, —SO.sub.2R.sup.6—, —SO.sub.2N(R.sup.7)R.sup.6—,

—NR.sup.5COR.sup.6—, —N(R.sup.8)C(O)N(R.sup.7)R.sup.6—, —NR.sup.8SOR.sup.6—, —

NR.sup.5SO.sub.2R.sup.6—, optionally substituted C.sub.1-C.sub.8alkylene, optionally substituted

C.sub.1—C heteroalkylene, optionally substituted C.sub.1-C.sub.8alkylene-O—C.sub.1-

C.sub.8alkylene, optionally substituted C.sub.1-C.sub.5 haloalkylene, optionally substituted

C.sub.1-C.sub.6 hydroxyalkylene, optionally substituted C.sub.1-C.sub.5alkylene-N(C.sub.1-

C.sub.8alkyl)-C.sub.1-C.sub.5alkylene, optionally substituted C.sub.3-C.sub.8 carbocyclylene,

optionally substituted 3-8 membered carbocyclyl, optionally substituted 3-8 membered

heterocyclyl, optionally substituted C.sub.2-C.sub.8 alkenylene, optionally substituted C.sub.2-

C.sub.8 alkynylene, optionally substituted aryl, and optionally substituted heteroaryl; R.sup.6 is

selected from null, optionally substituted C.sub.1-C.sub.6 alkylene, optionally substituted C.sub.1-

C.sub.8 heteroalkylene, optionally substituted C.sub.1-C.sub.8 alkylene-O—, optionally substituted

C.sub.1-C.sub.6 alkylene-N(C.sub.1-C.sub.8 alkyl)-, optionally substituted C.sub.1-C.sub.8 alkylene-O—C.sub.1-C.sub.8 alkylene, optionally substituted C.sub.1-C.sub.8 alkylene-N(C.sub.1-C.sub.8 alkyl)-C.sub.1-C.sub.5 alkylene, optionally substituted C.sub.3-C.sub.8 carbocyclyl, optionally substituted C.sub.3-C.sub.8 carbocyclyl-O—, optionally substituted 3-8 membered heterocyclyl, optionally substituted aryl, and optionally substituted heteroaryl; R^{sup.7} and R^{sup.8} are independently selected from null, hydrogen, optionally substituted C.sub.1-C.sub.8 alkyl,, optionally substituted C.sub.1-C.sub.5 heteroalkyl, optionally substituted C.sub.2-C.sub.8 alkenyl, optionally substituted C.sub.2-C.sub.8 alkynyl, optionally substituted C.sub.3-C.sub.8 carbocyclyl, optionally substituted heterocyclyl, optionally substituted arylalkyl, optionally substituted heteroarylalkyl, optionally substituted 3-8 membered carbocyclyl, optionally substituted 3-8 membered heterocyclyl, optionally substituted aryl, and optionally substituted heteroaryl; or R^{sup.6} and R^{sup.7} together with the atom to which they are connected optionally form a 3-8 membered carbocyclyl or heterocyclyl ring.

(134) In another embodiment, R^{sup.4} and R^{sup.5} are independently selected from null, optionally substituted

(135) ##STR00033##

(136) In another embodiment, R^{sup.4} and R^{sup.5} are independently selected from null, optionally substituted

(137) ##STR00034##

(138) In another embodiment, —R^{sup.4}—R^{sup.5}-is selected from null, optionally substituted

(139) ##STR00035##

(140) In another embodiment, —R^{sup.4}—R^{sup.5}-is optionally substituted

(141) ##STR00036##

(142) In another embodiment, Ar is selected from aryl and heteroaryl, each of which is optionally substituted with one or more substituents independently selected from F, Cl, CN, NO₂, OR^{sup.10}, NR^{sup.11}R^{sup.12}, COR^{sup.10}, CO₂R^{sup.10}, CONR^{sup.11}R^{sup.12}, SOR^{sup.10}, SO₂R^{sup.10}, SO₂NR^{sup.11}R^{sup.12}, NR^{sup.10}OR^{sup.12}, NR^{sup.10}C(O)NR^{sup.11}R^{sup.12}, NR^{sup.10}SOR^{sup.12}, NR^{sup.10}SO₂R^{sup.12}, optionally substituted C.sub.1-C.sub.6 alkyl, optionally substituted C.sub.1-C.sub.6 heteroalkyl, optionally substituted C.sub.1-C.sub.6 alkoxyalkyl, optionally substituted C.sub.1-C.sub.8 haloalkyl, optionally substituted C.sub.1-C.sub.6 hydroxyalkyl, optionally substituted C.sub.1-C.sub.5 alkylaminoC.sub.1-C.sub.8 alkyl, optionally substituted C.sub.3-C.sub.7 carbocyclyl, optionally substituted 3-7 membered heterocyclyl, optionally substituted C.sub.2-C.sub.6 alkenyl, optionally substituted C.sub.2-C.sub.6 alkynyl, optionally substituted aryl, and optionally substituted C.sub.4-C.sub.5 heteroaryl, wherein R^{sup.10}, R^{sup.11}, and R^{sup.12} are independently selected from null, hydrogen, optionally substituted C.sub.1-C.sub.6 alkyl, optionally substituted C.sub.1-C.sub.6 heteroalkyl, optionally substituted C.sub.2-C.sub.8 alkenyl, optionally substituted C.sub.2-C.sub.6 alkynyl, optionally substituted C.sub.3-C.sub.7 carbocyclyl, optionally substituted 3-7 membered heterocyclyl, optionally substituted aryl, and optionally substituted heteroaryl, or R^{sup.11} and R^{sup.12} together with the atom to which they are connected optionally form a 3-8 membered carbocyclyl or heterocyclyl ring.

(143) In another embodiment, Ar is aryl, which is optionally substituted with one or more substituents independently selected from F, Cl, Br, CN, and NO₂.

(144) In another embodiment, Ar is

(145) ##STR00037##

(146) In another embodiment, the TRK ligand comprises a moiety of FORMULA 7:

(147) ##STR00038##

wherein X^{sup.1} is selected from CR' and N; R' is selected from hydrogen, halogen, CN, NO₂, and optionally substituted C.sub.1-C.sub.6 alkyl, optionally substituted C.sub.1-C.sub.6 heteroalkyl, C.sub.3-C.sub.6 carbocyclyl, or 3-6 membered heterocyclyl; X^{sup.2} and X^{sup.3} are

selected from either C or N, with the proviso that only one of X^{sup.2} and X^{sup.3} is N; X is selected from null, a bond, C(R^{sup.2}).sub.2, C(R^{sup.2}).sub.2C(R^{sup.2}).sub.2, CO, C(R^{sup.2}).sub.2CO, NR^{sup.2}CO, OC(R^{sup.2}).sub.2, and NR^{sup.2}C(R^{sup.2}).sub.2; R^{sup.1} and each R^{sup.2} are independently selected from hydrogen, halogen, OH, NH₂, CN, NO₂, optionally substituted C₁-C₄ alkyl, optionally substituted C₁-C₄ heteroalkyl, optionally substituted C₁-C₄ alkoxy, optionally substituted C₁-C₄ alkylamino, optionally substituted C₁-C₄ alkoxyalkyl, optionally substituted C₁-C₄ haloalkyl, optionally substituted C₁-C₄ hydroxyalkyl, optionally substituted C₁-C₄alkylaminoC₁-C₄alkyl, optionally substituted C₃-C₆ carbocyclyl, optionally substituted C₃-C₆ cycloalkoxy, and optionally substituted 3-6 membered heterocyclyl; n is 1 to 4; R^{sup.3} is selected from hydrogen, optionally substituted C₁-C₆ alkyl, optionally substituted C₁-C₆ heteroalkyl, optionally substituted C₃-C₆ carbocyclyl, optionally substituted 3-6 membered heterocyclyl, optionally substituted C₁-C₆ alkoxyalkyl, optionally substituted C₁-C₆ haloalkyl, optionally substituted C₁-C₆ hydroxyalkyl, and optionally substituted C₁-C₈alkylaminoC₁-C₆alkyl; R^{sup.4} is connected to the linker moiety of the bivalent compound either directly or through R^{sup.5}; R^{sup.4} and R^{sup.5} are independently selected from null, —OR^{sup.6}—, —SR^{sup.6}—, —N(R^{sup.7})R^{sup.6}—, —COR^{sup.6}—, —CO.sub.2R^{sup.6}—, —CON(R^{sup.7})R^{sup.6}—, —SOR^{sup.6}—, —SO.sub.2R^{sup.6}—, —SO.sub.2N(R^{sup.7})R^{sup.6}—, —NR^{sup.8}COR^{sup.6}—, —N(R^{sup.8})C(O)N(R^{sup.7})R^{sup.6}—, —NR^{sup.8}SOR^{sup.6}—, —NR^{sup.8}SO.sub.2R^{sup.6}—, optionally substituted C₁-C₈ alkylene, optionally substituted C₁-C₅ heteroalkylene, optionally substituted C₁-C₅alkylene-O—C₁-C₈alkylene, optionally substituted C₁-C₅ haloalkylene, optionally substituted C₁-C₈ hydroxyalkylene, optionally substituted C₁-C₈alkylene-N(C₁-C₈alkyl)-C₁-C₅alkylene, optionally substituted C₃-C₈ carbocyclylene, optionally substituted 3-8 membered carbocyclyl, optionally substituted 3-8 membered heterocyclyl, optionally substituted C₂-C₈ alkenylene, optionally substituted C₂-C₈ alkynylene, optionally substituted aryl, and optionally substituted heteroaryl; R^{sup.6} is selected from null, optionally substituted C₁-C₆ alkylene, optionally substituted C₁-C₈ heteroalkylene, optionally substituted C₁-C₈ alkylene-O—, optionally substituted C₁-C₆ alkylene-N(C₁-C₈ alkyl)-, optionally substituted C₁-C₈ alkylene-O—C₁-C₈ alkylene, optionally substituted C₁-C₈alkylene-N(C₁-C₈ alkyl)-C₁-C₈alkylene, optionally substituted C₃-C₈ carbocyclyl, optionally substituted C₃-C₈ carbocyclyl-O—, optionally substituted 3-8 membered heterocyclyl, optionally substituted aryl, and optionally substituted heteroaryl; R^{sup.7} and R^{sup.8} are independently selected from null, hydrogen, optionally substituted C₁-C₈ alkyl, optionally substituted C₁-C₅ heteroalkyl, optionally substituted C₂-C₈ alkenyl, optionally substituted C₂-C₈ alkynyl, optionally substituted C₃-C₈ carbocyclyl, optionally substituted heterocyclyl, optionally substituted arylalkyl, optionally substituted heteroarylalkyl, optionally substituted 3-8 membered carbocyclyl, optionally substituted 3-8 membered heterocyclyl, optionally substituted aryl, and optionally substituted heteroaryl; or R^{sup.6} and R^{sup.7} together with the atom to which they are connected optionally form a 3-8 membered carbocyclyl or heterocyclyl ring; Ar is selected from aryl and heteroaryl, each of which is optionally substituted with one or more substituents independently selected from halogen, CN, NO₂, OR^{sup.10}, SR^{sup.10}, NR^{sup.11}R^{sup.12}, COR^{sup.10}, CO.sub.2R^{sup.10}, CONR^{sup.11}R^{sup.12}, SOR^{sup.10}, SO.sub.2R^{sup.10}, SO.sub.2NR^{sup.11}R^{sup.12}, NR^{sup.10}COR^{sup.12}, NR^{sup.10}C(O)NR^{sup.11}R^{sup.12}, NR^{sup.10}SOR^{sup.12}, NR^{sup.10}SO.sub.2R^{sup.12}, optionally substituted C₁-C₆ alkyl, optionally substituted C₁-C₅ heteroalkyl, optionally substituted C₁-C₈alkoxyC₁-C₈alkyl, optionally substituted C₁-C₅ haloalkyl, optionally

substituted C.sub.1-C.sub.8 hydroxyalkyl, optionally substituted C.sub.1-C.sub.5alkylaminoC.sub.1-C.sub.5alkyl, optionally substituted C.sub.3-C.sub.7 carbocyclyl, optionally substituted 3-7 membered heterocyclyl, optionally substituted C.sub.2-C.sub.8 alkenyl, optionally substituted C.sub.2-C.sub.8 alkynyl, optionally substituted aryl, and optionally substituted heteroaryl; R.sup.10, R.sup.11, and R.sup.12 are independently selected from null, hydrogen, optionally substituted C.sub.1-C.sub.8alkyl, optionally substituted C.sub.1-C.sub.5 heteroalkyl, optionally substituted C.sub.2-C.sub.8 alkenyl, optionally substituted C.sub.2-C.sub.8 alkynyl, optionally substituted C.sub.3-C.sub.7 carbocyclyl, optionally substituted 3-7 membered heterocyclyl, optionally substituted aryl, and optionally substituted heteroaryl; or R.sup.11 and R.sup.12 together with the atom to which they are connected optionally form a 3-8 membered carbocyclyl or heterocyclyl rings.

(148) In one embodiment, X.sup.1 and X.sup.3 are selected from CR' and N, and R' is selected from hydrogen, F, C.sub.1, CH.sub.3, CF.sub.3, and cyclopropyl.

(149) In one embodiment, X.sup.1 is N.

(150) In one embodiment, X.sup.1 is CR' and R' is selected from hydrogen, F, C.sub.1, CH.sub.3, CF.sub.3, and cyclopropyl. In one embodiment, X.sup.2 is C and X.sup.3 is N.

(151) In one embodiment, X.sup.3 is C and X.sup.2 is N.

(152) In another embodiment, X is selected from a bond, CH.sub.2, CH.sub.2CH.sub.2, CO, CH.sub.2CO, CONH, CONCH.sub.3,

(153) CH.sub.2O, CH.sub.2NH, and CH.sub.2NCH.sub.3.

(154) In another embodiment, X is CH.sub.2.

(155) In another embodiment, R.sup.1 and each R.sup.2 are independently selected from hydrogen, F, Cl, OH, optionally substituted C.sub.1-C.sub.4 alkyl, optionally substituted C.sub.1-C.sub.4 heteroalkyl, optionally substituted C.sub.1-C.sub.4 alkoxy, optionally substituted C.sub.1-C.sub.4 alkylamino, optionally substituted C.sub.1-C.sub.6 haloalkyl, optionally substituted C.sub.3-C.sub.6 carbocyclyl, optionally substituted C.sub.3-C.sub.6 cycloalkoxy, and optionally substituted 3-6 membered heterocyclyl.

(156) In another embodiment, R.sup.1 and R.sup.2 are hydrogen.

(157) In another embodiment, R.sup.3 is selected from hydrogen, CH.sub.3, CH.sub.2CH.sub.3, propyl, isopropyl, cyclopropyl, CH.sub.2F, CHF.sub.2, and CF.sub.3.

(158) In another embodiment, R.sup.3 is selected from hydrogen.

(159) In another embodiment, R.sup.4 is connected to the linker moiety of the bivalent compound directly, and R.sup.4 is selected from null, —OR.sup.6—, —SR.sup.6—, —N(R.sup.7)R.sup.6—, —COR.sup.6—, —CO.sub.2R.sup.6—, —CON(R.sup.7)R.sup.6—, —SOR.sup.6—, —SO.sub.2RE—, —SO.sub.2N(R.sup.7)R.sup.6—, —NR.sup.8COR.sup.6—, —N(R.sup.8)C(O)N(R.sup.7)R.sup.6—, —NR.sup.8SOR.sup.6—, —NR.sup.8SO.sub.2R.sup.6—, optionally substituted C.sub.1-C.sub.8 alkylene, optionally substituted C.sub.1-C.sub.5 heteroalkylene, optionally substituted C.sub.1-C.sub.5alkylene-O—C.sub.1-C.sub.8alkylene, optionally substituted C.sub.1-C.sub.5 haloalkylene, optionally substituted C.sub.1-C.sub.6 hydroxyalkylene, optionally substituted C.sub.1-C.sub.5alkylene-N(C.sub.1-C.sub.5alkyl)-C.sub.1-C.sub.6alkylene, optionally substituted C.sub.3-C.sub.8 carbocyclylene, optionally substituted 3-8 membered carbocyclyl, optionally substituted 3-8 membered heterocyclyl, optionally substituted C.sub.2-C.sub.8 alkenylene, optionally substituted C.sub.2-C.sub.8 alkynylene, optionally substituted aryl, and optionally substituted heteroaryl; R.sup.6 is selected from null, optionally substituted C.sub.1-C.sub.8alkylene, optionally substituted C.sub.1-C.sub.5 heteroalkylene, optionally substituted C.sub.1-C.sub.8 alkylene-O—, optionally substituted C.sub.1-C.sub.6 alkylene-N(C.sub.1-C.sub.8 alkyl)-, optionally substituted C.sub.1-C.sub.8alkylene-O—C.sub.1-C.sub.8 alkylene, optionally substituted C.sub.1-C.sub.5alkylene-N(C.sub.1-C.sub.8 alkyl)-C.sub.1-C.sub.8alkylene, optionally substituted C.sub.3-C.sub.8 carbocyclyl, optionally substituted C.sub.3-C.sub.8 carbocyclyl-O—, optionally substituted 3-8 membered heterocyclyl, optionally

substituted aryl, and optionally substituted heteroaryl; and R.sup.7 and R.sup.8 are independently selected from null, hydrogen, optionally substituted C.sub.1-C.sub.6 alkyl, optionally substituted C.sub.1-C.sub.5 heteroalkyl, optionally substituted C.sub.2-C.sub.8 alkenyl, optionally substituted C.sub.2-C.sub.8 alkynyl, optionally substituted C.sub.3-C.sub.8 carbocyclyl, optionally substituted heterocyclyl, optionally substituted arylalkyl, optionally substituted heteroarylalkyl, optionally substituted 3-8 membered carbocyclyl, optionally substituted 3-8 membered heterocyclyl, optionally substituted aryl, and optionally substituted heteroaryl; or R.sup.6 and R.sup.7 together with the atom to which they are connected optionally form a 3-8 membered carbocyclyl or heterocyclyl ring.

(160) In another embodiment, R.sup.4 is connected to the linker moiety of the bivalent compound through R.sup.5, and R.sup.4 and R.sup.5 are independently selected from null, —OR.sup.6—, —SR.sup.6—, —N(R.sup.7)R.sup.6—, —COR.sup.6—, —CO.sub.2R.sup.6—, —CON(R.sup.7)R.sup.6—, —SOR.sup.6—, —SO.sub.2R.sup.6—, —SO.sub.2N(R.sup.7)R.sup.6—, —NR.sup.8COR.sup.6—, —N(R.sup.8)C(O)N(R.sup.7)R.sup.6—, —NR.sup.8SOR.sup.6—, —NR.sup.8SO.sub.2R.sup.6—, optionally substituted C.sub.1-C.sub.8 alkylene, optionally substituted C.sub.1-C.sub.5 heteroalkylene, optionally substituted C.sub.1-C.sub.5alkylene-O—C.sub.1-C.sub.5alkylene, optionally substituted C.sub.1-C.sub.5 haloalkylene, optionally substituted C.sub.1-C.sub.5 hydroxyalkylene, optionally substituted C.sub.1-C.sub.5alkylene-N(C.sub.1-C.sub.8alkyl)-C.sub.1-C.sub.8alkylene, optionally substituted C.sub.3-C.sub.8 carbocyclylene, optionally substituted 3-8 membered carbocyclyl, optionally substituted 3-8 membered heterocyclyl, optionally substituted C.sub.2-C.sub.8 alkenylene, optionally substituted C.sub.2-C.sub.8 alkynylene, optionally substituted aryl, and optionally substituted heteroaryl; R.sup.6 is selected from null, optionally substituted C.sub.1-C.sub.6 alkylene, optionally substituted C.sub.1-C.sub.8 heteroalkylene, optionally substituted C.sub.1-C.sub.8 alkylene-O—, optionally substituted C.sub.1-C.sub.6 alkylene-N(C.sub.1-C.sub.8 alkyl)-, optionally substituted C.sub.1-C.sub.8alkylene-O—C.sub.1-C.sub.8alkylene, optionally substituted C.sub.1-C.sub.5alkylene-N(C.sub.1-C.sub.8alkyl)-C.sub.1-C.sub.5alkylene, optionally substituted C.sub.5-C.sub.8carbocyclyl, optionally substituted C.sub.3-C.sub.8 carbocyclyl-O—, optionally substituted 3-8 membered heterocyclyl, optionally substituted aryl, and optionally substituted heteroaryl; R.sup.7 and R.sup.8 are independently selected from null, hydrogen, optionally substituted C.sub.1-C.sub.8 alkyl, optionally substituted C.sub.1-C.sub.5 heteroalkyl, optionally substituted C.sub.2-C.sub.8 alkenyl, optionally substituted C.sub.2-C.sub.8 alkynyl, optionally substituted C.sub.3-C.sub.8 carbocyclyl, optionally substituted heterocyclyl, optionally substituted arylalkyl, optionally substituted heteroarylalkyl, optionally substituted 3-8 membered carbocyclyl, optionally substituted 3-8 membered heterocyclyl, optionally substituted aryl, and optionally substituted heteroaryl; or R.sup.6 and R.sup.7 together with the atom to which they are connected optionally form a 3-8 membered carbocyclyl or 3-8 membered heterocyclyl ring;

(161) ##STR00039##

(162) In another embodiment, R.sup.4 and R.sup.5 are independently selected from null, optionally substituted

(163) In another embodiment, R.sup.4 and R.sup.5 are independently selected from null, optionally substituted

(164) ##STR00040##

(165) In another embodiment, —R.sup.4—R.sup.5—is selected from null, optionally substituted

(166) ##STR00041##

(167) In another embodiment, —R.sup.4—R.sup.3—is optionally substituted

(168) ##STR00042##

(169) In another embodiment, Ar is selected from aryl and heteroaryl, each of which is optionally substituted with one or more substituents independently selected from F, Cl, CN, NO.sub.2, OR.sup.10, NR.sup.11R.sup.12, COR.sup.10, CO.sub.2R.sup.10, CONR.sup.11R.sup.12,

SOR.sup.10, SO.sub.2R.sup.10, SO.sub.2NR.sup.11R.sup.12, NR.sup.10COR.sup.12, NR.sup.10C(O)NR.sup.11R.sup.12, NR.sup.10SOR.sup.12, NR.sup.10SO.sub.2R.sup.12, optionally substituted C.sub.1-C.sub.6 alkyl, optionally substituted C.sub.1-C.sub.6 heteroalkyl, optionally substituted C.sub.1-C.sub.6 alkoxyalkyl, optionally substituted C.sub.1-C.sub.6 haloalkyl, optionally substituted C.sub.1-C.sub.6 hydroxyalkyl, optionally substituted C.sub.1-C.sub.6alkylaminoC.sub.1-C.sub.6alkyl, optionally substituted C.sub.3-C.sub.7 carbocyclyl, optionally substituted 3-7 membered heterocyclyl, optionally substituted C.sub.2-C.sub.6 alkenyl, optionally substituted C.sub.2-C.sub.6 alkynyl, optionally substituted aryl, and optionally substituted C.sub.4-C.sub.5 heteroaryl, wherein R.sup.10, R.sup.11, and R.sup.12 are independently selected from null, hydrogen, optionally substituted C.sub.1-C.sub.6 alkyl, optionally substituted C.sub.1-C.sub.6 heteroalkyl, optionally substituted C.sub.2-C.sub.6 alkenyl, optionally substituted C.sub.2-C.sub.6 alkynyl, optionally substituted C.sub.3-C.sub.7 carbocyclyl, optionally substituted 3-7 membered heterocyclyl, optionally substituted aryl, and optionally substituted heteroaryl, or R.sup.11 and R.sup.12 together with the atom to which they are connected optionally form a 3-8 membered carbocyclyl or heterocyclyl ring.

(170) In another embodiment, Ar is aryl, which is optionally substituted with one or more substituents independently selected from F, Cl, Br, CN, and NO.sub.2.

(171) In another embodiment, Ar is

(172) ##STR00043##

(173) In another embodiment, the TRK ligand comprises a moiety of FORMULAE 12-1 or 12-2;

(174) ##STR00044##

wherein X is selected from CR.sup.5, and N; Y is selected from O, S, and NR.sup.6; R.sup.1, R.sup.3 and R.sup.4 are independently selected from hydrogen, halogen, CN, NO.sub.2, OR.sup.7, SR.sup.7, NR.sup.8R.sup.9, COR.sup.7, CO.sub.2R.sup.7, CONR.sup.8R.sup.9, SOR.sup.7, SO.sub.2R.sup.7, SO.sub.2NR.sup.8R.sup.2, NR.sup.10COR.sup.9, NR.sup.10C(O)NR.sup.8R.sup.9, NR.sup.11SOR.sup.9, NR.sup.10SO.sub.2R.sup.9, optionally substituted C.sub.1-C.sub.8 alkyl, optionally substituted C.sub.1-C.sub.5 heteroalkyl, optionally substituted C.sub.1-C.sub.8alkoxyC.sub.1-C.sub.8alkyl, optionally substituted C.sub.1-C.sub.5 haloalkyl, optionally substituted C.sub.1-C.sub.8 hydroxyalkyl, optionally substituted C.sub.1-C.sub.5alkylaminoC.sub.1-C.sub.8alkyl, optionally substituted C.sub.3-C.sub.8 carbocyclyl, optionally substituted 3-8 membered carbocyclyl, optionally substituted 3-8 membered heterocyclyl, optionally substituted 3-8 membered carbocyclyl-C.sub.1-C.sub.5alkyl, optionally substituted 3-8 membered heterocyclyl-C.sub.1-C.sub.8alkyl, optionally substituted 3-8 membered carbocyclyl-O—, optionally substituted 3-8 membered heterocyclyl-O, optionally substituted 3-8 membered carbocyclyl-N(C.sub.1-C.sub.8alkyl)-, and optionally substituted 3-8 membered heterocyclyl-N(C.sub.1-C.sub.8alkyl)-, optionally substituted C.sub.2-C.sub.8 alkenyl, optionally substituted C.sub.2-C.sub.8 alkynyl, optionally substituted aryl, and optionally substituted heteroaryl; R.sup.7, R.sup.8, R.sup.9, and R.sup.10 are independently selected from null, hydrogen, optionally substituted C.sub.1-C.sub.5 alkyl, optionally substituted C.sub.1-C.sub.5 heteroalkyl, optionally substituted C.sub.2-C.sub.8 alkenyl, optionally substituted C.sub.2-C.sub.8 alkynyl, optionally substituted C.sub.3-C.sub.8 carbocyclyl, optionally substituted 3-8 membered carbocyclyl, optionally substituted heterocarbocyclyl, optionally substituted arylalkyl, optionally substituted heteroarylalkyl, optionally substituted 3-8 membered carbocyclyl, optionally substituted 3-8 membered heterocyclyl, optionally substituted 3-8 membered carbocyclyl-C.sub.1-C.sub.5alkyl, optionally substituted 3-8 membered heterocyclyl-C.sub.1-C.sub.8alkyl, optionally substituted 3-8 membered carbocyclyl-O—, optionally substituted 3-8 membered heterocyclyl-O, optionally substituted 3-8 membered carbocyclyl-N(C.sub.1-C.sub.8alkyl)-, and optionally substituted 3-8 membered heterocyclyl-N(C.sub.1-C.sub.5alkyl)-, optionally substituted aryl, and optionally substituted heteroaryl; or R.sup.8 and R.sup.9 together with the atom to which they are connected optionally form a 3-8 membered carbocyclyl or heterocyclyl ring; R.sup.1', R.sup.2, and

R.sup.4' are independently selected from null, ~OR.sup.11-, —SR.sup.11—, —NR.sup.12R.sup.11, —COR.sup.11—, CO.sub.2R.sup.11—, —CON(R.sup.12)R.sup.11, —SOR.sup.11, —SO.sub.2R.sup.11—, —SON(R.sup.12)R.sup.11—, —NR.sup.10OR.sup.11—, —NR.sup.13C(O)N(R.sup.12)R.sup.11, —NR.sup.13SOR.sup.11—, —NR.sup.13SO.sub.2R.sup.11—, optionally substituted C.sub.1-C.sub.8alkylene, optionally substituted C.sub.1-C.sub.8 heteroalkylene, optionally substituted C.sub.1-C.sub.8alkylene-O—C.sub.1-C.sub.5alkylene, optionally substituted C.sub.1-C.sub.8 haloalkylene, optionally substituted C.sub.1-C.sub.5 hydroxyalkylene, optionally substituted C.sub.1-C.sub.8alkylene-N(C.sub.1-C.sub.8alkyl)-C.sub.1-C.sub.5alkylene, optionally substituted C.sub.3-C.sub.8 carbocyclylene, optionally substituted 3-8 membered carbocyclyl, optionally substituted 3-8 membered heterocyclyl, optionally substituted 3-8 membered carbocyclyl-C.sub.1-C.sub.8alkylene, optionally substituted 3-8 membered heterocyclyl-C.sub.1-C.sub.5alkylene, optionally substituted 3-8 membered carbocyclyl-O—, optionally substituted 3-8 membered heterocyclyl-O, optionally substituted 3-8 membered carbocyclyl-N(C.sub.1-C.sub.5alkyl)-, and optionally substituted 3-8 membered heterocyclyl-N(C.sub.1-C.sub.5alkyl)-, optionally substituted C.sub.2-C.sub.8 alkenylene, optionally substituted C.sub.2-C.sub.8 alkynylene, optionally substituted aryl, and optionally substituted heteroaryl; R.sup.11 is selected from null, optionally substituted C.sub.1-C.sub.8 alkylene, optionally substituted C.sub.1-C.sub.5 heteroalkylene, optionally substituted C.sub.1-C.sub.6 alkylene-O—, optionally substituted C.sub.1-C.sub.8 alkylene-N(C.sub.1-C.sub.8alkyl)-, optionally substituted C.sub.1-C.sub.8 alkylene-O-alkylene, optionally substituted C.sub.1-C.sub.8alkylene-N(C.sub.1-C.sub.8 alkyl)-C.sub.1-C.sub.8alkylene, optionally substituted C.sub.3-C.sub.8 carbocyclylene, optionally substituted C.sub.3-C.sub.8 carbocyclylene-O—, optionally substituted 3-8 membered carbocyclyl-C.sub.1-C.sub.8alkylene, optionally substituted 3-8 membered heterocyclyl-C.sub.1-C.sub.8alkylene, optionally substituted 3-8 membered carbocyclyl, optionally substituted 3-8 membered heterocyclyl, optionally substituted aryl, and optionally substituted heteroaryl; R.sup.12 and R.sup.13 are independently selected from null, hydrogen, optionally substituted C.sub.1-C.sub.8 alkyl, optionally substituted C.sub.1-C.sub.5 heteroalkylene, optionally substituted C.sub.2-C.sub.8 alkenyl, optionally substituted C.sub.2-C.sub.8 alkynyl, optionally substituted C.sub.3-C.sub.8 carbocyclyl, optionally substituted heterocyclyl, optionally substituted arylalkyl, optionally substituted heteroarylalkyl, optionally substituted 3-8 membered carbocyclyl-C.sub.1-C.sub.5alkyl, optionally substituted 3-8 membered heterocyclyl-C.sub.1-C.sub.8alkyl, optionally substituted 3-8 membered carbocyclyl, optionally substituted 3-8 membered heterocyclyl, optionally substituted aryl, and optionally substituted heteroaryl; or R.sup.11 and R.sup.12 together with the atom to which they are connected optionally form a 3-8 membered carbocyclyl or heterocyclyl ring; R.sup.5 and R.sup.6, at each occurrence, are independently selected from hydrogen, halogen, CN, NO.sub.2, optionally substituted C.sub.1-C.sub.8 alkyl, optionally substituted C.sub.1-C.sub.5 heteroalkyl, optionally substituted C.sub.1-C.sub.5 alkoxy, optionally substituted C.sub.1-C.sub.8 alkylamino, optionally substituted C.sub.3-C.sub.8 carbocyclyl, optionally substituted C.sub.3-C.sub.8 carbocyclyl, and optionally substituted C.sub.3-C.sub.8 heterocyclyl; and n is selected from 0, 1, and 2.

(175) In another embodiment, X is selected from CH, CF, and N.

(176) In another embodiment, X is CH.

(177) In another embodiment, X is CF.

(178) In another embodiment, X is N.

(179) In another embodiment, Y is selected from O and S.

(180) In another embodiment, Y is O.

(181) In another embodiment, Y is S.

(182) In another embodiment, R.sup.1 is selected from optionally substituted acyclic amino, optionally substituted cyclic amino, optionally substituted phenyl, and optionally substituted heteroaryl.

(183) In another embodiment, R.sup.1 is selected from optionally substituted

(184) ##STR00045##

(185) In another embodiment, R.sup.1' is selected from null, optionally substituted acyclic amino, optionally substituted cyclic amino, optionally substituted phenyl, and optionally substituted heteroaryl.

(186) In another embodiment, R.sup.1' is selected from optionally substituted

(187) ##STR00046##

(188) In another embodiment, R.sup.2 is selected from null, —O—, —S—, —N(R.sup.12)—, —C(O)—, —CO.sub.2—, —CON(R.sup.12)—, —SO—, —SO.sub.2—, —SO.sub.2N(R.sup.12)—, —N(R.sup.13)CO—, —N(R.sup.13)C(O)N(R.sup.12)—, —N(R.sup.13)SO—, —N(R.sup.13)SO.sub.2—, optionally substituted optionally substituted C.sub.3-C.sub.8 carbocyclyl, and optionally substituted C.sub.3-C.sub.8 heterocyclyl, wherein R.sup.12, and R.sup.13 are independently selected from hydrogen, optionally substituted C.sub.1-C.sub.8 alkyl, optionally substituted C.sub.1-C.sub.5 heteroalkyl, optionally substituted C.sub.2-C.sub.8 alkenyl, optionally substituted C.sub.2-C.sub.8 alkynyl, optionally substituted C.sub.3-C.sub.8 carbocyclyl, optionally substituted heterocarbocyclyl, optionally substituted arylalkyl, optionally substituted heteroarylalkyl, optionally substituted 3-8 membered carbocyclyl, optionally substituted 3-8 membered heterocyclyl, optionally substituted aryl, and optionally substituted heteroaryl.

(189) In another embodiment, R.sup.2 is selected from null, —CONH—,

(190) ##STR00047##

(191) In another embodiment, R.sup.3 is selected from hydrogen, halogen, CN, NO.sub.2, OH, NH.sub.2, —CONH—,

(192) ##STR00048##

(193) In another embodiment, R.sup.2—R.sup.3 is selected from hydrogen, halogen, CN, NO.sub.2,

(194) ##STR00049##

(195) In another embodiment, R.sup.4 is selected from hydrogen, halogen, CN, NO.sub.2, optionally substituted C.sub.1-C.sub.8 alkyl, optionally substituted C.sub.1-C.sub.6 heteroalkyl, optionally substituted C.sub.1-C.sub.8alkoxyC.sub.1-C.sub.8alkyl, optionally substituted C.sub.1-C.sub.5 haloalkyl, optionally substituted C.sub.1-C.sub.5 hydroxyalkyl, optionally substituted C.sub.1-C.sub.5alkylaminoC.sub.1-C.sub.8alkyl, optionally substituted C.sub.3-C.sub.8 carbocyclyl, optionally substituted 3-8 membered carbocyclyl-C.sub.1-C.sub.5alkyl, optionally substituted 3-8 membered heterocyclyl-C.sub.1-C.sub.5alkyl, optionally substituted 3-8 membered carbocyclyl, and optionally substituted 3-8 membered heterocyclyl, optionally substituted 3-8 membered carbocyclyl-C.sub.1-C.sub.8alkyl, and optionally substituted 3-8 membered heterocyclyl-C.sub.1-C.sub.8alkyl, optionally substituted 3-8 membered carbocyclyl-O—, optionally substituted 3-8 membered heterocyclyl-O—, optionally substituted 3-8 membered carbocyclyl-N(C.sub.1-C.sub.8alkyl)-, and optionally substituted 3-8 membered heterocyclyl-N(C.sub.1-C.sub.8alkyl)-.

(196) In another embodiment, R.sup.4 is selected from hydrogen, halogen, CN, NO.sub.2, OCH.sub.3,

(197) ##STR00050##

(198) In another embodiment, R.sup.4' is selected from null, —O—, —S—, —N(R.sup.12')—, —C(O)—, —CO.sub.2—, —CON(R.sup.12')—, —SO—, —SO.sub.2—, —SO.sub.2N(R.sup.12')—, —N(R.sup.12')CO—, —N(R.sup.13')C(O)N(R.sup.12')—, —N(R.sup.13')SO—, —N(R.sup.13')SO.sub.2—, optionally substituted C.sub.1-C.sub.8alkylene, optionally substituted C.sub.1-C.sub.5 heteroalkylene, optionally substituted C.sub.1-C.sub.8alkylene-O—C.sub.1-C.sub.5alkylene, optionally substituted C.sub.1-C.sub.8 haloalkylene, optionally substituted C.sub.1-C.sub.8 hydroxyalkylene, optionally substituted C.sub.1-C.sub.5alkylene-N(C.sub.1-C.sub.5alkyl)-C.sub.1-C.sub.8alkylene, optionally substituted C.sub.3-C.sub.8 carbocyclene,

optionally substituted 3-8 membered carbocyclyl, optionally substituted 3-8 membered heterocyclyl, optionally substituted 3-8 membered carbocyclyl-C.sub.1-C.sub.5alkylene, and optionally substituted 3-8 membered heterocyclyl-C.sub.1-C.sub.8alkylene, optionally substituted 3-8 membered carbocyclyl-O—, optionally substituted 3-8 membered heterocyclyl-O—, optionally substituted 3-8 membered carbocyclyl-N(C.sub.1-C.sub.8alkyl)-, and optionally substituted 3-8 membered heterocyclyl-N(C.sub.1-C.sub.5alkyl)-, wherein R.sup.12', and R.sup.13' are independently selected from hydrogen, optionally substituted C.sub.1-C.sub.8 alkyl, optionally substituted C.sub.1-C.sub.5 heteroalkyl, optionally substituted C.sub.2-C.sub.8 alkenyl, optionally substituted C.sub.2-C.sub.8 alkynyl, optionally substituted C.sub.3-C.sub.8 carbocyclyl, optionally substituted heterocarbocyclyl, optionally substituted arylalkyl, optionally substituted heteroarylalkyl, optionally substituted 3-8 membered carbocyclyl, optionally substituted 3-8 membered heterocyclyl, optionally substituted aryl, and optionally substituted heteroaryl.

(199) In another embodiment, R.sup.4' is selected from null, —O—, optionally substituted C.sub.1-C.sub.8 alkylene, optionally substituted C.sub.1-C.sub.8 heteroalkylene, optionally substituted C.sub.1-C.sub.6alkylene-O—C.sub.1-C.sub.5alkylene, optionally substituted C.sub.1-C.sub.5alkylene-N(C.sub.1-C.sub.8alkyl)-C.sub.1-C.sub.5alkylene, optionally substituted 3-8 membered carbocyclyl, optionally substituted 3-8 membered heterocyclyl, optionally substituted 3-8 membered carbocyclyl-C.sub.1-C.sub.5alkylene, and optionally substituted 3-8 membered heterocyclyl-C.sub.1-C.sub.5alkylene, optionally substituted 3-8 membered carbocyclyl-O—, optionally substituted 3-8 membered heterocyclyl-O—, optionally substituted 3-8 membered carbocyclyl-N(C.sub.1-C.sub.8alkyl)-, and optionally substituted 3-8 membered heterocyclyl-N(C.sub.1-C.sub.8alkyl)-.

(200) In another embodiment, R.sup.4' is selected null, —O—,

(201) ##STR00051##

(202) In another embodiment, the TRK ligand comprises a moiety of FORMULA 13;

(203) ##STR00052##

wherein X is selected from CR.sup.9, and N; R.sup.9 is selected from from hydrogen, halogen, CN, NO.sub.2, optionally substituted C.sub.1-C.sub.8alkyl, optionally substituted C.sub.1-C.sub.5 heteroalkyl, optionally substituted C.sub.1-C.sub.8 alkoxy, optionally substituted C.sub.1-C.sub.8 alkylamino, optionally substituted C.sub.3-C.sub.8 carbocyclyl, optionally substituted C.sub.3-C.sub.8 carbocyclyl, and optionally substituted C.sub.3-C.sub.8 heterocyclyl; Y is selected from null, —O—, —N(optionally substituted C.sub.1-C.sub.6 alkyl)-, and optionally substituted C.sub.1-C.sub.8 alkylene; R.sup.1 and R.sup.2 are independently selected from hydrogen, halogen, CN, NO.sub.2, OH, NH.sub.2, optionally substituted C.sub.1-C.sub.8alkyl, optionally substituted C.sub.1-C.sub.5 heteroalkyl, optionally substituted C.sub.1-C.sub.8 alkoxy, optionally substituted C.sub.1-C.sub.8alkylamino, optionally substituted C.sub.3-C.sub.8 carbocyclyl, optionally substituted C.sub.3-C.sub.8 carbocyclyl, and optionally substituted C.sub.3-C.sub.8 heterocyclyl; R.sup.3, R.sup.4, R.sup.5, and R.sup.6 are independently selected from hydrogen, halogen, optionally substituted C.sub.1-C.sub.8 alkyl, optionally substituted C.sub.1-C.sub.5 heteroalkyl, optionally substituted C.sub.5-C.sub.8 carbocyclyl, optionally substituted C.sub.3-C.sub.8 carbocyclyl, and optionally substituted C.sub.3-C.sub.8 heterocyclyl; or R.sup.3 and R.sup.4; and/or R.sup.5 and R.sup.6, together with the atom to which they are connected optionally form a 3-8 membered carbocyclyl or heterocyclyl ring; m is selected from 1, 2, 3, and 4; n is selected from 0, 1, 2, 3, and 4; R.sup.7 and R.sup.8 are independently selected from null, —OR.sup.10—, —SR.sup.10—, —NR.sup.11R.sup.10—, —COR.sup.10—, —CO.sub.2R.sup.10—, —CONR.sup.11R.sup.10—, —SOR.sup.10—, —SO.sub.2R.sup.10—, —SO.sub.2NR.sup.11R.sup.10—, —NR.sup.12COR.sup.10—, —NR.sup.12C(O)NR.sup.11R.sup.10—, —NR.sup.12SOR.sup.10—NR.sup.12SO.sub.2R.sup.10—, optionally substituted C.sub.1-C.sub.8 alkylene, optionally substituted C.sub.1-C.sub.5 heteroalkylene, optionally substituted C.sub.1-C.sub.8alkylene-O—C.sub.1-C.sub.6alkylene, optionally substituted C.sub.1-C.sub.5

haloalkylene, optionally substituted C.sub.1-C.sub.5 hydroxyalkylene, optionally substituted C.sub.1-C.sub.5alkylene-N(C.sub.1-C.sub.5alkyl)-C.sub.1-C.sub.8alkylene, optionally substituted C.sub.3-C.sub.8 carbocyclylene, optionally substituted 3-8 membered carbocyclyl, optionally substituted 3-8 membered heterocyclyl, optionally substituted 3-8 membered carbocyclyl-C.sub.1-C.sub.8alkylene, optionally substituted 3-8 membered heterocyclyl-C.sub.1-C.sub.5alkylene, optionally substituted 3-8 membered carbocyclyl-O—, optionally substituted 3-8 membered heterocyclyl-O, optionally substituted 3-8 membered carbocyclyl-N(C.sub.1-C.sub.8alkyl)-, and optionally substituted 3-8 membered heterocyclyl-N(C.sub.1-C.sub.5alkyl)-, optionally substituted C.sub.2-C.sub.8 alkenylene, optionally substituted C.sub.4-C.sub.13 fused carbocyclyl, optionally substituted 5-13 membered fused heterocyclyl, optionally substituted C.sub.5-C.sub.13 bridged carbocyclyl, optionally substituted 5-13 membered bridged heterocyclyl, optionally substituted C.sub.5-C.sub.13 spiro carbocyclyl, optionally substituted 5-13 membered spiro heterocyclyl, optionally substituted C.sub.2-C.sub.8 alkynylene, optionally substituted aryl, and optionally substituted heteroaryl; R^{sup.10} is selected from null, optionally substituted C.sub.1-C.sub.8 alkylene, optionally substituted C.sub.1-C.sub.5 heteroalkylene, optionally substituted C.sub.1-C.sub.8 alkylene-O—, optionally substituted C.sub.1-C.sub.8 alkylene-N(C.sub.1-C.sub.8 alkyl)-, optionally substituted C.sub.1-C.sub.6 alkylene-O-alkylene, optionally substituted C.sub.1-C.sub.5alkylene-N(C.sub.1-C.sub.8 alkyl)-C.sub.1-C.sub.5alkylene, optionally substituted C.sub.3-C.sub.8 carbocyclylene, optionally substituted C.sub.3-C.sub.8 carbocyclylene-O—, optionally substituted 3-8 membered carbocyclyl-C.sub.1-C.sub.5alkylene, optionally substituted 3-8 membered heterocyclyl-C.sub.1-C.sub.5alkylene, optionally substituted 3-8 membered carbocyclyl, optionally substituted 3-8 membered heterocyclyl, optionally substituted C.sub.4-C.sub.13 fused carbocyclyl, optionally substituted 5-13 membered fused heterocyclyl, optionally substituted C.sub.5-C.sub.13 bridged carbocyclyl, optionally substituted 5-13 membered bridged heterocyclyl, optionally substituted C.sub.5-C.sub.13 spiro carbocyclyl, optionally substituted 5-13 membered spiro heterocyclyl, optionally substituted aryl, and optionally substituted heteroaryl; R^{sup.11} and R^{sup.12} are independently selected from null, hydrogen, optionally substituted C.sub.1-C.sub.6 alkyl, optionally substituted C.sub.2-C.sub.8 alkenyl, optionally substituted C.sub.2-C.sub.8 alkynyl, optionally substituted C.sub.3-C.sub.8 carbocyclyl, optionally substituted heterocarbocyclyl, optionally substituted arylalkyl, optionally substituted heteroarylalkyl, optionally substituted 3-8 membered carbocyclyl-C.sub.1-C.sub.6alkyl, optionally substituted 3-8 membered heterocyclyl-C.sub.1-C.sub.8alkyl, optionally substituted C.sub.3-C.sub.8 carbocyclyl-O—, optionally substituted 3-8 membered carbocyclyl, optionally substituted 3-8 membered heterocyclyl, optionally substituted C.sub.4-C.sub.13 fused carbocyclyl, optionally substituted 5-13 membered fused heterocyclyl, optionally substituted C.sub.5-C.sub.13 bridged carbocyclyl, optionally substituted 5-13 membered bridged heterocyclyl, optionally substituted C.sub.5-C.sub.13 spiro carbocyclyl, optionally substituted 5-13 membered spiro heterocyclyl, optionally substituted aryl, and optionally substituted heteroaryl; or R^{sup.10} and R^{sup.11} together with the atom to which they are connected optionally form a 3-8 membered carbocyclyl or heterocyclyl ring; and Ar is selected from aryl and heteroaryl, each of which is optionally substituted with one or more substituents independently selected from halogen, CN, NO₂, OR^{sup.14}, SR^{sup.14}, NR^{sup.15}R^{sup.16}, COR^{sup.14}, CO₂R^{sup.14}, CONR^{sup.15}R^{sup.16}, SOR^{sup.14}, SO₂R^{sup.14}, SO₂NR^{sup.15}R^{sup.16}, NR^{sup.14}COR^{sup.16}, NR^{sup.14}C(O)NR^{sup.15}R^{sup.16}, NR^{sup.14}SOR^{sup.16}, NR^{sup.14}SO₂R^{sup.16}, optionally substituted C.sub.1-C.sub.8 alkyl, optionally substituted C.sub.1-C.sub.5 heteroalkyl, optionally substituted C-C.sub.8alkoxyC.sub.1-C.sub.5alkyl, optionally substituted C.sub.1-C.sub.5 haloalkyl, optionally substituted C.sub.1-C.sub.6 hydroxyalkyl, optionally substituted C.sub.1-C.sub.5alkylaminoC.sub.1-C.sub.5alkyl, optionally substituted C.sub.3-C.sub.7 carbocyclyl, optionally substituted 3-7 membered heterocyclyl, optionally substituted C.sub.2-C.sub.8 alkenyl, optionally substituted C.sub.2-C.sub.8 alkynyl, optionally substituted aryl, and

optionally substituted heteroaryl; R.sup.14, R.sup.15, and R.sup.16 are independently selected from null, hydrogen, optionally substituted C.sub.1-C.sub.8 alkyl, optionally substituted C.sub.1-C.sub.5 heteroalkyl, optionally substituted C.sub.2-C.sub.8 alkenyl, optionally substituted C.sub.2-C.sub.8 alkynyl, optionally substituted C.sub.3-C.sub.7 carbocyclyl, optionally substituted 3-7 membered heterocyclyl, optionally substituted aryl, and optionally substituted heteroaryl; or R.sup.15 and R.sup.16 together with the atom to which they are connected optionally form a 3-8 membered carbocyclyl or heterocyclyl rings.

(204) In another embodiment, the TRK ligand comprises a moiety of FORMULA 13-1;

(205) ##STR00053##

wherein X, Y, R.sup.1, R.sup.2, R.sup.3, R.sup.4, R.sup.5, R.sup.6, R.sup.7, R.sup.8, Ar, m, and n are defined in FORMULA 13.

(206) In another embodiment, the TRK ligand comprises a moiety of FORMULA 13-2;

(207) ##STR00054##

wherein X, Y, R.sup.1, R.sup.2, R.sup.3, R.sup.4, R.sup.5, R.sup.6, R.sup.7, R.sup.8, and Ar are defined in FORMULA 13.

(208) In another embodiment, X is selected from CH, CF and N.

(209) In another embodiment, X is N.

(210) In another embodiment, Y is selected from—O—, —CH.sub.2—and-NH—.

(211) In another embodiment, Y is —O—.

(212) In another embodiment, R.sup.1 is selected from hydrogen and NH.sub.2.

(213) In another embodiment, R.sup.1 is NH.sub.2.

(214) In another embodiment, R.sup.2 is selected from hydrogen, halogen, CN, NO.sub.2, optionally substituted C.sub.1-C.sub.8 alkyl, optionally substituted C.sub.1-C.sub.5 heteroalkyl, optionally substituted C.sub.1-C.sub.8 carbocyclyl, and optionally substituted C.sub.1-C.sub.6 alkoxy.

(215) In another embodiment, R.sup.2 is selected from H, F, Cl, Br, OCH.sub.3, OCF.sub.3, and OCHF.sub.2.

(216) In another embodiment, R.sup.2 is OCH.sub.3.

(217) In another embodiment, R.sup.3, R.sup.4, R.sup.5, and R.sup.6 are independently selected from H, F, CH.sub.3, cyclopropyl, and cyclobutyl.

(218) In another embodiment, R.sup.3 and R.sup.4; and/or R.sup.5 and R.sup.6, together with the atom to which they are connected optionally form, cyclopropyl, cyclobutyl, cyclopentyl, and cyclohexyl.

(219) In another embodiment, R.sup.3, R.sup.4, R.sup.5, and R.sup.6 are H.

(220) In another embodiment, R.sup.7 is selected from null, optionally substituted 3-8 membered carbocyclyl, optionally substituted 3-8 membered heterocyclyl, optionally substituted C.sub.4-C.sub.13 fused carbocyclyl, optionally substituted 5-13 membered fused heterocyclyl, optionally substituted C.sub.5-C.sub.13 bridged carbocyclyl, optionally substituted 5-13 membered bridged heterocyclyl, optionally substituted C.sub.5-C.sub.13 spiro carbocyclyl, optionally substituted 5-13 membered spiro heterocyclyl, optionally substituted aryl, and optionally substituted heteroaryl.

(221) In another embodiment, R.sup.7 is selected from null,

(222) ##STR00055##

(223) In another embodiment, R.sup.5 is selected from null, —C(O)—, —C(O)—NH—, optionally substituted 3-8 membered carbocyclyl, optionally substituted 3-8 membered heterocyclyl, optionally substituted C.sub.4-C.sub.13 fused carbocyclyl, optionally substituted 5-13 membered fused heterocyclyl, optionally substituted C.sub.5-C.sub.13 bridged carbocyclyl, optionally substituted 5-13 membered bridged heterocyclyl, optionally substituted C.sub.5-C.sub.13 spiro carbocyclyl, optionally substituted 5-13 membered spiro heterocyclyl, optionally substituted aryl, and optionally substituted heteroaryl.

(224) In another embodiment, R.sup.8 is selected from null, —C(O)—, —C(O)—NH—,

(225) ##STR00056##

(226) In another embodiment, R.sup.7—R.sup.8 is selected from null,

(227) ##STR00057##

(228) In another embodiment, Ar is selected from optional substituted phenyl, and optional substituted pyridinyl.

(229) In another embodiment, Ar is selected from

(230) ##STR00058##

(231) In another embodiment, the TRK ligand is derived from any of the following:

(232) ##STR00059## ##STR00060##

(233) In another embodiment, the TRK ligand is derived from the following TRK kinase inhibitors: DS-6051b, F17752, PLX7486, AZD-6918, ASP7962, VM902A, PF-06273340, and ONO-4474.

(234) In another embodiment, the TRK ligand is selected from the group consisting of:

(235) ##STR00061## ##STR00062## ##STR00063## ##STR00064##

(236) In some embodiments, the degradation tag is a moiety selected from the group consisting of FORMULAE 5A, 5B, SC, and SD:

(237) ##STR00065##

wherein V, W, and X are independently selected from CR.sup.2 and N; Y is selected from CO, CR.sup.3R.sup.4, and N=N; Z is selected from null, CO, CR.sup.5R.sup.6, NR.sup.8, O, optionally substituted C.sub.1-C.sub.10 alkylene, optionally substituted C.sub.1-C.sub.5 heteroalkylene, optionally substituted C.sub.1-C.sub.10 alkenylene, optionally substituted C.sub.1-C.sub.10 alkynylene, optionally substituted 3-10 membered carbocyclyl, optionally substituted 3-10 membered heterocyclyl, optionally substituted C.sub.4-C.sub.13 fused carbocyclyl, optionally substituted 5-13 membered fused heterocyclyl, optionally substituted C.sub.5-C.sub.13 bridged carbocyclyl, optionally substituted 5-13 membered bridged heterocyclyl, optionally substituted C.sub.5-C.sub.13 spiro carbocyclyl, optionally substituted 5-13 membered spiro heterocyclyl, optionally substituted aryl, and optionally substituted heteroaryl; preferably, Z is selected from null, CH.sub.2, CH=CH, C=C, NH, O, optionally substituted 3-7 membered carbocyclyl, and optionally substituted 3-7 membered heterocyclyl; R.sup.1 is selected from hydrogen, halogen, cyano, nitro, optionally substituted C.sub.1-C.sub.6 alkyl, optionally substituted C.sub.1-C.sub.6 heteroalkyl, optionally substituted 3-6 membered carbocyclyl, and optionally substituted 3-6 membered heterocyclyl; R.sup.2 is selected from hydrogen, halogen, cyano, nitro, optionally substituted C.sub.1-C.sub.6 alkyl, optionally substituted C.sub.1-C.sub.6 heteroalkyl, optionally substituted C.sub.1-C.sub.8 alkoxy, optionally substituted C.sub.1-C.sub.6 alkylamino, optionally substituted 3-6 membered carbocyclyl, and optionally substituted 3-6 membered heterocyclyl; R.sup.3, and R.sup.4 are independently selected from hydrogen, halogen, cyano, nitro, optionally substituted C.sub.1-C.sub.6 alkyl, optionally substituted C.sub.1-C.sub.8 heteroalkyl, optionally substituted 3-6 membered carbocyclyl, and optionally substituted 3-6 membered heterocyclyl; or R.sup.3 and R.sup.4 together with the atom to which they are connected form a 3-6 membered carbocyclyl, or 3-6 membered heterocyclyl; and R.sup.5 and R.sup.6 are independently selected from null, hydrogen, halogen, oxo, hydroxyl, amino, cyano, nitro, optionally substituted C.sub.1-C.sub.6 alkyl, optionally substituted C.sub.1-C.sub.6 heteroalkyl, optionally substituted 3-6 membered carbocyclyl, and optionally substituted 3-6 membered heterocyclyl; or R.sup.5 and R.sup.6 together with the atom to which they are connected form a 3-6 membered carbocyclyl, or 3-6 membered heterocyclyl.

(238) In some embodiments, the degradation tag is a moiety selected from the group consisting of FORMULAE SE, 5F, 5G, 5H, SI, 5J, 5K, 5L, 5M, 5N, SO, 5P, and 5Q:

(239) ##STR00066## ##STR00067##

wherein U, V, W, X and X' are independently selected from CR.sup.2 and N; Y is selected from CR.sup.3R.sup.4, NR.sup.3 and O; preferably, Y is selected from CH.sub.2, NH, NCH.sub.3 and O; Y', Y'', and Y''' are independently selected from CR.sup.3R.sup.4; Z is selected from null, CO,

CR.sup.5R.sup.6, NR.sup.5, O, optionally substituted C.sub.1-C.sub.10 alkylene, optionally substituted C.sub.1-C.sub.10 heteroalkylene, optionally substituted C.sub.1-C.sub.10 alkenylene, optionally substituted C.sub.1-C.sub.10 alkynylene, optionally substituted 3-10 membered carbocyclyl, optionally substituted 3-10 membered heterocyclyl, optionally substituted C.sub.4-C.sub.13 fused carbocyclyl, optionally substituted 5-13 membered fused heterocyclyl, optionally substituted C.sub.5-C.sub.13 bridged carbocyclyl, optionally substituted 5-13 membered bridged heterocyclyl, optionally substituted C.sub.5-C.sub.13 spiro carbocyclyl, optionally substituted 5-13 membered spiro heterocyclyl, optionally substituted aryl, and optionally substituted heteroaryl; 20 preferably, Z is selected from null, CH.sub.2, CH=CH, C=C, NH, O, optionally substituted 3-7 membered carbocyclyl, and optionally substituted 3-7 membered heterocyclyl; R.sup.1 is independently selected from hydrogen, halogen, cyano, nitro, optionally substituted C.sub.1-C.sub.6 alkyl, optionally substituted C.sub.1-C.sub.6 heteroalkyl, optionally substituted 3-6 membered carbocyclyl, and optionally substituted 3-6 membered heterocyclyl; R.sup.2 is selected from hydrogen, halogen, cyano, nitro, optionally substituted C.sub.1-C.sub.6 alkyl, optionally substituted C.sub.1-C.sub.6 heteroalkyl, optionally substituted C.sub.1-C.sub.6 alkoxy, optionally substituted C.sub.1-C.sub.6 alkylamino, optionally substituted 3-6 membered carbocyclyl, and optionally substituted 3-6 membered heterocyclyl; R.sup.3, and R.sup.4 are independently selected from hydrogen, halogen, cyano, nitro, optionally substituted C.sub.1-C.sub.6 alkyl, optionally substituted C.sub.1-C.sub.8 heteroalkyl, optionally substituted 3 to 6 membered carbocyclyl, and optionally substituted 3-6 membered heterocyclyl, or R.sup.3 and R.sup.4 together with the atom to which they are connected optionally form a 3-6 membered carbocyclyl, or 3-6 membered heterocyclyl; and R.sup.5 and R.sup.6 are independently selected from null, hydrogen, halogen, oxo, hydroxyl, amino, cyano, nitro, optionally substituted C.sub.1-C.sub.6 alkyl, optionally substituted C.sub.1-C.sub.6 heteroalkyl, optionally substituted 3 to 6 membered carbocyclyl, and optionally substituted 3-6 membered heterocyclyl; or R.sup.5 and R.sup.6 together with the atom to which they are connected form a 3-6 membered carbocyclyl, or 3-6 membered heterocyclyl.

(240) In one embodiment, the degradation tag is a moiety of FORMULA 6A:

(241) ##STR00068##

wherein R.sup.1 and R.sup.2 are independently selected from hydrogen, optionally substituted C.sub.1-C.sub.8 alkyl, optionally substituted C.sub.1-C.sub.5 heteroalkyl, optionally substituted C.sub.1-C.sub.6alkoxyC.sub.1-C.sub.8alkyl, optionally substituted C.sub.1-C.sub.6 haloalkyl, optionally substituted C.sub.1-C.sub.5 hydroxyalkyl, optionally substituted C.sub.1-C.sub.5 aminoalkyl, optionally substituted C.sub.1-C.sub.6alkylaminoC.sub.1-C.sub.8alkyl, optionally substituted C.sub.3-C.sub.7 carbocyclyl, optionally substituted 3-7 membered heterocyclyl, optionally substituted C.sub.2-C.sub.8 alkenyl, and optionally substituted C.sub.2-C.sub.8 alkynyl; and R.sup.3 is hydrogen, optionally substituted C(O) C.sub.1-C.sub.8 alkyl, optionally substituted C(O) C.sub.1-C.sub.8 alkoxyC.sub.1-C.sub.8alkyl, optionally substituted C(O) C.sub.1-C.sub.8 haloalkyl, optionally substituted C(O) C.sub.1-C.sub.5 hydroxyalkyl, optionally substituted C(O) C.sub.1-C.sub.5 aminoalkyl, optionally substituted C(O) C.sub.1-C.sub.5alkylaminoC.sub.1-C.sub.8alkyl, optionally substituted C(O) C.sub.3-C.sub.7 carbocyclyl, optionally substituted C(O) (3-7 membered heterocyclyl), optionally substituted C(O) C.sub.2-C.sub.8 alkenyl, optionally substituted C(O) C.sub.2-C.sub.8 alkynyl, optionally substituted C(O)OC.sub.1-C.sub.8alkoxyC.sub.1-C.sub.5alkyl, optionally substituted C(O)OC.sub.1-C.sub.8 haloalkyl, optionally substituted C(O)OC.sub.1-C.sub.8 hydroxyalkyl, optionally substituted C(O)OC.sub.1-C.sub.5 aminoalkyl, optionally substituted C(O)OC.sub.1-C.sub.8alkylaminoC.sub.1-C.sub.8alkyl, optionally substituted C(O)OC.sub.3-C.sub.7 carbocyclyl, optionally substituted C(O)O(3-7 membered heterocyclyl), optionally substituted C(O)OC.sub.2-C.sub.8 alkenyl, optionally substituted C(O)OC.sub.2-C.sub.8 alkynyl, optionally substituted C(O)NC.sub.1-C.sub.8alkoxyC.sub.1-C.sub.5alkyl, optionally substituted C(O)NC.sub.1-C.sub.8 haloalkyl, optionally substituted C(O)NC.sub.1-C.sub.5 hydroxyalkyl, optionally substituted C(O)NC.sub.1-

C.sub.8 aminoalkyl, optionally substituted C(O)NC.sub.1-C.sub.8alkylaminoC.sub.1-C.sub.8alkyl, optionally substituted C(O)NC.sub.3-C.sub.7 carbocyclyl, optionally substituted C(O)N(3-7 membered heterocyclyl), optionally substituted C(O)NC.sub.2-C.sub.8 alkenyl, optionally substituted C(O)NC.sub.2-C.sub.8 alkynyl, optionally substituted P(O)(OH) 2, optionally substituted P(O)(OC.sub.1-C.sub.8alkyl).sub.2, and optionally substituted P(O)(OC.sub.1-C.sub.8 aryl).sub.2.

(242) In one embodiment, the degradation tag is a moiety selected from the group consisting of FORMULAE 6B, 6C, 6D, 6E and 6F:

(243) ##STR00069##

wherein R.sup.1 and R.sup.2 are independently selected from hydrogen, halogen, OH, NH.sub.2, CN, optionally substituted C.sub.1-C.sub.8 alkyl, optionally substituted C.sub.1-C.sub.6 heteroalkyl, optionally substituted C.sub.1-C.sub.8alkoxyC.sub.1-C.sub.5alkyl, optionally substituted C.sub.1—C haloalkyl, optionally substituted C.sub.1-C.sub.5 hydroxyalkyl, optionally substituted C.sub.1-C.sub.8 aminoalkyl, optionally substituted C.sub.1-C.sub.5alkylaminoC.sub.1-C.sub.5alkyl, optionally substituted C.sub.3-C.sub.7 carbocyclyl, optionally substituted 3-7 membered heterocyclyl, optionally substituted C.sub.2-C.sub.8 alkenyl, and optionally substituted C.sub.2-C.sub.8 alkynyl; (preferably, R.sup.1 is selected from iso-propyl or tert-butyl; and R.sup.2 is selected from hydrogen or methyl); R.sup.3 is hydrogen, optionally substituted C(O) C.sub.1-C.sub.8 alkyl, optionally substituted C(O) C.sub.1-C.sub.8alkoxyC.sub.1-C.sub.8alkyl, optionally substituted C(O) C.sub.1-C.sub.8 haloalkyl, optionally substituted C(O) C.sub.1-C.sub.8 hydroxyalkyl, optionally substituted C(O) C.sub.1-C.sub.5 aminoalkyl, optionally substituted C(O) C.sub.1-C.sub.8alkylaminoC.sub.1-C.sub.8alkyl, optionally substituted C(O) C.sub.3-C.sub.7 carbocyclyl, optionally substituted C(O)(3-7 membered heterocyclyl), optionally substituted C(O) C.sub.2-C.sub.8 alkenyl, optionally substituted C(O) C.sub.2-C.sub.8 alkynyl, optionally substituted C(O)OC.sub.1-C.sub.8alkoxyC.sub.1-C.sub.8alkyl, optionally substituted C(O)OC.sub.1-C.sub.8 haloalkyl, optionally substituted C(O)OC.sub.1-C.sub.8 hydroxyalkyl, optionally substituted C(O)OC.sub.1-C.sub.8 aminoalkyl, optionally substituted C(O)OC.sub.1-C.sub.8alkylaminoC.sub.1-C.sub.8alkyl, optionally substituted C(O)OC.sub.3-C.sub.7 carbocyclyl, optionally substituted C(O)O(3-7 membered heterocyclyl), optionally substituted C(O)OC.sub.2-C.sub.8 alkenyl, optionally substituted C(O)OC.sub.2-C.sub.8 alkynyl, optionally substituted C(O)NC.sub.1-C.sub.8alkoxyC.sub.1-C.sub.8alkyl, optionally substituted C(O)NC.sub.1-C.sub.8 haloalkyl, optionally substituted C(O)NC.sub.1-C.sub.8 hydroxyalkyl, optionally substituted C(O)NC.sub.1-C.sub.8 aminoalkyl, optionally substituted C(O)NC.sub.1-C.sub.8alkylaminoC.sub.1-C.sub.8alkyl, optionally substituted C(O)NC.sub.3-C.sub.7 carbocyclyl, optionally substituted C(O)N(3-7 membered heterocyclyl), optionally substituted C(O)NC.sub.2-C.sub.8 alkenyl, optionally substituted C(O)NC.sub.2-C.sub.8 alkynyl, optionally substituted P(O)(OH) 2, optionally substituted P(O)(OC.sub.1-C.sub.6 alkyl).sub.2, and optionally substituted P(O)(OC.sub.1-C.sub.8 aryl).sub.2; and R.sup.4 and R.sup.5 are independently selected from hydrogen, COR.sup.6, CO.sub.2R.sup.6, CONR.sup.6R.sup.7, SOR.sup.6, SO.sub.2R.sup.6, SO.sub.2NR.sup.6R.sup.7, optionally substituted C.sub.1-C.sub.8 alkyl, optionally substituted C.sub.1-C.sub.5 heteroalkyl, optionally substituted C.sub.1-C.sub.8alkoxyC.sub.1-C.sub.8alkyl, optionally substituted C.sub.1-C.sub.8alkylaminoC.sub.1-C.sub.6alkyl, optionally substituted 3-8 membered carbocyclyl, optionally substituted 3-8 membered heterocyclyl, optionally substituted aryl, and optionally substituted heteroaryl, wherein R.sup.6 and R.sup.7 are independently selected from hydrogen, optionally substituted C.sub.1-C.sub.8 alkyl, optionally substituted C.sub.1-C.sub.5 heteroalkyl, optionally substituted C.sub.1-C.sub.6 alkoxy, optionally substituted C.sub.1-C.sub.8alkoxyC.sub.1-C.sub.5alkyl, optionally substituted C.sub.1-C.sub.8alkylaminoC.sub.1-C.sub.5alkyl, optionally substituted 3-8 membered carbocyclyl, optionally substituted 3-8 membered heterocyclyl, optionally substituted aryl, and optionally substituted heteroaryl, or R.sup.4 and R.sup.5; or R.sup.6 and R.sup.7 together with the atom to which they are connected

form a 3-8 membered carbocyclyl or heterocyclyl ring; Ar is selected from aryl and heteroaryl, each of which is optionally substituted with one or more substituents independently selected from F, Cl, CN, NO₂, OR^{sup.8}, NR^{sup.8R^{sup.9}}, COR^{sup.8}, CO₂R^{sup.8}, CONR^{sup.8R^{sup.9}}, SOR^{sup.8}, SO₂R^{sup.8}, SO₂NR^{sup.8R^{sup.9}}, NR^{sup.10}COR^{sup.8}, NR^{sup.10}C(O)NR^{sup.8R^{sup.9}}, NR^{sup.10}SOR^{sup.8}, NR^{sup.10}SO₂R^{sup.8}, optionally substituted C_{sub.1}-C_{sub.8} alkyl, optionally substituted C_{sub.1}-C_{sub.6} heteroalkyl, optionally substituted C_{sub.1}-C_{sub.8} alkoxyalkyl, optionally substituted C_{sub.1}-C_{sub.6} haloalkyl, optionally substituted C_{sub.1}-C_{sub.6} hydroxyalkyl, optionally substituted C_{sub.1}-C_{sub.8}alkylaminoC_{sub.1}-C_{sub.8}alkyl, optionally substituted C_{sub.3}-C_{sub.7} carbocyclyl, optionally substituted 3-7 membered heterocyclyl, optionally substituted C_{sub.2}-C_{sub.6} alkenyl, optionally substituted C_{sub.2}-C_{sub.6} alkynyl, optionally substituted aryl, and optionally substituted C_{sub.4}-C_{sub.8} heteroaryl, wherein R^{sup.8}, R^{sup.9}, and R^{sup.10} are independently selected from null, hydrogen, optionally substituted C_{sub.1}-C_{sub.6} alkyl, optionally substituted C_{sub.1}-C_{sub.6} heteroalkyl, optionally substituted C_{sub.2}-C_{sub.8} alkenyl, optionally substituted C_{sub.2}-C_{sub.6} alkynyl, optionally substituted C_{sub.3}-C_{sub.7} carbocyclyl, optionally substituted 3-7 membered heterocyclyl, optionally substituted aryl, and optionally substituted heteroaryl, or R^{sup.8} and R^{sup.9} together with the atom to which they are connected optionally form a 3-8 membered carbocyclyl or heterocyclyl ring.

(244) In another embodiment, the degradation tag is a moiety of FORMULA 7A:

(245) ##STR00070##

wherein V, W, X, and Z are independently selected from CR^{sup.4} and N. R^{sup.1}, R^{sup.2}, R^{sup.3}, and R^{sup.4} are independently selected from hydrogen, optionally substituted C_{sub.1}-C_{sub.6} alkyl, optionally substituted C_{sub.1}-C_{sub.6} heteroalkyl, optionally substituted C_{sub.1}-C_{sub.6}alkoxyC_{sub.1}-C_{sub.5}alkyl, optionally substituted C_{sub.1}-C_{sub.5} haloalkyl, optionally substituted C_{sub.1}-C_{sub.5} hydroxyalkyl, optionally substituted C_{sub.3}-C_{sub.7} carbocyclyl, optionally substituted 3-7 membered heterocyclyl, optionally substituted C_{sub.2}-C_{sub.8} alkenyl, and optionally substituted C_{sub.2}-C_{sub.8} alkynyl.

(246) In another embodiment, the degradation tag is a moiety of FORMULA 7B:

(247) ##STR00071## wherein R^{sup.1}, R^{sup.2}, and R^{sup.3} are independently selected from hydrogen, halogene, optionally substituted C_{sub.1}-C_{sub.5} alkyl, optionally substituted C_{sub.1}-C_{sub.5} heteroalkyl, optionally substituted C_{sub.1}-C_{sub.8}alkoxyC_{sub.1}-C_{sub.8}alkyl, optionally substituted C_{sub.1}-C_{sub.6} haloalkyl, optionally substituted C_{sub.1}-C_{sub.5} hydroxyalkyl, optionally substituted C_{sub.3}-C_{sub.7} carbocyclyl, optionally substituted 3-7 membered heterocyclyl, optionally substituted C_{sub.2}-C_{sub.8} alkenyl, and optionally substituted C_{sub.2}-C_{sub.8} alkynyl; R^{sup.4} and R^{sup.5} are independently selected from hydrogen, COR^{sup.6}, CO₂R^{sup.6}, CONR^{sup.6R^{sup.7}}, SOR^{sup.6}, SO₂R^{sup.6}, SO₂NR^{sup.6R^{sup.7}}, optionally substituted C_{sub.1}-C_{sub.8}alkyl, optionally substituted C_{sub.1}-C_{sub.5} heteroalkyl, optionally substituted C_{sub.1}-C_{sub.8} alkoxyC_{sub.1}-C_{sub.5}alkyl, optionally substituted C_{sub.1}-C_{sub.5}alkylaminoC_{sub.1}-C_{sub.5}alkyl, optionally substituted aryl-C_{sub.1}-C_{sub.5}alkyl, optionally substituted 3-8 membered carbocyclyl, optionally substituted 3-8 membered heterocyclyl, optionally substituted aryl, and optionally substituted heteroaryl, wherein R^{sup.6} and R^{sup.7} are independently selected from hydrogen, optionally substituted C_{sub.1}-C_{sub.8} alkyl, optionally substituted C_{sub.1}-C_{sub.5} heteroalkyl, optionally substituted C_{sub.1}-C_{sub.8}alkoxyC_{sub.1}-C_{sub.5}alkyl, optionally substituted C_{sub.1}-C_{sub.5}alkylaminoC_{sub.1}-C_{sub.5}alkyl, optionally substituted 3-8 membered carbocyclyl, optionally substituted 3-8 membered heterocyclyl, optionally substituted aryl, and optionally substituted heteroaryl, or R^{sup.6} and R^{sup.7} together with the atom to which they are connected form a 3-8 membered carbocyclyl or heterocyclyl ring.

(248) In another embodiment, the degradation tag is derived from any of the following:

(249) ##STR00072## ##STR00073## ##STR00074## ##STR00075##

(250) In another embodiment, the degradation tag is derived from any of the following: thalidomide, pomalidomide, lenalidomide, CRBN-1, CRBN-2, CRBN-3, CRBN-4, CRBN-5, CRBN-6, CRBN-7, CRBN-8, CRBN-9, CRBN-10, and CRBN-11.

(251) In another embodiment, the degradation tag is selected from the group consisting of:

(252) ##STR00076## ##STR00077## ##STR00078## ##STR00079## ##STR00080##
##STR00081## ##STR00082## ##STR00083## ##STR00084## ##STR00085## ##STR00086##
##STR00087## ##STR00088## ##STR00089## ##STR00090## ##STR00091## ##STR00092##
##STR00093## ##STR00094## ##STR00095## ##STR00096## ##STR00097## ##STR00098##
(253) ##STR00099## ##STR00100## ##STR00101## ##STR00102## ##STR00103##
##STR00104## ##STR00105## ##STR00106## ##STR00107## ##STR00108## ##STR00109##
##STR00110## ##STR00111## ##STR00112## ##STR00113## ##STR00114## ##STR00115##
##STR00116## ##STR00117## ##STR00118## ##STR00119## ##STR00120## ##STR00121##
##STR00122## ##STR00123## ##STR00124## ##STR00125## ##STR00126## ##STR00127##
##STR00128## ##STR00129## ##STR00130## ##STR00131## ##STR00132##

(254) In another embodiment, the degradation tag is selected from the group consisting of:

FORMULA 8A, 8B, 8C, 8D, 8E, 8F, 8G, 8H, 8I, 8J, 8K, 8L, 8M, 8O, 8P, 8Q, 8R, 8AQ, 8AR, 8AS, 8AT, 8AU, 8AV, 8AW, 8AX, 8AY, 8AZ, 8BA, 8BB, 8BC, 8BD, 8BE, 8BF, 8BG, 8BH, 8B1, 8BJ, 8BK, 8BL, 8BM, and 8BN, 8BO, 8BP, 8BQ, 8BR, 8BS, 8CB, 8CC, 8CD, 8CE, 8CF, 8CG, 8CH, 8CI, 8CJ, 8CK, 8CL, 8 CM, 8CN, 8CO, 8CP, 8CQ, 8CR, 8CS, 8CT, 8CU, 8CV, 8CW, 8CX, 8CY, 8CZ, 8DA, 8 DB, 8DC, 8DD, 8DE, 8DF, 8DG, 8DH, 8D.sup.1, 8DJ, 8DK, 8DL, 8DM, 8DN, 8DO, 8DP, 8DQ, 8DR, 8DS, 8DT, 8DU, 8DV, 8DW, 8DX, 8DY, 8DZ, 8EA, 8EB, 8EC, 8ED, 8EE, 8EF, 8EG, 8EH, 8EI, 8EJ, 8EK, 8EL, 8EM, 8EN, 8EO, 8EP, 8EO, 8GU, 8GV, 8GW, 8GX, 8GY, 8GZ, 8HA, 8HB, 8HC, 8HD, 8HE, 8HF, 8HG, 8HH, 8HI, 8HJ, 8HK, 8HL, 8HM, 8HN, 8HO, 8HP, 8HQ, 8 HR, 8HS, 8HT, 8HU, 8HV, 8HW, 8HX, 8HY, 8HZ, 8IA, 8IB, 8IC, 8ID, 8IE, 8IF, 8IG, 8IH, 8II, 8IJ, 8IK, 8IL, 8IM, 8 IN, 8I0, 8IP, 8IQ, 8IR, 8IS, 8IT, 8IU, 8IV, 8IW, 8IX, 8IY, 8IZ, 8JA, 8JB, 8JC, 8JD, 8JE, 8JF, 8JG, 8JH, 8JI, 8JJ, 8JK, 8JL, 8JM, 8JN, 8JO, 8JP, 8JQ, 8JR, 8JS, 8JT, 8JU, 8JV, 8JW, 8JX, 8JY, 8JZ, 8KA, 8 KB, 8KC, 8KD, 8KE, 8KF, 8 KG, 8KH, 8KI, 8KJ, 8KK, 8KL, 8 KM, 8KN, 8KO, and 8KP.

(255) In some embodiments, the linker moiety is of FORMULA 9;

(256) ##STR00133##

wherein A, W and B, at each occurrence, are independently selected from null, or bivalent moiety selected from R.sup.r—R", R'COR", R'CO.sub.2R", R'C(O)N(R.sup.1)R", R'C(S)N(R.sup.1)R", R'OR", R'SR", R'SOR", R'SO.sub.2R", R'SO.sub.2N(R.sup.1)R", R'N(R.sup.1)R", R'N(R.sup.1)COR", R'N(R.sup.1)CON(R.sup.2)R", R'N(R.sup.1) C(S)R", optionally substituted C.sub.1-C.sub.6 alkylene, optionally substituted C.sub.1-C.sub.5 heteroalkylene, optionally substituted C.sub.2-C.sub.8 alkenylene, optionally substituted C.sub.2-C.sub.8 alkynylene, optionally substituted C.sub.1-C.sub.8alkoxyC.sub.1-C.sub.8alkylene, optionally substituted C.sub.1-C.sub.5 haloalkylene, optionally substituted C.sub.1-C.sub.6 hydroxyalkylene, optionally substituted C.sub.4-C.sub.13 fused carbocyclyl, optionally substituted 5-13 membered fused heterocyclyl, optionally substituted C.sub.5-C.sub.13 bridged carbocyclyl, optionally substituted 5-13 membered bridged heterocyclyl, optionally substituted C.sub.5-C.sub.13 spiro carbocyclyl, optionally substituted 5-13 membered spiro heterocyclyl, optionally substituted 3-10 membered carbocyclyl, optionally substituted 3-10 membered heterocyclyl, optionally substituted aryl, and optionally substituted heteroaryl, wherein R' and R" are independently selected from null, optionally substituted (C.sub.1-C.sub.8 alkylene)-R.sup.r (preferably, CH.sub.2—R.sup.r), optionally substituted R.sup.r—(C.sub.1-C.sub.8 alkylene), optionally substituted (C.sub.1-C.sub.8 alkylene)-R.sup.r—(C.sub.1-C.sub.8 alkyl), or a moiety comprising of optionally substituted C.sub.1-C.sub.8 alkyl, optionally substituted C.sub.1-C.sub.8 heteroalkyl, optionally substituted C.sub.2-C.sub.8 alkenyl, optionally substituted C.sub.2-C.sub.8 alkynyl, optionally substituted C.sub.1-C.sub.5 hydroxyalkyl, optionally substituted C.sub.1-C.sub.8alkoxyC.sub.1-C.sub.8alkyl, optionally

substituted C.sub.1-C.sub.8alkylaminoC.sub.1-C.sub.8alkyl, optionally substituted C.sub.1-C.sub.8 haloalkyl, optionally substituted C.sub.1-C.sub.8 alkylene, optionally substituted C.sub.1-C.sub.5 heteroalkylene, optionally substituted C.sub.2-C.sub.8 alkenylene, optionally substituted C.sub.2-C.sub.8 alkynylene, optionally substituted C.sub.1-C.sub.5 hydroxyalkylene, optionally substituted C.sub.1-C.sub.8alkoxyC.sub.1-C.sub.8alkylene, optionally substituted C.sub.1-C.sub.5alkylaminoC.sub.1-C.sub.8alkylene, optionally substituted C.sub.1-C.sub.8 haloalkylene, optionally substituted 3-10 membered carbocyclyl, optionally substituted 3-10 membered heterocyclyl, optionally substituted C.sub.4-C.sub.13 fused carbocyclyl, optionally substituted 5-13 membered fused heterocyclyl, optionally substituted C.sub.5-C.sub.15 bridged carbocyclyl, optionally substituted 5-13 membered bridged heterocyclyl, optionally substituted C.sub.5-C.sub.13 spiro carbocyclyl, optionally substituted 5-13 membered spiro heterocyclyl, optionally substituted aryl, and optionally substituted heteroaryl; R^{sup.r} is selected from optionally substituted 3-10 membered carbocyclyl, optionally substituted 3-10 membered heterocyclyl, optionally substituted C.sub.4-C.sub.13 fused carbocyclyl, optionally substituted 5-13 membered fused heterocyclyl, optionally substituted C.sub.5-C.sub.13 bridged carbocyclyl, optionally substituted 5-13 membered bridged heterocyclyl, optionally substituted C.sub.5-C.sub.13 spiro carbocyclyl, optionally substituted 5-13 membered spiro heterocyclyl, optionally substituted aryl, and optionally substituted heteroaryl; R^{sup.1} and R^{sup.2} are independently selected from hydrogen, optionally substituted C.sub.1-C.sub.6 alkyl, optionally substituted C.sub.1-C.sub.5 heteroalkyl, optionally substituted C.sub.2-C.sub.8 alkenyl, optionally substituted C.sub.2-C.sub.8 alkynyl, optionally substituted C.sub.1-C.sub.6 alkoxyalkyl, optionally substituted C.sub.1-C.sub.5 haloalkyl, optionally substituted C.sub.1-C.sub.6 hydroxyalkyl, optionally substituted C.sub.1-C.sub.8alkylaminoC.sub.1-C.sub.5alkyl, optionally substituted 3-10 membered carbocyclyl, optionally substituted 3-10 membered heterocyclyl, optionally substituted aryl, and optionally substituted heteroaryl; R' and R'', R^{sup.1} and R^{sup.2}, R' and R^{sup.1}, R' and R^{sup.2}, R'' and R^{sup.1}, or R'' and R^{sup.2} together with the atom to which they are connected optionally form a 3-20 membered carbocyclyl or 3-20 membered heterocyclyl ring; and m is 0 to 15.

(257) In one embodiment, the linker moiety is of FORMULA 9A:

(258) ##STR00134##

wherein R^{sup.1}, R^{sup.2}, R^{sup.3} and R^{sup.4}, at each occurrence, are independently selected from hydrogen, halogen, hydroxyl, amino, cyano, nitro, optionally substituted C.sub.1-C.sub.8 alkyl, optionally substituted C.sub.1-C.sub.5 heteroalkyl, optionally substituted C.sub.2-C.sub.8 alkenyl, optionally substituted C.sub.2-C.sub.8 alkynyl, optionally substituted C.sub.1-C.sub.8 alkoxy, optionally substituted C.sub.1-C.sub.8alkoxyalkyl, optionally substituted C.sub.1-C.sub.8 haloalkyl, optionally substituted C.sub.1-C.sub.5 hydroxyalkyl, optionally substituted C.sub.1-C.sub.8alkylamino, and optionally substituted C.sub.1-C.sub.6 alkylaminoC.sub.1-C.sub.8alkyl, optionally substituted 3-10 membered carbocyclyl, optionally substituted 3-8 membered cycloalkoxy, optionally substituted 3-10 membered carbocyclylamino, optionally substituted 3-8 membered membered heterocyclyl, optionally substituted aryl, and optionally substituted heteroaryl, or R^{sup.1} and R^{sup.2}, or R^{sup.3} and R^{sup.4} together with the atom to which they are connected optionally form a 3-20 membered carbocyclyl or 3-20 membered heterocyclyl ring; A, W and B, at each occurrence, are independently selected from null, or bivalent moiety selected from R^{sup.r}—R'', R'COR'', R'CO.sub.2R'', R'C(O)N(R^{sup.3})R'', R'C(S)N(R^{sup.5})R'', R'OR'', R'SR'', R'SOR'', R'SO.sub.2R'', R'SO.sub.2N(R^{sup.5})R'', R'N(R^{sup.5})R'', R'N(R^{sup.5})COR'', R'N(R^{sup.5})CON(RO)R'', R'N(R^{sup.5})C(S)R'', optionally substituted C.sub.1-C.sub.8 alkylene, optionally substituted C.sub.1-C.sub.5 heteroalkylene, optionally substituted C.sub.2-C.sub.8 alkenylene, optionally substituted C.sub.2-C.sub.8 alkynylene, optionally substituted C.sub.1-C.sub.8alkoxyC.sub.1-C.sub.5alkylene, optionally substituted C.sub.1-C.sub.5 haloalkylene, optionally substituted C.sub.1-C.sub.5 hydroxyalkylene, optionally substituted C.sub.4-C.sub.13 fused carbocyclyl, optionally substituted 5-13 membered fused heterocyclyl, optionally substituted

(259) C.sub.5-C.sub.13 bridged carbocyclyl, optionally substituted 5-13 membered bridged heterocyclyl, optionally substituted C.sub.5-C.sub.13 spiro carbocyclyl, optionally substituted 5-13 membered spiro heterocyclyl, optionally substituted 3-10 membered carbocyclyl, optionally substituted 3-10 membered heterocyclyl, optionally substituted aryl, and optionally substituted heteroaryl, wherein R' and R'' are independently selected from null, optionally substituted (C.sub.1-C.sub.6 alkylene)-R.sup.r (preferably, CH.sub.2—R.sup.r), optionally substituted R.sup.r—(C.sub.1-C.sub.8 alkylene), optionally substituted (C.sub.1-C.sub.8 alkylene)-R.sup.r—(C.sub.1-C.sub.8alkylene), or a moiety comprising of optionally substituted C.sub.1-C.sub.8alkyl, optionally substituted C.sub.1-C.sub.5 heteroalkyl, optionally substituted C.sub.2-C.sub.8 alkenyl, optionally substituted C.sub.2-C.sub.8 alkynyl, optionally substituted C.sub.1-C.sub.5 hydroxyalkyl, optionally substituted C.sub.1-C.sub.6alkoxyC.sub.1-C.sub.5alkyl, optionally substituted C.sub.1-C.sub.8alkylaminoC.sub.1-C.sub.5alkyl, optionally substituted C.sub.1-C.sub.8 haloalkyl, optionally substituted C.sub.1-C.sub.6 alkylene, optionally substituted C.sub.1-C.sub.8 heteroalkylene, optionally substituted C.sub.2-C.sub.8 alkenylene, optionally substituted C.sub.2-C.sub.8 alkynylene, optionally substituted C.sub.1-C.sub.5 hydroxyalkylene, optionally substituted C.sub.1-C.sub.8alkoxyC.sub.1-C.sub.8alkylene, optionally substituted C.sub.1-C.sub.8alkylaminoC.sub.1-C.sub.8alkylene, optionally substituted C.sub.1-C.sub.8 haloalkylene, optionally substituted 3-10 membered carbocyclyl, optionally substituted 3-10 membered heterocyclyl, optionally substituted C.sub.4-C.sub.13 fused carbocyclyl, optionally substituted 5-13 membered fused heterocyclyl, optionally substituted C.sub.5-C.sub.15 bridged carbocyclyl, optionally substituted 5-13 membered bridged heterocyclyl, optionally substituted C.sub.5-C.sub.13 spiro carbocyclyl, optionally substituted 5-13 membered spiro heterocyclyl, optionally substituted aryl, and optionally substituted heteroaryl; R.sup.r is selected from optionally substituted 3-10 membered carbocyclyl, optionally substituted 3-10 membered heterocyclyl, optionally substituted C.sub.4-C.sub.13 fused carbocyclyl, optionally substituted 5-13 membered fused heterocyclyl, optionally substituted C.sub.5-C.sub.15 bridged carbocyclyl, optionally substituted 5-13 membered bridged heterocyclyl, optionally substituted C.sub.5-C.sub.13 spiro carbocyclyl, optionally substituted 5-13 membered spiro heterocyclyl, optionally substituted aryl, and optionally substituted heteroaryl; R.sup.5 and R.sup.6 are independently selected from hydrogen, optionally substituted C.sub.1-C.sub.6 alkyl, optionally substituted C.sub.1-C.sub.5 heteroalkyl, optionally substituted C.sub.2-C.sub.8 alkenyl, optionally substituted C.sub.2-C.sub.8 alkynyl, optionally substituted C.sub.1-C.sub.6 alkoxyalkyl, optionally substituted C.sub.1-C.sub.5 haloalkyl, optionally substituted C.sub.1-C.sub.6 hydroxyalkyl, optionally substituted C.sub.1-C.sub.8alkylaminoC.sub.1-C.sub.5alkyl, optionally substituted 3-10 membered carbocyclyl, optionally substituted 3-10 membered heterocyclyl, optionally substituted aryl, and optionally substituted heteroaryl; R' and R'', R.sup.5 and R.sup.6, R' and R.sup.3, R' and R.sup.6, R'' and R.sup.5, or R'' and R.sup.6 together with the atom to which they are connected optionally form a 3-20 membered carbocyclyl or 3-20 membered heterocyclyl ring; m is 0 to 15; n, at each occurrence, is 0 to 15; and o is 0 to 15.

(260) In another embodiment, the linker moiety is of FORMULA 9B:

(261) ##STR00135##

wherein R.sup.1 and R.sup.2, at each occurrence, are independently selected from hydrogen, halogen, hydroxyl, amino, cyano, nitro, and optionally substituted C.sub.1-C.sub.8 alkyl, optionally substituted C.sub.1-C.sub.5 heteroalkyl, optionally substituted C.sub.1-C.sub.8 alkoxy, optionally substituted C.sub.1-C.sub.8 alkoxy C.sub.1-C.sub.6 alkyl, optionally substituted C.sub.1-C.sub.5 haloalkyl, optionally substituted C.sub.1-C.sub.6 hydroxyalkyl, optionally substituted C.sub.1-C.sub.8 alkylamino, C.sub.1-C.sub.8alkylaminoC.sub.1-C.sub.8alkyl, optionally substituted 3-10 membered carbocyclyl, optionally substituted 3-8 membered cycloalkoxy, optionally substituted 3-10 membered carbocyclylamino, optionally substituted 3-10 membered heterocyclyl, optionally substituted aryl, and optionally substituted heteroaryl, or R.sup.1 and R.sup.2 together with the

atom to which they are connected optionally form a 3-20 membered carbocyclyl or 3-20 membered heterocyclyl ring; A and B, at each occurrence, are independently selected from null, or bivalent moiety selected from R.sup.r—R'', R'COR'', R'CO.sub.2R'', R'C(O)N(R.sup.3)R'', R'C(S)N(R.sup.3)R'', R'OR'', R'SR'', R'SOR'', R'SOR'', R'SO.sub.2N(R.sup.3)R'', R'N(R.sup.3)R'', R'N(R.sup.3)COR'', R'N(R.sup.3)CON(R.sup.4)R'', R'N(R.sup.3)C(S)R'', optionally substituted C.sub.1-C.sub.6 alkylene, optionally substituted C.sub.1-C.sub.5 heteroalkylene, optionally substituted C.sub.2-C.sub.8 alkenylene, optionally substituted C.sub.2-C.sub.8 alkynylene, optionally substituted C.sub.1-C.sub.8alkoxyC.sub.1-C.sub.8alkylene, optionally substituted C.sub.1-C.sub.5 haloalkylene, optionally substituted C.sub.1-C.sub.5 hydroxyalkylene, optionally substituted C.sub.4-C.sub.13 fused carbocyclyl, optionally substituted 5-13 membered fused heterocyclyl, optionally substituted C.sub.5-C.sub.13 bridged carbocyclyl, optionally substituted 5-13 membered bridged heterocyclyl, optionally substituted C.sub.5-C.sub.13 spiro carbocyclyl, optionally substituted 5-13 membered spiro heterocyclyl, optionally substituted 3-10 membered carbocyclyl, optionally substituted 3-10 membered heterocyclyl, optionally substituted aryl, and optionally substituted heteroaryl, wherein R' and R'' are independently selected from null, optionally substituted (C.sub.1-C.sub.8 alkylene)-R.sup.r (preferably, CH.sub.2—R'), optionally substituted R.sup.r—(C.sub.1-C.sub.8 alkylene), optionally substituted (C.sub.1-C.sub.8 alkylene)-R.sup.r—(C.sub.1-C.sub.8 alkylene), or a moiety comprising of optionally substituted C.sub.1-C.sub.8 alkyl, optionally substituted C.sub.1-C.sub.6 heteroalkyl, optionally substituted C.sub.2-C.sub.8 alkenyl, optionally substituted C.sub.2-C.sub.8 alkynyl, optionally substituted C.sub.1-C.sub.8 hydroxyalkyl, optionally substituted C.sub.1-C.sub.6alkoxyC.sub.1-C.sub.5alkyl, optionally substituted C.sub.1-C.sub.8alkylaminoC.sub.1-C.sub.8alkyl, optionally substituted C.sub.1-C.sub.5 haloalkyl, optionally substituted C.sub.1-C.sub.6 alkylene, optionally substituted C.sub.1-C.sub.5 heteroalkylene, optionally substituted C.sub.2-C.sub.8 alkenylene, optionally substituted C.sub.2-C.sub.8 alkynylene, optionally substituted C.sub.1-C.sub.8 hydroxyalkylene, optionally substituted C.sub.1-C.sub.8alkoxy C.sub.1-C.sub.5alkylene, optionally substituted C.sub.1-C.sub.5alkylaminoC.sub.1-C.sub.5alkylene, optionally substituted C.sub.1-C.sub.8 haloalkylene, optionally substituted 3-10 membered carbocyclyl, optionally substituted 3-10 membered heterocyclyl, optionally substituted C.sub.4-C.sub.13 fused carbocyclyl, optionally substituted 5-13 membered fused heterocyclyl, optionally substituted C.sub.5-C.sub.15 bridged carbocyclyl, optionally substituted 5-13 membered bridged heterocyclyl, optionally substituted C.sub.5-C.sub.13 spiro carbocyclyl, optionally substituted 5-13 membered spiro heterocyclyl, optionally substituted aryl, and optionally substituted heteroaryl; R.sup.r is selected from optionally substituted 3-10 membered carbocyclyl, optionally substituted 3-10 membered heterocyclyl, optionally substituted C.sub.4-C.sub.13 fused carbocyclyl, optionally substituted 5-13 membered fused heterocyclyl, optionally substituted C.sub.5-C.sub.15 bridged carbocyclyl, optionally substituted 5-13 membered bridged heterocyclyl, optionally substituted C.sub.5-C.sub.13 spiro carbocyclyl, optionally substituted 5-13 membered spiro heterocyclyl, optionally substituted aryl, and optionally substituted heteroaryl; R.sup.3 and R.sup.4 are independently selected from hydrogen, optionally substituted C.sub.1-C.sub.8 alkyl, optionally substituted C.sub.1-C.sub.5 heteroalkyl, optionally substituted C.sub.2-C.sub.8 alkenyl, optionally substituted C.sub.2-C.sub.8 alkynyl, optionally substituted C.sub.1-C.sub.6 alkoxyalkyl, optionally substituted C.sub.1-C.sub.5 haloalkyl, optionally substituted C.sub.1-C.sub.5 hydroxyalkyl, optionally substituted C.sub.1-C.sub.5alkylaminoC.sub.1-C.sub.8alkyl, optionally substituted 3-10 membered carbocyclyl, optionally substituted 3-10 membered heterocyclyl, optionally substituted aryl, and optionally substituted heteroaryl; R' and R'', R.sup.3 and R.sup.4, R' and R.sup.3, R' and R.sup.4, R'' and R.sup.3, or R'' and R.sup.4 together with the atom to which they are connected optionally form a 3-20 membered carbocyclyl or 3-20 membered heterocyclyl ring; each m is 0 to 15; and n is 0 to 15.

(262) In another embodiment, the linker moiety is of FORMULA 9C:

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wherein X is selected from O, NH, and NR.^{sup.7}; R.^{sup.1}, R.^{sup.2}, R.^{sup.3}, R.^{sup.4}, R.^{sup.5}, and Ré, at each occurrence, are independently selected from hydrogen, halogen, hydroxyl, amino, cyano, nitro, optionally substituted C.sub.1-C.sub.6 alkyl, optionally substituted C.sub.1-C.sub.5 heteroalkyl, optionally substituted C.sub.2-C.sub.8 alkenyl, optionally substituted C.sub.2-C.sub.8 alkynyl, optionally substituted C.sub.1-C.sub.8 alkoxy, optionally substituted C.sub.1-C.sub.8 alkoxy C.sub.1-C.sub.8 alkyl, optionally substituted C.sub.1-C.sub.6 haloalkyl, optionally substituted C.sub.1-C.sub.8 hydroxyalkyl, optionally substituted C.sub.1-C.sub.6 alkylamino, optionally substituted C.sub.1-C.sub.6 alkylaminoC.sub.1-C.sub.6 alkyl, optionally substituted 3-10 membered carbocyclyl, optionally substituted 3-8 membered cycloalkoxy, optionally substituted 3-10 membered heterocyclyl, optionally substituted aryl, and optionally substituted heteroaryl; A and B are independently selected from null, or bivalent moiety selected from R.^{sup.r}—R", R'COR", R'CO.sub.2R", R'C(O)N(R.^{sup.3})R", R'C(S)N(R.^{sup.3})R", R'OR", R'SR", R'SOR", R'SO.sub.2R", R'SON(R.^{sup.8})R", R'N(R.^{sup.8})R", R'N(R.^{sup.8})COR", R'N(R.^{sup.8})CON(RO)R", R'N(R.^{sup.3})C(S)R", optionally substituted C.sub.1-C.sub.8 alkylene, optionally substituted C.sub.1-C.sub.8 heteroalkylene, optionally substituted C.sub.2-C.sub.8 alkenylene, optionally substituted C.sub.2-C.sub.8 alkynylene, optionally substituted C.sub.1-C.sub.8alkoxyC.sub.1-C.sub.5alkylene, optionally substituted C.sub.1-C.sub.8 haloalkylene, optionally substituted C.sub.1-C.sub.8 hydroxyalkylene, optionally substituted C.sub.4-C.sub.13 fused carbocyclyl, optionally substituted 5-13 membered fused heterocyclyl, optionally substituted C.sub.5-C.sub.13 bridged carbocyclyl, optionally substituted 5-13 membered bridged heterocyclyl, optionally substituted C.sub.5-C.sub.13 spiro carbocyclyl, optionally substituted 5-13 membered spiro heterocyclyl, optionally substituted 3-10 membered carbocyclyl, optionally substituted 3-10 membered heterocyclyl, optionally substituted aryl, and optionally substituted heteroaryl, wherein R' and R" are independently selected from null, optionally substituted (C.sub.1-C.sub.6 alkylene)-R.^{sup.r} (preferably, CH.sub.2—R'), optionally substituted R.^{sup.r}—(C.sub.1-C.sub.8 alkylene), optionally substituted (C.sub.1-C.sub.8 alkylene)-R.^{sup.r}—(C.sub.1-C.sub.8 alkylene), or a moiety comprising of optionally substituted C.sub.1-C.sub.8alkyl, optionally substituted C.sub.1-C.sub.8 heteroalkyl, optionally substituted C.sub.2-C.sub.8 alkenyl, optionally substituted C.sub.2-C.sub.8 alkynyl, optionally substituted C.sub.1-C.sub.8 hydroxyalkyl, optionally substituted C.sub.1-C.sub.6alkoxyC.sub.1-C.sub.5alkyl, optionally substituted C.sub.1-C.sub.8alkylaminoC.sub.1-C.sub.8alkyl, optionally substituted C.sub.1-C.sub.8 haloalkyl, optionally substituted C.sub.1-C.sub.6 alkylene, optionally substituted C.sub.1-C.sub.5 heteroalkyl, optionally substituted C.sub.2-C.sub.8 alkenylene, optionally substituted C.sub.2-C.sub.8 alkynylene, optionally substituted C.sub.1-C.sub.5 hydroxyalkylene, optionally substituted C.sub.1-C.sub.8alkoxyC.sub.1-C.sub.8alkylene, optionally substituted C.sub.1-C.sub.5alkylaminoC.sub.1-C.sub.8alkylene, optionally substituted C.sub.1-C.sub.5 haloalkylene, optionally substituted 3-10 membered carbocyclyl, optionally substituted 3-10 membered heterocyclyl, optionally substituted C.sub.4-C.sub.13 fused carbocyclyl, optionally substituted 5-13 membered fused heterocyclyl, optionally substituted C.sub.5-C.sub.13 bridged carbocyclyl, optionally substituted 5-13 membered bridged heterocyclyl, optionally substituted C.sub.5-C.sub.13 spiro carbocyclyl, optionally substituted 5-13 membered spiro heterocyclyl, optionally substituted aryl, and optionally substituted heteroaryl; R.^{sup.r} is selected from optionally substituted 3-10 membered carbocyclyl, optionally substituted 3-10 membered heterocyclyl, optionally substituted C.sub.4-C.sub.13 fused carbocyclyl, optionally substituted 5-13 membered fused heterocyclyl, optionally substituted C.sub.5-C.sub.15 bridged carbocyclyl, optionally substituted 5-13 membered bridged heterocyclyl, optionally substituted C.sub.5-C.sub.13 spiro carbocyclyl, optionally substituted 5-13 membered spiro heterocyclyl, optionally substituted aryl, and optionally substituted heteroaryl; R.^{sup.7}, R.^{sup.8} and R.^{sup.9} are independently selected from hydrogen, optionally substituted C.sub.1-C.sub.6 alkyl, optionally substituted C.sub.1-C.sub.5 heteroalkyl, optionally substituted C.sub.2-C.sub.8 alkenyl, optionally substituted C.sub.2-C.sub.8 alkynyl, optionally substituted C.sub.1-

C.sub.6 alkoxyalkyl, optionally substituted C.sub.1-C.sub.5 haloalkyl, optionally substituted C.sub.1-C.sub.5 hydroxyalkyl, optionally substituted C.sub.1-C.sub.5 alkylamino C.sub.1-C.sub.8 alkyl, optionally substituted 3-10 membered carbocyclyl, optionally substituted 3-10 membered heterocyclyl, optionally substituted aryl, and optionally substituted heteroaryl; R' and R'', R.sup.8 and R.sup.9, R' and R.sup.8, R' and R.sup.9, R'' and R.sup.8, or R'' and R' together with the atom to which they are connected optionally form a 3-20 membered carbocyclyl or 3-20 membered heterocyclyl ring; m, at each occurrence, is 0 to 15; n, at each occurrence, is 0 to 15; o is 0 to 15; and p is 0 to 15.

(264) In one embodiment, in FORMULA 9C, m and n is 0 or 1, and p is 0 to 15; In one embodiment, in FORMULA 9C, X is selected from O and NH; In one embodiment, in FORMULA 9C, R.sup.1, R.sup.2, R.sup.3, R.sup.4, R.sup.5, and R⁶, are independently selected from hydrogen, and optionally substituted C.sub.1-C.sub.6 alkyl.

(265) In another embodiment, the linker moiety comprises one or more rings selected from the group consisting of 3 to 13 membered rings, 4 to 13 membered fused rings, 5 to 13 membered bridged rings, and 5 to 13 membered spiro rings.

(266) In another embodiment, the linker moiety comprises a ring selected from the group consisting of Formula C.sub.1, C.sub.2, C.sub.3, C.sub.4 and C.sub.5;

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wherein X' and Y' are independently selected from N and CR.sup.b; A.sup.1, B.sup.1, C.sup.1 and D.sup.1, at each occurrence, are independently selected from null, O, CO, SO, SO.sub.2, NR.sup.9, and CRR.sup.c; A.sup.2, B.sup.2, C.sup.2, and D.sup.2, at each occurrence, are independently selected from N and CR.sup.b; A.sup.3, B.sup.3, C.sub.3, D.sup.3, and E.sup.3, at each occurrence, are independently selected from N, O, S, NR.sup.b, and CR.sup.b; R.sup.b and R.sup.c, at each occurrence, are independently selected from hydrogen, halogen, hydroxyl, amino, cyano, nitro, optionally substituted C.sub.1-C.sub.8 alkyl, optionally substituted C.sub.1-C.sub.5 heteroalkyl, optionally substituted C.sub.2-C.sub.8 alkenyl, optionally substituted C.sub.2-C.sub.8 alkynyl, optionally substituted C.sub.1-C.sub.8 alkoxy, optionally substituted C.sub.1-C.sub.8 alkoxyalkyl, optionally substituted C.sub.1-C.sub.5 haloalkyl, optionally substituted C.sub.1-C.sub.8 hydroxyalkyl, optionally substituted C.sub.1-C.sub.6 alkylamino, and optionally substituted C.sub.1-C.sub.6 alkylamino C.sub.1-C.sub.8 alkyl, optionally substituted 3-10 membered carbocyclyl, optionally substituted 3-8 membered cycloalkoxy, optionally substituted 3-10 membered carbocyclylamino, optionally substituted 3-8 membered membered heterocyclyl, optionally substituted aryl, and optionally substituted heteroaryl; R.sup.b and R.sup.b, or R.sup.b and R.sup.c together with the atom to which they are connected form a 3-8 membered carbocyclyl or 3-8 membered heterocyclyl ring; and m.sup.1, n.sup.1, o.sup.1 and p.sup.1 are independently selected from 0, 1, 2, 3, 4 and 5.

(268) In one embodiment, A, B and W, at each occurrence, are independently selected from null, optionally substituted —(CH.sub.2).sub.0-8—, optionally substituted —(CH.sub.2).sub.0-8—CO—(CH.sub.2).sub.0-8—, optionally substituted —(CH.sub.2).sub.0-8—NH—(CH.sub.2).sub.0-8—, optionally substituted —(CH.sub.2)-s-NH—CO—(CH.sub.2).sub.0-8—, optionally substituted —(CH.sub.2) 0-8—CO—NH—(CH.sub.2).sub.0-8—, optionally substituted —(CH.sub.2).sub.0-3—NH—(CH.sub.2).sub.0-3—CO—NH—(CH.sub.2).sub.0-8—, optionally substituted —(CH.sub.2).sub.0-3—NH—(CH.sub.2).sub.1-3—NH—CO—(CH.sub.2).sub.0-8—, optionally substituted —(CH.sub.2).sub.0-8—CO—NH—(CH.sub.2).sub.1-3—NH—(CH.sub.2).sub.0-3—, optionally substituted —(CH.sub.2).sub.0-8—R.sub.r—(CH.sub.2).sub.0-2—, optionally substituted —(CH.sub.2).sub.0-3—(CO)—(CH.sub.2)-3-R.sub.r—(CH.sub.2).sub.0-3—, optionally substituted —(CH.sub.2).sub.0-3—(CO—NH)—(CH.sub.2).sub.0-3—R.sub.r—(CH.sub.2).sub.0-3—, optionally substituted —(CH.sub.2).sub.0-3—(NH—CO)—(CH.sub.2).sub.0-3—R.sub.r—(CH.sub.2).sub.0-3—, and optionally substituted —(CH.sub.2).sub.0-3—(NH)—(CH.sub.2).sub.0-3—R.sub.r—(CH.sub.2).sub.0-3—

(269) In one embodiment, R' is of Formula C.sub.1, C.sub.2, C.sub.3, C.sub.4 or C.sub.5.

(270) In one embodiment, R.sub.r is selected from

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(272) In another embodiment, the length of the linker is 0 to 40 atoms.

(273) In another embodiment, the length of the linker is 0 to 20 atoms.

(274) In another embodiment, the length of the linker is 0 to 10 atoms.

(275) In another embodiment, the linker is selected from null, optionally substituted —(CO)—

(CH.sub.2).sub.0-8—, optionally substituted —(CH.sub.2).sub.0-9—, optionally substituted —

(CH.sub.2).sub.1-2—(CO)—NH—(CH.sub.2).sub.0-9—, optionally substituted —(CH.sub.2) 12-

(CO)—NH—(CH.sub.2).sub.0-3-(OCH.sub.2CH.sub.2).sub.1-7—, optionally substituted —

(CH.sub.2).sub.0-1—(CO)—(CH.sub.2).sub.1-3—(OCH.sub.2CH.sub.2).sub.1-7—, optionally

substituted —(CO)—(CH.sub.2).sub.0-3-(alkenylene)-(CH.sub.2).sub.0-3—, optionally

substituted —(CO)—(CH.sub.2).sub.0-3-(alkynylene)-(CH.sub.2).sub.0-3—, optionally substituted

—(CO)—(CH.sub.2).sub.0-3-(3-8 membered carbocyclyl)-(CH.sub.2).sub.0-3—, optionally

substituted —(CO)—(CH.sub.2).sub.0-3-(3-8 membered heterocyclyl)-(CH.sub.2)(-3-, optionally

substituted —(CH.sub.2).sub.0-3-(alkenylene)-(CH.sub.2).sub.0-3—, optionally substituted —

(CH.sub.2).sub.0-3-(alkynylene)-(CH).sub.0-3-, optionally substituted —(CH.sub.2).sub.0-3-(3-8

membered carbocyclyl)-(CH.sub.2) 3-, optionally substituted —(CH.sub.2).sub.0-3-(3-8 membered

heterocyclyl)-(CH.sub.2).sub.0-3—, optionally substituted —(CH.sub.2).sub.0-8—R.sub.r—

(CH.sub.2).sub.0-8-, optionally substituted —(CH.sub.2).sub.0-8—R.sub.5—(CO)—

(CH.sub.2).sub.1-8—, optionally substituted —(CH.sub.2).sub.0-8-RT-(CH.sub.2).sub.1-2—(CO)

—NH—(CH.sub.2).sub.2-9-, optionally substituted —(CH.sub.2).sub.0-8~RT-(CH.sub.2).sub.1-2

—(CO)—NH—(CH.sub.2).sub.1-3—(OCH.sub.2CH.sub.2).sub.1-7—, optionally substituted-

(CH.sub.2).sub.0-8~R.sub.r—(CH.sub.2).sub.0-1—(CO)—(CH.sub.2).sub.1-3—

(OCH.sub.2CH.sub.2).sub.1-7—, optionally substituted —(CH.sub.2).sub.0-8~R.sub.r—(CO)—

(CH.sub.2).sub.0-3-(alkenylene)-(CH.sub.2).sub.0-3—, optionally substituted —(CH.sub.2).sub.0-8

—R.sub.r—(CO)—(CH.sub.2).sub.0-3-(alkynylene)-(CH.sub.2).sub.0-3—, optionally substituted

—(CH.sub.2).sub.0-8—R.sub.r—(CO)—(CH.sub.2).sub.0-3-(3-8 membered carbocyclyl)-

(CH.sub.2).sub.0-3—, optionally substituted —(CH.sub.2).sub.0-8—R.sub.r—(CO)—

(CH.sub.2).sub.0-3-(3-8 membered heterocyclyl)-(CH.sub.2).sub.0-3—, optionally substituted-

(CH.sub.2).sub.0-8—R.sub.r—(CH.sub.2).sub.0-3-(alkenylene)-(CH.sub.2).sub.0-3—, optionally

substituted —(CH).sub.0-8—R.sub.r—(CH).sub.0-3-(alkynylene)-(CH.sub.2).sub.0-3—,

optionally substituted —(CH.sub.2).sub.0-8-R.sub.r—(CH.sub.2).sub.0-3-(3-8 membered

carbocyclyl)-(CH.sub.2).sub.0-3—, and optionally substituted —(CH.sub.2).sub.0-8—R.sub.r—

(CH.sub.2).sub.0-3-(3-8 membered heterocyclyl)-(CH.sub.2).sub.0-3—.

(276) In an aspect, provided herein is a bivalent compound comprising a Tropomyosin Receptor Kinase (TRK) ligand conjugated to a degradation tag via a linker, or a pharmaceutically acceptable salt or analog thereof, wherein the linker and the degradation tag are each independently selected from those disclosed herein, and the TRK ligand is a moiety of Formula 10 which connects to the linker via R.sub.4;

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wherein X.sub.1, X.sub.2, X.sub.3, X.sub.4, and X.sub.5 are each independently selected from C, CR.sub.1, and N; X is selected from CR.sub.1R.sub.2, CO, O, S, SO, SO.sub.2, and NR.sub.1; R is selected from optionally substituted C.sub.1-C.sub.8alkyl, optionally substituted C.sub.1-C.sub.5 heteroalkyl, optionally substituted C.sub.3-C.sub.10 carbocyclyl, optionally substituted 3-10 membered heterocyclyl, and Ar.sub.1; or X—R, taken together, stands for

(278) ##STR00144## U is selected from null, a bond, C(R.sub.2).sub.2,

C(R.sub.2).sub.2C(R.sub.2).sub.2, CO, C(R.sub.2).sub.2CO, CONR.sub.2, C(R.sub.2).sub.2O,

C(R.sub.2).sub.2NR.sub.2 and CH.sub.2NR.sub.2; R.sub.1 and R.sub.2, at each occurrence, are

independently selected from hydrogen, halogen, CN, NO.sub.2, OR.sub.6, SR.sub.6,

NR.sup.1R.sup.8, COR.sup.6, CO.sub.2R.sup.6, C(O)NR.sup.1R.sup.8, SOR.sup.6, SO.sub.2R.sup.6, SO.sub.2NR.sup.1R.sup.8, NR.sup.9C(O)R.sup.8, NR.sup.9C(O)NR.sup.7R.sup.8, NR.sup.6SOR.sup.8, NR.sup.6SO.sub.2R.sup.8, optionally substituted C.sub.1-C.sub.8 alkyl, optionally substituted C.sub.1-C.sub.5 heteroalkyl, optionally substituted C.sub.1-C.sub.8alkoxy, optionally substituted C.sub.1-C.sub.8alkoxyC.sub.1-C.sub.8alkyl, optionally substituted C.sub.1-C.sub.8alkylaminoC.sub.1-C.sub.8alkyl, optionally substituted C.sub.3-C.sub.10 carbocyclyl, optionally substituted C.sub.3-C.sub.10 cycloalkoxy, optionally substituted 3-10 membered heterocyclyl, optionally substituted C.sub.2-C.sub.8 alkenyl, and optionally substituted C.sub.2-C.sub.8 alkynyl; or R.sup.1 and R.sup.2, R.sup.1 and another R.sup.1, or R.sup.2 and another R.sup.2 together with the atom to which they are connected optionally form a 3-8 membered carbocyclyl or 3-8 membered heterocyclyl ring; R.sup.6, R.sup.7 and R.sup.8, at each occurrence, are independently selected from null, hydrogen, optionally substituted C.sub.1-C.sub.6 alkyl, optionally substituted C.sub.1-C.sub.5 heteroalkyl, optionally substituted C.sub.2-C.sub.8 alkenyl, optionally substituted C.sub.2-C.sub.8 alkynyl, optionally substituted C.sub.3-C.sub.6 carbocyclyl, optionally substituted 3-8 membered heterocyclyl, optionally substituted heterocarbocyclyl, optionally substituted arylalkyl, optionally substituted heteroarylalkyl, optionally substituted aryl, and optionally substituted heteroaryl, or R.sup.7 and R.sup.8, together with the atom to which they are connected, optionally form a 3-8 membered carbocyclyl or 3-8 membered heterocyclyl ring; n is 0, 1, 2, 3, or 4; n' is 0, 1, 2, or 3; L is Ar.sup.2, NR.sup.3CO or NR.sup.3COAr.sup.2; R.sup.3 is selected from hydrogen, optionally substituted C.sub.1-C.sub.6 alkyl, optionally substituted C.sub.1-C.sub.6 heteroalkyl, optionally substituted C.sub.3-C.sub.6 carbocyclyl, optionally substituted 3-6 membered heterocyclyl, optionally substituted C.sub.1-C.sub.6 alkoxyalkyl, optionally substituted C.sub.1-C.sub.6 haloalkyl, optionally substituted C.sub.1-C.sub.6 hydroxyalkyl, and optionally substituted C.sub.1-C.sub.8alkylaminoC.sub.1-C.sub.8alkyl; R.sup.4 is connected to the linker moiety of the bivalent compound either directly or through R.sup.5, wherein R.sup.4 and R.sup.5 are bivalent groups independently selected from null, —O—, —S—, —NR.sup.9—, —CO—, —CO.sub.2—, —CONR.sup.9—, —SO—, —SO.sub.2R.sup.9—, —SO.sub.2NR.sup.9—, —NR.sup.10CO—, —NR.sup.10C(O)NR.sup.9—, —NR.sup.10SO—, —NR.sup.10SO.sub.2—, optionally substituted C.sub.1-C.sub.8 alkylene, optionally substituted C.sub.1-C.sub.8 heteroalkylene, optionally substituted C.sub.1-C.sub.8alkoxyC.sub.1-C.sub.5alkylene, optionally substituted C.sub.1-C.sub.8 haloalkylene, optionally substituted C.sub.1-C.sub.8 hydroxyalkylene, optionally substituted C.sub.1-C.sub.5alkylaminoC.sub.1-C.sub.5alkylene, optionally substituted C.sub.5-C.sub.8 carbocyclylene, optionally substituted 3-8 membered heterocyclylene, optionally substituted C.sub.2-C.sub.8 alkenylene, optionally substituted C.sub.2-C.sub.8 alkynylene, optionally substituted aryl, and optionally substituted heteroaryl; R.sup.9 is selected from null, optionally substituted C.sub.1-C.sub.8 alkylene, optionally substituted C.sub.1-C.sub.8 heteroalkylene, optionally substituted C.sub.1-C.sub.6 alkylene-O—, optionally substituted C.sub.1-C.sub.8 alkylene-N(C.sub.1-C.sub.8 alkyl)-, optionally substituted C.sub.1-C.sub.8 alkylene-O—C.sub.1-C.sub.6 alkylene, optionally substituted C.sub.1-C.sub.8alkylene-N(C.sub.1—Ca alkyl)-C.sub.1-C.sub.5alkylene, optionally substituted C.sub.3-C.sub.8 carbocyclyl, optionally substituted C.sub.3-C.sub.8 carbocyclyl-O—, optionally substituted 3-8 membered heterocyclyl, optionally substituted aryl, and optionally substituted heteroaryl; R.sup.10 is selected from null, hydrogen, optionally substituted C.sub.1-C.sub.8 alkyl,, optionally substituted C.sub.1-C.sub.5 heteroalkyl, optionally substituted C.sub.2-C.sub.8 alkenyl, optionally substituted C.sub.2-C.sub.8 alkynyl, optionally substituted C.sub.3-C.sub.8 carbocyclyl, optionally substituted heterocyclyl, optionally substituted arylalkyl, optionally substituted heteroarylalkyl, optionally substituted 3-8 membered carbocyclyl, optionally substituted 3-8 membered heterocyclyl, optionally substituted aryl, and optionally substituted heteroaryl; Ar.sup.1 and Ar.sup.2 are independently selected from aryl and heteroaryl, each of which is optionally substituted with one or more substituents independently

selected from halogen, CN, NO.sub.2, OR.sup.1, SR.sup.11, NR.sup.12R.sup.13, COR.sup.11, CO.sub.2R.sup.1, CONR.sup.12R.sup.13, SOR.sup.11, SO.sub.2R.sup.11, SO.sub.2NR.sup.12R.sup.13, NR.sup.10OR.sup.1, NR.sup.11C(O)NR.sup.12R.sup.13, NR.sup.11SOR.sup.1, NR.sup.10SO.sub.2R.sup.13, optionally substituted C.sub.1-C.sub.8 alkyl, optionally substituted C.sub.1-C.sub.5 heteroalkyl, optionally substituted C.sub.1-C.sub.8alkoxyC.sub.1-C.sub.8alkyl, optionally substituted C.sub.1-C.sub.5 haloalkyl, optionally substituted C.sub.1-C.sub.5 hydroxyalkyl, optionally substituted C.sub.1-C.sub.5alkylaminoC.sub.1-C.sub.8alkyl, optionally substituted C.sub.3-C.sub.7 carbocyclyl, optionally substituted 3-7 membered heterocyclyl, optionally substituted C.sub.2-C.sub.8 alkenyl, optionally substituted C.sub.2-C.sub.8 alkynyl, optionally substituted aryl, and optionally substituted heteroaryl; and R.sup.11, R.sup.12, and R.sup.13 are independently selected from null, hydrogen, optionally substituted C.sub.1-C.sub.8alkyl, optionally substituted C.sub.1-C.sub.5 heteroalkyl, optionally substituted C.sub.2-C.sub.8 alkenyl, optionally substituted C.sub.2-C.sub.8 alkynyl, optionally substituted C.sub.3-C.sub.7 carbocyclyl, optionally substituted 3-7 membered heterocyclyl, optionally substituted aryl, optionally substituted heteroaryl, or R.sup.12 and R.sup.13 together with the atom to which they are connected optionally form a 3-8 membered carbocyclyl or heterocyclyl ring.

(279) In another aspect, provided herein is a bivalent compound comprising a Tropomyosin Receptor Kinase (TRK) ligand conjugated to a degradation tag via a linker, or a pharmaceutically acceptable salt or analog thereof, wherein TRK ligand and the linker are each independently selected from those disclosed herein, and the degradation tag is a moiety of Formula 11 which connects to the linker via Z or any of V.sup.1, V.sup.2, V.sup.3, V.sup.4, W.sup.1, W.sup.2, and W.sup.3;

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wherein V.sup.1, V.sup.2, V.sup.3, and V.sup.4 are each independently selected from CR.sup.2' and N; W.sup.1, W.sup.2, and W.sup.3 are each independently selected from CO, O, CR.sup.3'R.sup.4', and NR.sup.5', wherein R.sup.5' is selected from a hydrogen, halogen, cyano, nitro, optionally substituted C.sub.1-C.sub.6 alkyl, optionally substituted C.sub.1-C.sub.6 heteroalkyl, optionally substituted C.sub.3-C.sub.7 carbocyclyl, optionally substituted 3-7 membered heterocyclyl, and 2,6-dioxopiperidin-3-yl optionally substituted with R.sup.1', provided that there are no two adjacent CO groups; Z is selected from null, CO, CR.sup.3'R.sup.4', NR.sup.3', O, optionally substituted C.sub.1-C.sub.10 alkylene, optionally substituted C.sub.1-C.sub.10 heteroalkylene, optionally substituted C.sub.2-C.sub.10 alkenylene, optionally substituted C.sub.2-C.sub.10 alkynylene, optionally substituted 3-7 membered carbocyclyl, and optionally substituted 3-7 membered heterocyclyl; R.sup.1' is selected from hydrogen, halogen, cyano, nitro, optionally substituted C.sub.1-C.sub.6 alkyl, optionally substituted C.sub.1-C.sub.6 heteroalkyl, optionally substituted 3-6 membered carbocyclyl, and optionally substituted 3-6 membered heterocyclyl; R.sup.2' is selected from hydrogen, halogen, cyano, nitro, optionally substituted C.sub.1-C.sub.6 alkyl, optionally substituted C.sub.1-C.sub.6 heteroalkyl, optionally substituted C.sub.1-C.sub.6 alkoxy, optionally substituted C.sub.1-C.sub.6 alkylamino, optionally substituted 3-6 membered carbocyclyl, and optionally substituted 3-6 membered heterocyclyl; and R.sup.3' and R.sup.4' at each occurrence are independently selected from hydrogen, halogen, cyano, nitro, optionally substituted C.sub.1-C.sub.6 alkyl, optionally substituted C.sub.1-C.sub.6 heteroalkyl, optionally substituted 3-6 membered carbocyclyl, and optionally substituted 3-6 membered heterocyclyl; or R.sup.3' and R.sup.4' together with the atom to which they are connected optionally form a 3-8 membered carbocyclyl or heterocyclyl ring.

(281) In another aspect, provided herein is a bivalent compound comprising a Tropomyosin Receptor Kinase (TRK) ligand conjugated to a degradation tag via a linker, or a pharmaceutically acceptable salt or analog thereof, wherein the TRK ligand and the degradation tag are each independently selected from those disclosed herein, and the linker is a moiety of Formula 9;

(282) ##STR00146## wherein A and B, at each occurrence, are independently selected from null, or a bivalent moiety selected from R.sup.r—R", R'COR", R'CO.sub.2R", R'C(O)N(R.sup.1")R", R'C(S)N(R.sup.1")R", R'OR", R'SR", R'SOR", R'SO.sub.2R", R'SO.sub.2N(R.sup.1")R", R'N(R.sup.1")R", R"N(R.sup.1")COR", R"N(R.sup.1")CON(R.sup.2")R", R'N(R.sup.1")C(S)R", optionally substituted C.sub.1-C.sub.8alkylene, optionally substituted C.sub.1-C.sub.5 heteroalkylene, optionally substituted C.sub.2-C.sub.8 alkenylene, optionally substituted C.sub.2-C.sub.8 alkynylene, optionally substituted C.sub.1-C.sub.8alkoxyC.sub.1-C.sub.5alkylene, optionally substituted C.sub.1-C.sub.8 haloalkylene, optionally substituted C.sub.1-C.sub.8 hydroxyalkylene; each W is independently selected from null, optionally substituted C.sub.4-C.sub.13 fused carbocyclyl, optionally substituted 5-13 membered fused heterocyclyl, optionally substituted C.sub.5-C.sub.13 bridged carbocyclyl, optionally substituted 5-13 membered bridged heterocyclyl, optionally substituted C.sub.5-C.sub.13 spiro carbocyclyl, optionally substituted 5-13 membered spiro heterocyclyl, optionally substituted 3-10 membered carbocyclyl, optionally substituted 3-10 membered heterocyclyl, optionally substituted aryl, and optionally substituted heteroaryl; R' and R" are independently selected from null, optionally substituted C.sub.1-C.sub.8 alkyl, optionally substituted C.sub.1-C.sub.5 heteroalkyl, optionally substituted C.sub.2-C.sub.8 alkenyl, optionally substituted C.sub.2-C.sub.8 alkynyl, optionally substituted C.sub.1-C.sub.5 hydroxyalkyl, optionally substituted C.sub.1-C.sub.8alkoxyC.sub.1-C.sub.5alkyl, optionally substituted C.sub.1-C.sub.5alkylaminoC.sub.1-C.sub.5alkyl, optionally substituted C.sub.1-C.sub.5 haloalkyl, optionally substituted C.sub.1-C.sub.8 alkylene, optionally substituted C.sub.1-C.sub.5 heteroalkylene, optionally substituted C.sub.2-C.sub.8 alkenylene, optionally substituted C.sub.2-C.sub.8 alkynylene, optionally substituted C.sub.1-C.sub.5 hydroxyalkylene, optionally substituted C.sub.1-C.sub.8alkoxyC.sub.1-C.sub.8alkylene, optionally substituted C.sub.1-C.sub.5alkylaminoC.sub.1-C.sub.8alkylene, optionally substituted C.sub.1-C.sub.5 haloalkylene; R.sup.1" and R.sup.2" are independently selected from hydrogen, optionally substituted C.sub.1-C.sub.8alkyl, optionally substituted C.sub.1-C.sub.5 heteroalkyl, optionally substituted C.sub.2-C.sub.8 alkenyl, optionally substituted C.sub.2-C.sub.8 alkynyl, optionally substituted C.sub.1-C.sub.8 alkoxyalkyl, optionally substituted C.sub.1-C.sub.6 haloalkyl, optionally substituted C.sub.1-C.sub.6 hydroxyalkyl, optionally substituted C.sub.1-C.sub.8alkylaminoC.sub.1-C.sub.5alkyl, optionally substituted 3-10 membered carbocyclyl, optionally substituted 3-10 membered heterocyclyl, optionally substituted aryl, and optionally substituted heteroaryl; and m is 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, or 15.

(283) In another aspect, provided herein is a bivalent compound comprising a Tropomyosin Receptor Kinase (TRK) ligand conjugated to a degradation tag via a linker, or a pharmaceutically acceptable salt or analog thereof, wherein the TRK ligand is a moiety of Formula 10 which connects to the linker via R.sup.4;

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wherein X.sup.1, X.sup.2, X.sup.3, X.sup.4, and X.sup.3 are each independently selected from C, CR.sup.1, and N; X is selected from CR.sup.1R.sup.2, CO, O, S, SO, SO.sub.2, and NR.sup.1; R is selected from optionally substituted C.sub.1-C.sub.8alkyl, optionally substituted C.sub.1-C.sub.8 alkyl, optionally substituted C.sub.3-C.sub.10 carbocyclyl, optionally substituted 3-10 membered heterocyclyl, and Ar.sup.1; or X—R, taken together, stands for

(285) ##STR00148## U is selected from null, a bond, C(R.sup.2).sub.2, C(R.sup.2).sub.2C(R.sup.2).sub.2, CO, C(R.sup.2).sub.2CO, CONR.sup.2, C(R.sup.2), O, C(R.sup.2) NR.sup.2 and CH.sub.2NR.sup.2; R.sup.1 and R.sup.2, at each occurrence, are independently selected from hydrogen, halogen, CN, NO.sub.2, OR.sup.6, SR.sup.6, NR.sup.1R.sup.8, COR.sup.6, CO.sub.2R.sup.6, C(O)NR.sup.7R.sup.8, SOR.sup.6, SO.sub.2R.sup.6, SO.sub.2NR.sup.7R.sup.8, NR.sup.9C(O)R.sup.8, NR.sup.9C(O)NR.sup.7R.sup.8, NR.sup.6SOR.sup.8, NR.sup.6SO.sub.2R.sup.8, optionally substituted C.sub.1-C.sub.8 alkyl, optionally substituted C.sub.1-C.sub.5 heteroalkyl, optionally

substituted C.sub.1-C.sub.8alkoxy, optionally substituted C.sub.1-C.sub.8alkoxyC.sub.1-C.sub.5alkyl, optionally substituted C.sub.1-C.sub.5alkylaminoC.sub.1-C.sub.5alkyl, optionally substituted C.sub.3-C.sub.10 carbocyclyl, optionally substituted C.sub.3-C.sub.10 cycloalkoxy, optionally substituted 3-10 membered heterocyclyl, optionally substituted C.sub.2-C.sub.8 alkenyl, and optionally substituted C.sub.2-C.sub.8 alkynyl; or R.sup.1 and R.sup.2, R.sup.1 and another R.sup.1, or R.sup.2 and another R.sup.2 together with the atom to which they are connected optionally form a 3-8 membered carbocyclyl or 3-8 membered heterocyclyl ring; R.sup.6, R.sup.7 and R.sup.8, at each occurrence, are independently selected from null, hydrogen, optionally substituted C.sub.1-C.sub.8 alkyl, optionally substituted C.sub.1-C.sub.5 heteroalkyl, optionally substituted C.sub.2-C.sub.8 alkenyl, optionally substituted C.sub.2-C.sub.8 alkynyl, optionally substituted C.sub.3-C.sub.6 carbocyclyl, optionally substituted 3-8 membered heterocyclyl, optionally substituted heterocarbocyclyl, optionally substituted arylalkyl, optionally substituted heteroarylalkyl, optionally substituted aryl, and optionally substituted heteroaryl, or R.sup.7 and R.sup.8, together with the atom to which they are connected, optionally form a 3-8 membered carbocyclyl or 3-8 membered heterocyclyl ring; n is 0, 1, 2, 3, or 4; n' is 0, 1, 2, or 3; L is Ar.sup.2, NR.sup.3CO or NR.sup.3COAr.sup.2; R.sup.3 is selected from hydrogen, optionally substituted C.sub.1-C.sub.6 alkyl, optionally substituted C.sub.1-C.sub.6 heteroalkyl, optionally substituted C.sub.3-C.sub.6 carbocyclyl, optionally substituted 3-6 membered heterocyclyl, optionally substituted C.sub.1-C.sub.6 alkoxyalkyl, optionally substituted C.sub.1-C.sub.6 haloalkyl, optionally substituted C.sub.1-C.sub.8 hydroxyalkyl, and optionally substituted C.sub.1-C.sub.8alkylaminoC.sub.1-C.sub.8alkyl; R.sup.4 is connected to the linker moiety of the bivalent compound either directly or through R.sup.5, wherein R.sup.4 and R.sup.5 are bivalent groups independently selected from null, —O—, —S—, —NR.sup.9—, —CO—, —CO.sub.2—, —CONR.sup.9—, ~SO—, —SO.sub.2R.sup.9—, —SO.sub.2NR.sup.9—, —NR.sup.10CO—, —NR.sup.10C(O)NR.sup.9—, —NR.sup.10SO—, —NR.sup.10SO.sub.2—, optionally substituted C.sub.1-C.sub.8 alkylene, optionally substituted C.sub.1-C.sub.5 heteroalkylene, optionally substituted C.sub.1-C.sub.8alkoxyC.sub.1-C.sub.8alkylene, optionally substituted C.sub.1-C.sub.8 haloalkylene, optionally substituted C.sub.1-C.sub.5 hydroxyalkylene, optionally substituted C.sub.1-C.sub.8 alkylaminoC.sub.1-C.sub.8alkylene, optionally substituted C.sub.3-C.sub.8 carbocyclylene, optionally substituted 3-8 membered heterocyclylene, optionally substituted C.sub.2-C.sub.8 alkenylene, optionally substituted C.sub.2-C.sub.8 alkynylene, optionally substituted aryl, and optionally substituted heteroaryl; R.sup.9 is selected from null, optionally substituted C.sub.1-C.sub.8 alkylene, optionally substituted C.sub.1-C.sub.8 heteroalkylene, optionally substituted C.sub.1-C.sub.8alkylene-O—, optionally substituted C.sub.1-C.sub.8 alkylene-N(C.sub.1-C.sub.8 alkyl)-, optionally substituted C.sub.1-C.sub.8 alkylene-O—C.sub.1-C.sub.8 alkylene, optionally substituted C.sub.1-C.sub.5alkylene-N(C.sub.1-C.sub.8alkyl)-C.sub.1-C.sub.5alkylene, optionally substituted C.sub.3-C.sub.8 carbocyclyl, optionally substituted C.sub.3-C.sub.8 carbocyclyl-O—, optionally substituted 3-8 membered heterocyclyl, optionally substituted aryl, and optionally substituted heteroaryl; R.sup.10 is selected from null, hydrogen, optionally substituted C.sub.1-C.sub.8 alkyl, optionally substituted C.sub.1-C.sub.8 heteroalkyl, optionally substituted C.sub.2-C.sub.8 alkenyl, optionally substituted C.sub.2-C.sub.8 alkynyl, optionally substituted C.sub.3-C.sub.8 carbocyclyl, optionally substituted heterocyclyl, optionally substituted arylalkyl, optionally substituted heteroarylalkyl, optionally substituted 3-8 membered carbocyclyl, optionally substituted 3-8 membered heterocyclyl, optionally substituted aryl, and optionally substituted heteroaryl; Ar.sup.1 and Ar.sup.2 are independently selected from aryl and heteroaryl, each of which is optionally substituted with one or more substituents independently selected from halogen, CN, NO.sub.2, OR.sup.11, SR.sup.11, NR.sup.12R.sup.13, COR.sup.11, CO.sub.2R.sup.11, CONR.sup.12R.sup.13, SOR.sup.11, SO.sub.2R.sup.11, SO.sub.2NR.sup.12R.sup.13, NR.sup.11COR.sup.13, NR.sup.11C(O)NR.sup.12R.sup.13, NR.sup.11SOR.sup.13, NR.sup.11SO.sub.2R.sup.13, optionally substituted C.sub.1-C.sub.6 alkyl,

optionally substituted C.sub.1-C.sub.5 heteroalkyl, optionally substituted C.sub.1-C.sub.6alkoxyC.sub.1-C.sub.5alkyl, optionally substituted C.sub.1-C.sub.5 haloalkyl, optionally substituted C.sub.1-C.sub.8 hydroxyalkyl, optionally substituted C.sub.1-C.sub.5alkylaminoC.sub.1-C.sub.8alkyl, optionally substituted C.sub.3-C.sub.7 carbocyclyl, optionally substituted 3-7 membered heterocyclyl, optionally substituted C.sub.2-C.sub.8 alkenyl, optionally substituted C.sub.2-C.sub.8 alkynyl, optionally substituted aryl, and optionally substituted heteroaryl; and R.sup.11, R.sup.12, and R.sup.13 are independently selected from null, hydrogen, optionally substituted C.sub.1-C.sub.6 alkyl, optionally substituted C.sub.1-C.sub.5 heteroalkyl, optionally substituted C.sub.2-C.sub.8 alkenyl, optionally substituted C.sub.2-C.sub.8 alkynyl, optionally substituted C.sub.3-C.sub.7 carbocyclyl, optionally substituted 3-7 membered heterocyclyl, optionally substituted aryl, and optionally substituted heteroaryl, or R.sup.12 and R.sup.13 together with the atom to which they are connected optionally form a 3-8 membered carbocyclyl or 3-8 membered heterocyclyl ring; wherein the degradation tag is a moiety of Formula 11 which connects to the linker via Z or any of V.sup.1, V.sup.2, V.sup.3, V.sup.4, W.sup.1, W.sup.2, and W.sup.3;

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wherein V.sup.1, V.sup.2, V.sup.3, and V.sup.4 are each independently selected from CR.sup.2' and N; W.sup.1, W.sup.2, and W.sup.3 are each independently selected from CO, O, CR.sup.3'R.sup.4', and NR.sup.3', wherein R.sup.5' is selected from a hydrogen, halogen, cyano, nitro, optionally substituted C.sub.1-C.sub.6 alkyl, optionally substituted C.sub.1-C.sub.6 heteroalkyl, optionally substituted C.sub.3-C.sub.7 carbocyclyl, optionally substituted 3-7 membered heterocyclyl, and 2,6-dioxopiperidin-3-yl optionally substituted with R.sup.1', provided that there are no two adjacent CO groups; Z is selected from null, CO, CR.sup.3'R.sup.4', NR.sup.3', O, optionally substituted C.sub.1-C.sub.10 alkylene, optionally substituted C.sub.1-C.sub.10 heteroalkylene, optionally substituted C.sub.1-C.sub.10 alkenylene, optionally substituted C.sub.1-C.sub.10 alkynylene, optionally substituted 3-7 membered carbocyclyl, and optionally substituted 3-7 membered heterocyclyl; R.sup.1' is selected from hydrogen, halogen, cyano, nitro, optionally substituted C.sub.1-C.sub.6 alkyl, optionally substituted C.sub.1-C.sub.6 heteroalkyl, optionally substituted 3 to 6 membered carbocyclyl, and optionally substituted 3 to 6 membered heterocyclyl; R.sup.2' is selected from hydrogen, halogen, cyano, nitro, optionally substituted C.sub.1-C.sub.6 alkyl, optionally substituted C.sub.1-C.sub.6 heteroalkyl, optionally substituted C.sub.1-C.sub.8 alkoxy, optionally substituted C.sub.1-C.sub.6 alkylamino, optionally substituted 3 to 6 membered carbocyclyl, and optionally substituted 3 to 6 membered heterocyclyl; and R.sup.3' and R.sup.4' at each occurrence are independently selected from hydrogen, halogen, cyano, nitro, optionally substituted C.sub.1-C.sub.6 alkyl, optionally substituted C.sub.1-C.sub.6 heteroalkyl, optionally substituted 3 to 6 membered carbocyclyl, and optionally substituted 3 to 6 membered heterocyclyl; or R.sup.3' and R.sup.4' together with the atom to which they are connected optionally form a 3-8 membered carbocyclyl or heterocyclyl ring; and wherein the linker is a moiety of Formula 9;

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wherein A and B, at each occurrence, are independently selected from null, or a bivalent moiety selected from R.sup.r—R'', R'COR'', R'CO.sub.2R'', R'C(O)N(R.sup.1'')R'', R'C(S)N(R.sup.1'')R'', R'OR'', R'SR'', R'SOR'', R'SO.sub.2R'', R'SO.sub.2N(R.sup.1'')R'', R'N(R'')R'', R''N(R.sup.1'')COR'', R''N(R'')CON(R.sup.2'')R'', R'N(R.sup.1'')C(S)R'', optionally substituted C.sub.1-C.sub.8 alkylene, optionally substituted C.sub.1-C.sub.6 heteroalkylene, optionally substituted C.sub.2-C.sub.8 alkenylene, optionally substituted C.sub.2-C.sub.8 alkynylene, optionally substituted C.sub.1-C.sub.8alkoxyC.sub.1-C.sub.8alkylene, optionally substituted C.sub.1-C.sub.8 haloalkylene, optionally substituted C.sub.1-C.sub.5 hydroxyalkylene; each W is independently selected from null, optionally substituted C.sub.4-C.sub.13 fused carbocyclyl, optionally substituted 5-13 membered fused heterocyclyl, optionally substituted C.sub.5-C.sub.13

bridged carbocyclyl, optionally substituted 5-13 membered bridged heterocyclyl, optionally substituted C.sub.5-C.sub.13 spiro carbocyclyl, optionally substituted 5-13 membered spiro heterocyclyl, optionally substituted 3-10 membered carbocyclyl, optionally substituted 3-10 membered heterocyclyl, optionally substituted aryl, and optionally substituted heteroaryl; R' and R'' are independently selected from null, optionally substituted C.sub.1-C.sub.8 alkyl, optionally substituted C.sub.1-C.sub.5 heteroalkyl, optionally substituted C.sub.2-C.sub.8 alkenyl, optionally substituted C.sub.2-C.sub.8 alkynyl, optionally substituted C.sub.1-C.sub.8 hydroxyalkyl, optionally substituted C.sub.1-C.sub.8alkoxyC.sub.1-C.sub.8alkyl, optionally substituted C.sub.1-C.sub.5alkylaminoC.sub.1-C.sub.5alkyl, optionally substituted C.sub.1-C.sub.5 haloalkyl, optionally substituted C.sub.1-C.sub.8 alkylene, optionally substituted C.sub.1-C.sub.5 heteroalkylene, optionally substituted C.sub.2-C.sub.8 alkenylene, optionally substituted C.sub.2-C.sub.8 alkynylene, optionally substituted C.sub.1-C.sub.8 hydroxyalkylene, optionally substituted C.sub.1-C.sub.8 alkoxyC.sub.1-C.sub.5alkylene, optionally substituted C.sub.1-C.sub.8alkylaminoC.sub.1-C.sub.8alkylene, optionally substituted C.sub.1-C.sub.8 haloalkylene; R.sup.1'' and R.sup.2'' are independently selected from hydrogen, optionally substituted C.sub.1-C.sub.6 alkyl, optionally substituted C.sub.1-C.sub.5 heteroalkyl, optionally substituted C.sub.2-C.sub.8 alkenyl, optionally substituted C.sub.2-C.sub.8 alkynyl, optionally substituted C.sub.1-C.sub.6 alkoxyalkyl, optionally substituted C.sub.1-C.sub.5 haloalkyl, optionally substituted C.sub.1-C.sub.6 hydroxyalkyl, optionally substituted C.sub.1-C.sub.8alkylaminoC.sub.1-C.sub.5alkyl, optionally substituted 3-10 membered carbocyclyl, optionally substituted 3-10 membered heterocyclyl, optionally substituted aryl, and optionally substituted heteroaryl; and m is 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, or 15.

(288) In certain embodiments of the bivalent compounds disclosed herein, U is selected from a bond, CH.sub.2, CH.sub.2CH.sub.2, CO, CH.sub.2CO, CONH, CONCH.sub.3, CH.sub.2O, CH.sub.2NH, and CH.sub.2NCH.sub.3.

(289) In certain embodiments of the bivalent compounds disclosed herein, R.sup.1 and R.sup.2, at each occurrence, are independently selected from hydrogen, F, Cl, OH, optionally substituted C.sub.1-C.sub.4 alkyl, optionally substituted C.sub.1-C.sub.4 heteroalkyl, optionally substituted C.sub.1-C.sub.4 alkoxy, optionally substituted C.sub.1-C.sub.4 alkylamino, optionally substituted C.sub.1-C.sub.4 haloalkyl, optionally substituted C.sub.3-C.sub.6 carbocyclyl, optionally substituted C.sub.3-C.sub.6 cycloalkoxy, and optionally substituted 3-6 membered heterocyclyl.

(290) In certain embodiments of the bivalent compounds disclosed herein, R.sup.1 and R.sup.2, at each occurrence, are independently selected from hydrogen, F, Cl, CH.sub.3, CF.sub.3, and cyclopropyl.

(291) In certain embodiments of the bivalent compounds disclosed herein, Ar.sup.1 is selected from optionally substituted C.sub.6-C.sub.10 aryl and optionally substituted C.sub.5-C.sub.10 heteroaryl.

(292) In certain embodiments of the bivalent compounds disclosed herein, Art is selected from 3-fluorophenyl, 3,5-difluorophenyl, and 2,5-difluorophenyl.

(293) In certain embodiments of the bivalent compounds disclosed herein, L is Ar.sup.2 and R.sup.4 is connected to the linker moiety of the bivalent compound through R.sup.5.

(294) In certain embodiments of the bivalent compounds disclosed herein, L is Ar.sup.2 and R.sup.4 is connected to the linker moiety of the bivalent compound directly.

(295) In certain embodiments of the bivalent compounds disclosed herein, R.sup.4 is selected from

(296) ##STR00151##

(297) In certain embodiments of the bivalent compounds disclosed herein, L is Ar.sup.2, and Ar.sup.2 is selected from optionally substituted C.sub.1-C.sub.10 aryl and optionally substituted C.sub.5-C.sub.10 heteroaryl.

(298) In certain embodiments of the bivalent compounds disclosed herein, X—R, taken together, stands for

(299) ##STR00152##

(300) In certain embodiments of the bivalent compounds disclosed herein, X^{sup.1} is N; X^{sup.2} is N; X^{sup.3} is N; X^{sup.4} is CH; X^{sup.5} is C; X—R, taken together, stands for

(301) ##STR00153##

U is CH_{sub.2}; Ar^{sup.1} is 3-fluorophenyl; L is Ar^{sup.2}; and Ar^{sup.2} is 2-pyridyl.

(302) In certain embodiments of the bivalent compounds disclosed herein, L is Ar^{sup.2} or NR^{sup.3}COAr^{sup.2} and Ar^{sup.2}—R^{sup.4} is selected from a moiety of formulae B^{sup.1}, B^{sup.2}, and B^{sup.3};

(303) ##STR00154## wherein * indicates the connection to the linker moiety of the bivalent compound; Y^{sup.1}, Y^{sup.2}, Y^{sup.3}, and Y^{sup.4} are independently selected from CR^{sup.a} and N, with the proviso that up to 3 of Y^{sup.1}, Y^{sup.2}, Y^{sup.3}, and Y^{sup.4} are N; each R^{sup.a} is independently selected from hydrogen, halogen, CN, NO_{sub.2}, OR^{sup.14}, NR^{sup.15}R^{sup.16}, COR^{sup.14}, CO_{sub.2}R^{sup.14}, CONR^{sup.15}R^{sup.16}, SOR^{sup.14}, SO_{sub.2}R^{sup.14}, SO_{sub.2}NR^{sup.15}R^{sup.16}, NR^{sup.14}COR^{sup.15}, NR^{sup.14}C(O)NR^{sup.15}R^{sup.16}, NR^{sup.14}SOR^{sup.15}, NR^{sup.14}SO_{sub.2}R^{sup.15}, optionally substituted C_{sub.1}-C_{sub.8} alkyl, optionally substituted C_{sub.1}-C_{sub.5} heteroalkyl, optionally substituted C_{sub.1}-C_{sub.8}alkoxyC_{sub.1}-C_{sub.8}alkyl, optionally substituted C_{sub.1}-C_{sub.5} haloalkyl, optionally substituted C_{sub.1}-C_{sub.5} hydroxyalkyl, optionally substituted C_{sub.1}-C_{sub.5}alkylaminoC_{sub.1}-C_{sub.5}alkyl, optionally substituted C_{sub.3}-C_{sub.7} carbocyclyl, optionally substituted 3-7 membered heterocyclyl, optionally substituted C_{sub.2}-C_{sub.8} alkenyl, optionally substituted C_{sub.2}-C_{sub.8} alkynyl, optionally substituted aryl, and optionally substituted heteroaryl; and R^{sup.14}, R^{sup.15} and R^{sup.16}, at each occurrence, are independently selected from null, hydrogen, optionally substituted C_{sub.1}-C_{sub.8} alkyl, optionally substituted C_{sub.1}-C_{sub.5} heteroalkyl, optionally substituted C_{sub.2}-C_{sub.8} alkenyl, optionally substituted C_{sub.2}-C_{sub.8} alkynyl, optionally substituted C_{sub.3}-C_{sub.6} carbocyclyl, optionally substituted 3-8 membered heterocyclyl, optionally substituted heterocarbocyclyl, optionally substituted arylalkyl, optionally substituted heteroarylalkyl, optionally substituted aryl, and optionally substituted heteroaryl, or R^{sup.15} and R^{sup.16}, together with the atom to which they are connected, optionally form a 3-8 membered carbocyclyl or 3-8 membered heterocyclyl ring.

(304) In certain embodiments of the bivalent compounds disclosed herein, L is Ar^{sup.2} or NR^{sup.3}COAr^{sup.2} and Ar^{sup.2}—R^{sup.4} is selected from a moiety of formula B^{sup.4};

(305) ##STR00155## wherein * indicates the connection to the linker moiety of the bivalent compound; Y^{sup.1}, Y^{sup.2'}, Y^{sup.3'}, and Y^{sup.4'} are independently selected from CR^{sup.a}, N, O, and S, with the proviso that up to 3 of Y^{sup.1'}, Y^{sup.2'}, Y^{sup.3'}, and Y^{sup.4'} are N; each R^{sup.a} is independently selected from hydrogen, halogen, CN, NO_{sub.2}, OR^{sup.14}, NR^{sup.15}R^{sup.16}, COR^{sup.14}, CO_{sub.2}R^{sup.14}, CONR^{sup.15}R^{sup.16}, SOR^{sup.14}, SO_{sub.2}R^{sup.14}, SO_{sub.2}NR^{sup.15}R^{sup.16}, NR^{sup.14}COR^{sup.15}, NR^{sup.14}C(O)NR^{sup.15}R^{sup.16}, NR^{sup.14}SOR^{sup.15}, NR^{sup.14}SO_{sub.2}R^{sup.15}, optionally substituted C_{sub.1}-C_{sub.8} alkyl, optionally substituted C_{sub.1}-C_{sub.6} heteroalkyl, optionally substituted C_{sub.1}-C_{sub.8}alkoxyC_{sub.1}-C_{sub.5}alkyl, optionally substituted C_{sub.1}-C_{sub.8} haloalkyl, optionally substituted C_{sub.1}-C_{sub.5} hydroxyalkyl, optionally substituted C_{sub.1}-C_{sub.5}alkylaminoC_{sub.1}-C_{sub.8}alkyl, optionally substituted C_{sub.3}-C_{sub.7} carbocyclyl, optionally substituted 3-7 membered heterocyclyl, optionally substituted C_{sub.2}-C_{sub.8} alkenyl, optionally substituted C_{sub.2}-C_{sub.8} alkynyl, optionally substituted aryl, and optionally substituted heteroaryl; and R^{sup.14}, R^{sup.15} and R^{sup.16}, at each occurrence, are independently selected from null, hydrogen, optionally substituted C_{sub.1}-C_{sub.8} alkyl, optionally substituted C_{sub.1}-C_{sub.6} heteroalkyl, optionally substituted C_{sub.2}-C_{sub.8} alkenyl, optionally substituted C_{sub.2}-C_{sub.8} alkynyl, optionally substituted C_{sub.3}-C_{sub.6} carbocyclyl, optionally substituted 3-8 membered heterocyclyl, optionally substituted heterocarbocyclyl, optionally substituted arylalkyl, optionally substituted heteroarylalkyl, optionally substituted aryl,

and optionally substituted heteroaryl, or R.sup.15 and R.sup.16, together with the atom to which they are connected, optionally form a 3-8 membered carbocyclyl or 3-8 membered heterocyclyl ring.

(306) In certain embodiments of the bivalent compounds disclosed herein, L is NR.sup.3CO, and R.sup.3 is selected from hydrogen, CH.sub.3, CH.sub.2CH.sub.3, propyl, isopropyl, cyclopropyl, CH.sub.2F, CHF.sub.2, and CF.sub.3.

(307) In certain embodiments of the bivalent compounds disclosed herein, L is NR.sup.3CO and R.sup.4 is connected to the linker moiety of the bivalent compound directly.

(308) In certain embodiments of the bivalent compounds disclosed herein, L is NR.sup.3CO and R.sup.4 is connected to the linker moiety of the bivalent compound through R.sup.5.

(309) In certain embodiments of the bivalent compounds disclosed herein, L is NR.sup.3COAr.sup.2, wherein R.sup.3 is selected from hydrogen, CH.sub.3, CH.sub.2CH.sub.3, propyl, isopropyl, cyclopropyl, CH.sub.2F, CHF.sub.2, and CF.sub.3; and Ar.sup.2 is selected from optionally substituted C.sub.6-C.sub.10 aryl and optionally substituted C.sub.5-C.sub.10 heteroaryl.

(310) In certain embodiments of the bivalent compounds disclosed herein, L is NR.sup.3COAr.sup.2 and R.sup.4 is connected to the linker moiety of the bivalent compound directly.

(311) In certain embodiments of the bivalent compounds disclosed herein, L is NR.sup.3COAr.sup.2 and R.sup.4 is connected to the linker moiety of the bivalent compound through R.sup.5.

(312) In certain embodiments of the bivalent compounds disclosed herein, L is NR.sup.3COAr.sup.2 and Ar.sup.2—R.sup.4 is selected from a moiety of formulae B1, B2 and B3.

(313) In certain embodiments of the bivalent compounds disclosed herein, L is NR.sup.3COAr.sup.2 and Ar.sup.2—R.sup.4 is selected from a moiety of formula B1, where in Y.sup.1 is CH or N.

(314) In certain embodiments of the bivalent compounds disclosed herein, L is NR.sup.3COAr.sup.2 and Ar.sup.2—R.sup.4 is selected from a moiety of formula B2, where in Y.sup.1 is CR.sup.a, and R.sup.a is

(315) ##STR00156##

(316) In certain embodiments of the bivalent compounds disclosed herein, L is NR.sup.3COAr.sup.2 and Ar.sup.2—R.sup.4 is selected from a moiety of formula B3, wherein Y.sup.2 is CR.sup.a, and R.sup.a is

(317) ##STR00157##

(318) In certain embodiments of the bivalent compounds disclosed herein, L is NR.sup.3COAr.sup.2 and X—R is X—Ar.sup.1.

(319) In certain embodiments of the bivalent compounds disclosed herein, L is NR.sup.3COAr.sup.2 and X—R is X—Ar.sup.1, wherein X is CH.sub.2; and Ar.sup.1 is selected from 3-fluorophenyl, 3,5-di fluorophenyl, or 2,5-di fluorophenyl.

(320) In certain embodiments of the bivalent compounds disclosed herein, V.sup.1, V.sup.2, V.sup.3, and V.sup.4 are each independently CR.sup.2'.

(321) In certain embodiments of the bivalent compounds disclosed herein, at least one of W.sup.1, W.sup.2, and W.sup.3 is NR.sup.5'.

(322) In certain embodiments of the bivalent compounds disclosed herein, at least one of W.sup.1, W.sup.2, and W.sup.3 is NR.sup.5', and only one of R.sup.5' is 2,6-dioxopiperidin-3-yl optionally substituted with R'''. In certain embodiments of the bivalent compounds disclosed herein, W.sup.1 is CO.

(323) In certain embodiments of the bivalent compounds disclosed herein, W.sup.2 is CO.

(324) In certain embodiments of the bivalent compounds disclosed herein, W.sup.3 is CO.

(325) In certain embodiments of the bivalent compounds disclosed herein, W.sup.1 and W.sup.3 are

both CO. In certain embodiments of the bivalent compounds disclosed herein, W.sup.2 is CO and W.sup.1 and W.sup.3 are both NR.sup.5', and only one of R.sup.5' is 2,6-dioxopiperidin-3-yl optionally substituted with R.sup.1'.

(326) In certain embodiments of the bivalent compounds disclosed herein, W, at each occurrence, is independently selected from Formulae C.sub.1, C.sub.2, C.sub.3, C.sub.4 and CS:

(327) ##STR00158##

wherein X' and Y' are independently selected from N, and CR.sup.b; A.sup.1, B.sup.1, C.sub.1, and D.sup.1, at each occurrence, are independently selected from null, O, CO, SO, SO.sub.2, NR.sup.9, and CR R.sup.c; A.sup.2, B.sup.2, C.sub.2, and D.sup.2, at each occurrence, are independently selected from N, and CR.sup.b; A.sup.3, B.sup.3, C.sub.3, D.sup.3, and E.sup.3, at each occurrence, are independently selected from N, O, S, NR.sup.b, and CR.sup.b; R.sup.b and R.sup.c, at each occurrence, are independently selected from hydrogen, halogen, hydroxyl, amino, cyano, nitro, optionally substituted C.sub.1-C.sub.6 alkyl, optionally substituted C.sub.1-C.sub.5 heteroalkyl, optionally substituted C.sub.2-C.sub.8 alkenyl, optionally substituted C.sub.2-C.sub.8 alkynyl, optionally substituted C.sub.1-C.sub.8 alkoxy, optionally substituted C.sub.1-C.sub.8alkoxyalkyl, optionally substituted C.sub.1-C.sub.5 haloalkyl, optionally substituted C.sub.1-C.sub.8 hydroxyalkyl, optionally substituted C.sub.1-C.sub.8 alkylamino, and optionally substituted C.sub.1-C.sub.8 alkylaminoC.sub.1-C.sub.8 alkyl, optionally substituted 3-10 membered carbocyclyl, optionally substituted 3-8 membered cycloalkoxy, optionally substituted 3-10 membered carbocyclylamino, optionally substituted 3-8 membered membered heterocyclyl, optionally substituted aryl, and optionally substituted heteroaryl; or R.sup.b and R.sup.b, or R.sup.b and R.sup.c together with the atom to which they are connected form a 3-8 membered carbocyclyl or 3-8 membered heterocyclyl ring; and m.sup.1, n.sup.1, o.sup.1 and p.sup.1 are independently selected from 0, 1, 2, 3, 4 and 5.

(328) In certain embodiments of the bivalent compounds disclosed herein, the length of the linker is 3 to atoms.

(329) In certain embodiments of the bivalent compounds disclosed herein, the length of the linker is 3 to atoms.

(330) In certain embodiments of the bivalent compounds disclosed herein, the length of the linker is 3 to atoms.

(331) In certain embodiments of the bivalent compounds disclosed herein, the linker is selected from optionally substituted —(CO)—(CH.sub.2).sub.0-3—W—(CH.sub.2).sub.0-3—, —(CH.sub.2).sub.0-3—W—(CH.sub.2).sub.0-3—, optionally substituted —(CO)—(CH.sub.2).sub.0-8—, optionally substituted —(CH.sub.2).sub.0-9—, optionally substituted —(CH.sub.2).sub.1-2—(CO)—NH—(CH.sub.2).sub.2-9-, optionally substituted —(CH.sub.2).sub.1-2—(CO)—NH—(CH.sub.2).sub.1-3—(OCH.sub.2CH.sub.2).sub.1-7—, optionally substituted —(CH.sub.2).sub.0-1—(CO)—(CH.sub.2).sub.1-3—(OCH.sub.2CH.sub.2).sub.1-7—, optionally substituted —(CO)—(CH.sub.2).sub.0-3-(alkenylene)-(CH.sub.2).sub.0-3—, optionally substituted —(CO)—(CH.sub.2).sub.0-3-(alkynylene)-(CH.sub.2).sub.0-3—, optionally substituted —(CH.sub.2).sub.0-3-(alkenylene)-(CH.sub.2).sub.0-3—, optionally substituted (CH.sub.2).sub.0-3-(alkynylene)-(CH.sub.2).sub.0-3—, optionally substituted —(CH.sub.2).sub.0-3—(CO)—(CH.sub.2).sub.0-3—W—(CH.sub.2).sub.0-3—, optionally substituted —(CH.sub.2).sub.0-3—O—(CH.sub.2).sub.0-3—W—(CH.sub.2).sub.0-3—, optionally substituted —(CH.sub.2).sub.0-3—NH—(CH.sub.2).sub.0-3—W—(CH.sub.2).sub.0-3—, optionally substituted —(CH.sub.2).sub.0-3—O—(CH.sub.2).sub.0-3—W—(CH.sub.2).sub.0-3—O—(CH.sub.2).sub.0-3—, optionally substituted —(CH.sub.2).sub.0-3—NH—(CH.sub.2).sub.0-3—W—(CH.sub.2).sub.0-3—NH—(CH.sub.2).sub.0-3—, optionally substituted —(CH.sub.2).sub.0-3—O—(CH.sub.2).sub.0-3—W—(CH.sub.2).sub.0-3—NH—(CH.sub.2).sub.0-3—; and W is selected from optionally substituted 3-10 membered carbocyclylene, optionally substituted 3-10 membered heterocyclylene, optionally substituted C.sub.4-C.sub.13 fused carbocyclylene, optionally substituted 5-13 membered fused

heterocyclylene, optionally substituted C.sub.5-C.sub.13 bridged carbocyclylene, optionally substituted 5-13 membered bridged heterocyclylene, optionally substituted C.sub.5-C.sub.13 spiro carbocyclylene, optionally substituted 5-13 membered spiro heterocyclylene, optionally substituted arylene, and optionally substituted heteroarylene.

(332) In certain embodiments of the bivalent compounds disclosed herein, the linker is selected from optionally substituted —(CO)—(CH.sub.2).sub.0-3-(3-8 membered carbocyclyl)-(CH.sub.2).sub.0-3—, optionally substituted-(CO)—(CH.sub.2).sub.0-3-(3-8 membered heterocyclyl)-(CH.sub.2).sub.0-3—, optionally substituted —(CH.sub.2).sub.0-3-(3-8 membered carbocyclyl)-(CH.sub.2).sub.0-3—, and optionally substituted —(CH.sub.2).sub.0-3-(3-8 membered heterocyclyl)-(CH.sub.2).sub.0-3.

(333) In certain embodiments of the bivalent compounds disclosed herein, W is selected from

(334) ##STR00159## ##STR00160## ##STR00161## ##STR00162## ##STR00163##
##STR00164##

(335) In some embodiments, the bivalent compound is selected from the group consisting of CPD-001 to CPD-516, or a pharmaceutically acceptable salt or analog thereof.

(336) In some embodiments, the bivalent compound is selected from the group consisting of CPD-009, CPD-010, CPD-013, CPD-014, CPD-015, CPD-021, CPD-022, CPD-023, CPD-024, CPD-025, CPD-026, CPD-027, CPD-028, CPD-029, CPD-030, CPD-031, CPD-032, CPD-033, CPD-044, CPD-047, CPD-049, CPD-050, CPD-051, CPD-052, CPD-053, CPD-054, CPD-055, CPD-056, CPD-057, CPD-059, CPD-060, CPD-062, CPD-064, CPD-065, TR-104, TR-105, TR-106, TR-107, TR-108, TR-109, TR-113, TR-115, TR-116, TR-117, TR-118, TR-119, TR-120, TR-121, TR-122, TR-123, TR-124, TR-125, TR-127, TR-128, TR-129, TR-130, TR-131, TR-132, TR-134, TR-135, TR-137, TR-140, TR-141, TR-142, TR-143, TR-144, TR-145, TR-146, TR-147, TR-149, TR-151, TR-152, TR-153, TR-155, TR-156, TR-157, TR-158, TR-160, TR-161, TR-162, TR-163, TR-164, TR-165, TR-166, TR-167, TR-168, TR-169, TR-171, TR-172, TR-173, TR-176, TR-177, TR-181, TR-182, TR-184, TR-185, TR-186, TR-189, TR-190, TR-191, TR-194, TR-196, TR-198, TR-202, TR-203, TR-204, TR-208, TR-211, TR-216, TR-217, TR-220, TR-221, TR-223, TR-224, TR-225, TR-226, TR-TR-231, TR-232, TR-233, TR-235, TR-241, TR-247, TR-249, TR-250, TR-253, TR-254, TR-255, TR-258, TR-259, TR-260, TR-263, TR-264, TR-265, TR-266, TR-267, TR-268, TR-270, TR-275, TR-276, TR-279, TR-280, TR-281, TR-282, TR-284, TR-285, TR-286, TR-287, TR-288, TR-289, TR-290, TR-292, TR-293, TR-294, TR-301, TR-302, TR-303, TR-304, TR-305, TR-306, TR-308, TR-309, TR-315, TR-316, TR-317, TR-318, TR-319, TR-320, TR-321, TR-324, TR-325, TR-326, TR-327, TR-331, TR-332, TR-333, TR-335, TR-336, TR-337, TR-338, TR-339, TR-340, TR-341, TR-342, TR-343, TR-344, CPD-470, CPD-471, CPD-472, CPD-473, CPD-474, CPD-475, CPD-476, CPD-478, CPD-480, CPD-481, CPD-482, CPD-483, CPD-484, CPD-499, CPD-500, CPD-501, and a pharmaceutically acceptable salt or analog thereof.

(337) In some embodiments, the bivalent compound is selected from the group consisting of TR-106, TR-108, TR-109, TR-113, TR-115, TR-116, TR-117, TR-119, TR-121, TR-122, TR-123, TR-124, TR-125, TR-127, TR-128, TR-129, TR-130, TR-131, TR-132, TR-135, TR-137, TR-140, TR-141, TR-142, TR-143, TR-144, TR-145, TR-146, TR-149, TR-151, TR-152, TR-155, TR-156, TR-160, TR-161, TR-162, TR-165, TR-166, TR-167, TR-168, TR-169, TR-171, TR-172, TR-173, TR-176, TR-177, TR-181, TR-182, TR-184, TR-185, TR-186, TR-189, TR-190, TR-191, TR-194, TR-196, TR-198, TR-204, TR-208, TR-211, TR-216, TR-217, TR-220, TR-221, TR-224, TR-225, TR-226, TR-TR-231, TR-232, TR-233, TR-241, TR-247, TR-249, TR-250, TR-253, TR-254, TR-255, TR-258, TR-259, TR-260, TR-263, TR-264, TR-265, TR-266, TR-267, TR-270, TR-275, TR-276, TR-279, TR-280, TR-281, TR-282, TR-284, TR-285, TR-286, TR-287, TR-288, TR-289, TR-290, TR-292, TR-293, TR-301, TR-302, TR-304, TR-305, TR-306, TR-308, TR-309, TR-315, TR-316, TR-317, TR-318, TR-319, TR-320, TR-321, TR-324, TR-325, TR-331, TR-332, TR-335, TR-336, TR-337, TR-338, TR-339, TR-340, TR-341, TR-342, TR-343, TR-344, CPD-470, CPD-471, CPD-472, CPD-473, CPD-474, CPD-475, CPD-476, CPD-478, CPD-480, CPD-481, CPD-482, CPD-

483, CPD-484, CPD-499, CPD-500, CPD-501, and a pharmaceutically acceptable salt or analog thereof.

(338) In some embodiments, the bivalent compound is selected from the group consisting of TR-123, TR-172, TR-173, TR-181, TR-182, TR-184, TR-185, TR-186, TR-191, TR-196, TR-198, TR-204, TR-221, TR-224, TR-225, TR-226, TR-231, TR-233, TR-241, TR-249, TR-254, TR-258, TR-259, TR-260, TR-263, TR-264, TR-265, TR-266, TR-267, TR-270, TR-275, TR-276, TR-279, TR-280, TR-281, TR-282, TR-284, TR-285, TR-286, TR-287, TR-288, TR-290, TR-292, TR-293, TR-301, TR-302, TR-304, TR-306, TR-308, TR-309, TR-315, TR-316, TR-317, TR-318, TR-319, TR-320, TR-321, TR-324, TR-325,

(339) TR-331, TR-332, TR-335, TR-336, TR-337, TR-338, TR-339, TR-340, TR-341, TR-342, TR-343, TR-344, CPD-470, CPD-471, CPD-472, CPD-473, CPD-474, CPD-475, CPD-476, CPD-478, CPD-480, CPD-481, CPD-482, CPD-483, CPD-484, CPD-499, CPD-500, CPD-501, and a pharmaceutically acceptable salt or analog thereof.

(340) In some embodiments, the bivalent compound is not any of CPD-001 to CPD-246.

(341) In some embodiments, the bivalent compound is selected from the group consisting of CPD-247 to CPD-516, or a pharmaceutically acceptable salt or analog thereof.

(342) In some embodiments, the bivalent compound is selected from the group consisting of TR-247, TR-249, TR-250, TR-253, TR-254, TR-255, TR-258, TR-259, TR-260, TR-263, TR-264, TR-265, TR-266, TR-267, TR-268, TR-270, TR-275, TR-276, TR-279, TR-280, TR-281, TR-282, TR-284, TR-285, TR-286, TR-287, TR-288, TR-289, TR-290, TR-292, TR-293, TR-294, TR-301, TR-302, TR-303, TR-304, TR-305, TR-306, TR-308, TR-309, TR-315, TR-316, TR-317, TR-318, TR-319, TR-320, TR-321, TR-324, TR-325, TR-326, TR-327, TR-331, TR-332, TR-333, TR-335, TR-336, TR-337, TR-338, TR-339, TR-340, TR-341, TR-342, TR-343, TR-344, CPD-470, CPD-471, CPD-472, CPD-473, CPD-474, CPD-475, CPD-476, CPD-478, CPD-480, CPD-481, CPD-482, CPD-483, CPD-484, CPD-499, CPD-500, CPD-501, and a pharmaceutically acceptable salt or analog thereof.

(343) In some embodiments, the bivalent compound comprises an enantiomer of a compound described herein. In some embodiments, the bivalent compound comprises an(S) enantiomer. In some embodiments, the bivalent compound comprises an(R) enantiomer. Some embodiments comprise a composition comprising the bivalent compound. In some embodiments, the composition comprises or consists of the(S) enantiomer of the compound. In some embodiments, the composition comprises or consists of the(R) enantiomer of the compound. In some embodiments, the composition comprises or consists of a mixture of the(S) enantiomer and the(R) enantiomer. In some embodiments, the composition comprises or consists of a racemic mixture of the(S) enantiomer and the(R) enantiomer. In some embodiments, the composition is a pharmaceutical composition.

(344) In some embodiments, the bivalent compound is 2-(2,6-dioxopiperidin-3-yl)-5-((2-(4-(6-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)-2-oxoethyl)amino) isoindoline-1,3-dione (TR-123).

(345) In some embodiments, the bivalent compound is 2-(2,6-dioxopiperidin-3-yl)-5-(3-(4-(6-(6-((R)-2~ (3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)propyl)isoindoline-1,3-dione (TR-172).

(346) In some embodiments, the bivalent compound is 2~ (2,6-dioxopiperidin-3-yl)-5-(2-(4-(6-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)ethoxy) isoindoline-1,3-dione (TR-173).

(347) In some embodiments, the bivalent compound is 2-(2,6-dioxopiperidin-3-yl)-5-((2-(4-(6-(6-((R)-2~ (3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)ethyl)amino) isoindoline-1,3-dione (TR-181).

(348) In some embodiments, the bivalent compound is 2-(2,6-dioxopiperidin-3-yl)-5-(4-(4-(6-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-

yl)piperidin-1-yl)isoindoline-1,3-dione (TR-182).

(349) In some embodiments, the bivalent compound is 2-(2,6-dioxopiperidin-3-yl)-5-(3-(4-(6-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)azetidin-1-yl)isoindoline-1,3-dione (TR-184);

(350) In some embodiments, the bivalent compound is 3-(6-(3-(4-(6-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)propyl)-1-oxoisoindolin-2-yl)piperidine-2,6-dione (TR-185).

(351) In some embodiments, the bivalent compound is 3-(5-(3-(4-(6-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)propyl)-1-oxoisoindolin-2-yl)piperidine-2,6-dione (TR-186).

(352) In some embodiments, the bivalent compound is 3-(5-((2-(4-(6-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)ethyl)amino)-1-oxoisoindolin-2-yl)piperidine-2,6-dione (TR-191).

(353) In some embodiments, the bivalent compound is 3-(6-((2-(4-(6-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)ethyl)amino)-1-oxoisoindolin-2-yl)piperidine-2,6-dione (TR-196).

(354) In some embodiments, the bivalent compound is 2-(2,6-dioxopiperidin-3-yl)-5-(3-((4-(6-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)methyl)azetidin-1-yl)isoindoline-1,3-dione (TR-198).

(355) In some embodiments, the bivalent compound is 2-(2,6-dioxopiperidin-3-yl)-5-(2-(4-(6-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)ethyl)isoindoline-1,3-dione (TR-204).

(356) In some embodiments, the bivalent compound is 3-[5-[3-[4-[6-[6-[(2R)-2-(3-fluorophenyl)pyrrolidin-1-yl]imidazo[1,2-b]pyridazin-3-yl]-2-pyridyl]piperazin-1-yl]propylamino]-3-methyl-2-oxo-benzimidazol-1-yl]piperidine-2,6-dione (TR-221).

(357) In some embodiments, the bivalent compound is 3-((S)-5-(4-(3-(4-(6-(6-((S)-2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)prop-1-yn-1-yl)phenyl)-2-oxoxazolidin-3-yl)piperidine-2,6-dione (TR-224).

(358) In some embodiments, the bivalent compound is 2-(2,6-dioxopiperidin-3-yl)-5-(3-((4-(6-(6-((R)-2~ (3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)methyl)pyrrolidin-1-yl)isoindoline-1,3-dione (TR-225).

(359) In some embodiments, the bivalent compound is 2-(2,6-dioxopiperidin-3-yl)-5-(4-((4-(6-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)methyl)piperidin-1-yl)isoindoline-1,3-dione (TR-226).

(360) In some embodiments, the bivalent compound is N-[5-[(3,5-difluorophenyl)methyl]-1H-indazol-3-yl]-4-[4-[1-[2-(2,6-dioxo-3-piperidyl)-1,3-dioxo-isoindolin-5-yl]azetidin-3-yl]methyl]piperazin-1-yl]~2-(tetrahydropyran-4-ylamino)benzamide (TR-231).

(361) In some embodiments, the bivalent compound is N-(2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisoindolin-5-yl)-2-(4-(6-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)acetamide (TR-233).

(362) In some embodiments, the bivalent compound is 3-(4-(4-(4-(6-(6-((R)-2~ (3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)butyl)-3-methyl-2-oxo-2,3-dihydro-1H-benzo[d]imidazol-1-yl)piperidine-2,6-dione (TR-241).

(363) In some embodiments, the bivalent compound is N-(5-(3,5-difluorobenzyl)-1H-indazol-3-yl)-4-(4-((1-((2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisoindolin-5-yl)methyl)azetidin-3-yl)methyl)piperazin-1-yl)-2-((tetrahydro-2H-pyran-4-yl)amino)benzamide (TR-249).

(364) In some embodiments, the bivalent compound is N-(5-(3,5-difluorobenzyl)-1H-indazol-3-yl)-4-(4-(1-((2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisoindolin-5-yl)methyl)azetidin-3-yl)piperazin-1-yl)-2-((tetrahydro-2H-pyran-4-yl)amino)benzamide (TR-254).

(365) In some embodiments, the bivalent compound is 3-(4-(3-((4-(6-(6-((R)-2-(3-

fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)methyl)azetidin-1-yl)-3-methyl-2-oxo-2,3-dihydro-1H-benzo[d]imidazol-1-yl)piperidine-2,6-dione (TR-258),

(366) In some embodiments, the bivalent compound is 3-(5-(4-(4-(6-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)butyl)-3-methyl-2-oxo-2,3-dihydro-1H-benzo[d]imidazol-1-yl)piperidine-2,6-dione (TR-259).

(367) In some embodiments, the bivalent compound is 3-(5-(4-(4-(6-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)but-1-yn-1-yl)-3-methyl-2-oxo-2,3-dihydro-1H-benzo[d]imidazol-1-yl)piperidine-2,6-dione (TR-260).

(368) In some embodiments, the bivalent compound is N-(5-(3,5-difluorobenzyl)-1H-indazol-3-yl)-4-(4-(2-((2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-5-yl)amino)ethyl)piperazin-1-yl)benzamide (TR-263).

(369) In some embodiments, the bivalent compound is N-(5-(3,5-difluorobenzyl)-1H-indazol-3-yl)-4-(4-((2~ (2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-5-yl)glycyl)piperazin-1-yl)benzamide (TR-264).

(370) In some embodiments, the bivalent compound is N-(5~ (3,5-difluorobenzyl)-1H-indazol-3-yl)-4-(4-((1-(2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-5-yl)azetidin-3-yl)methyl)piperazin-1-yl)benzamide (TR-265).

(371) In some embodiments, the bivalent compound is N-(5-(3,5-difluorobenzyl)-1H-indazol-3-yl)-4-(4~ ((2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-5-yl)glycyl)piperazin-1-yl)-2-((2-fluoroethyl)amino)benzamide (TR-266).

(372) In some embodiments, the bivalent compound is N-(5-(3,5-difluorobenzyl)-1H-indazol-3-yl)-4-(4-((1-(2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-5-yl)azetidin-3-yl)methyl)piperazin-1-yl)-2-((2-fluoroethyl)amino)benzamide (TR-267).

(373) In some embodiments, the bivalent compound is N-(5-(3,5-difluorobenzyl)-1H-indazol-3-yl)-4-(4-((1-(2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-5-yl)pyrrolidin-3-yl)methyl)piperazin-1-yl)-2-((tetrahydro-2H-pyran-4-yl)amino)benzamide (TR-270).

(374) In some embodiments, the bivalent compound is N-(5-(3,5-difluorobenzyl)-1H-indazol-3-yl)-4-(4-(1-(2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-5-yl)azetidin-3-yl)piperazin-1-yl)-2-((tetrahydro-2H-pyran-4-yl)amino)benzamide (TR-275).

(375) In some embodiments, the bivalent compound is N-(5-(3,5-difluorobenzyl)-1H-indazol-3-yl)-4-(4-(3-(2~ (2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-5-yl)propyl)piperazin-1-yl)-2-((tetrahydro-2H-pyran-4-yl)amino)benzamide (TR-276).

(376) In some embodiments, the bivalent compound is N-(5-(3,5-difluorobenzyl)-1H-indazol-3-yl)-4-(4-(1-((2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-5-yl)methyl)piperidin-3-yl)piperazin-1-yl)-2-((tetrahydro-2H-pyran-4-yl)amino)benzamide (TR-279).

(377) In some embodiments, the bivalent compound is N-(5-(3,5-difluorobenzyl)-1H-indazol-3-yl)-4-(4-(1-(2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-5-yl)piperidin-4-yl)piperazin-1-yl)-2-((tetrahydro-2H-pyran-4-yl)amino)benzamide (TR-280).

(378) In some embodiments, the bivalent compound is N-(5-(3,5-difluorobenzyl)-1H-indazol-3-yl)-4-(4-(1-((2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-5-yl)methyl)piperidin-4-yl)piperazin-1-yl)-2-((tetrahydro-2H-pyran-4-yl)amino)benzamide (TR-281).

(379) In some embodiments, the bivalent compound is N-(5-(3,5-difluorobenzyl)-1H-indazol-3-yl)-4-(4-((4-(2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-5-yl)morpholin-2-yl)methyl)piperazin-1-yl)-2-((tetrahydro-2H-pyran-4-yl)amino)benzamide (TR-282).

(380) In some embodiments, the bivalent compound is N-(5-(3,5-difluorobenzyl)-1H-indazol-3-yl)-4-(4-(1-(2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-5-yl)pyrrolidin-3-yl)piperazin-1-yl)-2-((tetrahydro-2H-pyran-4-yl)amino)benzamide (TR-284).

(381) In some embodiments, the bivalent compound is N-(5-(3,5-difluorobenzyl)-1H-indazol-3-yl)-4-(4-(1-((2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-5-yl)methyl)pyrrolidin-3-

yl)piperazin-1-yl)-2-((tetrahydro-2H-pyran-4-yl)amino)benzamide (TR-285).

(382) In some embodiments, the bivalent compound is 2-(2,6-dioxopiperidin-3-yl)-5-(2-((4-(6-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)methyl)morpholino) isoindoline-1,3-dione (TR-286).

(383) In some embodiments, the bivalent compound is 2-(2,6-dioxopiperidin-3-yl)-5-(3-((1-(6-(6-((R)-2~ (3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperidin-4-yl)oxy) azetidin-1-yl)isoindoline-1,3-dione (TR-287).

(384) In some embodiments, the bivalent compound is 2-(2,6-dioxopiperidin-3-yl)-5-(3-((1-(6-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperidin-4-yl)amino) azetidin-1-yl)isoindoline-1,3-dione (TR-288).

(385) In some embodiments, the bivalent compound is N-(5-(3,5-difluorobenzyl)-1H-indazol-3-yl)-4-(4-(6-(2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisoindolin-5-yl)-6-azaspiro[3.4]octan-2-yl)piperazin-1-yl)-2-((tetrahydro-2H-pyran-4-yl)amino)benzamide (TR-290).

(386) In some embodiments, the bivalent compound is 2-(2,6-dioxopiperidin-3-yl)-5-(1-(2-(4-(6-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)ethyl)azetidin-3-yl)isoindoline-1,3-dione (TR-292).

(387) In some embodiments, the bivalent compound is 2-(2,6-dioxopiperidin-3-yl)-5-(1-((1-(6-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperidin-4-yl)methyl)azetidin-3-yl)isoindoline-1,3-dione (TR-293).

(388) In some embodiments, the bivalent compound is N-(5-(3,5-difluorobenzyl)-1H-indazol-3-yl)-4-(4-(1-(1-(2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisoindolin-5-yl)azetidin-3-yl)ethyl)piperazin-1-yl)-2-((tetrahydro-2H-pyran-4-yl)amino)benzamide (TR-301).

(389) In some embodiments, the bivalent compound is 3-(5-(1-(1-(6-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperidin-4-yl)azetidin-3-yl)-1-oxoisoindolin-2-yl)piperidine-2,6-dione (TR-302),

(390) In some embodiments, the bivalent compound is N-(5-(3,5-difluorobenzyl)-1H-indazol-3-yl)-4-(4-(4-(1-(2,6-dioxopiperidin-3-yl)-3-methyl-2-oxo-2,3-dihydro-1H-benzo[d]imidazol-4-yl)butyl)piperazin-1-yl)-2-((tetrahydro-2H-pyran-4-yl)amino)benzamide (TR-304).

(391) In some embodiments, the bivalent compound is N-(5-(3,5-difluorobenzyl)-1H-indazol-3-yl)-4-(4-(4-(1-(2,6-dioxopiperidin-3-yl)-3-methyl-2-oxo-2,3-dihydro-1H-benzo[d]imidazol-4-yl)but-3-yn-1-yl)piperazin-1-yl)-2-((tetrahydro-2H-pyran-4-yl)amino)benzamide (TR-306).

(392) In some embodiments, the bivalent compound is 2-(2,6-dioxopiperidin-3-yl)-5-(1-(1-(6-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperidin-4-yl)azetidin-3-yl)isoindoline-1,3-dione (TR-308).

(393) In some embodiments, the bivalent compound is 3-(6-(1-(1-(6-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperidin-4-yl)azetidin-3-yl)-1-oxoisoindolin-2-yl)piperidine-2,6-dione (TR-309).

(394) In some embodiments, the bivalent compound is N—(S-(3,5-difluorobenzyl)-1H-indazol-3-yl)-4-(4-(3-((1-(2,6-dioxopiperidin-3-yl)-3-methyl-2-oxo-2,3-dihydro-1H-benzo[d]imidazol-4-yl)amino)propyl)piperazin-1-yl)-2-((tetrahydro-2H-pyran-4-yl)amino)benzamide (TR-315).

(395) In some embodiments, the bivalent compound is N-(5-(3,5-difluorobenzyl)-1H-indazol-3-yl)-4-(4-(3-((1-(2,6-dioxopiperidin-3-yl)-3-methyl-2-oxo-2,3-dihydro-1H-benzo[d]imidazol-5-yl)amino)propyl)piperazin-1-yl)-2-((tetrahydro-2H-pyran-4-yl)amino)benzamide (TR-316).

(396) In some embodiments, the bivalent compound is 3-(6-(3-((4-(6-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)methyl)azetidin-1-yl)-1-oxoisoindolin-2-yl)piperidine-2,6-dione (TR-317).

(397) In some embodiments, the bivalent compound is N-(5-(3,5-difluorobenzyl)-1H-indazol-3-yl)-4-(4-((1-(2-(2,6-dioxopiperidin-3-yl)-3-oxoisoindolin-5-yl)azetidin-3-yl)methyl)piperazin-1-yl)-2-((tetrahydro-2H-pyran-4-yl)amino)benzamide (TR-318).

(398) In some embodiments, the bivalent compound is N-(5-(3,5-difluorobenzyl)-1H-indazol-3-

yl)-4-(4-(1-(2,6-dioxopiperidin-3-yl)-3-methyl-2-oxo-2,3-dihydro-1H-benzo[d]imidazol-5-yl)but-3-yn-1-yl)piperazin-1-yl)-2-((tetrahydro-2H-pyran-4-yl)amino)benzamide (TR-319).

(399) In some embodiments, the bivalent compound is N—(S-(3,5-difluorobenzyl)-1H-indazol-3-yl)-4-(4-(4-(1-(2,6-dioxopiperidin-3-yl)-3-methyl-2-oxo-2,3-dihydro-1H-benzo[d]imidazol-5-yl)butyl)piperazin-1-yl)-2-((tetrahydro-2H-pyran-4-yl)amino)benzamide (TR-320).

(400) In some embodiments, the bivalent compound is N-(5-(3,5-difluorobenzyl)-1H-indazol-3-yl)-4-(4-((1-(1-(2,6-dioxopiperidin-3-yl)-3-methyl-2-oxo-2,3-dihydro-1H-benzo[d]imidazol-5-yl)azetidin-3-yl)methyl)piperazin-1-yl)-2-((tetrahydro-2H-pyran-4-yl)amino)benzamide (TR-321).

(401) In some embodiments, the bivalent compound is N-(5-(3,5-difluorobenzyl)-1H-indazol-3-yl)-4-(4-(2-((1-(2,6-dioxopiperidin-3-yl)-3-methyl-2-oxo-2,3-dihydro-1H-benzo[d]imidazol-5-yl)amino)ethyl)piperazin-1-yl)-2-((tetrahydro-2H-pyran-4-yl)amino)benzamide (TR-324).

(402) In some embodiments, the bivalent compound is N-(5-(3,5-difluorobenzyl)-1H-indazol-3-yl)-4~(4-(2-((1-(2,6-dioxopiperidin-3-yl)-3-methyl-2-oxo-2,3-dihydro-1H-benzo[d]imidazol-4-yl)amino)ethyl)piperazin-1-yl)-2-((tetrahydro-2H-pyran-4-yl)amino)benzamide (TR-325).

(403) In some embodiments, the bivalent compound is 3-(5-(3-(4-(6-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)azetidin-1-yl)-1-oxoisindolin-2-yl)piperidine-2,6-dione (TR-331).

(404) In some embodiments, the bivalent compound is N-(5-(3,5-difluorobenzyl)-1H-indazol-3-yl)-4-(4-(1-(2-(2,6-dioxopiperidin-3-yl)-1-oxoisindolin-5-yl)azetidin-3-yl)piperazin-1-yl)-2-((tetrahydro-2H-pyran-4-yl)amino)benzamide (TR-332).

(405) In some embodiments, the bivalent compound is 3-(6-(3-(4-(6-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)azetidin-1-yl)-1-oxoisindolin-2-yl)piperidine-2,6-dione (TR-335).

(406) In some embodiments, the bivalent compound is N—(S-(3,5-difluorobenzyl)-1H-indazol-3-yl)-4-(4-(1-(2-(2,6-dioxopiperidin-3-yl)-3-oxoisindolin-5-yl)azetidin-3-yl)piperazin-1-yl)-2-((tetrahydro-2H-pyran-4-yl)amino)benzamide (TR-336).

(407) In some embodiments, the bivalent compound is N-(5-(3,5-difluorobenzyl)-1H-indazol-3-yl)-4-(4-((1-(2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-5-yl)azetidin-3-yl)amino)piperidin-1-yl)-2-((tetrahydro-2H-pyran-4-yl)amino)benzamide (TR-337).

(408) In some embodiments, the bivalent compound is N-(5-(3,5-difluorobenzyl)-1H-indazol-3-yl)-4-(4-(3-(2-(2,6-dioxopiperidin-3-yl)-1-oxoisindolin-5-yl)azetidin-1-yl)piperidin-1-yl)-2-((tetrahydro-2H-pyran-4-yl)amino)benzamide (TR-338).

(409) In some embodiments, the bivalent compound is N-(5-(3,5-difluorobenzyl)-1H-indazol-3-yl)-4-(4-(3-(2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-5-yl)azetidin-1-yl)piperidin-1-yl)-2-((tetrahydro-2H-pyran-4-yl)amino)benzamide (TR-339).

(410) In some embodiments, the bivalent compound is N-(5-(3,5-difluorobenzyl)-1H-indazol-3-yl)-4-(4-(6-(2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-5-yl)-2,6-diazaspiro[3.3]heptan-2-yl)piperidin-1-yl)-2-((tetrahydro-2H-pyran-4-yl)amino)benzamide (TR-340).

(411) In some embodiments, the bivalent compound is (S)—N-(5-(3,5-difluorobenzyl)-1H-indazol-3-yl)-4-(4-((1-(2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-5-yl)azetidin-3-yl)methyl)piperazin-1-yl)-2-((tetrahydro-2H-pyran-4-yl)amino)benzamide (TR-341).

(412) In some embodiments, the bivalent compound is (R)—N-(5-(3,5-difluorobenzyl)-1H-indazol-3-yl)-4-(4-((1-(2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-5-yl)azetidin-3-yl)methyl)piperazin-1-yl)-2-((tetrahydro-2H-pyran-4-yl)amino)benzamide (TR-342).

(413) In some embodiments, the bivalent compound is (S)—N-(5-(3,5-difluorobenzyl)-1H-indazol-3-yl)-4-(4-(1~(2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-5-yl)azetidin-3-yl)piperazin-1-yl)-2-((tetrahydro-2H-pyran-4-yl)amino)benzamide (TR-343).

(414) In some embodiments, the bivalent compound is (R)—N—(S-(3,5-difluorobenzyl)-1H-indazol-3-yl)-4-(4-(1-(2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-5-yl)azetidin-3-yl)piperazin-1-yl)-2-((tetrahydro-2H-pyran-4-yl)amino)benzamide (TR-344).

(415) In some embodiments, the bivalent compound is N-(5-((R)-2-(2,5-difluorophenyl)pyrrolidin-1-yl)pyrazolo[1,5-a]pyrimidin-3-yl)-6-((1-(4-((2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-5-yl)amino)butanoyl)piperidin-4-yl)amino)picolinamide (CPD-470).

(416) In some embodiments, the bivalent compound is N-(5-((R)-2-(2,5-difluorophenyl)pyrrolidin-1-yl)pyrazolo[1,5-a]pyrimidin-3-yl)-6-((1-(6-((2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-5-yl)amino)hexanoyl)piperidin-4-yl)amino)picolinamide (CPD-471).

(417) In some embodiments, the bivalent compound is N-(5-((R)-2-(2,5-difluorophenyl)pyrrolidin-1-yl)pyrazolo[1,5-a]pyrimidin-3-yl)-6-((1-(8-((2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-5-yl)amino)octanoyl)piperidin-4-yl)amino)picolinamide (CPD-472).

(418) In some embodiments, the bivalent compound is N-(5-((R)-2-(2,5-difluorophenyl)pyrrolidin-1-yl)pyrazolo[1,5-a]pyrimidin-3-yl)-6-((1-(3-(2-((2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-5-yl)amino)ethoxy)propanoyl)piperidin-4-yl)amino)picolinamide (CPD-473).

(419) In some embodiments, the bivalent compound is N-(5-((R)-2-(2,5-difluorophenyl)pyrrolidin-1-yl)pyrazolo[1,5-a]pyrimidin-3-yl)-6-((1-(3-(2-(2-((2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-5-yl)amino)ethoxy)ethoxy)propanoyl)piperidin-4-yl)amino)picolinamide (CPD-474).

(420) In some embodiments, the bivalent compound is N-(5-((R)-2-(2,5-difluorophenyl)pyrrolidin-1-yl)pyrazolo[1,5-a]pyrimidin-3-yl)-6-((1-(3-(2-(2-(2-((2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-5-yl)amino)ethoxy)ethoxy)ethoxy)propanoyl)piperidin-4-yl)amino)picolinamide (CPD-475).

(421) In some embodiments, the bivalent compound is N-(5-((R)-2-(2,5-difluorophenyl)pyrrolidin-1-yl)pyrazolo[1,5-a]pyrimidin-3-yl)-6-((1-(1-((2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-5-yl)amino)-3,6,9,12-tetraoxapentadecan-15-oyl)piperidin-4-yl)amino)picolinamide (CPD-476).

(422) In some embodiments, the bivalent compound is N-(2-(4-carbamoylpiperidin-1-yl)-4-((4-((2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-4-yl)glycyl)piperazin-1-yl)methyl)phenyl)-2-morpholinooxazole-4-carboxamide (CPD-478).

(423) In some embodiments, the bivalent compound is N-(2-(4-carbamoylpiperidin-1-yl)-4-((4-((2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-4-yl)amino)butanoyl)piperazin-1-yl)methyl)phenyl)-2-morpholinooxazole-4-carboxamide (CPD-480).

(424) In some embodiments, the bivalent compound is N-(2-(4-carbamoylpiperidin-1-yl)-4-((4-((2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-4-yl)amino)pentanoyl)piperazin-1-yl)methyl)phenyl)-2-morpholinooxazole-4-carboxamide (CPD-481).

(425) In some embodiments, the bivalent compound is N-(2-(4-carbamoylpiperidin-1-yl)-4-((4-((2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-4-yl)amino)hexanoyl)piperazin-1-yl)methyl)phenyl)-2-morpholinooxazole-4-carboxamide (CPD-482).

(426) In some embodiments, the bivalent compound is N-(2-(4-carbamoylpiperidin-1-yl)-4-((4-((2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-4-yl)amino)heptanoyl)piperazin-1-yl)methyl)phenyl)-2-morpholinooxazole-4-carboxamide (CPD-483).

(427) In some embodiments, the bivalent compound is N-(2-(4-carbamoylpiperidin-1-yl)-4-((4-((2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-4-yl)amino)octanoyl)piperazin-1-yl)methyl)phenyl)-2-morpholinooxazole-4-carboxamide (CPD-484).

(428) In some embodiments, the bivalent compound is 2-(4-(2-amino-3-(3-methoxy-4-((4-methoxybenzyl)oxy)benzyl)-3H-imidazo[4,5-b]pyridin-6-yl)-1H-pyrazol-1-yl)-N-(4-((2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-4-yl)amino)butyl)acetamide (CPD-499).

(429) In some embodiments, the bivalent compound is 2-(4-(2-amino-3-(3-methoxy-4-((4-methoxybenzyl)oxy)benzyl)-3H-imidazo[4,5-b]pyridin-6-yl)-1H-pyrazol-1-yl)-N-(6-((2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-4-yl)amino)hexyl)acetamide (CPD-500).

(430) In some embodiments, the bivalent compound is 2-(4-(2-amino-3-(3-methoxy-4-((4-methoxybenzyl)oxy)benzyl)-3H-imidazo[4,5-b]pyridin-6-yl)-1H-pyrazol-1-yl)-N-(8-((2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-4-yl)amino)octyl)acetamide (CPD-501).

(431) According to one aspect of the present disclosure, a composition disclosed herein comprises the bivalent compound or a pharmaceutically acceptable salt or analog thereof, and a pharmaceutically acceptable carrier or diluent.

(432) According to one aspect of the present disclosure, a method of treating a tropomyosin receptor kinase (TRK)-mediated disease disclosed herein comprises administering to a subject with a TRK-mediated disease the bivalent compound or a pharmaceutically acceptable salt or analog thereof.

(433) In one embodiment, the TRK-mediated disease results from TRK expression, mutation, or fusion.

(434) In one embodiment, wherein the subject with the TRK-mediated disease has an elevated TRK function relative to a healthy subject without the TRK-mediated disease.

(435) In one embodiment, the bivalent compound is selected from the group consisting of CPD-001 to CPD-516, or analogs thereof.

(436) In one embodiment, the bivalent compound is selected from the group consisting of CPD-247 to CPD-516, or analogs thereof.

(437) In one embodiment, the bivalent compound is administered to the subject orally, parenterally, intradermally, subcutaneously, topically, or rectally. In one embodiment, the method further comprises administering to the subject an additional therapeutic regimen for treating cancer.

(438) In one embodiment, the additional therapeutic regimen is selected from the group consisting of surgery, chemotherapy, radiation therapy, hormone therapy, and immunotherapy.

(439) In one embodiment, the TRK-mediated disease is selected from the group consisting of non-small cell lung cancer, colorectal cancer, gastric cancer, liver cancer, invasive breast cancer, lung adenocarcinoma, uterine cancer, adrenal cancer, pancreatic cancer, ovarian cancer, esophageal cancer, urinary bladder cancer, endometrial cancer, prostate cancer low-grade glioma, glioblastoma, Spitzoid cancer, soft tissue sarcoma, papillary thyroid carcinoma, head and neck squamous cell carcinoma, congenital fibrosarcoma, congenital mesoblastic nephroma, secretory breast carcinoma, mammary analogue secretory carcinoma, acute myeloid leukemia, ductal carcinoma, pulmonary neuroendocrine tumors, pheochromocytoma, and Wilms' tumor.

(440) In one embodiment, the TRK-mediated disease or condition comprises cancer, inflammatory diseases, acute and chronic pain, pruritus, bone-related diseases, neurodegenerative diseases, infectious diseases, and other diseases, including but not limited to neuroblastoma, prostate cancer, pancreatic cancer, melanoma, head and neck cancer, gastric carcinoma, lung carcinoma, liver cancer, uterine cancer, adrenal cancer, biliary tree cancer, intestinal cancer, colorectal cancer, ovarian cancer, lung carcinoma, small cell lung cancer, non-small cell lung cancer, gastric carcinoma, breast cancer, esophageal cancer, urinary bladder cancer, endometrial cancer, brain cancer, low-grade glioma, glioblastoma, medulloblastoma, secretory breast cancer, secretory breast carcinoma, salivary gland cancer, papillary thyroid carcinoma, ductal carcinoma, adult myeloid leukemia, acute myeloid leukemia, large cell neuroendocrine tumors, pulmonary neuroendocrine tumors, sarcomas, pheochromocytoma, fibrosarcoma, congenital fibrosarcoma, congenital mesoblastic nephroma, secretory breast carcinoma, malignant fibrous histiocytoma, embryonal rhabdomyosarcoma, leiomyosarcoma, neuro-fibrosarcoma, neoplasms of the central nervous systems, osteosarcoma, synovial sarcoma, liposarcoma, alveolar soft part sarcoma, Spitzoid cancer, Wilms' tumor, lymphomas (e.g. including Hodgkin's lymphoma, lymphoplasmacytoid lymphoma, follicular lymphoma, mucosa-associated lymphoid tissue lymphoma, mantle cell lymphoma, B-lineage large cell lymphoma, Burkitt's lymphoma, and T-cell anaplastic large cell lymphoma), inflammatory lung diseases (e.g. asthma), inflammatory bowel diseases, (e.g. ulcerative colitis, Crohn's disease), inflammatory skin diseases (e.g. atopic dermatitis, eczema and psoriasis), interstitial cystitis, rhinitis, acute pain, chronic pain, cancer pain, surgical pain, inflammatory pain, neuropathic pain, nociceptive pain, pain of osteoarthritis, chronic low back pain, low back pain of osteoporosis, pain of bone fracture, pain of rheumatoid arthritis, postherpetic pain, pain of diabetic

neuropathy, fibromyalgia, pain of pancreatitis, pain of interstitial cystitis, pain of endometriosis, pain of irritable bowel syndrome, migraine, pain of pulpitis, interstitial cystitis pain, painful bladder syndrome, central pain syndromes, postsurgical pain syndromes, bone and joint pain, repetitive motion pain, dental pain, myofascial pain, perioperative pain, dysmenorrhea, myofascial pain, angina pain, headache, primary hyperalgesia, secondary hyperalgesia, primary allodynia, secondary allodynia, other pain caused by central sensitization, systemic cutaneous pruritus, localized cutaneous pruritus, senile cutaneous pruritus, gestational pruritus, pruritus ani, vulvar pruritus, metastatic bone disease, treatment-induced bone loss, osteoporosis, rheumatoid arthritis, bone metastases, ankylosing spondylitis, Paget's disease, periodontal disease, osteolytic disease, multiple sclerosis, Parkinson's disease, Alzheimer's disease, Chagas disease, cachexia, anorexia, demyelination and dysmyelination. In certain embodiments, the disease or condition is a relapsed disease.

(441) In one embodiment, the TRK-mediated disease is a relapsed cancer.

(442) In one embodiment, the TRK-mediated disease is refractory to one or more previous treatments.

(443) According to one aspect of the present disclosure, a method for identifying a bivalent compound which mediates degradation or reduction of TRK is disclosed. The method comprises: providing a heterobifunctional test compound comprising an TRK ligand conjugated to a degradation tag through a linker; contacting the heterobifunctional test compound with a cell comprising a ubiquitin ligase and TRK; determining whether TRK level is decreased in the cell; and identifying the heterobifunctional test compound as a bivalent compound which mediates degradation or reduction of TRK.

(444) In one embodiment, the cell is a cancer cell.

(445) In one embodiment, the cancer cell is a TRK-mediated cancer cell.

(446) In one embodiment, the cell is a neuron,

INCORPORATION BY REFERENCE

(447) All publications, patents, and patent applications mentioned in this specification are herein incorporated by reference to the same extent as if each individual publication, patent, or patent application was specifically and individually indicated to be incorporated by reference.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The novel features of the invention are set forth with particularity in the appended claims. A better understanding of the features and advantages of the present invention will be obtained by reference to the following detailed description that sets forth illustrative embodiments, in which the principles of the invention are utilized, and the accompanying drawings of which:

(2) FIG. 1A shows an immunoblot of TPM3-TRKA fusion protein expressed by KM12 cells after treatment with Entrectinib or bivalent compounds CPD-001-CPD-022.

(3) FIG. 1B shows an immunoblot of TPM3-TRKA fusion protein expressed by KM12 cells after treatment with Entrectinib or bivalent compounds CPD-023-CPD-044.

(4) FIG. 1C shows an immunoblot of TPM3-TRKA fusion protein expressed by KM12 cells after treatment with Entrectinib or bivalent compounds CPD-045-CPD-065.

(5) FIG. 2 shows an immunoblot of TPM3-TRKA fusion protein expressed by KM12 cells after treatment with Entrectinib or bivalent compounds CPD-027, CPD-053, and CPD-060 at various time points.

(6) FIG. 3 shows immunoblots of TPM3-TRKA fusion protein expressed by KM12 cells in subcutaneous xenograft tumors after treatment with a dose range of CPD-027, CPD-053, and CPD-060.

- (7) FIG. 4A shows a graph of KM12 cell viability vs. concentration of bivalent compounds CPD-010, CPD-053, and CPD-057.
- (8) FIG. 4B shows KM12 and H358 cell viability vs. concentration of bivalent compound CPD-053.
- (9) FIG. 5A shows an immunoblot of TPM3-TRKA fusion protein expressed by KM12 cells after treatment with GNF-8625, LOXO101 or bivalent compounds TR-104-TR-129.
- (10) FIG. 5B shows an immunoblot of TPM3-TRKA fusion protein expressed by KM12 cells after treatment with a dose range of TR-115, TR-116, TR-119, TR-123, TR-124, TR-127 or TR-129.
- (11) FIG. 5C shows an immunoblot of TPM3-TRKA fusion protein expressed by KM12 cells after treatment with a dose range of TR-130, TR-131, TR-132, TR-140, TR-146, TR-150 or TR-168.
- (12) FIG. 6A shows an immunoblot of TPM3-TRKA fusion protein expressed by KM12 cells after treatment with a dose range of TR-171, TR-172, TR-173 or TR-176.
- (13) FIG. 6B shows an immunoblot of TPM3-TRKA fusion protein expressed by KM12 cells after treatment with a dose range of TR-177, TR-181, TR-182 or GNF-8625.
- (14) FIG. 6C shows an immunoblot of TPM3-TRKA fusion protein expressed by KM12 cells after treatment with a dose range of TR-186, TR-188, TR-189 or TR-190.
- (15) FIG. 6D shows an immunoblot of TPM3-TRKA fusion protein expressed by KM12 cells after treatment with a dose range of TR-191, TR-194, TR-196, TR-198 or GNF-8625.
- (16) FIG. 7 shows immunoblots of overexpressed TPM3-TRKA, AGL4-TRKB and ETV6-TRKC fusion protein in KM12 cells after treatment with a dose range of TR-123.
- (17) FIG. 8A shows an immunoblot of TPM3-TRKA fusion protein expressed by KM12 cells after treatment with a dose range of compound TR-123 or TR-123-neg.
- (18) FIG. 8B shows an immunoblot of wild type TRKA protein expressed by HEL cells after treatment with a dose range of compound TR-123 or TR-123-neg.
- (19) FIG. 9A shows an immunoblot of TPM3-TRKA fusion protein expressed by KM12 cells after treatment with a single dose of TR-123 or combinations with MG-132, Bortezomib or MLN4924.
- (20) FIG. 9B shows an immunoblot of wild type TRKA protein expressed by HEL cells after treatment with a single dose of TR-123 or combinations with MG-132, Bortezomib, MLN4924 or Pomalidomide.
- (21) FIG. 10A shows an immunoblot of TPM3-TRKA fusion protein in subcutaneous KM12 xenograft tumors after treatment with a dose range of TR-123.
- (22) FIG. 10B shows an immunoblot of TPM3-TRKA fusion protein expressed in subcutaneous KM12 xenograft tumors after treatment with TR-171, TR-172, TR-173, TR-177 or TR-181.
- (23) FIG. 11 shows a graph of plasma concentration of TR-123 vs. time points post dosing.
- (24) FIG. 12A shows a graph of subcutaneous KM12 xenograft tumor volume vs. days after treatment with a dose range of CPD-060.
- (25) FIG. 12B shows a graph of body weight vs. days after treatment with a dose range of CPD-060.
- (26) FIG. 13A shows an immunoblot of TPM3-TRKA fusion protein expressed by KM12 cells after treatment with a dose range of TR-202, TR-203 or TR-204.
- (27) FIG. 13B shows an immunoblot of TPM3-TRKA fusion protein expressed by KM12 cells after treatment with a dose range of TR-208, TR-210, TR-211 or TR-214.
- (28) FIG. 13C shows an immunoblot of TPM3-TRKA fusion protein expressed by KM12 cells after treatment with a dose range of TR-215, TR.sup.216, TR-217, TR.sup.218, TR-219 or TR-220.
- (29) FIG. 13D shows an immunoblot of TPM3-TRKA fusion protein expressed by KM12 cells after treatment with a dose range of TR-221, TR-222, TR-223, TR-224, TR-225, TR-226 or TR-227.
- (30) FIG. 13E shows an immunoblot of TPM3-TRKA fusion protein expressed by KM12 cells after treatment with a dose range of TR-231, TR-232, TR-233, TR-234 or TR-235.
- (31) FIG. 14 shows a graph of plasma concentration of TR-198 over time after dosing via

intravenous injection or oral gavage.

(32) FIG. 15A shows a graph of subcutaneous KM12 xenograft tumor volume as a function of days after treatment with a dose range of TR-181 or a single dose of TR-198.

(33) FIG. 15B shows a graph of body weight as a functional of days after treatment with a dose range of TR-181 or a single dose of TR-198.

(34) FIG. 16A shows a graph of the percentage of weight born on the injured limb following treatment with a single dose of vehicle (Veh), TR-181 or ibuprofen (Ibu) in rats.

(35) FIG. 16B shows a graph of the percentage of weight born on the injured limb following treatment with a single dose of TR-181 or ibuprofen (Ibu) in guinea pigs.

(36) FIG. 17A shows an immunoblot of TPM3-TRKA fusion protein expressed by KM12 cells after treatment with a dose range of TR-247, TR-248, TR-249, TR-250, TR-251, TR-252, or TR-253.

(37) FIG. 17B shows an immunoblot of TPM3-TRKA fusion protein expressed by KM12 cells after treatment with a dose range of TR-254, TR-255, TR-256, TR-257, TR-258, TR-259, or TR-260.

(38) FIG. 17C shows an immunoblot of TPM3-TRKA fusion protein expressed by KM12 cells after treatment with a dose range of TR-261, TR-262, TR-263, TR-264, TR-265, TR-256, or TR-267.

(39) FIG. 18A shows an immunoblot of TPM3-TRKA fusion protein expressed by KM12 cells after treatment with a dose range of TR-268, TR-269, Tr-270, TR-271, or TR-272.

(40) FIG. 18B shows an immunoblot of TPM3-TRKA fusion protein expressed by KM12 cells after treatment with a dose range of TR-273, TR-274, TR-275, TR-276, or TR-277.

(41) FIG. 18C shows an immunoblot of TPM3-TRKA fusion protein expressed by KM12 cells after treatment with a dose range of TR-279, Tr-280, TR-281, TR-282, or TR-283.

(42) FIG. 19A shows an immunoblot of TPM3-TRKA fusion protein expressed by KM12 cells after treatment with a dose range of TR-284, TR-285, TR-286, TR-287, or TR-288.

(43) FIG. 19B shows an immunoblot of TPM3-TRKA fusion protein expressed by KM12 cells after treatment with a dose range of TR-289, TR-290, TR-291, TR-292, TR-293, or TR-294.

(44) FIG. 19C shows an immunoblot of TPM3-TRKA fusion protein expressed by KM12 cells after treatment with a dose range of TR-295, TR-296, TR-297, TR-298, TR-300, or TR-301.

(45) FIG. 20A shows an immunoblot of TPM3-TRKA fusion protein expressed by KM12 cells after treatment with a dose range of TR-302, TR-303, TR-304, TR-305, TR-306, or TR-307.

(46) FIG. 20B shows an immunoblot of TPM3-TRKA fusion protein expressed by KM12 cells after treatment with a dose range of TR-308, TR-309, TR-310. TR-311, TR-312, or TR-313.

(47) FIG. 20C shows an immunoblot of TPM3-TRKA fusion protein expressed by KM12 cells after treatment with a dose range of TR-314, TR-315, TR-316, TR-317, TR-318, TR-319, or TR-320.

(48) FIG. 21A shows an immunoblot of TPM3-TRKA fusion protein expressed by KM12 cells after treatment with a dose range of TR-321, TR-322, TR-323, TR-324, TR-325, TR-326, or TR-327.

(49) FIG. 21B shows an immunoblot of TPM3-TRKA fusion protein expressed by KM12 cells after treatment with a dose range of TR-328, TR-329, TR-330, TR-331, TR-332, TR-333, or TR-334.

(50) FIG. 21C shows an immunoblot of TPM3-TRKA fusion protein expressed by KM12 cells after treatment with a dose range of TR-335, TR-336, TR-337, TR-338, TR-339, or TR-340.

(51) FIG. 22 shows an immunoblot of NPM-ALK fusion protein expressed in SU-DHL-1 cells after treatment with a dose range of entrectinib, CPD-032, CPD-037, or CPD-055.

(52) FIG. 23A shows graphs of plasma concentrations of TR-231 and TR-275 over time after dosing via intravenous injection or oral gavage.

(53) FIG. 23B shows an immunoblot of TPM3-TRKA fusion protein expressed in subcutaneous

KM12 xenograft tumors after treatment with TR-231 and TR-275 at different oral doses and at different time points

(54) FIG. 24A shows KM12 xenograft tumor growth curves in mice treated with vehicle or 40 mg/kg TR-231 twice per day.

(55) FIG. 24B shows KM12 xenograft tumor growth curves in mice treated with vehicle or 40 mg/kg TR-275 twice per day.

(56) FIG. 25A-25C shows immunoblots of TPM3-TRKA fusion protein expressed by KM12 cells after treatment with heterobifunctional compounds.

(57) FIG. 26 shows an immunoblot of wildtype TRKA protein expressed by HEL cells after treatment with heterobifunctional compounds.

(58) FIG. 27A-27C shows graphs of KM12 cell viability vs. concentration of heterobifunctional compounds CPD-470, CPD-471, CPD-474, CPD-480, CPD-481, CPD-482, CPD-499, CPD-500, and CPD-501.

(59) FIG. 28A-28B show immunoblots of TPM3-TRKA fusion protein expressed in a cell line after treatment with heterobifunctional compounds.

DETAILED DESCRIPTION OF THE INVENTION

(60) It is recognized in the present disclosure that tropomyosin receptor kinase (TRK) receptor family comprises three members, TRKA, TRKB and TRKC that are encoded by the NTRK1, NTRK2 and NTRK3 genes, respectively (Khotskaya et al., 2017). TRKs are receptor tyrosine kinases primarily implicated in development and functions of the neuronal tissues. The main ligands of TRKs include nerve growth factor (NGF) for TRKA, brain-derived growth factor (BDGF) for TRKB, and neurotrophins for TRKC (Vaishnavi et al., 2015). The binding of ligands to the extracellular domains of TRKs induces dimerization and activation of the receptors, which activates downstream signal transduction pathways, such as PI3K/AKT, RAF/MEK/ERK, and PLC γ pathways. These pathways have well established roles to support cellular proliferation, survival, and promote oncogenesis (Hanahan and Weinberg, 2011).

(61) It is further recognized herein that, like many other oncogenic receptor tyrosine kinases, TRKs are aberrantly activated in a variety of human malignancies. Interestingly, the primary molecular mechanism activating TRKs in cancer is not point mutations but in-frame fusions of NTRK genes (Vaishnavi et al., 2015). Typically, the 3' regions of the NTRK genes are joined with the 5' regions of a partner gene due to chromosomal rearrangement. The resulted chimeric proteins always retain the kinase domain of TRK proteins, indicating that the catalytic functions are crucial to the transforming activities. Loss of the 5' regions of the NTRK genes that encode the self-inhibitory domains renders these fusion kinases constitutively active. Additionally, expression of the chimeric proteins is driven by the promoters of the fusion partners, which often result in overexpression. The most common TRK fusions include LMNA-TRKA, TPM3-TRKA, and ETV6-TRKC (Amatu et al., 2016). Hence, genetic events lead to overexpressed and constitutively active TRK-fusion kinases. These fusions are oncogenic, as shown by their ability to transform mouse embryonic fibroblasts and normal epithelium (Russell et al., 2000; Vaishnavi et al., 2015).

(62) TRK fusion was first reported in a human colon carcinoma, which was named as oncD at that time (Martin-Zanca et al., 1986). Recent advances in high-throughput RNA sequencing greatly promote the efficiency of identifying chromosomal rearrangement events in patient samples. Consequently, TRK fusions have been found across a wide range of human malignancies, including but are not limited to non-small cell lung cancer, colorectal cancer, gastric cancer, low-grade glioma glioblastoma, Spitzoid cancer, soft tissue sarcoma, papillary thyroid carcinoma, head and neck squamous cell carcinoma, congenital fibrosarcoma, congenital mesoblastic nephroma, secretory breast carcinoma, mammary analogue secretory carcinoma, acute myeloid leukemia, and ductal carcinoma (Amatu et al., 2016; Khotskaya et al., 2017). The frequency of TRK fusions is relatively low. For example, approximately 0.5% to 2.7% colon cancers are affected by TRK fusions (Creancier et al., 2015; Lee et al., 2015). However, for certain cancer types, such as

secretory breast carcinoma, TRK fusions can be found in the vast majority of cases (Tognon et al., 2002).

(63) TRK mutations and deletions have been observed in additional human diseases, such as pulmonary neuroendocrine tumors, anhidrosis syndrome, obesity, congenital heart defects, and acute myeloid leukemia (Khotskaya et al., 2017). In addition, TRK amplification are associated with several human diseases, such as liver cancer, invasive breast cancer, lung adenocarcinoma, uterine cancer, adrenal cancer, pancreatic cancer, ovarian cancer, esophageal cancer, urinary bladder cancer, endometrial cancer, pheochromocytoma, Wilms' tumor, and prostate cancer (Khotskaya et al., 2017).

(64) The nerve growth factor (NGF) and its main receptor, tropomyosin receptor kinase A (TRKA), have long been recognized for their roles in central and peripheral pain (Denk et al., 2017). Nociceptive neurons express TRKA and mediate pain sensation by transmitting pain signals to the central nervous system. Multiple NGF-neutralizing antibodies, such as tanezumab, are undergoing clinical assessment in patients with osteoarthritis, lower back pain, cancer pain, neuropathic pain, and other pain conditions (Miller et al., 2017). The efficacy of NGF antibodies in pain relief has been clearly documented in clinics. However, administration of NGF neutralizing antibodies has been shown to result in rapidly progressed joint destruction in some patients that leads to total joint replacement (Schnitzer and Marks, 2015). These adverse events may be related to sustained exposure to NGF antibodies. Targeting TRK represents another promising therapeutic strategy blocking the NGF/TRK signaling pathway for pain management. However, currently available pan-TRK kinase inhibitors may induce significant on-target adverse effects through modulating TRK family members in the central nervous system. Peripherally restricted TRK bifunctional degraders are expected to selectively block the NGF/TRK pathway in peripheral nerves while spare these targets in the central nervous system.

(65) TRK is associated with cancer, inflammatory diseases, acute and chronic pain, pruritus, bone-related diseases, neurodegenerative diseases, infectious diseases, and other diseases, including but not limited to neuroblastoma, prostate cancer, pancreatic cancer, melanoma, head and neck cancer, gastric carcinoma, lung carcinoma, liver cancer, uterine cancer, adrenal cancer, biliary tree cancer, intestinal cancer, colorectal cancer, ovarian cancer, lung carcinoma, small cell lung cancer, non-small cell lung cancer, gastric carcinoma, breast cancer, esophageal cancer, urinary bladder cancer, endometrial cancer, brain cancer, low-grade glioma, glioblastoma, medulloblastoma, secretory breast carcinoma, salivary gland cancer, papillary thyroid carcinoma, ductal carcinoma, acute myeloid leukemia, large cell neuroendocrine tumors, pulmonary neuroendocrine tumors, sarcomas, pheochromocytoma, fibrosarcoma, congenital fibrosarcoma, congenital mesoblastic nephroma, secretory breast carcinoma, malignant fibrous histiocytoma, embryonal rhabdomyosarcoma, leiomyosarcoma, neuro-fibrosarcoma, neoplasms of the central nervous systems, osteosarcoma, synovial sarcoma, liposarcoma, alveolar soft part sarcoma, Spitzoid cancer, Wilms' tumor, lymphomas (e.g. including Hodgkin's lymphoma, lymphoplasmacytoid lymphoma, follicular lymphoma, mucosa-associated lymphoid tissue lymphoma, mantle cell lymphoma, B-lineage large cell lymphoma, Burkitt's lymphoma, and T-cell anaplastic large cell lymphoma), inflammatory lung diseases (e.g. asthma), inflammatory bowel diseases, (e.g. ulcerative colitis, Crohn's disease), inflammatory skin diseases (e.g. atopic dermatitis, eczema and psoriasis), interstitial cystitis, rhinitis, acute pain, chronic pain, cancer pain, surgical pain, inflammatory pain, neuropathic pain, nociceptive pain, pain of osteoarthritis, chronic low back pain, low back pain of osteoporosis, pain of bone fracture, pain of rheumatoid arthritis, postherpetic pain, pain of diabetic neuropathy, fibromyalgia, pain of pancreatitis, pain of interstitial cystitis, pain of endometriosis, pain of irritable bowel syndrome, migraine, pain of pulpitis, interstitial cystitis pain, painful bladder syndrome, central pain syndromes, postsurgical pain syndromes, bone and joint pain, repetitive motion pain, dental pain, myofascial pain, perioperative pain, dysmenorrhea, myofascial pain, angina pain, headache, primary hyperalgesia, secondary hyperalgesia, primary allodynia, secondary allodynia,

other pain caused by central sensitization, systemic cutaneous pruritus, localized cutaneous pruritus, senile cutaneous pruritus, gestational pruritus, pruritus ani, vulvar pruritus, metastatic bone disease, treatment-induced bone loss, osteoporosis, rheumatoid arthritis, bone metastases, ankylosing spondylitis, Paget's disease, periodontal disease, osteolytic disease, multiple sclerosis, Parkinson's disease, Alzheimer's disease, Chagas disease, cachexia, anorexia, demyelination and dysmyelination.

(66) TRK kinase inhibitors are currently undergoing clinical or pre-clinical development, including but are not limited to entrectinib (RXDX-101)(Menichincheri et al., 2016), GNF-8625 (Choi et al., 2015), larotrectinib (LOXO-101; ARRY-470)(Drilon et al., 2018), altiratinib (DCC2701, DCC-270, DP-5164) (Smith et al., 2015), sitravatinib (MGCD516)(Patwardhan et al., 2016), cabozantinib (XL-184, BMS-907351)(Fuse et al., 2017), dovitinib (TKI-258, CHIR-258)(Chong et al., 2017), milciclib (PHA-848125AC)(Brasca et al., 2009), belizatinib (TSR-011)(Ricciuti et al., 2017), GZ389988 (Bailey et al., 2017a, b), pegcantratinib (Cranston et al., 2017), AZD7451 (Tatematsu et al., 2014), TPX-0005 (Cui et al., 2016), LOXO-195 (Blake et al., 2016), regorafenib (Subbiah et al., 2017), DS-6051b (Fujiwara et al., 2018), F17752 (Amatu et al., 2016), PLX7486 (Amatu et al., 2016), AZD-6918 (Li et al., 2015), ASP7962 (Bailey et al., 2017a, b), VM902A (Bailey et al., 2017a, b), ONO-4474 (Bailey et al., 2017a, b), and PF-06273340 (Skerratt et al., 2016). The most advanced ones are entrectinib and larotrectinib (Khotskaya et al., 2017). These agents are tested in basket trials that recruit patients according to detection of TRK-fusions instead of histology. The phase 2 results of larotrectinib demonstrated that most patients (75%) responded to the therapy and that 55% patient remained progression-free at 1 year (Drilon et al., 2018). Phase 1 results of entrectinib also recorded marked and durable response in patients with TRK-fusion tumors (Drilon et al., 2017b). The remarkable efficacy of TRK inhibitors was independent of tumor types. These substantial results collectively highlight a role of TRK fusions as the sole oncogenic drivers in a subset of human malignancies, irrespective of tissue of origin.

(67) Non-specific side effects and the development of resistance to TRK kinase inhibitors remain a challenge in development of effective treatments. Thus, new small-molecule targeting TRK's functions through inhibition and/or degradation will be very useful.

(68) Without wishing to be bound by any theory, the present disclosure is believed to be based, at least in part, on the discovery that novel heterobivalent small molecules which degrade TRK, TRK fusion proteins, TRK splicing, and/or TRK mutant proteins are useful in the treatment of TRK-mediated diseases, particularly non-small cell lung cancer, colorectal cancer, gastric cancer, liver cancer, invasive breast cancer, lung adenocarcinoma, uterine cancer, adrenal cancer, pancreatic cancer, ovarian cancer, esophageal cancer, urinary bladder cancer, endometrial cancer, prostate cancer, low-grade glioma, glioblastoma, Spitzoid cancers, soft tissue sarcoma, papillary thyroid carcinoma, head and neck squamous cell carcinoma, congenital fibrosarcoma, congenital mesoblastic nephroma, secretory breast carcinoma, mammary analogue secretory carcinoma, acute myeloid leukemia, ductal carcinoma, pulmonary neuroendocrine tumors, pheochromocytoma, and Wilms' tumor (Amatu et al., 2016; Khotskaya et al., 2017). The disclosed novel bifunctional TRK degraders are useful in the treatment of TRK-mediated cancer, inflammatory diseases, acute and chronic pain, pruritus, bone-related diseases, neurodegenerative diseases, infectious diseases, and other diseases, including but not limited to neuroblastoma, prostate cancer, pancreatic cancer, melanoma, head and neck cancer, gastric carcinoma, lung carcinoma, liver cancer, uterine cancer, adrenal cancer, biliary tree cancer, intestinal cancer, colorectal cancer, ovarian cancer, lung carcinoma, small cell lung cancer, non-small cell lung cancer, gastric carcinoma, breast cancer, esophageal cancer, urinary bladder cancer, endometrial cancer, brain cancer, low-grade glioma, glioblastoma, medulloblastoma, secretory breast cancer, secretory breast carcinoma, salivary gland cancer, papillary thyroid carcinoma, ductal carcinoma, adult myeloid leukemia, acute myeloid leukemia, large cell neuroendocrine tumors, pulmonary neuroendocrine tumors, sarcomas, pheochromocytoma, fibrosarcoma, congenital fibrosarcoma, congenital mesoblastic nephroma,

secretory breast carcinoma, malignant fibrous histiocytoma, embryonal rhabdomyosarcoma, leiomyosarcoma, neuro-fibrosarcoma, neoplasms of the central nervous systems, osteosarcoma, synovial sarcoma, liposarcoma, alveolar soft part sarcoma, Spitzoid cancer, Wilms' tumor, lymphomas (e.g. including Hodgkin's lymphoma, lymphoplasmacytoid lymphoma, follicular lymphoma, mucosa-associated lymphoid tissue lymphoma, mantle cell lymphoma, B-lineage large cell lymphoma, Burkitt's lymphoma, and T-cell anaplastic large cell lymphoma), inflammatory lung diseases (e.g. asthma), inflammatory bowel diseases, (e.g. ulcerative colitis, Crohn's disease), inflammatory skin diseases (e.g. atopic dermatitis, eczema and psoriasis), interstitial cystitis, rhinitis, acute pain, chronic pain, cancer pain, surgical pain, inflammatory pain, neuropathic pain, nociceptive pain, pain of osteoarthritis, chronic low back pain, low back pain of osteoporosis, pain of bone fracture, pain of rheumatoid arthritis, postherpetic pain, pain of diabetic neuropathy, fibromyalgia, pain of pancreatitis, pain of interstitial cystitis, pain of endometriosis, pain of irritable bowel syndrome, migraine, pain of pulpitis, interstitial cystitis pain, painful bladder syndrome, central pain syndromes, postsurgical pain syndromes, bone and joint pain, repetitive motion pain, dental pain, myofascial pain, perioperative pain, dysmenorrhea, myofascial pain, angina pain, headache, primary hyperalgesia, secondary hyperalgesia, primary allodynia, secondary allodynia, other pain caused by central sensitization, systemic cutaneous pruritus, localized cutaneous pruritus, senile cutaneous pruritus, gestational pruritus, pruritus ani, vulvar pruritus, metastatic bone disease, treatment-induced bone loss, osteoporosis, rheumatoid arthritis, bone metastases, ankylosing spondylitis, Paget's disease, periodontal disease, osteolytic disease, multiple sclerosis, Parkinson's disease, Alzheimer's disease, Chagas disease, cachexia, anorexia, demyelination and dysmyelination.

(69) Selective degradation of a target protein induced by a small molecule may be achieved by recruiting an E₃ ubiquitin ligase and mimicking protein misfolding with a hydrophobic tag (Buckley and Crews, 2014). Additionally, the small molecule has one moiety that binds to an E₃ ubiquitin ligase and another moiety that binds the protein target of interest (Buckley and Crews, 2014). The induced proximity leads to ubiquitination of the target followed by its degradation via proteasome-mediated proteolysis. Several types of high affinity small-molecule E₃ ligase ligands have been identified or developed. They include (1) immunomodulatory drugs (IMiDs) such as thalidomide and pomalidomide, which bind cereblon (CRBN or CRL4CRBN), a component of a cullin-RING ubiquitin ligase (CRL) complex (Bondeson et al., 2015; Chamberlain et al., 2014; Fischer et al., 2014; Ito et al., 2010; Winter et al., 2015); (2) VHL-1, a hydroxyproline-containing ligand, which binds van Hippel-Lindau protein (VHL or CRL2VHL), a component of another CRL complex (Bondeson et al., 2015; Buckley et al., 2012a; Buckley et al., 2012b; Galdeano et al., 2014; Zengerle et al., 2015); (3) compound 7, which selectively binds KEBAP1, a component of a CRL3 complex (Davies et al., 2016); (4) AMG232, which selectively binds MDM2, a heterodimeric RING E3 ligase (Sun et al., 2014); and (5) LCL161, which selectively binds IAP, a homodimeric RING E3 ligase (Ohoka et al., 2017; Okuhira et al., 2011; Shibata et al., 2017). The E₃ ligase recruiting bifunctional degrader technology has been applied to degradation of several protein targets (Bondeson et al., 2015; Buckley et al., 2015; Lai et al., 2016; Lu et al., 2015; Winter et al., 2015; Zengerle et al., 2015). In addition, a hydrophobic tagging approach, which utilizes a bulky and hydrophobic adamantyl group, has been developed to mimic protein misfolding, leading to the degradation of the target protein by proteasome (Buckley and Crews, 2014). This approach has been applied to selective degradation of the pseudokinase HER3 (Xie et al., 2014). The inventors have not yet seen any efforts applying any of these approaches to degradation of TRK, TRK mutant, TRK deletion, TRK splicing or TRK fusion proteins.

(70) Currently available small molecules targeting TRK focus on inhibition of the kinase activity of TRK. A number of selective small-molecule TRK kinase inhibitors, such as entrectinib (RXDX-101) (Menichincheri et al., 2016), GNF-8625 (Choi et al., 2015), larotrectinib (LOXO-101; ARRY-

470) (Drilon et al., 2018), altiratinib (DCC.sub.2701, DCC-270, DP-5164)(Smith et al., 2015), sitravatinib (MGCD516)(Patwardhan et al., 2016), cabozantinib (XL-184, BMS-907351)(Fuse et al., 2017), dovitinib (TKI-258, CHIR-258)(Chong et al., 2017), milciclib (PHA-848125AC)(Brasca et al., 2009), belizatinib (TSR-011)(Ricciuti et al., 2017), GZ389988 (Bailey et al., 2017a, b), pegcantratinib (Cranston et al., 2017), AZD7451 (Tatematsu et al., 2014), TPX-0005 (Cui et al., 2016), LOXO-195 (Blake et al., 2016), regorafenib (Subbiah et al., 2017), DS-6051b (Fujiwara et al., 2018), F17752(Amatu et al., 2016), PLX7486 (Amatu et al., 2016), AZD-6918 (Li et al., 2015), ASP7962(Bailey et al., 2017a, b), VM902A (Bailey et al., 2017a, b), ONO-4474 (Bailey et al., 2017a, b), and PF-06273340 (Skerratt et al., 2016) have been reported.

(71) In the present disclosure, a novel approach is taken: to develop compounds that directly and selectively modulate not only the kinase activity of TRK, but also its protein level. Strategies for inducing protein degradation include recruiting E.sup.3 ubiquitin ligases, mimicking protein misfolding with hydrophobic tags, and inhibiting chaperones. Such an approach, based on the use of bivalent small molecule compounds, permits more flexible regulation of protein levels in vitro and in vivo compared with techniques such as gene knockout or short hairpin RNA-mediated (shRNA) knockdown. Unlike gene knockout or shRNA knockdown, a small molecule approach further provides an opportunity to study dose and time dependency in a disease model through modulating the administration routes, concentrations and frequencies of administration of the corresponding small molecule.

(72) Bivalent Compounds

(73) For the purpose of the present disclosure, the terms “bifunctional compound”, “bifunctional degrader”, “bifunctional TRK degrader”, “bivalent compound” and “heterobifunctional compound” are used interchangeably.

(74) In some aspects, the present disclosure provides bivalent compounds including a TRK ligand conjugated to a degradation tag, or a pharmaceutically acceptable salt or analog thereof. The TRK ligand may be conjugated to the degradation tag directly or via a linker moiety. In certain embodiments, the TRK ligand may be conjugated to the degradation tag directly. In certain embodiments, the TRK ligand may be conjugated to the degradation tag via a linker moiety.

(75) As used herein, the terms “tropomyosin receptor kinase ligand” and “TRK ligand”, or “TRK targeting moiety” are to be construed to encompass any molecules ranging from small molecules to large proteins that associate with or bind to TRK protein. In certain embodiments, the TRK ligand is capable of binding to a TRK protein comprising TRK, a TRK mutant, a TRK deletion, a TRK splicing, or a TRK fusion protein. The TRK ligand can be, for example but not limited to, a small molecule compound (i.e., a molecule of molecular weight less than about 1.5 kilodaltons (kDa)), a peptide or polypeptide, nucleic acid or oligonucleotide, carbohydrate such as oligosaccharides, or an antibody or fragment thereof.

(76) TRK Ligand

(77) The TRK ligand or targeting moiety can be a TRK kinase inhibitor or a portion of TRK kinase inhibitor. In certain embodiments, the TRK kinase inhibitor comprises one or more of (e.g., entrectinib (RXDX-101)(Menichincheri et al., 2016), GNF-8625 (Choi et al., 2015), larotrectinib (LOXO-101; ARRY-470)(Drilon et al., 2018), altiratinib (DCC.sub.2701, DCC-270, DP-5164) (Smith et al., 2015), sitravatinib (MGCD516)(Patwardhan et al., 2016), cabozantinib (XL-184, BMS-907351)(Fuse et al., 2017), dovitinib (TKI-258, CHIR-258)(Chong et al., 2017), milciclib (PHA-848125AC)(Brasca et al., 2009), belizatinib (TSR-011)(Ricciuti et al., 2017), GZ389988 (Bailey et al., 2017a, b), pegcantratinib (Cranston et al., 2017), AZD7451 (Tatematsu et al., 2014), TPX-0005 (Cui et al., 2016), LOXO-195 (Blake et al., 2016), regorafenib (Subbiah et al., 2017), DS-6051b (Fujiwara et al., 2018), F17752(Amatu et al., 2016), PLX7486 (Amatu et al., 2016), AZD-6918 (Li et al., 2015), ASP7962(Bailey et al., 2017a, b), VM902A (Bailey et al., 2017a, b), ONO-4474 (Bailey et al., 2017a, b), PF-06273340 (Skerratt et al., 2016) and analogs thereof), which is capable of inhibiting the kinase activity of TRK. As used herein, a “TRK kinase inhibitor”

refers to an agent that restrains, retards, or otherwise causes inhibition of a physiological, chemical or enzymatic action or function and causes a decrease in binding of at least 5%. An inhibitor can also or alternately refer to a drug, compound, or agent that prevents or reduces the expression, transcription, or translation of a gene or protein. An inhibitor can reduce or prevent the function of a protein, e.g., by binding to or activating/inactivating another protein or receptor.

(78) In certain embodiments, the TRK ligand is derived from a TRK kinase inhibitor comprising:

(79) ##STR00165## ##STR00166##

(80) In certain embodiments, the TRK ligand include, but are not limited to DS-6051b (Fujiwara et al., 2018), F17752 (Amatu et al., 2016), PLX7486 (Amatu et al., 2016), AZD-6918 (Li et al., 2015), ASP7962 (Bailey et al., 2017a, b), VM902A (Bailey et al., 2017a, b), PF-06273340 (Skerratt et al., 2016) and ONO-4474 (Bailey et al., 2017a, b). In certain embodiments, the TRK ligand is derived from any one or more of DS-6051b (Fujiwara et al., 2018), F17752 (Amatu et al., 2016), PLX7486 (Amatu et al., 2016), AZD-6918 (Li et al., 2015), ASP7962 (Bailey et al., 2017a, b), VM902A (Bailey et al., 2017a, b), PF-06273340 (Skerratt et al., 2016) and ONO-4474 (Bailey et al., 2017a, b).

(81) In one aspect, provided herein is a compound of Formula I:

(82) ##STR00167## or a pharmaceutically acceptable salt thereof, wherein X^{sup.1} and X^{sup.2} are independently selected from CH and N; X^{sup.3} and X^{sup.4} are independently selected from C(O) and CR^{sup.4}R^{sup.5}; R^{sup.1} is selected from H, —NR^{sup.2}R^{sup.3}, halogen, optionally substituted C_{sub.1-6}alkyl, optionally substituted C_{sub.1-6}heteroalkyl, optionally substituted C_{sub.3-10}carbocyclyl, optionally substituted 3- to 10-membered heterocyclyl, optionally substituted 5- to 10-membered heteroaryl, optionally substituted C: «heteroalkyl, optionally substituted C_{sub.1-6}haloalkyl, and optionally substituted C_{sub.1-6}alkoxy; R^{sup.2}, R^{sup.3}, R^{sup.4}, and R^{sup.5} are independently selected from H, optionally substituted C_{sub.1-6}alkyl, optionally substituted C_{sub.1-6}heteroalkyl, optionally substituted C_{sub.3-10}carbocyclyl, optionally substituted 3- to 10-membered heterocyclyl, optionally substituted 5- to 10-membered heteroaryl, optionally substituted C_{sub.1}heteroalkyl, optionally substituted C_{sub.1-6}haloalkyl, and optionally substituted C_{sub.1-6}alkoxy; and L is selected from a bond,

(83) ##STR00168## ##STR00169## ##STR00170##

or one of X^{sup.3} and X^{sup.4} is C(O) and the other is CR^{sup.4}R^{sup.5}; and L is

(84) ##STR00171##

(85) In some embodiments, X^{sup.1} and X^{sup.2} are each N.

(86) In some embodiments, X^{sup.3} is C(O) and X^{sup.4} is CR^{sup.4}R^{sup.5}. In some embodiments, X^{sup.3} is C(O) and X^{sup.4} is CR^{sup.4}R^{sup.5}.

(87) In some embodiments, X^{sup.3} and X^{sup.4} are both C(O). In some embodiments, X^{sup.3} and X^{sup.4} are both CR^{sup.4}R^{sup.5}.

(88) In some embodiments, R^{sup.1} is —NR^{sup.2}R^{sup.3}. In some embodiments, R^{sup.1} is

(89) ##STR00172##

(90) In one aspect, provided herein is a compound of Formula Ia:

(91) ##STR00173## or a pharmaceutically acceptable salt thereof, wherein X^{sup.1} and X^{sup.2} are independently selected from CH and N; X^{sup.3} and X^{sup.4} are independently selected from C(O), CR^{sup.4}R^{sup.5}, and NR^{sup.6}; R^{sup.1} is selected from H, —NR^{sup.2}R^{sup.3}, halogen, optionally substituted C_{sub.1-6}alkyl, optionally substituted C_{sub.1-6}heteroalkyl, optionally substituted C_{sub.3-10}carbocyclyl, optionally substituted 3- to 10-membered heterocyclyl, optionally substituted 5- to 10-membered heteroaryl, optionally substituted aryl, optionally substituted C_{sub.1}-heteroalkyl, optionally substituted C_{sub.1-6}haloalkyl, and optionally substituted C_{sub.1-6}alkoxy; R^{sup.2}, R^{sup.3}, R^{sup.4}, R^{sup.5}, and R^{sup.6} are independently selected from H, halogen, optionally substituted C_{sub.1-6}alkyl, optionally substituted C_{sub.1-6}heteroalkyl, optionally substituted C_{sub.3-10}carbocyclyl, optionally substituted 3- to 10-membered heterocyclyl, optionally substituted 5- to 10-membered heteroaryl,

optionally substituted C.sub.1-6heteroalkyl, optionally substituted C.sub.1-6haloalkyl, optionally substituted C.sub.1-6alkoxy, and optionally substituted 2,6-dioxopiperidin-3-yl; L is selected from a bond, R.sup.r—R'', R'COR'', R'CO.sub.2R'', R'C(O)N(R.sup.7)R'', R'C(S)N(R.sup.7)R'', R'OR'', R'SR'', R'SOR'', R'SOR'', R'SO.sub.2N(R.sup.1)R'', R'N(R.sup.7)R'', R'N(R.sup.7)COR'', R'N(R.sup.7)CON(R.sup.8)R'', R'N(R.sup.7)C(S)R'', optionally substituted C.sub.1-C.sub.6 alkylene, optionally substituted C.sub.1-C.sub.5 heteroalkylene, optionally substituted C.sub.2-C.sub.8 alkenylene, optionally substituted C.sub.2-C.sub.8 alkynylene, optionally substituted C.sub.1-C.sub.8heteroalkylene, optionally substituted C.sub.1-C.sub.8alkoxyC.sub.1-C.sub.5alkylene, optionally substituted C.sub.1-C.sub.8alkylaminoC.sub.1-C.sub.5alkylene, optionally substituted C.sub.1-C.sub.5 haloalkylene, optionally substituted C.sub.1-C.sub.5 hydroxyalkylene, optionally substituted C.sub.1-C.sub.5 aminoalkylene, optionally substituted C.sub.4-C.sub.13 fused carbocyclyl, optionally substituted 5-13 membered fused heterocyclyl, optionally substituted C.sub.5-C.sub.13 bridged carbocyclyl, optionally substituted 5-13 membered bridged heterocyclyl, optionally substituted C.sub.5-C.sub.13 spiro carbocyclyl, optionally substituted 5-13 membered spiro heterocyclyl, optionally substituted 3-10 membered carbocyclyl, optionally substituted 3-10 membered heterocyclyl, optionally substituted aryl, and optionally substituted heteroaryl; wherein L is optionally attached to X.sup.3 or X.sup.4; R' and R'' are independently selected from null, optionally substituted C.sub.1-C.sub.6 alkylene, optionally substituted C.sub.1-C.sub.6 heteroalkylene, optionally substituted C.sub.2-C.sub.8 alkenylene, optionally substituted C.sub.2-C.sub.8 alkynylene, optionally substituted C.sub.1-C.sub.8heteroalkylene, optionally substituted C.sub.1-C.sub.8 hydroxyalkylene, optionally substituted C.sub.1-C.sub.5 aminoalkylene, optionally substituted C.sub.1-C.sub.6alkoxyC.sub.1-C.sub.5alkylene, optionally substituted C.sub.1-C.sub.8alkylaminoC.sub.1-C.sub.5alkylene, optionally substituted C.sub.1-C.sub.6 haloalkylene, optionally substituted 3-10 membered carbocyclyl, optionally substituted 3-10 membered heterocyclyl, optionally substituted C.sub.4-C.sub.13 fused carbocyclyl, optionally substituted 5-13 membered fused heterocyclyl, optionally substituted C.sub.5-C.sub.13 bridged carbocyclyl, optionally substituted 5-13 membered bridged heterocyclyl, optionally substituted C.sub.5-C.sub.13 spiro carbocyclyl, optionally substituted 5-13 membered spiro heterocyclyl, optionally substituted aryl, and optionally substituted heteroaryl; and R.sup.7 and R.sup.8 are independently selected from hydrogen, optionally substituted C.sub.1-C.sub.8alkyl, optionally substituted C.sub.1-C.sub.5 heteroalkyl, optionally substituted C.sub.1-C.sub.6 heteroalkyl, optionally substituted C.sub.1-C.sub.8 alkoxy, optionally substituted C.sub.2-C.sub.8 alkenyl, optionally substituted C.sub.2-C.sub.8 alkynyl, optionally substituted C.sub.1-C.sub.8 alkoxy-C.sub.1-C.sub.8alkyl, optionally substituted C.sub.1-C.sub.5 haloalkyl, optionally substituted C.sub.1-C.sub.5 hydroxyalkyl, optionally substituted C.sub.1-C.sub.5 aminoalkyl, optionally substituted C.sub.1-C.sub.5alkylaminoC.sub.1-C.sub.8alkyl, optionally substituted 3-10 membered carbocyclyl, optionally substituted 3-10 membered heterocyclyl, optionally substituted aryl, and optionally substituted heteroaryl; or R' and R'', R.sup.7 and R.sup.8, R' and R.sup.7, R' and R.sup.8, R'' and R.sup.7, R'' and R.sup.5 together with the atom to which they are connected optionally form a 3-20 membered carbocyclyl or 3-20 membered heterocyclyl ring;

(92) In some embodiments, L is selected from

(93) ##STR00174##

(94) In some embodiments, X.sup.1 and X.sup.2 are each N.

(95) In some embodiments, at least one of X.sup.3 and X.sup.4 is NR.sup.6. In some embodiments, X.sup.3 and X.sup.4 are both NR.sup.6. In some embodiments, either X.sup.3 or X.sup.4 is —N-(2,6-dioxopiperidin-3-yl).

(96) In some embodiments, R.sup.1 is —NR.sup.2R.sup.3. In some embodiments, R.sup.1 is

(97) ##STR00175##

(98) In some embodiments, L is connected to X.sup.3. In some embodiments, L is connected to X.sup.4.

(99) In one aspect, provided herein is a compound of Formula II:

(100) ##STR00176## or a pharmaceutically acceptable salt thereof, wherein X^{sup.1} and X^{sup.2} are independently selected from CH and N; one of X^{sup.3} and X^{sup.4} is C(O) and the other is CR^{sup.1}R^{sup.2}; and L is selected from

(101) ##STR00177##

or X^{sup.3} and X^{sup.4} are each C(O); and; L is selected from

(102) ##STR00178##

and R^{sup.1} and R^{sup.2} are independently selected from H, halogen, optionally substituted C_{sub.1-6}alkyl, optionally substituted C_{sub.1}-C_{sub.6} heteroalkyl, optionally substituted C_{sub.3-10}carbocyclyl, optionally substituted 3- to 10-membered heterocyclyl, optionally substituted 5- to 10-membered heteroaryl, optionally substituted C_{sub.1} heteroalkyl, optionally substituted C_{sub.1-6}haloalkyl, and optionally substituted C_{sub.1-6}alkoxy.

(103) In some embodiments, X^{sup.1} and X^{sup.2} are each N.

(104) In one aspect, provided herein is a compound of Formula III:

(105) ##STR00179## or a pharmaceutically acceptable salt thereof, wherein X^{sup.1} and X^{sup.3} are independently selected from CR^{sup.1}, CR^{sup.1}R^{sup.2}, O, N, and NR^{sup.1}; X^{sup.2} is selected from N, CO, and CH; Y is selected from O, NR^{sup.2} and CR^{sup.8}R^{sup.9}; Ar is selected from C_{sub.6-10}aryl and 5- to 10-membered heteroaryl, each of which is optionally substituted with one or more substituents independently selected from hydrogen, halogen, CN, NO_{sub.2}, OR^{sup.17}, SR^{sup.17}, NR^{sup.18}R^{sup.19}, COR^{sup.17}, CO_{sub.2}R^{sup.17}, CONR^{sup.18}R^{sup.19}, SOR^{sup.17}, SO_{sub.2}R^{sup.17}, SO_{sub.2}NR^{sup.18}R^{sup.19}, NR^{sup.17}CORP, NR^{sup.17}C(O)NR^{sup.18}R^{sup.19}, NR^{sup.18}SOR^{sup.17}, NR^{sup.18}SO_{sub.2}R^{sup.17}, optionally substituted C_{sub.1-8}alkyl, optionally substituted C_{sub.1-8} heteroalkyl, optionally substituted C_{sub.2-8}alkenyl, optionally substituted C_{sub.2-8}alkynyl, optionally substituted C_{sub.1-8}heteroalkyl, optionally substituted C_{sub.1-8}alkoxy, optionally substituted C_{sub.1-8}alkyl amino, optionally substituted C_{sub.3-10}carbocyclyl, —O-(optionally substituted C_{sub.3-10}carbocyclyl), —NH-(optionally substituted C_{sub.3-10}carbocyclyl), optionally substituted 3- to 10-membered heterocyclyl, —O-(optionally substituted 3- to 10-membered heterocyclyl), —NH-(optionally substituted 3- to 10-membered heterocyclyl), optionally substituted C_{sub.6-10}aryl, and optionally substituted 5- to 10-membered heteroaryl; L is selected from a bond, R^{sup.r}—R^{sup.r}, R^{sup.r}COR^{sup.r}, R^{sup.r}CO_{sub.2}R^{sup.r}, R^{sup.r}C(O)N(R^{sup.21})R^{sup.r}, R^{sup.r}C(S)N(R^{sup.21})R^{sup.r}, R^{sup.r}OR^{sup.r}, R^{sup.r}SR^{sup.r}, R^{sup.r}SOR^{sup.r}, R^{sup.r}SO_{sub.2}R^{sup.r}, R^{sup.r}SO_{sub.2}N(R^{sup.21})R^{sup.r}, R^{sup.r}N(R^{sup.21})R^{sup.r}, R^{sup.r}N(R^{sup.21})COR^{sup.r}, R^{sup.r}N(R^{sup.21})CON(R^{sup.22})R^{sup.r}, R^{sup.r}N(R^{sup.21})C(S)R^{sup.r}, optionally substituted C_{sub.1-8}alkylene, optionally substituted C_{sub.1-8} heteroalkylene, optionally substituted C_{sub.2-8} alkenylene, optionally substituted C_{sub.2-8} alkynylene, optionally substituted C_{sub.1-8}heteroalkylene, optionally substituted C_{sub.1-8}alkoxyC_{sub.1-8}alkylene, optionally substituted C_{sub.1-8}alkylaminoC_{sub.1-8}alkylene, optionally substituted C_{sub.1-8} haloalkylene, optionally substituted C_{sub.1-8} hydroxyalkylene, optionally substituted C_{sub.1-8} aminoalkylene, optionally substituted C_{sub.4-13} fused carbocyclyl, optionally substituted 5-13 membered fused heterocyclyl, optionally substituted C_{sub.5-13} bridged carbocyclyl, optionally substituted 5-13 membered bridged heterocyclyl, optionally substituted C_{sub.5-13} spiro carbocyclyl, optionally substituted 5-13 membered spiro heterocyclyl, optionally substituted 3-10 membered carbocyclyl, optionally substituted 3-10 membered heterocyclyl, optionally substituted aryl, and optionally substituted heteroaryl; wherein L is optionally attached to X^{sup.1} or X^{sup.3}; R^{sup.r} and R^{sup.r} are independently selected from null, optionally substituted C_{sub.1-8}alkylene, optionally substituted C_{sub.1-8} heteroalkylene, optionally substituted C_{sub.2-8} alkenylene, optionally substituted C_{sub.2-8} alkynylene, optionally substituted C_{sub.1-8} heteroalkylene, optionally substituted C_{sub.1-8} hydroxyalkylene, optionally substituted

C.sub.1-C.sub.5 aminoalkylene, optionally substituted C.sub.1-C.sub.8alkoxyC.sub.1-C.sub.5alkylene, optionally substituted C.sub.1-C.sub.5alkylaminoC.sub.1-C.sub.5alkylene, optionally substituted C.sub.1-C.sub.5 haloalkylene, optionally substituted 3-10 membered carbocyclyl, optionally substituted 3-10 membered heterocyclyl, optionally substituted C.sub.4-C.sub.13 fused carbocyclyl, optionally substituted 5-13 membered fused heterocyclyl, optionally substituted C.sub.5-C.sub.13 bridged carbocyclyl, optionally substituted 5-13 membered bridged heterocyclyl, optionally substituted C.sub.5-C.sub.13 spiro carbocyclyl, optionally substituted 5-13 membered spiro heterocyclyl, optionally substituted aryl, and optionally substituted heteroaryl; R.sup.1 and R.sup.2 are independently selected at each occurrence from H, halogen, optionally substituted C.sub.1-8 alkyl, optionally substituted C.sub.1-heteroalkyl, optionally substituted C.sub.3-10carbocyclyl, optionally substituted 3- to 10-membered heterocyclyl, optionally substituted 5- to 10-membered heteroaryl, optionally substituted C.sub.1-6heteroalkyl, optionally substituted C.sub.1-6haloalkyl, optionally substituted Ct alkoxy, and optionally substituted 2,6-dioxopiperidin-3-yl; R.sup.3 is selected from a bond, ~OR.sup.14-, —SR.sup.14—, —N(R.sup.15)R.sup.14—, —COR.sup.14, —CO.sub.2R.sup.14—, —CON(R.sup.15)R.sup.14—, —SOR.sup.14—, —SO.sub.2R.sup.14—, —SO.sub.2N(R.sup.15)R.sup.14—, —N(R.sup.16)COR.sup.14—, —N(R.sup.16)CON(R.sup.15)R.sup.14—, N(R.sup.16)SOR.sup.14—, —N(R.sup.16)SO.sub.2R.sup.14—, optionally substituted C.sub.1-8alkylene, optionally substituted C.sub.1-C.sub.8 heteroalkylene, optionally substituted C.sub.2-C.sub.8alkenylene, optionally substituted C.sub.2-C.sub.8alkynylene, optionally substituted C.sub.1-8heteroalkylene, optionally substituted C.sub.3-10carbocyclyl, optionally substituted 3- to 10-membered heterocyclyl, optionally substituted C.sub.6-10aryl, and optionally substituted 5- to 10-membered heteroaryl; R.sup.4, R.sup.5, and R.sup.6 are independently selected from hydrogen, halogen, CN, NO.sub.2, OR.sup.10, SR.sup.11, NR.sup.12R.sup.13, COR.sup.10, CO.sub.2R.sup.10, C(O)NR.sup.12R.sup.13, SOR.sup.10, SO.sub.2R.sup.10, SO.sub.2NR.sup.12R.sup.13, NR.sup.10C(O)R.sup.13, NR.sup.10C(O)NR.sup.12R.sup.13, NR.sup.10SOR.sup.13, NR.sup.10SO.sub.2R.sup.13, optionally substituted C.sub.1-8alkyl, optionally substituted C.sub.1-C.sub.8 heteroalkyl, optionally substituted C.sub.2-C.sub.8alkenyl, optionally substituted C.sub.2-C.sub.8alkynyl, optionally substituted C.sub.1-8 heteroalkyl, optionally substituted C.sub.1-8 alkoxy, optionally substituted C.sub.3-10carbocyclyl, and optionally substituted 3- to 10-membered heterocyclyl; R.sup.7 is selected from optionally substituted C.sub.1-8alkyl, optionally substituted C.sub.1-8 heteroalkyl, optionally substituted C.sub.3-10carbocyclyl, optionally substituted 3- to 10-membered heterocyclyl, optionally substituted C.sub.6-10aryl, and optionally substituted 5- to 10-membered heteroaryl; R.sup.8 and R.sup.9 are independently selected from hydrogen, halogen, OH, optionally substituted C.sub.1-8alkyl, optionally substituted C.sub.1-8heteroalkyl, optionally substituted C.sub.1-8alkoxy, optionally substituted C.sub.3-10 carbocyclyl, —O-(optionally substituted C.sub.3-10carbocyclyl), optionally substituted C.sub.1-8alkylamino, —NH-(optionally substituted C.sub.3-10carbocyclyl), and optionally substituted 3- to 10-membered heterocyclyl; or R.sup.8 and R.sup.9 are taken together with the atom to which they are connected to form an optionally substituted C.sub.3-10carbocyclyl or an optionally substituted 3- to 10-membered heterocyclyl; R.sup.10, R.sup.11, R.sup.12, and R.sup.13 are independently selected from hydrogen, optionally substituted C.sub.1-8alkyl, optionally substituted C.sub.1-C.sub.8 heteroalkyl, optionally substituted C.sub.2-C.sub.8 alkenyl, optionally substituted C.sub.2-C.sub.8 alkynyl, optionally substituted C.sub.1-8heteroalkyl, optionally substituted C.sub.3-10carbocyclyl, optionally substituted 3- to 10-membered heterocyclyl, optionally substituted C.sub.6-10aryl, and optionally substituted 5- to 10-membered heteroaryl; or R.sup.12 and R.sup.13 are taken together with the atom to which they are connected to form an optionally substituted 3- to 10-membered heterocyclyl; R.sup.14 is selected from null, optionally substituted C.sub.1-8alkylene, optionally substituted C.sub.1-C.sub.8 heteroalkylene, optionally substituted C.sub.2-C.sub.8 alkenylene, optionally substituted C.sub.2-C.sub.8 alkynylene, optionally substituted C.sub.1-8heteroalkylene,

optionally substituted C.sub.1-8alkoxy, optionally substituted C.sub.3-10 carbocyclyl, —O-(optionally substituted C.sub.3-10 carbocyclyl), optionally substituted C.sub.1-8alkylamino, —NH-(optionally substituted C.sub.3-10 carbocyclyl), optionally substituted C.sub.3-10carbocyclyl, optionally substituted 3- to 10-membered heterocyclyl, optionally substituted C.sub.6-10aryl, and optionally substituted 5- to 10-membered heteroaryl; R.sup.15 and R.sup.16 are independently selected from hydrogen, optionally substituted C.sub.1-8alkyl, optionally substituted C.sub.1-C.sub.5 heteroalkyl, optionally substituted C.sub.2-C.sub.8 alkenyl, optionally substituted C.sub.2-C.sub.8 alkynyl, optionally substituted C.sub.1-8heteroalkyl, optionally substituted C.sub.1-8alkoxy, optionally substituted C.sub.3-10 carbocyclyl, optionally substituted 3- to 10-membered heterocyclyl, optionally substituted C.sub.6-10aryl, and optionally substituted 5- to 10-membered heteroaryl; or R.sup.14 and R.sup.15, together with the atom to which they are connected, optionally form an optionally substituted C.sub.3-10carbocyclyl or an optionally substituted 3- to 10-membered heterocyclyl; R.sup.17, R.sup.18, and R.sup.19 are independently selected from hydrogen, optionally substituted C.sub.1-8alkyl, optionally substituted C.sub.1-C.sub.5 heteroalkyl, optionally substituted C.sub.2-C.sub.8 alkenyl, optionally substituted C.sub.2-C.sub.8 alkynyl, optionally substituted C.sub.1-8heteroalkyl, optionally substituted C.sub.1-8alkoxy, optionally substituted C.sub.3-10carbocyclyl, —O-(optionally substituted C.sub.3-10carbocyclyl), optionally substituted 3- to 10-membered heterocyclyl, optionally substituted C.sub.6-10aryl, and optionally substituted 5- to 10-membered heteroaryl; or R.sup.18 and R.sup.19 are together with the atom to which they are connected to form an optionally substituted C.sub.3-10carbocyclyl or an optionally substituted 3- to 10-membered heterocyclyl; and R.sup.21 and R.sup.22 are independently selected from hydrogen, optionally substituted C.sub.1-C.sub.8 alkyl, optionally substituted C.sub.1-C.sub.6 heteroalkyl, optionally substituted C.sub.1-C.sub.8 alkoxy, optionally substituted C.sub.2-C.sub.8 alkenyl, optionally substituted C.sub.2-C.sub.8 alkynyl, optionally substituted C.sub.1-C.sub.8 alkoxy-C.sub.1-C.sub.5alkyl, optionally substituted C.sub.1-C.sub.5 haloalkyl, optionally substituted C.sub.1-C.sub.5 hydroxyalkyl, optionally substituted C.sub.1-C.sub.5 aminoalkyl, optionally substituted C.sub.5-C.sub.8alkylaminoC.sub.1-C.sub.8alkyl, optionally substituted C.sub.3-10 carbocyclyl, optionally substituted 3-10 membered heterocyclyl, optionally substituted aryl, and optionally substituted heteroaryl; or R' and R'', R.sup.21 and R.sup.22, R' and R.sup.21, R' and R.sup.22, R'' and R.sup.21, or R'' and R.sup.22 together with the atom to which they are connected optionally form a 3-20 membered carbocyclyl or 3-20 membered heterocyclyl ring.

(106) In some embodiments, L is selected from

(107) ##STR00180##

(108) In some embodiments, R.sup.4, R.sup.5, and R.sup.6 are each hydrogen.

(109) In some embodiments, Y is CR.sup.8R.sup.9. In some embodiments, Y is CH.sub.2.

(110) In some embodiments, R.sup.7 is optionally substituted C.sub.6-10aryl. In some embodiments, R.sup.7 is

(111) ##STR00181##

(112) In some embodiments, Ar is C.sub.6-10aryl substituted with NR.sup.18R.sup.19. In some embodiments, Ar is

(113) ##STR00182##

(114) In some embodiments, R.sup.3 is optionally substituted 3- to 10-membered heterocyclyl. In some embodiments, R.sup.3 is

(115) ##STR00183##

(116) In some embodiments, X.sup.1 is CR.sup.1, X.sup.2 is CH, and X.sup.3 is NR.sup.1. In some embodiments, X.sup.1 is NR.sup.1, X.sup.2 is CH, and X.sup.3 is CR.sup.1. In some embodiments, X.sup.1 is CR.sup.1, X.sup.2 is N, and X.sup.3 is NR.sup.1. In some embodiments, X.sup.1 is NR.sup.1, X.sup.2 is N, and X.sup.3 is CR.sup.1. In some embodiments, X.sup.1 is NR.sup.1, X.sup.2 is CH, and X.sup.3 is N. In some embodiments, X.sup.1 is N, X.sup.2 is CH, and X.sup.3 is NR.sup.1. In some embodiments, X.sup.1 is CR.sup.1R.sup.2, X.sup.2 is CO, and X.sup.3 is

NR.sup.1. In some embodiments, X.sup.1 is NR.sup.1, X.sup.2 is CO, and X.sup.3 is NR.sup.1. In some embodiments, X.sup.1 is O, X.sup.2 is

(117) CO, and X.sup.3 is NR.sup.1. In some embodiments, X.sup.1 is CR.sup.1, X.sup.2 is CO, and X.sup.3 is NR.sup.1. In some embodiments, X.sup.1 is N, X.sup.2 is CO, and X.sup.3 is NR.sup.1.

(118) In some embodiments, R.sup.1 is

(119) ##STR00184##

(120) In one aspect, provided herein is a compound of Formula IV:

(121) ##STR00185## or a pharmaceutically acceptable salt thereof, wherein X.sup.1 and X.sup.3 are independently selected from CR.sup.1, N, and NR.sup.1; X.sup.2 is selected from N and CH; Y.sup.1 is selected from N and CR.sup.6; Y.sup.2, Y.sup.3, and Y.sup.4 are independently selected from N and C, with the proviso that only one of Y.sup.2, Y.sup.3, and Y.sup.4 is N; Z is selected from null, a bond, C(R.sup.5).sub.2, C(R.sup.5),C(R.sup.5).sub.2, CO, C(R.sup.5), CO, CONR.sup.3, C(R.sup.5), O, C(R.sup.5).sub.2NR.sup.5 and CH.sub.2NR.sup.5; Ar.sup.1 and Ar.sup.2 are independently selected from C.sub.6-10aryl and 5- to 10-membered heteroaryl, each of which is optionally substituted with one or more substituents independently selected from halogen, CN, NO.sub.2, OR.sup.10, SR.sup.10, NR.sup.11R.sup.12, COR.sup.11, CO.sub.2R.sup.10, CONR.sup.11R.sup.12, SOR.sup.10, SO.sub.2R.sup.10, SO.sub.2NR.sup.11R.sup.12, NR.sup.10 COR.sup.12, NR.sup.10C(O)NR.sup.11R.sup.12, NR.sup.10SOR.sup.12, NR.sup.10SO.sub.2R.sup.12, optionally substituted C.sub.1-8alkyl, optionally substituted C.sub.1-C.sub.8 heteroalkyl, optionally substituted C.sub.2-8alkenyl, optionally substituted C.sub.2-8alkynyl, optionally substituted C.sub.1-8 heteroalkyl, optionally substituted C.sub.1-8haloalkyl, optionally substituted C.sub.3-10carbocyclyl, optionally substituted 3- to 10-membered heterocyclyl, optionally substituted C.sub.6-10aryl, and optionally substituted 5- to 10-membered heteroaryl; L is selected from a bond, R.sup.r—R", R'COR", R'CO.sub.2R", R'C(O)N(R.sup.13)R", R'C(S)N(R.sup.13)R", R'OR", R'SR", R'SOR", R'SO.sub.2R", R'SON(R.sup.13)R", R'N(R.sup.13)R", R"N(R.sup.13)COR", R'N(R.sup.13)CON(R.sup.14)R", R'N(R.sup.13)C(S)R", optionally substituted C.sub.1-C.sub.8 alkylene, optionally substituted C.sub.1-C.sub.5 heteroalkylene, optionally substituted C.sub.2-C.sub.8 alkenylene, optionally substituted C.sub.2-C.sub.8 alkynylene, optionally substituted C.sub.1-C.sub.8heteroalkylene, optionally substituted C.sub.1-C.sub.8alkoxyC.sub.1-C.sub.8alkylene, optionally substituted C.sub.1-C.sub.8alkylaminoC.sub.1-C.sub.8 alkylene, optionally substituted C.sub.1-C.sub.5 haloalkylene, optionally substituted C.sub.1-C.sub.8 hydroxyalkylene, optionally substituted C.sub.1-C.sub.5 aminoalkylene, optionally substituted C.sub.4-C.sub.13 fused carbocyclyl, optionally substituted 5-13 membered fused heterocyclyl, optionally substituted C.sub.5-C.sub.13 bridged carbocyclyl, optionally substituted 5-13 membered bridged heterocyclyl, optionally substituted C.sub.5-C.sub.13 spiro carbocyclyl, optionally substituted 5-13 membered spiro heterocyclyl, optionally substituted 3-10 membered carbocyclyl, optionally substituted 3-10 membered heterocyclyl, optionally substituted aryl, and optionally substituted heteroaryl; wherein L is optionally attached to X.sup.1 or X.sup.3; R' and R" are independently selected from null, optionally substituted C.sub.1-C.sub.8 alkylene, optionally substituted C.sub.1-C.sub.8 heteroalkylene, optionally substituted C.sub.2-C.sub.8 alkenylene, optionally substituted C.sub.2-C.sub.8 alkynylene, optionally substituted C.sub.1-C.sub.8heteroalkylene, optionally substituted C.sub.1-C.sub.8 hydroxyalkylene, optionally substituted C.sub.1-C.sub.5 aminoalkylene, optionally substituted C.sub.1-C.sub.8alkoxyC.sub.1-C.sub.5alkylene, optionally substituted C.sub.1-C.sub.5alkylaminoC.sub.1-C.sub.8alkylene, optionally substituted C.sub.1-C.sub.5 haloalkylene, optionally substituted 3-10 membered carbocyclyl, optionally substituted 3-10 membered heterocyclyl, optionally substituted C.sub.4-C.sub.13 fused carbocyclyl, optionally substituted 5-13 membered fused heterocyclyl, optionally substituted C.sub.5-C.sub.13 bridged carbocyclyl, optionally substituted 5-13 membered bridged heterocyclyl, optionally substituted

C.sub.5-C.sub.13 spiro carbocyclyl, optionally substituted 5-13 membered spiro heterocyclyl, optionally substituted aryl, and optionally substituted heteroaryl; R.sup.1 is selected at each occurrence from H, halogen, optionally substituted C.sub.1-6alkyl, optionally substituted C.sub.1-heteroalkyl, optionally substituted C.sub.3-10carbocyclyl, optionally substituted 3- to 10-membered heterocyclyl, optionally substituted 5- to 10-membered heteroaryl, optionally substituted C.sub.1 heteroalkyl, optionally substituted C.sub.1-6haloalkyl, optionally substituted C.sub.1-8alkoxy, and optionally substituted 2,6-dioxopiperidin-3-yl; R.sup.3 is selected from a bond, —OR.sup.7—, —SR.sup.7—, —N(R.sup.8)R.sup.7—, —COR.sup.7—, —CO.sub.2R.sup.7—, —CON(R.sup.8)R.sup.7—, —SOR.sup.7—, —SO.sub.2R.sup.7—, —SO.sub.2N(R.sup.8)R.sup.7—, —N(R.sup.9)COR.sup.7—, —N(R.sup.9)CON(R.sup.8)R.sup.7—, N(R.sup.9)SOR.sup.7—, —N(R.sup.9)SO.sub.2R.sup.7—, optionally substituted C.sub.1-8alkylene, optionally substituted C.sub.1-C.sub.5 heteroalkylene, optionally substituted C.sub.2-C.sub.8alkenylene, optionally substituted C.sub.2-C.sub.8alkynylene, optionally substituted C.sub.1-8heteroalkylene, optionally substituted C.sub.3-10carbocyclyl, optionally substituted 3- to 10-membered heterocyclyl, optionally substituted C.sub.6-10aryl, and optionally substituted 5- to 10-membered heteroaryl; R.sup.4 and R.sup.5 are independently selected at each occurrence from hydrogen, halogen, OH, NH.sub.2, CN, NO.sub.2, optionally substituted C.sub.1-4alkyl, optionally substituted C.sub.1-C.sub.4 heteroalkyl, optionally substituted C.sub.1-4alkoxy, optionally substituted C.sub.1-4heteroalkyl, optionally substituted C.sub.1-4haloalkyl, optionally substituted C.sub.3-10carbocyclyl, —O-(optionally substituted C.sub.3-10 carbocyclyl), —NH-(optionally substituted C.sub.3-10carbocyclyl), and optionally substituted 3- to 10-membered heterocyclyl; R.sup.6 is selected from hydrogen, halogen, CN, NO.sub.2, optionally substituted C.sub.1-6alkyl, optionally substituted C.sub.1-C.sub.6 heteroalkyl, optionally substituted C.sub.3-10carbocyclyl, and optionally substituted 3- to 10-membered heterocyclyl; R.sup.7 is selected from null, optionally substituted C.sub.1-8alkylene, optionally substituted C.sub.1-C.sub.8 heteroalkylene, optionally substituted C.sub.2-C.sub.8 alkenylene, optionally substituted C.sub.2-C.sub.8 alkynylene, optionally substituted C.sub.1-heteroalkylene, optionally substituted C.sub.1-8alkoxy, optionally substituted C.sub.3-10 carbocyclyl, —O-(optionally substituted C.sub.3-10 carbocyclyl), optionally substituted C.sub.1-8alkylamino, —NH-(optionally substituted C.sub.3-10 carbocyclyl), optionally substituted C.sub.3-10carbocyclyl, optionally substituted 3- to 10-membered heterocyclyl, optionally substituted C.sub.6-10aryl, and optionally substituted 5- to 10-membered heteroaryl; R.sup.8 and R.sup.9 are independently selected from hydrogen, optionally substituted C.sub.1-8alkyl, optionally substituted C.sub.1-C.sub.5 heteroalkyl, optionally substituted C.sub.2-C.sub.8 alkenyl, optionally substituted C.sub.2-C.sub.8 alkynyl, optionally substituted C.sub.1-8heteroalkyl, optionally substituted C.sub.1-8alkoxy, optionally substituted C.sub.3-10 carbocyclyl, optionally substituted 3- to 10-membered heterocyclyl, optionally substituted C.sub.6-10aryl, and optionally substituted 5- to 10-membered heteroaryl; or R.sup.7 and R.sup.8, together with the atom to which they are connected, optionally form an optionally substituted C.sub.3-10carbocyclyl or an optionally substituted 3- to 10-membered heterocyclyl; R.sup.10, R.sup.11, and R.sup.12 are independently selected from hydrogen, optionally substituted C.sub.1-8alkyl, optionally substituted C.sub.1-C.sub.5 heteroalkyl, optionally substituted C.sub.2-8alkenyl, optionally substituted C.sub.2-8-alkynyl, optionally substituted C.sub.3-10carbocyclyl, optionally substituted 3- to 10-membered heterocyclyl, optionally substituted C.sub.6-10aryl, and optionally substituted 5- to 10-membered heteroaryl; or R.sup.11 and R.sup.12 are together with the atom to which they are connected to form an optionally substituted C.sub.3-C.sub.10carbocyclyl or an optionally substituted 3- to 10-membered heterocyclyl; R.sup.13 and R.sup.14 are independently selected from hydrogen, optionally substituted C.sub.1-C.sub.8 alkyl, optionally substituted C.sub.1-C.sub.5 heteroalkyl, optionally substituted C.sub.1-C.sub.8 alkoxy, optionally substituted C.sub.2-C.sub.8 alkenyl, optionally substituted C.sub.2-C.sub.8 alkynyl, optionally substituted C.sub.1-C.sub.8 alkoxy-C.sub.1-C.sub.8alkyl, optionally substituted C.sub.1-C.sub.5 haloalkyl, optionally

substituted C.sub.1-C.sub.5 hydroxyalkyl, optionally substituted C.sub.1-C.sub.5 aminoalkyl, optionally substituted C.sub.1-C.sub.5 alkylamino C.sub.1-C.sub.8 alkyl, optionally substituted 3-10 membered carbocyclyl, optionally substituted 3-10 membered heterocyclyl, optionally substituted aryl, and optionally substituted heteroaryl; or R' and R'', R.sup.13 and R.sup.14, R' and R.sup.13, R' and R.sup.14, R'' and R.sup.13, R'' and R.sup.14 together with the atom to which they are connected optionally form a 3-20 membered carbocyclyl or 3-20 membered heterocyclyl ring;; and n is 0, 1, 2, 3, or 4.

(122) In some embodiments, L is selected from

(123) ##STR00186##

(124) In some embodiments, Y.sup.1 is N, Y.sup.2 is N, Y.sup.3 is C, and Y.sup.4 is C.

(125) In some embodiments, Ar.sup.1 is C.sub.6-10 aryl optionally substituted with halogen. In some embodiments, Ar.sup.1 is

(126) ##STR00187##

(127) In some embodiments, Ar.sup.2 is C.sub.6-10 aryl optionally substituted with NR.sup.11 R.sup.12. In some embodiments, Ar.sup.2 is

(128) ##STR00188##

(129) In some embodiments, R.sup.3 is optionally substituted 3- to 10-membered heterocyclyl. In some embodiments, R.sup.3 is

(130) ##STR00189##

(131) In some embodiments, R.sup.4 is hydrogen.

(132) In some embodiments, Z is C(R.sup.5).sub.2. In some embodiments, Z is CH.sub.2.

(133) In some embodiments, n is 0.

(134) In some embodiments, X.sup.1 is CR.sup.1, X.sup.2 is CH, and X.sup.3 is NR.sup.1. In some embodiments, X.sup.1 is NR.sup.1, X.sup.2 is CH, and X.sup.3 is CR.sup.1. In some embodiments, X.sup.1 is CR.sup.1, X.sup.2 is N, and X.sup.3 is NR.sup.1. In some embodiments, X.sup.1 is NR.sup.1, X.sup.2 is N, and X.sup.3 is CR.sup.1. In some embodiments, X.sup.1 is NR.sup.1, X.sup.2 is CH, and X.sup.3 is N. In some embodiments, XL is N, X.sup.2 is CH, and X.sup.3 is NR.sup.1.

(135) In some embodiments, R.sup.1 is methyl. In some embodiments, R' is

(136) ##STR00190##

(137) In another embodiment, the TRK ligand comprises a moiety of Formula 1;

(138) Wherein,

(139) ##STR00191## R.sup.1, R.sup.2, R.sup.3, R.sup.4, Ar, and X are defined as before.

(140) In another embodiment, the TRK ligand comprises a moiety of Formula 1

(141) ##STR00192## wherein X is selected from CR'R'', CO, O, S, SO, SO.sub.2, and NR', wherein R' and R'' are independently selected from hydrogen, halogen, OH, optionally substituted C.sub.1-C.sub.8 alkyl, optionally substituted C.sub.1-C.sub.5 heteroalkyl, optionally substituted C.sub.1-8 alkoxy, optionally substituted C.sub.1-C.sub.8 alkoxy C.sub.1-C.sub.8 alkyl, optionally substituted C.sub.1-C.sub.8 alkylamino, optionally substituted C.sub.1-C.sub.8 alkylamino C.sub.1-C.sub.8 alkyl, optionally substituted C.sub.3-C.sub.10 carbocyclyl, optionally substituted C.sub.3-C.sub.10 cycloalkoxy, and optionally substituted 3-10 membered heterocyclyl; or R' and R'' together with the atom to which they are connected optionally form an optionally substituted 3-8 membered carbocyclyl or heterocyclyl ring; R is selected from optionally substituted C.sub.1-C.sub.8 alkyl, optionally substituted C.sub.1-C.sub.8 heteroalkyl, optionally substituted C.sub.3-C.sub.10 carbocyclyl, optionally substituted 3-10 membered heterocyclyl, optionally substituted aryl, and optionally substituted heteroaryl; R.sup.1, R.sup.2, and R.sup.3 are independently selected from hydrogen, halogen, CN, NO.sub.2, OR.sup.3, SR.sup.6, NR.sup.1 R.sup.8, COR.sup.5, CO.sub.2 R.sup.5, C(O)NR.sup.1 R.sup.8, SOR.sup.8, SO.sub.2 R.sup.5, SO.sub.2 NR.sup.7 R.sup.8, NR.sup.9 C(O)R.sup.8, NR.sup.3 C(O)NR.sup.7 R.sup.8, NR.sup.10 SOR.sup.9 NR.sup.7 SO.sub.2 R.sup.8, optionally substituted C.sub.1-C.sub.8 alkyl,

optionally substituted C.sub.1-C.sub.5 heteroalkyl, optionally substituted C.sub.1-C.sub.6 alkoxy, optionally substituted C.sub.1-C.sub.8alkoxyC.sub.1-C.sub.5alkyl, optionally substituted C.sub.1-C.sub.8alkylaminoC.sub.1-C.sub.8alkyl, optionally substituted C.sub.3-C.sub.10 carbocyclyl, optionally substituted C.sub.3-C.sub.10 cycloalkoxy, optionally substituted 3-10 membered heterocyclyl, optionally substituted C.sub.2-C.sub.8 alkenyl, and optionally substituted C.sub.2-C.sub.8 alkynyl, wherein R.sup.5, R.sup.6, R.sup.7, and R.sup.8 are independently selected from hydrogen, optionally substituted C.sub.1-C.sub.8alkyl, optionally substituted C.sub.1-C.sub.6 heteroalkyl, optionally substituted C.sub.1-C.sub.8alkoxyC.sub.1-C.sub.6alkyl, optionally substituted C.sub.1-C.sub.5alkylaminoC.sub.1-C.sub.5alkyl, optionally substituted C.sub.3-C.sub.10 carbocyclyl, optionally substituted 3-10 membered heterocyclyl, optionally substituted C.sub.2-C.sub.8 alkenyl, optionally substituted C.sub.2-C.sub.8 alkynyl, optionally substituted aryl, or optionally substituted heteroaryl, or R.sup.7 and R.sup.8 together with the atom to which they are connected optionally form an optionally substituted 3-8 membered heterocyclyl ring; R.sup.4 is connected to the linker moiety of the bivalent compound, and is selected from a bond, OR.sup.9, SR.sup.9, NR.sup.10R.sup.11, COR.sup.9, CO.sub.2R.sup.9, CONR.sup.10R.sup.11, SOR.sup.9, SO.sub.2R', SO.sub.2NR.sup.10R.sup.11, NR.sup.10OR.sup.11, NR.sup.9 CONR.sup.10R.sup.11 NR.sup.10SOR.sup.11, NR.sup.10SO.sub.2R.sup.11, optionally substituted C.sub.1-C.sub.8 alkyl, optionally substituted C.sub.1-C.sub.5 heteroalkyl, optionally substituted C.sub.1-C.sub.8alkoxy, optionally substituted C.sub.1-C.sub.6alkoxyC.sub.1-C.sub.6alkyl, optionally substituted C.sub.1-C.sub.8alkylaminoC.sub.1-C.sub.8alkyl, optionally substituted C.sub.3-C.sub.8 carbocyclyl, optionally substituted C.sub.3-C.sub.8 cycloalkoxy, optionally substituted 3-8 membered heterocyclyl, optionally substituted C.sub.2-C.sub.8 alkenyl, optionally substituted C.sub.2-C.sub.8 alkynyl, aryl, and optionally substituted heteroaryl, wherein R.sup.9, R.sup.10, and R.sup.11 are independently selected from null, a bond, hydrogen, optionally substituted C.sub.1-C.sub.8 alkyl, optionally substituted C.sub.1-C.sub.6 heteroalkyl, optionally substituted C.sub.1-C.sub.8 alkoxy, optionally substituted C.sub.1-C.sub.6alkoxyC.sub.1-C.sub.5alkyl, optionally substituted C.sub.1-C.sub.5alkylaminoC.sub.1-C.sub.5alkyl, optionally substituted C.sub.3-C.sub.8 carbocyclyl, optionally substituted C.sub.3-C.sub.8 cycloalkoxy, optionally substituted 3-8 membered heterocyclyl, optionally substituted C.sub.2-C.sub.8 alkenyl, optionally substituted C.sub.2-C.sub.8 alkynyl, optionally substituted aryl, and optionally substituted heteroaryl, or R.sup.10 and R.sup.11 together with the atom to which they are connected optionally form a 3-8 membered carbocyclyl or heterocyclyl ring; and Ar is selected from aryl and heteroaryl group, each of which is optionally substituted with one or more substituents independently selected from hydrogen, halogen, CN, NO.sub.2, OR.sup.12, SR.sup.12, NR.sup.13R.sup.14, COR.sup.12, CO.sub.2R.sup.12, CONR.sup.13R.sup.14, SOR.sup.12, SO.sub.2R.sup.12, SO.sub.2NR.sup.13R.sup.14, NR.sup.13COR.sup.14, NR.sup.15C(O)NR.sup.13R.sup.14, NR.sup.13SOR.sup.14, (142) NR.sup.13SO.sub.2R.sup.14, optionally substituted C.sub.1-C.sub.8 alkyl, optionally substituted C.sub.1-C.sub.8 heteroalkyl, optionally substituted C.sub.1-C.sub.8 alkoxy, optionally substituted C.sub.1-C.sub.8alkoxyC.sub.1-C.sub.5alkyl, optionally substituted C.sub.1-C.sub.8alkylaminoC.sub.1-C.sub.6alkyl, optionally substituted C.sub.3-C.sub.8 carbocyclyl, optionally substituted C.sub.3-C.sub.8 cycloalkoxy, optionally substituted 3-8 membered heterocyclyl, optionally substituted C.sub.2-C.sub.8 alkenyl, optionally substituted C.sub.2-C.sub.8 alkynyl, optionally substituted aryl, and optionally substituted heteroaryl, wherein R.sup.12, R.sup.13, R.sup.14, and R.sup.15 are independently selected from hydrogen, optionally substituted C.sub.1-C.sub.6 alkyl, optionally substituted C.sub.1-C.sub.5 heteroalkyl, optionally substituted C.sub.1-C.sub.6 alkoxy, optionally substituted C.sub.1-C.sub.8alkoxyC.sub.1-C.sub.8alkyl, optionally substituted C.sub.1-C.sub.5alkylaminoC.sub.1-C.sub.5alkyl, optionally substituted C.sub.3-C.sub.8 carbocyclyl, optionally substituted C.sub.3-C.sub.8 cycloalkoxy, optionally substituted 3-8 membered heterocyclyl, optionally substituted C.sub.2-C.sub.8 alkenyl, optionally substituted C.sub.2-C.sub.8 alkynyl, optionally substituted aryl, and optionally substituted

heteroaryl, or R.sup.13 and R.sup.14 together with the atom to which they are connected optionally form a 3-8 membered carbocyclyl or heterocyclyl ring.

(143) In one embodiment, X is selected from CR'R'', O, and NR', wherein R' and R'' are independently selected from hydrogen, F, OH, optionally substituted C.sub.1-C.sub.8 alkyl, optionally substituted C.sub.1-C.sub.3 heteroalkyl, and optionally substituted C.sub.1-C.sub.3 alkoxy, or R' and R'' together with the atom to which they are connected form an optionally substituted 3-6 membered carbocyclyl or heterocyclyl ring.

(144) In another embodiment, X is selected from CH.sub.2, cyclopropylene, CHF, CF.sub.2, O, NH, NCH.sub.3, NCH.sub.2CH.sub.3, and N-isopropyl.

(145) In another embodiment, R is selected from optionally substituted C.sub.3-C.sub.8 carbocyclyl, optionally substituted 3-8 membered heterocyclyl, optionally substituted aryl, and optionally substituted heteroaryl.

(146) In another embodiment, R is selected from optionally substituted phenyl and optionally substituted heteroaryl.

(147) In another embodiment, X is CH.sub.2; and R is 3,5-difluorophenyl.

(148) In another embodiment, R.sup.1, R.sup.2, and R.sup.3 are independently selected from hydrogen, F, Cl, and OH. In another embodiment, R.sup.4—Ar is selected from a moiety of formulae A.sup.1, A.sup.2, A.sup.3, and A.sup.4;

(149) ##STR00193##

wherein * indicates the connection to the linker moiety of the bivalent compound; and R.sup.a is selected from hydrogen, halogen, CN, NO.sub.2, OR.sup.12, SR.sup.12, NR.sup.13R.sup.14, COR.sup.12, CO.sub.2R.sup.12, CONR.sup.13R.sup.14, SOR.sup.12, SO.sub.2R.sup.12, SO.sub.2NR.sup.13R.sup.14, NR.sup.13COR.sup.14, NR.sup.15C(O)NR.sup.13R.sup.14, NR.sup.13SOR.sup.14, NR.sup.13SO.sub.2R.sup.14, optionally substituted C.sub.1-C.sub.8 alkyl, optionally substituted C.sub.1-C.sub.5 heteroalkyl, optionally substituted C.sub.1-C.sub.6 alkoxy, optionally substituted C.sub.1-C.sub.8 alkoxyC.sub.1-C.sub.8alkyl, optionally substituted C.sub.1-C.sub.5alkylaminoC.sub.1-C.sub.5alkyl, optionally substituted C.sub.3-C.sub.8 carbocyclyl, optionally substituted C.sub.3-C.sub.8 cycloalkoxy, optionally substituted 3-8 membered heterocyclyl, optionally substituted C.sub.2-C.sub.8 alkenyl, optionally substituted C.sub.2-C.sub.8 alkynyl, optionally substituted aryl, and optionally substituted heteroaryl, wherein R.sup.12, R.sup.13, R.sup.14, and R.sup.15 are independently selected from hydrogen, optionally substituted C.sub.1-C.sub.8 alkyl, optionally substituted C.sub.1-C.sub.5 heteroalkyl, optionally substituted C.sub.1-C.sub.8 alkoxy, optionally substituted C.sub.1-C.sub.8alkoxyC.sub.1-C.sub.8alkyl, optionally substituted C.sub.1-C.sub.5alkylaminoC.sub.1-C.sub.8alkyl, optionally substituted C.sub.3-C.sub.8 carbocyclyl, optionally substituted C.sub.3-C.sub.8 cycloalkoxy, optionally substituted 3-8 membered heterocyclyl, optionally substituted C.sub.2-C.sub.8 alkenyl, optionally substituted C.sub.2-C.sub.8 alkynyl, aryl, and optionally substituted heteroaryl, or R.sup.13 and R.sup.14 together with the atom to which they are connected optionally form a 3-8 membered carbocyclyl or heterocyclyl ring.

(150) In another embodiment, R.sup.4—Ar is selected from a moiety of formulae A.sup.1, A.sup.3, A.sup.3 and A.sup.4;

(151) ##STR00194##

wherein * indicates the connection to the linker moiety of the bivalent compound; and R.sup.a is selected from hydrogen, halogen, NR.sup.13R.sup.14, and NR.sup.13COR.sup.14, wherein R.sup.13 and R.sup.14 are independently selected from hydrogen, optionally substituted C.sub.1-C.sub.8 alkyl, optionally substituted C.sub.1-C.sub.5 heteroalkyl, optionally substituted C.sub.1-C.sub.8alkoxyC.sub.1-C.sub.5alkyl, optionally substituted C.sub.1-C.sub.5alkylaminoC.sub.1-C.sub.8alkyl, optionally substituted arylalkyl, optionally substituted heteroarylalkyl, optionally substituted C.sub.3-C.sub.8 carbocyclyl, optionally substituted C.sub.3-C.sub.8 cycloalkoxy, optionally substituted 3-8 membered heterocyclyl, optionally substituted C.sub.2-C.sub.8 alkenyl,

optionally substituted C.sub.2-C.sub.8 alkynyl, phenyl, and optionally substituted C.sub.5-C.sub.6 heteroaryl, or R.sup.13 and R.sup.14 together with the atom to which they are connected optionally form a 3-8 membered carbocyclyl or heterocyclyl ring.

(152) In another embodiment, R.sup.a is selected from H, (tetrahydro-2H-pyran-4-yl)amino, and 2-fluoroethyl)amino.

(153) In another embodiment, R.sup.4 is selected from

(154) ##STR00195##

(155) In another embodiment, the TRK ligand comprises a moiety of Formula 2;

(156) ##STR00196## wherein, R.sup.1, R.sup.2, R.sup.3, R.sup.4, Ar.sup.1, Ar.sup.2, X, X.sup.1, X.sup.2, X.sup.3, X.sup.4 and n are defined as before.

(157) In another embodiment, the TRK ligand comprises a moiety of Formula 2;

(158) ##STR00197##

wherein X.sup.1, X.sup.2, X.sup.3, and X.sup.4 are independently selected from C, CR', and N (preferly, X.sup.1 is selected from CR' and N, X.sup.2, X.sup.3, and X.sup.4 are independently selected from C and N), wherein R' is selected from hydrogen, halogen, CN, NO.sub.2, optionally substituted C.sub.1-C.sub.6 alkyl, optionally substituted C.sub.1-C.sub.6 heteroalkyl, optionally substituted C.sub.3-C.sub.6 carbocyclyl, and optionally substituted 3-6 membered heterocyclyl; X is selected from null, a bond, C(R.sup.2).sub.2, C(R.sup.2).sub.2C(R.sup.2).sub.2, CO, C(R.sup.2), CO, CONR.sup.2, C(R.sup.2).sub.2O,

(159) C(R.sup.2).sub.2NR.sup.2, and CH.sub.2NR.sup.2; R.sup.1 and R.sup.2, at each occurrence, are independently selected from hydrogen, halogen, OH, NH.sub.2, CN, NO.sub.2, optionally substituted C.sub.1-C.sub.4 alkyl, optionally substituted C.sub.1-C.sub.4 heteroalkyl, optionally substituted C.sub.1-C.sub.4 alkoxy, optionally substituted C.sub.1-C.sub.4 alkylamino, optionally substituted C.sub.1-C.sub.4 alkoxyalkyl, optionally substituted C.sub.1-C.sub.4 haloalkyl, optionally substituted C.sub.1-C.sub.4 hydroxyalkyl, optionally substituted C.sub.1-C.sub.4alkylaminoC.sub.1-C.sub.8alkyl, optionally substituted C.sub.3-C.sub.6 carbocyclyl, optionally substituted C.sub.3-C.sub.6 cycloalkoxy, and optionally substituted 3-6 membered heterocyclyl; n is 1 to 4; R.sup.3 is connected to the linker moiety of the bivalent compound either directly or through R.sup.4; R.sup.3 and R.sup.4 are independently selected from null, a bond, OR.sup.5, SR.sup.5, NR.sup.6R.sup.7, COR.sup.5, CO.sub.2R.sup.5,

(160) CONR.sup.6R.sup.7, SOR.sup.8, SOR.sup.5, SO.sub.2NR.sup.6R.sup.7, NR.sup.9 COR.sup.7, NR.sup.9C(O)NR.sup.6R.sup.7, NR.sup.6SOR.sup.7, NR.sup.9SO.sub.2R.sup.7, optionally substituted C.sub.1-C.sub.8 alkyl, optionally substituted C.sub.1-C.sub.5 heteroalkyl, optionally substituted C.sub.1-C.sub.8alkoxyC.sub.1-C.sub.8alkyl, optionally substituted C.sub.1-C.sub.8 haloalkyl, optionally substituted C.sub.1-C.sub.8 hydroxyalkyl, optionally substituted C.sub.1-C.sub.8alkylaminoC.sub.1-C.sub.8alkyl, optionally substituted C.sub.3-C.sub.8 carbocyclyl, optionally substituted C.sub.3-C.sub.8 cycloalkoxy, optionally substituted 3-8 membered heterocyclyl, optionally substituted C.sub.2-C.sub.8 alkenyl, optionally substituted C.sub.2-C.sub.8 alkynyl, optionally substituted aryl, and optionally substituted heteroaryl, wherein R.sup.5, R.sup.6 and R.sup.7 are independently selected from null, a bond, hydrogen, optionally substituted C.sub.1-C.sub.8 alkyl, optionally substituted C.sub.1-C.sub.5 heteroalkyl, optionally substituted C.sub.2-C.sub.8 alkenyl, optionally substituted C.sub.2-C.sub.8 alkynyl, optionally substituted C.sub.3-C.sub.6 carbocyclyl, optionally substituted 3-8 membered heterocyclyl, optionally substituted heterocarbocyclyl, optionally substituted arylalkyl, optionally substituted heteroarylalkyl, optionally substituted aryl, and optionally substituted heteroaryl, or R.sup.6 and R.sup.7 together with the atom to which they are connected optionally form a 3-8 membered carbocyclyl or 3-8 membered heterocyclyl ring; and Ar.sup.1 and Ar.sup.2 are independently selected from aryl and heteroaryl, each of which is optionally substituted with one or more substituents independently selected from halogen, CN, NO.sub.2, OR.sup.10, SR.sup.10, NR.sup.11R.sup.12, COR.sup.10, CO.sub.2R.sup.10, CONR.sup.11R.sup.12, SOR.sup.10,

SO.sub.2R.sup.10, SO.sub.2NR.sup.11R.sup.12, NR.sup.10OR.sup.12, NR.sup.10C(O)NR.sup.11R.sup.12, NR.sup.10SOR.sup.12, NR.sup.10SO.sub.2R.sup.12, optionally substituted C.sub.1-C.sub.8 alkyl, optionally substituted C.sub.1-C.sub.6 heteroalkyl, optionally substituted C.sub.1-C.sub.6alkoxyC.sub.1-C.sub.6alkyl, optionally substituted C.sub.1-C.sub.5 haloalkyl, optionally substituted C.sub.1-C.sub.5 hydroxyalkyl, optionally substituted C.sub.1-C.sub.5alkylaminoC.sub.1-C.sub.8alkyl, optionally substituted C.sub.3-C.sub.7 carbocyclyl, optionally substituted 3-7 membered heterocyclyl, optionally substituted C.sub.2-C.sub.8 alkenyl, optionally substituted C.sub.2-C.sub.8 alkynyl, optionally substituted aryl, and optionally substituted heteroaryl, wherein R.sup.10, R.sup.11, and R.sup.12, at each occurrence, are independently selected from null, hydrogen, optionally substituted C.sub.1-C.sub.8 alkyl, optionally substituted C.sub.1-C.sub.5 heteroalkyl, optionally substituted C.sub.2-C.sub.8 alkenyl, optionally substituted C.sub.2-C.sub.8 alkynyl, optionally substituted C.sub.3-C.sub.7 carbocyclyl, optionally substituted 3-7 membered heterocyclyl, optionally substituted aryl, and optionally substituted heteroaryl, or R.sup.11 and R.sup.12 together with the atom to which they are connected optionally form a 3-8 membered carbocyclyl or heterocyclyl ring.

(161) In one embodiment, X.sup.1 is selected from CR' and N, wherein R' is selected from hydrogen, F, C.sub.1,

(162) CH.sub.3, CF.sub.3, and cyclopropyl

(163) In another embodiment, X.sup.2, X.sup.3, and X.sup.4 are independently selected from C and N.

(164) In another embodiment, X is selected from a bond, CH.sub.2, CH.sub.2CH.sub.2, CO, CH.sub.2CO, CONH, CONCH.sub.3, CH.sub.2O, CH.sub.2NH, and CH.sub.2NCH.sub.3.

(165) In another embodiment, R.sup.1 and R.sup.2, at each occurrence, are independently selected from hydrogen, F, Cl, OH, optionally substituted C.sub.1-C.sub.6 alkyl, optionally substituted C.sub.1-C.sub.4 heteroalkyl, optionally substituted C.sub.1-C.sub.4 alkoxy, optionally substituted C.sub.1-C.sub.4 alkylamino, optionally substituted C.sub.1-C.sub.4 haloalkyl, optionally substituted C.sub.3-C.sub.6 carbocyclyl, optionally substituted C.sub.3-C.sub.8 cycloalkoxy, and optionally substituted 3-6 membered heterocyclyl.

(166) In another embodiment, X is CH.sub.2; and Ar.sup.1 is 3-fluorophenyl.

(167) In another embodiment, R.sup.3 is connected to the linker moiety of the bivalent compound directly, and R.sup.3 is selected from null, a bond, —OR.sup.5—, —SR.sup.5—, —NR.sup.6R.sup.7—, —COR.sup.5—, —CO.sub.2R.sup.5—, —CONR.sup.6R.sup.7—, —SOR.sup.5—, —SO.sub.2R.sup.5—, —SO.sub.2NR.sup.6R.sup.7—, —NR.sup.5COR.sup.7—, —NR.sup.9C(O)NR.sup.6R.sup.7—, —NR.sup.5SOR.sup.7—, —NR.sup.5SO.sub.2R.sup.7—, optionally substituted C.sub.1-C.sub.8 alkyl, optionally substituted C.sub.1-C.sub.5 heteroalkyl, optionally substituted C.sub.1-C.sub.6alkoxyC.sub.1-C.sub.5alkyl, optionally substituted C.sub.1—C: haloalkyl, optionally substituted C.sub.1-C.sub.5 hydroxyalkyl, optionally substituted C.sub.1-C.sub.8alkylaminoC.sub.1-C.sub.5alkyl, optionally substituted C.sub.3-C.sub.8 carbocyclyl, optionally substituted 3-8 membered heterocyclyl, optionally substituted C.sub.2-C.sub.8 alkenyl, optionally substituted C.sub.2-C.sub.8 alkynyl, optionally substituted aryl, and optionally substituted heteroaryl, wherein R.sup.5, R.sup.6 and R.sup.7 are independently selected from null, a bond, hydrogen, optionally substituted C.sub.1-C.sub.8 alkyl, optionally substituted C.sub.1-C.sub.5 heteroalkyl, optionally substituted C.sub.2-C.sub.8 alkenyl, optionally substituted C.sub.2-C.sub.8 alkynyl, optionally substituted C.sub.3-C.sub.6 carbocyclyl, optionally substituted 3-8 membered heterocyclyl, optionally substituted heterocarbocyclyl, optionally substituted arylalkyl, optionally substituted heteroarylalkyl, optionally substituted aryl, and optionally substituted heteroaryl, or R.sup.6 and R.sup.7 together with the atom to which they are connected optionally form a 3-8 membered carbocyclyl or heterocyclyl ring.

(168) In another embodiment, R.sup.3 is connected to the linker moiety of the bivalent compound through R.sup.4, and R.sup.3 and R.sup.4 are independently selected from null, a bond, —

OR.sup.5—, —SR.sup.5—, —NR.sup.6R.sup.7—, —COR.sup.5—, —CO.sub.2R.sup.5—, —CONR.sup.6R.sup.7—, —SOR.sup.5—, —SO.sub.2R.sup.5—, —SOZNR.sup.6R.sup.7—, —NR.sup.5COR.sup.7—, —NR.sup.5C(O)NR.sup.6R.sup.7—, —NR.sup.5SOR.sup.7—, —NR.sup.5SO.sub.2R.sup.7—, optionally substituted C.sub.1-C.sub.6 alkyl, optionally substituted C.sub.1-C.sub.5 heteroalkyl, optionally substituted C.sub.1-C.sub.8alkoxyC.sub.1-C.sub.5alkyl, optionally substituted C.sub.1-C.sub.5 haloalkyl, optionally substituted C.sub.1-C.sub.5 hydroxyalkyl, optionally substituted C.sub.1-C.sub.5alkylaminoC.sub.1-C.sub.5alkyl, optionally substituted C.sub.3-C.sub.8 carbocyclyl, optionally substituted 3-8 membered heterocyclyl, optionally substituted C.sub.2-C.sub.8 alkenyl, optionally substituted C.sub.2-C.sub.8 alkynyl, optionally substituted aryl, and optionally substituted heteroaryl, wherein R.sup.5, R.sup.6 and R.sup.7 are independently selected from null, a bond, hydrogen, optionally substituted C.sub.1-C.sub.8 alkyl, optionally substituted C.sub.1-C.sub.5 heteroalkyl, optionally substituted C.sub.2-C.sub.8 alkenyl, optionally substituted C.sub.2-C.sub.8 alkynyl, optionally substituted C.sub.3-C.sub.6 carbocyclyl, optionally substituted 3-8 membered heterocyclyl, optionally substituted heterocarbocyclyl, optionally substituted arylalkyl, optionally substituted heteroarylalkyl, optionally substituted aryl, and optionally substituted heteroaryl, or R.sup.6 and R.sup.7 together with the atom to which they are connected optionally form a 3-8 membered carbocyclyl or heterocyclyl ring.

(169) In another embodiment, Ar.sup.1 is selected from C.sub.6-C.sub.10 aryl and C.sub.5-C.sub.10 heteroaryl, each of which is optionally substituted with one or more substituents independently selected from F, Cl, CN, NO.sub.2, OR.sup.10, NR.sup.11R.sup.12, COR.sup.10, CO.sub.2R.sup.10, CONR.sup.11R.sup.12, SOR.sup.10, SO.sub.2R.sup.10, SO.sub.2NR.sup.11R.sup.12, NR.sup.10OR.sup.12, NR.sup.10C(O)NR.sup.11R.sup.12, NR.sup.10SOR.sup.12, NR.sup.10SO R.sup.12, optionally substituted C.sub.1-C.sub.6 alkyl, optionally substituted C.sub.1-C.sub.6 heteroalkyl, optionally substituted C.sub.1-C.sub.6 alkoxyalkyl, optionally substituted C.sub.1-C.sub.6 haloalkyl, optionally substituted C.sub.1-C.sub.8 hydroxyalkyl, optionally substituted C.sub.1-C.sub.8alkylaminoC.sub.1-C.sub.5alkyl, optionally substituted C.sub.3-C.sub.7 carbocyclyl, optionally substituted 3-7 membered heterocyclyl, optionally substituted C.sub.2-C.sub.6 alkenyl, optionally substituted C.sub.2-C.sub.6 alkynyl, optionally substituted aryl, and optionally substituted C.sub.4-C.sub.5 heteroaryl, wherein R.sup.10, R.sup.11, and R.sup.12, at each occurrence, are independently selected from null, hydrogen, optionally substituted C.sub.1-C.sub.6 alkyl, optionally substituted C.sub.1-C.sub.6 heteroalkyl, optionally substituted C.sub.2-C.sub.6 alkenyl, optionally substituted C.sub.2-C.sub.6 alkynyl, optionally substituted C.sub.3-C.sub.7 carbocyclyl, optionally substituted 3-7 membered heterocyclyl, optionally substituted aryl, and optionally substituted heteroaryl, or R.sup.11 and R.sup.12 together with the atom to which they are connected optionally form a 3-8 membered carbocyclyl or heterocyclyl ring.

(170) In another embodiment, Ar.sup.2 is selected from C.sub.1-C.sub.10 aryl and C.sub.5-C.sub.10 heteroaryl, each of which is optionally substituted with one or more substituents independently selected from F, Cl, CN, NO.sub.2, OR.sup.13, NR.sup.14R.sup.15, COR.sup.13, CO.sub.2R.sup.13, CONR.sup.14R.sup.15, SOR.sup.13, SO.sub.2R.sup.13, SO.sub.2NR.sup.14R.sup.15, NR.sup.13COR.sup.14, NR.sup.13C(O)NR.sup.14R.sup.15, NR.sup.13SOR.sup.14, NR.sup.13SO.sub.2R.sup.14, optionally substituted C.sub.1-C.sub.6 alkyl, optionally substituted C.sub.1-C.sub.6 heteroalkyl, optionally substituted C.sub.1-C.sub.6 alkoxyalkyl, optionally substituted C.sub.1-C.sub.8 haloalkyl, optionally substituted C.sub.1-C.sub.6 hydroxyalkyl, optionally substituted C.sub.1-C.sub.8alkylaminoC.sub.1-C.sub.8alkyl, optionally substituted C.sub.3-C.sub.7 carbocyclyl, optionally substituted 3-7 membered heterocyclyl, optionally substituted C.sub.2-C.sub.6 alkenyl, optionally substituted C.sub.2-C.sub.6 alkynyl, optionally substituted aryl, and optionally substituted C.sub.4-C.sub.5 heteroaryl, wherein R.sup.13, R.sup.14, and R.sup.15, at each occurrence, are independently selected from null,

hydrogen, optionally substituted C.sub.1-C.sub.6 alkyl, optionally substituted C.sub.1-C.sub.6 heteroalkyl, optionally substituted C.sub.2-C.sub.6 alkenyl, optionally substituted C.sub.2-C.sub.6 alkynyl, optionally substituted C.sub.3-C.sub.7 carbocyclyl, optionally substituted 3-7 membered heterocyclyl, optionally substituted aryl, and optionally substituted heteroaryl, or R.sup.14 and R.sup.15 together with the atom to which they are connected optionally form a 3-8 membered carbocyclyl or heterocyclyl ring.

(171) In another embodiment, R.sup.3—Ar.sup.2 is selected from a moiety of formulae B1 and B2;

(172) ##STR00198##

wherein * indicates the connection to the linker moiety of the bivalent compound; Y.sup.1, Y.sup.2, Y.sup.3, and Y.sup.4 are independently selected from CH and N, with the proviso that up to 3 of Y.sup.1, Y.sup.2, Y.sup.3, and Y.sup.4 are N; each R.sup.a is independently selected from hydrogen, halogen, CN, NO.sub.2, OR.sup.13, NR.sup.14R.sup.15, COR.sup.13, CO.sub.2R.sup.13, CONR.sup.14R.sup.15, SOR.sup.13, SO.sub.2R.sup.13, SO.sub.2NR.sup.14R.sup.15, NR.sup.13COR.sup.14, NR.sup.13C(O)NR.sup.14R.sup.15, NR.sup.13SOR.sup.14, NR.sup.13SO.sub.2R.sup.14, optionally substituted C.sub.1-C.sub.8 alkyl, optionally substituted C.sub.1-C.sub.6 heteroalkyl, optionally substituted C.sub.1-C.sub.8 alkoxy, optionally substituted C.sub.1-C.sub.8 alkoxyC.sub.1-C.sub.8 alkyl, optionally substituted C.sub.1-C.sub.8alkylaminoC.sub.1-C.sub.8alkyl, optionally substituted C.sub.3-C.sub.8 carbocyclyl, optionally substituted C.sub.3-C.sub.8 cycloalkoxy, optionally substituted 3-8 membered heterocyclyl, optionally substituted C.sub.2-C.sub.8 alkenyl, optionally substituted C.sub.2-C.sub.8 alkynyl, optionally substituted aryl, and optionally substituted heteroaryl, wherein R.sup.13, R.sup.14, and R.sup.15, at each occurrence, are independently selected from hydrogen, optionally substituted C.sub.1-C.sub.8alkyl, optionally substituted C.sub.1-C.sub.5 heteroalkyl, optionally substituted C.sub.1-C.sub.8 alkoxy, optionally substituted C.sub.1-C.sub.8alkoxyC.sub.1-C.sub.5alkyl, optionally substituted C.sub.1-C.sub.5alkylaminoC.sub.1-C.sub.5alkyl, optionally substituted C.sub.3-C.sub.8 carbocyclyl, optionally substituted C.sub.3-C.sub.8 cycloalkoxy, optionally substituted 3-8 membered heterocyclyl, optionally substituted C.sub.2-C.sub.8 alkenyl, optionally substituted C.sub.2-C.sub.8 alkynyl, optionally substituted aryl, and optionally substituted heteroaryl, or R.sup.14 and R.sup.15 together with the atom to which they are connected optionally form a 3-8 membered carbocyclyl or heterocyclyl ring; m is 0 to 4; and R.sup.3 is the same as defined in Formula 2.

(173) In another embodiment, R.sup.3—Ar.sup.2 is selected from a moiety of formula B3;

(174) ##STR00199##

wherein * indicates the connection to the linker moiety of the bivalent compound; Y.sup.1, Y.sup.2, Y.sup.3, and Y.sup.4 are independently selected from CR.sup.a, N, O, and S, with the proviso that up to 3 of Y.sup.1, Y.sup.2, Y.sup.3, and Y.sup.4 are N; each R.sup.a is independently selected from hydrogen, halogen, CN, NO.sub.2, OR.sup.13, NR.sup.14R.sup.15, COR.sup.13, CO.sub.2R.sup.13, CONR.sup.14R.sup.15, SOR.sup.13, SO.sub.2R.sup.13, SO.sub.2NR.sup.14R.sup.15, NR.sup.13COR.sup.14, NR.sup.13C(O)NR.sup.14R.sup.15, NR.sup.13SOR.sup.14, NR.sup.13SO.sub.2R.sup.14, optionally substituted C.sub.1-C.sub.8 alkyl, optionally substituted C.sub.1-C.sub.6 heteroalkyl, optionally substituted C.sub.1-C.sub.6 alkoxy, optionally substituted C.sub.1-C.sub.6alkoxyC.sub.1-C.sub.5alkyl, optionally substituted C.sub.1-C.sub.8alkylaminoC.sub.1-C.sub.5alkyl, optionally substituted C.sub.3-C.sub.8 carbocyclyl, optionally substituted C.sub.3-C.sub.8 cycloalkoxy, optionally substituted 3-8 membered heterocyclyl, optionally substituted C.sub.2-C.sub.8 alkenyl, optionally substituted C.sub.2-C.sub.8 alkynyl, optionally substituted aryl, and optionally substituted heteroaryl, wherein R.sup.13, R.sup.14, and R.sup.15 are independently selected from hydrogen, optionally substituted C.sub.1-C.sub.8 alkyl, optionally substituted C.sub.1-C.sub.5 heteroalkyl, optionally substituted C.sub.1-C.sub.8 alkoxy, optionally substituted C.sub.1-C.sub.8alkoxyC.sub.1-C.sub.5alkyl, optionally substituted C.sub.1-C.sub.5alkylaminoC.sub.1-C.sub.5alkyl, optionally substituted C.sub.3-C.sub.8

carbocyclyl, optionally substituted C.sub.3-C.sub.8 cycloalkoxy, optionally substituted 3-8 membered heterocyclyl, optionally substituted C.sub.2-C.sub.8 alkenyl, optionally substituted C.sub.2-C.sub.8 alkynyl, optionally substituted aryl, and optionally substituted heteroaryl, or R.sup.14 and R is together with the atom to which they are connected optionally form a 3-8 membered carbocyclyl or heterocyclyl ring; m is 0 to 4; and R.sup.3 is the same as defined in Formula 2.

(175) In another embodiment, X.sup.1 is N; X.sup.2 is N; X.sup.3 is C; X.sup.4 is C; and X is CH.sub.2.

(176) In another embodiment, Ar.sup.1 is 3-fluorophenyl.

(177) In another embodiment, Ar.sup.2 is 2-pyridyl.

(178) In another embodiment, R.sup.3 is selected from

(179) ##STR00200##

(180) In another embodiment, the TRK ligand comprises a moiety of Formula 3;

(181) ##STR00201##

Wherein, R.sup.1, R.sup.2, R.sup.3, R.sup.4, Ar, X, X.sup.1, X.sup.2, X.sup.3, X.sup.4 and n are defined as before.

(182) In another embodiment, the TRK ligand comprises a moiety of FORMULA 3;

(183) ##STR00202##

wherein X.sup.1, X.sup.2, X.sup.3, and X.sup.4 are independently selected from C, CR', and N (preferably, X.sup.1 and X.sup.4 are independently selected from CR' and N; X.sup.2 and X.sup.3 are independently selected from C and N), wherein R' is selected from hydrogen, halogen, CN, NO.sub.2, and optionally substituted C.sub.1-C.sub.8 alkyl, optionally substituted C.sub.1-C.sub.6 heteroalkyl, C.sub.3-C.sub.6 carbocyclyl, or 3-6 membered heterocyclyl; X is selected from null, a bond, C(R.sup.2).sub.2, C(R.sup.2); C(R.sup.2),, CO, C(R.sup.2), CO, NR.sup.2CO, OC(R.sup.2).sub.2, and NR.sup.2C(R.sup.2).sub.2; R.sup.1 and each R.sup.2 are independently selected from hydrogen, halogen, OH, NH.sub.2, CN, NO.sub.2, optionally substituted C.sub.1-C.sub.4 alkyl, optionally substituted C.sub.1-C.sub.4 heteroalkyl, optionally substituted C.sub.1-C.sub.4 alkoxy, optionally substituted C.sub.1-C.sub.4 alkylamino, optionally substituted C.sub.1-C.sub.4 alkoxyalkyl, optionally substituted C.sub.1-C.sub.6 haloalkyl, optionally substituted C.sub.1-C.sub.4 hydroxyalkyl, optionally substituted C.sub.1-C.sub.6alkylaminoC.sub.1-C.sub.8alkyl, optionally substituted C.sub.3-C.sub.6 carbocyclyl, optionally substituted C.sub.3—C cycloalkoxy, and optionally substituted 3-6 membered heterocyclyl; n is 1 to 4; R.sup.3 is selected from hydrogen, optionally substituted C.sub.1-C.sub.6 alkyl, optionally substituted C.sub.1-C.sub.6 heteroalkyl, optionally substituted C.sub.3-C.sub.6 carbocyclyl, optionally substituted 3-6 membered heterocyclyl, optionally substituted C.sub.1-C.sub.6 alkoxyalkyl, optionally substituted C.sub.1-C.sub.8 haloalkyl, optionally substituted C.sub.1-C.sub.6 hydroxyalkyl, and optionally substituted C.sub.1-C.sub.6alkylaminoC.sub.1-C.sub.8alkyl; R.sup.4 is connected to the linker moiety of the bivalent compound either directly or through R.sup.5, and R.sup.4 and R.sup.5 are independently selected from null, —OR.sup.6—, —SR.sup.6—, —N(R.sup.7)R.sup.6—, —COR.sup.6—, —CO.sub.2R.sup.6—, —CON(R.sup.7)R.sup.6—, —SOR.sup.6—, —SO.sub.2R.sup.6—, —SO.sub.2N(R.sup.7)R.sup.6—, —NR.sup.5COR.sup.6—, —N(R.sup.8)C(O)N(R.sup.7)R.sup.6—, —NR.sup.8SOR.sup.6—, —NR.sup.5SO.sub.2R.sup.6— optionally substituted C.sub.1-C.sub.8alkylene, optionally substituted C.sub.1-C.sub.5alkylene-O—C.sub.1-C.sub.5alkylene, optionally substituted C.sub.1-C.sub.5 haloalkylene, optionally substituted C.sub.1-C.sub.5 hydroxyalkylene, optionally substituted C.sub.1-C.sub.8alkylene-N(C.sub.1-C.sub.8 alkyl)-C.sub.1-C.sub.5alkylene, optionally substituted C.sub.3-C.sub.8 carbocyclylene, optionally substituted 3-8 membered carbocyclyl, optionally substituted 3-8 membered heterocyclyl, optionally substituted C.sub.2-C.sub.8 alkenylene, optionally substituted C.sub.2-C.sub.8 alkynylene, optionally substituted aryl, and optionally substituted heteroaryl; R.sup.6 is selected from null, optionally substituted C.sub.1-C.sub.8alkylene, optionally substituted

C.sub.1-C.sub.5 heteroalkylene, optionally substituted C.sub.1-C.sub.8 alkylene-O—, optionally substituted C.sub.1-C.sub.8 alkylene-N(C.sub.1-C.sub.6 alkyl)-, optionally substituted C.sub.1-C.sub.8alkylene-O—C.sub.1-C.sub.8alkylene, optionally substituted C.sub.1-C.sub.5alkylene-N(C.sub.1-C.sub.8 alkyl)-C.sub.1-C.sub.5alkylene, optionally substituted C.sub.3-C.sub.8 carbocyclyl, optionally substituted C.sub.3-C.sub.8 carbocyclyl-O—, optionally substituted 3-8 membered heterocyclyl, optionally substituted aryl, and optionally substituted heteroaryl; R.sup.7 and R.sup.8 are independently selected from null, hydrogen, optionally substituted C.sub.1-C.sub.8 alkyl, optionally substituted C.sub.1-C.sub.5 heteroalkyl, optionally substituted C.sub.2-C.sub.8 alkenyl], optionally substituted C.sub.2-C.sub.8 alkynyl, optionally substituted C.sub.3-C.sub.8 carbocyclyl, optionally substituted heterocyclyl, optionally substituted arylalkyl, optionally substituted heteroarylalkyl, optionally substituted 3-8 membered carbocyclyl, optionally substituted 3-8 membered heterocyclyl, optionally substituted aryl, and optionally substituted heteroaryl; or R.sup.6 and R.sup.7 together with the atom to which they are connected optionally form a 3-8 membered carbocyclyl or heterocyclyl ring; Ar is selected from aryl and heteroaryl, each of which is optionally substituted with one or more substituents independently selected from halogen, CN, NO.sub.2, OR.sup.10, SR.sup.10, NR.sup.11R.sup.12, COR.sup.1, CO.sub.2R.sup.10, CONR.sup.11R.sup.12, SOR.sup.10, SO.sub.2R.sup.10, SO.sub.2NR.sup.11R.sup.12, NR.sup.10OR.sup.12, NR.sup.10C(O)NR.sup.11R.sup.12, NR.sup.10SOR.sup.12, NR.sup.10SO.sub.2R.sup.12, optionally substituted C.sub.1-C.sub.8 alkyl, optionally substituted C.sub.1-C.sub.5 heteroalkyl, optionally substituted C.sub.1-C.sub.8alkoxyC.sub.1-C.sub.5alkyl, optionally substituted C.sub.1-C.sub.5 haloalkyl, optionally substituted C.sub.1-C.sub.5 hydroxyalkyl, optionally substituted C.sub.1-C.sub.8 alkylaminoC.sub.1-C.sub.8 alkyl, optionally substituted C.sub.3-C.sub.7 carbocyclyl, optionally substituted 3-7 membered heterocyclyl, optionally substituted C.sub.2-C.sub.8 alkenyl, optionally substituted C.sub.2-C.sub.8 alkynyl, optionally substituted aryl, and optionally substituted heteroaryl, wherein R.sup.10, R.sup.11, and R.sup.12 are independently selected from null, hydrogen, optionally substituted C.sub.1-C.sub.8 alkyl, optionally substituted C.sub.1-C.sub.8 heteroalkyl, optionally substituted C.sub.2-C.sub.8 alkenyl, optionally substituted C.sub.2-C.sub.8 alkynyl, optionally substituted C.sub.3-C.sub.7 carbocyclyl, optionally substituted 3-7 membered heterocyclyl, optionally substituted aryl, and optionally substituted heteroaryl, or R.sup.11 and R.sup.12 together with the atom to which they are connected optionally form a 3-8 membered carbocyclyl or heterocyclyl rings.

(184) In one embodiment, X.sup.1 and X.sup.4 is selected from CR' and N, and R' is selected from hydrogen, F, C.sub.1, CH.sub.3, CF.sub.3, and cyclopropyl.

(185) In another embodiment, X.sup.2 and X.sup.3 are independently selected from C and N.

(186) In one embodiment, X.sup.2 is C and X.sup.3 is N.

(187) In one embodiment, X.sup.3 is C and X.sup.2 is N.

(188) In another embodiment, X is selected from a bond, CH.sub.2, CH.sub.2CH.sub.2, CO, CH.sub.2CO, CONH, CONCH.sub.3, CH.sub.2O, CH.sub.2NH, and CH.sub.2NCH.sub.3.

(189) In another embodiment, R.sup.1 and each R.sup.2 are independently selected from hydrogen, F, Cl, OH, optionally substituted C.sub.1-C.sub.4 alkyl, optionally substituted C.sub.1-C.sub.4 heteroalkyl, optionally substituted C.sub.1-C.sub.4 alkoxy, optionally substituted C.sub.1-C.sub.4 alkylamino, optionally substituted C.sub.1-C.sub.4 haloalkyl, optionally substituted C.sub.3-C.sub.6 carbocyclyl, optionally substituted C.sub.3-C.sub.6 cycloalkoxy, and optionally substituted 3-6 membered heterocyclyl.

(190) In another embodiment, R.sup.3 is selected from hydrogen, CH.sub.3, CH.sub.2CH.sub.3, propyl, isopropyl, cyclopropyl, CH.sub.2F, CHF.sub.2, and CF.sub.3.

(191) In another embodiment, R.sup.4 is connected to the linker moiety of the bivalent compound directly, and R.sup.4 is selected from null, —OR.sup.6—, —SR.sup.6—, —N(R.sup.7)R.sup.6—, —COR.sup.6—, —CO.sub.2R.sup.6—, —CON(R.sup.7)R.sup.6—, —SOR.sup.6—, —SO.sub.2RE—, —SO.sub.2N(R.sup.7)R.sup.6—, —NR.sup.8COR.sup.6—, —

N(R.sup.8)C(O)N(R.sup.7)R.sup.6—, —NR.sup.8SOR.sup.6—, —NR.sup.8SO.sub.2R.sup.6—, optionally substituted C.sub.1-C.sub.5 alkylene, optionally substituted C.sub.1-C.sub.5 heteroalkylene, optionally substituted C.sub.1-C.sub.5alkylene-O—C.sub.1-C.sub.8alkylene, optionally substituted C.sub.1-C.sub.5 haloalkylene, optionally substituted C.sub.1-C.sub.8 hydroxyalkylene, optionally substituted C.sub.1-C.sub.5alkylene-N(C.sub.1-C.sub.5alkyl)-C.sub.1-C.sub.6alkylene, optionally substituted C.sub.3-C.sub.8 carbocyclylene, optionally substituted 3-8 membered carbocyclyl, optionally substituted 3-8 membered heterocyclyl, optionally substituted C.sub.2-C.sub.8 alkenylene, optionally substituted C.sub.2-C.sub.8 alkynylene, optionally substituted aryl, and optionally substituted heteroaryl; R.sup.6 is selected from null, optionally substituted C.sub.1-C.sub.6 alkylene, optionally substituted C.sub.1-C.sub.8 heteroalkylene, optionally substituted C.sub.1-C.sub.8alkylene-O—, optionally substituted C.sub.1-C.sub.6 alkylene-N(C.sub.1-C.sub.8alkyl)-, optionally substituted C.sub.1-C.sub.8alkylene-O—C.sub.1-C.sub.8 alkylene, optionally substituted C.sub.1-C.sub.8alkylene-N(C.sub.1-C.sub.8 alkyl)-C.sub.1-C.sub.8alkylene, optionally substituted C.sub.3-C.sub.8 carbocyclyl, optionally substituted C.sub.3-C.sub.8 carbocyclyl-O—, optionally substituted 3-8 membered heterocyclyl, optionally substituted aryl, and optionally substituted heteroaryl; R.sup.7 and R.sup.8 are independently selected from null, hydrogen, optionally substituted C.sub.1-C.sub.8 alkyl,, optionally substituted C.sub.1-C.sub.5 heteroalkyl, optionally substituted C.sub.2-C.sub.8 alkenyl, optionally substituted C.sub.2-C.sub.8 alkynyl, optionally substituted C.sub.3-C.sub.8 carbocyclyl, optionally substituted heterocyclyl, optionally substituted arylalkyl, optionally substituted heteroarylalkyl, optionally substituted 3-8 membered carbocyclyl, optionally substituted 3-8 membered heterocyclyl, optionally substituted aryl, and optionally substituted heteroaryl; or R.sup.6 and R.sup.7 together with the atom to which they are connected optionally form a 3-8 membered carbocyclyl or heterocyclyl ring; In another embodiment, R.sup.4 is connected to the linker moiety of the bivalent compound through R.sup.5, and R.sup.4 and R.sup.5 are independently selected from null, —OR.sup.6—, —SR.sup.6—, —N(R.sup.7)R.sup.6—, —COR.sup.6—, —CO.sub.2R.sup.6—, —CON(R.sup.7)R.sup.6—, —SOR.sup.6—, —SO.sub.2R.sup.6—, —SO.sub.2N(R.sup.7)R.sup.6—, —NR.sup.5COR.sup.6—, —N(R.sup.2)C(O)N(R.sup.7)R.sup.6—, —NR.sup.8SOR.sup.6—, —NR.sup.8SO.sub.2R.sup.6—, optionally substituted C.sub.1-C.sub.8alkylene, optionally substituted C.sub.1-C.sub.5 heteroalkylene, optionally substituted C.sub.1-C.sub.5alkylene-O—C.sub.1-C.sub.5alkylene, optionally substituted C.sub.1-C.sub.5 haloalkylene, optionally substituted C.sub.1-C.sub.5 hydroxyalkylene, optionally substituted C.sub.1-C.sub.5alkylene-N(C.sub.1-C.sub.8alkyl)-C.sub.1-C.sub.8alkylene, optionally substituted C.sub.3-C.sub.8 carbocyclylene, optionally substituted 3-8 membered carbocyclyl, optionally substituted 3-8 membered heterocyclyl, optionally substituted C.sub.2-C.sub.8 alkenylene, optionally substituted C.sub.2-C.sub.8 alkynylene, optionally substituted aryl, and optionally substituted heteroaryl; R.sup.6 is selected from null, optionally substituted C.sub.1-C.sub.6 alkylene, optionally substituted C.sub.1-C.sub.8 heteroalkylene, optionally substituted C.sub.1-C.sub.8 alkylene-O—, optionally substituted C.sub.1-C.sub.8 alkylene-N(C.sub.1-C.sub.6 alkyl)-, optionally substituted C.sub.1-C.sub.8 alkylene-O—C.sub.1-C.sub.8 alkylene, optionally substituted C.sub.1-C.sub.5alkylene-N(C.sub.1-C.sub.8 alkyl)-C.sub.1-C.sub.8alkylene, optionally substituted C.sub.3-C.sub.8 carbocyclyl, optionally substituted C.sub.3-C.sub.8 carbocyclyl-O—, optionally substituted 3-8 membered heterocyclyl, optionally substituted aryl, and optionally substituted heteroaryl; R.sup.7 and R.sup.8 are independently selected from null, hydrogen, optionally substituted C.sub.1-C.sub.6 alkyl,, optionally substituted C.sub.1-C.sub.5 heteroalkyl, optionally substituted C.sub.2-C.sub.8 alkenyl, optionally substituted C.sub.2-C.sub.8 alkynyl, optionally substituted C.sub.3-C.sub.8 carbocyclyl, optionally substituted heterocyclyl, optionally substituted arylalkyl, optionally substituted heteroarylalkyl, optionally substituted 3-8 membered carbocyclyl, optionally substituted 3-8 membered heterocyclyl, optionally substituted aryl, and optionally substituted heteroaryl; or R.sup.6 and R.sup.7 together with the atom to which they are connected optionally form a 3-8

membered carbocyclyl or heterocyclyl ring; In another embodiment, Ar is selected from aryl and heteroaryl, each of which is optionally substituted with one or more substituents independently selected from F, Cl, CN, NO₂, OR¹⁰, NR¹¹R¹², COR¹⁰, CO₂R¹⁰, CONR¹¹R¹², SOR¹⁰, SO₂RIG, SO₂NR¹¹R¹², NR¹⁰COR¹², NR¹⁰C(O)NR¹¹R¹², NR¹⁰SOR¹², NR¹⁰SO₂R¹², optionally substituted C₁-C₆ alkyl, optionally substituted C₁-C₆ heteroalkyl, optionally substituted C₁-C₆ alkoxyalkyl, optionally substituted C₁-C₆ haloalkyl, optionally substituted C₁-C₆ hydroxyalkyl, optionally substituted C₁-C₆alkylaminoC₁-C₅alkyl, optionally substituted C₃-C₇ carbocyclyl, optionally substituted 3-7 membered heterocyclyl, optionally substituted C₂-C₆ alkenyl, optionally substituted C₂-C₆ alkynyl, optionally substituted aryl, and optionally substituted C₄-C₅ heteroaryl, wherein R¹⁰, R¹¹, and R¹² are independently selected from null, hydrogen, optionally substituted C₁-C₆ alkyl, optionally substituted C₁-C₆ heteroalkyl, optionally substituted C₂-C₆ alkenyl, optionally substituted C₂-C₆ alkynyl, optionally substituted C₃-C₇ carbocyclyl, optionally substituted 3-7 membered heterocyclyl, optionally substituted aryl, and optionally substituted heteroaryl, or R¹¹ and R¹² together with the atom to which they are connected form a 3-8 membered carbocyclyl or heterocyclyl ring.

(192) In another embodiment, the TRK ligand comprises a moiety of FORMULA 3A:

(193) ##STR00203##

wherein X¹ is selected from CR' and N; R' is selected from hydrogen, halogen, CN, NO₂, and optionally substituted C₁-C₆ alkyl, optionally substituted C₁-C₆ heteroalkyl, optionally substituted C₃-C₆ carbocyclyl, or optionally substituted 3-6 membered heterocyclyl; X² and X³ are selected from either C or N, with the proviso that only one of X² and X³ is N; X is selected from null, a bond, C(R²)₂, C(R²)₂CO, C(R²)₂, CO, NR²CO, OC(R²)₂, and NR²C(R²)₂; R¹ and each R² are independently selected from hydrogen, halogen, OH, NH₂, CN, NO₂, optionally substituted C₁-C₄ alkyl, optionally substituted C₁-C₄ heteroalkyl, optionally substituted C₁-C₄ alkoxy, optionally substituted C₁-C₄ alkylamino, optionally substituted C₁-C₄ alkoxyalkyl, optionally substituted C₁-C₄ haloalkyl, optionally substituted C₁-C₄ hydroxyalkyl, optionally substituted C₁-C₄alkylaminoC₁-C₄alkyl, optionally substituted C₃-C₆ carbocyclyl, optionally substituted C₃-C₈ cycloalkoxy, and optionally substituted 3-6 membered heterocyclyl; n is 1 to 4; R³ is selected from hydrogen, optionally substituted C₁-C₆ alkyl, optionally substituted C₁-C₆ heteroalkyl, optionally substituted C₃-C₆ carbocyclyl, optionally substituted 3-6 membered heterocyclyl, optionally substituted C₁-C₆ alkoxyalkyl, optionally substituted C₁-C₈ haloalkyl, optionally substituted C₁-C₆ hydroxyalkyl, and optionally substituted C₁-C₆alkylaminoC₁-C₆alkyl; R⁴ is connected to the linker moiety of the bivalent compound either directly or through R⁵; R⁴ and R⁵ are independently selected from null, ~OR⁶-, ~SR⁶-, ~N(R⁷)R⁶-, ~COR⁶-, ~CO₂R⁶-, ~CON(R⁷)R⁶-, ~SOR⁶-, ~SO₂R⁶-, ~SO₂N(R⁷)R⁶-, ~NR⁸COR⁶-, ~N(R⁸)C(O)N(R⁷)R⁶-, ~NR⁸SOR⁶-, ~NR⁸SO₂R⁶-, optionally substituted C₁-C₈alkylene, optionally substituted C₁-C₆ heteroalkylene, optionally substituted C₁-C₅alkylene-O-C₁-C₅alkylene, optionally substituted C₁-C₅ haloalkylene, optionally substituted C₁-C₅ hydroxyalkylene, optionally substituted C₁-C₅alkylene-N(C₁-C₅alkyl)-C₁-C₅alkylene, optionally substituted C₃-C₈ carbocyclylene, optionally substituted 3-8 membered carbocyclyl, optionally substituted 3-8 membered heterocyclyl, optionally substituted C₂-C₈ alkenylene,

optionally substituted C.sub.2-C.sub.8 alkynylene, optionally substituted aryl, and optionally substituted heteroaryl; R.sup.6 is selected from null, optionally substituted C.sub.1-C.sub.8 alkylene, optionally substituted C.sub.1-C.sub.8 heteroalkylene, optionally substituted C.sub.1-C.sub.6 alkylene-O—, optionally substituted C.sub.1-C.sub.8 alkylene-N(C.sub.1-C.sub.8alkyl)-, optionally substituted C.sub.1-C.sub.8 alkylene-O—C.sub.1-C.sub.6 alkylene, optionally substituted C.sub.1-C.sub.8alkylene-N(C.sub.1-C.sub.8 alkyl)-C.sub.1-C.sub.8alkylene, optionally substituted C.sub.3-C.sub.8 carbocyclyl, optionally substituted C.sub.3-C.sub.8 carbocyclyl-O—, optionally substituted 3-8 membered heterocyclyl, optionally substituted aryl, and optionally substituted heteroaryl; R.sup.7 and R.sup.8 are independently selected from null, hydrogen, optionally substituted C.sub.1-C.sub.8 alkyl, optionally substituted C.sub.1-C.sub.5 heteroalkyl, optionally substituted C.sub.2-C.sub.8 alkenyl, optionally substituted C.sub.2-C.sub.8 alkynyl, optionally substituted C.sub.3-C.sub.8 carbocyclyl, optionally substituted heterocyclyl, optionally substituted arylalkyl, optionally substituted heteroarylalkyl, optionally substituted 3-8 membered carbocyclyl, optionally substituted 3-8 membered heterocyclyl, optionally substituted aryl, and optionally substituted heteroaryl; or R.sup.6 and R.sup.7 together with the atom to which they are connected optionally form a 3-8 membered carbocyclyl or heterocyclyl ring; Ar is selected from aryl and heteroaryl, each of which is optionally substituted with one or more substituents independently selected from halogen, CN, NO.sub.2, OR.sup.10, SR.sup.10, NR.sup.11R.sup.12, COR.sup.10, CO.sub.2R.sup.10, CONR.sup.11R.sup.12, SOR.sup.10, SO.sub.2R.sup.10, SO.sub.2NR.sup.11R.sup.12, NR.sup.10OR.sup.12, NR.sup.10C(O)NR.sup.11R.sup.12, NR.sup.10SOR.sup.12, NR.sup.10SO.sub.2R.sup.12, optionally substituted C.sub.1-C.sub.8 alkyl, optionally substituted C.sub.1-C.sub.5 heteroalkyl, optionally substituted C.sub.1-C.sub.8alkoxyC.sub.1-C.sub.8alkyl, optionally substituted C.sub.1-C.sub.5 haloalkyl, optionally substituted C.sub.1-C.sub.5 hydroxyalkyl, optionally substituted C.sub.1-C.sub.5alkylaminoC.sub.1-C.sub.5alkyl, optionally substituted C.sub.3-C.sub.7 carbocyclyl, optionally substituted 3-7 membered heterocyclyl, optionally substituted C.sub.2-C.sub.8 alkenyl, optionally substituted C.sub.2-C.sub.8 alkynyl, optionally substituted aryl, and optionally substituted heteroaryl; R.sup.10, R.sup.11, and R.sup.12 are independently selected from null, hydrogen, optionally substituted C.sub.1-C.sub.8 alkyl, optionally substituted C.sub.1-C.sub.5 heteroalkyl, optionally substituted C.sub.2-C.sub.8 alkenyl, optionally substituted C.sub.2-C.sub.8 alkynyl, optionally substituted C.sub.3-C.sub.7 carbocyclyl, optionally substituted 3-7 membered heterocyclyl, optionally substituted aryl, and optionally substituted heteroaryl; or R.sup.11 and R.sup.12 together with the atom to which they are connected optionally form a 3-8 membered carbocyclyl or heterocyclyl rings.

(194) In one embodiment, X.sup.1 and X.sup.3 are selected from CR' and N, and R' is selected from hydrogen, F, Cl, CH.sub.3, CF.sub.3, and cyclopropyl.

(195) In one embodiment, X.sup.1 is N.

(196) In one embodiment, X.sup.1 is CR' and R' is selected from hydrogen, F, Cl, CH.sub.3, CF.sub.3, and cyclopropyl. In one embodiment, X.sup.2 is C and X.sup.3 is N.

(197) In one embodiment, X.sup.3 is C and X.sup.2 is N.

(198) In another embodiment, X is selected from a bond, CH.sub.2, CH.sub.2CH.sub.2, CO, CH.sub.2CO, CONH, CONCH.sub.3, CH.sub.2O, CH.sub.2NH, and CH.sub.2NCH.sub.3.

(199) In another embodiment, X is CH.sub.2.

(200) In another embodiment, R.sup.1 and each R.sup.2 are independently selected from hydrogen, F, Cl, OH, optionally substituted C.sub.1-C.sub.4 alkyl, optionally substituted C.sub.1-C.sub.4 heteroalkyl, optionally substituted C.sub.1-C.sub.4 alkoxy, optionally substituted C.sub.1-C.sub.4 alkylamino, optionally substituted C.sub.1-C.sub.4 haloalkyl, optionally substituted C.sub.3-C.sub.6 carbocyclyl, optionally substituted C.sub.3-C.sub.6 cycloalkoxy, and optionally substituted 3-6 membered heterocyclyl.

(201) In another embodiment, R.sup.1 and R.sup.2 are hydrogen.

(202) In another embodiment, R.sup.3 is selected from hydrogen, CH.sub.3, CH.sub.2CH.sub.3, propyl, isopropyl, cyclopropyl, CH.sub.2F, CHF.sub.2, and CF.sub.3.

(203) In another embodiment, R.sup.3 is selected from hydrogen.

(204) In another embodiment, R.sup.4 is connected to the linker moiety of the bivalent compound directly, and R.sup.4 is selected from null, —OR.sup.6—, —SR.sup.6—, —N(R.sup.7)R.sup.6—, —COR.sup.6—, —CO.sub.2R.sup.6—, —CON(RO)R.sup.6—, —SOR.sup.6—, —SOR.sup.6—, —SO.sub.2N(R.sup.7)R.sup.6—, —NR.sup.8COR.sup.6—, —N(R.sup.8)C(O)N(R.sup.7)R.sup.6—, —NR.sup.5SOR.sup.6—, —NR.sup.8SO.sub.2R.sup.6—, optionally substituted C.sub.1-C.sub.5 alkylene, optionally substituted C.sub.1-C.sub.5 heteroalkylene, optionally substituted C.sub.1-C.sub.5alkylene-O—C-C.sub.8alkylene, optionally substituted C.sub.1-C.sub.8 haloalkylene, optionally substituted C.sub.1-C.sub.5 hydroxyalkylene, optionally substituted C.sub.1-C.sub.8alkylene-N(C.sub.1-C.sub.5alkyl)-C.sub.1-C.sub.8alkylene, optionally substituted C.sub.3-C.sub.8 carbocyclene, optionally substituted 3-8 membered carbocyclyl, optionally substituted 3-8 membered heterocyclyl, optionally substituted C.sub.2-C.sub.8 alkenylene, optionally substituted C.sub.2-C.sub.8 alkynylene, optionally substituted aryl, and optionally substituted heteroaryl; R.sup.6 is selected from null, optionally substituted C.sub.1-C.sub.8 alkylene, optionally substituted C.sub.1-C.sub.8 heteroalkylene, optionally substituted C.sub.1-C.sub.8alkylene-O—, optionally substituted C.sub.1-C.sub.8 alkylene-N(C.sub.1-C.sub.8 alkyl)-, optionally substituted C.sub.1-C.sub.8 alkylene-O—C.sub.1-C.sub.6 alkylene, optionally substituted C.sub.1-C.sub.8alkylene-N(C.sub.1-C.sub.8 alkyl)-C.sub.1-C.sub.5alkylene, optionally substituted C.sub.3-C.sub.8 carbocyclyl, optionally substituted C.sub.3-C.sub.8 carbocyclyl-O—, optionally substituted 3-8 membered heterocyclyl, optionally substituted aryl, and optionally substituted heteroaryl; and R.sup.7 and R.sup.8 are independently selected from null, hydrogen, optionally substituted C.sub.1-C.sub.8 alkyl, optionally substituted C.sub.1-C.sub.5 heteroalkyl, optionally substituted C.sub.2-C.sub.8 alkenyl, optionally substituted C.sub.2-C.sub.8 alkynyl, optionally substituted C.sub.3-C.sub.8 carbocyclyl, optionally substituted heterocyclyl, optionally substituted arylalkyl, optionally substituted heteroarylalkyl, optionally substituted 3-8 membered carbocyclyl, optionally substituted 3-8 membered heterocyclyl, optionally substituted aryl, and optionally substituted heteroaryl; or R.sup.6 and R.sup.7 together with the atom to which they are connected optionally form a 3-8 membered carbocyclyl or heterocyclyl ring.

(205) In another embodiment, R.sup.4 is connected to the linker moiety of the bivalent compound through R.sup.5, and R.sup.4 and R.sup.5 are independently selected from null, —OR.sup.6—, —SR.sup.6—, —N(R.sup.7)R.sup.6—, —COR.sup.6—, —CO.sub.2R.sup.6—, —CON(R.sup.7)R.sup.6—, —SOR.sup.6—, —SO.sub.2R.sup.6—, —SO.sub.2N(R.sup.7)R.sup.6—, —NR.sup.8COR.sup.6—, —N(R.sup.8)C(O)N(R.sup.7)R.sup.6—, —NR.sup.8SOR.sup.6—, —NR.sup.8SO.sub.2R.sup.6—, optionally substituted C.sub.1-C.sub.8alkylene, optionally substituted C.sub.1-C.sub.5 heteroalkylene, optionally substituted C.sub.1-C.sub.8 alkylene-O—C.sub.1-C.sub.5alkylene, optionally substituted C.sub.1-C.sub.5 haloalkylene, optionally substituted C.sub.1-C.sub.6 hydroxyalkylene, optionally substituted C.sub.1-C.sub.8alkylene-N(C.sub.1-C.sub.8alkyl)-C.sub.1-C.sub.5alkylene, optionally substituted C.sub.3-C.sub.8 carbocyclene, optionally substituted 3-8 membered carbocyclyl, optionally substituted 3-8 membered heterocyclyl, optionally substituted C.sub.2-C.sub.8 alkenylene, optionally substituted C.sub.2-C.sub.8 alkynylene, optionally substituted aryl, and optionally substituted heteroaryl; R.sup.6 is selected from null, optionally substituted C.sub.1-C.sub.8 alkylene, optionally substituted C.sub.1-C.sub.5 heteroalkylene, optionally substituted C.sub.1-C.sub.8 alkylene-O—, optionally substituted C.sub.1-C.sub.8 alkylene-N(C.sub.1-C.sub.6 alkyl)-, optionally substituted C.sub.1-C.sub.8 alkylene-O—C.sub.1-C.sub.8alkylene, optionally substituted C.sub.1-C.sub.8alkylene-N(C.sub.1-C.sub.8 alkyl)-C.sub.1-C.sub.8alkylene, optionally substituted C.sub.3-C.sub.8 carbocyclyl, optionally substituted C.sub.3-C.sub.8 carbocyclyl-O—, optionally substituted 3-8 membered heterocyclyl, optionally substituted aryl, and optionally substituted heteroaryl; R.sup.7 and R.sup.8

are independently selected from null, hydrogen, optionally substituted C.sub.1-C.sub.6 alkyl,, optionally substituted C.sub.1-C.sub.5 heteroalkyl, optionally substituted C.sub.2-C.sub.8 alkenyl, optionally substituted C.sub.2-C.sub.8 alkynyl, optionally substituted C.sub.3-C.sub.8 carbocyclyl, optionally substituted heterocyclyl, optionally substituted arylalkyl, optionally substituted heteroarylalkyl, optionally substituted 3-8 membered carbocyclyl, optionally substituted 3-8 membered heterocyclyl, optionally substituted aryl, and optionally substituted heteroaryl; or R.sup.6 and R.sup.7 together with the atom to which they are connected optionally form a 3-8 membered carbocyclyl or 3-8 membered heterocyclyl ring.

(206) In another embodiment, R.sup.4 and R.sup.5 are independently selected from null, optionally substituted

(207) ##STR00204##

(208) In another embodiment, R.sup.4 and R.sup.5 are independently selected from null, optionally substituted

(209) ##STR00205##

(210) In another embodiment, —R.sup.4—R.sup.5—is selected from null, optionally substituted

(211) ##STR00206##

(212) In another embodiment, —R.sup.4—R.sup.5—is optionally substituted

(213) ##STR00207##

(214) In another embodiment, Ar is selected from aryl and heteroaryl, each of which is optionally substituted with one or more substituents independently selected from F, Cl, CN, NO.sub.2, OR.sup.10, NR.sup.11R.sup.12, COR.sup.10, CO.sub.2R.sup.10, CONR.sup.11R.sup.12, SOR.sup.10, SO.sub.2R.sup.10, SO.sub.2NR.sup.11R.sup.12, NR.sup.10COR.sup.12, NR.sup.10C(O)NR.sup.11R.sup.12, NR.sup.10SOR.sup.12, NR.sup.10SO.sub.2R.sup.12, optionally substituted C.sub.1-C.sub.6 alkyl, optionally substituted C.sub.1-C.sub.6 heteroalkyl, optionally substituted C.sub.1-C.sub.6 alkoxyalkyl, optionally substituted C.sub.1-C.sub.6 haloalkyl, optionally substituted C.sub.1-C.sub.6 hydroxyalkyl, optionally substituted C.sub.1-C.sub.8alkylaminoC.sub.1-C.sub.8alkyl, optionally substituted C.sub.3-C.sub.7 carbocyclyl, optionally substituted 3-7 membered heterocyclyl, optionally substituted C.sub.2-C.sub.6 alkenyl, optionally substituted C.sub.2-C.sub.6 alkynyl, optionally substituted aryl, and optionally substituted C.sub.4-C.sub.5 heteroaryl, wherein R.sup.10, R.sup.11, and R.sup.12 are independently selected from null, hydrogen, optionally substituted C.sub.1-C.sub.6 alkyl, optionally substituted C.sub.1-C.sub.6 heteroalkyl, optionally substituted C.sub.2-C.sub.6 alkenyl, optionally substituted C.sub.2-C.sub.6 alkynyl, optionally substituted C.sub.3-C.sub.7 carbocyclyl, optionally substituted 3-7 membered heterocyclyl, optionally substituted aryl, and optionally substituted heteroaryl, or R.sup.11 and R.sup.12 together with the atom to which they are connected optionally form a 3-8 membered carbocyclyl or heterocyclyl ring.

(215) In another embodiment, Ar is aryl, which is optionally substituted with one or more substituents independently selected from F, Cl, Br, CN, and NO.sub.2.

(216) In another embodiment, Ar is

(217) ##STR00208##

(218) In another embodiment, the TRK ligand comprises a moiety of FORMULAE 12-1 or 12-2;

(219) ##STR00209##

Wherein X is selected from CR.sup.5, and N; Y is selected from O, S, and NR.sup.6; R.sup.1, R.sup.3 and R.sup.4 are independently selected from hydrogen, halogen, CN, NO.sub.2, OR.sup.7, SR.sup.7, NR.sup.8R.sup.9, COR.sup.7, —CO.sub.2R.sup.7, CONR.sup.8R.sup.9, SOR.sup.7, SO.sub.2R.sup.7, SO.sub.2NR.sup.8R.sup.9, NR.sup.10COR.sup.9, NR.sup.10C(O)NR.sup.8R.sup.9, NR.sup.10SOR', NR.sup.10SO.sub.2R', optionally substituted C.sub.1-C.sub.8 alkyl, optionally substituted C.sub.1-C.sub.6 heteroalkyl, optionally substituted C.sub.1-C.sub.8alkoxyC.sub.1-C.sub.5alkyl, optionally substituted C.sub.1-C.sub.6 haloalkyl, optionally substituted C.sub.1-C.sub.8 hydroxyalkyl, optionally substituted C.sub.1-

C.sub.5alkylaminoC.sub.1-C.sub.5alkyl, optionally substituted C.sub.3-C.sub.8 carbocyclyl, optionally substituted 3-8 membered carbocyclyl, optionally substituted 3-8 membered heterocyclyl, optionally substituted 3-8 membered carbocyclyl-C.sub.1-C.sub.8alkyl, optionally substituted 3-8 membered heterocyclyl-C.sub.1-C.sub.8alkyl, optionally substituted 3-8 membered carbocyclyl-O—, optionally substituted 3-8 membered heterocyclyl-O, optionally substituted 3-8 membered carbocyclyl-N(C.sub.1-C.sub.8alkyl)-, and optionally substituted 3-8 membered heterocyclyl-N(C.sub.1-C.sub.5alkyl)-, optionally substituted C.sub.2-C.sub.8 alkenyl, optionally substituted C.sub.2-C.sub.8 alkynyl, optionally substituted aryl, and optionally substituted heteroaryl; R.sup.7, R.sup.8, R.sup.9, and R.sup.10 are independently selected from null, hydrogen, optionally substituted C.sub.1-C.sub.8 alkyl, optionally substituted C.sub.1-C.sub.6 heteroalkyl, optionally substituted C.sub.2-C.sub.8 alkenyl, optionally substituted C.sub.2-C.sub.8 alkynyl, optionally substituted C.sub.3-C.sub.8 carbocyclyl, optionally substituted 3-8 membered carbocyclyl, optionally substituted heterocarbocyclyl, optionally substituted arylalkyl, optionally substituted heteroarylalkyl, optionally substituted 3-8 membered carbocyclyl, optionally substituted 3-8 membered heterocyclyl, optionally substituted 3-8 membered carbocyclyl-C.sub.1-C.sub.5alkyl, optionally substituted 3-8 membered heterocyclyl-C.sub.1-C.sub.8alkyl, optionally substituted 3-8 membered carbocyclyl-O—, optionally substituted 3-8 membered heterocyclyl-O, optionally substituted 3-8 membered carbocyclyl-N(C.sub.1-C.sub.5alkyl)-, and optionally substituted 3-8 membered heterocyclyl-N(C.sub.1-C.sub.8alkyl)-, optionally substituted aryl, and optionally substituted heteroaryl; or R.sup.8 and R.sup.o together with the atom to which they are connected optionally form a 3-8 membered carbocyclyl or heterocyclyl ring; R.sup.1', R.sup.2, and R.sup.4' are independently selected from null, —OR.sup.11—, —SR.sup.9—, —NR.sup.12R.sup.11—, —COR.sup.11—, —CO.sub.2R.sup.11—, —CON(R.sup.12)R.sup.11—, —SOR.sup.11—, —SO.sub.2R.sup.11—, —SON(R.sup.12)R.sup.11—, —NR.sup.13COR.sup.11—, —NR.sup.13C(O)N(R.sup.12)R.sup.11—, —NR.sup.13SOR.sup.11—, —NR.sup.13SO.sub.2R.sup.11—, optionally substituted C.sub.1-C.sub.8 alkylene, optionally substituted C.sub.1-C.sub.8 heteroalkylene, optionally substituted C.sub.1-C.sub.5alkylene-O—C.sub.1-C.sub.8alkylene, optionally substituted C.sub.1-C.sub.8 haloalkylene, optionally substituted C.sub.1-C.sub.5 hydroxyalkylene, optionally substituted C.sub.1-C.sub.5alkylene-N(C.sub.1-C.sub.8alkyl)-C.sub.1-C.sub.5alkylene, optionally substituted C.sub.3-C.sub.8 carbocyclylene, optionally substituted 3-8 membered carbocyclyl, optionally substituted 3-8 membered heterocyclyl, optionally substituted 3-8 membered carbocyclyl-C.sub.1-C.sub.8alkylene, optionally substituted 3-8 membered heterocyclyl-C.sub.1-C.sub.8alkylene, optionally substituted 3-8 membered carbocyclyl-O—, optionally substituted 3-8 membered heterocyclyl-O, optionally substituted 3-8 membered carbocyclyl-N(C.sub.1-C.sub.8alkyl)-, and optionally substituted 3-8 membered heterocyclyl-N(C.sub.1-C.sub.8alkyl)-, optionally substituted C.sub.2-C.sub.8 alkenylene, optionally substituted C.sub.2-C.sub.8 alkynylene, optionally substituted aryl, and optionally substituted heteroaryl; R.sup.11 is selected from null, optionally substituted C.sub.1-C.sub.8 alkylene, optionally substituted C.sub.1-C.sub.8 heteroalkylene, optionally substituted C.sub.1-C.sub.8alkylene-O—, optionally substituted C.sub.1-C.sub.8 alkylene-N(C.sub.1-C.sub.8alkyl)-, optionally substituted C.sub.1-C.sub.8alkylene-O-alkylene, optionally substituted C.sub.1-C.sub.8alkylene-N(C.sub.1-C.sub.8alkyl)-C.sub.1-C.sub.5alkylene, optionally substituted C.sub.3-C.sub.8 carbocyclylene, optionally substituted C.sub.3-C.sub.8 carbocyclylene-O—, optionally substituted 3-8 membered carbocyclyl-C.sub.1-C.sub.8alkylene, optionally substituted 3-8 membered heterocyclyl-C.sub.1-C.sub.5alkylene, optionally substituted 3-8 membered carbocyclyl, optionally substituted 3-8 membered heterocyclyl, optionally substituted aryl, and optionally substituted heteroaryl; R.sup.12, R.sup.13, and R.sup.14 are independently selected from null, hydrogen, optionally substituted C.sub.1-C.sub.8 alkyl, optionally substituted C.sub.1-C.sub.5 heteroalkylene, optionally substituted C.sub.2-C.sub.8 alkenyl, optionally substituted C.sub.2-C.sub.8 alkynyl, optionally substituted C.sub.3-C.sub.8 carbocyclyl, optionally substituted

heterocyclyl, optionally substituted arylalkyl, optionally substituted heteroarylalkyl, optionally substituted 3-8 membered carbocyclyl-C.sub.1-C.sub.8alkyl, optionally substituted 3-8 membered heterocyclyl-C.sub.1-C.sub.6alkyl, optionally substituted 3-8 membered carbocyclyl, optionally substituted 3-8 membered heterocyclyl, optionally substituted aryl, and optionally substituted heteroaryl; or R.sup.11 and R.sup.12 together with the atom to which they are connected optionally form a 3-8 membered carbocyclyl or 3-8 membered heterocyclyl ring; R.sup.5 and R.sup.6, at each occurrence, are independently selected from hydrogen, halogen, CN, NO.sub.2, optionally substituted C.sub.1-C.sub.8alkyl, optionally substituted C.sub.1-C.sub.5 heteroalkyl, optionally substituted C.sub.1-C.sub.5 alkoxy, optionally substituted C.sub.1-C.sub.8 alkylamino, optionally substituted C.sub.3-C.sub.8 carbocyclyl, optionally substituted C.sub.3-C.sub.8 carbocyclyl, and optionally substituted C.sub.3-C.sub.8 heterocyclyl; and n is selected from 0, 1, and 2.

(220) In another embodiment, X is selected from CH, CF, and N.

(221) In another embodiment, X is CH.

(222) In another embodiment, X is CF.

(223) In another embodiment, X is N.

(224) In another embodiment, Y is selected from O and S.

(225) In another embodiment, Y is O.

(226) In another embodiment, Y is S.

(227) In another embodiment, R.sup.1 is selected from optionally substituted acyclic amino, optionally substituted cyclic amino, optionally substituted phenyl, and optionally substituted heteroaryl.

(228) In another embodiment, R.sup.1 is selected from optionally substituted

(229) ##STR00210##

(230) In another embodiment, R.sup.1' is selected from null, optionally substituted acyclic amino, optionally substituted cyclic amino, optionally substituted phenyl, and optionally substituted heteroaryl.

(231) In another embodiment, R.sup.1' is selected from optionally substituted

(232) ##STR00211##

In another embodiment, R.sup.2 is selected from null, —O—, —S—, —N(R.sup.12)—, —C(O)—, —CO.sub.2—, —CON(R.sup.12)—, —SO—, —SO.sub.2—, ~SO.sub.2N(R.sup.12)—, —N(R.sup.13)CO—, —N(R.sup.13)C(O)N(R.sup.12)—, —N(R.sup.13)SO—, —N(R.sup.13)SO.sub.2—, optionally substituted optionally substituted C.sub.3-C.sub.8 carbocyclyl, and optionally substituted C.sub.3-C.sub.8 heterocyclyl, wherein R.sup.12 and R.sup.13 are independently selected from hydrogen, optionally substituted C.sub.1-C.sub.8 alkyl, optionally substituted C.sub.1-C.sub.5 heteroalkyl, optionally substituted C.sub.2-C.sub.8 alkenyl, optionally substituted C.sub.2-C.sub.8 alkynyl, optionally substituted C.sub.3-C.sub.8 carbocyclyl, optionally substituted heterocarbocyclyl, optionally substituted arylalkyl, optionally substituted heteroarylalkyl, optionally substituted 3-8 membered carbocyclyl, optionally substituted 3-8 membered heterocyclyl, optionally substituted aryl, and optionally substituted heteroaryl.

(233) In another embodiment, R.sup.2 is selected from null, —CONH—,

(234) ##STR00212##

(235) In another embodiment, R.sup.3 is selected from hydrogen, halogen, CN, NO.sub.2, OH, NH.sub.2, —CONH—

(236) ##STR00213##

(237) In another embodiment, R.sup.2—R.sup.3 is selected from hydrogen, halogen, CN, NO.sub.2,

(238) ##STR00214##

(239) In another embodiment, R.sup.4 is selected from hydrogen, halogen, CN, NO.sub.2, optionally substituted C.sub.1-C.sub.5 alkyl, optionally substituted C.sub.1-C.sub.5 heteroalkyl, optionally substituted C.sub.1-C.sub.8alkoxyC.sub.1-C.sub.8alkyl, optionally substituted C.sub.1

—C: haloalkyl, optionally substituted C.sub.1-C.sub.5 hydroxyalkyl, optionally substituted C.sub.1-C.sub.8alkylaminoC.sub.1-C.sub.8alkyl, optionally substituted C.sub.3-C.sub.8 carbocyclyl, optionally substituted 3-8 membered carbocyclyl-C.sub.1-C.sub.8 alkyl, optionally substituted 3-8 membered heterocyclyl-C.sub.1-C.sub.5alkyl, optionally substituted 3-8 membered carbocyclyl, and optionally substituted 3-8 membered heterocyclyl, optionally substituted 3-8 membered carbocyclyl-C.sub.1-C.sub.8alkyl, and optionally substituted 3-8 membered heterocyclyl-C.sub.1-C.sub.5alkyl, optionally substituted 3-8 membered carbocyclyl-O—, optionally substituted 3-8 membered heterocyclyl-O—, optionally substituted 3-8 membered carbocyclyl-N(C.sub.1-C.sub.5alkyl)-, and optionally substituted 3-8 membered heterocyclyl-N(C.sub.1-C.sub.8 alkyl)-.

(240) In another embodiment. R.sup.4 is selected from hydrogen, halogen, CN, NO.sub.2, OCH.sub.3,

(241) ##STR00215##

(242) In another embodiment, R.sup.4' is selected from null, —O—, —S—, —N(R.sup.12')—, —C(O)—, —CO.sub.2—, —CON(R.sup.12')—, —SO—, —SO.sub.2—, —SO.sub.2N(R.sup.12')—, —N(R.sup.12')CO—, —N(R.sup.13;) C(O)N(R.sup.12')—, —N(R.sup.13')SO—, —N(R.sup.13')SO.sub.2—, optionally substituted C.sub.1-C.sub.8alkylene, optionally substituted C.sub.1-C.sub.8 heteroalkylene, optionally substituted C.sub.1-C.sub.8alkylene-O—C.sub.1-C.sub.8alkylene, optionally substituted C.sub.1-C.sub.8 haloalkylene, optionally substituted C.sub.1-C.sub.8 hydroxyalkylene, optionally substituted C.sub.1-C.sub.5alkylene-N(C.sub.1-C.sub.5alkyl)-C.sub.1-C.sub.8alkylene, optionally substituted C.sub.3-C.sub.8 carbocyclene, optionally substituted 3-8 membered carbocyclyl, optionally substituted 3-8 membered heterocyclyl, optionally substituted 3-8 membered carbocyclyl-C.sub.1-C.sub.8alkylene, and optionally substituted 3-8 membered heterocyclyl-C.sub.1-C.sub.5alkylene, optionally substituted 3-8 membered carbocyclyl-O—, optionally substituted 3-8 membered heterocyclyl-O—, optionally substituted 3-8 membered carbocyclyl-N(C.sub.1-C.sub.8alkyl)-, and optionally substituted 3-8 membered heterocyclyl-N(C.sub.1-C.sub.8 alkyl)-, wherein R.sup.12' and R.sup.13' are independently selected from hydrogen, optionally substituted C.sub.1-C.sub.8 alkyl, optionally substituted C.sub.1-C.sub.5 heteroalkyl, optionally substituted C.sub.2-C.sub.8 alkenyl, optionally substituted C.sub.2-C.sub.8 alkynyl, optionally substituted C.sub.5-C.sub.8carbocyclyl, optionally substituted heterocarbocyclyl, optionally substituted arylalkyl, optionally substituted heteroarylalkyl, optionally substituted 3-8 membered carbocyclyl, optionally substituted 3-8 membered heterocyclyl, optionally substituted aryl, and optionally substituted heteroaryl.

(243) In another embodiment, R.sup.4' is selected from null, —O—, optionally substituted C.sub.1-C.sub.6 alkylene, optionally substituted C.sub.1-C.sub.6 heteroalkylene, optionally substituted C.sub.1-C.sub.5alkylene-O—C.sub.1-C.sub.8alkylene, optionally substituted C.sub.1-C.sub.5alkylene-N(C.sub.1-C.sub.8alkyl)-C.sub.1-C.sub.8alkylene, optionally substituted 3-8 membered carbocyclyl, optionally substituted 3-8 membered heterocyclyl, optionally substituted 3-8 membered carbocyclyl-C.sub.1-C.sub.8alkylene, and optionally substituted 3-8 membered heterocyclyl-C.sub.1-C.sub.5alkylene, optionally substituted 3-8 membered carbocyclyl-O—, optionally substituted 3-8 membered heterocyclyl-O—, optionally substituted 3-8 membered carbocyclyl-N(C.sub.1-C.sub.5alkyl)-, and optionally substituted 3-8 membered heterocyclyl-N(C.sub.1-C.sub.5alkyl)-.

(244) In another embodiment, R.sup.4' is selected null, —O—,

(245) ##STR00216##

(246) In another embodiment, the TRK ligand comprises a moiety of FORMULA 13:

(247) ##STR00217##

wherein X is selected from CR.sup.9, and N; R.sup.9 is selected from from hydrogen, halogen, CN, NO.sub.2, optionally substituted C.sub.1-C.sub.8 alkyl, optionally substituted C.sub.1-C.sub.5 heteroalkyl, optionally substituted C.sub.1-C.sub.8 alkoxy, optionally substituted C.sub.1-C.sub.8

alkylamino, optionally substituted C.sub.3-C.sub.8 carbocyclyl, optionally substituted C.sub.3-C.sub.8 carbocyclyl, and optionally substituted C.sub.3-C.sub.8 heterocyclyl; Y is selected from null, —O—, —N(optionally substituted C.sub.1-C.sub.8 alkyl)-, and optionally substituted C.sub.1-C.sub.6 alkylene; R.sup.1 and R.sup.2 are independently selected from hydrogen, halogen, CN, NO.sub.2, OH, NH.sub.2, optionally substituted C.sub.1-C.sub.8 alkyl, optionally substituted C.sub.1-C.sub.5 heteroalkyl, optionally substituted C.sub.1-C.sub.8 alkoxy, optionally substituted C.sub.1-C.sub.8 alkylamino, optionally substituted C.sub.3-C.sub.8 carbocyclyl, optionally substituted C.sub.3-C.sub.8 carbocyclyl, and optionally substituted C.sub.3-C.sub.8 heterocyclyl; R.sup.3, R.sup.4, R.sup.5, and R.sup.6 are independently selected from hydrogen, halogen, optionally substituted C.sub.1-C.sub.8 alkyl, optionally substituted C.sub.1-C.sub.8 heteroalkyl, optionally substituted C.sub.3-C.sub.8 carbocyclyl, optionally substituted C.sub.3-C.sub.8 carbocyclyl, and optionally substituted C.sub.3-C.sub.8 heterocyclyl; or R.sup.3 and R.sup.4; and/or R.sup.5 and R.sup.6, together with the atom to which they are connected optionally form a 3-8 membered carbocyclyl or heterocyclyl ring; m is selected from 1, 2, 3, and 4; n is selected from 0, 1, 2, 3, and 4; R.sup.7 and R.sup.8 are independently selected from null, —OR.sup.10—, —SR.sup.10—, —NR.sup.11R.sup.10—, —COR.sup.10—, —CO.sub.2R.sup.10—, —CONR.sup.11R.sup.10—, —SOR.sup.10—, —SO.sub.2R.sup.10—, —SO.sub.2NR.sup.11R.sup.11, —NR.sup.12COR.sup.10—, —NR.sup.12C(O)NR.sup.11R.sup.10—, —NR.sup.12SOR.sup.10—, —NR.sup.12SO.sub.2R.sup.10—, optionally substituted C.sub.1-C.sub.8alkylene, optionally substituted C.sub.1-C.sub.5 heteroalkyl, optionally substituted C.sub.1-C.sub.8alkylene-O—C.sub.1-C.sub.8alkylene, optionally substituted C.sub.1-C.sub.5 haloalkylene, optionally substituted C.sub.1-C.sub.5 hydroxyalkylene, optionally substituted C.sub.1-C.sub.5alkylene-N(C.sub.1-C.sub.8alkyl)-C.sub.1-C.sub.5alkylene, optionally substituted C.sub.3-C.sub.8 carbocyclylene, optionally substituted 3-8 membered carbocyclyl, optionally substituted 3-8 membered heterocyclyl, optionally substituted 3-8 membered carbocyclyl-C.sub.1-C.sub.8alkylene, optionally substituted 3-8 membered heterocyclyl-C.sub.1-C.sub.5alkylene, optionally substituted 3-8 membered carbocyclyl-O—, optionally substituted 3-8 membered heterocyclyl-O, optionally substituted 3-8 membered carbocyclyl-N(C.sub.1-C.sub.5alkyl)-, and optionally substituted 3-8 membered heterocyclyl-N(C.sub.1-C.sub.8alkyl)-, optionally substituted C.sub.2-C.sub.8 alkenylene, optionally substituted C.sub.4-C.sub.13 fused carbocyclyl, optionally substituted 5-13 membered fused heterocyclyl, optionally substituted C.sub.5-C.sub.13 bridged carbocyclyl, optionally substituted 5-13 membered bridged heterocyclyl, optionally substituted C.sub.5-C.sub.13 spiro carbocyclyl, optionally substituted 5-13 membered spiro heterocyclyl, optionally substituted C.sub.2-C.sub.8 alkynylene, optionally substituted aryl, and optionally substituted heteroaryl; R.sup.10 is selected from null, optionally substituted C.sub.1-C.sub.6 alkylene, optionally substituted C.sub.1-C.sub.5 heteroalkylene, optionally substituted C.sub.1-C.sub.8 alkylene-O—, optionally substituted C.sub.1-C.sub.8 alkylene-N(C.sub.1-C.sub.6 alkyl)-, optionally substituted C.sub.1-C.sub.6 alkylene-O-alkylene, optionally substituted C.sub.1-C.sub.5alkylene-N(C.sub.1-C.sub.6 alkyl)-C.sub.1-C.sub.5alkylene, optionally substituted C.sub.3-C.sub.8 carbocyclylene, optionally substituted C.sub.3-C.sub.8 carbocyclylene-O—, optionally substituted 3-8 membered carbocyclyl-C.sub.1-C.sub.5alkylene, optionally substituted 3-8 membered heterocyclyl-C.sub.1-C.sub.8alkylene, optionally substituted 3-8 membered carbocyclyl, optionally substituted 3-8 membered heterocyclyl, optionally substituted C.sub.4-C.sub.13 fused carbocyclyl, optionally substituted 5-13 membered fused heterocyclyl, optionally substituted C.sub.5-C.sub.15 bridged carbocyclyl, optionally substituted 5-13 membered bridged heterocyclyl, optionally substituted C.sub.5-C.sub.13 spiro carbocyclyl, optionally substituted 5-13 membered spiro heterocyclyl, optionally substituted aryl, and optionally substituted heteroaryl; R.sup.11 and R.sup.12 are independently selected from null, hydrogen, optionally substituted C.sub.1-C.sub.6 alkyl, optionally substituted C.sub.2-C.sub.8 alkenyl, optionally substituted C.sub.2-C.sub.8 alkynyl, optionally substituted C.sub.3-C.sub.8 carbocyclyl, optionally substituted

heterocarbocyclyl, optionally substituted arylalkyl, optionally substituted heteroarylalkyl, optionally substituted 3-8 membered carbocyclyl-C.sub.1-C.sub.6alkyl, optionally substituted 3-8 membered heterocyclyl-C.sub.1-C.sub.8alkyl, optionally substituted C.sub.3-C.sub.8 carbocyclyl-O—, optionally substituted 3-8 membered carbocyclyl, optionally substituted 3-8 membered heterocyclyl, optionally substituted C.sub.4-C.sub.13 fused carbocyclyl, optionally substituted 5-13 membered fused heterocyclyl, optionally substituted C.sub.5-C.sub.13 bridged carbocyclyl, optionally substituted 5-13 membered bridged heterocyclyl, optionally substituted C.sub.5-C.sub.13 spiro carbocyclyl, optionally substituted 5-13 membered spiro heterocyclyl, optionally substituted aryl, and optionally substituted heteroaryl; or R.sup.10 and R.sup.11 together with the atom to which they are connected optionally form a 3-8 membered carbocyclyl or heterocyclyl ring; and Ar is selected from aryl and heteroaryl, each of which is optionally substituted with one or more substituents independently selected from halogen, CN, NO.sub.2, OR.sup.14, SR.sup.14, NR.sup.15R.sup.16, COR.sup.14, CO.sub.2R.sup.14, CONR.sup.15R.sup.16, SOR.sup.14, SO.sub.2R.sup.14, SO.sub.2NR.sup.15R.sup.16, NR.sup.14COR.sup.16, NR.sup.14C(O)NR.sup.15R.sup.16, NR.sup.14SOR.sup.16, NR.sup.14SO.sub.2R.sup.16, optionally substituted C.sub.1-C.sub.8alkyl, optionally substituted C.sub.1-C.sub.5 heteroalkyl, optionally substituted C.sub.1-C.sub.8alkoxyC.sub.1-C.sub.5alkyl, optionally substituted C.sub.1-C.sub.5 haloalkyl, optionally substituted C.sub.1-C.sub.5 hydroxyalkyl, optionally substituted C.sub.1-C.sub.8alkylaminoC.sub.1-C.sub.8alkyl, optionally substituted C.sub.3-C.sub.7 carbocyclyl, optionally substituted 3-7 membered heterocyclyl, optionally substituted C.sub.2-C.sub.8 alkenyl, optionally substituted C.sub.2-C.sub.8 alkynyl, optionally substituted aryl, and optionally substituted heteroaryl; R.sup.14, R.sup.15, and R.sup.16 are independently selected from null, hydrogen, optionally substituted C.sub.1-C.sub.8 alkyl, optionally substituted C.sub.1-C.sub.5 heteroalkyl, optionally substituted C.sub.2-C.sub.8 alkenyl, optionally substituted C.sub.2-C.sub.8 alkynyl, optionally substituted C.sub.3-C.sub.7 carbocyclyl, optionally substituted 3-7 membered heterocyclyl, optionally substituted aryl, and optionally substituted heteroaryl; or R.sup.15 and R.sup.16 together with the atom to which they are connected optionally form a 3-8 membered carbocyclyl or 3-8 membered heterocyclyl rings.

(248) In another embodiment, the TRK ligand comprises a moiety of FORMULA 13-1;

(249) ##STR00218##

wherein X, Y, R.sup.1, R.sup.2, R.sup.3, R.sup.4, R.sup.5, R.sup.6, R.sup.7, R.sup.8, Ar, m, and n are defined in FORMULA 13.

(250) In another embodiment, the TRK ligand comprises a moiety of FORMULA 13-2;

(251) ##STR00219##

wherein X, Y, R.sup.1, R.sup.2, R.sup.3, R.sup.4, R.sup.5, R.sup.6, R.sup.7, R.sup.3, and Ar are defined in FORMULA 13.

(252) In another embodiment, X is selected from CH, CF and N.

(253) In another embodiment, X is N.

(254) In another embodiment, Y is selected from—O—, —CH.sub.2—and-NH—.

(255) In another embodiment, Y is —O—.

(256) In another embodiment, R.sup.1 is selected from hydrogen and NH.sub.2.

(257) In another embodiment, R.sup.1 is NH.sub.2.

(258) In another embodiment, R.sup.2 is selected from hydrogen, halogen, CN, NO.sub.2, optionally substituted C.sub.1-C.sub.6 alkyl, optionally substituted C.sub.1-C.sub.5 heteroalkyl, optionally substituted C.sub.1-C.sub.8 carbocyclyl, and optionally substituted C.sub.1-C.sub.6 alkoxy.

(259) In another embodiment, R.sup.2 is selected from H, F, C.sub.1, Br, OCH.sub.3, OCF.sub.3, and OCHF.sub.2.

(260) In another embodiment, R.sup.2 is OCH.sub.3.

(261) In another embodiment, R.sup.3, R.sup.4, R.sup.5, and R.sup.6 are independently selected

from H, F, CH.sub.3, cyclopropyl, and cyclobutyl.

(262) In another embodiment, R.sup.3 and R.sup.4; and/or R.sup.5 and R.sup.6, together with the atom to which they are connected optionally form, cyclopropyl, cyclobutyl, cyclopentyl, and cyclohexyl.

(263) In another embodiment, R.sup.3, R.sup.4, R.sup.5, and R.sup.6 are H.

(264) In another embodiment, R.sup.7 is selected from null, optionally substituted 3-8 membered carbocyclyl, optionally substituted 3-8 membered heterocyclyl, optionally substituted C.sub.4-C.sub.13 fused carbocyclyl, optionally substituted 5-13 membered fused heterocyclyl, optionally substituted C.sub.5-C.sub.13 bridged carbocyclyl, optionally substituted 5-13 membered bridged heterocyclyl, optionally substituted C.sub.5-C.sub.13 spiro carbocyclyl, optionally substituted 5-13 membered spiro heterocyclyl, optionally substituted aryl, and optionally substituted heteroaryl

(265) In another embodiment, R.sup.7 is selected from null,

(266) ##STR00220##

(267) In another embodiment, R.sup.8 is selected from null, —C(O)—, —C(O)—NH—, optionally substituted 3-8 membered carbocyclyl, optionally substituted 3-8 membered heterocyclyl, optionally substituted C.sub.4-C.sub.13 fused carbocyclyl, optionally substituted 5-13 membered fused heterocyclyl, optionally substituted C.sub.5-Cis bridged carbocyclyl, optionally substituted 5-13 membered bridged heterocyclyl, optionally substituted C.sub.5-C.sub.13 spiro carbocyclyl, optionally substituted 5-13 membered spiro heterocyclyl, optionally substituted aryl, and optionally substituted heteroaryl.

(268) In another embodiment, R.sup.8 is selected from null, —C(O)—, —C(O)—NH—,

(269) ##STR00221##

(270) In another embodiment, R.sup.7—R.sup.8 is selected from null,

(271) ##STR00222##

(272) In another embodiment, Ar is selected from optional substituted phenyl, and optional substituted pyridinyl.

(273) In another embodiment, Ar is selected from

(274) ##STR00223##

(275) In some embodiments, the TRK ligand is a moiety of Formula 10 which connects to the linker via R.sup.4

(276) ##STR00224##

wherein X.sup.1, X.sup.2, X.sup.3, X.sup.4, and X.sup.5 are each independently selected from C, CR.sup.1, and N; X is selected from CR.sup.1R.sup.2, CO, O, S, SO, SO.sub.2, and NR.sup.1; R is selected from optionally substituted C.sub.1-C.sub.8alkyl, optionally substituted C.sub.1-C.sub.5 heteroalkyl, optionally substituted C.sub.5-C.sub.10 carbocyclyl, optionally substituted 3-10 membered heterocyclyl, and Ar.sup.1; or X—R, taken together, stands for

(277) ##STR00225## U is selected from null, a bond, C(R.sup.2).sub.2, C(R.sup.2).sub.2C(R.sup.2).sub.2, CO, C(R.sup.2).sub.2CO, CONR.sup.2, C(R.sup.2), O, C(R.sup.2)—NR.sup.2 and CH.sub.2NR.sup.2; R.sup.1 and R.sup.2, at each occurrence, are independently selected from hydrogen, halogen, CN, NO.sub.2, OR", SR.sup.6, NR.sup.7R.sup.8, COR.sup.6, CO.sub.2R.sup.6, C(O)NR.sup.1R.sup.8, SOR.sup.6, SO.sub.2R.sup.6, SO.sub.2NR.sup.1R.sup.8, NR.sup.6C(O)R.sup.8, NR.sup.9C(O)NR.sup.7R.sup.8, NR.sup.6SOR.sup.8, NR.sup.6SO.sub.2R.sup.8, optionally substituted C.sub.1-C.sub.8alkyl, optionally substituted C.sub.1-C.sub.5 heteroalkyl, optionally substituted C.sub.1-C.sub.8 alkoxy, optionally substituted C.sub.1-C.sub.8alkoxyC.sub.1-C.sub.8alkyl, optionally substituted C.sub.1-C.sub.8alkylaminoC.sub.1-C.sub.8alkyl, optionally substituted C.sub.3-C.sub.10 carbocyclyl, optionally substituted C.sub.3-C.sub.10 cycloalkoxy, optionally substituted 3-10 membered heterocyclyl, optionally substituted C.sub.2-C.sub.8 alkenyl, optionally substituted C.sub.2-C.sub.8 alkynyl, or R.sup.1 and R.sup.2, R' and another R.sup.1, or R.sup.2 and another R.sup.2 together with the atom to which they are connected form a 3-8 membered carbocyclyl or 3-8 membered

heterocyclyl ring; R.sup.6, R.sup.7 and R.sup.8, at each occurrence, are independently selected from null, hydrogen, optionally substituted C.sub.1-C.sub.8 alkyl, optionally substituted C.sub.1-C.sub.5 heteroalkyl, optionally substituted C.sub.2-C.sub.8 alkenyl, optionally substituted C.sub.2-C.sub.8 alkynyl, optionally substituted C.sub.3-C.sub.6 carbocyclyl, optionally substituted 3-8 membered heterocyclyl, optionally substituted heterocarbocyclyl, optionally substituted arylalkyl, optionally substituted heteroarylalkyl, optionally substituted aryl, and optionally substituted heteroaryl, or R.sup.7 and R.sup.8, together with the atom to which they are connected, optionally form a 3-8 membered carbocyclyl or 3-8 membered heterocyclyl ring; n is 0, 1, 2, 3, or 4; n' is 0, 1, 2, or 3; L is Ar.sup.2, NR.sup.3CO or NR.sup.3COAr.sup.2; R.sup.3 is selected from hydrogen, optionally substituted C.sub.1-C.sub.6 alkyl, optionally substituted C.sub.1-C.sub.6 heteroalkyl, optionally substituted C.sub.3-C.sub.6 carbocyclyl, optionally substituted 3-6 membered heterocyclyl, optionally substituted C.sub.1-C.sub.6 alkoxyalkyl, optionally substituted C.sub.1-C.sub.6 haloalkyl, optionally substituted C.sub.1-C.sub.6 hydroxyalkyl, and optionally substituted C.sub.1-C.sub.6alkylaminoC.sub.1-C.sub.6alkyl; R.sup.4 is connected to the linker moiety of the bivalent compound either directly or through R.sup.5, wherein R.sup.4 and R.sup.5 are bivalent groups independently selected from null, —O—, —S—, —NR.sup.9—, —CO—, —CO.sub.2—, —CONR.sup.9—, —SO—, —SOR.sup.9—, —SO.sub.2NR.sup.9—, —NR.sup.10 CO—, —NR.sup.10C(O)NR.sup.9—, —NR.sup.10SO—, —NR.sup.10SO.sub.2—, optionally substituted C.sub.1-C.sub.8 alkylene, optionally substituted C.sub.1-C.sub.5 heteroalkylene, optionally substituted C.sub.1-C.sub.8alkoxyC.sub.1-C.sub.5alkylene, optionally substituted C.sub.1-C.sub.8 haloalkylene, optionally substituted C.sub.1-C.sub.5 hydroxyalkylene, optionally substituted C.sub.1-C.sub.6alkylaminoC.sub.1-C.sub.5alkylene, optionally substituted C.sub.3-C.sub.8 carbocyclylene, optionally substituted 3-8 membered heterocyclylene, optionally substituted C.sub.2-C.sub.8 alkonylene, optionally substituted C.sub.2-C.sub.8 alkynylene, optionally substituted aryl, and optionally substituted heteroaryl; R.sup.9 is selected from null, optionally substituted C.sub.1-C.sub.6 alkylene, optionally substituted C.sub.1-C.sub.8 heteroalkylene, optionally substituted C.sub.1-C.sub.8alkylene-O—, optionally substituted C.sub.1-C.sub.6 alkylene-N(C.sub.1-C.sub.8alkyl)-, optionally substituted C.sub.1-C.sub.8 alkylene-O—C.sub.1-C.sub.8 alkylene, optionally substituted C.sub.1-C.sub.5alkylene-N(C.sub.1-C.sub.8 alkyl)-C.sub.1-C.sub.8alkylene, optionally substituted C.sub.3-C.sub.8 carbocyclyl, optionally substituted C.sub.3-C.sub.8 carbocyclyl-O—, optionally substituted 3-8 membered heterocyclyl, optionally substituted aryl, and optionally substituted heteroaryl; R.sup.10 is selected from null, hydrogen, optionally substituted C.sub.1-C.sub.8 alkyl,, optionally substituted C.sub.1-C.sub.5 heteroalkyl, optionally substituted C.sub.2-C.sub.8 alkenyl, optionally substituted C.sub.2-C.sub.8 alkynyl, optionally substituted C.sub.3-C.sub.8 carbocyclyl, optionally substituted heterocyclyl, optionally substituted arylalkyl, optionally substituted heteroarylalkyl, optionally substituted 3-8 membered carbocyclyl, optionally substituted 3-8 membered heterocyclyl, optionally substituted aryl, and optionally substituted heteroaryl; Ar.sup.1 and Ar.sup.2 are independently selected from aryl and heteroaryl, each of which is optionally substituted with one or more substituents independently selected from halogen, CN, NO.sub.2, OR.sup.11, SR.sup.11, NR.sup.12R.sup.13, COR.sup.11, CO.sub.2R.sup.11, CONR.sup.12R.sup.13, SOR.sup.11, SO.sub.2R.sup.11, SO.sub.2NR.sup.12R.sup.13, NR.sup.10OR.sup.13, NR.sup.11C(O)NR.sup.12R.sup.13, NR.sup.10SOR.sup.13, NR.sup.11SO.sub.2R.sup.13, optionally substituted C.sub.1-C.sub.8 alkyl, optionally substituted C.sub.1-C.sub.5 heteroalkyl, optionally substituted C.sub.1-C.sub.8alkoxyC.sub.1-C.sub.8alkyl, optionally substituted C.sub.1-C.sub.5 haloalkyl, optionally substituted C.sub.1-C.sub.5 hydroxyalkyl, optionally substituted C.sub.1-C.sub.8alkylaminoC.sub.1-C.sub.8alkyl, optionally substituted C.sub.3-C.sub.7 carbocyclyl, optionally substituted 3-7 membered heterocyclyl, optionally substituted C.sub.2-C.sub.8 alkenyl, optionally substituted C.sub.2-C.sub.8 alkynyl, optionally substituted aryl, and optionally substituted heteroaryl; and R.sup.11, R.sup.12, and R.sup.13 are independently selected from null,

hydrogen, optionally substituted C.sub.1-C.sub.8 alkyl, optionally substituted C.sub.1-C.sub.5 heteroalkyl, optionally substituted C.sub.2-C.sub.8 alkenyl, optionally substituted C.sub.2-C.sub.8 alkynyl, optionally substituted C.sub.3-C.sub.7 carbocyclyl, optionally substituted 3-7 membered heterocyclyl, optionally substituted aryl, and optionally substituted heteroaryl, or R.sup.12 and R.sup.13 together with the atom to which they are connected form a 3-8 membered carbocyclyl or heterocyclyl ring.

(278) In certain embodiments, U is selected from a bond, CH.sub.2, CH.sub.2CH.sub.2, CO, CH.sub.2CO, CONH, CONCH.sub.3, CH.sub.2O, CH.sub.2NH, and CH.sub.2NCH.sub.3.

(279) In certain embodiments, R.sup.1 and R.sup.2, at each occurrence, are independently selected from hydrogen, F, Cl, OH, optionally substituted C.sub.1-C.sub.6 alkyl, optionally substituted C.sub.1-C.sub.4 heteroalkyl, optionally substituted C.sub.1-C.sub.4 alkoxy, optionally substituted C.sub.1-C.sub.4 alkylamino, optionally substituted C.sub.1-C.sub.4 haloalkyl, optionally substituted C.sub.3-C.sub.6 carbocyclyl, optionally substituted C.sub.3-C.sub.6 cycloalkoxy, and optionally substituted 3-6 membered heterocyclyl.

(280) In certain embodiments, R.sup.1 and R.sup.2, at each occurrence, are independently selected from hydrogen, F, C.sub.1, CH.sub.3, CF.sub.3, and cyclopropyl.

(281) In certain embodiments, Ar.sup.1 is selected from optionally substituted C.sub.1-C.sub.10 aryl and optionally substituted C.sub.5-C.sub.10 heteroaryl.

(282) In certain embodiments, Ar.sup.1 is selected from 3-fluorophenyl, 3,5-di fluorophenyl, and 2,5-di fluorophenyl.

(283) In certain embodiments, L is Ar.sup.2 and R.sup.4 is connected to the linker moiety of the bivalent compound through R.sup.3.

(284) In certain embodiments, L is Ar.sup.2 and R.sup.4 is connected to the linker moiety of the bivalent compound directly.

(285) In certain embodiments, R.sup.4 is selected from

(286) ##STR00226##

(287) In certain embodiments, L is Ar.sup.2, and Ar.sup.2 is selected from optionally substituted C.sub.6-C.sub.10 aryl and optionally substituted C.sub.5-C.sub.10 heteroaryl.

(288) In certain embodiments, X—R, taken together, stands for

(289) ##STR00227##

(290) In certain embodiments, X.sup.1 is N; X.sup.2 is N; X.sup.3 is N; X.sup.4 is CH; X.sup.5 is C; X—R, taken together, stands for

(291) ##STR00228##

U is CH.sub.2; Art is 3-fluorophenyl; L is Ar.sup.2; and Ar.sup.2 is 2-pyridyl.

(292) In certain embodiments, L is Ar.sup.2 or NR.sup.3COAr.sup.2 and Ar.sup.2—R.sup.4 is selected from a moiety of formulae B1, B2, and B3;

(293) ##STR00229## wherein * indicates the connection to the linker moiety of the bivalent compound; Y.sup.1, Y.sup.2, Y.sup.3, and Y.sup.4 are independently selected from CR.sup.a and N, with the proviso that up to 3 of Y.sup.1, Y.sup.2, Y.sup.3, and Y.sup.4 are N; each R.sup.a is independently selected from hydrogen, halogen, CN, NO.sub.2, OR.sup.14, NR.sup.15R.sup.16, COR.sup.14, CO.sub.2R.sup.14, CONR.sup.18R.sup.16, SOR.sup.14, SO.sub.2R.sup.14, SO.sub.2NR.sup.1RIG, NR.sup.14COR.sup.15, NR.sup.14C(O)NR.sup.15R.sup.16, NR.sup.14SOR.sup.15, NR.sup.14SO.sub.2R.sup.15, optionally substituted C.sub.1-C.sub.8 alkyl, optionally substituted C.sub.1-C.sub.5 heteroalkyl, optionally substituted C.sub.1-C.sub.8alkoxyC.sub.1-C.sub.5alkyl, optionally substituted C.sub.1-C.sub.5 haloalkyl, optionally substituted C.sub.1-C.sub.8 hydroxyalkyl, optionally substituted C.sub.1-C.sub.5alkylaminoC.sub.1-C.sub.5alkyl, optionally substituted C.sub.3-C.sub.7 carbocyclyl, optionally substituted 3-7 membered heterocyclyl, optionally substituted C.sub.2-C.sub.8 alkenyl, optionally substituted C.sub.2-C.sub.8 alkynyl, optionally substituted aryl, and optionally substituted heteroaryl; and R.sup.14, R.sup.15 and R.sup.16, at each occurrence, are independently

selected from null, hydrogen, optionally substituted C.sub.1-C.sub.8 alkyl, optionally substituted C.sub.1-C.sub.5 heteroalkyl, optionally substituted C.sub.2-C.sub.8 alkenyl, optionally substituted C.sub.2-C.sub.8 alkynyl, optionally substituted C.sub.3-C.sub.6 carbocyclyl, optionally substituted 3-8 membered heterocyclyl, optionally substituted heterocarbocyclyl, optionally substituted arylalkyl, optionally substituted heteroarylalkyl, optionally substituted aryl, and optionally substituted heteroaryl, or R.sup.15 and R.sup.16, together with the atom to which they are connected, optionally form a 3-8 membered carbocyclyl or 3-8 membered heterocyclyl ring.

(294) In certain embodiments, L is Ar.sup.2 or NR.sup.3COAr.sup.2 and Ar.sup.2—R.sup.4 is selected from a moiety of formula B4;

(295) ##STR00230## wherein * indicates the connection to the linker moiety of the bivalent compound; Y.sup.1, Y.sup.2', Y.sup.3', and Y.sup.4' are independently selected from CR.sup.a, N, O, and S, with the proviso that up to 3 of Y.sup.1'', Y.sup.2'', Y.sup.3'', and Y.sup.4'' are N; each R.sup.a is independently selected from hydrogen, halogen, CN, NO.sub.2, OR.sup.14, NR.sup.11R.sup.16, COR.sup.14, CO.sub.2R.sup.14, CONR.sup.15R.sup.16, SOR.sup.14, SO.sub.2R.sup.14, SO.sub.2NR.sup.15R.sup.16, NR.sup.14COR.sup.15, NR.sup.14C(O)NR.sup.15R.sup.16, NR.sup.14SOR.sup.15, NR.sup.14SO.sub.2R.sup.15, optionally substituted C.sub.1-C.sub.8 alkyl, optionally substituted C.sub.1-C.sub.5 heteroalkyl, optionally substituted C.sub.1-C.sub.8alkoxyC.sub.1-C.sub.8alkyl, optionally substituted C.sub.1-C.sub.5 haloalkyl, optionally substituted C.sub.1-C.sub.6 hydroxyalkyl, optionally substituted C.sub.1-C.sub.8alkylaminoC.sub.1-C.sub.8alkyl, optionally substituted C.sub.3-C.sub.7 carbocyclyl, optionally substituted 3-7 membered heterocyclyl, optionally substituted C.sub.2-C.sub.8 alkenyl, optionally substituted C.sub.2-C.sub.8 alkynyl, optionally substituted aryl, and optionally substituted heteroaryl; and R.sup.14, R.sup.15 and R.sup.16, at each occurrence, are independently selected from null, hydrogen, optionally substituted C.sub.1-C.sub.8 alkyl, optionally substituted C.sub.1-C.sub.5 heteroalkyl, optionally substituted C.sub.2-C.sub.8 alkenyl, optionally substituted C.sub.2-C.sub.8 alkynyl, optionally substituted C.sub.3-C.sub.6 carbocyclyl, optionally substituted 3-8 membered heterocyclyl, optionally substituted heterocarbocyclyl, optionally substituted arylalkyl, optionally substituted heteroarylalkyl, optionally substituted aryl, and optionally substituted heteroaryl, or R.sup.15 and R.sup.16, together with the atom to which they are connected, optionally form a 3-8 membered carbocyclyl or 3-8 membered heterocyclyl ring.

(296) In certain embodiments, L is NR.sup.3CO, and R.sup.3 is selected from hydrogen, CH.sub.3, CH.sub.2CH.sub.3, propyl, isopropyl, cyclopropyl, CH.sub.2F, CHF.sub.2, and CF.sub.3.

(297) In certain embodiments, L is NR.sup.3CO and R.sup.4 is connected to the linker moiety of the bivalent compound directly

(298) In certain embodiments, L is NR.sup.3CO and R.sup.4 is connected to the linker moiety of the bivalent compound through R.sup.5.

(299) In certain embodiments, L is NR.sup.3COAr.sup.2, wherein R.sup.3 is selected from hydrogen, CH.sub.3, CH.sub.2CH.sub.3, propyl, isopropyl, cyclopropyl, CH.sub.2F, CHF.sub.2, and CF.sub.3; and Ar.sup.2 is selected from optionally substituted C.sub.1-C.sub.10 aryl and optionally substituted C.sub.5-C.sub.10 heteroaryl.

(300) In certain embodiments, L is NR.sup.3COAr.sup.2 and R.sup.4 is connected to the linker moiety of the bivalent compound directly.

(301) In certain embodiments, L is NR.sup.3COAr.sup.2 and R.sup.4 is connected to the linker moiety of the bivalent compound through R.sup.5.

(302) In certain embodiments, L is NR.sup.3COAr.sup.2 and Ar.sup.2—R.sup.4 is selected from a moiety of formulae B1, B2 and B3.

(303) In certain embodiments of the bivalent compounds disclosed herein, L is NR.sup.3COAr.sup.2 and Ar.sup.2—R.sup.4 is selected from a moiety of formula B1, where in Y.sup.1 is CH or N.

(304) In certain embodiments, L is NR.sup.3COAr.sup.2 and Ar.sup.2—R.sup.4 is selected from a moiety of formula B2, wherein Y.sup.1 is CR.sup.a, and R.sup.a is H

(305) ##STR00231##

(306) In certain embodiments, L is NR.sup.3COAr.sup.2 and Ar.sup.2—R.sup.4 is selected from a moiety of formula B3,

(307) wherein Y.sup.2 is CR.sup.a, and R.sup.a is

(308) ##STR00232##

(309) In certain embodiments, L is NR.sup.3COAr.sup.2 and X—R is X—Ar.sup.1.

(310) In certain embodiments, L is NR.sup.3COAr.sup.2 and X—R is X—Ar.sup.1, wherein X is CH.sub.2; and Ar.sup.1 is selected from 3-fluorophenyl, 3,5-di fluorophenyl, or 2,5-di fluorophenyl.

(311) In another embodiment, the TRK ligand is selected from the group consisting of:

(312) ##STR00233## ##STR00234## ##STR00235## ##STR00236## ##STR00237##

Degradation Tag

(313) As used herein, the term “degradation tag” refers to a compound, which associates with or binds to an ubiquitin ligase for recruitment of the corresponding ubiquitination machinery to TRK or is a hydrophobic group or a tag that leads to misfolding of the TRK protein and subsequent degradation at the proteasome or loss of function.

(314) In some embodiments, the degradation tag is a moiety selected from the group consisting of FORMULAE 5A, 5B, 5C, and 5D:

(315) ##STR00238##

wherein V, W, and X are independently selected from CR.sup.2 and N; Y is selected from CO, CR.sup.3R.sup.4, and N=N; Z is selected from null, CO, CR.sup.5R.sup.6, NR.sup.5, O, optionally substituted C.sub.1-C.sub.10 alkylene, optionally substituted C.sub.1-C.sub.10 heteroalkyl, optionally substituted C.sub.1-C.sub.10 alkenylene, optionally substituted C.sub.1-C.sub.10 alkynylene, optionally substituted 3-10 membered carbocyclyl, optionally substituted 3-10 membered heterocyclyl, optionally substituted C.sub.4-C.sub.13 fused carbocyclyl, optionally substituted 5-13 membered fused heterocyclyl, optionally substituted C.sub.5-C.sub.13 bridged carbocyclyl, optionally substituted 5-13 membered bridged heterocyclyl, optionally substituted C.sub.5-C.sub.13 spiro carbocyclyl, optionally substituted 5-13 membered spiro heterocyclyl, optionally substituted aryl, and optionally substituted heteroaryl; preferably, Z is selected from null, CH.sub.2, CH—CH, C=C, NH, O, optionally substituted 3-7 membered carbocyclyl, and optionally substituted 3-7 membered heterocyclyl; R.sup.1 is selected from hydrogen, halogen, cyano, nitro, optionally substituted C.sub.1-C.sub.6 alkyl, optionally substituted C.sub.1-C.sub.6 heteroalkyl, optionally substituted 3 to 6 membered carbocyclyl, and optionally substituted 3 to 6 membered heterocyclyl; R.sup.2 is selected from hydrogen, halogen, cyano, nitro, optionally substituted C.sub.1-C.sub.6 alkyl, optionally substituted C.sub.1-C.sub.6 heteroalkyl, optionally substituted C.sub.1-C.sub.6 alkoxy, optionally substituted C.sub.1-C.sub.6 alkylamino, optionally substituted 3-6 membered carbocyclyl, and optionally substituted 3-6 membered heterocyclyl; R.sup.3, and R.sup.4 are independently selected from hydrogen, halogen, cyano, nitro, optionally substituted C.sub.1-C.sub.6 alkyl, optionally substituted C.sub.1-C.sub.6 heteroalkyl, optionally substituted 3-6 membered carbocyclyl, and optionally substituted 3-6 membered heterocyclyl; or R.sup.3 and R.sup.4 together with the atom to which they are connected form a 3-6 membered carbocyclyl, or 3-6 membered heterocyclyl; and R.sup.5 and R.sup.6 are independently selected from null, hydrogen, halogen, oxo, hydroxyl, amino, cyano, nitro, optionally substituted C.sub.1-C.sub.6 alkyl, optionally substituted C.sub.1-C.sub.6 heteroalkyl, optionally substituted 3-6 membered carbocyclyl, and optionally substituted 3-6 membered heterocyclyl; or R.sup.5 and R.sup.6 together with the atom to which they are connected form a 3-6 membered carbocyclyl, or 3-6 membered heterocyclyl.

(316) In some embodiments, the degradation tag is a moiety selected from the group consisting of

FORMULAE SE, 5F, 5G, 5H, 5I, 5J, 5K, 5L, 5M, 5N, 5O, 5P, and 5Q:

(317) ##STR00239## ##STR00240## ##STR00241##

wherein U, V, W, X and X' are independently selected from CR.sup.2 and N; Y is selected from CR.sup.3R.sup.4, NR.sup.3 and O; preferably, Y is selected from CH.sub.2, NH, NCH.sub.3 and O; Y', Y'', and Y''' are independently selected from CR.sup.3R.sup.4; Z is selected from null, CO, CR.sup.5R.sup.6, NR.sup.8, O, optionally substituted C.sub.1-C.sub.10 alkylene, optionally substituted C.sub.1-C.sub.5 heteroalkylene, optionally substituted C.sub.1-C.sub.10 alkenylene, optionally substituted C.sub.1-C.sub.10 alkynylene, optionally substituted 3-10 membered carbocyclyl, optionally substituted 3-10 membered heterocyclyl, optionally substituted C.sub.4-C.sub.13 fused carbocyclyl, optionally substituted 5-13 membered fused heterocyclyl, optionally substituted C.sub.5-C.sub.13 bridged carbocyclyl, optionally substituted 5-13 membered bridged heterocyclyl, optionally substituted C.sub.5-C.sub.13 spiro carbocyclyl, optionally substituted 5-13 membered spiro heterocyclyl, optionally substituted aryl, and optionally substituted heteroaryl; preferably, Z is selected from null, CH.sub.2, CH=CH, C=C, NH, O, optionally substituted 3-7 membered carbocyclyl, and optionally substituted 3-7 membered heterocyclyl; R.sup.1 is selected from hydrogen, halogen, cyano, nitro, optionally substituted C.sub.1-C.sub.6 alkyl, optionally substituted C.sub.1-C.sub.6 heteroalkyl, optionally substituted 3 to 6 membered carbocyclyl, and optionally substituted 3-6 membered heterocyclyl; R.sup.2 is selected from hydrogen, halogen, cyano, nitro, optionally substituted C.sub.1-C.sub.6 alkyl, optionally substituted C.sub.1-C.sub.6 heteroalkyl, optionally substituted C.sub.1-C.sub.6 alkoxy, optionally substituted C.sub.1-C.sub.6 alkylamino, optionally substituted 3 to 6 membered carbocyclyl, and optionally substituted 3-6 membered heterocyclyl; R.sup.3, and R.sup.4 are independently selected from hydrogen, halogen, cyano, nitro, optionally substituted C.sub.1-C.sub.6 alkyl, optionally substituted C.sub.1-C.sub.8 heteroalkyl, optionally substituted 3 to 6 membered carbocyclyl, and optionally substituted 3-6 membered heterocyclyl; or R.sup.3 and R.sup.4 together with the atom to which they are connected form a 3-6 membered carbocyclyl, or 3-6 membered heterocyclyl; and R.sup.5 and R.sup.6 are independently selected from null, hydrogen, halogen, oxo, hydroxyl, amino, cyano, nitro, optionally substituted C.sub.1-C.sub.6 alkyl, optionally substituted C.sub.1-C.sub.6 heteroalkyl, optionally substituted 3 to 6 membered carbocyclyl, and optionally substituted 3-6 membered heterocyclyl; or R.sup.5 and R.sup.6 together with the atom to which they are connected form a 3-6 membered carbocyclyl, or 3-6 membered heterocyclyl.

(318) In one embodiment, the degradation tag is a moiety of FORMULA 6A:

(319) ##STR00242##

wherein R.sup.1 and R.sup.2 are independently selected from hydrogen, optionally substituted C.sub.1-C.sub.8 alkyl, optionally substituted C.sub.1-C.sub.5 heteroalkyl, optionally substituted C.sub.1-C.sub.8alkoxyC.sub.1-C.sub.5alkyl, optionally substituted C.sub.1-C.sub.5 haloalkyl, optionally substituted C.sub.1-C.sub.8 hydroxyalkyl, optionally substituted C.sub.1-C.sub.8 aminoalkyl, optionally substituted C.sub.1-C.sub.5alkylaminoC.sub.1-C.sub.6alkyl, optionally substituted C.sub.3-C.sub.7 carbocyclyl, optionally substituted 3-7 membered heterocyclyl, optionally substituted C.sub.2-C.sub.8 alkenyl, and optionally substituted C.sub.2-C.sub.8 alkynyl; and R.sup.3 is hydrogen, optionally substituted C(O) C.sub.1-C.sub.6 alkyl, optionally substituted C(O) C.sub.1-C.sub.8alkoxyC.sub.1-C.sub.8alkyl, optionally substituted C(O) C.sub.1-C.sub.8 haloalkyl, optionally substituted C(O) C.sub.1-C.sub.5 hydroxyalkyl, optionally substituted C(O) C.sub.1-C.sub.5 aminoalkyl, optionally substituted C(O) C.sub.1-C.sub.8alkylaminoC.sub.1-C.sub.8alkyl, optionally substituted C(O) C.sub.3-C.sub.7 carbocyclyl, optionally substituted C(O) (3-7 membered heterocyclyl), optionally substituted C(O) C.sub.2-C.sub.8 alkenyl, optionally substituted C(O) C.sub.2-C.sub.8 alkynyl, optionally substituted C(O)OC.sub.1-C.sub.8alkoxyC.sub.1-C.sub.5alkyl, optionally substituted C(O)OC.sub.1-C.sub.5 haloalkyl, optionally substituted C(O)OC.sub.1-C.sub.8 hydroxyalkyl, optionally substituted C(O)OC.sub.1-C.sub.6 aminoalkyl, optionally substituted C(O)OC.sub.1-C.sub.8alkylaminoC.sub.1-C.sub.8alkyl,

optionally substituted C(O)OC.sub.3—C; carbocyclyl, optionally substituted C(O)O(3-7 membered heterocyclyl), optionally substituted C(O)OC.sub.2-C.sub.8 alkenyl, optionally substituted C(O)OC.sub.2-C.sub.8 alkynyl, optionally substituted C(O)NC.sub.1-C.sub.8alkoxyC.sub.1-C.sub.8alkyl, optionally substituted C(O)NC.sub.1-C.sub.8 haloalkyl, optionally substituted C(O)NC.sub.1-C.sub.5 hydroxyalkyl, optionally substituted C(O)NC.sub.1-C.sub.8 aminoalkyl, optionally substituted C(O)NC.sub.1-C.sub.8alkylaminoC.sub.1-C.sub.8alkyl, optionally substituted C(O)NC.sub.3-C.sub.7 carbocyclyl, optionally substituted C(O)N(3-7 membered heterocyclyl), optionally substituted C(O)NC.sub.2-C.sub.8 alkenyl, optionally substituted C(O)NC.sub.2-C.sub.8 alkynyl, optionally substituted P(O)(OH) 2, optionally substituted P(O)(OC.sub.1-C.sub.8 alkyl).sub.2, and optionally substituted P(O)(OC.sub.1-C.sub.8 aryl).sub.2. (320) In one embodiment, the degradation tag is a moiety selected from the group consisting of FORMULAE 6B, 6C, 6D, 6E and 6F:

(321) ##STR00243##

wherein R.sup.1 and R.sup.2 are independently selected from hydrogen, halogen, OH, NH.sub.2, CN, optionally substituted C.sub.1-C.sub.8 alkyl, optionally substituted C.sub.1-C.sub.6 heteroalkyl, optionally substituted C.sub.1-C.sub.8alkoxyC.sub.1-C.sub.8alkyl, optionally substituted C.sub.1-C.sub.5 haloalkyl, optionally substituted C.sub.1-C.sub.5 hydroxyalkyl, optionally substituted C.sub.1-C.sub.5 aminoalkyl, optionally substituted C.sub.1-C.sub.5alkylaminoC.sub.1-C.sub.6alkyl, optionally substituted C.sub.3-C.sub.7 carbocyclyl, optionally substituted 3-7 membered heterocyclyl, optionally substituted C.sub.2-C.sub.8 alkenyl, and optionally substituted C.sub.2-C.sub.8 alkynyl; (preferably, R.sup.1 is selected from iso-propyl or tert-butyl; and R.sup.2 is selected from hydrogen or methyl); R.sup.3 is hydrogen, optionally substituted C(O) C.sub.1-C.sub.6 alkyl, optionally substituted C(O) C.sub.1-C.sub.8alkoxyC.sub.1-C.sub.8alkyl, optionally substituted C(O) C.sub.1-C.sub.8 haloalkyl, optionally substituted C(O) C.sub.1-C.sub.5 hydroxyalkyl, optionally substituted C(O) C.sub.1-C.sub.5 aminoalkyl, optionally substituted C(O) C.sub.1-C.sub.5alkylaminoC.sub.1-C.sub.8alkyl, optionally substituted C(O) C.sub.3-C.sub.7 carbocyclyl, optionally substituted C(O)(3-7 membered heterocyclyl), optionally substituted C(O) C.sub.2-C.sub.8 alkenyl, optionally substituted C(O) C.sub.2-C.sub.8 alkynyl, optionally substituted C(O)OC.sub.1-C.sub.8alkoxyC.sub.1-C.sub.8 alkyl, optionally substituted C(O)OC.sub.1-C.sub.5 haloalkyl, optionally substituted C(O)OC.sub.1-C.sub.8 hydroxyalkyl, optionally substituted C(O)OC.sub.1-C.sub.8 aminoalkyl, optionally substituted C(O)OC.sub.1-C.sub.8alkylaminoC.sub.1-C.sub.8alkyl, optionally substituted C(O)OC.sub.3—C; carbocyclyl, optionally substituted C(O)O(3-7 membered heterocyclyl), optionally substituted C(O)OC.sub.2-C.sub.8 alkenyl, optionally substituted C(O)OC.sub.2-C.sub.8 alkynyl, optionally substituted C(O)NC.sub.1-C.sub.8alkoxyC.sub.1-C.sub.8alkyl, optionally substituted C(O)NC.sub.1-C.sub.8 haloalkyl, optionally substituted C(O)NC.sub.1-C.sub.8 hydroxyalkyl, optionally substituted C(O)NC.sub.1-C.sub.8 aminoalkyl, optionally substituted C(O)NC.sub.1-C.sub.8alkylaminoC.sub.1-C.sub.8alkyl, optionally substituted C(O)NC.sub.3-C.sub.7 carbocyclyl, optionally substituted C(O)N(3-7 membered heterocyclyl), optionally substituted C(O)NC.sub.2-C.sub.8 alkenyl, optionally substituted C(O)NC.sub.2-C.sub.8 alkynyl, optionally substituted P(O)(OH) 2, optionally substituted P(O)(OC.sub.1-C.sub.8 alkyl).sub.2, and optionally substituted P(O)(OC.sub.1-C.sub.8 aryl).sub.2; and R.sup.4 and R.sup.5 are independently selected from hydrogen, COR.sup.6, CO.sub.2R.sup.6, CONR.sup.6R.sup.7, SOR.sup.6, SO.sub.2R.sup.6,

(322) SO.sub.2NR.sup.6R.sup.7, optionally substituted C.sub.1-C.sub.8 alkyl, optionally substituted C.sub.1-C.sub.6 heteroalkyl, optionally substituted C.sub.1-C.sub.8alkoxyC.sub.1-C.sub.8alkyl, optionally substituted C.sub.1-C.sub.8 alkylaminoC.sub.1-C.sub.5alkyl, optionally substituted 3-8 membered carbocyclyl, optionally substituted 3-8 membered heterocyclyl, optionally substituted aryl, and optionally substituted heteroaryl, wherein R.sup.6 and R.sup.7 are independently selected from hydrogen, optionally substituted C.sub.1-C.sub.8 alkyl, optionally substituted C.sub.1-C.sub.6 heteroalkyl, optionally substituted C.sub.1-C.sub.8 alkoxy, optionally

substituted C.sub.1-C.sub.8alkoxyC.sub.1-C.sub.5alkyl, optionally substituted C.sub.1-C.sub.5alkylaminoC.sub.1-C.sub.5alkyl, optionally substituted 3-8 membered carbocyclyl, optionally substituted 3-8 membered heterocyclyl, optionally substituted aryl, and optionally substituted heteroaryl, or R.sup.4 and R.sup.5; or R.sup.6 and R.sup.7 together with the atom to which they are connected form a 3-8 membered carbocyclyl or 3-8 membered heterocyclyl ring; Ar is selected from aryl and heteroaryl, each of which is optionally substituted with one or more substituents independently selected from F, Cl, CN, NO.sub.2, OR.sup.3, NR.sup.8R.sup.9, COR.sup.8, CO.sub.2R.sup.8, CONR.sup.8R.sup.9, SOR.sup.8, SO.sub.2R.sup.5, SO.sub.2NR.sup.8R.sup.9, NR.sup.10OR.sup.9, NR.sup.10C(O)NR.sup.8R.sup.9, NR.sup.10SOR.sup.8, NR.sup.10SO.sub.2R.sup.8, optionally substituted C.sub.1-C.sub.6 alkyl, optionally substituted C.sub.1-C.sub.6 heteroalkyl, optionally substituted C.sub.1-C.sub.6 alkoxyalkyl, optionally substituted C.sub.1-C.sub.6 haloalkyl, optionally substituted C.sub.1-C.sub.6 hydroxyalkyl, optionally substituted C.sub.1-C.sub.6alkylaminoC.sub.1-C.sub.6alkyl, optionally substituted C.sub.3-C.sub.7 carbocyclyl, optionally substituted 3-7 membered heterocyclyl, optionally substituted C.sub.2-C.sub.6 alkenyl, optionally substituted C.sub.2-C.sub.6 alkynyl, optionally substituted aryl, and optionally substituted C.sub.4-C.sub.8 heteroaryl, wherein R.sup.8, R.sup.9 and R.sup.10 are independently selected from null, hydrogen, optionally substituted C.sub.1-C.sub.6 alkyl, optionally substituted C.sub.1-C.sub.6 heteroalkyl, optionally substituted C.sub.2-C.sub.6 alkenyl, optionally substituted C.sub.2-C.sub.8 alkynyl, optionally substituted C.sub.3-C.sub.7 carbocyclyl, optionally substituted 3-7 membered heterocyclyl, optionally substituted aryl, and optionally substituted heteroaryl, or R.sup.8 and R' together with the atom to which they are connected form a 3-8 membered carbocyclyl or 3-8 membered heterocyclyl ring.

(323) In another embodiment, the degradation tag is a moiety of FORMULA 7A:

(324) ##STR00244## wherein V, W, X, and Z are independently selected from CR.sup.4 and N.

(325) R.sup.1, R.sup.2, R.sup.3, and R.sup.4 are independently selected from hydrogen, optionally substituted C.sub.1-C.sub.6 alkyl, optionally substituted C.sub.1-C.sub.6 heteroalkyl, optionally substituted C.sub.1-C.sub.6alkoxyC.sub.1-C.sub.5alkyl, optionally substituted C.sub.1-C.sub.6 haloalkyl, optionally substituted C.sub.1-C.sub.5 hydroxyalkyl, optionally substituted C.sub.3-C.sub.7 carbocyclyl, optionally substituted 3-7 membered heterocyclyl, optionally substituted C.sub.2-C.sub.8 alkenyl, and optionally substituted C.sub.2-C.sub.8 alkynyl.

(326) In another embodiment, the degradation tag is a moiety of FORMULA 7B:

(327) ##STR00245## wherein R.sup.1, R.sup.2, and R.sup.3 are independently selected from hydrogen, halogene, optionally substituted C.sub.1-C.sub.8 alkyl, optionally substituted C.sub.1-C.sub.6 heteroalkyl, optionally substituted C.sub.1-C.sub.8alkoxyC.sub.1-C.sub.5alkyl, optionally substituted C.sub.1-C.sub.5 haloalkyl, optionally substituted C.sub.1-C.sub.5 hydroxyalkyl, optionally substituted C.sub.3-C.sub.7 carbocyclyl, optionally substituted 3-7 membered heterocyclyl, optionally substituted C.sub.2-C.sub.8 alkenyl, and optionally substituted C.sub.2-C.sub.8 alkynyl; R.sup.4 and R.sup.5 are independently selected from hydrogen, COR.sup.6, CO.sub.2R.sup.6, CONR.sup.6R.sup.7, SOR.sup.6, SO.sub.2R.sup.6, SO.sub.2NR.sup.6R.sup.7, optionally substituted C.sub.1-C.sub.8 alkyl, optionally substituted C.sub.1-C.sub.6 heteroalkyl, optionally substituted C.sub.1-C.sub.8alkoxyC.sub.1-C.sub.5alkyl, optionally substituted C.sub.1-C.sub.5alkylaminoC.sub.1-C.sub.5alkyl, optionally substituted aryl-C.sub.1-C.sub.8alkyl, optionally substituted 3-8 membered carbocyclyl, optionally substituted 3-8 membered heterocyclyl, optionally substituted aryl, and optionally substituted heteroaryl, wherein R.sup.6 and R.sup.7 are independently selected from hydrogen, optionally substituted C.sub.1-C.sub.8alkyl, optionally substituted C.sub.1-C.sub.6 heteroalkyl, optionally substituted C.sub.1-C.sub.8alkoxyC.sub.1-C.sub.5alkyl, optionally substituted C.sub.1-C.sub.5alkylaminoC.sub.1-C.sub.8alkyl, optionally substituted 3-8 membered carbocyclyl, optionally substituted 3-8 membered heterocyclyl, optionally substituted aryl, and optionally substituted heteroaryl, or

R.sup.6 and R.sup.7 together with the atom to which they are connected form a 3-8 membered carbocyclyl or heterocyclyl ring.

(328) In certain embodiments, the degradation tag is a moiety of Formula 11 which connects to the linker via Z or any of V.sup.1, V.sup.2, V.sup.3, V.sup.4, W.sup.1, W.sup.2, and W.sup.3;

(329) ##STR00246## wherein V.sup.1, V.sup.2, V.sup.3, and V.sup.4 are each independently selected from CR.sup.2' and N; W.sup.1, W.sup.2, and W.sup.3 are each independently selected from CO, O, CR.sup.3'R.sup.4', and NR.sup.5', wherein R.sup.5' is selected from a hydrogen, halogen, cyano, nitro, optionally substituted C.sub.1-C.sub.6 alkyl, optionally substituted C.sub.1-C.sub.6 heteroalkyl, optionally substituted C.sub.3-C.sub.7 carbocyclyl, optionally substituted 3-7 membered heterocyclyl, and 2,6-dioxopiperidin-3-yl optionally substituted with R.sup.1', provided that there are no two adjacent CO groups; Z is selected from null, CO, CR.sup.3'R.sup.4', NR.sup.3', O, optionally substituted C.sub.1-C.sub.10 alkylene, optionally substituted C.sub.1-C.sub.10 heteroalkylene, optionally substituted C.sub.1-C.sub.10 alkenylene, optionally substituted C.sub.1-C.sub.10 alkynylene, optionally substituted 3-7 membered carbocyclyl, and optionally substituted 3-7 membered heterocyclyl; R.sup.1' is selected from hydrogen, halogen, cyano, nitro, optionally substituted C.sub.1-C.sub.6 alkyl, optionally substituted C.sub.1-C.sub.6 heteroalkyl, optionally substituted 3-6 membered carbocyclyl, and optionally substituted 3-6 membered heterocyclyl; R.sup.2' is selected from hydrogen, halogen, cyano, nitro, optionally substituted C.sub.1-C.sub.6 alkyl, optionally substituted C.sub.1-C.sub.6 heteroalkyl, optionally substituted C.sub.1-C.sub.6 alkoxy, optionally substituted C.sub.1-C.sub.6 alkylamino, optionally substituted 3-6 membered carbocyclyl, and optionally substituted 3-6 membered heterocyclyl; and R.sup.3' and R.sup.4' at each occurrence are independently selected from hydrogen, halogen, cyano, nitro, optionally substituted C.sub.1-C.sub.6 alkyl, optionally substituted C.sub.1-C.sub.6 heteroalkyl, optionally substituted 3-6 membered carbocyclyl, and optionally substituted 3-6 membered heterocyclyl; or R.sup.3' and R.sup.4' together with the atom to which they are connected form a 3-8 membered carbocyclyl or 3-8 membered heterocyclyl ring

(330) In certain embodiments, V.sup.1, V.sup.2, V.sup.3, and V.sup.4 are each independently CR.sup.2'.

(331) In certain embodiments, at least one of W.sup.1, W.sup.2, and W.sup.3 is NR.sup.5'.

(332) In certain embodiments, at least one of W.sup.1, W.sup.2, and W.sup.3 is NR.sup.5', and only one of R.sup.5' is 2,6-dioxopiperidin-3-yl optionally substituted with R.sup.1.

(333) In certain embodiments, W.sup.1 is CO.

(334) In certain embodiments, W.sup.2 is CO.

(335) In certain embodiments, W.sup.3 is CO.

(336) In certain embodiments, W.sup.1 and W.sup.3 are both CO.

(337) In certain embodiments, W.sup.2 is CO and W.sup.1 and W.sup.3 are both NR.sup.5', and only one of R.sup.5' is 2,6-dioxopiperidin-3-yl optionally substituted with R.sup.1.

(338) In another embodiment, the degradation tag is derived from any of the following:

(339) ##STR00247## ##STR00248## ##STR00249## ##STR00250## ##STR00251##

(340) In certain embodiments, the degradation tag is an E.sup.3 ligase. In certain embodiments, the degradation tag comprises one or more of cereblon E.sup.3 ligase, a VHL E.sup.3 ligase, an IAP ligase, a MDM2 ligase, a TRIM24 ligase, a TRIM21 ligase, a KEAP1 ligase, DCAF16 ligase, RNF4 ligase, RNF114 ligase, and AhR ligase. In certain embodiments, the degradation tags of the present disclosure include, e.g., pomalidomide (Fischer et al., 2014), thalidomide (Fischer et al., 2014), lenalidomide (Fischer et al., 2014), VH032 (Galdeano et al., 2014; Maniaci et al., 2017), adamantane (Xie et al., 2014), 1-((4,4,5,5,5-pentafluoropentyl)sulfinyl)nonane (E. Wakeling, 1995), nutlin-3a (Vassilev et al., 2004), RG7112 (Vu et al., 2013), RG7338, AMG 232 (Sun et al., 2014), AA-115 (Aguilar et al., 2017), bestatin (Hiroyuki Suda et al., 1976), MV1 (Varfolomeev et al., 2007), LCL161 (Weisberg et al., 2010), CPD36, GDC-0152, CRBN-1, CRBN-2, CRBN-3, CRBN-4, CRBN-5, CRBN-6, CRBN-7, CRBN-8, CRBN-9, CRBN-10, CRBN-11, and/or analogs

thereof. In certain embodiments, the degradation tag is derived from a compound comprising pomalidomide (Fischer et al., 2014), thalidomide (Fischer et al., 2014), lenalidomide (Fischer et al., 2014), VH032 (Galdeano et al., 2014; Maniaci et al., 2017), adamantane (Xie et al., 2014), 1-((4,4,5,5,5-pentafluoropentyl)sulfinyl)nonane (E. Wakeling, 1995), nutlin-3a (Vassilev et al., 2004), RG7112 (Vu et al., 2013), RG7338, AMG 232 (Sun et al., 2014), AA-115 (Aguilar et al., 2017), bestatin (Hiroyuki Suda et al., 1976), MV1 (Varfolomeev et al., 2007), LCL161 (Weisberg et al., 2010), CPD36, GDC-0152, CRBN-1, CRBN-2, CRBN-3, CRBN-4, CRBN-5, CRBN-6, CRBN-7, CRBN-8, CRBN-9, CRBN-10, CRBN-11, and/or analogs thereof.

(341) In another embodiment, the degradation tag is selected from the group consisting of:

(342) ##STR00252## ##STR00253## ##STR00254## ##STR00255## ##STR00256##
##STR00257## ##STR00258## ##STR00259## ##STR00260## ##STR00261## ##STR00262##
##STR00263## ##STR00264## ##STR00265## ##STR00266## ##STR00267## ##STR00268##
##STR00269## ##STR00270## ##STR00271## ##STR00272## ##STR00273## ##STR00274##
##STR00275## ##STR00276## ##STR00277## ##STR00278##

(343) ##STR00279## ##STR00280## ##STR00281## ##STR00282## ##STR00283##
##STR00284## ##STR00285## ##STR00286## ##STR00287## ##STR00288## ##STR00289##
##STR00290## ##STR00291## ##STR00292## ##STR00293## ##STR00294## ##STR00295##
##STR00296## ##STR00297## ##STR00298## ##STR00299## ##STR00300## ##STR00301##
##STR00302## ##STR00303## ##STR00304## ##STR00305## ##STR00306## ##STR00307##
##STR00308## ##STR00309## ##STR00310## ##STR00311## ##STR00312##

(344) In another embodiment, the degradation tag is selected from the group consisting of:

FORMULA 8A, 8B, 8C, 8D, 8E, 8F, 8G, 8H, 8I, 8J, 8K, 8L, 8M, 8O, 8P, 8Q, 8R, 8AQ, 8AR, 8AS, 8AT, 8AU, 8AV, 8AW, 8AX, 8AY, 8AZ, 8BA, 8BB, 8BC, 8BD, 8BE, 8BF, 8BG, 8BH, 8B1, 8BJ, 8BK, 8BL, 8BM, and 8BN, 8BO, 8BP, 8BQ, 8BR, 8BS, 8CB, 8CC, 8CD, 8CE, 8CF, 8CG, 8CH, 8CI, 8CJ, 8CK, 8CL, 8 CM, 8CN, 8CO, 8CP, 8CQ, 8CR, 8CS, 8CT, 8CU, 8CV, 8CW, 8CX, 8CY, 8CZ, 8DA, 8 DB, 8DC, 8DD, 8DE, 8DF, 8DG, 8DH, 8D^{sup.1}, 8DJ, 8DK, 8DL, 8DM, 8DN, 8DO, 8DP, 8DQ, 8DR, 8DS, 8DT, 8DU, 8DV, 8DW, 8DX, 8DY, 8DZ, 8EA, 8EB, 8EC, 8ED, SEE, 8EF, 8EG, 8EH, 8EI, 8EJ, 8EK, SEL, 8EM, 8EN, 8EO, 8EP, SEO, 8GU, 8GV, 8GW, 8GX, 8GY, 8GZ, 8HA, 8HB, 8HC, 8HD, 8HE, 8HF, 8HG, 8HH, 8HI, 8HJ, 8HK, 8HL, 8HM, 8HN, 8HO, 8HP, 8HQ, 8 HR, 8HS, 8HT, 8HU, 8HV, 8HW, 8HX, 8HY, 8HZ, 8IA, 8IB, 8IC, 8ID, SIE, 8IF, 8IG, 8IH, 8II, 8IJ, 8IK, 8IL, 8IM, 8 IN, 8I0, 8IP, SIQ, 8IR, 8IS, 8IT, 8IU, 8IV, 8IW, 8IX, 8IY, 8IZ, 8JA, 8JB, 8JC, 8JD, 8JE, 8JF, 8JG, 8JH, 8JI, 8JJ, 8JK, 8JL, 8JM, 8JN, 8JO, 8JP, 8JQ, 8JR, 8JS, and 8JT.

(345) Linker Moiety

(346) As used herein, a “linker” or “linker moiety” is a bond, molecule, or group of molecules that binds two separate entities to one another. Linkers provide for optimal spacing of the two entities. The term “linker” in some aspects refers to any agent or molecule that bridges the TRK ligand to the degradation tag. One of ordinary skill in the art recognizes that sites on the TRK ligand or the degradation tag, which are not necessary for the function of the degraders of the present disclosure, are ideal sites for attaching a linker, provided that the linker, once attached to the conjugate of the present disclosures, does not interfere with the function of the TRK ligand, i.e., its ability to bind TRK, or the function of the degradation tag, i.e., its ability to recruit a ubiquitin ligase.

(347) The length of the linker of the bivalent compound can be adjusted to minimize the molecular weight of the bivalent compounds, avoid the clash of the TRK ligand or targeting moiety with the ubiquitin ligase and/or induce TRK misfolding by the hydrophobic tag. In certain embodiments, the linker comprises acyclic or cyclic saturated or unsaturated carbon, ethylene glycol, amide, amino, ether, urea, carbamate, aromatic, heteroaromatic, heterocyclic or carbonyl groups. In certain embodiments, the length of the linker is 0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20 or more atoms.

(348) In some embodiments, the linker moiety is of FORMULA 9;

(349) ##STR00313##

wherein A, W and B, at each occurrence, are independently selected from null, or bivalent moiety selected from R^{sup.r}—R'', R'COR'', R'CO.sub.2R'', R'C(O)N(R^{sup.1})R'', R'C(S)N(R^{sup.1})R'', R'OR'', R'SR'', R'SOR'', R'SO.sub.2R'', R'SO.sub.2N(R^{sup.1})R'', R'N(R^{sup.1})R'', R'N(R^{sup.1})COR'', R'N(R^{sup.1})CON(R^{sup.2})R'', R'N(R^{sup.1})C(S)R'', optionally substituted C.sub.1—C: alkylene, optionally substituted C.sub.1-C.sub.8 heteroalkylene, optionally substituted C.sub.2-C.sub.8 alkenylene, optionally substituted C.sub.2-C.sub.8 alkynylene, optionally substituted C.sub.1-C.sub.8alkoxyC.sub.1-C.sub.8alkylene, optionally substituted C.sub.1-C.sub.6 haloalkylene, optionally substituted C.sub.1-C.sub.5 hydroxyalkylene, optionally substituted C.sub.4-C.sub.13 fused carbocyclyl, optionally substituted 5-13 membered fused heterocyclyl, optionally substituted C.sub.5-C.sub.13 bridged carbocyclyl, optionally substituted 5-13 membered bridged heterocyclyl, optionally substituted C.sub.5-C.sub.13 spiro carbocyclyl, optionally substituted 5-13 membered spiro heterocyclyl, optionally substituted 3-10 membered carbocyclyl, optionally substituted 3-10 membered heterocyclyl, optionally substituted aryl, and optionally substituted heteroaryl, wherein R' and R'' are independently selected from null, optionally substituted (C.sub.1-C.sub.8 alkylene)-R^{sup.r} (preferably, CH.sub.2—R^{sup.1}), optionally substituted R^{sup.r}—(C.sub.1-C.sub.8 alkylene), optionally substituted (C.sub.1-C.sub.8 alkylene)-R^{sup.r}—(C.sub.1-C.sub.8alkylene), or a moiety comprising of optionally substituted C.sub.1-C.sub.8alkyl, optionally substituted C.sub.1-C.sub.5 heteroalkyl, optionally substituted C.sub.2-C.sub.8 alkenyl, optionally substituted C.sub.2-C.sub.8 alkynyl, optionally substituted C.sub.1-C.sub.5 hydroxyalkyl, optionally substituted C.sub.1-C.sub.6alkoxyC.sub.1-C.sub.5alkyl, optionally substituted C.sub.1-C.sub.8alkylaminoC.sub.1-C.sub.5alkyl, optionally substituted C.sub.1-C.sub.5 haloalkyl, optionally substituted C.sub.1-C.sub.8 alkylene, optionally substituted C.sub.1-C.sub.5 heteroalkylene, optionally substituted C.sub.2-C.sub.8 alkenylene, optionally substituted C.sub.2-C.sub.8 alkynylene, optionally substituted C.sub.1-C.sub.5 hydroxyalkylene, optionally substituted C.sub.1-C.sub.8alkoxy C.sub.1-C.sub.8alkylene, optionally substituted C.sub.1-C.sub.5alkylaminoC.sub.1-C.sub.5alkylene, optionally substituted C.sub.1-C.sub.8 haloalkylene, optionally substituted 3-10 membered carbocyclyl, optionally substituted 3-10 membered heterocyclyl, optionally substituted C.sub.4-C.sub.13 fused carbocyclyl, optionally substituted 5-13 membered fused heterocyclyl, optionally substituted C.sub.5-C.sub.15 bridged carbocyclyl, optionally substituted 5-13 membered bridged heterocyclyl, optionally substituted C.sub.5-C.sub.13 spiro carbocyclyl, optionally substituted 5-13 membered spiro heterocyclyl, optionally substituted aryl, and optionally substituted heteroaryl; R^{sup.r} is selected from optionally substituted 3-10 membered carbocyclyl, optionally substituted 3-10 membered heterocyclyl, optionally substituted C.sub.4-C.sub.13 fused carbocyclyl, optionally substituted 5-13 membered fused heterocyclyl, optionally substituted C.sub.5-C.sub.15 bridged carbocyclyl, optionally substituted 5-13 membered bridged heterocyclyl, optionally substituted C.sub.5-C.sub.13 spiro carbocyclyl, optionally substituted 5-13 membered spiro heterocyclyl, optionally substituted aryl, and optionally substituted heteroaryl; R^{sup.1} and R^{sup.2} are independently selected from hydrogen, optionally substituted C.sub.1-C.sub.8 alkyl, optionally substituted C.sub.1-C.sub.5 heteroalkyl, optionally substituted C.sub.2-C.sub.8 alkenyl, optionally substituted C.sub.2-C.sub.8 alkynyl, optionally substituted C.sub.1-C.sub.6 alkoxyalkyl, optionally substituted C.sub.1-C.sub.5 haloalkyl, optionally substituted C.sub.1-C.sub.8 hydroxyalkyl, optionally substituted C.sub.1-C.sub.8alkylaminoC.sub.1-C.sub.5alkyl, optionally substituted 3-10 membered carbocyclyl, optionally substituted 3-10 membered heterocyclyl, optionally substituted aryl, and optionally substituted heteroaryl; R' and R'', R^{sup.1} and R^{sup.2}, R' and R^{sup.1}, R' and R^{sup.2}, R'' and R^{sup.1}, R'' and R^{sup.2} together with the atom to which they are connected form a 3-20 membered carbocyclyl or 3-20 membered heterocyclyl ring; and m is 0 to 15.

(350) In one embodiment, the linker moiety is of FORMULA 9A:

(351) ##STR00314##

Wherein R^{sup.1}, R^{sup.2}, R^{sup.3} and R^{sup.4}, at each occurrence, are independently selected

from hydrogen, halogen, hydroxyl, amino, cyano, nitro, optionally substituted C.sub.1-C.sub.8 alkyl, optionally substituted C.sub.1-C.sub.8 heteroalkyl, optionally substituted C.sub.2-C.sub.8 alkenyl, optionally substituted C.sub.2-C.sub.8 alkynyl, optionally substituted C.sub.1-C.sub.5 alkoxy, optionally substituted C.sub.1-C.sub.8alkoxyalkyl, optionally substituted C.sub.1-C.sub.8 haloalkyl, optionally substituted C.sub.1-C.sub.5 hydroxyalkyl, optionally substituted C.sub.1-C.sub.8 alkylamino, and optionally substituted C.sub.1-C.sub.5 alkylaminoC.sub.1-C.sub.8 alkyl, optionally substituted 3-10 membered carbocyclyl, optionally substituted 3-8 membered cycloalkoxy, optionally substituted 3-10 membered carbocyclylamino, optionally substituted 3-8 membered membered heterocyclyl, optionally substituted aryl, and optionally substituted heteroaryl, or R.sup.1 and R.sup.2, or R.sup.3 and R.sup.4 together with the atom to which they are connected form a 3-20 membered carbocyclyl or 3-20 membered heterocyclyl ring; A, W and B, at each occurrence, are independently selected from null, or bivalent moiety selected from R.sup.r—R'', R'COR'', R'CO.sub.2R'', R'C(O)N(RO)R'', R'C(S)N(RO)R'', R'OR'', R'SR'', R'SOR'', R'SO.sub.2R'', R'SO.sub.2N(R.sup.5)R'', R'N(R.sup.5)R'', R'N(R.sup.5) COR'', R'N(R.sup.5)CON(RO)R'', R'N(R.sup.5)C(S)R'', optionally substituted C.sub.1-C.sub.6 alkylene, optionally substituted C.sub.1-C.sub.5 heteroalkylene, optionally substituted C.sub.2-C.sub.8 alkenylene, optionally substituted C.sub.2-C.sub.8 alkynylene, optionally substituted C.sub.1-C.sub.8alkoxyC.sub.1-C.sub.8alkylene, optionally substituted C.sub.1-C.sub.5 haloalkylene, optionally substituted C.sub.1-C.sub.5 hydroxyalkylene, optionally substituted C.sub.4-C.sub.13 fused carbocyclyl, optionally substituted 5-13 membered fused heterocyclyl, optionally substituted C.sub.5-C.sub.13 bridged carbocyclyl, optionally substituted 5-13 membered bridged heterocyclyl, optionally substituted C.sub.5-C.sub.13 spiro carbocyclyl, optionally substituted 5-13 membered spiro heterocyclyl, optionally substituted 3-10 membered carbocyclyl, optionally substituted 3-10 membered heterocyclyl, optionally substituted aryl, and optionally substituted heteroaryl, wherein R' and R'' are independently selected from null, optionally substituted (C.sub.1-C.sub.8 alkyl)-RE (preferably, CH.sub.2—R.sup.r), optionally substituted R.sup.r—(C.sub.1-C.sub.8alkylene), optionally substituted (C.sub.1-C.sub.8 alkylene)-R.sup.r—(C.sub.1-C.sub.8 alkylene), or a moiety comprising of optionally substituted C.sub.1-C.sub.8 alkyl, optionally substituted C.sub.1-C.sub.8 heteroalkyl, optionally substituted C.sub.2-C.sub.8 alkenyl, optionally substituted C.sub.2-C.sub.8 alkynyl, optionally substituted C.sub.1-C.sub.5 hydroxyalkyl, optionally substituted C.sub.1-C.sub.6alkoxyC.sub.1-C.sub.5alkyl, optionally substituted C.sub.1-C.sub.8alkylaminoC.sub.1-C.sub.5alkyl, optionally substituted C.sub.1-C.sub.5 haloalkyl, optionally substituted C.sub.1-C.sub.8 alkylene, optionally substituted C.sub.1-C.sub.5 heteroalkylene, optionally substituted C.sub.2-C.sub.8 alkenylene, optionally substituted C.sub.2-C.sub.8 alkynylene, optionally substituted C.sub.1-C.sub.5 hydroxyalkylene, optionally substituted C.sub.1-C.sub.8alkoxyC.sub.1-C.sub.5alkylene, optionally substituted C.sub.1-C.sub.8 alkylaminoC.sub.1-C.sub.8alkylene, optionally substituted C.sub.1-C.sub.8 haloalkylene, optionally substituted 3-10 membered carbocyclyl, optionally substituted 3-10 membered heterocyclyl, optionally substituted C.sub.4-C.sub.13 fused carbocyclyl, optionally substituted 5-13 membered fused heterocyclyl, optionally substituted C.sub.5-C.sub.13 bridged carbocyclyl, optionally substituted 5-13 membered bridged heterocyclyl, optionally substituted C.sub.5-C.sub.13 spiro carbocyclyl, optionally substituted 5-13 membered spiro heterocyclyl, optionally substituted aryl, and optionally substituted heteroaryl; R.sup.r is selected from optionally substituted 3-10 membered carbocyclyl, optionally substituted 3-10 membered heterocyclyl, optionally substituted C.sub.4-C.sub.13 fused carbocyclyl, optionally substituted 5-13 membered fused heterocyclyl, optionally substituted C.sub.5-C.sub.15 bridged carbocyclyl, optionally substituted 5-13 membered bridged heterocyclyl, optionally substituted C.sub.5-C.sub.13 spiro carbocyclyl, optionally substituted 5-13 membered spiro heterocyclyl, optionally substituted aryl, and optionally substituted heteroaryl; R.sup.5 and R.sup.6 are independently selected from hydrogen, optionally substituted C.sub.1-C.sub.8 alkyl, optionally substituted C.sub.1-C.sub.5 heteroalkyl, optionally substituted C.sub.2-C.sub.8 alkenyl,

optionally substituted C.sub.2-C.sub.8 alkynyl, optionally substituted C.sub.1-C.sub.8 alkoxyalkyl, optionally substituted C.sub.1-C.sub.6 haloalkyl, optionally substituted C.sub.1-C.sub.5 hydroxyalkyl, optionally substituted C.sub.1-C.sub.8 alkylamino C.sub.1-C.sub.5 alkyl, optionally substituted 3-10 membered carbocyclyl, optionally substituted 3-10 membered heterocyclyl, optionally substituted aryl, and optionally substituted heteroaryl; R' and R'', R.sup.5 and R.sup.6, R' and R.sup.3, R' and R.sup.6, R'' and R.sup.5, or R'' and R.sup.6 together with the atom to which they are connected form a 3-20 membered carbocyclyl or 3-20 membered heterocyclyl ring; m is 0 to 15; n, at each occurrence, is 0 to 15; and o is 0 to 15.

(352) In another embodiment, the linker moiety is of FORMULA 9B:

(353) ##STR00315##

wherein R.sup.1 and R.sup.2, at each occurrence, are independently selected from hydrogen, halogen, hydroxyl, amino, cyano, nitro, and optionally substituted C.sub.1-C.sub.8 alkyl, optionally substituted C.sub.1-C.sub.5 heteroalkyl, optionally substituted C.sub.1-C.sub.8 alkoxy, optionally substituted C.sub.1-C.sub.8 alkoxy C.sub.1-C.sub.8 alkyl, optionally substituted C.sub.1-C.sub.6 haloalkyl, optionally substituted C.sub.1-C.sub.6 hydroxyalkyl, optionally substituted C.sub.1-C.sub.8 alkylamino, C.sub.2-C.sub.8 alkylamino C.sub.1-C.sub.8 alkyl, optionally substituted 3-10 membered carbocyclyl, optionally substituted 3-8 membered cycloalkoxy, optionally substituted 3-10 membered carbocyclylamino, optionally substituted 3-10 membered heterocyclyl, optionally substituted aryl, and optionally substituted heteroaryl, or R.sup.1 and R.sup.2 together with the atom to which they are connected form a 3-20 membered carbocyclyl or 3-20 membered heterocyclyl ring; A and B, at each occurrence, are independently selected from null, or bivalent moiety selected from R.sup.r—R'', R'COR'', R'CO.sub.2R'', R'C(O)N(R.sup.3)R'', R'C(S)N(R.sup.3)R'', R'OR'', R'SR'', R'SOR'', R'SO.sub.2R'', R'SO.sub.2N(R.sup.3)R'', R'N(R.sup.3)R'', R'N(R.sup.3)COR'', R'N(R.sup.3)CON(R.sup.4)R'', R'N(R.sup.3)C(S)R'', optionally substituted C.sub.1-C.sub.6 alkylene, optionally substituted C.sub.1-C.sub.5 heteroalkylene, optionally substituted C.sub.2-C.sub.8 alkenylene, optionally substituted C.sub.2-C.sub.8 alkynylene, optionally substituted C.sub.1-C.sub.8 alkoxy C.sub.1-C.sub.5 alkylene, optionally substituted C.sub.1-C.sub.5 haloalkylene, optionally substituted C.sub.1-C.sub.8 hydroxyalkylene, optionally substituted C.sub.4-C.sub.13 fused carbocyclyl, optionally substituted 5-13 membered fused heterocyclyl, optionally substituted C.sub.5-C.sub.13 bridged carbocyclyl, optionally substituted 5-13 membered bridged heterocyclyl, optionally substituted C.sub.5-C.sub.13 spiro carbocyclyl, optionally substituted 5-13 membered spiro heterocyclyl, optionally substituted 3-10 membered carbocyclyl, optionally substituted 3-10 membered heterocyclyl, optionally substituted aryl, and optionally substituted heteroaryl, wherein.

(354) R' and R'' are independently selected from null, optionally substituted (C.sub.1-C.sub.6 alkylene)-R' (preferably, CH.sub.2—R.sup.r), optionally substituted R.sup.r—(C.sub.1-C.sub.8 alkylene), optionally substituted (C.sub.1-C.sub.8 alkylene)-RT-(C.sub.1-C.sub.8 alkylene), or a moiety comprising of optionally substituted C.sub.1-C.sub.8 alkyl, optionally substituted C.sub.1-C.sub.8 heteroalkyl, optionally substituted C.sub.2-C.sub.8 alkenyl, optionally substituted C.sub.2-C.sub.8 alkynyl, optionally substituted C.sub.1-C.sub.8 hydroxyalkyl, optionally substituted C.sub.1-C.sub.6 alkoxy C.sub.1-C.sub.5 alkyl, optionally substituted C.sub.1-C.sub.8 alkylamino C.sub.1-C.sub.8 alkyl, optionally substituted C.sub.1-C.sub.5 haloalkyl, optionally substituted C.sub.1-C.sub.6 alkylene, optionally substituted C.sub.1-C.sub.5 heteroalkylene, optionally substituted C.sub.2-C.sub.8 alkenylene, optionally substituted C.sub.2-C.sub.8 alkynylene, optionally substituted C.sub.1-C.sub.6 hydroxyalkylene, optionally substituted C.sub.1-C.sub.8 alkoxy C.sub.1-C.sub.5 alkylene, optionally substituted C.sub.1-C.sub.5 alkylamino C.sub.1-C.sub.5 alkylene, optionally substituted C.sub.1-C.sub.8 haloalkylene, optionally substituted 3-10 membered carbocyclyl, optionally substituted 3-10 membered heterocyclyl, optionally substituted C.sub.4-C.sub.13 fused carbocyclyl, optionally substituted 5-13 membered fused heterocyclyl, optionally substituted C.sub.5-C.sub.13 bridged carbocyclyl,

optionally substituted 5-13 membered bridged heterocyclyl, optionally substituted C.sub.5-C.sub.13 spiro carbocyclyl, optionally substituted 5-13 membered spiro heterocyclyl, optionally substituted aryl, and optionally substituted heteroaryl; R.sup.r is selected from optionally substituted 3-10 membered carbocyclyl, optionally substituted 3-10 membered heterocyclyl, optionally substituted C.sub.4-C.sub.13 fused carbocyclyl, optionally substituted 5-13 membered fused heterocyclyl, optionally substituted C.sub.5-C.sub.15 bridged carbocyclyl, optionally substituted 5-13 membered bridged heterocyclyl, optionally substituted C.sub.5-C.sub.13 spiro carbocyclyl, optionally substituted 5-13 membered spiro heterocyclyl, optionally substituted aryl, and optionally substituted heteroaryl; R.sup.3 and R.sup.4 are independently selected from hydrogen, optionally substituted C.sub.1-C.sub.8alkyl, optionally substituted C.sub.1-C.sub.5 heteroalkyl, optionally substituted C.sub.2-C.sub.8 alkenyl, optionally substituted C.sub.2-C.sub.8 alkynyl, optionally substituted C.sub.1-C.sub.8 alkoxyalkyl, optionally substituted C.sub.1-C.sub.6 haloalkyl, optionally substituted C.sub.1-C.sub.8 hydroxyalkyl, optionally substituted C.sub.1-C.sub.8alkylaminoC.sub.1-C.sub.5alkyl, optionally substituted 3-10 membered carbocyclyl, optionally substituted 3-10 membered heterocyclyl, optionally substituted aryl, and optionally substituted heteroaryl; R' and R'', R.sup.3 and R.sup.4, R' and R.sup.3, R' and R.sup.4, R'' and R.sup.3, or R'' and R.sup.4 together with the atom to which they are connected form a 3-20 membered carbocyclyl or 3-20 membered heterocyclyl ring; each m is 0 to 15; and n is 0 to 15. (355) In another embodiment, the linker moiety is of FORMULA 9C:

(356) ##STR00316##

wherein X is selected from O, NH, and NR.sup.7; R.sup.1, R.sup.2, R.sup.3, R.sup.4, R.sup.5, and R.sup.6, at each occurrence, are independently selected from hydrogen, halogen, hydroxyl, amino, cyano, nitro, optionally substituted C.sub.1-C.sub.8alkyl, optionally substituted C.sub.1-C.sub.8 heteroalkyl, optionally substituted C.sub.2-C.sub.8 alkenyl, optionally substituted C.sub.2-C.sub.8 alkynyl, optionally substituted C.sub.1-C.sub.6 alkoxy, optionally substituted C.sub.1-C.sub.8 alkoxy C.sub.1-C.sub.8 alkyl, optionally substituted C.sub.1-C.sub.5 haloalkyl, optionally substituted C.sub.1-C.sub.5 hydroxyalkyl, optionally substituted C.sub.1-C.sub.8 alkylamino, optionally substituted C.sub.1-C.sub.6 alkylaminoC.sub.1-C.sub.6 alkyl, optionally substituted 3-10 membered carbocyclyl, optionally substituted 3-8 membered cycloalkoxy, optionally substituted 3-10 membered heterocyclyl, optionally substituted aryl, and optionally substituted heteroaryl; A and B are independently selected from null, or bivalent moiety selected from R.sup.r—R'', R'COR'', R'CO.sub.2R'', R'C(O)N(R.sup.8)R'', R'C(S)N(R.sup.3)R'', R'OR'', R'SR'', R'SOR'', R'SO.sub.2R'', R'SON(R.sup.5)R'', R'N(R.sup.8)R'', R'N(R.sup.8)COR'', R'N(R.sup.8)CON(RO)R'', R'N(R.sup.3)C(S)R'', optionally substituted C.sub.1-C.sub.6 alkylene, optionally substituted C.sub.1-C.sub.6 heteroalkylene, optionally substituted C.sub.2-C.sub.8 alkenylene, optionally substituted C.sub.2-C.sub.8 alkynylene, optionally substituted C.sub.1-C.sub.8alkoxyC.sub.1-C.sub.5alkylene, optionally substituted C.sub.1-C.sub.8 haloalkylene, optionally substituted C.sub.1-C.sub.5 hydroxyalkylene, optionally substituted C.sub.4-C.sub.13 fused carbocyclyl, optionally substituted 5-13 membered fused heterocyclyl, optionally substituted C.sub.5-C.sub.13 bridged carbocyclyl, optionally substituted 5-13 membered bridged heterocyclyl, optionally substituted C.sub.5-C.sub.13 spiro carbocyclyl, optionally substituted 5-13 membered spiro heterocyclyl, optionally substituted 3-10 membered carbocyclyl, optionally substituted 3-10 membered heterocyclyl, optionally substituted aryl, and optionally substituted heteroaryl, wherein R' and R'' are independently selected from null, optionally substituted (C.sub.1-C.sub.6 alkylene)-R.sup.r (preferably, CH.sub.2—R'), optionally substituted R.sup.r—(C.sub.1-C.sub.8 alkylene), optionally substituted (C.sub.1-C.sub.8 alkylene)-R.sup.r—(C.sub.1-C.sub.6 alkylene), or a moiety comprising of optionally substituted C.sub.1-C.sub.8 alkyl, optionally substituted C.sub.1-C.sub.8 heteroalkyl, optionally substituted C.sub.2-C.sub.8 alkenyl, optionally substituted C.sub.2-C.sub.8 alkynyl, optionally substituted C.sub.1-C.sub.8 hydroxyalkyl, optionally substituted C.sub.1-C.sub.8alkoxyC.sub.1-C.sub.5alkyl, optionally substituted C.sub.1-C.sub.8alkylaminoC.sub.1-

C.sub.8alkyl, optionally substituted C.sub.1-C.sub.8 haloalkyl, optionally substituted C.sub.1-C.sub.6 alkylene, optionally substituted C.sub.1-C.sub.5 heteroalkylene, optionally substituted C.sub.2-C.sub.8 alkenylene, optionally substituted C.sub.2-C.sub.8 alkynylene, optionally substituted C.sub.1-C.sub.5 hydroxyalkylene, optionally substituted C.sub.1-C.sub.8alkoxyC.sub.1-C.sub.5alkylene, optionally substituted C.sub.1-C.sub.8 alkylaminoC.sub.1-C.sub.5alkylene, optionally substituted C.sub.1-C.sub.8 haloalkylene, optionally substituted 3-10 membered carbocyclyl, optionally substituted 3-10 membered heterocyclyl, optionally substituted C.sub.4-C.sub.13 fused carbocyclyl, optionally substituted 5-13 membered fused heterocyclyl, optionally substituted C.sub.5-C.sub.13 bridged carbocyclyl, optionally substituted 5-13 membered bridged heterocyclyl, optionally substituted C.sub.5-C.sub.13 spiro carbocyclyl, optionally substituted 5-13 membered spiro heterocyclyl, optionally substituted aryl, and optionally substituted heteroaryl; R^{sup.r} is selected from optionally substituted 3-10 membered carbocyclyl, optionally substituted 3-10 membered heterocyclyl, optionally substituted C.sub.4-C.sub.13 fused carbocyclyl, optionally substituted 5-13 membered fused heterocyclyl, optionally substituted C.sub.5-C.sub.13 bridged carbocyclyl, optionally substituted 5-13 membered bridged heterocyclyl, optionally substituted C.sub.5-C.sub.13 spiro carbocyclyl, optionally substituted 5-13 membered spiro heterocyclyl, optionally substituted aryl, and optionally substituted heteroaryl; R^{sup.7}, R^{sup.8} and R^{sup.9} are independently selected from hydrogen, optionally substituted C.sub.1-C.sub.8alkyl, optionally substituted C.sub.1-C.sub.5 heteroalkyl, optionally substituted C.sub.2-C.sub.8 alkenyl, optionally substituted C.sub.2-C.sub.8 alkynyl, optionally substituted C.sub.1-C.sub.8 alkoxyalkyl, optionally substituted C.sub.1-C.sub.6 haloalkyl, optionally substituted C.sub.1-C.sub.8 hydroxyalkyl, optionally substituted C.sub.1-C.sub.8alkylaminoC.sub.1-C.sub.5alkyl, optionally substituted 3-10 membered carbocyclyl, optionally substituted 3-10 membered heterocyclyl, optionally substituted aryl, and optionally substituted heteroaryl; R' and R'', R^{sup.8} and R^{sup.9}, R' and R^{sup.5}, R' and R^{sup.9}, R'' and R^{sup.8}, R'' and R^{sup.9} together with the atom to which they are connected form a 3-20 membered carbocyclyl or 3-20 membered heterocyclyl ring; m, at each occurrence, is 0 to 15; n, at each occurrence, is 0 to 15; o is 0 to 15; and p is 0 to 15.

(357) In one embodiment, in FORMULA 9C, m and n is 0 or 1, and p is 0 to 15; In one embodiment, in FORMULA 9C, X is selected from O and NH; In one embodiment, in FORMULA 9C, R^{sup.1}, R^{sup.2}, R^{sup.3}, R^{sup.4}, R^{sup.5}, and Ré, are independently selected from hydrogen, and optionally substituted C.sub.1-C.sub.6 alkyl.

(358) In another embodiment, the linker moiety comprises one more rings selected from the group consisting of 3 to 13 membered rings, 4 to 13 membered fused rings, 5 to 13 membered bridged rings, and 5 to 13 membered spiro rings.

(359) In certain embodiments, the linker moiety comprises a ring selected from the group consisting of

(360) Formula C.sub.1, C.sub.2, C.sub.3, C.sub.4 and C.sub.5;

(361) ##STR00317##

wherein X' and Y' are independently selected from N and CR^{sup.b}; A^{sup.1}, B^{sup.1}, C^{sup.1} and D^{sup.1}, at each occurrence, are independently selected from null, O, CO, SO, SO.sub.2, NR^{sup.9}, and CR R^{sup.c}; A^{sup.2}, B^{sup.2}, C.sub.2, and D^{sup.2}, at each occurrence, are independently selected from N, and CR^{sup.b}; A^{sup.3}, B^{sup.3}, C.sub.3, D^{sup.3}, and E^{sup.3}, at each occurrence, are independently selected from N, O, S, NR^{sup.b}, and CR^{sup.b}; R^{sup.b} and R^{sup.c}, at each occurrence, are independently selected from hydrogen, halogen, hydroxyl, amino, cyano, nitro, optionally substituted C.sub.1-C.sub.8alkyl, optionally substituted C.sub.1-C.sub.8 heteroalkyl, optionally substituted C.sub.2-C.sub.8 alkenyl, optionally substituted C.sub.2-C.sub.8 alkynyl, optionally substituted C.sub.1-C.sub.8 alkoxy, optionally substituted C.sub.1-C.sub.8alkoxyalkyl, optionally substituted C.sub.1-C.sub.5 haloalkyl, optionally substituted C.sub.1-C.sub.8 hydroxyalkyl, optionally substituted C.sub.1-C.sub.8 alkylamino, and optionally

substituted C.sub.1-C.sub.8 alkylamino C.sub.1-C.sub.8 alkyl, optionally substituted 3-10 membered carbocyclyl, optionally substituted 3-8 membered cycloalkoxy, optionally substituted 3-10 membered carbocyclylamino, optionally substituted 3-8 membered membered heterocyclyl, optionally substituted aryl, and optionally substituted heteroaryl; or R.sup.b and R.sup.b, or R.sup.b and R.sup.c together with the atom to which they are connected form a 3-8 membered carbocyclyl or 3-8 membered heterocyclyl ring; and m.sup.1, n.sup.1, o.sup.1 and p.sup.1 are independently selected from 0, 1, 2, 3, 4 and 5.

(362) In one embodiment, A, B and W, at each occurrence, are independently selected from null, optionally substituted $-(CH_{sub.2})_{sub.0-8}-$, optionally substituted $-(CH_{sub.2})_{sub.0-8}-CO-(CH_{sub.2})_{sub.0-8}-$, optionally substituted $-(CH_{sub.2})_{sub.0-8}-NH-(CH_{sub.2})_{sub.0-8}-$, $-(CH_{sub.2})_{sub.0-8}-NH-CO-(CH_{sub.2})_{sub.0-8}-$, optionally substituted $-(CH_{sub.2})_{sub.0-8}-CO-NH-(CH_{sub.2})_{sub.0-8}-$, optionally substituted $-(CH_{sub.2})_{sub.0-3}-NH-(CH_{sub.2})_{sub.0-3}-CO-NH-(CH_{sub.2})_{sub.0-8}-$, optionally substituted $-(CH_{sub.2})_{sub.0-3}-NH-(CH_{sub.2})_{sub.1-3}-NH-CO-(CH_{sub.2})_{sub.0-8}-$, optionally substituted $-(CH_{sub.2})_{sub.0-8}-CO-NH-(CH_{sub.2})_{sub.1-3}-NH-(CH_{sub.2})_{sub.0-3}-$, optionally substituted $-(CH_{sub.2})_{sub.0-8}-R^f-(CH_{sub.2})_{sub.0-8}-$, optionally substituted $-(CH_{sub.2})_{sub.0-3}-(CO)-(CH_{sub.2})_{sub.0-3}-R^{sup.r}-(CH_{sub.2})_{sub.0-3}-$, optionally substituted $-(CH_{sub.2})_{sub.0-3}-(CO-NH)-(CH_{sub.2})_{sub.0-3}-R^{sup.r}-(CH_{sub.2})_{sub.0-3}-$, optionally substituted $-(CH_{sub.2})_{sub.0-3}-(NH-CO)-(CH_{sub.2})_{sub.0-3}-R^{sup.r}-(CH_{sub.2})_{sub.0-3}-$, and optionally substituted $-(CH_{sub.2})_{sub.0-3}-(NH)-(CH_{sub.2})_{sub.0-3}-RT-(CH_{sub.2})_{sub.0-3}-$.

(363) In one embodiment, R.sup.r is of Formula C.sub.1, C.sub.2, C.sub.3, C.sub.4 or C.sub.5.

(364) In one embodiment, R' is selected from

(365) ##STR00318## ##STR00319## ##STR00320## ##STR00321## ##STR00322## ##STR00323##

(366) In another embodiment, the length of the linker is 0 to 40 atoms.

(367) In another embodiment, the length of the linker is 0 to 20 atoms.

(368) In another embodiment, the length of the linker is 0 to 10 atoms.

(369) In another embodiment, the linker is selected from null, optionally substituted $-(CO)-(CH_{sub.2})_{sub.0-8}-$; optionally substituted $-(CH_{sub.2})_{sub.0-9}-$, optionally substituted $-(CH_{sub.2})_{sub.1-2}-(CO)-NH-(CH_{sub.2})_{sub.0-9}-$, optionally substituted $-(CH_{sub.2})_{sub.1-2}-(CO)-NH-(CH_{sub.2})_{sub.0-3}-(OCH_{sub.2}CH_{sub.2})_{sub.0-7}-$, optionally substituted $-(CH_{sub.2})_{sub.0-1}-(CO)-(CH_{sub.2})_{sub.1-3}-(OCH_{sub.2}CH_{sub.2})_{sub.1-7}-$, optionally substituted $-(CO)-(CH_{sub.2})_{sub.0-3}-(alkenylene)-(CH_{sub.2})_{sub.0-3}-$, optionally substituted $-(CO)-(CH_{sub.2})_{sub.0-3}-(alkynylene)-(CH_{sub.2})_{sub.0-3}-$, optionally substituted $-(CO)-(CH_{sub.2})_{sub.0-3}-(3-8 \text{ membered carbocyclyl})-(CH_{sub.2})_{sub.0-3}-$, optionally substituted $-(CO)-(CH_{sub.2})_{sub.0-3}-(3-8 \text{ membered heterocyclyl})-(CH_{sub.2})_{sub.0-3}-$, optionally substituted $-(CH_{sub.2})_{sub.0-3}-(alkenylene)-(CH_{sub.2})_{sub.0-3}-$, optionally substituted $-(CH_{sub.2})_{sub.0-3}-(alkynylene)-(CH_{sub.2})_{sub.0-3}-$, optionally substituted $-(CH_{sub.2})_{sub.0-3}-(3-8 \text{ membered carbocyclyl})-(CH_{sub.2})_{sub.0-3}-$, optionally substituted $-(CH_{sub.2})_{sub.0-3}-(3-8 \text{ membered heterocyclyl})-(CH_{sub.2})_{sub.0-3}-$, optionally substituted $-(CH_{sub.2})_{sub.0-8}-R^{sup.r}-(CH_{sub.2})_{sub.0-8}-$, optionally substituted $-(CH_{sub.2})_{sub.0-8}-R^{sup.r}-(CO)-(CH_{sub.2})_{sub.1-8}-$, optionally substituted $-(CH_{sub.2})_{sub.0-8}-R^{sup.r}-(CH_{sub.2})_{sub.1-2}-(CO)-NH-(CH_{sub.2})_{sub.2-9}-$, optionally substituted $-(CH_{sub.2})_{sub.0-8}-R^{sup.r}-(CH_{sub.2})_{sub.1-2}-(CO)-NH-(CH_{sub.2})_{sub.0-3}-(OCH_{sub.2}CH_{sub.2})_{sub.1-7}-$, optionally substituted $-(CH_{sub.2})_{sub.0-8}-RT-(CH_{sub.2})_{sub.1}-(CO)-(CH_{sub.2})_{sub.1-3}-(OCH_{sub.2}CH_{sub.2})_{sub.1-7}-$, optionally substituted $-(CH_{sub.2})_{sub.0-8}-RT-(CO)-(CH_{sub.2})_{sub.0-3}-(alkenylene)-(CH_{sub.2})_{sub.0-3}-$, optionally substituted $-(CH_{sub.2})_{sub.0-8}-R^{sup.r}-(CO)-(CH_{sub.2})_{sub.0-3}-(alkynylene)-(CH_{sub.2})_{sub.0-3}-$, optionally substituted $-(CH_{sub.2})_{sub.0-8}-R^{sup.r}-(CO)-(CH_{sub.2})_{sub.0-3}-(3-8 \text{ membered carbocyclyl})-$

(CH.sub.2).sub.0-3—, optionally substituted —(CH.sub.2) 0.8—R.sup.r—(CO)—
(CH.sub.2).sub.0-3-(3-8 membered heterocyclyl)-(CH.sub.2).sub.0-3—, optionally substituted-
(CH.sub.2).sub.0-8—R.sup.r—(CH.sub.2).sub.0-3-(alkenylene)-(CH.sub.2).sub.0-3—, optionally
substituted —(CH.sub.2).sub.0-8—R.sub.r—(CH.sub.2).sub.0-3-(alkynylene)-(CH.sub.2).sub.0-3
—, optionally substituted —(CH.sub.2).sub.0-2-RE-(CH.sub.2).sub.0-3~ (3-8 membered
carbocyclyl)-(CH.sub.2).sub.0-3—, and optionally substituted —(CH.sub.2).sub.0-8~RE-
(CH.sub.2).sub.0-3-(3-8 membered heterocyclyl)-(CH.sub.2).sub.0-3—.

(370) In certain embodiments, the linker is a moiety of Formula 9;

(371) ##STR00324##

wherein A and B, at each occurrence, are independently selected from null, or a bivalent moiety
selected from R.sup.r—R", R'COR", R'CO.sub.2R", R'C(O)N(R.sup.1")R", R'C(S)N(R.sup.1")R",
R'OR", R'SR", R'SOR", R'SOR", R'SO.sub.2N(R.sup.1")R", R'N(R")R", R"N(R.sup.1")COR",
R"N(R")CON(R.sup.2")R", R'N(R.sup.1")C(S)R", optionally substituted C.sub.1-C.sub.8
alkylene, optionally substituted C.sub.1-C.sub.5 heteroalkylene, optionally substituted C.sub.2-
C.sub.8 alkenylene, optionally substituted C.sub.2-C.sub.8 alkynylene, optionally substituted
C.sub.1-C.sub.8alkoxyC.sub.1-C.sub.8alkylene, optionally substituted C.sub.1-C.sub.6
haloalkylene, optionally substituted C.sub.1-C.sub.5 hydroxyalkylene; each W is independently
selected from null, optionally substituted C.sub.4-C.sub.13 fused carbocyclyl, optionally
substituted 5-13 membered fused heterocyclyl, optionally substituted C.sub.5-C.sub.13 bridged
carbocyclyl, optionally substituted 5-13 membered bridged heterocyclyl, optionally substituted
C.sub.5-C.sub.13 spiro carbocyclyl, optionally substituted 5-13 membered spiro heterocyclyl,
optionally substituted 3-10 membered carbocyclyl, optionally substituted 3-10 membered
heterocyclyl, optionally substituted aryl, and optionally substituted heteroaryl; R' and R" are
independently selected from null, optionally substituted C.sub.1-C.sub.6 alkyl, optionally
substituted C.sub.1-C.sub.5 heteroalkyl, optionally substituted C.sub.2-C.sub.8 alkenyl, optionally
substituted C.sub.2-C.sub.8 alkynyl, optionally substituted C.sub.1-C.sub.5 hydroxyalkyl,
optionally substituted C.sub.1-C.sub.8alkoxyC.sub.1-C.sub.5alkyl, optionally substituted C.sub.1-
C.sub.5alkylaminoC.sub.1-C.sub.5alkyl, optionally substituted C.sub.1-C.sub.5 haloalkyl,
optionally substituted C.sub.1-C.sub.8 alkylene, optionally substituted C.sub.1-C.sub.5
heteroalkylene, optionally substituted C.sub.2-C.sub.8 alkenylene, optionally substituted C.sub.2-
C.sub.8 alkynylene, optionally substituted C.sub.1-C.sub.8 hydroxyalkylene, optionally substituted
C.sub.1-C.sub.8alkoxyC.sub.1-C.sub.8alkylene, optionally substituted C.sub.1-
C.sub.5alkylaminoC.sub.1-C.sub.8alkylene, optionally substituted C.sub.1-C.sub.5 haloalkylene;
R.sup.1" and R.sup.2" are independently selected from hydrogen, optionally substituted C.sub.1-
C.sub.6 alkyl, optionally substituted C.sub.1-C.sub.5 heteroalkyl, optionally substituted C.sub.2-
C.sub.8 alkenyl, optionally substituted C.sub.2-C.sub.8 alkynyl, optionally substituted C.sub.1-
C.sub.6 alkoxyalkyl, optionally substituted C.sub.1-C.sub.5 haloalkyl, optionally substituted
C.sub.1-C.sub.6 hydroxyalkyl, optionally substituted C.sub.1-C.sub.8alkylaminoC.sub.1-
C.sub.5alkyl, optionally substituted 3-10 membered carbocyclyl, optionally substituted 3-10
membered heterocyclyl, optionally substituted aryl, and optionally substituted heteroaryl; and m is
1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, or 15.

(372) In certain embodiments, the length of the linker is 3 to 40 atoms.

(373) In certain embodiments, the length of the linker is 3 to 20 atoms.

(374) In certain embodiments, the length of the linker is 3 to 10 atoms.

(375) In certain embodiments, the linker is selected from optionally substituted —(CO)—
(CH.sub.2).sub.0-3—W—(CH.sub.2) 0-3-, optionally substituted —(CH.sub.2).sub.0-3—W—
(CH.sub.2).sub.0-3—, optionally substituted —(CO)—(CH.sub.2).sub.0-8—, optionally substituted
—(CH.sub.2).sub.0-9—, optionally substituted —(CH.sub.2).sub.1-2—(CO)—NH—
(CH.sub.2).sub.2-9-, optionally substituted —(CH.sub.2) 1-2-(CO)—NH—(CH.sub.2).sub.1-3—
(OCH.sub.2CH.sub.2).sub.1-7—, optionally substituted —(CH.sub.2).sub.0-1—(CO)—

(CH.sub.2).sub.1-3—(OCH.sub.2CH.sub.2).sub.1-7—, optionally substituted —(CO)—
 (CH.sub.2).sub.0-3-(alkenylene)-(CH.sub.2).sub.0-3—, optionally substituted —(CO)—
 (CH.sub.2).sub.0-3-(alkynylene)-(CH.sub.2).sub.0-3—, optionally substituted —(CH.sub.2).sub.0-3-(alkenylene)-(CH.sub.2).sub.0-3—, optionally substituted —(CH.sub.2).sub.0-3-(alkynylene)-(CH.sub.2).sub.0-3—, optionally substituted —(CH.sub.2).sub.0-3—(CO)—(CH.sub.2).sub.0-3—W—(CH.sub.2).sub.0-3—, optionally substituted —(CH.sub.2).sub.0-3—O—(CH.sub.2).sub.0-3—W—(CH.sub.2).sub.0-3—, optionally substituted —(CH.sub.2).sub.0-3—NH—(CH.sub.2).sub.0-3—W—(CH.sub.2).sub.0-3—, optionally substituted —(CH.sub.2).sub.0-3—O—(CH.sub.2).sub.0-3—W—(CH.sub.2).sub.0-3—NH—(CH.sub.2).sub.0-3—W—(CH.sub.2).sub.0-3—NH—(CH.sub.2).sub.0-3—; and W is selected from optionally substituted 3-10 membered carbocyclylene, optionally substituted 3-10 membered heterocyclylene, optionally substituted C.sub.4-C.sub.13 fused carbocyclylene, optionally substituted 5-13 membered fused heterocyclylene, optionally substituted C.sub.5-C.sub.13 bridged carbocyclylene, optionally substituted 5-13 membered bridged heterocyclylene, optionally substituted C.sub.5-C.sub.13 spiro carbocyclylene, optionally substituted 5-13 membered spiro heterocyclylene, optionally substituted arylene, and optionally substituted heteroarylene.

(376) In certain embodiments, the linker is selected from optionally substituted —(CO)—(CH.sub.2).sub.0-3-(3-8 membered carbocyclyl)-(CH.sub.2).sub.0-3—, optionally substituted —(CO)—(CH.sub.2).sub.0-3-(3-8 membered heterocyclyl)-(CH.sub.2).sub.0-3—, optionally substituted —(CH.sub.2).sub.0-3-(3-8 membered carbocyclyl)-(CH.sub.2).sub.0-3—, and optionally substituted —(CH.sub.2).sub.0-3-(3-8 membered heterocyclyl)-(CH.sub.2).sub.0-3—.

(377) In certain embodiments, W is selected from

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(379) Without wishing to be bound by any particular theory, it is contemplated herein that, in some embodiments, attaching pomalidomide or VHL-1 to either portion of the molecule can recruit the cereblon E.sub.3 ligase or VHL E.sub.3 ligase to TRK.

(380) The bivalent compounds disclosed herein can selectively affect TRK-mediated disease cells compared to WT (wild type) cells (i.e., an bivalent compound able to kill or inhibit the growth of an TRK-mediated disease cell while also having a relatively low ability to lyse or inhibit the growth of a WT cell), e.g., possess a GI.sub.50 for one or more TRK-mediated disease cells more than 1.5-fold lower, more than 2-fold lower, more than 2.5-fold lower, more than 3-fold lower, more than 4-fold lower, more than 5-fold lower, more than 6-fold lower, more than 7-fold lower, more than 8-fold lower, more than 9-fold lower, more than 10-fold lower, more than 15-fold lower, or more than 20-fold lower than its GI.sub.50 for one or more WT cells, e.g., WT cells of the same species and tissue type as the TRK-mediated disease cells.

(381) In some aspects, provided herein is a method for identifying a bivalent compound which mediates degradation or reduction of TRK, the method comprising: providing a heterobifunctional test compound comprising an TRK ligand conjugated to a degradation tag through a linker; contacting the heterobifunctional test compound with a cell comprising a ubiquitin ligase and TRK; determining whether TRK level is decreased in the cell; and identifying the heterobifunctional test compound as a bivalent compound which mediates degradation or reduction of TRK. In certain embodiments, the cell is a cancer cell. In certain embodiments, the cancer cell is a TRK-mediated cancer cell.

(382) Cross-reactivity with Protein Kinases

(383) In some aspects, the TRK ligand can be bound to TRK, TRK splicing, TRK fusion proteins, and/or TRK mutant proteins. In some aspects, the TRK ligand can be bound to ROS1, ROS1 fusion proteins, ROS1 splicing proteins and/or ROS1 mutant proteins. In some aspects, the TRK ligand

can be bound to ALK, ALK fusion proteins, ALK splicing proteins, and/or ALK mutant proteins. In some aspects, the TRK ligand can be bound to TRK, ROS1, or ALK. In some aspects, the TRK ligand can be bound to TRK or ROS1. In some aspects, the TRK ligand can be bound to TRK or ALK. In some aspects, the TRK ligand can be bound to ROS1 or ALK.

(384) In some aspect, the disclosed TRK degraders could degrade TRK, TRK splicing proteins, TRK fusion proteins, TRK mutant proteins, ALK, ALK splicing proteins, ALK fusion proteins, ALK mutant proteins, ROS1, ROS1 splicing proteins, ROS1 fusion proteins, or ROS1 mutant proteins,

(385) Synthesis and Testing of Bivalent Compounds

(386) The binding affinity of novel synthesized bivalent compounds can be assessed using standard biophysical assays known in the art (e.g., isothermal titration calorimetry (ITC), surface plasmon resonance (SPR)). Cellular assays can then be used to assess the bivalent compound's ability to induce TRK degradation and inhibit cancer cell proliferation. Besides evaluating a bivalent compound's induced changes in the protein levels of TRK, TRK mutants, TRK splicing or TRK fusion proteins, enzymatic activity can also be assessed. Assays suitable for use in any or all of these steps are known in the art, and include, e.g., western blotting, quantitative mass spectrometry (MS) analysis, flow cytometry, enzymatic activity assay, ITC, SPR, cell growth inhibition, xenograft, orthotopic, and patient-derived xenograft models. Suitable cell lines for use in any or all of these steps are known in the art and include, cancer cell lines: 1); KM12, 2); CUTO3.29, 3); MO91, 4); HEL. Suitable mouse models for use in any or all of these steps are known in the art and include subcutaneous xenograft models, orthotopic models, patient-derived xenograft models, and patient-derived orthotopic models.

(387) By way of non-limiting example, detailed synthesis protocols are described in the Examples for specific exemplary bivalent compounds.

(388) Pharmaceutically acceptable isotopic variations of the compounds disclosed herein are contemplated and can be synthesized using conventional methods known in the art or methods corresponding to those described in the Examples (substituting appropriate reagents with appropriate isotopic variations of those reagents). Specifically, an isotopic variation is a compound in which at least one atom is replaced by an atom having the same atomic number, but an atomic mass different from the atomic mass usually found in nature. Useful isotopes are known in the art and include, for example, isotopes of hydrogen, carbon, nitrogen, oxygen, phosphorus, sulfur, fluorine, and chlorine. Exemplary isotopes thus include, e.g., ²H, ³H, ¹³C, ¹⁴C, ¹⁵N, ¹⁷O, ¹⁸O, ³²P, ³⁵S, ¹⁸F, and ³⁶Cl.

(389) Isotopic variations (e.g., isotopic variations containing ²H) can provide therapeutic advantages resulting from greater metabolic stability, e.g., increased in vivo half-life or reduced dosage requirements.

(390) In addition, certain isotopic variations (particularly those containing a radioactive isotope) can be used in drug or substrate tissue distribution studies. The radioactive isotopes tritium (³H) and carbon-14 (¹⁴C) are particularly useful for this purpose in view of their ease of incorporation and ready means of detection.

(391) Pharmaceutically acceptable solvates of the compounds disclosed herein are contemplated. A solvate can be generated, e.g., by substituting a solvent used to crystallize a compound disclosed herein with an isotopic variation (e.g., D²O in place of H₂O, d₆-acetone in place of acetone, or de-DMSO in place of DMSO).

(392) Pharmaceutically acceptable fluorinated variations of the compounds disclosed herein are contemplated and can be synthesized using conventional methods known in the art or methods corresponding to those described in the Examples (substituting appropriate reagents with appropriate fluorinated variations of those reagents). Specifically, a fluorinated variation is a compound in which at least one hydrogen atom is replaced by a fluoro atom. Fluorinated variations can provide therapeutic advantages resulting from greater metabolic stability, e.g., increased in vivo

half-life or reduced dosage requirements.

(393) Pharmaceutically acceptable prodrugs of the compounds disclosed herein are contemplated and can be synthesized using conventional methods known in the art or methods corresponding to those described in the Examples (e.g., converting hydroxyl groups or carboxylic acid groups to ester groups). As used herein, a “prodrug” refers to a compound that can be converted via some chemical or physiological process (e.g., enzymatic processes and metabolic hydrolysis) to a therapeutic agent. Thus, the term “prodrug” also refers to a precursor of a biologically active compound that is pharmaceutically acceptable. A prodrug may be inactive when administered to a subject, i.e. an ester, but is converted in vivo to an active compound, for example, by hydrolysis to the free carboxylic acid or free hydroxyl. The prodrug compound often offers advantages of solubility, tissue compatibility or delayed release in an organism. The term “prodrug” is also meant to include any covalently bonded carriers, which release the active compound in vivo when such prodrug is administered to a subject. Prodrugs of an active compound may be prepared by modifying functional groups present in the active compound in such a way that the modifications are cleaved, either in routine manipulation or in vivo, to the parent active compound. Prodrugs include compounds wherein a hydroxy, amino or mercapto group is bonded to any group that, when the prodrug of the active compound is administered to a subject, cleaves to form a free hydroxy, free amino or free mercapto group, respectively. Examples of prodrugs include, but are not limited to, acetate, formate and benzoate derivatives of an alcohol or acetamide, formamide and benzamide derivatives of an amine functional group in the active compound and the like.

DEFINITION OF TERMS

(394) Unless defined otherwise, all technical and scientific terms used herein have the same meaning as is commonly understood by one of skill in the art to which this disclosure belongs.

(395) As used herein, the singular form “a”, “an” and “the” includes plural references unless the context clearly dictates otherwise.

(396) The term “C.sub.x-y” when used in conjunction with a chemical moiety, such as alkyl, haloalkyl, or heteroalkyl, is meant to include groups that contain from x to y carbons in the chain. For example, the term “C.sub.1-6alkyl” refers to substituted or unsubstituted saturated hydrocarbon groups, including straight-chain alkyl and branched-chain alkyl groups that contain from 1 to 6 carbons. The term-C.sub.x-yalkylene-refers to a substituted or unsubstituted alkylene chain with from x to y carbons in the alkylene chain. For example—C.sub.1-6alkylene—may be selected from methylene, ethylene, propylene, butylene, pentylene, and hexylene, any one of which is optionally substituted.

(397) As used herein, the terms “comprising” and “including” are used in their open, non-limiting sense.

(398) “Alkyl” refers to a straight or branched hydrocarbon chain radical consisting solely of carbon and hydrogen atoms, containing no unsaturation. An alkyl may comprise one, two, three, four, five, six, seven, eight, nine, ten, eleven, twelve, thirteen, fourteen, fifteen, or sixteen carbon atoms. In certain embodiments, an alkyl comprises one to fifteen carbon atoms (e.g., C.sub.1-C.sub.15 alkyl). In certain embodiments, an alkyl comprises one to thirteen carbon atoms (e.g., C.sub.1-C.sub.13 alkyl). In certain embodiments, an alkyl comprises one to eight carbon atoms (e.g., C.sub.1-C.sub.8 alkyl). In other embodiments, an alkyl comprises five to fifteen carbon atoms (e.g., C.sub.5-C.sub.15 alkyl). In other embodiments, an alkyl comprises five to eight carbon atoms (e.g., C.sub.5-C.sub.8 alkyl). The alkyl is attached to the rest of the molecule by a single bond, for example, methyl (Me), ethyl (Et), n-propyl, 1-methylethyl (iso-propyl), n-butyl, n-pentyl, 1,1-dimethylethyl(+butyl), pentyl, 3-methylhexyl, 2-methylhexyl, and the like.

(399) “Alkenyl” refers to a straight or branched hydrocarbon chain radical group consisting solely of carbon and hydrogen atoms, containing at least one double bond. An alkenyl may comprise two, three, four, five, six, seven, eight, nine, ten, eleven, twelve, thirteen, fourteen, fifteen, or sixteen carbon atoms. In certain embodiments, an alkenyl comprises two to twelve carbon atoms (e.g.,

C.sub.2-C.sub.12 alkenyl). In certain embodiments, an alkenyl comprises two to eight carbon atoms (e.g., C.sub.2-C.sub.8 alkenyl). In certain embodiments, an alkenyl comprises two to six carbon atoms (e.g., C.sub.2-C.sub.6 alkenyl). In other embodiments, an alkenyl comprises two to four carbon atoms (e.g., C.sub.2-C.sub.4 alkenyl). The alkenyl is attached to the rest of the molecule by a single bond, for example, ethenyl (i.e., vinyl), prop-1-enyl (i.e., allyl), but-1-enyl, pent-1-enyl, penta-1,4-dienyl, and the like.

(400) The term “allyl,” as used herein, means a $\text{—CH}_2\text{CH=CH}_2$ group.

(401) As used herein, the term “alkynyl” refers to a straight or branched hydrocarbon chain radical group consisting solely of carbon and hydrogen atoms, containing at least one triple bond. An alkynyl may comprise two, three, four, five, six, seven, eight, nine, ten, eleven, twelve, thirteen, fourteen, fifteen, or sixteen carbon atoms. In certain embodiments, an alkynyl comprises two to twelve carbon atoms (e.g., C.sub.2-C.sub.12 alkynyl). In certain embodiments, an alkynyl comprises two to eight carbon atoms (e.g., C.sub.2-C.sub.8 alkynyl). In other embodiments, an alkynyl has two to six carbon atoms (e.g., C.sub.2-C.sub.6 alkynyl). In other embodiments, an alkynyl has two to four carbon atoms (e.g., C.sub.2-C.sub.4 alkynyl). The alkynyl is attached to the rest of the molecule by a single bond. Examples of such groups include, but are not limited to, ethynyl, propynyl, 1-butylnyl, 2-butylnyl, 1-pentylnyl, 2-pentylnyl, 1-hexynyl, 2-hexynyl, 3-hexynyl, and the like.

(402) The term “alkoxy”, as used herein, means an alkyl group as defined herein which is attached to the rest of the molecule via an oxygen atom. Examples of such groups include, but are not limited to, methoxy, ethoxy, n-propyloxy, iso-propyloxy, n-butoxy, iso-butoxy, tert-butoxy, pentyloxy, hexyloxy, and the like.

(403) “Heteroalkyl” refers to a substituted or unsubstituted alkyl group which has one or more skeletal chain atoms selected from an atom other than carbon. Exemplary skeletal chain atoms selected from an atom other than carbon include, e.g., O, N, P, Si, S, or combinations thereof, wherein the nitrogen, phosphorus, and sulfur atoms may optionally be oxidized and the nitrogen heteroatom may optionally be quaternized. If given, a numerical range refers to the chain length in total. For example, a 3- to 8-membered heteroalkyl has a chain length of 3 to 8 atoms. Connection to the rest of the molecule may be through either a heteroatom or a carbon in the heteroalkyl chain. Unless stated otherwise specifically in the specification, a heteroalkyl group is optionally substituted by one or more substituents such as those substituents described herein.

(404) The term “aryl”, as used herein, “refers to a radical derived from an aromatic monocyclic or polycyclic hydrocarbon ring system by removing a hydrogen atom from a ring carbon atom. The aromatic monocyclic or polycyclic hydrocarbon ring system contains only hydrogen and carbon atoms. An aryl may comprise from six to eighteen carbon atoms, where at least one of the rings in the ring system is fully unsaturated, i.e., it contains a cyclic, delocalized $(4n+2)$ π -electron system in accordance with the Hückel theory. In certain embodiments, an aryl comprises six to fourteen carbon atoms (C.sub.6-C.sub.14 aryl). In certain embodiments, an aryl comprises six to ten carbon atoms (C.sub.6-C.sub.10 aryl). Examples of such groups include, but are not limited to, phenyl, fluorenyl and naphthyl. The terms “Ph” and “phenyl,” as used herein, mean a $\text{—C}_6\text{H}_5$ group.

(405) The term “heteroaryl”, refers to a radical derived from a 3- to 18-membered aromatic ring radical that comprises two to seventeen carbon atoms and from one to six heteroatoms selected from nitrogen, oxygen and sulfur. As used herein, the heteroaryl radical may be a monocyclic, bicyclic, tricyclic or tetracyclic ring system, wherein at least one of the rings in the ring system is fully unsaturated, i.e., it contains a cyclic, delocalized $(4n+2)$ π -electron system in accordance with the Hückel theory. Heteroaryl includes fused or bridged ring systems. In certain embodiments, a heteroaryl refers to a radical derived from a 3- to 10-membered aromatic ring radical (3-10 membered heteroaryl). In certain embodiments, a heteroaryl refers to a radical derived from 5- to 7-membered aromatic ring (5-7 membered heteroaryl). Heteroaryl includes fused or bridged ring systems. The heteroatom(s) in the heteroaryl radical is optionally oxidized. One or more nitrogen

atoms, if present, are optionally quaternized. The heteroaryl is attached to the rest of the molecule through any atom of the ring(s). Examples of such groups include, but not limited to, pyridinyl, imidazolyl, pyrimidinyl, pyrazolyl, triazolyl, pyrazinyl, tetrazolyl, furyl, thienyl, isoxazolyl, thiazolyl, oxazolyl, isothiazolyl, pyrrolyl, quinolinyl, isoquinolinyl, indolyl, benzimidazolyl, benzofuranyl, cinnolinyl, indazolyl, indoliziny, phthalazinyl, pyridazinyl, triazinyl, isoindolyl, pteridinyl, purinyl, oxadiazolyl, thiadiazolyl, furazanyl, benzofurazanyl, benzothiophenyl, benzothiazolyl, benzoxazolyl, quinazolinyl, quinoxalinyl, naphthyridinyl, furopyridinyl, and the like. In certain embodiments, an heteroaryl is attached to the rest of the molecule via a ring carbon atom. In certain embodiments, an heteroaryl is attached to the rest of the molecule via a nitrogen atom (N-attached) or a carbon atom (C-attached). For instance, a group derived from pyrrole may be pyrrol-1-yl (N-attached) or pyrrol-3-yl (C-attached). Further, a group derived from imidazole may be imidazol-1-yl (N-attached) or imidazol-3-yl (C-attached).

(406) The term “heterocyclyl” or “heterocarboyclyl”, as used herein, means a non-aromatic, monocyclic, bicyclic, tricyclic, or tetracyclic radical having a total of from 4, 5, 6, 7, 8, 9, 10, 11, 12, or 13 atoms in its ring system, and containing from 3 to 12 carbon atoms and from 1 to 4 heteroatoms each independently selected from O, S and N, and with the proviso that the ring of said group does not contain two adjacent O atoms or two adjacent S atoms. A heterocyclyl group may include fused, bridged or spirocyclic ring systems. In certain embodiments, a heterocyclyl group comprises 3 to 10 ring atoms (3-10 membered heterocyclyl). In certain embodiments, a heterocyclyl group comprises 3 to 8 ring atoms (3-8 membered heterocyclyl). In certain embodiments, a heterocyclyl group comprises 4 to 8 ring atoms (4-8 membered heterocyclyl). In certain embodiments, a heterocyclyl group comprises 3 to 6 ring atoms (3-6 membered heterocyclyl). A heterocyclyl group may contain an oxo substituent at any available atom that will result in a stable compound. For example, such a group may contain an oxo atom at an available carbon or nitrogen atom. Such a group may contain more than one oxo substituent if chemically feasible. In addition, it is to be understood that when such a heterocyclyl group contains a sulfur atom, said sulfur atom may be oxidized with one or two oxygen atoms to afford either a sulfoxide or sulfone. An example of a 4 membered heterocyclyl group is azetidiny (derived from azetidine). An example of a 5 membered cycloheteroalkyl group is pyrrolidinyl. An example of a 6 membered cycloheteroalkyl group is piperidinyl. An example of a 9 membered cycloheteroalkyl group is indolinyl. An example of a 10 membered cycloheteroalkyl group is 4H-quinolizinyl. Further examples of such heterocyclyl groups include, but are not limited to, tetrahydrofuranyl, dihydrofuranyl, tetrahydrothienyl, tetrahydropyranyl, dihydropyranyl, tetrahydrothiopyranyl, piperidino, morpholino, thiomorpholino, thioxanyl, piperazinyl, azetidiny, oxetanyl, thietanyl, homopiperidinyl, oxepanyl, thiepanyl, oxazepiny, diazepiny, thiazepiny, 1,2,3,6-tetrahydropyridinyl, 2-pyrrolinyl, 3-pyrrolinyl, indolinyl, 2H-pyranyl, 4H-pyranyl, dioxanyl, 1,3-dioxolanyl, pyrazolinyl, dithianyl, dithiolanyl, dihydropyranyl, dihydrothienyl, dihydrofuranyl, pyrazolidinyl, imidazolinyl, imidazolidinyl, 3-azabicyclo[3.1.0]hexanyl, 3-azabicyclo[4.1.0]heptanyl, 3H-indolyl, quinolizinyl, 3-oxopiperazinyl, 4-methylpiperazinyl, 4-ethylpiperazinyl, and 1-oxo-2,8, diazaspiro[4.5]dec-8-yl. A heteroaryl group may be attached to the rest of molecular via a carbon atom (C-attached) or a nitrogen atom (N-attached). For instance, a group derived from piperazine may be piperazin-1-yl (N-attached) or piperazin-2-yl (C-attached).

(407) The term “carbocyclyl” means a saturated, monocyclic, bicyclic, tricyclic, or tetracyclic radical having a total of from 4, 5, 6, 7, 8, 9, 10, 11, 12, or 13 carbon atoms in its ring system. A carbocyclyl may be fused, bridged or spirocyclic. In certain embodiments, a carbocyclyl comprises 3 to 8 carbon ring atoms (C.sub.3-C.sub.8 carbocyclyl). In certain embodiments, a carbocyclyl comprises 3 to 6 carbon ring atoms (C.sub.3-C.sub.6 carbocyclyl). Examples of such groups include, but are not limited to, cyclopropyl, cyclobutyl, cyclopentyl, cyclopentenyl, cyclohexyl, cycloheptyl, adamantyl, and the like.

(408) The term “carbocyclylene” is a bidentate radical obtained by removing a hydrogen atom from

a carbocyclyl ring as defined above. Examples of such groups include, but are not limited to, cyclopropylene, cyclobutylene, cyclopentylene, cyclopentenylene, cyclohexylene, cycloheptylene, and the like.

(409) The term “spirocyclic” as used herein has its conventional meaning, that is, any ring system containing two or more rings wherein two of the rings have one ring carbon in common. Each ring of the spirocyclic ring system, as herein defined, independently comprises 3 to 20 ring atoms. Preferably, they have 3 to 10 ring atoms. Non-limiting examples of a spirocyclic system include spiro[3.3]heptane, spiro[3.4]octane, and spiro[4.5]decane.

(410) The term cyano” refers to a —C≡N group.

(411) An “aldehyde” group refers to a —C(O) H group.

(412) An “alkoxy” group refers to both an—O-alkyl, as defined herein.

(413) An “alkoxycarbonyl” refers to a —C(O)-alkoxy, as defined herein.

(414) An “alkylaminoalkyl” group refers to an-alkyl-NR-alkyl group, as defined herein.

(415) An “alkylsulfonyl” group refers to a —SO.sub.2alkyl, as defined herein.

(416) An “amino” group refers to an optionally substituted-NH.sub.2.

(417) An “aminoalkyl” group refers to an-alkyl-amino group, as defined herein.

(418) An “aminocarbonyl” refers to a —C(O)-amino, as defined herein.

(419) An “arylalkyl” group refers to-alkylaryl, where alkyl and aryl are defined herein.

(420) An “aryloxy” group refers to both an-O-aryl and an-O-heteroaryl group, as defined herein.

(421) An “aryloxycarbonyl” refers to—C(O)-aryloxy, as defined herein.

(422) An “arylsulfonyl” group refers to a —SO.sub.2aryl, as defined herein.

(423) A “carbonyl” group refers to a —C(O)-group, as defined herein.

(424) A “carboxylic acid” group refers to a —C(O)OH group.

(425) A “cycloalkoxy” refers to a —O-carbocyclyl group, as defined herein.

(426) A “halo” or “halogen” group refers to fluorine, chlorine, bromine or iodine.

(427) A “haloalkyl” group refers to an alkyl group substituted with one or more halogen atoms.

(428) A “hydroxy” group refers to an-OH group.

(429) A “nitro” group refers to a —NO.sub.2 group.

(430) An “oxo” group refers to the =O substituent.

(431) A “trihalomethyl” group refers to a methyl substituted with three halogen atoms.

(432) The term “length” when refers to a moiety means the smallest number of carbon and/or heteroatoms from one end to the other end of the moiety. When it refers to the linker, it means the smallest number of atoms from the end connects to the TRK ligand and the end connects to the degradation tag. It applies to both situations where the linker is linear or branched, and where the linker comprises a ring system.

(433) The term “substituted,” means that the specified group or moiety bears one or more substituents independently selected from C.sub.1-C.sub.4 alkyl, aryl, heteroaryl, aryl-C.sub.1-C.sub.4 alkyl-, heteroaryl-C.sub.1-C.sub.4 alkyl-,

(434) C.sub.1-C.sub.6 haloalkyl, —OC.sub.1-C.sub.4 alkyl, —OC.sub.1-C.sub.4 alkylphenyl, —C.sub.1-C.sub.4 alkyl-OH, —OC.sub.1-C.sub.4 haloalkyl, halo, —OH, —NH.sub.2, —C.sub.1-C.sub.4 alkyl-NH.sub.2, —N(C.sub.1-C.sub.4 alkyl)(C.sub.1-C.sub.4 alkyl), —NH(C.sub.1-C.sub.4 alkyl), —N(C.sub.1-C.sub.4 alkyl)(C.sub.1-C.sub.4 alkylphenyl), —NH(C.sub.1-C.sub.4 alkylphenyl), cyano, nitro, oxo, —CO.sub.2H, —C(O)OC.sub.1-C.sub.4 alkyl, —CON(C.sub.1-C.sub.4 alkyl)(C.sub.1-C.sub.4 alkyl), —CONH(C.sub.1-C.sub.4 alkyl), —CONH.sub.2, —NHC(O)(C.sub.1-C.sub.4 alkyl), —NHC(O)(phenyl), —N(C.sub.1-C.sub.4 alkyl)C(O)(C.sub.1-C.sub.4 alkyl), —N(C.sub.1-C.sub.4 alkyl)C(O)(phenyl), —C(O) C.sub.1-C.sub.4 alkyl, —C(O) C.sub.1-C.sub.4 alkylphenyl, —C(O) C.sub.1-C.sub.4 haloalkyl, —OC(O) C.sub.1-C.sub.6 alkyl, —SO.sub.2(C.sub.1-C.sub.4 alkyl), —SO.sub.2 (phenyl), —SO.sub.2(C.sub.1-C.sub.4 haloalkyl), —SO.sub.2NH.sub.2, —SO.sub.2NH(C.sub.1-C.sub.4 alkyl), —SO.sub.2NH (phenyl), —NHSO.sub.2(C.sub.1-C.sub.4 alkyl), —NHSO.sub.2(phenyl), and —NHSO.sub.2(C.sub.1-C.sub.4

haloalkyl).

(435) The term “null” means the absence of an atom or moiety, and there is a bond between adjacent atoms in the structure.

(436) The term “optionally substituted” means that the specified group may be either unsubstituted or substituted by one or more substituents as defined herein. It is to be understood that in the compounds of the present invention when a group is said to be “unsubstituted,” or is “substituted” with fewer groups than would fill the valencies of all the atoms in the compound, the remaining valencies on such a group are filled by hydrogen. For example, if a Ce aryl group, also called “phenyl” herein, is substituted with one additional substituent, one of ordinary skill in the art would understand that such a group has 4 open positions left on carbon atoms of the Ce aryl ring (6 initial positions, minus one at which the remainder of the compound of the present invention is attached to and an additional substituent, remaining 4 positions open). In such cases, the remaining 4 carbon atoms are each bound to one hydrogen atom to fill their valencies. Similarly, if a C.sub.6 aryl group in the present compounds is said to be “disubstituted,” one of ordinary skill in the art would understand it to mean that the Ce aryl has 3 carbon atoms remaining that are unsubstituted. Those three unsubstituted carbon atoms are each bound to one hydrogen atom to fill their valencies. Unless otherwise specified, an optionally substituted radical may be a radical unsubstituted or substituted with one or more substituents selected from halogen, CN, NO.sub.2, OR.sup.m, SR.sup.m, NR.sup.nR.sup.o, COR.sup.m, CO.sub.2R.sup.m, CONR.sub.nR.sup.o, SOR[”], SO.sub.2R.sup.m, SO.sub.2NR[”]R.sup.9, NR[”]COR.sup.9, NR.sup.9C(O)NR[”]R.sup.9, NR[”]SOR.sup.9, NR.sup.nSO.sub.2R.sup.9, C.sub.1-C.sub.8 alkyl, C.sub.1-C.sub.8alkoxyC.sub.1-C.sub.5alkyl, C.sub.1-C.sub.8 haloalkyl, C.sub.1-C.sub.5 hydroxyalkyl, C.sub.1-C.sub.8alkylaminoC.sub.1-C.sub.8alkyl, C.sub.3-C.sub.7 carbocyclyl, 3-7 membered heterocyclyl, C.sub.2-C.sub.8 alkenyl, C.sub.2-C.sub.8 alkynyl, aryl, and heteroaryl, wherein R.sup.m, Re, and R.sup.o are independently selected from null, hydrogen, C.sub.1-C.sub.8 alkyl, C.sub.2-C.sub.8 alkenyl, C.sub.2-C.sub.8 alkynyl, C.sub.3-C.sub.7 carbocyclyl, 3-7 membered heterocyclyl, aryl, and heteroaryl, or Rn and R.sup.o together with the atom to which they are connected form a 3-8 membered carbocyclyl or heterocyclyl ring.

(437) As used herein, the same symbol in different FORMULA means different definition, for example, the definition of R.sup.1 in FORMULA 1 is as defined with respect to FORMULA 1 and the definition of R.sup.1 in FORMULA 6 is as defined with respect to FORMULA 6.

(438) As used herein, when m (or n or o or p) is defined by a range, for example, “m is 0 to 15” or “m=0-3” mean that m is an integer from 0 to 15 (i.e. m is 0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, or 15) or m is an integer from 0 to 3 (i.e. m is 0, 1, 2, or 3) or is any integer in the defined range.

(439) “Pharmaceutically acceptable salt” includes both acid and base addition salts. A pharmaceutically acceptable salt of any one of the bivalent compounds described herein is intended to encompass any and all pharmaceutically suitable salt forms. Preferred pharmaceutically acceptable salts of the compounds described herein are pharmaceutically acceptable acid addition salts and pharmaceutically acceptable base addition salts.

(440) “Pharmaceutically acceptable acid addition salt” refers to those salts which retain the biological effectiveness and properties of the free bases, which are not biologically or otherwise undesirable, and which are formed with inorganic acids such as hydrochloric acid, hydrobromic acid, sulfuric acid, nitric acid, phosphoric acid, hydroiodic acid, hydrofluoric acid, phosphorous acid, and the like. Also included are salts that are formed with organic acids such as aliphatic mono- and dicarboxylic acids, phenyl-substituted alkanolic acids, hydroxy alkanolic acids, alkanedioic acids, aromatic acids, aliphatic and aromatic sulfonic acids, etc. and include, for example, acetic acid, trifluoroacetic acid, propionic acid, glycolic acid, pyruvic acid, oxalic acid, maleic acid, malonic acid, succinic acid, fumaric acid, tartaric acid, citric acid, benzoic acid, cinnamic acid, mandelic acid, methanesulfonic acid, ethanesulfonic acid, p-toluenesulfonic acid, salicylic acid, and the like. Exemplary salts thus include sulfates, pyrosulfates, bisulfates, sulfites,

bisulfites, nitrates, phosphates, monohydrogenphosphates, dihydrogenphosphates, metaphosphates, pyrophosphates, chlorides, bromides, iodides, acetates, trifluoroacetates, propionates, caprylates, isobutyrate, oxalates, malonates, succinate suberates, sebacates, fumarates, maleates, mandelates, benzoates, chlorobenzoates, methylbenzoates, dinitrobenzoates, phthalates, benzenesulfonates, toluenesulfonates, phenylacetates, citrates, lactates, malates, tartrates, methanesulfonates, and the like. Also contemplated are salts of amino acids, such as arginates, gluconates, and galacturonates (see, for example, Berge S. M. et al., "Pharmaceutical Salts," *Journal of Pharmaceutical Science*, 66:1-19 (1997), which is hereby incorporated by reference in its entirety). Acid addition salts of basic compounds may be prepared by contacting the free base forms with a sufficient amount of the desired acid to produce the salt according to methods and techniques with which a skilled artisan is familiar. "Pharmaceutically acceptable base addition salt" refers to those salts that retain the biological effectiveness and properties of the free acids, which are not biologically or otherwise undesirable. These salts are prepared from addition of an inorganic base or an organic base to the free acid. Pharmaceutically acceptable base addition salts may be formed with metals or amines, such as alkali and alkaline earth metals or organic amines. Salts derived from inorganic bases include, but are not limited to, sodium, potassium, lithium, ammonium, calcium, magnesium, iron, zinc, copper, manganese, aluminum salts and the like. Salts derived from organic bases include, but are not limited to, salts of primary, secondary, and tertiary amines, substituted amines including naturally occurring substituted amines, cyclic amines and basic ion exchange resins, for example, isopropylamine, trimethylamine, diethylamine, triethylamine, tripropylamine, ethanolamine, diethanolamine, 2-dimethylaminoethanol, 2-diethylaminoethanol, dicyclohexylamine, lysine, arginine, histidine, caffeine, procaine, N,N-dibenzylethylenediamine, chloroprocaine, hydrabamine, choline, betaine, ethylenediamine, ethylenedianiline, N-methylglucamine, glucosamine, methylglucamine, theobromine, purines, piperazine, piperidine, N-ethylpiperidine, polyamine resins and the like. See Berge et al., supra.

(441) Pharmaceutical Compositions

(442) In some aspects, the compositions and methods described herein include the manufacture and use of pharmaceutical compositions and medicaments that include one or more bivalent compounds as disclosed herein. Also included are the pharmaceutical compositions themselves.

(443) In some aspects, the compositions disclosed herein can include other compounds, drugs, or agents used for the treatment of cancer. For example, in some instances, pharmaceutical compositions disclosed herein can be combined with one or more (e.g., one, two, three, four, five, or less than ten) compounds. Such additional compounds can include, e.g., conventional chemotherapeutic agents or any other cancer treatment known in the art. When co-administered, bivalent compounds disclosed herein can operate in conjunction with conventional chemotherapeutic agents or any other cancer treatment known in the art to produce mechanistically additive or synergistic therapeutic effects.

(444) In some aspects, the pH of the compositions disclosed herein can be adjusted with pharmaceutically acceptable acids, bases, or buffers to enhance the stability of the bivalent compound or its delivery form.

(445) Pharmaceutical compositions typically include a pharmaceutically acceptable excipient, adjuvant, or vehicle. As used herein, the phrase "pharmaceutically acceptable" refers to molecular entities and compositions that are generally believed to be physiologically tolerable and do not typically produce an allergic or similar untoward reaction, such as gastric upset, dizziness and the like, when administered to a human. A pharmaceutically acceptable excipient, adjuvant, or vehicle is a substance that can be administered to a patient, together with a compound of the invention, and which does not compromise the pharmacological activity thereof and is nontoxic when administered in doses sufficient to deliver a therapeutic amount of the compound. Exemplary conventional nontoxic pharmaceutically acceptable excipients, adjuvants, and vehicles include, but not limited to, saline, solvents, dispersion media, coatings, antibacterial and antifungal agents,

isotonic and absorption delaying agents, and the like, compatible with pharmaceutical administration.

(446) In particular, pharmaceutically acceptable excipients, adjuvants, and vehicles that can be used in the pharmaceutical compositions of this invention include, but are not limited to, ion exchangers, alumina, aluminum stearate, lecithin, self-emulsifying drug delivery systems (SEDDS) such as d- α -tocopherol polyethylene glycol 1000 succinate, surfactants used in pharmaceutical dosage forms such as Tweens or other similar polymeric delivery matrices, serum proteins, such as human serum albumin, buffer substances such as phosphates, glycine, sorbic acid, potassium sorbate, partial glyceride mixtures of saturated vegetable fatty acids, water, salts or electrolytes, such as protamine sulfate, disodium hydrogen phosphate, potassium hydrogen phosphate, sodium chloride, zinc salts, colloidal silica, magnesium trisilicate, polyvinyl pyrrolidone, cellulose-based substances, polyethylene glycol, sodium carboxymethylcellulose, polyacrylates, waxes, polyethylene-polyoxypropylene-block polymers, polyethylene glycol and wool fat. Cyclodextrins such as α -, β -, and γ -cyclodextrin, may also be advantageously used to enhance delivery of compounds of the formulae described herein.

(447) Depending on the dosage form selected to deliver the bivalent compounds disclosed herein, different pharmaceutically acceptable excipients, adjuvants, and vehicles may be used. In the case of tablets for oral use, pharmaceutically acceptable excipients, adjuvants, and vehicles may be used include lactose and corn starch. Lubricating agents, such as magnesium stearate, are also typically added. For oral administration in a capsule form, useful diluents include lactose and dried corn starch. When aqueous suspensions or emulsions are administered orally, the active ingredient may be suspended or dissolved in an oily phase is combined with emulsifying or suspending agents. If desired, certain sweetening, flavoring, or coloring agents can be added.

(448) As used herein, the bivalent compounds disclosed herein are defined to include pharmaceutically acceptable derivatives or prodrugs thereof. A "pharmaceutically acceptable derivative" means any pharmaceutically acceptable salt, solvate, or prodrug, e.g., carbamate, ester, phosphate ester, salt of an ester, or other derivative of a compound or agent disclosed herein, which upon administration to a recipient is capable of providing (directly or indirectly) a compound described herein, or an active metabolite or residue thereof. Particularly favored derivatives and prodrugs are those that increase the bioavailability of the compounds disclosed herein when such compounds are administered to a subject (e.g., by allowing an orally administered compound to be more readily absorbed into the blood) or which enhance delivery of the parent compound to a biological compartment (e.g., the brain or lymphatic system) relative to the parent species. Preferred prodrugs include derivatives where a group that enhances aqueous solubility or active transport through the gut membrane is appended to the structure of formulae described herein. Such derivatives are recognizable to those skilled in the art without undue experimentation.

Nevertheless, reference is made to the teaching of Burger's Medicinal Chemistry and Drug Discovery, 5th Edition, Vol. 1; Principles and Practice, which is incorporated herein by reference to the extent of teaching such derivatives.

(449) The bivalent compounds disclosed herein include pure enantiomers, mixtures of enantiomers, pure diastereoisomers, mixtures of diastereoisomers, diastereoisomeric racemates, mixtures of diastereoisomeric racemates and the meso-form and pharmaceutically acceptable salts, solvent complexes, morphological forms, or deuterated derivatives thereof. The single enantiomers or diastereoisomers, i.e., optically active forms, can be obtained by asymmetric synthesis or by resolution of the racemates. Resolution of the racemates can be accomplished, for example, by conventional methods such as crystallization in the presence of a resolving agent, or chromatography, using, for example a chiral high-pressure liquid chromatography (HPLC) column. In addition, compounds include Z- and E-forms (or cis - and trans-forms) of compounds with carbon-carbon double bonds. Where compounds described herein exist in various tautomeric forms, the term "compound" is intended to include all tautomeric forms of the compound.

(450) The bivalent compounds disclosed herein also include crystalline and amorphous forms of those compounds, including, for example, polymorphs, pseudopolymorphs, solvates (including hydrates), unsolvated polymorphs (including anhydrides), conformational polymorphs, and amorphous forms of the compounds, as well as mixtures thereof. “Crystalline form,” “polymorph,” and “novel form” may be used interchangeably herein, and are meant to include all crystalline and amorphous forms of the compound, including, for example, polymorphs, pseudopolymorphs, solvates (including hydrates), unsolvated polymorphs (including anhydrides), conformational polymorphs, and amorphous forms, as well as mixtures thereof, unless a particular crystalline or amorphous form is referred to. Similarly, “pharmaceutically acceptable salts” of the bivalent compounds also include crystalline and amorphous forms of those compounds, including, for example, polymorphs, pseudopolymorphs, solvates (including hydrates), unsolvated polymorphs (including anhydrides), conformational polymorphs, and amorphous forms of the pharmaceutically acceptable salts, as well as mixtures thereof.

(451) A “solvate” is formed by the interaction of a solvent and a compound. The term “compound” is intended to include solvates of compounds. Similarly, “pharmaceutically acceptable salts” includes solvates of pharmaceutically acceptable salts. Suitable solvates are pharmaceutically acceptable solvates, such as hydrates, including monohydrates and hemi-hydrates.

(452) In some aspects, the pharmaceutical compositions disclosed herein can include an effective amount of one or more bivalent compounds. The terms “effective amount” and “effective to treat,” as used herein, refer to an amount or a concentration of one or more compounds or a pharmaceutical composition described herein utilized for a period of time (including acute or chronic administration and periodic or continuous administration) that is effective within the context of its administration for causing an intended effect or physiological outcome (e.g., treatment or prevention of cell growth, cell proliferation, or cancer). In some aspects, pharmaceutical compositions can further include one or more additional compounds, drugs, or agents used for the treatment of cancer (e.g., conventional chemotherapeutic agents) in amounts effective for causing an intended effect or physiological outcome (e.g., treatment or prevention of cell growth, cell proliferation, or cancer).

(453) In some aspects, the pharmaceutical compositions disclosed herein can be formulated for sale in the United States, import into the United States, or export from the United States.

(454) Administration of Pharmaceutical Compositions

(455) The pharmaceutical compositions disclosed herein can be formulated or adapted for administration to a subject via any route, e.g., any route approved by the Food and Drug Administration (FDA). Exemplary methods are described in the FDA Data Standards Manual (DSM). In particular, the pharmaceutical compositions can be formulated for and administered via oral, parenteral, or transdermal delivery. The term “parenteral” as used herein includes subcutaneous, intracutaneous, intravenous, intramuscular, intraperitoneal, intra-articular, intra-arterial, intrasynovial, intrasternal, intrathecal, intralesional, and intracranial injection or infusion techniques.

(456) For example, the pharmaceutical compositions disclosed herein can be administered, e.g., topically, rectally, nasally (e.g., by inhalation spray or nebulizer), buccally, vaginally, subdermally (e.g., by injection or via an implanted reservoir), or ophthalmically.

(457) For example, pharmaceutical compositions of this invention can be orally administered in any orally acceptable dosage form including, but not limited to, capsules, tablets, emulsions and aqueous suspensions, dispersions and solutions.

(458) For example, the pharmaceutical compositions of this invention can be administered in the form of suppositories for rectal administration. These compositions can be prepared by mixing a compound of this invention with a suitable non-irritating excipient which is solid at room temperature but liquid at the rectal temperature and therefore will melt in the rectum to release the active components. Such materials include, but are not limited to, cocoa butter, beeswax, and

polyethylene glycols.

(459) For example, the pharmaceutical compositions of this invention can be administered by nasal aerosol or inhalation. Such compositions are prepared according to techniques well-known in the art of pharmaceutical formulation and can be prepared as solutions in saline, employing benzyl alcohol or other suitable preservatives, absorption promoters to enhance bioavailability, fluorocarbons, or other solubilizing or dispersing agents known in the art.

(460) For example, the pharmaceutical compositions of this invention can be administered by injection (e.g., as a solution or powder). Such compositions can be formulated according to techniques known in the art using suitable dispersing or wetting agents (such as, for example, Tween 80) and suspending agents. The sterile injectable preparation may also be a sterile injectable solution or suspension in a non-toxic parenterally acceptable diluent or solvent, e.g., as a solution in 1,3-butanediol. Among the acceptable vehicles and solvents that may be employed are mannitol, water, Ringer's solution, and isotonic sodium chloride solution. In addition, sterile, fixed oils are conventionally employed as a solvent or suspending medium. For this purpose, any bland fixed oil can be employed, including synthetic mono- or diglycerides. Fatty acids, such as oleic acid and its glyceride derivatives are useful in the preparation of injectables, as are natural pharmaceutically-acceptable oils, e.g., olive oil or castor oil, especially in their polyoxyethylated versions. These oil solutions or suspensions can also contain a long-chain alcohol diluent or dispersant, or carboxymethyl cellulose or similar dispersing agents which are commonly used in the formulation of pharmaceutically acceptable dosage forms such as emulsions and or suspensions. Other commonly used surfactants such as Tweens, Spans, or other similar emulsifying agents or bioavailability enhancers which are commonly used in the manufacture of pharmaceutically acceptable solid, liquid, or other dosage forms can also be used for the purposes of formulation.

(461) In some aspects, an effective dose of a pharmaceutical composition of this invention can include, but is not limited to, e.g., about 0.00001, 0.0001, 0.001, 0.01, 0.02, 0.03, 0.04, 0.05, 0.06, 0.07, 0.08, 0.09, 0.1, 0.15, 0.2, 0.25, 0.3, 0.35, 0.4, 0.45, 0.5, 0.55, 0.6, 0.65, 0.7, 0.75, 0.8, 0.85, 0.9, 0.95, 1, 1.25, 1.5, 1.75, 2, 2.5, 3, 4, 5, 6, 7, 8, 9, 10, 15, 20, 30, 40, 50, 60, 70, 80, 90, 100, 200, 300, 400, 500, 600, 700, 800, 900, 1000, 2500, 5000, or 10000 mg/kg/day, or according to the requirements of the particular pharmaceutical composition.

(462) When the pharmaceutical compositions disclosed herein include a combination of the bivalent compounds described herein and one or more additional compounds (e.g., one or more additional compounds, drugs, or agents used for the treatment of cancer or any other condition or disease, including conditions or diseases known to be associated with or caused by cancer), both the bivalent compounds and the additional compounds may be present at dosage levels of between about 1 to 100%, and more preferably between about 5 to 95% of the dosage normally administered in a monotherapy regimen. The additional agents can be administered separately, as part of a multiple dose regimen, from the compounds of this invention. Alternatively, those agents can be part of a single dosage form, mixed together with the compounds of this invention in a single composition.

(463) In some aspects, the pharmaceutical compositions disclosed herein can be included in a container, pack, or dispenser together with instructions for administration.

(464) Methods of Treatment

(465) The methods disclosed herein contemplate administration of an effective amount of a compound or composition to achieve the desired or stated effect. Typically, the compounds or compositions of the invention will be administered from about 1 to about 6 times per day or, alternately or in addition, as a continuous infusion. Such administration can be used as a chronic or acute therapy. The amount of active ingredient that can be combined with the carrier materials to produce a single dosage form will vary depending upon the host treated and the particular mode of administration. A typical preparation will contain from about 5% to about 95% active compound (w/w). Alternatively, such preparations can contain from about 20% to about 80% active

compound.

(466) In some aspects, provided herein are a bivalent compound described herein for preventing or treating a disease or condition.

(467) In some aspects, provided herein are a bivalent compound described herein for treating or preventing one or more diseases or conditions disclosed herein in a subject in need thereof. In certain embodiments, the disease or condition is a TRK-mediated disease or condition. In certain embodiments, the disease or condition is resulted from TRK expression, mutation, or fusion. In certain embodiments, the disease or condition comprises non-small cell lung cancer, colorectal cancer, gastric cancer, liver cancer, invasive breast cancer, lung adenocarcinoma, uterine cancer, adrenal cancer, pancreatic cancer, ovarian cancer, esophageal cancer, urinary bladder cancer, endometrial cancer, prostate cancer low-grade glioma, glioblastoma, Spitzoid cancer, soft tissue sarcoma, papillary thyroid carcinoma, head and neck squamous cell carcinoma, congenital fibrosarcoma, congenital mesoblastic nephroma, secretory breast carcinoma, mammary analogue secretory carcinoma, acute myeloid leukemia, ductal carcinoma, pulmonary neuroendocrine tumors, pheochromocytoma, and Wilms' tumor. In certain embodiments, the disease or condition comprises cancer, inflammatory diseases, acute and chronic pain, pruritus, bone-related diseases, neurodegenerative diseases, infectious diseases, and other diseases, including but not limited to neuroblastoma, prostate cancer, pancreatic cancer, melanoma, head and neck cancer, gastric carcinoma, lung carcinoma, liver cancer, uterine cancer, adrenal cancer, biliary tree cancer, intestinal cancer, colorectal cancer, ovarian cancer, lung carcinoma, small cell lung cancer, non-small cell lung cancer, gastric carcinoma, breast cancer, esophageal cancer, urinary bladder cancer, endometrial cancer, brain cancer, low-grade glioma, glioblastoma, medulloblastoma, secretory breast cancer, secretory breast carcinoma, salivary gland cancer, papillary thyroid carcinoma, ductal carcinoma, adult myeloid leukemia, acute myeloid leukemia, large cell neuroendocrine tumors, pulmonary neuroendocrine tumors, sarcomas pheochromocytoma, fibrosarcoma, congenital fibrosarcoma, congenital mesoblastic nephroma, secretory breast carcinoma, malignant fibrous histiocytoma, embryonal rhabdomyosarcoma, leiomyosarcoma, neuro-fibrosarcoma, neoplasms of the central nervous systems, osteosarcoma, synovial sarcoma, liposarcoma, alveolar soft part sarcoma, Spitzoid cancer, Wilms' tumor, lymphomas (e.g. including Hodgkin's lymphoma, lymphoplasmacytoid lymphoma, follicular lymphoma, mucosa-associated lymphoid tissue lymphoma, mantle cell lymphoma, B-lineage large cell lymphoma, Burkitt's lymphoma, and T-cell anaplastic large cell lymphoma), inflammatory lung diseases (e.g. asthma), inflammatory bowel diseases, (e.g. ulcerative colitis, Crohn's disease), inflammatory skin diseases (e.g. atopic dermatitis, eczema and psoriasis), interstitial cystitis, rhinitis, acute pain, chronic pain, cancer pain, surgical pain, inflammatory pain, neuropathic pain, nociceptive pain, pain of osteoarthritis, chronic low back pain, low back pain of osteoporosis, pain of bone fracture, pain of rheumatoid arthritis, postherpetic pain, pain of diabetic neuropathy, fibromyalgia, pain of pancreatitis, pain of interstitial cystitis, pain of endometriosis, pain of irritable bowel syndrome, migraine, pain of pulpitis, interstitial cystitis pain, painful bladder syndrome, central pain syndromes, postsurgical pain syndromes, bone and joint pain, repetitive motion pain, dental pain, myofascial pain, perioperative pain, dysmenorrhea, myofascial pain, angina pain, headache, primary hyperalgesia, secondary hyperalgesia, primary allodynia, secondary allodynia, other pain caused by central sensitization, systemic cutaneous pruritus, localized cutaneous pruritus, senile cutaneous pruritus, gestational pruritus, pruritus ani, vulvar pruritus, metastatic bone disease, treatment-induce bone loss, osteoporosis, rheumatoid arthritis, bone metastases, ankylosing spondylitis, Paget's disease, periodontal disease, osteolytic disease, multiple sclerosis, Parkinson's disease, Alzheimer's disease, Chagas disease, cachexia, anorexia, demyelination and dysmyelination. In certain embodiments, the disease or condition is a relapsed disease. In certain embodiments, the disease or condition is a relapsed cancer. In certain embodiments, the disease or condition is refractory to one or more previous treatments. Some embodiments include a method of treating any of these cancers,

comprising administering to a subject in need thereof a compound described herein. The subject may be a vertebrate subject. The subject may be a mammal such as a human. An example of a TRK-mediated cancer that may be treated with compounds described herein includes colon cancer. An example of a TRK-mediated cancer that may be treated with compounds described herein includes lymphoma. Examples of lymphoma include anaplastic lymphoma and anaplastic large cell lymphoma.

(468) In some aspects, provided herein are use of a bivalent compound in manufacture of a medicament for preventing or treating one or more diseases or conditions disclosed herein.

(469) In some aspects, the methods disclosed include the administration of a therapeutically effective amount of one or more of the compounds or compositions described herein to a subject (e.g., a mammalian subject, e.g., a human subject) who is in need of, or who has been determined to be in need of, such treatment. In some aspects, the methods disclosed include selecting a subject and administering to the subject an effective amount of one or more of the compounds or compositions described herein, and optionally repeating administration as required for the prevention or treatment of cancer.

(470) In some aspects, subject selection can include obtaining a sample from a subject (e.g., a candidate subject) and testing the sample for an indication that the subject is suitable for selection. In some aspects, the subject can be confirmed or identified, e.g. by a health care professional, as having had, having an elevated risk to have, or having a condition or disease. In some aspects, suitable subjects include, for example, subjects who have or had a condition or disease but that resolved the disease or an aspect thereof, present reduced symptoms of disease (e.g., relative to other subjects (e.g., the majority of subjects) with the same condition or disease), or that survive for extended periods of time with the condition or disease (e.g., relative to other subjects (e.g., the majority of subjects) with the same condition or disease), e.g., in an asymptomatic state (e.g., relative to other subjects (e.g., the majority of subjects) with the same condition or disease). In some aspects, exhibition of a positive immune response towards a condition or disease can be made from patient records, family history, or detecting an indication of a positive immune response. In some aspects, multiple parties can be included in subject selection. For example, a first party can obtain a sample from a candidate subject and a second party can test the sample. In some aspects, subjects can be selected or referred by a medical practitioner (e.g., a general practitioner). In some aspects, subject selection can include obtaining a sample from a selected subject and storing the sample or using the in the methods disclosed herein. Samples can include, e.g., cells or populations of cells.

(471) In some aspects, methods of treatment can include a single administration, multiple administrations, and repeating administration of one or more compounds disclosed herein as required for the prevention or treatment of the disease or condition disclosed herein (e.g., an TRK-mediated disease). In some aspects, methods of treatment can include assessing a level of disease in the subject prior to treatment, during treatment, or after treatment. In some aspects, treatment can continue until a decrease in the level of disease in the subject is detected. The treatment or administration may reduce tumor growth. The treatment or administration may reduce tumor size. Treatment of cancer may include inhibition of cancer growth or development.

(472) The term “subject,” as used herein, refers to any animal. In some instances, the subject is a mammal. In some instances, the term “subject,” as used herein, refers to a human (e.g., a man, a woman, or a child).

(473) The terms “administer,” “administering,” or “administration,” as used herein, refer to implanting, ingesting, injecting, inhaling, or otherwise absorbing a compound or composition, regardless of form. For example, the methods disclosed herein include administration of an effective amount of a compound or composition to achieve the desired or stated effect.

(474) The terms “treat,” “treating,” or “treatment,” as used herein, refer to partially or completely alleviating, inhibiting, ameliorating, or relieving the disease or condition from which the subject is

suffering. This means any manner in which one or more of the symptoms of a disease or disorder (e.g., cancer) are ameliorated or otherwise beneficially altered. As used herein, amelioration of the symptoms of a particular disorder (e.g., cancer) refers to any lessening, whether permanent or temporary, lasting or transient that can be attributed to or associated with treatment by the bivalent compounds, compositions and methods of the present invention. In some embodiments, treatment can promote or result in, for example, a decrease in the number of tumor cells (e.g., in a subject) relative to the number of tumor cells prior to treatment; a decrease in the viability (e.g., the average/mean viability) of tumor cells (e.g., in a subject) relative to the viability of tumor cells prior to treatment; a decrease in the rate of growth of tumor cells; a decrease in the rate of local or distant tumor metastasis; or reductions in one or more symptoms associated with one or more tumors in a subject relative to the subject's symptoms prior to treatment.

(475) The terms “prevent,” “preventing,” and “prevention,” as used herein, shall refer to a decrease in the occurrence of a disease or decrease in the risk of acquiring a disease or its associated symptoms in a subject. The prevention may be complete, e.g., the total absence of disease or pathological cells in a subject. The prevention may also be partial, such that the occurrence of the disease or pathological cells in a subject is less than, occurs later than, or develops more slowly than that which would have occurred without the present invention. In certain embodiments, the subject has an elevated risk of developing one or more TRK-mediated diseases. Exemplary TRK-mediated diseases that can be treated with bivalent compounds include, for example, non-small cell lung cancer, colorectal cancer, gastric cancer, liver cancer, invasive breast cancer, lung adenocarcinoma, uterine cancer, adrenal cancer, pancreatic cancer, ovarian cancer, esophageal cancer, urinary bladder cancer, endometrial cancer, prostate cancer low-grade glioma, glioblastoma, spitzoid cancer, soft tissue sarcoma, papillary thyroid carcinoma, head and neck squamous cell carcinoma, congenital fibrosarcoma, congenital mesoblastic nephroma, secretory breast carcinoma, mammary analogue secretory carcinoma, acute myeloid leukemia, ductal carcinoma, pulmonary neuroendocrine tumors, pheochromocytoma, and Wilms' tumor. Exemplary TRK-mediated diseases that can be treated with bivalent compounds include, for example, cancer, inflammatory diseases, acute and chronic pain, pruritus, bone-related diseases, neurodegenerative diseases, infectious diseases, and other diseases, including but not limited to neuroblastoma, prostate cancer, pancreatic cancer, melanoma, head and neck cancer, gastric carcinoma, lung carcinoma, liver cancer, uterine cancer, adrenal cancer, biliary tree cancer, intestinal cancer, colorectal cancer, ovarian cancer, lung carcinoma, small cell lung cancer, non-small cell lung cancer, gastric carcinoma, breast cancer, esophageal cancer, urinary bladder cancer, endometrial cancer, brain cancer, low-grade glioma, glioblastoma, medulloblastoma, secretory breast cancer, secretory breast carcinoma, salivary gland cancer, papillary thyroid carcinoma, ductal carcinoma, adult myeloid leukemia, acute myeloid leukemia, large cell neuroendocrine tumors, pulmonary neuroendocrine tumors, sarcomas, pheochromocytoma, fibrosarcoma, congenital fibrosarcoma, congenital mesoblastic nephroma, secretory breast carcinoma, malignant fibrous histiocytoma, embryonal rhabdomyosarcoma, leiomyosarcoma, neuro-fibrosarcoma, neoplasms of the central nervous systems, osteosarcoma, synovial sarcoma, liposarcoma, alveolar soft part sarcoma, Spitzoid cancer, Wilms' tumor, lymphomas (e.g. including Hodgkin's lymphoma, lymphoplasmacytoid lymphoma, follicular lymphoma, mucosa-associated lymphoid tissue lymphoma, mantle cell lymphoma, B-lineage large cell lymphoma, Burkitt's lymphoma, and T-cell anaplastic large cell lymphoma), inflammatory lung diseases (e.g. asthma), inflammatory bowel diseases, (e.g. ulcerative colitis, Crohn's disease), inflammatory skin diseases (e.g. atopic dermatitis, eczema and psoriasis), interstitial cystitis, rhinitis, acute pain, chronic pain, cancer pain, surgical pain, inflammatory pain, neuropathic pain, nociceptive pain, pain of osteoarthritis, chronic low back pain, low back pain of osteoporosis, pain of bone fracture, pain of rheumatoid arthritis, postherpetic pain, pain of diabetic neuropathy, fibromyalgia, pain of pancreatitis, pain of interstitial cystitis, pain of endometriosis, pain of irritable bowel syndrome, migraine, pain of pulpitis, interstitial cystitis pain, painful bladder syndrome,

central pain syndromes, postsurgical pain syndromes, bone and joint pain, repetitive motion pain, dental pain, myofascial pain, perioperative pain, dysmenorrhea, myofascial pain, angina pain, headache, primary hyperalgesia, secondary hyperalgesia, primary allodynia, secondary allodynia, other pain caused by central sensitization, systemic cutaneous pruritus, localized cutaneous pruritus, senile cutaneous pruritus, gestational pruritus, pruritus ani, vulvar pruritus, metastatic bone disease, treatment-induced bone loss, osteoporosis, rheumatoid arthritis, bone metastases, ankylosing spondylitis, Paget's disease, periodontal disease, osteolytic disease, multiple sclerosis, Parkinson's disease, Alzheimer's disease, Chagas disease, cachexia, anorexia, demyelination and dysmyelination.

(476) Specific dosage and treatment regimens for any particular patient will depend upon a variety of factors, including the activity of the specific compound employed, the age, body weight, general health status, sex, diet, time of administration, rate of excretion, drug combination, the severity and course of the disease, condition or symptoms, the patient's disposition to the disease, condition or symptoms, and the judgment of the treating physician.

(477) An effective amount can be administered in one or more administrations, applications or dosages. A therapeutically effective amount of a therapeutic compound (i.e., an effective dosage) depends on the therapeutic compounds selected. Moreover, treatment of a subject with a therapeutically effective amount of the compounds or compositions described herein can include a single treatment or a series of treatments. For example, effective amounts can be administered at least once. The compositions can be administered from one or more times per day to one or more times per week; including once every other day. The skilled artisan will appreciate that certain factors can influence the dosage and timing required to effectively treat a subject, including but not limited to the severity of the disease or disorder, previous treatments, the general health or age of the subject, and other diseases present.

(478) Following administration, the subject can be evaluated to detect, assess, or determine their level of disease. In some instances, treatment can continue until a change (e.g., reduction) in the level of disease in the subject is detected. Upon improvement of a patient's condition (e.g., a change (e.g., decrease) in the level of disease in the subject), a maintenance dose of a compound, or composition disclosed herein can be administered, if necessary. Subsequently, the dosage or frequency of administration, or both, can be reduced, e.g., as a function of the symptoms, to a level at which the improved condition is retained. Patients may, however, require intermittent treatment on a long-term basis upon any recurrence of disease symptoms.

(479) The present disclosure is also described and demonstrated by way of the following examples. However, the use of these and other examples anywhere in the specification is illustrative only and in no way limits the scope and meaning of the invention or of any exemplified term. Likewise, the invention is not limited to any particular preferred embodiment or aspect described herein. Indeed, many modifications and variations may be apparent to those skilled in the art upon reading this specification, and such variations can be made without departing from the invention in spirit or in scope. The invention is therefore to be limited only by the terms of the appended claims along with the full scope of equivalents to which those claims are entitled.

EXAMPLES

Example 1: 4-((2-Aminoethyl)amino)-2-(2,6-dioxopiperidin-3-yl)isoindoline-1,3-dione (Linker 1)

(480) ##STR00332##

(481) A solution of 2-(2,6-dioxopiperidin-3-yl)-4-fluoroisoindoline-1,3-dione (1.66 g, 6.0 mmol), tert-butyl (2-aminoethyl)carbamate (1.25 g, 6.6 mmol) and N,N-diisopropylethylamine (2.32 g, 18 mmol) in DMF (12 mL) was heated to 85° C. in a microwave reactor for 50 min. Three batches were combined and diluted with EtOAc (200 mL). The reaction was washed with water and brine. The separated organic layer was dried over anhydrous sodium sulfate and concentrated under reduced pressure. The resulting residue was purified by silica gel chromatography (eluted with hexanes/EtOAc=1:1) to give tert-butyl 2-((2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisoindolin-4-

yl)amino)ethyl)carbamate (1.3 g, yield: 16%) as a yellow solid. MS (ESI) $m/z=317.1$ [M-100+H].sup.+ . A solution of tert-butyl (2-((2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-4-yl)amino)ethyl)carbamate (2.0 g, 4.5 mmol) in DCM (10 mL) and TFA (5 mL) was stirred at room temperature for 2 b. The reaction was concentrated and triturated with EtOAc. The solid precipitate was filtered. And the solid was washed with MTBE, and dried to give 4-((2-aminoethyl)amino)-2-(2,6-dioxopiperidin-3-yl)isoindoline-1,3-dione as a yellow solid (Linker 1)(1.3 g, yield: 98%). .sup.1H NMR (400 MHZ, DMSO-d.sub.6) δ 11.14 (s, 1H), 7.85 (s, 3H), 7.45 (t, J=7.2 Hz, 1H), 7.19 (d, J=7.2 Hz, 1H), 7.10 (d, J=7.2 Hz, 1H), 6.84 (t, J=6.4 Hz, 1H), 5.07 (dd, J=5.2, 12.8 Hz, 1H), 3.58 (q, J=6.4 Hz, 2H), 3.00 (s, 2H), 2.94-2.85 (m, 1H), 2.62-2.50 (m, 2H), 2.05-2.00 (m, 1H). MS (ESI) $m/z=317.1$ [M+H].sup.+.

Example 2:4-((3-Aminopropyl)amino)-2-(2,6-dioxopiperidin-3-yl)isoindoline-1,3-dione (Linker 2) (482) ##STR00333##

(483) Linker 2 was synthesized following the same procedures as Linker 1 as described in Example 1. (1.2 g, yield: 11% over 2 steps). .sup.1H NMR (400 MHZ, DMSO-d.sub.6) 11.11 (s, 1H), 7.74 (s, 3H), 7.62-7.58 (m, 1H), 7.15 (d, J=8.4 Hz, 1H), 7.05 (d, J=7.2 Hz, 1H), 6.78-6.75 (m, 1H), 5.08-5.04 (m, 1H), 3.43-3.36 (m, 2H), 2.90-2.86 (m, 3H), 2.62-2.51 (m, 2H), 2.08-2.01 (m, 1H), 1.86-1.80 (m, 2H). MS (ESI) $m/z=331.1$ [M+H].sup.+.

Example 3:4-((4-Aminobutyl)amino)-2-(2,6-dioxopiperidin-3-yl)isoindoline-1,3-dione (Linker 3) (484) ##STR00334##

(485) Linker 3 was synthesized following the same procedures as Linker 1 as described in Example 1. (1.4 g, yield: 15% over 2 steps). .sup.1H NMR (400 MHZ, DMSO-d.sub.6) 11.11 (s, 1H), 7.84 (s, 3H), 7.62-7.57 (m, 1H), 7.13 (d, J=8.4 Hz, 1H), 7.04 (d, J=6.8 Hz, 1H), 6.62(s, 1H), 5.08-5.04 (m, 1H), 3.34 (s, 2H), 2.90-2.83 (m, 3H), 2.62-2.51 (m, 2H), 2.06-2.01 (m, 1H), 1.65-1.60 (m, 4H). MS (ESI) $m/z=345.1$ [M+H].sup.+.

Example 4:4-((5-Aminopentyl)amino)-2-(2,6-dioxopiperidin-3-yl)isoindoline-1,3-dione (Linker 4) (486) ##STR00335##

(487) Linker 4 was synthesized following the same procedures as Linker 1 as described in Example 1. (2.3 g, yield: 26% over 2 steps). .sup.1H NMR (400 MHZ, DMSO-d.sub.6) δ 11.14 (s, 1H), 7.72(s, 3H), 7.61-7.57 (m, 1H), 7.10 (d, J=8.4 Hz, 1H), 7.03 (d, J=7.2 Hz, 1H), 6.56-6.53 (m, 1H), 5.07-5.03 (m, 1H), 3.32-3.28 (m, 2H), 2.90-2.78 (m, 3H), 2.62-2.51 (m, 2H), 2.05-1.90 (m, 1H), 1.62-1.54 (m, 4H), 1.41-1.37 (m, 2H). MS (ESI) $m/z=359.1$ [M+H].sup.+.

Example 5:4-((6-Aminohexyl)amino)-2-(2,6-dioxopiperidin-3-yl)isoindoline-1,3-dione (Linker 5) (488) ##STR00336##

(489) Linker 5 was synthesized following the same procedures as Linker 1 as described in Example 1. (1.8 g, yield: 20% over 2 steps). .sup.1H NMR (400 MHZ, DMSO-d.sub.6) δ 11.10 (s, 1H), 7.76 (s, 3H), 7.58 (t, J=7.2 Hz, 1H), 7.10 (d, J=8.4 Hz, 1H), 7.03 (d, J=7.2 Hz, 1H), 6.54 (t, J=6.0 Hz, 1H), 5.07-5.03 (m, 1H), 3.37-3.27 (m, 2H), 2.88-2.78 (m, 3H), 2.61-2.50 (m, 2H), 2.04-2.01 (m, 1H), 1.57-1.52(m, 4H), 1.40-1.30 (m, 4H). MS (ESI) $m/z=373.1$ [M+H].sup.+.

Example 6:4-((7-Aminoheptyl)amino)-2-(2,6-dioxopiperidin-3-yl)isoindoline-1,3-dione (Linker 6) (490) ##STR00337##

(491) Linker 6 was synthesized following the same procedures as Linker 1 as described in Example 1. (2.0 g, yield: 25% over 2 steps). .sup.1H NMR (400 MHZ, DMSO-d6) δ 11.05 (br, 1H), 7.94-7.56 (m, 4H), 7.10-7.02(m, 2H), 6.52(t, J=6.0 Hz, 1H), 5.07-5.02(m, 1H), 3.32-3.27 (m, 2H), 2.88-2.77 (m, 1H), 2.75-2.61 (m, 2H), 2.60-2.50 (m, 2H), 2.04-2.02(m, 1H), 1.59-1.50 (m, 4H), 1.35-1.30 (m, 6H). MS (ESI) $m/z=387.2$ [M+H].sup.+.

Example 7:4-((8-Amino-octyl)amino)-2-(2,6-dioxopiperidin-3-yl)isoindoline-1,3-dione (Linker 7) (492) ##STR00338##

(493) Linker 7 was synthesized following the same procedures as Linker 1 as described in Example 1. (1.1 g, yield: 18% over 2 steps). .sup.1H NMR (400 MHZ, DMSO-d6) δ 11.10 (s, 1H), 7.69-7.56 (m, 4H), 7.09 (d, J=8.4 Hz, 1H), 7.03 (d, J=6.8 Hz, 1H), 6.52(t, J=6.0 Hz, 1H), 5.07-5.03 (m, 1H),

3.34-3.26 (m, 2H), 2.89-2.85 (m, 1H), 2.76 (s, 2H), 2.61-2.56 (m, 2H), 2.04-2.00 (m, 1H), 1.59-1.49 (m, 4H), 1.35-1.27 (m, 8H). MS (ESI) m/z =401.2 [M+H].sup.+.

Example 8:4-((2-(2-Aminoethoxy)ethyl)amino)-2-(2,6-dioxopiperidin-3-yl)isoindoline-1,3-dione (Linker 8)

(494) ##STR00339##

(495) Linker 8 was synthesized following the same procedures as Linker 1 as described in Example 1. (2.0 g, yield: 23% over 2 steps). .sup.1H NMR (400 MHz, DMSO-d.sub.6) δ 10.10 (s, 1H), 7.88 (s, 3H), 7.60 (t, J=8.0 Hz, 1H), 7.17 (d, J=8.4 Hz, 1H), 7.06 (d, J=6.8 Hz, 1H), 6.40 (d, J=5.6 Hz, 1H), 5.05 (dd, J=5.2, 12.8 Hz, 1H), 3.67-3.62 (m, 4H), 3.54-3.50 (m, 2H), 3.00 (s, 2H), 2.90-2.85 (m, 1H), 2.62-2.50 (m, 2H), 2.03 (t, J=7.6 Hz, 1H). MS (ESI) m/z =361.1 [M+H].sup.+.

Example 9:4-((2-(2-(2-Aminoethoxy)ethoxy)ethyl)amino)-2-(2,6-dioxopiperidin-3-yl)isoindoline-1,3-dione (Linker 9)

(496) ##STR00340##

(497) Linker 9 was synthesized following the same procedures as Linker 1 as described in Example 1. (1.1 g, yield: 17% over 2 steps). .sup.1H NMR (400 MHz, DMSO-d.sub.6) δ 11.11 (s, 1H), 7.84 (s, 3H), 7.62-7.58 (m, 1H), 7.15 (d, J=8.8 Hz, 1H), 7.05 (d, J=6.8 Hz, 1H), 6.62-6.59 (m, 1H), 5.08-5.04 (m, 1H), 3.65-3.59 (m, 8H), 3.50-3.46 (m, 2H), 2.97-2.86 (m, 3H), 2.62-2.51 (m, 2H), 2.05-1.99 (m, 1H). MS (ESI) m/z =405.2 [M+H].sup.+.

Example 10:4-((2-(2-(2-(2-Aminoethoxy)ethoxy)ethoxy)ethyl)amino)-2-(2,6-dioxopiperidin-3-yl)isoindoline-1,3-dione (Linker 10)

(498) ##STR00341##

(499) Linker 10 was synthesized following the same procedures as Linker 1 as described in Example 1. (1.3 g, yield: 17% over 2 steps). .sup.1H NMR (400 MHz, DMSO-d.sub.6) δ 11.11 (s, 1H), 7.83 (s, 3H), 7.61-7.57 (m, 1H), 7.15 (d, J=8.8 Hz, 1H), 7.05 (d, J=6.8 Hz, 1H), 6.62-6.59 (m, 1H), 5.08-5.04 (m, 1H), 3.64-3.45 (m, 14H), 2.97-2.86 (m, 3H), 2.62-2.51 (m, 2H), 2.08-2.01 (m, 1H). MS (ESI) m/z =449.2 [M+H].sup.+.

Example 11:4-((14-Amino-3,6,9,12-tetraoxatetradecyl)amino)-2-(2,6-dioxopiperidin-3-yl)isoindoline-1,3-dione (Linker 11)

(500) ##STR00342##

(501) Linker 11 was synthesized following the same procedures as Linker 1 as described in Example 1. (1.2 g, yield: 16% over 2 steps). .sup.1H NMR (400 MHz, DMSO-d₆) δ 11.11 (s, 1H), 7.84 (s, 3H), 7.61-7.57 (m, 1H), 7.15 (d, J=8.8 Hz, 1H), 7.05 (d, J=6.8 Hz, 1H), 6.61 (s, 1H), 5.08-5.04 (m, 1H), 3.64-3.47 (m, 18H), 2.99-2.86 (m, 3H), 2.62-2.51 (m, 2H), 2.08-2.01 (m, 1H). MS (ESI) m/z =493.2 [M+H].sup.+.

Example 12:4-((17-Amino-3,6,9,12,15-pentaoxaheptadecyl)amino)-2-(2,6-dioxopiperidin-3-yl)isoindoline-1,3-dione (Linker 12)

(502) ##STR00343##

(503) Linker 12 was synthesized following the same procedures as Linker 1 as described in Example 1. (1.2 g, yield: 15% over 2 steps). .sup.1H NMR (400 MHz, DMSO-d₆) δ 11.11 (s, 1H), 7.82 (s, 3H), 7.61-7.57 (m, 1H), 7.15 (d, J=8.4 Hz, 1H), 7.05 (d, J=7.2 Hz, 1H), 6.61-6.59 (m, 1H), 5.08-5.03 (m, 1H), 3.64-3.47 (m, 22H), 3.00-2.86 (m, 3H), 2.62-2.51 (m, 2H), 2.05-2.02 (m, 1H). MS (ESI) m/z =537.2 [M+H].sup.+.

Example 13: (2-(2,6-Dioxopiperidin-3-yl)-1,3-dioxisoindolin-4-yl)glycine (Linker 13)

(504) ##STR00344##

(505) Linker 13 was synthesized following the same procedures as Linker 1 as described in Example 1. (840 mg, yield: 16% over 2 steps). .sup.1H NMR (400 MHz, DMSO-d₆) δ 11.07 (s, 1H), 7.52 (t, J=7.6 Hz, 1H), 6.99-6.88 (m, 3H), 5.04 (dd, J=5.2, 12.8 Hz, 1H), 3.73 (s, 2H), 2.93-2.83 (m, 1H), 2.61-2.50 (m, 2H), 2.02 (t, J=5.6 Hz, 1H). MS (ESI) m/z =330.1 [M-H].sup.-.

Example 14:3-((2-(2,6-Dioxopiperidin-3-yl)-1,3-dioxisoindolin-4-yl)amino)propanoic acid (Linker 14)

(506) ##STR00345##

(507) Linker 14 was synthesized following the same procedures as Linker 1 as described in Example 1. (1.42 g, yield: 24% over 2 steps). ¹H NMR (400 MHz, DMSO-d₆) δ 11.61 (br, 1H), 11.08 (s, 1H), 7.58 (dd, J=7.2, 8.8 Hz, 1H), 7.15 (d, J=8.8 Hz, 1H), 7.04 (d, J=7.2 Hz, 1H), 6.64 (s, 1H), 5.05 (dd, J=5.2, 12.8 Hz, 1H), 3.53 (t, J=6.4 Hz, 2H), 2.92-2.83 (m, 1H), 2.61-2.50 (m, 4H), 2.05-2.00 (m, 1H). MS (ESI) m/z=346.1 [M+H]⁺.

Example 15: 4-((2-(2,6-Dioxopiperidin-3-yl)-1,3-dioxoisindolin-4-yl)amino)butanoic acid (Linker 15)

(508) ##STR00346##

(509) Linker 15 was synthesized following the same procedures as Linker 1 as described in Example 1. (1.27 g, yield: 13% over 2 steps), ¹H NMR (400 MHz, DMSO-d₆) δ 12.12 (br, 1H), 11.08 (s, 1H), 7.58 (dd, J=7.2, 8.8 Hz, 1H), 7.13 (d, J=8.8 Hz, 1H), 7.03 (d, J=7.2 Hz, 1H), 6.64 (t, J=6.0 Hz, 1H), 5.05 (dd, J=5.6, 12.8 Hz, 1H), 3.33 (q, J=6.8 Hz, 2H), 2.93-2.83 (m, 1H), 2.61-2.50 (m, 2H), 2.31 (t, J=6.8 Hz, 2H), 2.07-2.00 (m, 1H), 1.83-1.75 (m, 2H). MS (ESI) m/z=360.1 [M+H]⁺.

Example 16: 5-((2-(2,6-Dioxopiperidin-3-yl)-1,3-dioxoisindolin-4-yl)amino)pentanoic acid (Linker 16)

(510) ##STR00347##

(511) Linker 16 was synthesized following the same procedures as Linker 1 as described in Example 1. (1.4 g, yield: 15% over 2 steps). ¹H NMR (400 MHz, DMSO-d₆) δ 12.02 (br, 1H), 11.08 (s, 1H), 7.58 (dd, J=8.8, 7.2 Hz, 1H), 7.10 (d, J=8.4 Hz, 1H), 7.02 (d, J=7.2 Hz, 1H), 6.64 (t, J=5.6 Hz, 1H), 5.07-5.03 (m, 1H), 3.32-3.02 (m, 2H), 2.93-2.84 (m, 1H), 2.61-2.54 (m, 2H), 2.28-2.25 (m, 2H), 2.05-2.01 (m, 1H), 1.60-1.51 (m, 4H). MS (ESI) m/z=374.1 [M+H]⁺.

Example 17: 6-((2-(2,6-Dioxopiperidin-3-yl)-1,3-dioxoisindolin-4-yl)amino)hexanoic acid (Linker 17)

(512) ##STR00348##

(513) Linker 17 was synthesized following the same procedures as Linker 1 as described in Example 1. (1.43 g, yield: 18% over 2 steps). ¹H NMR (400 MHz, DMSO-d₆) δ 11.97 (s, 1H), 11.08 (s, 1H), 7.57 (dd, J=7.2, 8.8 Hz, 1H), 7.08 (d, J=8.8 Hz, 1H), 7.02 (d, J=7.2 Hz, 1H), 6.52 (t, J=6.0 Hz, 1H), 5.05 (dd, J=5.6, 12.8 Hz, 1H), 3.30 (q, J=6.8 Hz, 2H), 2.93-2.83 (m, 1H), 2.61-2.50 (m, 2H), 2.32 (t, J=7.2 Hz, 2H), 2.07-2.00 (m, 1H), 1.61-1.50 (m, 4H), 1.39-1.33 (m, 2H). MS (ESI) m/z=388.1 [M+H]⁺.

Example 18: 7-((2-(2,6-Dioxopiperidin-3-yl)-1,3-dioxoisindolin-4-yl)amino)heptanoic acid (Linker 18)

(514) ##STR00349##

(515) Linker 18 was synthesized following the same procedures as Linker 1 as described in Example 1. (2.3 g, yield: 24% over 2 steps). ¹H NMR (400 MHz, DMSO-d₆) δ 11.92 (br, 1H), 11.08 (s, 1H), 7.57 (t, J=8.0 Hz, 1H), 7.13 (d, J=8.8 Hz, 1H), 7.03 (d, J=6.8 Hz, 1H), 6.52 (t, J=5.6 Hz, 1H), 5.05 (dd, J=5.6, 12.8 Hz, 1H), 3.30 (q, J=6.4 Hz, 2H), 2.93-2.83 (m, 1H), 2.61-2.50 (m, 2H), 2.31 (t, J=7.2 Hz, 2H), 2.07-2.00 (m, 1H), 1.58-1.48 (m, 4H), 1.34-1.31 (m, 4H). MS (ESI) m/z=402.1 [M+H]⁺.

Example 19: 8-((2-(2,6-Dioxopiperidin-3-yl)-1,3-dioxoisindolin-4-yl)amino)octanoic acid (Linker 19)

(516) ##STR00350##

(517) Linker 19 was synthesized following the same procedures as Linker 1 as described in Example 1. (1.14 g, yield: 35% over 2 steps). ¹H NMR (400 MHz, DMSO-d₆) δ 11.94 (s, 1H), 11.08 (s, 1H), 7.57 (t, J=8.0 Hz, 1H), 7.08 (d, J=8.4 Hz, 1H), 7.02 (d, J=6.8 Hz, 1H), 6.52 (t, J=5.6 Hz, 1H), 5.05 (dd, J=5.6, 12.8 Hz, 1H), 3.31-3.26 (m, 2H), 2.93-2.83 (m, 1H), 2.61-2.50 (m, 2H), 2.19 (t, J=7.2 Hz, 2H), 2.05-2.00 (m, 1H), 1.58-1.47 (m, 4H), 1.35-1.25 (s, 6H). MS (ESI) m/z=416.1 [M+H]⁺.

Example 20:3-(2-((2-(2,6-Dioxopiperidin-3-yl)-1,3-dioxoisindolin-4-yl)amino)ethoxy) propanoic acid (Linker 20)

(518) ##STR00351##

(519) Linker 20 was synthesized following the same procedures as Linker 1 as described in Example 1. (3.5 g, yield: 18% over 2 steps). ¹H NMR (400 MHz, DMSO-d₆) δ 12.18 (s, 1H), 11.08 (s, 1H), 7.58 (dd, J=7.2 Hz, 8.8 Hz, 1H), 7.13 (d, J=8.4 Hz, 1H), 7.04 (d, J=7.2 Hz, 1H), 6.58 (t, J=5.6 Hz, 1H), 5.05 (dd, J=6.4 Hz, 12.8 Hz, 1H), 3.67-3.58 (m, 4H), 3.47-3.43 (m, 2H), 2.93-2.84 (m, 1H), 2.61-2.45 (m, 4H), 2.07-2.01 (m, 1H). MS (ESI) m/z=390.1 [M+H].⁺

Example 21:3-(2-(2-((2-(2,6-Dioxopiperidin-3-yl)-1,3-dioxoisindolin-4-yl)amino)ethoxy)ethoxy)propanoic acid (Linker 21)

(520) ##STR00352##

(521) Linker 21 was synthesized following the same procedures as Linker 1 as described in Example 1. (2.0 g, yield: 24% over 2 steps). ¹H NMR (400 MHz, DMSO-d₆) δ 12.13 (s, 1H), 11.08 (s, 1H), 7.58 (dd, J=7.2 Hz, 8.4 Hz, 1H), 7.14 (d, J=8.4 Hz, 1H), 7.04 (d, J=6.8 Hz, 1H), 6.60 (t, J=6.0 Hz, 1H), 5.05 (dd, J=5.2 Hz, 12.4 Hz, 1H), 3.63-3.44 (m, 10H), 2.88-2.85 (m, 1H), 2.61-2.49 (m, 2H), 2.44-2.41 (m, 2H), 2.04-2.01 (m, 1H). MS (ESI) m/z=434.1 [M+H].⁺

Example 22:3-(2-(2-(2-((2-(2,6-Dioxopiperidin-3-yl)-1,3-dioxoisindolin-4-yl)amino)ethoxy)ethoxy)ethoxy)propanoic acid (Linker 22)

(522) ##STR00353##

(523) Linker 22 was synthesized following the same procedures as Linker 1 as described in Example 1. (3.2 g, yield: 42% over 2 steps). ¹H NMR (400 MHz, DMSO-d₆) δ 12.14 (s, 1H), 11.08 (s, 1H), 7.58 (dd, J=7.2 Hz, 8.4 Hz, 1H), 7.14 (d, J=8.8 Hz, 1H), 7.04 (d, J=6.8 Hz, 1H), 6.60 (t, J=6.0 Hz, 1H), 5.05 (dd, J=5.2 Hz, 12.8 Hz, 1H), 3.63-3.45 (m, 14H), 2.88-2.85 (m, 1H), 2.61-2.49 (m, 2H), 2.44-2.40 (m, 2H), 2.04-2.01 (m, 1H). MS (ESI) m/z=478.2 [M+H].⁺

Example 23:1-((2-(2,6-Dioxopiperidin-3-yl)-1,3-dioxoisindolin-4-yl)amino)-3,6,9,12-tetraoxapentadecan-15-oic acid (Linker 23)

(524) ##STR00354##

(525) Linker 23 was synthesized following the same procedures as Linker I as described in Example 1. (2.3 g, yield: 31% over 2 steps). ¹H NMR (400 MHz, DMSO-d₆) δ 12.14 (s, 1H), 11.08 (s, 1H), 7.58 (dd, J=7.2 Hz, 8.8 Hz, 1H), 7.14 (d, J=8.4 Hz, 1H), 7.04 (d, J=7.2 Hz, 1H), 6.60 (t, J=6.0 Hz, 1H), 5.05 (dd, J=5.2 Hz, 12.8 Hz, 1H), 3.63-3.48 (m, 18H), 2.898-2.85 (m, 1H), 2.61-2.49 (m, 2H), 2.44-2.41 (m, 2H), 2.04-2.01 (m, 1H). MS (ESI) m/z=522.2 [M+H].⁺

Example 24:1-((2-(2,6-Dioxopiperidin-3-yl)-1,3-dioxoisindolin-4-yl)amino)-3,6,9,12,15-pentaoxaoctadecan-18-oic acid (Linker 24)

(526) ##STR00355##

(527) Linker 24 was synthesized following the same procedures as Linker 1 as described as Example 1. (2.4 g, yield: 36% over 2 steps). ¹H NMR (400 MHz, DMSO-d₆) δ 11.09 (s, 1H), 7.58 (dd, J=7.2, 8.4 Hz, 1H), 7.13 (d, J=8.4 Hz, 1H), 7.04 (d, J=7.2 Hz, 1H), 6.60 (t, J=5.6 Hz, 1H), 5.05 (dd, J=5.6, 12.8 Hz, 1H), 3.64-3.46 (m, 22H), 2.93-2.83 (m, 1H), 2.61-2.50 (m, 2H), 2.44-2.40 (m, 2H), 2.02 (t, J=6.4 Hz, 1H). MS (ESI) m/z=566.2 [M+H].⁺

Example 25: (2S,4R)-1-((S)-2-(2-Aminoacetamido)-3,3-dimethylbutanoyl)-4-hydroxy-N-(4-(4-methylthiazol-5-yl)benzyl)pyrrolidine-2-carboxamide (Linker 25)

(528) ##STR00356##

(529) Step 1: To a solution of (2S,4R)-1-((S)-2-amino-3,3-dimethylbutanoyl)-4-hydroxy-N-(4-(4-methylthiazol-5-yl)benzyl)pyrrolidine-2-carboxamide (2.00 g, 4.67 mmol), 2-((tert-butoxycarbonyl)amino) acetic acid (900 mg, 5.14 mmol) and triethylamine (TEA)(3.2 mL, 23.35 mmol) in DCM/DMF (225 mL/11 mL) were added EDCl (1.07 g, 5.60 mmol), HOBT (756 mg, 5.60 mmol) at 0° C. The mixture was stirred at room temperature for 16 hours. The mixture was poured into water and extracted with DCM. The combined organic layers were concentrated and the residue was purified by chromatography on a silica gel column (DCM/MeOH=20/1, v/v) to

give the desired product tert-butyl (2-(((S)-1-((2S,4R)-4-hydroxy-2-((4-(4-methylthiazol-5-yl)benzyl)carbamoyl)pyrrolidin-1-yl)-3,3-dimethyl-1-oxobutan-2-yl)amino)-2-oxoethyl)carbamate (1.5 g, yield: 55%). MS (ESI) m/z =588.2 [M+H].sup.+.

(530) Step 2; To a solution of tert-butyl (2-(((S)-1-((2S,4R)-4-hydroxy-2-((4-(4-methylthiazol-5-yl)benzyl)carbamoyl)pyrrolidin-1-yl)-3,3-dimethyl-1-oxobutan-2-yl)amino)-2-oxoethyl)carbamate (1.50 g, 2.56 mmol) in ethylacetate (EtOAc)(30 mL) was added HCV/EtOAc (100 mL). The mixture was stirred at room temperature for 3 hours and filtered to give the desired product which was dissolved in water (100 mL) and lyophilized to give (2S,4R)-1-((S)-2-(2-aminoacetamido)-3,3-dimethylbutanoyl)-4-hydroxy-N-(4-(4-methylthiazol-5-yl)benzyl)pyrrolidine-2-carboxamide hydrochloride (Linker 25)(1.07 g, yield: 80%). .sup.1H NMR (400 MHz, DMSO-d₆) δ 9.29 (s, 1H), 8.72(s, 1H), 8.56 (d, J=9.2 Hz, 1H), 8.26 (s, 3H), 7.38-7.47 (m, 4H), 4.61 (d, J=9.2 Hz, 1H), 4.36-4.47 (m, 3H), 4.20-4.25 (m, 1H), 3.60 -3.70 (m, 4H), 2.46 (s, 3H), 2.10-2.05 (m, 1H), 1.97-1.89 (m, 1H), 0.95 (s, 9H). MS (ESI) m/z =488.3 [M+H].sup.+.

Example 26: (2S,4R)-1-((S)-2-(3-Aminopropanamido)-3,3-dimethylbutanoyl)-4-hydroxy-N-(4-(4-methylthiazol-5-yl)benzyl)pyrrolidine-2-carboxamide (Linker 26)

(531) ##STR00357##

(532) Linker 26 was synthesized following the same procedures as Linker 25 as described in Example 25. (1.38 g, yield: 37% over 2 steps). .sup.1H NMR (400 MHz, DMSO-d₆) δ 9.36 (s, 1H), 8.68 (s, 1H), 8.26 (d, J=9.2 Hz, 1H), 8.16 (s, 3H), 7.49-7.39 (m, 4H), 4.53 (d, J=9.2 Hz, 1H), 4.47-4.35 (m, 3H), 4.24-4.19 (m, 1H), 3.69-3.60 (m, 2H), 2.94-2.93 (m, 2H), 2.64 (t, J=7.2 Hz, 2H), 2.48 (s, 3H), 2.06-2.01 (m, 1H), 1.92-1.85 (m, 1H), 0.95 (s, 9H). MS (ESI) m/z =502.3 [M+H].sup.+.

Example 27: (2S,4R)-1-((S)-2-(4-Aminobutanamido)-3,3-dimethylbutanoyl)-4-hydroxy-N-(4-(4-methylthiazol-5-yl)benzyl)pyrrolidine-2-carboxamide (Linker 27)

(533) ##STR00358##

(534) Linker 27 was synthesized following the same procedures as Linker 25 as described in Example 25. (1.38 g, yield: 46% over 2 steps). .sup.1H NMR (400 MHz, DMSO-d₆) δ 9.66 (s, 1H), 8.74 (t, J=6.0, 1H), 8.25 (s, 3H), 8.03 (d, J=9.2 Hz, 1H), 7.49-7.41 (m, 4H), 4.53 (d, J=9.2 Hz, 1H), 4.51-4.35 (m, 3H), 4.29-4.24 (m, 1H), 3.71-3.65 (m, 2H), 2.79-2.77 (m, 2H), 2.52(s, 3H), 2.45-2.27 (m, 2H), 2.12-2.07 (m, 1H), 1.94-1.80 (m, 3H), 0.94 (s, 9H). MS (ESI) m/z =516.0 [M+H].sup.+.

Example 28: (2S,4R)-1-((S)-2-(5-Aminopentanamido)-3,3-dimethylbutanoyl)-4-hydroxy-N-(4-(4-methylthiazol-5-yl)benzyl)pyrrolidine-2-carboxamide (Linker 28)

(535) ##STR00359##

(536) Linker 28 was synthesized following the same procedures as Linker 25 as described in Example 25. (1.50 g, yield: 57% over 2 steps). .sup.1H NMR (400 MHz, DMSO-d₆) δ 9.52(s, 1H), 8.73 (t, J=11.6 Hz, 1H), 8.20 (s, 3H), 7.95 (d, J=9.6 Hz, 1H), 7.43-7.50 (m, 4H), 4.55 (d, J=9.2 Hz, 1H), 4.38-4.50 (m, 3H), 4.23-4.29 (m, 1H), 3.64-3.71 (m, 2H), 2.74-2.78 (m, 2H), 2.51 (s, 3H), 2.30-2.35 (m, 1H), 2.18-2.23 (m, 1H), 2.07-2.12(m, 1H), 1.88-1.95 (m, 1H), 1.58 (d, J=4.4 Hz, 4H), 0.96 (s, 9H). MS (ESI) m/z =530.1 [M+H].sup.+.

Example 29: (2S,4R)-1-((S)-2-(6-Aminohexanamido)-3,3-dimethylbutanoyl)-4-hydroxy-N-(4-(4-methylthiazol-5-yl)benzyl)pyrrolidine-2-carboxamide (Linker 29)

(537) ##STR00360##

(538) Linker 29 was synthesized following the same procedures as Linker 25 as described in Example 25. (2.70 g, yield: 87% over 2 steps). .sup.1H NMR (400 MHz, DMSO-d₆): δ 9.36 (s, 1H), 8.69 (t, J=6.4 Hz, 1H), 8.12(brs, 3H), 7.92(d, J=9.6 Hz, 1H), 7.44 (dd, J=13.6, 8.4 Hz, 4H), 4.54 (d, J=9.6 Hz, 1H), 4.48-4.39 (m, 2H), 4.36 (brs, 1H), 4.28-4.19 (m, 1H), 3.72-3.60 (m, 2H), 2.79-2.67 (m, 2H), 2.49 (s, 3H), 2.31-2.21 (m, 1H), 2.20-2.12(m, 1H), 2.10-2.01 (m, 1H), 1.94-1.85 (m, 1H), 1.62-1.54 (m, 2H), 1.53-1.44 (m, 2H), 1.34-1.22(m, 2H), 0.94 (s, 9H). MS (ESI) m/z =544.3 [M+H].sup.+.

Example 30: (2S,4R)-1-((S)-2-(7-Aminoheptanamido)-3,3-dimethylbutanoyl)-4-hydroxy-N-(4-(4-methylthiazol-5-yl)benzyl)pyrrolidine-2-carboxamide (Linker 30)

(539) ##STR00361##

(540) Linker 30 was synthesized following the same procedures as Linker 25 as described in Example 25. (2.13 g, yield: 76% over 2 steps). ¹H NMR (400 MHz, DMSO-d₆): δ 9.45 (s, 1H), 8.70 (t, J=6.0 Hz, 1H), 8.14 (brs, 3H), 7.86 (d, J=9.2 Hz, 1H), 7.44 (dd, J=12.8, 8.4 Hz, 4H), 4.54 (d, J=9.2 Hz, 1H), 4.49-4.40 (m, 2H), 4.36 (brs, 1H), 4.29-4.20 (m, 1H), 3.71-3.61 (m, 2H), 2.78-2.67 (m, 2H), 2.50 (s, 3H), 2.31-2.22 (m, 1H), 2.21-2.13 (m, 1H), 2.11-2.03 (m, 1H), 1.95-1.85 (m, 1H), 1.60-1.44 (m, 4H), 1.35-1.18 (m, 4H), 0.94 (s, 9H). MS (ESI) m/z=558.3 [M+H]⁺.

Example 31: (2S,4R)-1-((S)-2-(8-Aminooctanamido)-3,3-dimethylbutanoyl)-4-hydroxy-N-(4-(4-methylthiazol-5-yl)benzyl)pyrrolidine-2-carboxamide (Linker 31)

(541) ##STR00362##

(542) Linker 31 was synthesized following the same procedures as Linker 25 as described in Example 25. (1.81 g, yield: 65% over 2 steps). ¹H NMR (400 MHz, DMSO-d₆): δ 9.35 (s, 1H), 8.69 (t, J=6.0 Hz, 1H), 8.11 (brs, 3H), 7.88 (d, J=9.2 Hz, 1H), 7.44 (dd, J=14.0, 8.4 Hz, 4H), 4.54 (d, J=9.6 Hz, 1H), 4.48-4.39 (m, 2H), 4.36 (brs, 1H), 4.27-4.20 (m, 1H), 3.71-3.60 (m, 2H), 2.78-2.68 (m, 2H), 2.49 (s, 3H), 2.31-2.22 (m, 1H), 2.18-2.11 (m, 1H), 2.09-2.01 (m, 1H), 1.94-1.85 (m, 1H), 1.58-1.44 (m, 4H), 1.32-1.19 (m, 6H), 0.94 (s, 9H). MS (ESI) m/z=572.3 [M+H]⁺.

Example 32: (2S,4R)-1-((S)-2-(9-Aminononanamido)-3,3-dimethylbutanoyl)-4-hydroxy-N-(4-(4-methylthiazol-5-yl)benzyl)pyrrolidine-2-carboxamide (Linker 32)

(543) ##STR00363##

(544) Linker 32 was synthesized following the same procedures as Linker 25 as described in Example 25. (2.32 g, yield: 80% over 2 steps). ¹H NMR (400 MHz, DMSO-d₆): δ 9.30 (s, 1H), 8.67 (t, J=6.4 Hz, 1H), 8.10 (brs, 3H), 7.88 (d, J=9.2 Hz, 1H), 7.43 (dd, J=14.0, 8.8 Hz, 4H), 4.55 (d, J=9.2 Hz, 1H), 4.48-4.39 (m, 2H), 4.35 (brs, 1H), 4.28-4.19 (m, 1H), 3.71-3.60 (m, 2H), 2.77-2.67 (m, 2H), 2.48 (s, 3H), 2.31-2.22 (m, 1H), 2.17-2.10 (m, 1H), 2.09-2.01 (m, 1H), 1.94-1.85 (m, 1H), 1.60-1.40 (m, 4H), 1.33-1.19 (m, 8H), 0.94 (s, 9H). MS (ESI) m/z=586.3 [M+H]⁺.

Example 33: (2S,4R)-1-((S)-2-(10-Aminodecanamido)-3,3-dimethylbutanoyl)-4-hydroxy-N-(4-(4-methylthiazol-5-yl)benzyl)pyrrolidine-2-carboxamide (Linker 33)

(545) ##STR00364##

(546) Linker 33 was synthesized following the same procedures as Linker 25 as described as Example 25. (2.29 g, yield: 77% over 2 steps). ¹H NMR (400 MHz, DMSO-d₆): δ 9.41 (s, 1H), 8.67 (t, J=6.0 Hz, 1H), 8.14 (brs, 3H), 7.85 (d, J=8.8 Hz, 1H), 7.44 (dd, J=13.6, 8.8 Hz, 4H), 4.54 (d, J=8.8 Hz, 1H), 4.48-4.39 (m, 2H), 4.36 (brs, 1H), 4.29-4.20 (m, 1H), 3.71-3.60 (m, 2H), 2.78-2.67 (m, 2H), 2.49 (s, 3H), 2.32-2.22 (m, 1H), 2.17-2.11 (m, 1H), 2.10-2.01 (m, 1H), 1.95-1.86 (m, 1H), 1.62-1.40 (m, 4H), 1.34-1.16 (m, 10H), 0.94 (s, 9H). MS (ESI) m/z=600.4 [M+H]⁺.

Example 34: (2S,4R)-1-((S)-2-(11-Aminoundecanamido)-3,3-dimethylbutanoyl)-4-hydroxy-N-(4-(4-methylthiazol-5-yl)benzyl)pyrrolidine-2-carboxamide (Linker 34)

(547) ##STR00365##

(548) Linker 34 was synthesized following the same procedures as Linker 25 as described as Example 25. (1.10 g, yield: 37% over 2 steps). ¹H NMR (400 MHz, DMSO-d₆): δ 8.99 (s, 1H), 8.61 (t, J=6.4 Hz, 1H), 7.87 (d, J=8.8 Hz, 1H), 7.41 (dd, J=17.6, 8.0 Hz, 4H), 4.55 (d, J=9.6 Hz, 1H), 4.49-4.40 (m, 2H), 4.36 (brs, 1H), 4.26-4.17 (m, 1H), 3.70-3.64 (m, 2H), 2.59-2.52 (m, 2H), 2.45 (s, 3H), 2.31-2.22 (m, 1H), 2.16-2.08 (m, 1H), 2.06-1.99 (m, 1H), 1.96-1.86 (m, 1H), 1.56-1.42 (m, 2H), 1.39-1.30 (m, 2H), 1.28-1.19 (m, 12H), 0.94 (s, 9H). MS (ESI) m/z=614.4 [M+H]⁺.

Example 35: (2S,4R)-1-((S)-2-(2-(2-Aminoethoxy) acetamido)-3,3-dimethylbutanoyl)-4-hydroxy-

N-(4-(4-methylthiazol-5-yl)benzyl)pyrrolidine-2-carboxamide (Linker 35)

(549) ##STR00366##

(550) Linker 35 was synthesized following the same procedures as Linker 25 as described in Example 25. (1.35 g, yield: 55% over 2 steps), ¹H NMR (400 MHz, DMSO-d₆) δ 9.23 (s, 1H), 8.70 (t, J=6.0 Hz, 1H), 8.35-8.14 (m, 3H), 7.78 (d, J=9.6 Hz, 1H), 7.47-7.38 (m, 4H), 4.61 (d, J=9.6 Hz, 1H), 4.49-4.34 (m, 3H), 4.30-4.21 (m, 1H), 4.09-3.99 (m, 2H), 3.75-3.58 (m, 4H), 3.06-2.94 (m, 2H), 2.48 (s, 3H), 2.13-2.03 (m, 1H), 1.95-1.85 (m, 1H), 0.95 (s, 9H). MS (ESI) m/z=532.0 [M+H]⁺.

Example 36: (2S,4R)-1-((S)-2-(3-(2-Aminoethoxy)propanamido)-3,3-dimethylbutanoyl)-4-hydroxy-N-(4-(4-methylthiazol-5-yl)benzyl)pyrrolidine-2-carboxamide (Linker 36)

(551) ##STR00367##

(552) Linker 36 was synthesized following the same procedures as Linker 25 as described in Example 25. (1.32 g, yield: 49% over 2 steps). ¹H NMR (400 MHz, DMSO-d₆) δ 8.99 (s, 1H), 8.57 (t, J=6.0 Hz, 1H), 8.03 (d, J=8 Hz, 1H), 7.85 (s, 3H), 7.43-7.37 (m, 4H), 4.57 (d, J=9.2 Hz, 1H), 4.46-4.31 (m, 3H), 4.26-4.20 (m, 1H), 3.69-3.55 (m, 6H), 3.99-2.95 (m, 2H), 2.60-2.56 (m, 1H), 2.46-2.42 (m, 4H), 2.05-2.03 (m, 1H), 1.93-1.92 (m, 1H), 0.95 (s, 9H). MS (ESI) m/z=546.0 [M+H]⁺.

Example 37: (2S,4R)-1-((S)-2-(2-(2-(2-Aminoethoxy)ethoxy) acetamido)-3,3-dimethylbutanoyl)-4-hydroxy-N-(4-(4-methylthiazol-5-yl)benzyl)pyrrolidine-2-carboxamide (Linker 37)

(553) ##STR00368##

(554) Linker 37 was synthesized following the same procedures as Linker 25 as described as Example 25. (1.2 g, yield: 49% over 2 steps). ¹H NMR (400 MHz, DMSO-d₆) δ 9.38 (s, 1H), 8.78 (t, J=6.0 Hz, 1H), 8.18 (s, 3H), 7.59-7.37 (m, 5H), 4.58 (d, J=9.6 Hz, 1H), 4.49 (t, J=8.2 Hz, 1H), 4.42-4.26 (m, 3H), 4.09-3.95 (m, 2H), 3.72-3.55 (m, 8H), 2.99-2.92 (m, 2H), 2.49 (s, 3H), 2.15-2.04 (m, 1H), 1.95-1.85 (m, 1H), 0.95 (s, 9H). MS (ESI) m/z=576.1 [M+H]⁺.

Example 38: (2S,4R)-1-((S)-2-(3~(2-(2-Aminoethoxy)ethoxy)propanamido)-3,3-dimethylbutanoyl)-4-hydroxy-N-(4-(4-methylthiazol-5-yl)benzyl)pyrrolidine-2-carboxamide (Linker 38)

(555) ##STR00369##

(556) Linker 38 was synthesized following the same procedures as Linker 25 as described as Example 25. (1.34 g, yield: 49% over 2 steps). ¹H NMR (400 MHz, DMSO-d₆) δ 9.02 (s, 1H), 8.58 (t, J=6.0 Hz, 1H), 7.94 (d, J=8 Hz, 1H), 7.82 (s, 3H), 7.42-7.30 (m, 4H), 4.58 (d, J=9.2 Hz, 1H), 4.60-4.37 (m, 3H), 4.25-4.31 (m, 1H), 3.70-3.50 (m, 10H), 3.00-2.96 (m, 2H), 2.57-2.55 (m, 1H), 2.45 (s, 3H), 2.41-2.38 (m, 1H), 2.06-2.04 (m, 1H), 1.95-1.93 (m, 1H), 0.95 (s, 9H). MS (ESI) m/z=590.1 [M+H]⁺.

Example 39: (2S,4R)-1-((S)-14-Amino-2-(tert-butyl)-4-oxo-6,9,12-trioxa-3-azatetradecanoyl)-4-hydroxy-N-(4-(4-methylthiazol-5-yl)benzyl)pyrrolidine-2-carboxamide (Linker 39)

(557) ##STR00370##

(558) Linker 39 was synthesized following the same procedures as Linker 25 as described as Example 25. (1.53 g, yield: 56% over 2 steps). ¹H NMR (400 MHz, DMSO-d₆) δ 9.01 (s, 1H), 8.59 (t, J=6.0 Hz, 1H), 7.81 (s, 3H), 7.48-7.41 (m, 5H), 4.58 (d, J=9.6 Hz, 1H), 4.47-4.26 (m, 4H), 3.99 (s, 2H), 3.70-3.58 (m, 12H), 3.0-2.96 (m, 2H), 2.46 (s, 3H), 2.11-2.06 (m, 1H), 1.95-1.88 (m, 1H), 0.96 (s, 9H). MS (ESI) m/z=621.1 [M+H]⁺.

Example 40: (2S,4R)-1-((S)-1-Amino-14-(tert-butyl)-12-oxo-3,6,9-trioxa-13-azapentadecan-15-oyl)-4-hydroxy-N-(4-(4-methylthiazol-5-yl)benzyl)pyrrolidine-2-carboxamide (Linker 40)

(559) ##STR00371##

(560) Linker 40 was synthesized following the same procedures as Linker 25 as described as Example 25. (1.52 g, yield: 51% over 2 steps). ¹H NMR (400 MHz, DMSO-d₆) δ 9.01 (s, 1H), 8.57 (t, J=6.0 Hz, 1H), 7.91 (d, J=9.2 Hz, 1H), 7.81 (s, 3H), 7.44-7.38 (m, 4H), 4.58-4.55 (m, 1H), 4.45-4.36 (m, 3H), 4.25-4.21 (m, 1H), 3.70-3.48 (m, 14H), 3.00-2.97 (m, 2H), 2.59-

2.52(m, 1H), 2.46 (s, 3H), 2.39-2.34 (m, 1H), 2.08-2.03 (m, 1H), 1.95-1.88 (m, 1H), 0.94 (s, 9H). MS (ESI) m/z=633.8 [M+H].sup.+.

Example 41: (2S,4R)-1-((S)-1-Amino-17-(tert-butyl)-15-oxo-3,6,9,12-tetraoxa-16-azaoctadecan-18-oyl)-4-hydroxy-N-(4-(4-methylthiazol-5-yl)benzyl)pyrrolidine-2-carboxamide (Linker 41) (561) ##STR00372##

(562) Linker 41 was synthesized following the same procedures as Linker 25 as described as Example 25. (1.12 g, yield: 37% over 2 steps). .sup.1H NMR (400 MHZ, DMSO-d.sub.6) δ 8.98 (s, 1H), 8.58 (t, J=5.6 Hz, 1H), 7.92(d, J=9.2 Hz, 1H), 7.44-7.38 (m, 4H), 4.56 (d, J=9.2 Hz, 1H), 4.47-4.41 (m, 2H), 4.38-4.34 (m, 1H), 4.26-4.19 (m, 1H), 3.70-3.55 (m, 5H), 3.53-3.45 (m, 14H), 3.35 (t, J=5.6 Hz, 2H), 2.64 (t, J=5.6 Hz, 2H), 2.58-2.50 (m, 1H), 2.45 (s, 3H), 2.40-2.35 (m, 1H), 2.08-2.00 (m, 1H), 1.94--1.91 (m, 1H), 0.94 (s, 9H). MS (ESI) m/z=678.1 [M+H].sup.+.

Example 42: (2S,4R)-1-((S)-1-Amino-20-(tert-butyl)-18-oxo-3,6,9,12,15-pentaoxa-19-azahenicosan-21-oyl)-4-hydroxy-N-(4-(4-methylthiazol-5-yl)benzyl)pyrrolidine-2-carboxamide (Linker 42)

(563) ##STR00373##

(564) Linker 42 was synthesized following the same procedures as Linker 25 as described as Example 25. (1.1 g, 1.52 mmol, yield: 32% over 2 steps), .sup.1H NMR (400 MHZ, DMSO-d.sub.6) δ 9.38 (s, 1H), 8.67 (t, J=16 Hz, 1H), 8.14 (br, 3H), 7.91 (d, J=9.2 Hz, 1H), 7.39-7.48 (m, 4H), 4.53 (d, J=9.2 Hz, 1H), 4.39-4.46 (m, 2H), 4.36-4.34 (m, 1H), 4.20-4.25 (m, 1H), 3.45-3.68 (m, 22H), 2.91-2.95 (m, 2H), 2.52-2.58 (m, 1H), 2.47 (s, 3H), 2.32-2.39 (m, 1H), 2.03-2.08 (m, 1H), 1.85-1.92(m, 1H), 0.92(s, 9H). MS (ESI) m/z=722.4 [M+H].sup.+.

Example 43:4—(((S)-1-((2S,4R)-4-Hydroxy-2-((4-(4-methylthiazol-5-yl)benzyl)carbamoyl)pyrrolidin-1-yl)-3,3-dimethyl-1-oxobutan-2-yl)amino)-4-oxobutanoic acid (Linker 43)

(565) ##STR00374##

(566) A mixture of (2S,4R)-1-((S)-2-amino-3,3-dimethylbutanoyl)-4-hydroxy-N-(4-(4-methylthiazol-5-yl)benzyl)pyrrolidine-2-carboxamide (1.0 g, 2.3 mmol) and succinic anhydride (465 mg, 4.65 mmol) in pyridine (5 mL) was stirred at room temperature for overnight. The mixture was concentrated. The residue was purified by flash chromatography (reversed-phase, MeCN/H.sub.2O) to give the title compound Linker 43 (1.05 g, yield: 86%). .sup.1H NMR (400 MHZ, DMSO-d.sub.6): δ 12.02(s, 1H), 8.99 (s, 1H), 8.58 (t, J=6.0 Hz, 1H), 7.96 (d, J=9.2 Hz, 1H), 7.43-7.37 (m, 4H), 5.13 (d, J=3.6 Hz, 1H), 4.53 (d, J=9.2 Hz, 1H), 4.46-4.40 (m, 2H), 4.34 (s, 1H), 4.21 (dd, J=16.0, 5.2 Hz, 1H), 3.69-3.60 (m, 2H), 2.45 (s, 3H), 2.44-2.33 (m, 4H), 2.06-2.01 (m, 1H), 1.93-1.87 (m, 1H), 0.93 (s, 9H). .sup.13C NMR (100 MHZ, DMSO-d.sub.6): δ 173.83, 171.92, 170.86, 169.56, 151.41, 147.70, 139.48, 131.15, 129.63, 128.62, 127.41, 68.87, 58.70, 56.44, 56.34, 41.65, 37.91, 35.35, 29.74, 29.25, 26.35, 15.92. MS (ESI) m/z=531.2 [M+H].sup.+.

Example 44:5—(((S)-1-((2S,4R)-4-Hydroxy-2-((4-(4-methylthiazol-5-yl)benzyl)carbamoyl)pyrrolidin-1-yl)-3,3-dimethyl-1-oxobutan-2-yl)amino)-S-oxopentanoic acid (Linker 44)

(567) ##STR00375##

(568) Linker 44 was synthesized following the same procedures as Linker 43 as described as Example 43. (1.5 g, yield: 79%). .sup.1H NMR (400 MHZ, DMSO-d.sub.6): δ 8.99 (s, 1H), 8.59 (t, J=6.0 Hz, 1H), 7.91 (d, J=9.2 Hz, 1H), 7.44-7.37 (m, 4H), 5.16 (brs, 1H), 4.54 (d, J=9.2 Hz, 1H), 4.47-4.42(m, 2H), 4.36 (S, 1H), 4.21 (dd, J=16.0, 5.2 Hz, 1H), 3.7-3.64 (m, 2H), 2.45 (s, 3H), 2.31-2.14 (m, 4H), 2.07-2.02 (m, 1H), 1.94-1.81 (m, 1H), 1.74-1.68 (m, 2H), 0.94 (s, 9H). .sup.13C NMR (100 MHZ, DMSO-d.sub.6): δ 174.18, 171.94, 171.63, 169.66, 151.41, 147.70, 139.46, 131.15, 129.61, 128.62, 127.41, 68.86, 58.69, 56.38, 41.65, 37.91, 35.16, 34.03, 33.10, 26.35, 20.89, 15.92. MS (ESI) m/z=543.2 [M-H].sup.-.

Example 45: 6—(((S)-1-(2S,4R)-4-Hydroxy-2-(4-(4-methylthiazol-5-yl)benzyl)carbamoyl)pyrrolidin-1-yl)-3,3-dimethyl-1-oxobutan-2-yl)amino)-6-oxohexanoic acid

(Linker 45)

(569) ##STR00376##

(570) Linker 45 was synthesized following the same procedures as Linker 25 as described as Example 25. (1.2 g, yield: 55% over 2 steps). ¹H NMR (400 MHz, CDCl₃) δ 8.68 (s, 1H), 7.75 (s, 1H), 7.32-7.27 (m, 5H), 4.64-4.57 (m, 3H), 4.56-4.50 (m, 1H), 4.28-4.25 (m, 1H), 4.02-3.99 (m, 1H), 3.71-3.68 (m, 1H), 2.47 (s, 3H), 2.24-2.18 (m, 6H), 1.59-1.48 (m, 4H), 0.96 (s, 9H). MS (ESI) m/z=559.3 [M+H]⁺.

Example 46:7—(((S)-1-((2S,4R)-4-Hydroxy-2-((4-(4-methylthiazol-5-yl)benzyl)carbamoyl)pyrrolidin-1-yl)-3,3-dimethyl-1-oxobutan-2-yl)amino)-7-oxoheptanoic acid (Linker 46)

(571) ##STR00377##

(572) Linker 46 was synthesized following the same procedures as Linker 45 as described as Example 45. (1.1 g, yield: 33% over 2 steps). ¹H NMR (400 MHz, CDCl₃) δ 8.67 (s, 1H), 7.56-7.55 (m, 1H), 7.34-7.30 (m, 5H), 4.68-4.59 (m, 3H), 4.59-4.51 (m, 1H), 4.25 (dd, J=4.8 Hz, 15.2 Hz, 1H), 4.06-4.03 (m, 1H), 3.70-3.68 (m, 1H), 2.46 (s, 3H), 2.31-2.11 (m, 6H), 1.55-1.51 (m, 4H), 1.29-1.24 (m, 2H), 0.94 (s, 9H). MS (ESI) m/z=573.1 [M+H]⁺.

Example 47:8—(((S)-1-((2S,4R)-4-Hydroxy-2-((4-(4-methylthiazol-5-yl)benzyl)carbamoyl)pyrrolidin-1-yl)-3,3-dimethyl-1-oxobutan-2-yl)amino)-8-oxooctanoic acid (Linker 47)

(573) ##STR00378##

(574) Linker 47 was synthesized following the same procedures as Linker 45 as described as Example 45. (1.08 g, yield: 52% over 2 steps), ¹H NMR (400 MHz, DMSO-d₆) δ 8.99 (s, 1H), 8.55 (t, J=2.4 Hz, 1H), 7.83 (d, J=9.2 Hz, 1H), 7.44-7.38 (m, 4H), 4.55 (d, J=9.6 Hz, 1H), 4.52-4.41 (m, 2H), 4.36 (s, 1H), 4.25-4.21 (m, 1H), 3.67-3.66 (m, 2H), 2.45 (s, 3H), 2.30-1.91 (m, 6H), 1.49-1.47 (m, 4H), 1.26-1.24 (m, 4H), 0.92 (s, 9H). MS (ESI) m/z=587.3 [M+H]⁺.

Example 48:9—(((S)-1-((2S,4R)-4-Hydroxy-2-((4-(4-methylthiazol-5-yl)benzyl)carbamoyl)pyrrolidin-1-yl)-3,3-dimethyl-1-oxobutan-2-yl)amino)-9-oxononanoic acid (Linker 48)

(575) ##STR00379##

(576) Linker 48 was synthesized following the same procedures as Linker 45 as described in Example 45. (1.16 g, yield: 44% over 2 steps). ¹H NMR (400 MHz, CDCl₃) δ 8.70 (s, 1H), 7.55 (s, 1H), 7.33-7.27 (m, 4H), 7.08 (d, J=8.0 Hz, 1H), 4.68-4.52 (m, 4H), 4.31-4.27 (m, 1H), 4.08-4.05 (m, 1H), 3.69-3.67 (m, 1H), 2.48 (s, 3H), 2.33-2.11 (m, 6H), 1.60-1.47 (m, 4H), 1.29-1.20 (m, 6H), 0.96 (s, 9H). MS (ESI) m/z=601.1 [M+H]⁺.

Example 49:10—(((S)-1-((2S,4R)-4-Hydroxy-2-((4-(4-methylthiazol-5-yl)benzyl)carbamoyl)pyrrolidin-1-yl)-3,3-dimethyl-1-oxobutan-2-yl)amino)-10-oxodecanoic acid (Linker 49)

(577) ##STR00380##

(578) Linker 49 was synthesized following the same procedure as Linker 45 as described as Example 45. (1.1 g, yield: 35%). ¹H NMR (400 MHz, DMSO-d₆): 8.99 (s, 1H), 8.58 (t, J=6.0 Hz, 1H), 7.85 (d, J=9.2 Hz, 1H), 7.43-7.37 (m, 4H), 4.54 (d, J=9.2 Hz, 1H), 4.47-4.41 (m, 2H), 4.35 (s, 1H), 4.21 (dd, J=16.0, 5.6 Hz, 1H), 3.69-3.63 (m, 2H), 2.45 (s, 3H), 2.29-2.09 (m, 4H), 2.03-2.01 (m, 1H), 1.94-1.88 (m, 1H), 1.47 (m, 4H), 1.24 (b, 8H), 0.94 (s, 9H). ¹³C NMR (100 MHz, DMSO-d₆): δ 172.07, 171.92, 169.69, 151.41, 147.70, 139.48, 131.14, 129.62, 128.61, 127.40, 68.84, 58.67, 56.32, 56.26, 41.64, 37.93, 35.18, 34.85, 28.62, 26.36, 25.39, 15.93. MS (ESI) m/z=615.3 [M+H]⁺.

Example 50:11—(((S)-1-((2S,4R)-4-Hydroxy-2-((4-(4-methylthiazol-5-yl)benzyl)carbamoyl)pyrrolidin-1-yl)-3,3-dimethyl-1-oxobutan-2-yl)amino)-11-oxoundecanoic acid (Linker 50)

(579) ##STR00381##

(580) Linker 50 was synthesized following the same procedure as Linker 45 as described as Example 45 (1.1 g, yield: 50%). ¹H NMR (400 MHz, DMSO-d₆): δ 8.99 (s, 1H), 8.58 (t, J=6.0 Hz, 1H), 7.85 (t, J=9.2 Hz, 1H), 7.37-7.43 (m, 4H), 4.56-4.19 (m, 5H), 3.70-3.60 (m, 2H), 2.45 (s, 3H), 2.27-1.90 (m, 6H), 1.49-1.45 (m, 4H), 1.23 (m, 10H), 0.93 (s, 9H). ¹³C NMR (100 MHz, DMSO-d₆): δ 174.59, 172.07, 171.92, 169.69, 151.42, 147.70, 139.49, 131.14, 129.62, 128.61, 127.41, 68.84, 58.67, 56.32, 56.25, 41.64, 37.93, 35.19, 34.85, 33.80, 28.82, 28.70, 28.68, 28.62, 28.55, 26.37, 25.42, 24.55, 15.93. MS (ESI) m/z=629.4 [M+H]⁺.

Example 51: 3-(3-(((S)-1-((2S,4R)-4-Hydroxy-2-((4-(4-methylthiazol-5-yl)benzyl)carbamoyl)pyrrolidin-1-yl)-3,3-dimethyl-1-oxobutan-2-yl)amino)-3-oxopropoxy)propanoic acid (Linker 51)

(581) ##STR00382##

(582) Linker 51 was synthesized following the same procedure as Linker 45 as described in Example 45 (1.1 g, yield: 42%). ¹H NMR (400 MHz, DMSO-d₆) δ 8.98 (s, 1H), 8.55 (t, J=6.0 Hz, 1H), 7.91 (d, J=9.2 Hz, 1H), 7.43-7.37 (m, 4H), 4.55-4.53 (m, 1H), 4.45-4.40 (m, 2H), 4.35 (s, 1H), 4.24-4.19 (m, 1H), 3.68-3.52 (m, 6H), 2.54-2.56 (m, 1H), 2.45-2.37 (m, 5H), 2.34-2.30 (m, 1H), 2.05-2.00 (m, 1H), 1.93-1.86 (m, 1H), 0.93 (s, 9H). MS (ESI) m/z=575 [M+H]⁺.

Example 52: 2-(2-(((S)-1-((2S,4R)-4-Hydroxy-2-((4-(4-methylthiazol-5-yl)benzyl)carbamoyl)pyrrolidin-1-yl)-3,3-dimethyl-1-oxobutan-2-yl)amino)-2-oxoethoxy)acetic acid (Linker 52)

(583) ##STR00383##

(584) Linker 52 was synthesized following the same procedure as Linker 43 as described as Example 43. (1.2 g, yield: 63%). ¹H NMR (400 MHz, DMSO-d₆) δ 12.81 (br s, 1H), 8.98 (s, 1H), 8.58 (t, J=6.0 Hz, 1H), 7.60 (d, J=9.6 Hz, 1H), 7.45-7.35 (m, 4H), 5.14 (br, 1H), 4.58-4.55 (m, 1H), 4.46-4.36 (m, 3H), 4.28-4.26 (m, 1H), 4.14 (s, 2H), 4.04 (s, 2H), 3.69-3.60 (m, 2H), 2.44 (s, 3H), 2.08-2.03 (m, 1H), 1.93-1.87 (m, 1H), 0.95 (s, 9H). MS (ESI) m/z=547 [M+H]⁺.

Example 53: 3-(2-(3-(((S)-1-((2S,4R)-4-Hydroxy-2-((4-(4-methylthiazol-5-yl)benzyl)carbamoyl)pyrrolidin-1-yl)-3,3-dimethyl-1-oxobutan-2-yl)amino)-3-oxopropoxy)ethoxy)propanoic acid (Linker 53)

(585) ##STR00384##

(586) Linker 53 was synthesized following the same procedures as Linker 45 as described as Example 45. (1.4 g, yield 23% over 2 steps). ¹H NMR (400 MHz, DMSO-d₆): δ 8.98 (s, 1H), 8.56 (t, J=6.0 Hz, 1H), 7.91 (d, J=9.2 Hz, 1H), 7.43-7.37 (m, 4H), 4.55 (d, J=9.6 Hz, 1H), 4.46-4.41 (m, 2H), 4.35 (s, 1H), 4.29-4.20 (m, 1H), 3.70-3.57 (m, 7H), 3.50-3.45 (m, 5H), 2.57-2.55 (m, 1H), 2.45 (s, 3H), 2.43-2.41 (m, 1H), 2.37-2.32 (m, 1H), 2.09-2.01 (m, 1H), 1.94-1.87 (m, 1H), 0.94 (s, 9H). MS (ESI) m/z=619.3 [M+H]⁺.

Example 54: 2-(2-(2-(((S)-1-((2S,4R)-4-Hydroxy-2-((4-(4-methylthiazol-5-yl)benzyl)carbamoyl)pyrrolidin-1-yl)-3,3-dimethyl-1-oxobutan-2-yl)amino)-2-oxoethoxy)ethoxy)acetic acid (Linker 54)

(587) ##STR00385##

(588) Linker 54 was synthesized following the same procedures as Linker 53 as described as Example 53. (1.13 g, yield 20% over 2 steps). ¹H NMR (400 MHz, DMSO-d₆): δ 8.98 (s, 1H), 8.60 (t, J=6.0 Hz, 1H), 7.49 (d, J=9.2 Hz, 1H), 7.40 (s, 4H), 4.57 (d, J=9.2 Hz, 1H), 4.47-4.36 (m, 3H), 4.28-4.23 (m, 1H), 4.05-3.93 (m, 4H), 3.69-3.61 (m, 6H), 2.45 (s, 3H), 2.08-2.03 (m, 1H), 1.94-1.87 (m, 1H), 0.94 (s, 9H). MS (ESI) m/z=591.2 [M+H]⁺.

Example 55: (S)-15-((2S,4R)-4-Hydroxy-2-((4-(4-methylthiazol-5-yl)benzyl)carbamoyl)pyrrolidine-1-carbonyl)-16,16-dimethyl-13-oxo-4,7,10-trioxa-14-azaheptadecanoic acid (Linker 55)

(589) ##STR00386##

(590) Linker 55 was synthesized following the same procedure as Linker 45 as described in Example 45. (1.7 g, yield 37%). ¹H NMR (400 MHz, DMSO-d₆): δ 8.99 (s, 1H), 8.56 (t,

J=6.0 Hz, 1H), 7.91 (d, J=9.6 Hz, 1H), 7.44-7.38 (m, 4H), 4.56 (d, J=9.2 Hz, 1H), 4.47-4.42(m, 2H), 4.36 (s, 1H), 4.25-4.20 (m, 1H), 3.70-3.55 (m, 6H), 3.50-3.46 (m, 8H), 2.58-2.51 (m, 3H), 2.45-2.42(m, 5H), 2.40-2.33 (m, 1H), 2.07-2.02(m, 1H), 1.94-1.88 (m, 1H), 0.94 (s, 9H). LCMS (ESI) m/z=661.0 [M-H].sup.-.

Example 56: (S)-13-((25,4R)-4-Hydroxy-2-((4-(4-methylthiazol-5-yl)benzyl)carbamoyl)pyrrolidine-1-carbonyl)-14,14-dimethyl-11-oxo-3,6,9-trioxa-12-azapentadecanoic acid (Linker 56)

(591) ##STR00387##

(592) Linker 56 was synthesized following the same procedures as Linker 45 as described as Example 45. (1.21 g, yield 31% over 2 steps). .sup.1H NMR (400 MHz, CDCl.sub.3): δ 8.68 (s, 1H), 7.80-7.71 (m, 11H), 7.41-7.33 (m, 5H), 4.71-7.65 (m, 1H), 4.61-4.50 (m, 3H), 4.37-4.33 (m, 1H), 4.07-3.94 (m, 5H), 3.77-3.58 (m, 10H), 2.51 (s, 3H), 2.38-2.30 (m, 1H), 2.24-2.19 (m, 1H), 0.98 (s, 9H). LCMS (ESI) m/z=635.0 [M+H].sup.+.

Example 57: (S)-18-((28,4R)-4-Hydroxy-2-((4-(4-methylthiazol-5-yl)benzyl)carbamoyl)pyrrolidine-1-carbonyl)-19,19-dimethyl-16-oxo-4,7, 10,13-tetraoxa-17-azaicosanoic acid (Linker 57)

(593) ##STR00388##

(594) Linker 57 was synthesized following the same procedure as Linker 45 as described as Example 45. (1.6 g, yield 43%). .sup.1H NMR (400 MHz, CDCl.sub.3): δ 8.69 (s, 1H), 7.55-7.52(m, 1H), 7.47-7.45 (m, 1H), 7.36 (s, 4H), 4.70-4.66 (m, 1H), 4.62-4.57 (m, 2H), 4.50 (s, 1H), 4.34-4.29 (m, 1H), 4.12-4.09 (m, 1H), 3.75-3.48 (m, 18H), 2.56-2.47 (m, 7H), 2.40-2.33 (m, 1H), 2.23-2.18 (m, 1H), 0.96 (s, 9H). MS (ESI) m/z=707.1 [M+H].sup.+.

Example 58: (S)-21-((28,4R)-4-Hydroxy-2-((4-(4-methylthiazol-5-yl)benzyl)carbamoyl)pyrrolidine-1-carbonyl)-22,22-dimethyl-19-oxo-4,7,10, 13,16-pentaoxa-20-azatricosanoic acid (Linker 58)

(595) ##STR00389##

(596) Linker 58 was synthesized following the same procedure as Linker 45 as described as Example 45. (1.2 g, yield: 23%). .sup.1H NMR (400 MHz, DMSO-d.sub.6) δ 8.98 (s, 1H), 8.57 (t, J=6.0 Hz, 1H), 7.91 (d, J=9.6 Hz, 1H), 7.43-7.31 (m, 4H), 4.56-4.53 (m, 1H), 4.45-4.35 (m, 3H), 4.24-4.19 (m, 1H), 3.69-3.55 (m, 6H), 3.49-3.47 (m, 16H), 2.57-2.53 (m, 1H), 2.45 (s, 3H), 2.39-2.32(m, 3H), 2.06-2.01 (m, 1H), 1.93-1.86 (m, 1H), 0.95 (s, 9H). MS (ESI) m/z=751 [M+H].sup.+.

Example 59: (S)-19-((2,4R)-4-Hydroxy-2-((4-(4-methylthiazol-5-yl)benzyl)carbamoyl)pyrrolidine-1-carbonyl)-20,20-dimethyl-17-oxo-3,6,9,12,15-pentaoxa-18-azahenicosanoic acid (Linker 59)

(597) ##STR00390##

(598) Linker 59 was synthesized following the same procedure as Linker 45 as described as Example 45. (1.3 g, yield: 39%). .sup.1H NMR (400 MHz, DMSO-d.sub.6) δ 8.98 (s, 1H), 8.69 (t, J=6.0 Hz, 1H), 7.45 (d, J=9.6 Hz, 1H), 7.43-7.37 (m, 4H), 4.57-4.55 (m, 1H), 4.47-4.34 (m, 3H), 4.27-4.22(m, 1H), 3.97 (s, 2H), 3.68-3.65 (m, 2H), 3.61-3.48 (m, 18H), 2.45 (s, 3H), 2.09-2.04 (m, 1H), 1.92-1.86 (m, 1H), 0.94 (s, 9H). MS (ESI) m/z=723 [M+H].sup.+.

Example 60. Synthesis of TRKi-1

(599) ##STR00391## ##STR00392## ##STR00393##

Step 1

(600) ##STR00394##

(601) To a solution of 3-cyano-4-fluorophenylboronic acid (1)(3.3 g, 20 mmol) in toluene (30 mL) were added potassium phosphate (8.5 g, 40 mmol) and tetrakis(triphenylphosphine) palladium (462 mg, 0.4 mmol), followed by 3,5-difluorobenzyl bromide (2)(4.2 g, 10 mmol). The reaction mixture was heated at 100° C. for 2 hours. After the reaction was completed, the resulting black mixture was diluted with ether (200 mL), washed with saturated aqueous ammonium chloride (2×50 mL) and brine (3×50 mL). The organic layer was dried over sodium sulphate, evaporated and purified by silica gel flash chromatography (n-hexane/ethyl acetate 95:5) to yield 5-(3,5-difluorobenzyl)-2-

fluorobenzonitrile (3)(2.9 g, yield 59%) as a white solid. ¹H NMR (400 MHz, DMSO-d₆) 7.90 (dd, J=6.0 Hz, 2.0 Hz, 1H), 7.73-7.69 (m, 1H), 7.46 (t, J=8.8 Hz, 1H), 7.09-7.04 (m, 3H), 4.01 (s, 2H). MS (ESI) m/z=248.2 [M+H]⁺.

Step 2

(602) ##STR00395##

(603) A mixture of 5-(3,5-difluoro-benzyl)-2-fluoro-benzonitrile (3)(2.9 g, 11.74 mmol) and hydrazine hydrate (1.76 mL, 35.22 mmol) in n-butanol (200 mL) was heated at 120° C. overnight. The reaction mixture was diluted with water and ethyl acetate. The organic phase was washed twice with brine, dried and concentrated. The resulting residue was purified by silica gel chromatography (DCM/MeOH=95:5) to afford 5-(3,5-difluorobenzyl)-1H-indazol-3-amine (4)(2.7 g, yield 89%) as a white solid. ¹H NMR (400 MHz, DMSO-d₆) δ 11.32(s, 1H), 7.52(s, 1H), 7.18-7.11 (m, 2H), 7.04 (t, J=9.6 Hz, 1H), 6.95-6.93 (m, 2H), 5.26 (s, 2H), 4.00 (s, 2H). MS (ESI) m/z=260.0 [M+H]⁺.

Step 3

(604) ##STR00396##

(605) A solution of 4-fluoro-2-nitro-benzoic acid (5)(10 g, 54 mmol), di-tert-butyl-dicarbonate (23.6 g, 108 mmol) and 4-dimethylaminopyridine (1.98 g, 16.2 mmol) in tert-butanol (100 mL) and dichloromethane (100 mL) was stirred at room temperature overnight. The reaction mixture was then diluted with ethyl acetate (500 mL), washed with 1N hydrochloric acid (500 mL), water (500 mL) and brine (500 mL). The organic phase was dried over sodium sulfate, concentrated and purified by silica gel chromatography column (DCM/MeOH=20/1) to afford tert-butyl 4-fluoro-2-nitrobenzoate (6) as a yellow solid (10.7 g, yield 82%). ¹H NMR (400 MHz, DMSO-d₆): δ 8.04 (dd, J=8.4 Hz, 2.8 Hz, 1H), 7.94 (dd, J=8.8 Hz, 1.6 Hz, 1H), 7.71 (ddd, J=8.4 Hz, 2.4 Hz, 1H), 1.50 (s, 9H). MS (ESI) m/z=242.2 [M+H]⁺.

Step 4

(606) ##STR00397##

(607) To a solution of piperazine (13.7 g, 159.75 mmol) in tetrahydrofuran (150 mL) was added tert-butyl 4-fluoro-2-nitrobenzoate (6)(7.7 g, 31.95 mmol). The mixture was stirred at 70° C. for 16h, before being poured into water and extracted with ethyl acetate (100 mL×3). The combined organic layers were washed with water, brine, dried over sodium sulfate and concentrated to give crude tert-butyl 2-nitro-4-(piperazin-1-yl)benzoate (7)(9.7 g, yield 99%) as yellow oil, which was used in the next step without further purification. MS (ESI) m/z=308.1 [M+H]⁺.

Step 5

(608) ##STR00398##

(609) To a solution of 2-nitro-4-piperazin-1-yl-benzoic acid tert-butyl ester (7)(13.5 g, 44.12 mmol) in dichloromethane (200 mL) were added triethylamine (13.4 g, 132.35 mmol) and trifluoroacetic anhydride (18.5 g, 88.24 mmol) at 0° C. The mixture was stirred at room temperature for 1 h. The solvent was evaporated to give a residue, which was purified by flash chromatography silica gel column (petroleum ether/ethyl acetate=1/1) to give tert-butyl 2-nitro-4-(4-(2,2,2-trifluoroacetyl)piperazin-1-yl)benzoate (8)(16.5 g, yield: 93%) as a yellow solid. ¹H NMR (400 MHz, DMSO-d₆): δ 7.72(d, J=9.2 Hz, 1H), 7.32(d, J=2.8 Hz, 1H), 7.16 (dd, J=2.8, 9.2 Hz, 1H), 3.72-3.70 (m, 4H), 3.56-3.52(m, 4H), 1.45 (s, 9H). MS (ESI) m/z=404.3 [M+H]⁺, Step 6

(610) ##STR00399##

(611) tert-butyl 2-nitro-4-(4-(2,2,2-trifluoroacetyl)piperazin-1-yl)benzoate (8)(8.0 g, 19.85 mmol) was dissolved in methanol (150 mL), before Pd/C (1.0 g) was added. The resulting mixture was stirred under hydrogen at atmosphere pressure for 16 h. The solution was filtered over a pad of celite and washed with methanol several times. The filtrate was concentrated under vacuum to afford tert-butyl 2-amino-4-(4-(2,2,2-trifluoroacetyl)piperazin-1-yl)benzoate (9)(6.3 g, yield 85%) as a yellow solid. ¹H NMR (400 MHz, DMSO-d₆): δ 7.51 (d, J=9.2 Hz, 1H), 6.47 (br, 2H), 6.20 (dd, J=2.8, 9.2 Hz, 1H), 6.13 (d, J=2.8 Hz, 1H), 3.71-3.69 (m, 4H), 3.31-3.29 (m, 4H),

1.50 (s, 9H). MS (ESI) m/z =374.0 [M+H].sup.+.

Step 7
(612) ##STR00400##

(613) To a solution of tert-butyl 2-amino-4-(4-(2,2,2-trifluoroacetyl)piperazin-1-yl)benzoate (6.3 g, 16.89 mmol) in dichloromethane (9)(150 mL) were added tetrahydro-pyran-4-one (2.1 g, 21.11 mmol), trifluoroacetic acid (3.5 mL) and tetramethylammonium triacetoxymethylborohydride (6.7 g, 25.34 mmol). The mixture was stirred at room temperature for 16 h, before being washed sequentially with 0.5N hydrochloric acid, 0.5N sodium hydroxide, and a saturated solution of sodium bicarbonate. The organic layer was dried over sodium sulfate and concentrated to afford tert-butyl 2-((tetrahydro-2H-pyran-4-yl)amino)-4-(4-(2,2,2-trifluoroacetyl)piperazin-1-yl)benzoate (10)(3.5 g, yield 50%) as a pale yellow solid. .sup.1HNMR (400 MHz, DMSO-d.sub.6): δ 7.72(d, J=7.6 Hz, 1H), 7.60 (d, J=9.2 Hz, 1H), 6.20 (dd, J=2.4, 9.2 Hz, 1H), 6.09 (d, J=2.0 Hz, 1H), 3.86-3.82(m, 2H), 3.70-3.69 (m, 5H), 3.52-3.46 (m, 2H), 3.39-3.38 (m, 4H), 1.97-1.94 (m, 2H), 1.50 (s, 9H), 1.43-1.34 (m, 2H). MS (ESI) m/z =458.1 [M+H].sup.+.

Step 8

(614) ##STR00401##

(615) tert-butyl 2-((tetrahydro-2H-pyran-4-yl)amino)-4-(4-(2,2,2-trifluoroacetyl)piperazin-1-yl)benzoate (10)(3.8 g, 8.32 mmol) was dissolved in dichloromethane (100 ml) and cooled to 0° C. To the resulting solution was added triethylamine (1.3 g, 12.47 mmol) followed by a slow addition of trifluoroacetic anhydride (2.3 g, 10.81 mmol). After being stirred for 1 h, the reaction was quenched with water and diluted with DCM. The organic layer was washed with a saturated solution of aqueous sodium bicarbonate and brine. The organic layer was dried over sodium sulfate, concentrated and purified by silica gel chromatography column (petroleum ether/ethyl acetate=2/1) to give tert-butyl 2-(2,2,2-trifluoro-N-(tetrahydro-2H-pyran-4-yl)acetamido)-4-(4-(2,2,2-trifluoroacetyl)piperazin-1-yl)benzoate (11)(4.2 g, yield 91%) as a yellow solid. .sup.1HNMR (400 MHz, DMSO-d.sub.6): δ 7.85 (d, J=8.8 Hz, 1H), 7.08 (dd, J=2.4, 8.8 Hz, 1H), 6.85 (d, J=2.4 Hz, 1H), 4.52-4.44 (m, 1H), 3.89-3.77 (m, 2H), 3.75-3.72 (m, 4H), 3.55-3.49 (m, 4H), 3.45-3.32(m, 2H), 1.99-1.97 (m, 1H), 1.65-1.53 (m, 1H), 1.48-1.45 (m, 1H), 1.45 (s, 9H), 1.08-0.96 (m, 1H). MS (ESI) m/z =554.1 [M+H].sup.+.

Step 9

(616) ##STR00402##

(617) To a solution of tert-butyl 2-(2,2,2-trifluoro-N-(tetrahydro-2H-pyran-4-yl)acetamido)-4-(4-(2,2,2-trifluoroacetyl)piperazin-1-yl)benzoate (11)(4.2 g, 7.59 mmol) in DCM (50 ml) was added TFA (50 mL) at 0° C. After the reaction was stirred at room temperature for 16 h, the solvent was removed under vacuum. The residue was washed with diethyl ether to give 2-(2,2,2-trifluoro-N-(tetrahydro-2H-pyran-4-yl)acetamido)-4-(4-(2,2,2-trifluoroacetyl)piperazin-1-yl)benzoic acid (12) (3.5 g, yield 93%) as a white solid. .sup.1H NMR (400 MHz, DMSO-d.sub.6) δ 12.70 (s, 1H), 7.90 (d, J=8.8 Hz, 1H), 7.06 (dd, J=2.8, 9.2 Hz, 1H), 6.86 (d, J=2.0 Hz, 1H), 4.51-4.43 (m, 1H), 3.88-3.79 (m, 2H), 3.75-3.72(m, 4H), 3.55-3.41 (m, 6H), 1.97-1.94 (m, 1H), 1.64-1.49 (m, 2H), 1.12-1.02(m, 1H). MS (ESI) m/z =498.0 [M+H].sup.+.

Step 10

(618) ##STR00403##

(619) To a suspension of 2-(2,2,2-trifluoro-N-(tetrahydro-2H-pyran-4-yl)acetamido)-4-(4-(2,2,2-trifluoroacetyl)piperazin-1-yl)benzoic acid (12)(3.5 g, 7.04 mmol) in dry dichloromethane (150 mL) were added catalytic amount of N,N-dimethylformamide, oxalyl chloride (2.7 g, 21.13 mmol) at 0° C. The mixture was stirred for about 1.5 h before being concentrated. The residue was azeotroped with dry dichloromethane twice. The acyl chloride was dissolved in dry dichloromethane (50 mL). The resulting suspension was added slowly to a solution of 5-(3,5-difluoro-benzyl)-1H-indazol-3-ylamine (1.86 g, 7.04 mol) and triethylamine (2.2 g, 21.13 mmol) in dry tetrahydrofuran (100 mL) at -20° C. The mixture was stirred at room temperature for 16 h before being concentrated. The resulting residue was purified by silica gel chromatography column

(petroleum ether/ethyl acetate=1/1) to give N-(5-(3,5-difluorobenzyl)-1H-indazol-3-yl)-2-(2,2,2-trifluoro-N-(tetrahydro-2H-pyran-4-yl)acetamido)-4-(4-(2,2,2-trifluoroacetyl)piperazin-1-yl)benzamide (13)(4.0 g, yield 77%) as a yellow solid. ¹H NMR (400 MHz, DMSO-d₆) δ 12.70 (s, 1H), 10.58 (s, 1H), 7.86 (d, J=8.4 Hz, 1H), 7.43-7.41 (m, 2H), 7.27 (d, J=9.2 Hz, 1H), 7.18-7.11 (m, 1H), 7.04-6.99 (m, 1H), 6.95-6.93 (m, 3H), 4.47-4.41 (m, 1H), 4.01 (s, 2H), 3.80-3.72 (m, 4H), 3.22-3.17 (m, 4H), 3.51-3.47 (m, 4H), 1.93-1.90 (m, 1H), 1.67-1.64 (m, 1H), 1.60-1.50 (m, 1H), 1.37-1.26 (m, 1H). MS (ESI) m/z=739.0 [M+H]⁺.

Step 11

(620) ##STR00404##

(621) To a solution of N-(5-(3,5-difluorobenzyl)-1H-indazol-3-yl)-2-(2,2,2-trifluoro-N-(tetrahydro-2H-pyran-4-yl)acetamido)-4-(4-(2,2,2-trifluoroacetyl)piperazin-1-yl)benzamide (13)(4.0 g, 5.42 mmol) in methanol (100 mL) was added potassium carbonate (3.7 g, 27.1 mmol). The mixture was stirred at room temperature for 2 h before being filtered. The filtrate was evaporated and the residue was purified by silica gel chromatography (DCM/MeOH=10:1) to give N-(5-(3,5-difluorobenzyl)-1H-indazol-3-yl)-4-(piperazin-1-yl)-2-((tetrahydro-2H-pyran-4-yl)amino)benzamide (TRKi-1)(1.9 g, yield: 64%) as a blue solid. ¹H NMR (400 MHz, DMSO-d₆) δ 12.68 (s, 1H), 10.12 (s, 1H), 8.31 (d, J=6.8 Hz, 1H), 7.81 (d, J=8.4 Hz, 1H), 7.50 (s, 1H), 7.41 (d, J=8.4 Hz, 1H), 7.26 (d, J=8.4 Hz, 1H), 7.03-6.98 (m, 3H), 6.23 (d, J=8.4 Hz, 1H), 6.13 (s, 1H), 4.04 (s, 2H), 3.83-3.80 (m, 2H), 3.68-3.62 (m, 1H), 3.52-3.47 (m, 2H), 3.22-3.17 (m, 4H), 2.87-2.80 (m, 4H), 1.95-1.92 (m, 2H), 1.36-1.34 (m, 2H). MS (ESI) m/z=547.2 [M+H]⁺.

Example 61. Synthesis of TRKi-2

(622) ##STR00405##

Step 1

(623) ##STR00406##

(624) To a solution of 3-bromo-6-chloroimidazo[1,2-b]pyridazine (14)(4.6 g, 20.0 mmol) in dimethylsulfoxide (40 mL) were added potassium fluoride (20 g, 362 mmol) and (R)-2-(3-fluorophenyl)pyrrolidine (15)(3 g, 18.2 mmol). The resulting mixture was stirred at 100° C. for 12 h. The mixture was diluted with ethyl acetate, washed with water. The organic layer was concentrated and the residue was purified by column chromatography (ethyl acetate) to give (R)-3-bromo-6-(2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazine (16)(1.8 g, yield 28%) as a yellow solid. MS (ESI) m/z=360.9 [M+H]⁺.

Step 2

(625) ##STR00407##

(626) To a solution of (R)-3-bromo-6-(2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazine (16) (2.17 g, 6.03 mmol) in toluene (50 mL) were added 2-fluoro-6-(tributylstannyl)pyridine (17) (3.5 g, 9.04 mmol) and tetrakis(triphenylphosphine) palladium (566 mg, 0.49 mmol). The resulting mixture was stirred at 110° C. for 12 h under nitrogen atmosphere before being poured into ethyl acetate and sat. potassium fluoride. After stirring at room temperature for 2 h, the mixture was extracted with ethyl acetate. The combined organic layers were concentrated and purified by column chromatography (hexanes: ethyl acetate=1:1 to 100% ethyl acetate) to give (R)-6-(2-(3-fluorophenyl)pyrrolidin-1-yl)-3-(6-fluoropyridin-2-yl)imidazo[1,2-b]pyridazine (18)(2.2 g, yield 97%) as yellow oil. MS (ESI) m/z=378.0 [M+H]⁺.

Step 3

(627) ##STR00408##

(628) To a solution of (R)-6-(2-(3-fluorophenyl)pyrrolidin-1-yl)-3-(6-fluoropyridin-2-yl)imidazo[1,2-b]pyridazine (18, 1.4 g, 3.7 mmol) in dimethylsulfoxide (40 mL) was added piperazine (6.4 g, 74 mmol), followed by potassium fluoride (8.6 g, 148 mmol). The resulting mixture was stirred at 130° C. for 12 h before being poured into water and extracted with ethyl acetate. The combined organic layers were washed with water, concentrated and purified by column chromatography (dichloromethane:methanol=10:1 to 5:1) to give desired product as a

yellow oil, which was dissolved in hydrochloric acid/ethyl acetate (4 M), and stirred for 1h, concentrated to give (R)-6-(2-(3-fluorophenyl)pyrrolidin-1-yl)-3-(6-(piperazin-1-yl)pyridin-2-yl)imidazo[1,2-b]pyridazine hydrochloride (TRKi-2)(1.168 g, yield 66%) as a yellow solid. ¹H NMR (400 MHz, DMSO-*d*₆): δ 9.62(s, 2H), 8.63 (s, 1H), 8.21 (s, 1H), 7.62-7.19 (m, 6H), 7.06-7.01 (m, 2H), 5.26-5.25 (m, 1H), 4.07-4.02(m, 1H), 3.86-3.85 (m, 4H), 3.74-3.72(m, 1H), 3.16-3.15 (m, 4H), 2.08-2.07 (m, 2H), 1.92-1.91 (m, 2H). MS (ESI) *m/z*=444.2 [M+H].⁺.

Example 62. Synthesis of TRKi-3

(629) ##STR00409##

(630) To a solution of (R)-6-(2-(3-fluorophenyl)pyrrolidin-1-yl)-3-(6-(piperazin-1-yl)pyridine-2-yl)imidazo[1,2-b]pyridazine (TRKi-2)(1 g, 2.25 mmol) in DMF (40 ml) were added K₂CO₃ (621 mg, 4.50 mmol) and tert-butyl 2-bromoacetate (510 mg, 2.60 mmol). The resulting mixture was stirred at room temperature for 3 hours. After the amine was totally consumed, the reaction was poured into water (300 mL) and extracted with ethyl acetate (3×50 mL). The combined organic layers were washed with saturated brine (100 mL) and dried over anhydrous sodium sulfate. After filtration, the filtrate was concentrated under reduced pressure. The resulting residue was purified by silica gel column chromatography to give tert-butyl (R)-2-(4-(6-(2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)acetate (1.05g, yield: 84%) as a light yellow solid. MS (ESI) *m/z*: 558.7 [M+H].⁺.

(631) To a solution of tert-butyl (R)-2-(4-(6-(2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)acetate (1 g, 1.79 mmol) in dichloromethane (20 ml) was added trifluoroacetic acid (20 mL). The resulting mixture was stirred at room temperature for 3 hours. After the starting material was totally consumed, the reaction was evaporated under reduced pressure. The resulting residue was purified by reverse-phase chromatography to give (R)-2-(4-(6-(2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)acetic acid (TRKi-3)(860 mg, yield: 79% over two steps) as a light yellow solid. MS (ESI) *m/z*: 502.6 [M+H].⁺.

Example 63. Synthesis of TRKi-4

(632) ##STR00410##

(633) To a solution of N-(5-(3,5-difluorobenzyl)-1H-indazol-3-yl)-4-(piperazin-1-yl)-2-((tetrahydro-2H-pyran-4-yl)amino)benzamide (TRKi-1)(1.0 g, 1.83 mmol) in DMF (40 ml) were added K₂CO₃ (505 mg, 3.66 mmol) and tert-butyl 2-bromoacetate (357 mg, 1.83 mmol). The resulting mixture was stirred at room temperature for 3 hours. After the amine was totally consumed, the reaction was poured into water (300 mL) and extracted with ethyl acetate (3×50 mL). The combined organic layers were washed with saturated brine (100 mL) and dried over anhydrous sodium sulfate. After filtration, the filtrate was concentrated under reduced pressure. The resulting residue was purified by silica gel column chromatography to give tert-butyl 2-(4-(4-((5-(3,5-difluorobenzyl)-1H-indazol-3-yl)carbamoyl)-3-((tetrahydro-2H-pyran-4-yl)amino)phenyl)piperazin-1-yl)acetate (1.03g, yield: 85%) as a light yellow solid. MS (ESI) *m/z*: 661.3 [M+H].⁺.

(634) To a solution of tert-butyl 2-(4-(4-((5-(3,5-difluorobenzyl)-1H-indazol-3-yl)carbamoyl)-3-((tetrahydro-2H-pyran-4-yl)amino)phenyl)piperazin-1-yl)acetate (1 g, 1.51 mmol) in dichloromethane (20 ml) was added trifluoroacetic acid (20 mL). The resulting mixture was stirred at room temperature for 3 hours. After the starting material was totally consumed, the reaction was evaporated under reduced pressure. The resulting residue was purified by reverse-phase chromatography to give 2-(4-(4-((5-(3,5-difluorobenzyl)-1H-indazol-3-yl)carbamoyl)-3-((tetrahydro-2H-pyran-4-yl)amino)phenyl)piperazin-1-yl)acetic acid (TRKi-4)(790 mg, yield 73%) as a light yellow solid. MS (ESI) *m/z*: 605.3 [M+H].⁺.

Example 64: N-(5-(3,5-Difluorobenzyl)-1H-indazol-3-yl)-4-(4-(4-((2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-4-yl)amino)butanoyl)piperazin-1-yl)-2-((tetrahydro-2H-pyran-4-yl)amino)benzamide (CPD-053)

(635) ##STR00411##

(636) To a solution of 4-((2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-4-yl)amino)butanoic acid (657 mg, 1.83 mmol) in DMSO (30 mL) were added NMM (926 mg, 9.15 mmol), HOAT (373 mg, 2.74 mmol), EDCI (526 mg, 2.74 mmol) and N-(5-(3,5-difluorobenzyl)-1H-indazol-3-yl)-4-(piperazin-1-yl)-2-((tetrahydro-2H-pyran-4-yl)amino)benzamide (TRKi-1) (1 g, 1.83 mmol) sequentially. The resulting solution was stirred at room temperature for 16 hours, before the reaction was poured into water (200 mL) and extracted with ethyl acetate (3×50 mL). The combined organic layers were washed with saturated brine (100 mL) and dried over anhydrous sodium sulfate. After filtration, the filtrate was concentrated under reduced pressure. The resulting residue was purified by reverse-phase chromatography to give N-(5-(3,5-difluorobenzyl)-1H-indazol-3-yl)-4-(4-((2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-4-yl)amino)butanoyl)piperazin-1-yl)-2-((tetrahydro-2H-pyran-4-yl)amino)benzamide (CPD-053) (1.2 g, yield 74%) as a light yellow solid. ¹H NMR (400 MHz, DMSO-d₆) δ 11.09 (s, 1H), 10.28 (s, 1H), 7.89 (d, J=9.0 Hz, 1H), 7.59 (dd, J=8.4, 7.2 Hz, 1H), 7.52 (s, 1H), 7.42 (d, J=8.6 Hz, 1H), 7.26 (d, J=9.9 Hz, 1H), 7.19 (d, J=8.6 Hz, 1H), 7.08-6.94 (m, 4H), 6.47-6.28 (m, 2H), 5.06 (dd, J=12.9, 5.3 Hz, 1H), 4.04 (s, 2H), 3.84 (s, 2H), 3.63 (s, 5H), 3.57 (s, 1H), 3.46 (t, J=10.5 Hz, 3H), 3.40-3.24 (m, 6H), 3.17 (s, 1H), 2.87 (d, J=12.0 Hz, 1H), 2.57 (dd, J=19.7, 10.5 Hz, 1H), 2.47 (d, J=6.8 Hz, 2H), 2.08-1.76 (m, 5H), 1.48-1.34 (m, 2H), 1.23 (s, 1H). MS (ESI) m/z: 888.6 [M+H].⁺

Example 65: (2S,4R)-1-((S)-2-(6-(4-(6-(6-((R)-2-(3-Fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)-6-oxohexanamido)-3,3-dimethylbutanoyl)-4-hydroxy-N-(4-(4-methylthiazol-5-yl)benzyl)pyrrolidine-2-carboxamide (CPD-001)

(637) ##STR00412##

(638) CPD-001 was synthesized following the standard procedure for preparing CPD-053 (12 mg, yield 76%). MS (ESI) m/z: 984.7 [M+H].⁺

Example 66: (2S,4R)-1-((S)-2-(2-(2-(2-(4-(6-(6-((R)-2-(3-Fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)-2-oxoethoxy)ethoxy)acetamido)-3,3-dimethylbutanoyl)-4-hydroxy-N-(4-(4-methylthiazol-5-yl)benzyl)pyrrolidine-2-carboxamide (CPD-002)

(639) ##STR00413##

(640) CPD-002 was synthesized following the standard procedure for preparing CPD-053 (11 mg, yield 69%). MS (ESI) m/z: 1016.6 [M+H].⁺

Example 67: (2S,4R)-1-((S)-2-(tert-butyl)-20-(4-(6-(6-((R)-2-(3-Fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)-4,20-dioxo-6,9,12,15,18-pentaoxa-3-azaicosanoyl)-4-hydroxy-N-(4-(4-methylthiazol-5-yl)benzyl)pyrrolidine-2-carboxamide (CPD-003)

(641) ##STR00414##

(642) CPD-003 was synthesized following the standard procedure for preparing CPD-053 (14 mg, yield 72%). MS (ESI) m/z: 1148.8 [M+H].⁺

Example 68: (2S,4R)-1-((S)-2-(7-(4-(6-(6-((R)-2-(3-Fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)-7-oxoheptanamido)-3,3-dimethylbutanoyl)-4-hydroxy-N-(4-(4-methylthiazol-5-yl)benzyl)pyrrolidine-2-carboxamide (CPD-004)

(643) ##STR00415##

(644) CPD-004 was synthesized following the standard procedure for preparing CPD-053 (11 mg, yield 70%). MS (ESI) m/z: 997.8 [M+H].⁺

Example 69: (2S,4R)-1-((S)-2-(tert-butyl)-22-(4-(6-(6-((R)-2-(3-Fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)-4,22-dioxo-7,10,13,16,19-pentaoxa-3-azadocosanoyl)-4-hydroxy-N-(4-(4-methylthiazol-5-yl)benzyl)pyrrolidine-2-carboxamide (CPD-005)

(645) ##STR00416##

(646) CPD-005 was synthesized following the standard procedure for preparing CPD-053 (13 mg, yield 76%). MS (ESI) m/z: 1176.6 [M+H].sup.+.

Example 70: (2S,4R)-1-((S)-2-(4-(4-(6-(6-((R)-2-(3-Fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)-4-oxobutanamido)-3,3-dimethylbutanoyl)-4-hydroxy-N-(4-(4-methylthiazol-5-yl)benzyl)pyrrolidine-2-carboxamide (CPD-006)

(647) ##STR00417##

(648) CPD-006 was synthesized following the standard procedure for preparing CPD-053 (9 mg, yield 63%). MS (ESI) m/z: 956.4 [M+H].sup.+.

Example 71: (2S,4R)-1-(S)-2-(3-(3-(4-(6-(6-((R)-2-(3-Fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)-3-oxopropoxy)propanamido)-3,3-dimethylbutanoyl)-4-hydroxy-N-(4-(4-methylthiazol-5-yl)benzyl)pyrrolidine-2-carboxamide (CPD-007)

(649) ##STR00418##

(650) CPD-007 was synthesized following the standard procedure for preparing CPD-053 (12 mg, yield 73%). MS (ESI) m/z: 1000.5 [M+H].sup.+.

Example 72: (2S,4R)-1-((S)-2-(2-(2-(4-(6-(6-((R)-2-(3-Fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)-2-oxoethoxy) acetamido)-3,3-dimethylbutanoyl)-4-hydroxy-N-(4-(4-methylthiazol-5-yl)benzyl)pyrrolidine-2-carboxamide (CPD-008)

(651) ##STR00419##

(652) CPD-008 was synthesized following the standard procedure for preparing CPD-053 (11 mg, yield 75%). MS (ESI) m/z: 972.4 [M+H].sup.+.

Example 73: 2-(2,6-Dioxopiperidin-3-yl)-4-((7-(4-(6-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)-7-oxoheptyl)amino) isoindoline-1,3-dione (CPD-009)

(653) ##STR00420##

(654) CPD-009 was synthesized following the standard procedure for preparing CPD-053 (13 mg, yield 79%). MS (ESI) m/z: 827.4 [M+H].sup.+.

Example 74: 2-(2,6-Dioxopiperidin-3-yl)-4-((2-(2-(2-(3-(4-(6-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)-3-oxopropoxy)ethoxy)ethoxy)ethyl)amino) isoindoline-1,3-dione (CPD-010)

(655) ##STR00421##

(656) CPD-010 was synthesized following the standard procedure for preparing CPD-053 (11 mg, yield 73%). MS (ESI) m/z: 903.4 [M+H].sup.+.

Example 75: (2S,4R)-1-((S)-2-(tert-butyl)-14-(4-(6-(6-((R)-2-(3-Fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)-4, 14-dioxo-6,9,12-trioxo-3-azatetradecanoyl)-4-hydroxy-N-(4-(4-methylthiazol-5-yl)benzyl)pyrrolidine-2-carboxamide (CPD-011)

(657) ##STR00422##

(658) CPD-011 was synthesized following the standard procedure for preparing CPD-053 (15 mg, yield 81%). MS (ESI) m/z: 1060.5 [M+H].sup.+.

Example 76: (2S,4R)-1-((S)-2-(5-(4-(6-(6-((R)-2-(3-Fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)-5-oxopentanamido)-3,3-dimethylbutanoyl)-4-hydroxy-N-(4-(4-methylthiazol-5-yl)benzyl)pyrrolidine-2-carboxamide (CPD-012)

(659) ##STR00423##

(660) CPD-012 was synthesized following the standard procedure for preparing CPD-053 (11 mg, yield 72%). MS (ESI) m/z: 970.4 [M+H].sup.+.

Example 77: 2-(2,6-Dioxopiperidin-3-yl)-4-((4-(4-(6-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)-4-oxobutyl)amino) isoindoline-1,3-dione (CPD-013)

(661) ##STR00424##

(662) CPD-013 was synthesized following the standard procedure for preparing CPD-053 (8 mg, yield 69%). MS (ESI) m/z: 785.3 [M+H].sup.+.

Example 78: 2-(2,6-Dioxopiperidin-3-yl)-4-((5-(4-(6-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)-5-oxopentyl)amino) isoindoline-1,3-dione (CPD-014)

(663) ##STR00425##

(664) CPD-014 was synthesized following the standard procedure for preparing CPD-053 (11 mg, yield 73%). MS (ESI) m/z: 799.3 [M+H].sup.+.

Example 79: 2-(2,6-Dioxopiperidin-3-yl)-4-((6-(4-(6-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)-6-oxohexyl)amino) isoindoline-1,3-dione (CPD-015)

(665) ##STR00426##

(666) CPD-015 was synthesized following the standard procedure for preparing CPD-053 (11 mg, yield 74%). MS (ESI) m/z: 813.4 [M+H].sup.+.

Example 80: (2S,4R)-1-((S)-2-(tert-butyl)-16-(4-(6-(6-((R)-2-(3-Fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)-4, 16-dioxo-7,10,13-trioxa-3-azahexadecanoyl)-4-hydroxy-N-(4-(4-methylthiazol-5-yl)benzyl)pyrrolidine-2-carboxamide (CPD-016)

(667) ##STR00427##

(668) CPD-016 was synthesized following the standard procedure for preparing CPD-053 (13 mg, yield 73%). MS (ESI) m/z: 1088.5 [M+H].sup.+.

Example 81: (2S,4R)-1-((S)-2-(10-(4-(6-(6-((R)-2-(3-Fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)-10-oxodecanamido)-3,3-dimethylbutanoyl)-4-hydroxy-N-(4-(4-methylthiazol-5-yl)benzyl)pyrrolidine-2-carboxamide (CPD-017)

(669) ##STR00428##

(670) CPD-017 was synthesized following the standard procedure for preparing CPD-053 (11 mg, yield 70%). ¹H NMR (400 MHz, DMSO-d₆) δ 11.09 (s, 1H), 8.21 (s, 1H), 7.97 (d, J=9.9 Hz, 1H), 7.54 (dd, J=20.1, 12.4 Hz, 2H), 7.39 (dd, J=14.1, 7.9 Hz, 1H), 7.26-7.08 (m, 2H), 7.02 (dt, J=15.3, 7.8 Hz, 2H), 6.76 (d, J=8.6 Hz, 1H), 6.58 (s, 1H), 5.19 (d, J=5.9 Hz, 1H), 5.03 (dd, J=12.8, 5.2 Hz, 1H), 4.01 (d, J=5.0 Hz, 1H), 3.76-3.66 (m, 2H), 3.65-3.50 (m, 8H), 3.45 (s, 3H), 2.94-2.79 (m, 1H), 2.77 (d, J=4.9 Hz, 1H), 2.64 (t, J=6.4 Hz, 1H), 2.56 (d, J=7.2 Hz, 1H), 2.53 (s, 1H), 2.47-2.43 (m, 1H), 2.33 (s, 1H), 2.02 (dd, J=15.7, 8.4 Hz, 3H), 1.90 (s, 1H), 1.27 (d, J=25.3 Hz, 3H), 0.84 (d, J=6.8 Hz, 1H). MS (ESI) m/z: 1040.5 [M+H].sup.+.

Example 82: (2S,4R)-1-((S)-2-(9-(4-(6-(6-((R)-2-(3-Fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)-9-oxononanamido)-3,3-dimethylbutanoyl)-4-hydroxy-N-(4-(4-methylthiazol-5-yl)benzyl)pyrrolidine-2-carboxamide (CPD-018)

(671) ##STR00429##

(672) CPD-018 was synthesized following the standard procedure for preparing CPD-053 (9 mg, yield 72%). MS (ESI) m/z: 1026.5 [M+H].sup.+.

Example 83: (2S,4R)-1-((S)-2-(tert-butyl)-19-(4-(6-(6-((R)-2-(3-Fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)-4,19-dioxo-7,10,13,16-tetraoxa-3-azanonadecanoyl)-4-hydroxy-N-(4-(4-methylthiazol-5-yl)benzyl)pyrrolidine-2-carboxamide (CPD-019)

(673) ##STR00430##

(674) CPD-019 was synthesized following the standard procedure for preparing CPD-053 (12 mg, yield 78%). MS (ESI) m/z: 1132.5 [M+H].sup.+.

Example 84: (2S,4R)-1-((S)-2-(3-(2-(3-(4-(6-(6-((R)-2-(3-Fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)-3-oxopropoxy)ethoxy)propanamido)-3,3-dimethylbutanoyl)-4-hydroxy-N-(4-(4-methylthiazol-5-yl)benzyl)pyrrolidine-2-carboxamide (CPD-020)

(675) ##STR00431##

(676) CPD-020 was synthesized following the standard procedure for preparing CPD-053 (10 mg, yield 71%). MS (ESI) m/z: 1044.5 [M+H].sup.+.

Example 85: 2-(2,6-Dioxopiperidin-3-yl)-4-((15-(4-(6-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)-15-oxo-3,6,9,12-tetraoxapentadecyl)amino) isoindoline-1,3-dione (CPD-021)

(677) ##STR00432##

(678) CPD-021 was synthesized following the standard procedure for preparing CPD-053 (12 mg, yield 76%). MS (ESI) m/z: 947.4 [M+H].sup.+.

Example 86: (2S,4R)-1-((S)-2-(11-(4-(6-(6-((R)-2-(3-Fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)-11-oxoundecanamido)-3,3-dimethylbutanoyl)-4-hydroxy-N-(4-(4-methylthiazol-5-yl)benzyl)pyrrolidine-2-carboxamide (CPD-022)

(679) ##STR00433##

(680) CPD-022 was synthesized following the standard procedure for preparing CPD-053 (13 mg, yield 78%). MS (ESI) m/z: 1054.5 [M+H].sup.+.

Example 87: 2-(2,6-Dioxopiperidin-3-yl)-4-((2~(4-(6-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)-2-oxoethyl)amino) isoindoline-1,3-dione (CPD-023)

(681) ##STR00434##

(682) CPD-023 was synthesized following the standard procedure for preparing CPD-053 (11 mg, yield 72%). MS (ESI) m/z: 757.3 [M+H].sup.+.

Example 88: 2-(2,6-Dioxopiperidin-3-yl)-4-((8-(4-(6-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)-8-oxooctyl)amino) isoindoline-1,3-dione (CPD-024)

(683) ##STR00435##

(684) CPD-024 was synthesized following the standard procedure for preparing CPD-053 (13 mg, yield 75%). MS (ESI) m/z: 841.4 [M+H].sup.+.

Example 89: 2-(2,6-Dioxopiperidin-3-yl)-4-((18-(4-(6-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)-18-oxo-3,6,9,12,15-pentaoxaoctadecyl)amino) isoindoline-1,3-dione (CPD-025)

(685) ##STR00436##

(686) CPD-025 was synthesized following the standard procedure for preparing CPD-053 (14 mg, yield 76%). MS (ESI) m/z: 991.4 [M+H].sup.+.

Example 90: 2-(2,6-Dioxopiperidin-3-yl)-4-((3-(4-(6-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)-3-oxopropyl)amino) isoindoline-1,3-dione (CPD-026)

(687) ##STR00437##

(688) CPD-026 was synthesized following the standard procedure for preparing CPD-053 (12 mg, yield 71%). MS (ESI) m/z: 771.3 [M+H].sup.+.

Example 91: 2-(2,6-Dioxopiperidin-3-yl)-4~((2-(3-(4~(6-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)-3-oxopropoxy)ethyl)amino) isoindoline-1,3-dione (CPD-027)

(689) ##STR00438##

(690) CPD-027 was synthesized following the standard procedure for preparing CPD-053 (14 mg, yield 77%). MS (ESI) m/z: 815.3 [M+H].sup.+.

Example 92: 2-(2,6-Dioxopiperidin-3-yl)-4-((2-(2-(3-(4-(6-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)-3-oxopropoxy)ethoxy)ethyl)amino) isoindoline-1,3-dione (CPD-028)

(691) ##STR00439##

(692) CPD-028 was synthesized following the standard procedure for preparing CPD-053 (11 mg,

yield 72%). MS (ESI) m/z: 859.4 [M+H].sup.+.

Example 93: (2S,4R)-1-((S)-2-(8-(4-(6-(6-((R)-2-(3-Fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)-8-oxooctanamido)-3,3-dimethylbutanoyl)-4-hydroxy-N-(4-(4-methylthiazol-5-yl)benzyl)pyrrolidine-2-carboxamide (CPD-029)

(693) ##STR00440##

(694) CPD-029 was synthesized following the standard procedure for preparing CPD-053 (14 mg, yield 76%). MS (ESI) m/z: 1012.5 [M+H].sup.+.

Example 94: N-(5-(3,5-Difluorobenzyl)-1H-indazol-3-yl)-4-(4-(3-(2-((2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-4-yl)amino)ethoxy)propanoyl)piperazin-1-yl)-2-((tetrahydro-2H-pyran-4-yl)amino)benzamide (CPD-030)

(695) ##STR00441##

(696) CPD-030 was synthesized following the standard procedure for preparing CPD-053 (10 mg, yield 72%). MS (ESI) m/z: 918.4 [M+H].sup.+.

Example 95: N-(5-(3,5-Difluorobenzyl)-1H-indazol-3-yl)-4-(4-(1-((2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-4-yl)amino)-3,6,9,12-tetraoxapentadecan-15-oyl)piperazin-1-yl)-2-((tetrahydro-2H-pyran-4-yl)amino)benzamide (CPD-031)

(697) ##STR00442##

(698) CPD-031 was synthesized following the standard procedure for preparing CPD-053 (13 mg, yield 76%). MS (ESI) m/z: 1050.4 [M+H].sup.+.

Example 96: N-(5-(3,5-Difluorobenzyl)-1H-indazol-3-yl)-4-(4-(3-(2-(2-((2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-4-yl)amino)ethoxy)ethoxy)propanoyl)piperazin-1-yl)-2-((tetrahydro-2H-pyran-4-yl)amino)benzamide (CPD-032)

(699) ##STR00443##

(700) CPD-032 was synthesized following the standard procedure for preparing CPD-053 (10 mg, yield 71%). MS (ESI) m/z: 962.4 [M+H].sup.+.

Example 97: (2S,4R)-1-((S)-2-(8-(4-(4-((5-(3,5-Difluorobenzyl)-1H-indazol-3-yl)carbamoyl)-3-((tetrahydro-2H-pyran-4-yl)amino)phenyl)piperazin-1-yl)-8-oxooctanamido)-3,3-dimethylbutanoyl)-4-hydroxy-N-(4-(4-methylthiazol-5-yl)benzyl)pyrrolidine-2-carboxamide (CPD-033)

(701) ##STR00444##

(702) CPD-033 was synthesized following the standard procedure for preparing CPD-053 (13 mg, yield 73%). MS (ESI) m/z: 1115.5 [M+H].sup.+.

Example 98: (2S,4R)-1-((S)-2-(10-(4-(4-((5-(3,5-Difluorobenzyl)-1H-indazol-3-yl)carbamoyl)-3-((tetrahydro-2H-pyran-4-yl)amino)phenyl)piperazin-1-yl)-10-oxodecanamido)-3,3-dimethylbutanoyl)-4-hydroxy-N-(4-(4-methylthiazol-5-yl)benzyl)pyrrolidine-2-carboxamide (CPD-034)

(703) ##STR00445##

(704) CPD-034 was synthesized following the standard procedure for preparing CPD-053 (12 mg, yield 71%). MS (ESI) m/z: 1143.6 [M+H].sup.+.

Example 99: (2S,4R)-1-((S)-2-(11-(4-(4-((5-(3,5-Difluorobenzyl)-1H-indazol-3-yl)carbamoyl)-3-((tetrahydro-2H-pyran-4-yl)amino)phenyl)piperazin-1-yl)-11-oxoundecanamido)-3,3-dimethylbutanoyl)-4-hydroxy-N-(4-(4-methylthiazol-5-yl)benzyl)pyrrolidine-2-carboxamide (CPD-035)

(705) ##STR00446##

(706) CPD-035 was synthesized following the standard procedure for preparing CPD-053 (13 mg, yield 75%). MS (ESI) m/z: 1157.6 [M+H].sup.+.

Example 100: (2S,4R)-1-((S)-2-(5-(4-(4-((5-(3,5-Difluorobenzyl)-1H-indazol-3-yl)carbamoyl)-3-((tetrahydro-2H-pyran-4-yl)amino)phenyl)piperazin-1-yl)-5-oxopentanamido)-3,3-dimethylbutanoyl)-4-hydroxy-N-(4-(4-methylthiazol-5-yl)benzyl)pyrrolidine-2-carboxamide (CPD-036)

(707) ##STR00447##

(708) CPD-036 was synthesized following the standard procedure for preparing CPD-053 (10 mg, yield 71%). MS (ESI) m/z: 1073.5 [M+H].sup.+.

Example 101: (2S,4R)-1-((S)-2-(9-(4-(4-((5-(3,5-Difluorobenzyl)-1H-indazol-3-yl)carbamoyl)-3-((tetrahydro-2H-pyran-4-yl)amino)phenyl)piperazin-1-yl)-9-oxononanamido)-3,3-dimethylbutanoyl)-4-hydroxy-N-(4-(4-methylthiazol-5-yl)benzyl)pyrrolidine-2-carboxamide (CPD-037)

(709) ##STR00448##

(710) CPD-037 was synthesized following the standard procedure for preparing CPD-053 (13 mg, yield 76%). MS (ESI) m/z: 1129.5 [M+H].sup.+.

Example 102: (2S,4R)-1-((S)-2-(3-(2-(3-(4-(4-((5-(3,5-Difluorobenzyl)-1H-indazol-3-yl)carbamoyl)-3-((tetrahydro-2H-pyran-4-yl)amino)phenyl)piperazin-1-yl)-3-oxopropoxy)ethoxy)propanamido)-3,3-dimethylbutanoyl)-4-hydroxy-N-(4-(4-methylthiazol-5-yl)benzyl)pyrrolidine-2-carboxamide (CPD-038)

(711) ##STR00449##

(712) CPD-038 was synthesized following the standard procedure for preparing CPD-053 (10 mg, yield 72%). ¹H NMR (400 MHz, DMSO-d₆) δ 11.09 (s, 1H), 9.65 (s, 1H), 8.06 (d, J=99.8 Hz, 1H), 7.87-7.50 (m, 1H), 7.42-7.18 (m, 1H), 7.18 (s, 1H), 7.24-7.01 (m, 3H), 6.87 (dd, J=73.2, 17.7 Hz, 1H), 6.96-6.71 (m, 1H), 6.96-6.61 (m, 1H), 6.52 (s, 1H), 5.32 (s, 1H), 5.11 (dd, J=53.0, 7.1 Hz, 1H), 4.00 (s, 1H), 3.68-3.57 (m, 10H), 3.30 (m, 4H), 2.86-2.77 (m, 4H), 2.58 (d, J=16.2 Hz, 1H), 2.35 (s, 1H), 1.97 (d, J=59.9 Hz, 2H), 1.54 (d, J=22.7 Hz, 4H), 1.28 (d, J=37.7 Hz, 6H), 0.91 (s, 1H), 0.85 (s, 1H). MS (ESI) m/z: 1147.5 [M+H].sup.+.

Example 103: (2S,4R)-1-((S)-2-(tert-butyl)-16-(4-(4-((5-(3,5-Difluorobenzyl)-1H-indazol-3-yl)carbamoyl)-3-((tetrahydro-2H-pyran-4-yl)amino)phenyl)piperazin-1-yl)-4,16-dioxo-7,10,13-trioxa-3-azahexadecanoyl)-4-hydroxy-N-(4-(4-methylthiazol-5-yl)benzyl)pyrrolidine-2-carboxamide (CPD-039)

(713) ##STR00450##

(714) CPD-039 was synthesized following the standard procedure for preparing CPD-053 (13 mg, yield 77%). MS (ESI) m/z: 1191.5 [M+H].sup.+.

Example 104: (2S,4R)-1-((S)-2-(tert-butyl)-20-(4-(4-((5-(3,5-Difluorobenzyl)-1H-indazol-3-yl)carbamoyl)-3-((tetrahydro-2H-pyran-4-yl)amino)phenyl)piperazin-1-yl)-4,20-dioxo-6,9,12,15,18-pentaoxa-3-azaicosanoyl)-4-hydroxy-N-(4-(4-methylthiazol-5-yl)benzyl)pyrrolidine-2-carboxamide (CPD-040)

(715) ##STR00451##

(716) CPD-040 was synthesized following the standard procedure for preparing CPD-053 (10 mg, yield 71%). MS (ESI) m/z: 1251.6 [M+H].sup.+.

Example 105: (2S,4R)-1-((S)-2-(tert-butyl)-19-(4-(4-((5-(3,5-Difluorobenzyl)-1H-indazol-3-yl)carbamoyl)-3-((tetrahydro-2H-pyran-4-yl)amino)phenyl)piperazin-1-yl)-4,19-dioxo-7,10,13,16-tetraoxa-3-azanonadecanoyl)-4-hydroxy-N-(4-(4-methylthiazol-5-yl)benzyl)pyrrolidine-2-carboxamide (CPD-041)

(717) ##STR00452## ##STR00453##

(718) CPD-041 was synthesized following the standard procedure for preparing CPD-053 (12 mg, yield 78%). MS (ESI) m/z: 1235.6 [M+H].sup.+.

Example 106: (2S,4R)-1-((S)-2-(2-(2-(2-(4-(4-((5-(3,5-Difluorobenzyl)-1H-indazol-3-yl)carbamoyl)-3-((tetrahydro-2H-pyran-4-yl)amino)phenyl)piperazin-1-yl)-2-oxoethoxy)ethoxy)acetamido)-3,3-dimethylbutanoyl)-4-hydroxy-N-(4-(4-methylthiazol-5-yl)benzyl)pyrrolidine-2-carboxamide (CPD-042)

(719) ##STR00454##

(720) CPD-042 was synthesized following the standard procedure for preparing CPD-053 (13 mg, yield 76%). MS (ESI) m/z: 1119.5 [M+H].sup.+.

Example 107: (28,4R)-1-((S)-2-(tert-butyl)-22-(4-(4-((5-(3,5-Difluorobenzyl)-1H-indazol-3-yl)carbamoyl)-3-((tetrahydro-2H-pyran-4-yl)amino)phenyl)piperazin-1-yl)-4,22-dioxo-7,10,13,16,19-pentaoxa-3-azadocosanoyl)-4-hydroxy-N-(4-(4-methylthiazol-5-yl)benzyl)pyrrolidine-2-carboxamide (CPD-043)

(721) ##STR00455##

(722) CPD-043 was synthesized following the standard procedure for preparing CPD-053 (10 mg, yield 72%). MS (ESI) m/z: 1279.6 [M+H].sup.+.

Example 108: (25,4R)-1-(S)-2-(6-(4-4-(5-(3,5-Difluorobenzyl)-1H-indazol-3-yl)carbamoyl)-3,3-((tetrahydro-2H-pyran-4-yl)amino)phenyl)piperazin-1-yl)-6-oxohexanamido)-3,3-dimethylbutanoyl)-4-hydroxy-N-(4-(4-methylthiazol-5-yl)benzyl)pyrrolidine-2-carboxamide (CPD-044)

(723) ##STR00456##

(724) CPD-044 was synthesized following the standard procedure for preparing CPD-053 (13 mg, yield 76%). .sup.1H NMR (400 MHz, DMSO-d.sub.6) δ 11.09 (s, 1H), 8.09 (s, 1H), 7.88 (d, J=9.8 Hz, 1H), 7.68-7.54 (m, 2H), 7.49 (s, 1H), 7.38 (dd, J=14.1, 7.9 Hz, 1H), 7.17 (t, J=8.9 Hz, 3H), 7.03 (t, J=9.2 Hz, 2H), 6.78 (t, J=19.4 Hz, 2H), 5.17 (d, J=7.9 Hz, 1H), 5.06 (dd, J=12.8, 5.3 Hz, 1H), 4.00 (dd, J=9.9, 5.2 Hz, 1H), 3.63 (dt, J=43.4, 13.8 Hz, 9H), 2.87 (dd, J=21.7, 9.7 Hz, 1H), 2.67-2.53 (m, 3H), 2.47 (t, J=6.3 Hz, 3H), 2.04 (s, 3H), 1.93-1.75 (m, 3H), 1.34-1.15 (m, 2H). MS (ESI) m/z: 1087.5 [M+H].sup.+.

Example 109: (28,4R)-1-((S)-2-(2-(2-(4-(4-((5-(3,5-Difluorobenzyl)-1H-indazol-3-yl)carbamoyl)-3-((tetrahydro-2H-pyran-4-yl)amino)phenyl)piperazin-1-yl)-2-oxoethoxy)acetamido)-3,3-dimethylbutanoyl)-4-hydroxy-N-(4-(4-methylthiazol-5-yl)benzyl)pyrrolidine-2-carboxamide (CPD-045)

(725) ##STR00457##

(726) CPD-045 was synthesized following the standard procedure for preparing CPD-053 (15 mg, yield 79%). MS (ESI) m/z: 1075.5 [M+H].sup.+.

Example 110: (25,4R)-1-((S)-2-(4-(4-(4-((S-(3,5-Difluorobenzyl)-1H-indazol-3-yl)carbamoyl)-3,3-((tetrahydro-2H-pyran-4-yl)amino)phenyl)piperazin-1-yl)-4-oxobutanamido)-3,3-dimethylbutanoyl)-4-hydroxy-N-(4-(4-methylthiazol-5-yl)benzyl)pyrrolidine-2-carboxamide (CPD-046)

(727) ##STR00458##

(728) CPD-046 was synthesized following the standard procedure for preparing CPD-053 (11 mg, yield 73%). MS (ESI) m/z: 1059.5 [M+H].sup.+.

Example 111: (28,4R)-1-((S)-2-(7-(4-(4-((5-(3,5-Difluorobenzyl)-1H-indazol-3-yl)carbamoyl)-3-((tetrahydro-2H-pyran-4-yl)amino)phenyl)piperazin-1-yl)-7-oxoheptanamido)-3,3-dimethylbutanoyl)-4-hydroxy-N-(4-(4-methylthiazol-5-yl)benzyl)pyrrolidine-2-carboxamide (CPD-047)

(729) ##STR00459##

(730) CPD-047 was synthesized following the standard procedure for preparing CPD-053 (15 mg, yield 78%). MS (ESI) m/z: 1101.5 [M+H].sup.+.

Example 112: (28,4R)-1-((S)-2-(3-(3-(4-(4-((5-(3,5-Difluorobenzyl)-1H-indazol-3-yl)carbamoyl)-3-((tetrahydro-2H-pyran-4-yl)amino)phenyl)piperazin-1-yl)-3-oxopropoxy)propanamido)-3,3-dimethylbutanoyl)-4-hydroxy-N-(4-(4-methylthiazol-5-yl)benzyl)pyrrolidine-2-carboxamide (CPD-048)

(731) ##STR00460##

(732) CPD-048 was synthesized following the standard procedure for preparing CPD-053 (11 mg, yield 72%). MS (ESI) m/z: 1103.5 [M+H].sup.+.

Example 113: N—(S-(3,5-Difluorobenzyl)-1H-indazol-3-yl)-4-(4-(7-((2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-4-yl)amino) heptanoyl)piperazin-1-yl)-2-((tetrahydro-2H-pyran-4-yl)amino)benzamide (CPD-049)

(733) ##STR00461##

(734) CPD-049 was synthesized following the standard procedure for preparing CPD-053 (15 mg, yield 81%). MS (ESI) m/z: 930.4 [M+H].sup.+.

Example 114: N-(5-(3,5-Difluorobenzyl)-1H-indazol-3-yl)-4-(4-(5-((2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-4-yl)amino)pentanoyl)piperazin-1-yl)-2-((tetrahydro-2H-pyran-4-yl)amino)benzamide (CPD-050)

(735) ##STR00462##

(736) CPD-050 was synthesized following the standard procedure for preparing CPD-053 (12 mg, yield 76%). MS (ESI) m/z: 902.4 [M+H].sup.+.

Example 115: N-(5-(3,5-Difluorobenzyl)-1H-indazol-3-yl)-4-(4-(3-((2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-4-yl)amino)propanoyl)piperazin-1-yl)-2-((tetrahydro-2H-pyran-4-yl)amino)benzamide (CPD-051)

(737) ##STR00463##

(738) CPD-051 was synthesized following the standard procedure for preparing CPD-053 (11 mg, yield 72%). ¹H NMR (400 MHz, DMSO-d₆) δ 12.64 (s, 1H), 11.09 (s, 1H), 10.11 (s, 1H), 8.29 (s, 1H), 7.82 (s, 1H), 7.60 (d, J=7.8 Hz, 1H), 7.49 (s, 1H), 7.41 (d, J=8.5 Hz, 1H), 7.25 (d, J=8.7 Hz, 1H), 7.18 (d, J=8.5 Hz, 1H), 7.01 (dd, J=24.2, 7.5 Hz, 4H), 6.24 (d, J=8.8 Hz, 1H), 6.13 (s, 1H), 5.04 (d, J=8.3 Hz, 1H), 3.81 (s, 2H), 3.59 (m, 8H), 3.49 (t, J=10.0 Hz, 3H), 3.29 (s, 4H), 2.86 (d, J=13.9 Hz, 1H), 2.73 (s, 2H), 1.94 (d, J=12.9 Hz, 4H), 1.35 (d, J=10.2 Hz, 2H), 1.24 (s, 1H). MS (ESI) m/z: 874.3 [M+H].sup.+.

Example 116: N-(5-(3,5-Difluorobenzyl)-1H-indazol-3-yl)-4-(4-(8-((2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-4-yl)amino) octanoyl)piperazin-1-yl)-2-((tetrahydro-2H-pyran-4-yl)amino)benzamide (CPD-052)

(739) ##STR00464##

(740) CPD-052 was synthesized following the standard procedure for preparing CPD-053 (14 mg, yield 77%). MS (ESI) m/z: 944.4 [M+H].sup.+.

Example 117: N-(5-(3,5-Difluorobenzyl)-1H-indazol-3-yl)-4-(4-(1-((2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-4-yl)amino)-3,6,9,12, 15-pentaoxaoctadecan-18-oyl)piperazin-1-yl)-2-((tetrahydro-2H-pyran-4-yl)amino)benzamide (CPD-054)

(741) ##STR00465##

(742) CPD-054 was synthesized following the standard procedure for preparing CPD-053 (12 mg, yield 73%). MS (ESI) m/z: 1094.5 [M+H].sup.+.

Example 118: N-(5-(3,5-Difluorobenzyl)-1H-indazol-3-yl)-4-(4-((2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-4-yl)glycyl)piperazin-1-yl)-2-((tetrahydro-2H-pyran-4-yl)amino)benzamide (CPD-055)

(743) ##STR00466##

(744) CPD-055 was synthesized following the standard procedure for preparing CPD-053 (14 mg, yield 79%). MS (ESI) m/z: 860.3 [M+H].sup.+.

Example 119: N-(5-(3,5-Difluorobenzyl)-1H-indazol-3-yl)-4-(4-(6-((2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-4-yl)amino)hexanoyl)piperazin-1-yl)-2-((tetrahydro-2H-pyran-4-yl)amino)benzamide (CPD-056)

(745) ##STR00467##

(746) CPD-056 was synthesized following the standard procedure for preparing CPD-053 (11 mg, yield 71%). ¹H NMR (400 MHz, DMSO-d₆) δ 12.65 (s, 1H), 11.10 (s, 1H), 10.12 (s, 1H), 8.30 (s, 1H), 7.82 (d, J=9.0 Hz, 1H), 7.67-7.54 (m, 1H), 7.49 (s, 1H), 7.41 (d, J=8.6 Hz, 1H), 7.25 (d, J=8.6 Hz, 1H), 7.17-6.90 (m, 3H), 6.60 (d, J=42.5 Hz, 1H), 6.25 (d, J=8.4 Hz, 1H), 6.14 (s, 1H), 5.33 (d, J=4.8 Hz, 1H), 5.05 (dd, J=12.8, 5.2 Hz, 1H), 4.04 (s, 1H), 3.82 (d, J=11.5 Hz, 1H), 3.74-3.44 (m, 5H), 3.26 (dd, J=39.1, 19.2 Hz, 5H), 3.00-2.52 (m, 3H), 2.37 (t, J=7.2 Hz, 2H), 2.12-1.81 (m, 3H), 1.58 (dd, J=14.2, 7.0 Hz, 3H), 1.51-1.08 (m, 8H), 0.84 (d, J=6.9 Hz, 1H). MS (ESI) m/z: 916.4 [M+H].sup.+.

Example 120: N—(S-(3,5-Difluorobenzyl)-1H-indazol-3-yl)-4-(4-(3-(2-(2-(2-((2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-4-yl)amino)ethoxy)ethoxy)ethoxy)propanoyl)piperazin-1-yl)-2-((tetrahydro-2H-pyran-4-yl)amino)benzamide (CPD-057)

(747) ##STR00468##

(748) CPD-057 was synthesized following the standard procedure for preparing CPD-053 (14 mg, yield 78%). ¹H NMR (400 MHz, DMSO-d₆) δ 12.75 (d, J=68.8 Hz, 1H), 11.09 (s, 1H), 10.20 (d, J=59.5 Hz, 1H), 7.82 (d, J=9.0 Hz, 1H), 7.64-7.53 (m, 1H), 7.46 (d, J=17.1 Hz, 1H), 7.41 (d, J=8.6 Hz, 1H), 7.29-7.20 (m, 1H), 7.20-7.07 (m, 1H), 6.99 (dt, J=38.9, 17.1 Hz, 3H), 6.63 (d, J=20.8 Hz, 1H), 6.24 (d, J=7.8 Hz, 1H), 6.15 (s, 1H), 5.32 (t, J=4.7 Hz, 1H), 5.06 (dd, J=12.9, 5.3 Hz, 1H), 4.04 (s, 2H), 3.89-3.74 (m, 2H), 3.72-3.56 (m, 8H), 3.24 (dd, J=28.3, 24.5 Hz, 3H), 2.99-2.81 (m, 1H), 2.77 (d, J=4.8 Hz, 1H), 2.70-2.51 (m, 9H), 2.11-1.84 (m, 3H), 1.53-1.10 (m, 7H), 0.85 (t, J=6.6 Hz, 1H). MS (ESI) m/z: 1006.4 [M+H]⁺.

Example 121: (2S,4R)-1-((S)-2-(tert-butyl)-14-(4-(4-((5-(3,5-Difluorobenzyl)-1H-indazol-3-yl)carbamoyl)-3-((tetrahydro-2H-pyran-4-yl)amino)phenyl)piperazin-1-yl)-4,14-dioxo-6,9,12-trioxa-3-azatetradecanoyl)-4-hydroxy-N-(4-(4-methylthiazol-5-yl)benzyl)pyrrolidine-2-carboxamide (CPD-058)

(749) ##STR00469##

(750) CPD-058 was synthesized following the standard procedure for preparing CPD-053 (11 mg, yield 72%). MS (ESI) m/z: 1163.5 [M+H]⁺.

Example 122: N-(2-(2-(2-(2-((2-(2,6-Dioxopiperidin-3-yl)-1,3-dioxoisindolin-4-yl)amino)ethoxy)ethoxy)ethoxy)ethyl)-2-(4-(6-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)acetamide (CPD-059)

(751) ##STR00470##

(752) CPD-059 was synthesized following the standard procedure for preparing CPD-053 (14 mg, yield 76%). MS (ESI) m/z: 932.4 [M+H]⁺.

Example 123: N-(2-((2[~] (2,6-Dioxopiperidin-3-yl)-1,3-dioxoisindolin-4-yl)amino)ethyl)-2-(4-(6-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)acetamide (CPD-060)

(753) ##STR00471##

(754) CPD-060 was synthesized following the standard procedure for preparing CPD-053 (13 mg, yield 78%). ¹H NMR (400 MHz, DMSO-d₆) δ 11.12 (d, J=8.8 Hz, 1H), 10.50 (s, 1H), 8.94 (t, J=5.4 Hz, 1H), 8.51 (s, 1H), 8.13 (d, J=10.9 Hz, 1H), 7.61 (dt, J=13.9, 7.0 Hz, 2H), 7.45-7.35 (m, 1H), 7.26-7.12 (m, 3H), 7.05 (t, J=7.2 Hz, 2H), 6.95 (d, J=8.9 Hz, 1H), 6.76 (s, 1H), 5.23 (d, J=6.3 Hz, 1H), 5.07 (dd, J=12.8, 5.5 Hz, 1H), 4.47 (s, 2H), 4.14-3.93 (m, 4H), 3.72 (dd, J=17.8, 8.2 Hz, 3H), 3.42-3.36 (m, 4H), 3.21 (s, 3H), 2.95-2.80 (m, 1H), 2.65-2.53 (m, 2H), 2.04 (dd, J=31.3, 12.8 Hz, 3H), 1.91 (s, 2H), 1.25 (d, J=10.1 Hz, 1H). MS (ESI) m/z: 800.3 [M+H]⁺.

Example 124: (2S,4R)-1-((S)-2-(8-(2-(4-(4-((5-(3,5-Difluorobenzyl)-1H-indazol-3-yl)carbamoyl)-3-((tetrahydro-2H-pyran-4-yl)amino)phenyl)piperazin-1-yl)acetamido)octanamido)-3,3-dimethylbutanoyl)-4-hydroxy-N-(4-(4-methylthiazol-5-yl)benzyl)pyrrolidine-2-carboxamide (CPD-061)

(755) ##STR00472##

(756) CPD-061 was synthesized following the standard procedure for preparing CPD-053 (10 mg, yield 76%). MS (ESI) m/z: 1158.6 [M+H]⁺.

Example 125: N-(5-(3,5-Difluorobenzyl)-1H-indazol-3-yl)-4-(4-(2-((2-((2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-4-yl)amino)ethyl)amino)-2-oxoethyl)piperazin-1-yl)-2-((tetrahydro-2H-pyran-4-yl)amino)benzamide (CPD-062)

(757) ##STR00473##

(758) CPD-062 was synthesized following the standard procedure for preparing CPD-053 (12 mg, yield 72%). MS (ESI) m/z: 903.4 [M+H]⁺.

Example 126: (2S,4R)-1-((S)-2-(2-(2-(4-(4-((5-(3,5-Difluorobenzyl)-1H-indazol-3-

yl)carbamoyl)-3-((tetrahydro-2H-pyran-4-yl)amino)phenyl)piperazin-1-yl)acetamido)acetamido)-3,3-dimethylbutanoyl)-4-hydroxy-N-(4-(4-methylthiazol-5-yl)benzyl)pyrrolidine-2-carboxamide (CPD-063)

(759) ##STR00474##

(760) CPD-063 was synthesized following the standard procedure for preparing CPD-053 (15 mg, yield 76%). MS (ESI) m/z: 1073.5 [M+H].sup.+.

Example 127: N—(S-(3,5-Difluorobenzyl)-1H-indazol-3-yl)-4-(4-(14-((2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-4-yl)amino)-2-oxo-6,9,12-trioxa-3-azatetradecyl)piperazin-1-yl)-2-((tetrahydro-2H-pyran-4-yl)amino)benzamide (CPD-064)

(761) ##STR00475##

(762) CPD-064 was synthesized following the standard procedure for preparing CPD-053 (11 mg, yield 77%). MS (ESI) m/z: 1035.4 [M+H].sup.+.

Example 128: N-(5-(3,5-Difluorobenzyl)-1H-indazol-3-yl)-4-(4-(20-((2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-4-yl)amino)-2-oxo-6,9,12,15,18-pentaoxa-3-azaicosyl)piperazin-1-yl)-2-((tetrahydro-2H-pyran-4-yl)amino)benzamide (CPD-065)

(763) ##STR00476##

(764) CPD-065 was synthesized following the standard procedure for preparing CPD-053 (13 mg, yield 79%). MS (ESI) m/z: 1123.5 [M+H].sup.+.

Example 129: 5-((2-(2-Aminoethoxy)ethyl)amino)-2-(2,6-dioxopiperidin-3-yl)isoindoline-1,3-dione (Linker 60)

(765) ##STR00477##

(766) A mixture of 5-fluoroisobenzofuran-1,3-dione (87 g, 524 mmol), 3-aminopiperidine-2,6-dione (85.7 g, 524 mmol) and NaOAc (85.9 g, 1050 mmol) in acetic acid (500 mL) was stirred at 130° C. overnight. After cooling down to room temperature, the mixture was concentrated. The resulting residue was poured into ice water, and filtered. The filter cake was washed with water (500 mL×2), EtOH (500 mL×2), MeOH (500 mL) and DCM (500 mL) to afford a solid which was dried in vacuum to give 2-(2,6-dioxopiperidin-3-yl)-5-fluoroisindoline-1,3-dione (120 g, yield: 83%) as yellow solid. MS (ESI) m/z=277.1 [M+H].sup.+.

(767) A mixture of 2-(2,6-dioxopiperidin-3-yl)-5-fluoroisindoline-1,3-dione (6.9 g, 25.0 mmol), tert-butyl (2-(2-aminoethoxy)ethyl)carbamate (5.6 g, 27.5 mmol) and DIEA (9.7 g, 75 mmol) in NMP (75 mL) was stirred at 130° C. in microwave reactor for 50 min. After cooling down to room temperature, the mixture was poured into EtOAc (200 mL), and washed with water (200 mL×2) followed by brine (200 mL). The organic phase was dried over anhydrous Na.sub.2SO.sub.4, filtered and concentrated to give a crude product which was purified by silica gel chromatography (petroleum ether:EtOAc=2:1 to 1:2) to give tert-butyl (2-(2-((2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-5-yl)amino)ethoxy)ethyl) carbamate (2.4 g, yield: 21%) as yellow oil. MS (ESI) m/z=361.1 [M+H].sup.+.

(768) To a solution of tert-butyl (2-(2-((2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-5-yl)amino)ethoxy)ethyl)carbamate (2.4 g, 5.2 mmol) in DCM (10 mL) was added TFA (5 mL) in one portion. The reaction mixture was stirred at room temperature for 2 h. After concentration, the resulting residue was dissolved in water (20 mL), washed with EtOAc (40 mL) and MTBE (40 mL). The aqueous phase was lyophilized to afford TFA salt of 5-((2-(2-aminoethoxy)ethyl)amino)-2-(2,6-dioxopiperidin-3-yl)isoindoline-1,3-dione (1.9 g, yield: 77%) as yellow solid. .sup.1H NMR (400 MHz, DMSO-d.sub.6) δ 11.06 (s, 1H), 8.01 (s, 3H), 7.58 (d, J=8.4 Hz, 1H), 7.12(br, s, 1H), 7.02(d, J=2.0 Hz, 1H), 6.91 (dd, J=2.0 Hz, 8.8 Hz, 1H), 5.04 (dd, J=5.6 Hz, 13.2 Hz, 1H), 3.64 (t, J=5.6 Hz, 4H), 3.40 (t, J=5.2 Hz, 2H), 3.01 (br, 2H), 2.89-2.83 (m, 1H), 2.60-2.50 (m, 2H), 2.03-1.97 (m, 1H). MS (ESI) m/z=361.1 [M+H].sup.+.

Example 130: 5-((2-(2-(2-Aminoethoxy)ethoxy)ethyl)amino)-2-(2,6-dioxopiperidin-3-yl)isoindoline-1,3-dione (Linker 61)

(769) ##STR00478##

(770) Linker 61 was synthesized following the same procedure as Linker 60 as described for Example 60. (1.4 g, yield: 71%). ¹H NMR (400 MHz, DMSO-d₆) δ 11.05 (s, 1H), 7.94 (br, 3H), 7.56 (d, J=8.4 Hz, 1H), 7.01 (s, 1H), 6.90 (d, J=8.0 Hz, 1H), 5.03 (dd, J=5.2 Hz, 12.8 Hz, 1H), 3.58 (br, 8H), 3.36 (s, 2H), 2.97-2.92(m, 2H), 2.91-2.83 (m, 1H), 2.60-2.50 (m, 2H), 2.01-1.99 (m, 1H). MS (ESI) m/z=405.1 [M+H]⁺.

Example 131:5-((2-(2-(2-(2-Aminoethoxy)ethoxy)ethoxy)ethyl)amino)-2-(2,6-dioxopiperidin-3-yl)isoindoline-1,3-dione (Linker 62)

(771) ##STR00479##

(772) Linker 62 was synthesized following the same procedure as Linker 60 as described for Example 60. (1.19 g, yield: 59%). ¹H NMR (400 MHz, DMSO-d₆) δ 11.05 (s, 1H), 7.79 (br, 3H), 7.57 (d, J=8.4 Hz, 1H), 7.15 (br, s, 1H), 7.00 (d, J=2.0 Hz, 1H), 6.90 (dd, J=2.0 Hz, 8.4 Hz, 1H), 5.03 (dd, J=5.6 Hz, 12.8 Hz, 1H), 3.61-3.55 (m, 12H), 3.36 (t, J=5.6 Hz, 2H), 2.99-2.94 (m, 2H), 2.88-2.84 (m, 1H), 2.60-2.52(m, 2H) 2.01-1.98 (m, 1H). MS (ESI) m/z=449.1 [M+H]⁺.

Example 132:5-((14-Amino-3,6,9,12-tetraoxatetradecyl)amino)-2-(2,6-dioxopiperidin-3-yl)isoindoline-1,3-dione (Linker 63)

(773) ##STR00480##

(774) Linker 63 was synthesized following the same procedure as Linker 60 as described for Example 60. (1.2 g, yield: 73%). ¹H NMR (400 MHz, DMSO-d₆) δ 11.05 (s, 1H), 7.79 (br, J=1.6 Hz, 3H), 7.56 (d, J=8.4 Hz, 1H), 7.14 (br, s, 1H), 7.01 (d, J=2.0 Hz, 1H), 6.90 (dd, J=2.0 Hz, 8.4 Hz, 1H), 5.03 (dd, J=5.6 Hz, 13.2 Hz, 1H), 3.61-3.56 (m, 16H), 3.36 (t, J=5.2 Hz, 2H), 2.99-2.95 (m, 2H), 2.89-2.83 (m, 1H), 2.60-2.53 (m, 2H) 2.01-1.97 (m, 1H). MS (ESI) m/z=493.1 [M+H]⁺.

Example 133:5-((17-Amino-3,6,9,12,15-pentaoxaheptadecyl)amino)-2-(2,6-dioxopiperidin-3-yl)isoindoline-1,3-dione (Linker 64)

(775) ##STR00481##

(776) Linker 64 was synthesized following the same procedure as Linker 60 as described for Example 60. (1.73 g, yield: 88%). ¹H NMR (400 MHz, DMSO-d₆) δ 11.05 (s, 1H), 7.79 (s, 3H), 7.55 (d, J=8.4 Hz, 1H), 7.18 (br, s, 1H), 7.01 (s, 1H), 6.90 (d, J=8.4 Hz, 1H), 5.03 (dd, J=5.2 Hz, 12.8 Hz, 1H), 3.61-3.54 (m, 20H), 3.35 (s, 2H), 2.98 (s, 2H), 2.92-2.83 (m, 1H), 2.61-2.54 (m, 2H), 2.02-1.98 (m, 1H). MS (ESI) m/z=537.2 [M+H]⁺.

Example 134: (2-(2,6-Dioxopiperidin-3-yl)-1,3-dioxoisindolin-5-yl)glycine (Linker 65)

(777) ##STR00482##

(778) Linker 65 was synthesized following the same procedure as Linker 60 as described for Example 60. (1.0 g, yield: 84%). ¹H NMR (400 MHz, DMSO-d₆) δ 12.80 (br, 1H), 11.06 (s, 1H), 7.59 (d, J=8.4 Hz, 1H), 7.32(br, s, 1H), 6.98 (d, J=1.2 Hz, 1H), 6.89 (dd, J=2.0 Hz, 8.4 Hz, 1H), 5.04 (dd, J=5.6 Hz, 13.2 Hz, 1H), 4.03 (s, 2H), 2.92-2.83 (m, 1H), 2.60-2.52(m, 2H), 2.03-1.98 (m, 1H). MS (ESI) m/z=332.0 [M+H]⁺.

Example 135:3-((2-(2,6-Dioxopiperidin-3-yl)-1,3-dioxoisindolin-5-yl)amino)propanoic acid (Linker 66)

(779) ##STR00483##

(780) Linker 66 was synthesized following the same procedure as Linker 60 as described for Example 60. (1.24 g, yield: 60%). ¹H NMR (400 MHz, DMSO-d₆) δ 11.05 (s, 1H), 7.57 (d, J=8.4 Hz, 1H), 6.97 (d, J=2.0 Hz, 1H), 6.87 (dd, J=2.0 Hz, 8.4 Hz, 1H), 5.02(dd, J=5.2 Hz, 12.8 Hz, 1H), 3.41 (t, J=6.8 Hz, 2H), 2.89-2.83 (m, 1H), 2.60-2.52(m, 4H), 2.02-1.97 (m, 1H). MS (ESI) m/z=346.0 [M+H]⁺.

Example 136:4-((2-(2,6-Dioxopiperidin-3-yl)-1,3-dioxoisindolin-5-yl)amino)butanoic acid (Linker 67)

(781) ##STR00484##

(782) Linker 67 was synthesized following the same procedure as Linker 60 as described for

Example 60: (0.52 g, yield: 25%). ¹H NMR (400 MHz, DMSO-d₆) δ 12.12(s, 1H), 11.05 (s, 1H), 7.55 (d, J=8.4 Hz, 1H), 7.14 (t, J=4.8 Hz, 1H), 6.95 (d, J=2.0 Hz, 1H), 6.85 (dd, J=2.0 Hz, 8.4 Hz, 1H), 5.02(dd, J=5.6 Hz, 12.8 Hz, 1H), 3.21-3.16 (m, 2H), 2.91-2.83 (m, 1H), 2.60-2.51 (m, 2H), 2.34 (t, J=7.2 Hz, 2H), 2.01-1.97 (m, 1H), 1.82-1.75 (m, 2H). MS (ESI) m/z=360.1

[M+H].sup.+.

Example 137:5-((2-(2,6-Dioxopiperidin-3-yl)-1,3-dioxoisindolin-5-yl)amino)pentanoic acid (Linker 68)

(783) ##STR00485##

(784) Linker 68 was synthesized following the same procedure as Linker 60 as described for Example 60. (0.66 g, yield: 51%). ¹H NMR (400 MHz, DMSO-d₆) δ 12.03 (br, 1H), 11.05 (s, 1H), 7.55 (d, J=8.4 Hz, 1H), 7.10 (t, J=5.2 Hz, 1H), 6.94 (s, 1H), 6.83 (dd, J=1.6 Hz, 8.4 Hz, 1H), 5.02(dd, J=5.6 Hz, 12.8 Hz, 1H), 3.17-3.16 (m, 2H), 2.92-2.83 (m, 1H), 2.60-2.53 (m, 2H), 2.26-2.25 (m, 2H), 2.01-1.98 (m, 1H), 1.60-1.59 (m, 4H). MS (ESI) m/z=374. 1 [M+H].sup.+.

Example 138:6-((2-(2,6-Dioxopiperidin-3-yl)-1,3-dioxoisindolin-5-yl)amino)hexanoic acid (Linker 69)

(785) ##STR00486##

(786) Linker 69 was synthesized following the same procedure as Linker 60 as described for Example 60. (1.33 g, yield: 66%). ¹H NMR (400 MHz, DMSO-d₆) δ 11.98 (s, 1H), 11.05 (s, 1H), 7.55 (d, J=8.4 Hz, 1H), 7.08 (t, J=5.2 Hz, 1H), 6.95 (s, 1H), 6.83 (dd, J=1.2 Hz, 8.4 Hz, 1H), 5.03 (dd, J=5.2 Hz, 12.8 Hz, 1H), 3.17-3.12(m, 2H), 2.92-2.83 (m, 1H), 2.60-2.53 (m, 2H), 2.22(t, J=7.2 Hz, 2H), 2.01-1.98 (m, 1H), 1.61-1.51 (m, 4H), 1.41-1.33 (m, 2H). MS (ESI) m/z=388.1 [M+H].sup.+.

Example 139:7-((2-(2,6-Dioxopiperidin-3-yl)-1,3-dioxoisindolin-5-yl)amino) heptanoic acid (Linker 70)

(787) ##STR00487##

(788) Linker 70 was synthesized following the same procedure as Linker 60 as described for Example 60. (1.06 g, yield: 39%). ¹H NMR (400 MHz, DMSO-d₆) δ 11.94 (s, 1H), 11.04 (s, 1H), 7.55 (d, J=8.4 Hz, 1H), 7.09 (t, J=5.6 Hz, 1H), 6.94 (d, J=2.0 Hz, 1H), 6.84 (dd, J=2.0 Hz, 8.4 Hz, 1H), 5.02(dd, J=5.6 Hz, 13.2 Hz, 1H), 3.17-3.12(m, 2H), 2.88-2.83 (m, 1H), 2.60-2.53 (m, 2H), 2.21 (t, J=7.2 Hz, 2H), 2.01-1.97 (m, 1H), 1.58-1.48 (m, 4H), 1.39-1.29 (m, 4H). MS (ESI) m/z=402. 1 [M+H].sup.+.

Example 140:8-((2-(2,6-Dioxopiperidin-3-yl)-1,3-dioxoisindolin-5-yl)amino) octanoic acid (Linker 71)

(789) ##STR00488##

(790) Linker 71 was synthesized following the same procedure as Linker 60 as described for Example 60. (1.66 g, yield: 51%). ¹H NMR (400 MHz, DMSO-d₆) δ 11.95 (s, 1H), 11.05 (s, 1H), 7.55 (d, J=8.4 Hz, 1H), 7.09 (t, J=5.6 Hz, 1H), 6.94 (d, J=2.0 Hz, 1H), 6.84 (dd, J=2.0 Hz, 8.4 Hz, 1H), 5.02(dd, J=5.6 Hz, 13.2 Hz, 1H), 3.17-3.12(m, 2H), 2.88-2.83 (m, 1H), 2.60-2.53 (m, 2H), 2.19 (t, J=7.2 Hz, 2H), 2.02-1.98 (m, 1H), 1.58-1.47 (m, 4H), 1.36-1.29 (m, 6H). MS (ESI) m/z=416. 1 [M+H].sup.+.

Example 141:5-((2-Aminoethyl)amino)-2-(2,6-dioxopiperidin-3-yl)isoindoline-1,3-dione (Linker 72)

(791) ##STR00489##

(792) Linker 72 was synthesized following the same procedure as Linker 60 as described for Example 60. (1.74 g, yield: 80%). ¹H NMR (400 MHz, DMSO-d₆) δ 11.08 (s, 1H), 8.10 (s, 3H), 7.62(d, J=8.4 Hz, 1H), 7.33 (t, J=5.2 Hz, 1H), 7.05 (s, 1H), 6.94 (d, J=8.0 Hz, 1H), 5.07 (dd, J=5.2 Hz, 12.8 Hz, 1H), 3.50-3.49 (m, 2H), 3.03 (t, J=6.0 Hz, 2H), 2.95-2.86 (m, 1H), 2.63-2.57 (m, 2H), 2.05-2.02(m, 1H). MS (ESI) m/z=317.1 [M+H].sup.+.

Example 142:5-((3-Aminopropyl)amino)-2-(2,6-dioxopiperidin-3-yl)isoindoline-1,3-dione (Linker 73)

(793) ##STR00490##

(794) Linker 73 was synthesized following the same procedure as Linker 60 as described for Example 60. (1.3 g, yield: 57%). ¹H NMR (400 MHz, DMSO-d₆) δ 11.07 (s, 1H), 7.85 (br, 3H), 7.59 (d, J=8.4 Hz, 1H), 7.22(t, J=5.2 Hz, 1H), 6.98 (d, J=2.0 Hz, 1H), 6.88 (dd, J=2.0 Hz, 8.4 Hz, 1H), 5.04 (dd, J=5.6 Hz, 13.2 Hz, 1H), 3.29-3.25 (m, 2H), 2.91-2.85 (m, 3H), 2.60-2.53 (m, 2H), 2.02-1.98 (m, 1H), 1.87-1.81 (m, 2H). MS (ESI) m/z=331.1 [M+H]⁺.

Example 143:5-((4-Aminobutyl)amino)-2-(2,6-dioxopiperidin-3-yl)isoindoline-1,3-dione (Linker 74)

(795) ##STR00491##

(796) Linker 74 was synthesized following the same procedure as Linker 60 as described for Example 60. (2.9 g, yield: 85%). ¹H NMR (400 MHz, DMSO-d₆) δ 11.08 (s, 1H), 7.97 (br, 3H), 7.58 (d, J=8.4 Hz, 1H), 7.22(br, s, 1H), 6.99 (s, 1H), 6.89 (d, J=8.0 Hz, 1H), 5.05 (dd, J=5.2 Hz, 12.8 Hz, 1H), 3.22(s, 2H), 2.93-2.84 (m, 3H), 2.63-2.53 (m, 2H), 2.04-2.00 (m, 1H), 1.66 (s, 4H). MS (ESI) m/z=345.1 [M+H]⁺.

Example 144:5-((5-Aminopentyl)amino)-2-(2,6-dioxopiperidin-3-yl)isoindoline-1,3-dione (Linker 75)

(797) ##STR00492##

(798) Linker 75 was synthesized following the same procedure as Linker 60 as described for Example 60. (1.8 g, yield: 78%). ¹H NMR (400 MHz, DMSO-d₆) δ 11.09 (s, 1H), 7.89 (br, 3H), 7.57 (d, J=6.8 Hz, 1H), 7.17 (br, s, 1H), 6.96 (s, 1H), 6.86 (d, J=6.0 Hz, 1H), 5.05 (d, J=7.2 Hz, 1H), 3.19-3.15 (m, 2H), 2.89-2.70 (m, 3H), 2.61-2.51 (m, 2H), 2.01-1.90 (m, 1H), 1.62-1.56 (m, 4H), 1.45-1.40 (m, 2H). MS (ESI) m/z=359.1 [M+H]⁺.

Example 145:5-((6-Aminohexyl)amino)-2-(2,6-dioxopiperidin-3-yl)isoindoline-1,3-dione (Linker 76)

(799) ##STR00493##

(800) Linker 76 was synthesized following the same procedure as Linker 60 as described for Example 60. (1.8 g, yield: 62%). ¹H NMR (400 MHz, DMSO-d₆) δ 11.05 (s, 1H), 7.71 (br, 3H), 7.57 (d, J=8.4 Hz, 1H), 7.12(t, J=5.2 Hz, 1H), 6.94 (d, J=2.0 Hz, 1H), 6.85 (dd, J=2.0 Hz, 8.4 Hz, 1H), 5.03 (dd, J=5.2 Hz, 12.8 Hz, 1H), 3.17-3.16 (m, 2H), 2.88-2.77 (m, 3H), 2.60-2.53 (m, 2H), 2.01-1.98 (m, 1H), 1.59-1.51 (m, 4H), 1.37-1.36 (m, 4H). MS (ESI) m/z=373.1 [M+H]⁺.

Example 146:5-((7-Aminoheptyl)amino)-2-(2,6-dioxopiperidin-3-yl)isoindoline-1,3-dione (Linker 77)

(801) ##STR00494##

(802) Linker 77 was synthesized following the same procedure as Linker 60 as described for Example 60. (1.3 g, yield: 70%). ¹H NMR (400 MHz, DMSO-d₆) δ 11.05 (s, 1H), 7.72(br, 3H), 7.56 (d, J=8.4 Hz, 1H), 7.12(t, J=5.6 Hz, 1H), 6.94 (d, J=2.0 Hz, 1H), 6.85 (dd, J=2.4 Hz, 8.8 Hz, 1H), 5.03 (dd, J=5.6 Hz, 12.8 Hz, 1H), 3.18-3.14 (m, 2H), 2.92-2.76 (m, 3H), 2.60-2.51 (m, 2H), 2.01-1.98 (m, 1H), 1.59-1.51 (m, 4H), 1.36-1.32(m, 6H). MS (ESI) m/z=387.1 [M+H]⁺.

Example 147:5-(8-Amino-octyl)amino)-2-(2,6-dioxopiperidin-3-yl)isoindoline-1,3-dione (Linker 78)

(803) ##STR00495##

(804) Linker 78 was synthesized following the same procedure as Linker 60 as described for Example 60. (1.6 g, yield: 62%). ¹H NMR (400 MHz, DMSO-d₆) δ 11.05 (s, 1H), 7.73 (br, 3H), 7.56 (d, J=8.4 Hz, 1H), 7.14 (br, 1H), 6.94 (d, J=1.6 Hz, 1H), 6.85 (dd, J=2.0 Hz, 8.8 Hz, 1H), 5.03 (dd, J=5.6 Hz, 12.8 Hz, 1H), 3.15 (t, J=7.2 Hz, 2H), 2.89-2.83 (m, 1H), 2.80-2.75 (m, 2H), 2.60-2.54 (m, 2H), 2.02-1.98 (m, 1H), 1.59-1.51 (m, 4H), 1.37-1.30 (m, 8H), MS (ESI) m/z=401.1 [M+H]⁺.

Example 148:3-(2-((2-(2,6-Dioxopiperidin-3-yl)-1,3-dioxoisindolin-5-yl)amino)ethoxy)propanoic acid (Linker 79)

(805) ##STR00496##

(806) Linker 79 was synthesized following the same procedure as Linker 60 as described for Example 60. (1.7 g, yield: 60%). ¹H NMR (400 MHz, DMSO-d₆) δ 12.19 (br, 1H), 11.06 (s, 1H), 7.57 (d, J=8.4 Hz, 1H), 7.09 (br, 1H), 7.01 (d, J=2.0 Hz, 1H), 6.90 (dd, J=2.0 Hz, 8.4 Hz, 1H), 5.04 (dd, J=5.6 Hz, 13.2 Hz, 1H), 3.66 (t, J=6.4 Hz, 2H), 3.59 (t, J=5.6 Hz, 2H), 3.35 (t, J=5.2 Hz, 2H), 2.93-2.84 (m, 1H), 2.62-2.56 (m, 2H), 2.52-2.47 (m, 2H), 2.03-1.99 (m, 1H). MS (ESI) m/z=390.1 [M+H].⁺

Example 149: 3-(2-(2-((2-(2,6-Dioxopiperidin-3-yl)-1,3-dioxoisindolin-5-yl)amino)ethoxy)ethoxy)propanoic acid (Linker 80)

(807) ##STR00497##

(808) Linker 80 was synthesized following the same procedure as Linker 60 as described for Example 60. (2.3 g, yield: 78%). ¹H NMR (400 MHz, DMSO-d₆) δ 11.06 (s, 1H), 7.57 (d, J=8.4 Hz, 1H), 7.02 (d, J=2.0 Hz, 1H), 6.90 (dd, J=2.0 Hz, 8.4 Hz, 1H), 5.04 (dd, J=5.6 Hz, 13.2 Hz, 1H), 3.63-3.59 (m, 4H), 3.57-3.51 (m, 4H), 3.36 (t, J=5.6 Hz, 2H), 2.90-2.84 (m, 1H), 2.61-2.55 (m, 2H), 2.44 (t, J=6.4 Hz, 2H), 2.04-1.99 (m, 1H). MS (ESI) m/z=434.1 [M+H].⁺

Example 150: 3-(2-(2-(2-((2-(2,6-Dioxopiperidin-3-yl)-1,3-dioxoisindolin-5-yl)amino)ethoxy)ethoxy)ethoxy)propanoic acid (Linker 81)

(809) ##STR00498##

(810) Linker 81 was synthesized following the same procedure as Linker 60 as described for Example 60. (1.2 g, yield: 52%). ¹H NMR (400 MHz, DMSO-d₆) δ 7.59 (d, J=11.2 Hz, 1H), 7.23 (t, J=6.8 Hz, 1H), 7.04 (d, J=1.6 Hz, 1H), 7.04 (dd, J=2.4 Hz, 11.2 Hz, 1H), 5.06 (dd, J=7.2 Hz, 16.8 Hz, 1H), 3.64-3.57 (m, 8H), 3.54-3.48 (m, 4H), 3.40-3.38 (m, 2H), 2.92-2.89 (m, 1H), 2.64-2.54 (m, 2H), 2.42-2.38 (m, 2H), 2.05-2.01 (m, 1H). MS (ESI) m/z=478.1 [M+H].⁺

Example 151: 1-((2-(2,6-Dioxopiperidin-3-yl)-1,3-dioxoisindolin-5-yl)amino)-3,6,9,12-tetraoxapentadecan-15-oic acid (Linker 82)

(811) ##STR00499##

(812) Linker 82 was synthesized following the same procedure as Linker 60 as described for Example 60. (1.3 g, yield: 55%). ¹H NMR (400 MHz, DMSO-d₆) δ 12.17 (br, 1H), 11.07 (s, 1H), 7.56 (d, J=8.4 Hz, 1H), 7.17 (t, J=5.6 Hz, 1H), 7.01 (d, J=1.2 Hz, 1H), 6.90 (dd, J=1.6 Hz, 8.4 Hz, 1H), 5.03 (dd, J=5.6 Hz, 12.8 Hz, 1H), 3.61-3.48 (m, 18H), 2.92-2.83 (m, 1H), 2.60-2.54 (m, 2H), 2.43 (t, J=6.4 Hz, 2H), 2.03-1.98 (m, 1H). MS (ESI) m/z=522.1 [M+H].⁺

Example 152: 1-((2-(2,6-Dioxopiperidin-3-yl)-1,3-dioxoisindolin-5-yl)amino)-3,6,9,12,15-pentaoxaoctadecan-18-oic acid (Linker 83)

(813) ##STR00500##

(814) Linker 83 was synthesized following the same procedure as Linker 60 as described for Example 60. (1.0 g, yield: 50%). ¹H NMR (400 MHz, DMSO-d₆) δ 12.17 (br, s, 1H), 11.07 (s, 1H), 7.56 (d, J=8.0 Hz, 1H), 7.17 (t, J=5.6 Hz, 1H), 7.01 (s, 1H), 6.90 (dd, J=1.6 Hz, 8.4 Hz, 1H), 5.03 (dd, J=5.6 Hz, 13.2 Hz, 1H), 3.60-3.48 (m, 22H), 2.89-2.83 (m, 1H), 2.60-2.54 (m, 2H), 2.43 (t, J=6.4 Hz, 2H), 2.01-1.98 (m, 1H). MS (ESI) m/z=566.1 [M+H].⁺

Example 153: N-(8-((2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-4-yl)amino) octyl)-2-(4-(6-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)acetamide (TR-102)

(815) ##STR00501##

(816) TR-102 was synthesized following the standard procedure for preparing TR-059 (12 mg, yield 65%). MS (ESI) m/z: 884.7 [M+H].⁺

Example 154: N-(5-((2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-4-yl)amino)pentyl)-2-(4-(6-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)acetamide (TR-103)

(817) ##STR00502##

(818) TR-103 was synthesized following the standard procedure for preparing TR-059 (11 mg,

yield 63%). MS (ESI) m/z: 842.7 [M+H].sup.+.

Example 155: N-(17-((2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-4-yl)amino)-3,6,9,12,15-pentaoxaheptadecyl)-2-(4-(6-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)acetamide (TR-104)

(819) ##STR00503##

(820) TR-104 was synthesized following the standard procedure for preparing TR-059 (12 mg, yield 60%). MS (ESI) m/z: 1020.9 [M+H].sup.+.

Example 156: N-(2-(2-(2-((2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-4-yl)amino)ethoxy)ethoxy)ethyl)-2-(4-(6-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)acetamide (TR-105)

(821) ##STR00504##

(822) TR-105 was synthesized following the standard procedure for preparing TR-059 (11 mg, yield 63%). MS (ESI) m/z: 888.8 [M+H].sup.+.

Example 157: N-(2-(2-((2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-4-yl)amino)ethoxy)ethyl)-2-(4-(6-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)acetamide (TR-106)

(823) ##STR00505##

(824) TR-106 was synthesized following the standard procedure for preparing TR-059 (10 mg, yield 55%). MS (ESI) m/z: 844.6 [M+H].sup.+.

Example 158: N-(4-((2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-4-yl)amino)butyl)-2-(4-(6-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)acetamide (TR-107)

(825) ##STR00506##

(826) TR-107 was synthesized following the standard procedure for preparing TR-059 (11.5 mg, yield 62%). MS (ESI) m/z: 828.6 [M+H].sup.+.

Example 159: N-(14-((2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-4-yl)amino)-3,6,9,12-tetraoxatetradecyl)-2-(4-(6-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)acetamide (TR-108)

(827) ##STR00507##

(828) TR-108 was synthesized following the standard procedure for preparing TR-059 (13 mg, yield 64%). MS (ESI) m/z: 976.8 [M+H].sup.+.

Example 160: N-(3-((2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-4-yl)amino)propyl)-2-(4-(6-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)acetamide (TR-109)

(829) ##STR00508##

(830) TR-109 was synthesized following the standard procedure for preparing TR-059 (12 mg, yield 63%). MS (ESI) m/z: 814.6 [M+H].sup.+.

Example 161: N-(7-((2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-4-yl)amino)heptyl)-2-(4-(6-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)acetamide (TR-110)

(831) ##STR00509##

(832) TR-110 was synthesized following the standard procedure for preparing TR-059 (10 mg, yield 61%). MS (ESI) m/z: 870.7 [M+H].sup.+.

Example 162: N-(6-((2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-4-yl)amino)hexyl)-2-(4-(6-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)acetamide (TR-111)

(833) ##STR00510##

(834) TR-111 was synthesized following the standard procedure for preparing TR-059 (8.6 mg, yield 60%). MS (ESI) m/z: 856.7 [M+H].sup.+.

Example 163: 2-(2,6-Dioxopiperidin-3-yl)-5-((2-(2-(3-(4-(6-(6-((R)-2-(3-fluorophenyl)pyrrolidin-

1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)-3-oxopropoxy)ethoxy)ethyl)amino) isoindoline-1,3-dione (TR-113)

(835) ##STR00511##

(836) TR-113 was synthesized following the standard procedure for preparing TR-053 (7.7 mg, yield 59%). MS (ESI) m/z: 859.8 [M+H].sup.+.

Example 164:2-(2,6-Dioxopiperidin-3-yl)-5-((2-(2-(2-(3-(4-(6-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)-3-oxopropoxy)ethoxy)ethoxy)ethyl)amino) isoindoline-1,3-dione (TR-114)

(837) ##STR00512##

(838) TR-114 was synthesized following the standard procedure for preparing TR-053 (8.6 mg, yield 61%). MS (ESI) m/z: 903.9 [M+H].sup.+.

Example 165:2-(2,6-Dioxopiperidin-3-yl)-5-((5-(4-(6-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)-5-oxopentyl)amino) isoindoline-1,3-dione (TR-115)

(839) ##STR00513##

(840) TR-115 was synthesized following the standard procedure for preparing TR-053 (8.1 mg, yield 66%). MS (ESI) m/z: 799.8 [M+H].sup.+.

Example 166:2-(2,6-Dioxopiperidin-3-yl)-5-((18-(4-(6-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)-18-oxo-3,6,9,12,15-pentaoxaoctadecyl)amino) isoindoline-1,3-dione (TR-116)

(841) ##STR00514##

(842) TR-116 was synthesized following the standard procedure for preparing TR-053 (8.6 mg, yield 64%). MS (ESI) m/z: 991.6 [M+H].sup.+.

Example 167:2-(2,6-Dioxopiperidin-3-yl)-5-((7-(4-(6-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)-7-oxoheptyl)amino) isoindoline-1,3-dione (TR-117)

(843) ##STR00515##

(844) TR-117 was synthesized following the standard procedure for preparing TR-053 (8.8 mg, yield 61%). MS (ESI) m/z: 827.6 [M+H].sup.+.

Example 168:2-(2,6-Dioxopiperidin-3-yl)-5-((8-(4-(6-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)-8-oxooctyl)amino) isoindoline-1,3-dione (TR-118)

(845) ##STR00516##

(846) TR-118 was synthesized following the standard procedure for preparing TR-053 (8.8 mg, yield 61%). MS (ESI) m/z: 841.6 [M+H].sup.+.

Example 169:2-(2,6-Dioxopiperidin-3-yl)-5-((2-(3-(4-(6-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)-3-oxopropoxy)ethyl)amino) isoindoline-1,3-dione (TR-119)

(847) ##STR00517##

(848) TR-119 was synthesized following the standard procedure for preparing TR-053 (8.9 mg, yield 61%). MS (ESI) m/z: 815.5 [M+H].sup.+.

Example 170:2-(2,6-Dioxopiperidin-3-yl)-5-((15-(4-(6-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)-15-oxo-3,6,9,12-tetraoxapentadecyl)amino) isoindoline-1,3-dione (TR-120)

(849) ##STR00518##

(850) TR-120 was synthesized following the standard procedure for preparing TR-053 (8.9 mg, yield 61%). MS (ESI) m/z: 947.6 [M+H].sup.+.

Example 171:2-(2,6-Dioxopiperidin-3-yl)-5-((3-(4-(6-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)-3-oxopropyl)amino) isoindoline-1,3-dione (TR-121)

(851) ##STR00519##

(852) TR-121 was synthesized following the standard procedure for preparing TR-053 (10.2 mg, yield 63%). MS (ESI) m/z: 1542.5 [M+H].sup.+.

Example 172: 2-(2,6-Dioxopiperidin-3-yl)-5-((6-(4-(6-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)-6-oxohexyl)amino) isoindoline-1,3-dione (TR-122)

(853) ##STR00520##

(854) TR-122 was synthesized following the standard procedure for preparing TR-053 (10.6 mg, yield 62%). MS (ESI) m/z: 813.9 [M+H].sup.+.

Example 173: 2~ (2,6-Dioxopiperidin-3-yl)-5-((2~ (4-(6-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)-2-oxoethyl)amino) isoindoline-1,3-dione (TR-123)

(855) ##STR00521##

(856) TR-123 was synthesized following the standard procedure for preparing TR-053 (10.6 mg, yield 62%). MS (ESI) m/z: 1514.8 [M+H].sup.+.

Example 174: N-(4-((2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisoindolin-5-yl)amino)butyl)-2-(4-(6-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)acetamide (TR-124)

(857) ##STR00522##

(858) TR-124 was synthesized following the standard procedure for preparing TR-059 (9.3 mg, yield 61%). MS (ESI) m/z: 828.5 [M+H].sup.+.

Example 175: N-(14-((2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisoindolin-5-yl)amino)-3,6,9,12-tetraoxatetradecyl)-2-(4-(6-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)acetamide (TR-125)

(859) ##STR00523##

(860) TR-125 was synthesized following the standard procedure for preparing TR-059 (9.6 mg, yield 62%). MS (ESI) m/z: 976.8 [M+H].sup.+.

Example 176: N-(7-((2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisoindolin-5-yl)amino) heptyl)-2-(4-(6-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)acetamide (TR-126)

(861) ##STR00524##

(862) TR-126 was synthesized following the standard procedure for preparing TR-059 (9.9 mg, yield 63%). MS (ESI) m/z: 870.7 [M+H].sup.+.

Example 177: N-(3-((2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisoindolin-5-yl)amino)propyl)-2-(4-(6-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)acetamide (TR-127)

(863) ##STR00525##

(864) TR-127 was synthesized following the standard procedure for preparing TR-059 (9.3 mg, yield 61%). MS (ESI) m/z: 814.6 [M+H].sup.+.

Example 178: N-(6-((2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisoindolin-5-yl)amino)hexyl)-2-(4-(6-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)acetamide (TR-128)

(865) ##STR00526##

(866) TR-128 was synthesized following the standard procedure for preparing TR-059 (9.7 mg, yield 62%). MS (ESI) m/z: 856.7 [M+H].sup.+.

Example 179: N-(2-(2-(2-(2-((2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisoindolin-5-yl)amino)ethoxy)ethoxy)ethyl)-2-(4-(6-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)acetamide (TR-129)

(867) ##STR00527##

(868) TR-129 was synthesized following the standard procedure for preparing TR-059 (9.7 mg,

yield 62%). MS (ESI) m/z: 932.6 [M+H].sup.+.

Example 180: N-(2-((2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-5-yl)amino)ethyl)-2-(4-(6-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)acetamide (TR-130)

(869) ##STR00528##

(870) TR-130 was synthesized following the standard procedure for preparing TR-059 (9.3 mg, yield 62%). MS (ESI) m/z: 800.6 [M+H].sup.+.

Example 181: N-(5-((2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-5-yl)amino)pentyl)-2-(4-(6-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)acetamide (TR-131)

(871) ##STR00529##

(872) TR-131 was synthesized following the standard procedure for preparing TR-059 (10.5 mg, yield 65%). MS (ESI) m/z: 842.6 [M+H].sup.+.

Example 182: N-(17-((2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-5-yl)amino)-3,6,9,12,15-pentaoxaheptadecyl)-2-(4-(6-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)acetamide (TR-132)

(873) ##STR00530##

(874) TR-132 was synthesized following the standard procedure for preparing TR-059 (7.5 mg, yield 55%). MS (ESI) m/z: 1020.9 [M+H].sup.+.

Example 183: N-(8-((2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-5-yl)amino) octyl)-2-(4-(6-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)acetamide (TR-133)

(875) ##STR00531##

(876) TR-133 was synthesized following the standard procedure for preparing TR-059 (8.6 mg, yield 55%). MS (ESI) m/z: 884.7 [M+H].sup.+.

Example 184: N-(2-(2-(2-((2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-5-yl)amino)ethoxy)ethoxy)ethyl)-2-(4-(6-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)acetamide (TR-134)

(877) ##STR00532##

(878) TR-134 was synthesized following the standard procedure for preparing TR-059 (8.8 mg, yield 59%). MS (ESI) m/z: 888.7 [M+H].sup.+.

Example 185: N-(2-(2-((2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-5-yl)amino)ethoxy)ethyl)-2-(4-(6-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)acetamide (TR-135)

(879) ##STR00533##

(880) TR-135 was synthesized following the standard procedure for preparing TR-059 (8.1 mg, yield 56%). MS (ESI) m/z: 844.7 [M+H].sup.+.

Example 186: N-(5-(3,5-Difluorobenzyl)-1H-indazol-3-yl)-4-(4-(8-((2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-5-yl)amino) octanoyl)piperazin-1-yl)-2-((tetrahydro-2H-pyran-4-yl)amino)benzamide (TR-136)

(881) ##STR00534##

(882) TR-136 was synthesized following the standard procedure for preparing TR-053 (11 mg, yield 59%). MS (ESI) m/z: 944.4 [M+H].sup.+.

Example 187: N-(5-(3,5-Difluorobenzyl)-1H-indazol-3-yl)-4-(4-(1-((2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-5-yl)amino)-3,6,9, 12, 15-pentaoxaoctadecan-18-oyl)piperazin-1-yl)-2-((tetrahydro-2H-pyran-4-yl)amino)benzamide (TR-137)

(883) ##STR00535##

(884) TR-137 was synthesized following the standard procedure for preparing TR-053 (13 mg, yield 64%). MS (ESI) m/z: 1094.5 [M+H].sup.+.

Example 188: N-(5-(3,5-Difluorobenzyl)-1H-indazol-3-yl)-4-(4-(6-((2-(2,6-dioxopiperidin-3-

yl)-1,3-dioxoisindolin-5-yl)amino)hexanoyl)piperazin-1-yl)-2-((tetrahydro-2H-pyran-4-yl)amino)benzamide (TR-138)

(885) ##STR00536##

(886) TR-138 was synthesized following the standard procedure for preparing TR-053 (14 mg, yield 55%). MS (ESI) m/z: 916.4 [M+H].sup.+.

Example 189: N-(5-(3,5-Difluorobenzyl)-1H-indazol-3-yl)-4-(4-(7-((2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-5-yl)amino) heptanoyl)piperazin-1-yl)-2-((tetrahydro-2H-pyran-4-yl)amino)benzamide (TR-139)

(887) ##STR00537##

(888) TR-139 was synthesized following the standard procedure for preparing TR-053 (16 mg, yield 62%). MS (ESI) m/z: 930.4 [M+H].sup.+.

Example 190: N-(5-(3,5-Difluorobenzyl)-1H-indazol-3-yl)-4-(4-(1-((2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-5-yl)amino)-3,6,9,12-tetraoxapentadecan-15-oyl)piperazin-1-yl)-2-((tetrahydro-2H-pyran-4-yl)amino)benzamide (TR-140)

(889) ##STR00538##

(890) TR-140 was synthesized following the standard procedure for preparing TR-053 (12 mg, yield 56%). MS (ESI) m/z: 1050.4 [M+H].sup.+.

Example 191: N-(5-(3,5-Difluorobenzyl)-1H-indazol-3-yl)-4-(4-(3-(2-(2-((2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-5-yl)amino)ethoxy)ethoxy)propanoyl)piperazin-1-yl)-2-((tetrahydro-2H-pyran-4-yl)amino)benzamide (TR-141)

(891) ##STR00539##

(892) TR-141 was synthesized following the standard procedure for preparing TR-053 (15 mg, yield 65%). MS (ESI) m/z: 962.4 [M+H].sup.+.

Example 192: N-(5-(3,5-Difluorobenzyl)-1H-indazol-3-yl)-4-(4-(5-((2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-5-yl)amino)pentanoyl)piperazin-1-yl)-2-((tetrahydro-2H-pyran-4-yl)amino)benzamide (TR-142)

(893) ##STR00540##

(894) TR-142 was synthesized following the standard procedure for preparing TR-053 (12 mg, yield 62%). MS (ESI) m/z: 902.4 [M+H].sup.+.

Example 193: N-(5-(3,5-Difluorobenzyl)-1H-indazol-3-yl)-4-(4-(8-((2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-5-yl)amino) octanoyl)piperazin-1-yl)-2-((tetrahydro-2H-pyran-4-yl)amino)benzamide (TR-143)

(895) ##STR00541##

(896) TR-143 was synthesized following the standard procedure for preparing TR-053 (16 mg, yield 67%). MS (ESI) m/z: 918.4 [M+H].sup.+.

Example 194: N-(5-(3,5-Difluorobenzyl)-1H-indazol-3-yl)-4-(4-((2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-5-yl)glycyl)piperazin-1-yl)-2-((tetrahydro-2H-pyran-4-yl)amino)benzamide (TR-144)

(897) ##STR00542##

(898) TR-144 was synthesized following the standard procedure for preparing TR-053 (14 mg, yield 61%). MS (ESI) m/z: 860.3 [M+H].sup.+.

Example 195: N-(5-(3,5-Difluorobenzyl)-1H-indazol-3-yl)-4-(4-(3-((2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-5-yl)amino)propanoyl)piperazin-1-yl)-2-((tetrahydro-2H-pyran-4-yl)amino)benzamide (TR-145)

(899) ##STR00543##

(900) TR-145 was synthesized following the standard procedure for preparing TR-053 (15 mg, yield 62%). MS (ESI) m/z: 874.3 [M+H].sup.+.

Example 196: N-(5-(3,5-Difluorobenzyl)-1H-indazol-3-yl)-4-(4-(3-(2-(2-(2-((2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-5-yl)amino)ethoxy)ethoxy)ethoxy)propanoyl)piperazin-1-yl)-2-((tetrahydro-2H-pyran-4-yl)amino)benzamide (TR-146)

(901) ##STR00544##

(902) TR-146 was synthesized following the standard procedure for preparing TR-053 (16 mg, yield 65%). MS (ESI) m/z: 1006.4 [M+H].sup.+.

Example 197: N—(S-(3,5-Difluorobenzyl)-1H-indazol-3-yl)-4-(4-(2-((6-((2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-4-yl)amino)hexyl)amino)-2-oxoethyl)piperazin-1-yl)-2-((tetrahydro-2H-pyran-4-yl)amino)benzamide (TR-147)

(903) ##STR00545##

(904) TR-147 was synthesized following the standard procedure for preparing TR-053 (12 mg, yield 56%). MS (ESI) m/z: 959.4 [M+H].sup.+.

Example 198: N-(5-(3,5-Difluorobenzyl)-1H-indazol-3-yl)-4-(4-(2-((8-((2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-4-yl)amino) octyl)amino)-2-oxoethyl)piperazin-1-yl)-2-((tetrahydro-2H-pyran-4-yl)amino)benzamide (TR-148)

(905) ##STR00546##

(906) TR-148 was synthesized following the standard procedure for preparing TR-053 (16 mg, yield 62%). MS (ESI) m/z: 987.5 [M+H].sup.+.

Example 199: N-(5-(3,5-Difluorobenzyl)-1H-indazol-3-yl)-4-(4-(2-((5-((2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-4-yl)amino)pentyl)amino)-2-oxoethyl)piperazin-1-yl)-2-((tetrahydro-2H-pyran-4-yl)amino)benzamide (TR-149)

(907) ##STR00547##

(908) TR-149 was synthesized following the standard procedure for preparing TR-053 (14 mg, yield 52%). MS (ESI) m/z: 945.4 [M+H].sup.+.

Example 200: N-(5-(3,5-Difluorobenzyl)-1H-indazol-3-yl)-4-(4-(17-((2~ (2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-4-yl)amino)-2-oxo-6,9,12,15-tetraoxa-3-azaheptadecyl)piperazin-1-yl)-2~ ((tetrahydro-2H-pyran-4-yl)amino)benzamide (TR-150)

(909) ##STR00548##

(910) TR-150 was synthesized following the standard procedure for preparing TR-053 (14 mg, yield 58%). MS (ESI) m/z: 1079.5 [M+H].sup.+.

Example 201: N-(5-(3,5-Difluorobenzyl)-1H-indazol-3-yl)-4-(4-(2-((2-(2-((2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-4-yl)amino)ethoxy)ethyl)amino)-2-oxoethyl)piperazin-1-yl)-2-((tetrahydro-2H-pyran-4-yl)amino)benzamide (TR-151)

(911) ##STR00549##

(912) TR-151 was synthesized following the standard procedure for preparing TR-053 (12 mg, yield 53%). MS (ESI) m/z: 947.4 [M+H].sup.+.

Example 202: N-(5-(3,5-Difluorobenzyl)-1H-indazol-3-yl)-4-(4-(2-((3-((2-(2,6-dioxopiperidin-3,3-yl)-1,3-dioxoisindolin-4-yl)amino)propyl)amino)-2-oxoethyl)piperazin-1-yl)-2-((tetrahydro-2H-pyran-4-yl)amino)benzamide (TR-152)

(913) ##STR00550##

(914) TR-152 was synthesized following the standard procedure for preparing TR-053 (13 mg, yield 57%). MS (ESI) m/z: 917.4 [M+H].sup.+.

Example 203: N-(5-(3,5-Difluorobenzyl)-1H-indazol-3-yl)-4-(4-(2-((2-(2-(2-((2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-4-yl)amino)ethoxy)ethoxy)ethyl)amino)-2-oxoethyl)piperazin-1-yl)-2-((tetrahydro-2H-pyran-4-yl)amino)benzamide (TR-153)

(915) ##STR00551##

(916) TR-153 was synthesized following the standard procedure for preparing TR-053 (12 mg, yield 53%). MS (ESI) m/z: 991.4 [M+H].sup.+.

Example 204: N-(5-(3,5-Difluorobenzyl)-1H-indazol-3-yl)-4-(4-(2-((7-((2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-4-yl)amino) heptyl)amino)-2-oxoethyl)piperazin-1-yl)-2-((tetrahydro-2H-pyran-4-yl)amino)benzamide (TR-154)

(917) ##STR00552##

(918) TR-154 was synthesized following the standard procedure for preparing TR-053 (15 mg,

yield 57%). MS (ESI) m/z: 973.4 [M+H].sup.+.

Example 205: N-(5-(3,5-Difluorobenzyl)-1H-indazol-3-yl)-4-(4-(2-((4-((2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-4-yl)amino)butyl)amino)-2-oxoethyl)piperazin-1-yl)-2-((tetrahydro-2H-pyran-4-yl)amino)benzamide (TR-155)

(919) ##STR00553##

(920) TR-155 was synthesized following the standard procedure for preparing TR-053 (13 mg, yield 56%). MS (ESI) m/z: 931.4 [M+H].sup.+.

Example 206: N-(5-(3,5-Difluorobenzyl)-1H-indazol-3-yl)-4-(4-(2-((4-((2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-5-yl)amino)butyl)amino)-2-oxoethyl)piperazin-1-yl)-2-((tetrahydro-2H-pyran-4-yl)amino)benzamide (TR-156)

(921) ##STR00554##

(922) TR-156 was synthesized following the standard procedure for preparing TR-053 (12 mg, yield 53%). MS (ESI) m/z: 931.4 [M+H].sup.+.

Example 207: N-(5-(3,5-Difluorobenzyl)-1H-indazol-3-yl)-4-(4-(2-((7-((2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-5-yl)amino) heptyl)amino)-2-oxoethyl)piperazin-1-yl)-2-((tetrahydro-2H-pyran-4-yl)amino)benzamide (TR-157)

(923) ##STR00555##

(924) TR-157 was synthesized following the standard procedure for preparing TR-053 (14 mg, yield 62%). MS (ESI) m/z: 973.4 [M+H].sup.+.

Example 208: N—(S-(3,5-Difluorobenzyl)-1H-indazol-3-yl)-4-(4-(8-((2~(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-5-yl)amino) octanoyl)piperazin-1-yl)-2-((tetrahydro-2H-pyran-4-yl)amino)benzamide (TR-158)

(925) ##STR00556##

(926) TR-158 was synthesized following the standard procedure for preparing TR-053 (12 mg, yield 56%). MS (ESI) m/z: 944.4 [M+H].sup.+.

Example 209: N—(S-(3,5-Difluorobenzyl)-1H-indazol-3-yl)-4-(4-(2-((8-((2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-5-yl)amino) octyl)amino)-2-oxoethyl)piperazin-1-yl)-2-(tetrahydro-2H-pyran-4-yl)amino)benzamide (TR-159)

(927) ##STR00557##

(928) TR-159 was synthesized following the standard procedure for preparing TR-053 (14 mg, yield 58%). MS (ESI) m/z: 987.5 [M+H].sup.+.

Example 210: N-(5-(3,5-Difluorobenzyl)-1H-indazol-3-yl)-4-(4-(17-((2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-5-yl)amino)-2-oxo-6,9,12,15-tetraoxa-3-azaheptadecyl)piperazin-1-yl)-2-((tetrahydro-2H-pyran-4-yl)amino)benzamide (TR-160)

(929) ##STR00558##

(930) TR-160 was synthesized following the standard procedure for preparing TR-053 (13 mg, yield 56%). MS (ESI) m/z: 987.5 [M+H].sup.+.

Example 211: N-(5-(3,5-Difluorobenzyl)-1H-indazol-3-yl)-4-(4-(2-((2-(2-((2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-5-yl)amino)ethoxy)ethyl)amino)-2-oxoethyl)piperazin-1-yl)-2-((tetrahydro-2H-pyran-4-yl)amino)benzamide (TR-161)

(931) ##STR00559##

(932) TR-161 was synthesized following the standard procedure for preparing TR-053 (13 mg, yield 58%). MS (ESI) m/z: 947.4 [M+H].sup.+.

Example 212: N-(5-(3,5-Difluorobenzyl)-1H-indazol-3-yl)-4-(4-(2-((2-(2-(2-((2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-5-yl)amino)ethoxy)ethoxy)ethyl)amino)-2-oxoethyl)piperazin-1-yl)-2-((tetrahydro-2H-pyran-4-yl)amino)benzamide (TR-162)

(933) ##STR00560##

(934) TR-162 was synthesized following the standard procedure for preparing TR-053 (11 mg, yield 52%). MS (ESI) m/z: 9881.4 [M+H].sup.+.

Example 213: N—(S-(3,5-Difluorobenzyl)-1H-indazol-3-yl)-4-(4-(14-((2-(2,6-dioxopiperidin-3-

yl)-1,3-dioxoisindolin-5-yl)amino)-2-oxo-6,9,12-trioxa-3-azatetradecyl)piperazin-1-yl)-2~
((tetrahydro-2H-pyran-4-yl)amino)benzamide (TR-163)
(935) ##STR00561##

(936) TR-163 was synthesized following the standard procedure for preparing TR-053 (15 mg, yield 58%). MS (ESI) m/z: 1035.4 [M+H].sup.+.

Example 214: N-(5-(3,5-Difluorobenzyl)-1H-indazol-3-yl)-4-(4-(2-((5-((2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-5-yl)amino)pentyl)amino)-2-oxoethyl)piperazin-1-yl)-2-((tetrahydro-2H-pyran-4-yl)amino)benzamide (TR-164)

(937) ##STR00562##

(938) TR-164 was synthesized following the standard procedure for preparing TR-053 (13 mg, yield 56%). MS (ESI) m/z: 945.4 [M+H].sup.+.

Example 215: N-(5-(3,5-Difluorobenzyl)-1H-indazol-3-yl)-4-(4-(2-((2-((2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-5-yl)amino)ethyl)amino)-2-oxoethyl)piperazin-1-yl)-2-((tetrahydro-2H-pyran-4-yl)amino)benzamide (TR-165)

(939) ##STR00563##

(940) TR-165 was synthesized following the standard procedure for preparing TR-053 (12 mg, yield 53%). MS (ESI) m/z: 903.4 [M+H].sup.+.

Example 216: N-(5-(3,5-Difluorobenzyl)-1H-indazol-3-yl)-4-(4-(20-((2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-5-yl)amino)-2-oxo-6,9,12,15,18-pentaoxa-3-azaicosyl)piperazin-1-yl)-2-((tetrahydro-2H-pyran-4-yl)amino)benzamide (TR-166)

(941) ##STR00564##

(942) TR-166 was synthesized following the standard procedure for preparing TR-053 (14 mg, yield 56%). MS (ESI) m/z: 1223.5 [M+H].sup.+.

Example 217: N-(5-(3,5-Difluorobenzyl)-1H-indazol-3-yl)-4-(4-(2-((3-((2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-5-yl)amino)propyl)amino)-2-oxoethyl)piperazin-1-yl)-2-((tetrahydro-2H-pyran-4-yl)amino)benzamide (TR-167)

(943) ##STR00565##

(944) TR-167 was synthesized following the standard procedure for preparing TR-053 (14 mg, yield 55%). MS (ESI) m/z: 917.4 [M+H].sup.+.

Example 218: 2-(2,6-Dioxopiperidin-3-yl)-5-((4-(4-(6-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)-4-oxobutyl)amino) isoindoline-1,3-dione (TR-168)

(945) ##STR00566##

(946) TR-168 was synthesized following the standard procedure for preparing TR-053 (16 mg, yield 58%). MS (ESI) m/z: 785.3 [M+H].sup.+.

Example 219: N-(5-(3,5-Difluorobenzyl)-1H-indazol-3-yl)-4-(4-(4-((2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-5-yl)amino)butanoyl)piperazin-1-yl)-2-((tetrahydro-2H-pyran-4-yl)amino)benzamide (TR-169)

(947) ##STR00567##

(948) TR-169 was synthesized following the standard procedure for preparing TR-053 (14 mg, yield 56%). MS (ESI) m/z: 888.4 [M+H].sup.+.

Example 220: 2-(2,6-Dioxopiperidin-3-yl)-5-(4-(6-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)isoindoline-1,3-dione (TR-170)

(949) ##STR00568##

(950) A mixture of (R)-6-(2-(3-fluorophenyl)pyrrolidin-1-yl)-3-(6-(piperazin-1-yl)pyridine-2-yl)imidazo[1,2-b]pyridazine (30 mg, 0.06 mmol), 2-(2,6-dioxopiperidin-3-yl)-5-fluoroisoindoline-1,3-dione (15.8 mg, 0.06 mmol), DIEA (0.037 mL, 0.23 mmol) and NMP (0.5 mL) was heated to 120° C. under microwave for 1.5 h. The reaction mixture was purified by reverse phase chromatography to give the desired product (8.6 mg, 32% yield) as a light yellow solid. MS (ESI) m/z: 700.3 [M+H].sup.+.

Example 221: 2~ (2,6-Dioxopiperidin-3-yl)-5-((3-(4-(6-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)propyl)amino) isoindoline-1,3-dione (TR-171)

(951) ##STR00569##

(952) Step 1. Synthesis of tert-butyl (R)-(3-(4-(6-(6-(2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)propyl)carbamate

(953) ##STR00570##

(954) To a solution of (R)-6-(2-(3-fluorophenyl)pyrrolidin-1-yl)-3-(6-(piperazin-1-yl)pyridin-2-yl)imidazo[1,2-b]pyridazine (50 mg, 0.11 mmol) in DMAC (5 mL) were added Na.sub.2CO₃ (24 mg, 0.22 mmol) and tert-butyl (3-bromopropyl)carbamate (39 mg, 0.16 mmol), the resulting mixture was stirred at 80° C. for 8 h. The reaction was cooled to room temperature and H.sub.2O (50 mL) was added. The mixture was extracted with EtOAc (10 mL×3), the combined organic layers were concentrated and the residue was purified by reverse phase chromatography to the desired product (43 mg, 65% yield) as a pale brown solid. MS (ESI) m/z: 601.3 [M+H].sup.+.

(955) Step 2. Synthesis of (R)-3-(4-(6-(6-(2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)propan-1-amine hydrochloride

(956) ##STR00571##

(957) To a solution tert-butyl (R)-(3-(4-(6-(6-(2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)propyl)carbamate (40 mg, 0.07 mmol) in methanol (2 mL) was added HCl/dioxane (4 M, 2 mL) at room temperature, then it was stirred at room temperature for 5 h. The mixture was concentrated to get the crude product (35 mg, 98% yield) as a pale brown solid which was used directly in the next step. MS (ESI) m/z: 501.3 [M+H].sup.+.

(958) Step 3. Synthesis of 2-(2,6-dioxopiperidin-3-yl)-5-((3-(4-(6-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)propyl)amino) isoindoline-1,3-dione (TR-171)

(959) ##STR00572##

(960) A mixture of (R)-3-(4-(6-(6-(2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)propan-1-amine hydrochloride (35 mg, 0.07 mmol), 2-(2,6-dioxopiperidin-3-yl)-5-fluoroisoindoline-1,3-dione (15.8 mg, 0.06 mmol), DIEA (0.037 mL, 0.23 mmol) and NMP (0.5 mL) was heated to 120° C. under microwave for 1 h. The reaction mixture was purified by reverse phase chromatography to give the desired product (11.2 mg, 22% yield) as a light yellow solid. MS (ESI) m/z: 757.3 [M+H].sup.+.

Example 222: 2-(2,6-Dioxopiperidin-3-yl)-5-(3-(4-(6-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)propyl)isoindoline-1,3-dione (TR-172)

(961) ##STR00573##

(962) Step 1. Synthesis of 2-(2,6-dioxopiperidin-3-yl)-5-(3-hydroxyprop-1-yn-1-yl)isoindoline-1,3-dione

(963) ##STR00574##

(964) A mixture of 5-bromo-2-(2,6-dioxopiperidin-3-yl)isoindoline-1,3-dione (100 mg, 0.30 mmol), prop-2-yn-1-ol (25 mg, 0.45 mmol), Pd(dppf) C.sub.1 (22 mg, 0.03 mmol), CuI (5.7 mg, 0.03 mmol) and DIPEA (77 mg, 0.60 mmol) in THF (10 mL) was stirred at reflux for 8 h. The reaction mixture was concentrated and the residue was purified by reverse phase chromatography to give the desired product (68 mg, 72% yield) as a pale brown solid. MS (ESI) m/z: 313.3 [M+H].sup.+.

(965) Step 2. Synthesis of 2-(2,6-dioxopiperidin-3-yl)-5-(3-hydroxypropyl)isoindoline-1,3-dione

(966) ##STR00575##

(967) A mixture of 2-(2,6-dioxopiperidin-3-yl)-5-(3-hydroxyprop-1-yn-1-yl)isoindoline-1,3-dione (68 mg, 0.22 mmol) and Pd/C (10 mg) in methanol (5 mL) was stirred under H.sub.2(1 atm, balloon) at room temperature for 5 h. The reaction mixture was purified by reverse phase chromatography to give the desired product (55 mg, 81% yield) as a light yellow solid. MS (ESI)

m/z: 317.1 [M+H].sup.+,

(968) Step 3. Synthesis of 3-(2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-5-yl)propyl 4-methylbenzenesulfonate

(969) ##STR00576##

(970) To a solution 2-(2,6-dioxopiperidin-3-yl)-5-(3-hydroxypropyl)isoindoline-1,3-dione (55 mg, 0.17 mmol) and TEA (35 mg, 0.34 mmol) in DCM (5 mL) was added 4-methylbenzenesulfonyl chloride (39 mg, 0.20 mmol) at room temperature, then it was stirred at room temperature for 4 h. The mixture was concentrated and purified by reverse phase chromatography to give the desired product (42 mg, 52% yield) as a light yellow solid. MS (ESI) m/z: 471.1 [M+H].sup.+.

(971) Step 4. Synthesis of 2-(2,6-dioxopiperidin-3-yl)-5-(3-(4-(6-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)propyl)isoindoline-1,3-dione

(972) ##STR00577##

(973) A mixture of 3-(2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-5-yl)propyl 4-methylbenzenesulfonate (40 mg, 0.08 mmol), (R)-6-(2-(3-fluorophenyl)pyrrolidin-1-yl)-3-(6-(piperazin-1-yl)pyridin-2-yl)imidazo[1,2-b]pyridazine (37 mg, 0.08 mmol), K.sub.2CO.sub.3 (22 mg, 0.16 mmol) and NaI (1.5 mg, 0.01 mmol) in CH.sub.3CN (3 mL) were stirred at 80° C. for 5 h. LCMS showed the reaction was completed. The mixture was concentrated and purified by reverse phase chromatography to give the desired product (21 mg, 36% yield) as a light yellow solid. MS (ESI) m/z: 742.3 [M+H].sup.+.

Example 223: 2-(2,6-Dioxopiperidin-3-yl)-5-(2-(4-(6-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)ethoxy) isoindoline-1,3-dione (TR-173)

(974) ##STR00578##

(975) Step 1. Synthesis of 2-(2,6-dioxopiperidin-3-yl)-5-(2-hydroxyethoxy) isoindoline-1,3-dione

(976) ##STR00579##

(977) A mixture of 2-(2,6-dioxopiperidin-3-yl)-5-hydroxyisoindoline-1,3-dione (1 g, 3.65 mmol), 2-bromoethan-1-ol (452 mg, 3.65 mmol), NaHCO.sub.3 (613 mg, 7.30 mmol) and KI (605 mg, 3.65 mmol) in DMF (20 mL) were stirred at 100° C. for 16 h. The mixture was filtered and purified by reverse phase chromatography to give the desired product (340 mg, 29% yield) as a light yellow solid. MS (ESI) m/z: 319.1 [M+H].sup.+.

Step 3. Synthesis of 2-((2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-5-yl)oxy)ethyl 4-methylbenzenesulfonate

(978) ##STR00580##

(979) To a solution 2-(2,6-dioxopiperidin-3-yl)-5-(2-hydroxyethoxy) isoindoline-1,3-dione (200 mg, 0.63 mmol) and TEA (1.26 g, 1.26 mmol) in DCM (15 mL) was added 4-methylbenzenesulfonyl chloride (143 mg, 0.76 mmol) at room temperature, then it was stirred at room temperature for 4 h. The mixture was concentrated and purified by reverse phase chromatography to give the desired product (172 mg, 58% yield) as a light yellow solid. MS (ESI) m/z: 473.1 [M+H].sup.+.

Step 4. Synthesis of 2-(2,6-dioxopiperidin-3-yl)-5-(2-(4-(6-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)ethoxy) isoindoline-1,3-dione

(980) ##STR00581##

(981) A mixture of 2-((2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-5-yl)oxy)ethyl 4-methylbenzenesulfonate (40 mg, 0.08 mmol), (R)-6-(2-(3-fluorophenyl)pyrrolidin-1-yl)-3-(6-(piperazin-1-yl)pyridin-2-yl)imidazo[1,2-b]pyridazine (37 mg, 0.08 mmol), K.sub.2CO.sub.3 (22 mg, 0.16 mmol) and NaI (1.5 mg, 0.01 mmol) in CH.sub.3CN (3 mL) were stirred at 80° C. for 5 h. LCMS showed the reaction was completed. The mixture was concentrated and purified by reverse phase chromatography to give the desired product (16 mg, 42% yield) as a light yellow solid. MS (ESI) m/z: 744.3 [M+H].sup.+.

Example 224: Synthesis of 2-(2,6-dioxopiperidin-3-yl)-5-((1~(6-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperidin-4-yl)amino) isoindoline-1,3-dione (TR-174)

(982) ##STR00582##

(983) TR-174 was synthesized following the standard procedure for preparing TR-175. MS (ESI) m/z: 714.3 [M+H].sup.+.

Example 225: 2-(2,6-Dioxopiperidin-3-yl)-5-((1-(6-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)azetidin-3-yl)amino) isoindoline-1,3-dione (TR-175)

(984) ##STR00583##

Step 1. Synthesis of tert-butyl 3-((2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisoindolin-5-yl)amino) azetidine-1-carboxylate

(985) ##STR00584##

(986) To a solution of 2-(2,6-dioxopiperidin-3-yl)-5-fluoroisoindoline-1,3-dione (200 mg, 724.6 umol) in DMSO (3 mL) were added KF (126 mg, 2.17 mmol) and tert-butyl 3-aminoazetidine-1-carboxylate (373.9 mg, 2.17 mmol). The resulting mixture was stirred at 130° C. for 1 hr. After the Imide was totally consumed, the reaction was poured into water (20 mL) and extracted with ethyl acetate (3×10 mL). The combined organic layers were washed with saturated brine (20 mL), dried over anhydrous sodium sulfate, filtered and evaporated under reduced pressure. The resulting residue was purified by silica gel column chromatography to afford the desired product tert-butyl 3-((2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisoindolin-5-yl)amino) azetidine-1-carboxylate (80 mg, 26% yield) as a light yellow solid. MS (ESI) m/z: 429.2 [M+H].sup.+.

Step 2. Synthesis of 5-(azetidin-3-ylamino)-2-(2,6-dioxopiperidin-3-yl)isoindoline-1,3-dione

(987) ##STR00585##

(988) To a solution of tert-butyl 3-((2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisoindolin-5-yl)amino) azetidine-1-carboxylate (80 mg, 186.9 umol) in DCM (2 mL) was added TFA (2 mL). The resulting mixture was stirred at 25° C. for 5 hr. After the starting material was totally consumed, the reaction was evaporated under reduced pressure. The resulting residue was purified by reverse-phase chromatography to yield the desired product 5-(azetidin-3-ylamino)-2-(2,6-dioxopiperidin-3-yl)isoindoline-1,3-dione (56 mg, 91% yield) as a light yellow solid. MS (ESI) m/z: 329.2 [M+H].sup.+.

Step 3. Synthesis of 2-(2,6-dioxopiperidin-3-yl)-5-((1-(6-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)azetidin-3-yl)amino) isoindoline-1,3-dione

(989) ##STR00586##

(990) To a solution of 5-(azetidin-3-ylamino)-2-(2,6-dioxopiperidin-3-yl)isoindoline-1,3-dione (50 mg, 152.4 umol) in DMSO (3 mL) were added KF (26.5 mg, 457.3 umol) and (R)-6-(2-(3-fluorophenyl)pyrrolidin-1-yl)-3-(6-fluoropyridin-2-yl)imidazo[1,2-b]pyridazine (57.4 mg, 152.4 umol). The resulting mixture was stirred at 130° C. for 1 hr. After the Imide was totally consumed, the reaction was poured into water (20 mL) and extracted with ethyl acetate (3×10 mL). The combined organic layers were washed with saturated brine (20 mL), dried over anhydrous sodium sulfate, filtered and evaporated under reduced pressure. The resulting residue was purified by reverse-phase chromatography to afford the desired product 2-(2,6-dioxopiperidin-3-yl)-5-((1-(6-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)azetidin-3-yl)amino) isoindoline-1,3-dione (10 mg, 10% yield) as a light yellow solid. MS (ESI) m/z: 686.3 [M+H].sup.+.

Example 226: Synthesis of 2-(2,6-Dioxopiperidin-3-yl)-5-((2-(2-(4-(6-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)ethoxy)ethyl)amino) isoindoline-1,3-dione (TR-176)

(991) ##STR00587## ##STR00588##

(992) TR-176 was synthesized following the standard procedure for preparing TR-177. MS (ESI) m/z: 787.3 [M+H].sup.+.

Example 227: Synthesis of 2-(2,6-Dioxopiperidin-3-yl)-5-((2-(2-(2-(4-(6-(6-((S)-2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)ethoxy)ethoxy)ethyl)amino) isoindoline-1,3-dione (TR-177)

(993) ##STR00589## ##STR00590##

(994) Step 1. Synthesis of 2,2-dimethyl-4-oxo-3,8,11-trioxa-5-azatridecan-13-yl 4-methylbenzenesulfonate

(995) ##STR00591##

(996) To a solution of tert-butyl (2-(2-(2-hydroxyethoxy)ethoxy)ethyl)carbamate (200 mg, 803.2 umol) in DCM (5 mL) were added DMAP (9.7 mg, 80.3 umol), DIPEA (311 mg, 2.4 mmol) and TsCl (183.7 mg, 963.8 mmol). The resulting mixture was stirred at 25° C. for 15 hr. The reaction was poured into water (20 mL) and extracted with ethyl acetate (3×10 mL). The combined organic layers were washed with saturated brine (20 mL), dried over anhydrous sodium sulfate, filtered and evaporated under reduced pressure. The resulting residue was purified by reverse-phase chromatography to afford the desired product 2,2-dimethyl-4-oxo-3,8,11-trioxa-5-azatridecan-13-yl 4-methylbenzenesulfonate (230 mg, 71.2% yield) as a light yellow solid. MS (ESI) m/z: 404.2 [M+H].sup.+,

Step 2. Synthesis of tert-butyl(S)-(2-(2-(2-(4-(6-(6-(2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)ethoxy)ethoxy)ethyl)carbamate

(997) ##STR00592##

(998) To a solution of (S)-6-(2-(3-fluorophenyl)pyrrolidin-1-yl)-3-(6-(piperazin-1-yl)pyridin-2-yl)imidazo[1,2-b]pyridazine (50 mg, 112.8 umol) in CH.sub.3CN (5 mL) were added K2003 (46.7 mg, 338.6 umol) and NaI (1.5 mg, 11.2 mmol). The resulting mixture was stirred at 80° C. for 15 hr. The reaction was poured into water (20 mL) and extracted with ethyl acetate (3×10 mL). The combined organic layers were washed with saturated brine (20 mL), dried over anhydrous sodium sulfate, filtered and evaporated under reduced pressure. The resulting residue was purified by reverse-phase chromatography to afford the desired product tert-butyl(S)-(2-(2-(2-(4-(6-(6-(2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)ethoxy)ethoxy)ethyl)carbamate (30 mg, 39.6% yield) as a light yellow solid. MS (ESI) m/z: 675.4 [M+H].sup.+,

Step 3. Synthesis of (S)-2-(2-(2-(4-(6-(6-(2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)ethoxy)ethoxy)ethan-1-amine

(999) ##STR00593##

(1000) To a solution of tert-butyl(S)-(2-(2-(2-(4-(6-(6-(2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)ethoxy)ethoxy)ethyl)carbamate (20 mg, 29.6 umol) in DCM (2 mL) was added TFA (2 mL). The resulting mixture was stirred at 25° C. for 5 hr. After the starting material was totally consumed, the reaction was evaporated under reduced pressure. The resulting residue was purified by reverse-phase chromatography to yield the desired product (S)-2-(2-(2-(4-(6-(6-(2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)ethoxy)ethoxy)ethan-1-amine (15 mg, 88% yield) as a light yellow solid. MS (ESI) m/z: 575.3 [M+H].sup.+,

Step 4. Synthesis of 2-(2,6-dioxopiperidin-3-yl)-5-((2-(2-(2-(4-(6-(6-((S)-2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)ethoxy)ethoxy)ethyl)amino) isoindoline-1,3-dione (TR-177)

(1001) ##STR00594##

(1002) To a solution of tert-butyl(S)-2-(2-(2-(4-(6-(6-(2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)ethoxy)ethoxy)ethan-1-amine (20 mg, 34.9 umol) in NMP (1 mL) were added 2-(2,6-dioxopiperidin-3-yl)-5-fluoroisoindoline-1,3-dione (28.9 mg, 104.7 umol) and DIPEA (13 mg, 104.7 umol). Microwave reaction was stirred at 120° C. for 50 min. The resulting residue was purified by reverse-phase chromatography to yield the desired product 2-(2,6-dioxopiperidin-3-yl)-5-((2-(2-(2-(4-(6-(6-((S)-2-(3-fluorophenyl)pyrrolidin-

1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)ethoxy)ethoxy)ethyl)amino) isoindoline-1,3-dione (10 mg, 34.6% yield) as a light yellow solid. MS (ESI) m/z: 831.4 [M+H].sup.+.

Example 228: Synthesis of 2-(2,6-Dioxopiperidin-3-yl)-5-(((1-(6-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperidin-4-yl)methyl)amino) isoindoline-1,3-dione (TR-178)

(1003) ##STR00595##

(1004) TR-178 was synthesized following the standard procedure for preparing TR-175. MS (ESI) m/z: 728.3 [M+H].sup.+.

Example 229: Synthesis of 2-(2,6-Dioxopiperidin-3-yl)-5-(((1-(6-(6-((R)-2~ (3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)azetidin-3-yl)methyl)amino) isoindoline-1,3-dione (TR-179)

(1005) ##STR00596##

(1006) TR-179 was synthesized following the standard procedure for preparing TR-175. MS (ESI) m/z: 700.3 [M+H].sup.+.

Example 230: Synthesis of 2-(2,6-Dioxopiperidin-3-yl)-5-((2-(1-(6-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperidin-4-yl)ethyl)amino) isoindoline-1,3-dione (TR-180)

(1007) ##STR00597##

(1008) TR-180 was synthesized following the standard procedure for preparing TR-175. MS (ESI) m/z: 742.3 [M+H].sup.+.

Example 231: 2-(2,6-Dioxopiperidin-3-yl)-5-((2-(4-(6-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-ylethyl)amino) isoindoline-1,3-dione (TR-181)

(1009) ##STR00598##

(1010) Step 1. Synthesis of tert-butyl (R)-(2-(4-(6-(6-(2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)ethyl)carbamate

(1011) ##STR00599##

(1012) To a solution of (R)-6-(2-(3-fluorophenyl)pyrrolidin-1-yl)-3-(6-(piperazin-1-yl)pyridin-2-yl)imidazo[1,2-b]pyridazine hydrochloride (200 mg, 0.42 mmol) in DMAC (5 mL) were added Na.sub.2CO.sub.3 (133.56 mg, 1.26 mmol), NaI (126 mg, 0.84 mmol) and tert-butyl (2-bromoethyl)carbamate (186 mg, 0.83 mmol), the resulting mixture was stirred at 60° C. for 3 h. The reaction was cooled to room temperature and H.sub.2O (20 mL) was added. The mixture was extracted with EtOAc (10 mL*3), the combined organic layers were concentrated and the residue was purified by reverse phase chromatography to desired product (225 mg, 91% yield) as a white solid. MS (ESI) m/z: 587.9 [M+H].sup.+.

(1013) Step 2. Synthesis of (R)-2-(4-(6-(6-(2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)ethan-1-amine

(1014) ##STR00600##

(1015) To a solution tert-butyl (R)-(2-(4-(6-(6-(2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)ethyl)carbamate (225 mg, 0.38 mmol) in methanol (5 mL) was added HCl/dioxane (4 M, 4 mL) at room temperature, then it was stirred at room temperature for 12 h. The mixture was concentrated to get crude product (220 mg, 99% yield) as a white solid which was used directly in the next step. MS (ESI) m/z: 487.7 [M+H].sup.+.

(1016) Step 3. Synthesis of 2-(2,6-Dioxopiperidin-3-yl)-5-((2-(4-(6-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)ethyl)amino) isoindoline-1,3-dione (TR-181)

(1017) ##STR00601##

(1018) A mixture of (R)-2-(4-(6-(6-(2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)ethan-1-amine (30 mg, 0.06 mmol), 2-(2,6-dioxopiperidin-3-yl)-5-

fluoroisoindoline-1,3-dione (15.8 mg, 0.06 mmol), DIEA (0.037 mL, 0.23 mmol) and NMP (0.3 mL) was heated to 120° C. under microwave for 2 h. The reaction mixture was purified by reverse phase chromatography to give desired product (6.5 mg, 15% yield) as a white solid. MS (ESI) m/z: 743.8 [M+H].sup.+.

Example 232: 2-(2,6-Dioxopiperidin-3-yl)-5-(4-(4-(6-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)piperidin-1-yl)isoindoline-1,3-dione (TR-182)

(1019) ##STR00602##

(1020) Step 1. Synthesis of tert-butyl (R)-4-(4-(6-(6-(2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)piperidine-1-carboxylate

(1021) ##STR00603##

(1022) The mixture of (R)-6-(2-(3-fluorophenyl)pyrrolidin-1-yl)-3-(6-(piperazin-1-yl)pyridin-2-yl)imidazo[1,2-b]pyridazine (0.050 g, 112.73 umol), tert-butyl 4-oxopiperidine-1-carboxylate (44.92 mg, 225.47 umol) and AcOH (one drop) in DCE (3 mL) was stirred for 1 h. Then it was added NaBH₃CN (22.62 mg, 338.20 umol) and stirred for another 16 h. The solvent was removed under vacuum to give the crude product, which was purified by Prep-TLC (MeOH/DCM=5/100) to give the desired product (8 mg, 11% yield) as a yellow oil. MS (ESI) m/z: 627.8 [M+H].sup.+.

Step 2. Synthesis of (R)-6-(2-(3-fluorophenyl)pyrrolidin-1-yl)-3-(6-(4-(piperidin-4-yl)piperazin-1-yl)pyridin-2-yl)imidazo[1,2-b]pyridazine

(1023) ##STR00604##

(1024) To a solution of tert-butyl (R)-4-(4-(6-(6-(2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)piperidine-1-carboxylate (8 mg, 12.76 umol) in DCM (5 mL) was added TFA (14.55 mg, 127.64 umol), then the reaction was stirred for 2 h. LCMS showed the reaction was completed. The solvent was removed under vacuum to give the crude product (6.72 mg, 99% yield) which was used for next step without further purification. MS (ESI) m/z: 527.6 [M+H].sup.+.

Step 3. Synthesis of 2-(2,6-Dioxopiperidin-3-yl)-5-(4-(4-(6-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)piperidin-1-yl)isoindoline-1,3-dione (TR-182)

(1025) ##STR00605##

(1026) The mixture of (R)-6-(2-(3-fluorophenyl)pyrrolidin-1-yl)-3-(6-(4-(piperidin-4-yl)piperazin-1-yl)pyridin-2-yl)imidazo[1,2-b]pyridazine (6.72 mg, 11.39 umol), 2-(2,6-dioxo-3-piperidyl)-5-fluoro-isoindoline-1,3-dione (3.15 mg, 11.39 umol) and DIEA (2.94 mg, 22.79 umol) in NMP (0.5 mL) were heated at 120° C. for 0.5 h under microwave. The mixture was purified by Prep-TLC (DCM/MeOH=100/5) to give the desired product (1.8 mg, 20% yield) as a white solid. MS (ESI) m/z: 783.8 [M+H].sup.+.

Example 233: 2-(2,6-Dioxopiperidin-3-yl)-5-(3-(4-(6-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)prop-1-yn-1-yl)isoindoline-1,3-dione (TR-183)

(1027) ##STR00606##

Step 1. Synthesis of 3-(2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisoindolin-5-yl)prop-2-yn-1-yl 4-methylbenzenesulfonate

(1028) ##STR00607##

(1029) To a solution 2-(2,6-dioxopiperidin-3-yl)-5-(3-hydroxyprop-1-yn-1-yl)isoindoline-1,3-dione (50 mg, 0.16 mmol) and TEA (32 mg, 0.32 mmol) in DCM (3 mL) was added 4-methylbenzenesulfonyl chloride (36 mg, 0.19 mmol) at room temperature, then it was stirred at room temperature for 4 h. The mixture was concentrated and purified by reverse phase chromatography to give the desired product (43 mg, 58% yield) as a light yellow solid. MS (ESI) m/z: 467.1 [M+H].sup.+.

Step 2. Synthesis of 2-(2,6-Dioxopiperidin-3-yl)-5-(3-(4-(6-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-

yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)prop-1-yn-1-yl)isoindoline-1,3-dione (1030) ##STR00608##

(1031) A mixture of 3-(2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-5-yl)prop-2-yn-1-yl 4-methylbenzenesulfonate (40 mg, 0.08 mmol), (R)-6-(2-(3-fluorophenyl)pyrrolidin-1-yl)-3-(6-(piperazin-1-yl)pyridin-2-yl)imidazo[1,2-b]pyridazine (37 mg, 0.08 mmol), K₂CO₃ (22 mg, 0.16 mmol) and NaI (1.5 mg, 0.01 mmol) in CH₃CN (3 mL) were stirred at 80° C. for 5 h. LCMS showed the reaction was completed. The mixture was concentrated and purified by reverse phase chromatography to give the desired product (18 mg, 28% yield) as a light yellow solid. MS (ESI) m/z 738.3 [M+H]⁺.

Example 234: 2-(2,6-dioxopiperidin-3-yl)-5-(3-(4-(6-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)azetidin-1-yl)isoindoline-1,3-dione (TR-184)

(1032) ##STR00609##

Step 1. Synthesis of tert-butyl (R)-3-(4-(6-(6-(2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)azetidine-1-carboxylate

(1033) ##STR00610##

(1034) To a solution of (R)-6-(2-(3-fluorophenyl)pyrrolidin-1-yl)-3-(6-(piperazin-1-yl)pyridin-2-yl)imidazo[1,2-b]pyridazine (50 mg, 112.73 μmol) and tert-butyl 3-oxoazetidine-1-carboxylate (38.60 mg, 225.47 μmol) in MeOH (1 mL), AcOH (1 mL) and DCE (2 mL) was added NaBH₃CN (14.20 mg, 225.47 μmol) at room temperature. The resulting reaction mixture was stirred at room temperature for 48 h, concentrated and purified by silica gel chromatography (DCM/MeOH=30/1) to give desired product (40 mg, 74% yield) as a yellow oil. MS (ESI) m/z: 599.6 [M+H]⁺.

Step 2. Synthesis of (R)-3-(6-(4-(azetidin-3-yl)piperazin-1-yl)pyridin-2-yl)-6-(2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazine

(1035) ##STR00611##

(1036) To a solution of tert-butyl (R)-3-(4-(6-(6-(2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)azetidine-1-carboxylate (60 mg, 100.22 μmol) in DCM (2 mL) was added TFA (228.53 mg, 2.00 mmol) at room temperature. The reaction mixture was stirred at room temperature for 2 h, and then it was concentrated. The residue was dissolved in DCM (10 mL), washed with Na₂CO₃ (aq), dried over Na₂SO₄, concentrated to give desired product (50 mg, 99% yield) as a yellow resin. MS (ESI) m/z: 499.5 [M+H]⁺.

Step 3. Synthesis of 2-(2,6-Dioxopiperidin-3-yl)-5-(3-(4-(6-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)azetidin-1-yl)isoindoline-1,3-dione (TR-184)

(1037) ##STR00612##

(1038) To a solution of (R)-3-(6-(4-(azetidin-3-yl)piperazin-1-yl)pyridin-2-yl)-6-(2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazine (50 mg, 100.28 μmol) and 2-(2,6-dioxopiperidin-3-yl)-5-fluoroisoindoline-1,3-dione (27.70 mg, 100.28 μmol) in DMSO (2 mL) was added DIEA (38.81 mg, 300.84 μmol) at room temperature. The reaction mixture was warmed to 110° C. and stirred for 5 h. After cooling to room temperature, the reaction mixture was purified by prep-HPLC to give desired product (60 mg, 79% yield) as a yellow solid. MS (ESI) m/z: 755.7 [M+H]⁺.

Example 235: 3-(6-(3-(4-(6-(6-((R)-2-(3-Fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)propyl)-1-oxoisindolin-2-yl)piperidine-2,6-dione (TR-185)

(1039) ##STR00613##

Step 1. Synthesis of 3-(6-(3-hydroxypropyl)-1-oxoisindolin-2-yl)piperidine-2,6-dione

(1040) ##STR00614##

(1041) A mixture of 3-(6-(3-hydroxyprop-1-yn-1-yl)-1-oxoisindolin-2-yl)piperidine-2,6-dione (100 mg, 0.34 mmol) and Pd/C (20 mg) in methanol (5 mL) was stirred under H₂ (1 atm, balloon) at room temperature for 3 h. The reaction mixture was purified by reverse phase

chromatography to give the desired product (86 mg, 86% yield) as a light yellow solid. MS (ESI) m/z: 303.1 [M+H].sup.+.

Step 2. Synthesis of 3-(2-(2,6-dioxopiperidin-3-yl)-3-oxoisindolin-5-yl)propyl 4-methylbenzenesulfonate

(1042) ##STR00615##

(1043) To a solution 3-(6-(3-hydroxypropyl)-1-oxoisindolin-2-yl)piperidine-2,6-dione (80 mg, 0.26 mmol) and TEA (52 mg, 0.52 mmol) in DCM (5 mL) was added 4-methylbenzenesulfonyl chloride (59 mg, 0.31 mmol) at room temperature, then it was stirred at room temperature for 2 b. The mixture was concentrated and purified by reverse phase chromatography to give the desired product (61 mg, 51% yield) as a light yellow solid. MS (ESI) m/z: 457.1 [M+H].sup.+.

Step 3. Synthesis of 3-(6-(3-(4-(6-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)propyl)-1-oxoisindolin-2-yl)piperidine-2,6-dione

(1044) ##STR00616##

(1045) A mixture of 3-(2-(2,6-dioxopiperidin-3-yl)-3-oxoisindolin-5-yl)propyl 4-methylbenzenesulfonate (40 mg, 0.08 mmol), (R)-6-(2-(3-fluorophenyl)pyrrolidin-1-yl)-3-(6-(piperazin-1-yl)pyridin-2-yl)imidazo[1,2-b]pyridazine (37 mg, 0.08 mmol), K.sub.2CO.sub.3 (22 mg, 0.16 mmol) and NaI (1.5 mg, 0.01 mmol) in CH.sub.3CN (3 mL) were stirred at 80° C. for 5 h. LCMS showed the reaction was completed. The mixture was concentrated and purified by reverse phase chromatography to give the desired product (23 mg, 36% yield) as a light yellow solid. MS (ESI) m/z: 728.3 [M+H].sup.+.

Example 236:3-(5-(3-(4-(6-(6-((R)-2-(3-Fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)propyl)-1-oxoisindolin-2-yl)piperidine-2,6-dione (TR-186)

(1046) ##STR00617##

Step 1. Synthesis of 3-(5-(3-hydroxypropyl)-1-oxoisindolin-2-yl)piperidine-2,6-dione

(1047) ##STR00618##

(1048) A mixture of 3-(5-(3-hydroxyprop-1-yn-1-yl)-1-oxoisindolin-2-yl)piperidine-2,6-dione (100 mg, 0.34 mmol) and Pd/C (20 mg) in methanol (5 mL) was stirred under H.sub.2(1 atm, balloon) at room temperature for 3 h. The reaction mixture was purified by reverse phase chromatography to give the desired product (78 mg, 78% yield) as a light yellow solid. MS (ESI) m/z: 303.1 [M+H].sup.+.

Step 2. Synthesis of 3-(2-(2,6-dioxopiperidin-3-yl)-1-oxoisindolin-5-yl)propyl 4-methylbenzenesulfonate

(1049) ##STR00619##

(1050) To a solution 3-(5-(3-hydroxypropyl)-1-oxoisindolin-2-yl)piperidine-2,6-dione (80 mg, 0.26 mmol) and TEA (52 mg, 0.52 mmol) in DCM (5 mL) was added 4-methylbenzenesulfonyl chloride (59 mg, 0.31 mmol) at room temperature, then it was stirred at room temperature for 2 h. The mixture was concentrated and purified by reverse phase chromatography to give the desired product (67 mg, 54% yield) as a light yellow solid. MS (ESI) m/z: 457.1 [M+H].sup.+.

Step 3. Synthesis of 3-(5-(3-(4-(6-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)propyl)-1-oxoisindolin-2-yl)piperidine-2,6-dione

(1051) ##STR00620##

(1052) A mixture of 3-(2-(2,6-dioxopiperidin-3-yl)-1-oxoisindolin-5-yl)propyl 4-methylbenzenesulfonate (40 mg, 0.08 mmol), (R)-6-(2-(3-fluorophenyl)pyrrolidin-1-yl)-3-(6-(piperazin-1-yl)pyridin-2-yl)imidazo[1,2-b]pyridazine (37 mg, 0.08 mmol), K.sub.2CO.sub.3 (22 mg, 0.16 mmol) and NaI (1.5 mg, 0.01 mmol) in CH.sub.3CN (3 mL) were stirred at 80° C. for 5 h. LCMS showed the reaction was completed. The mixture was concentrated and purified by reverse phase chromatography to give the desired product (26 mg, 38% yield) as a light yellow solid. MS (ESI) m/z: 728.3 [M+H].sup.+.

Example 237:2-(2,6-Dioxopiperidin-3-yl)-5-(3-(4-(6-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)prop-1~yn-1-yl)isoindoline-1,3-dione (TR-187)

(1053) ##STR00621##

Step 1. Synthesis of 3-(2-(2,6-dioxopiperidin-3-yl)-3-oxoisindolin-5-yl)prop-2-yn-1-yl 4-methylbenzenesulfonate

(1054) ##STR00622##

(1055) To a solution 3-(6-(3-hydroxyprop-1-yn-1-yl)-1-oxoisindolin-2-yl)piperidine-2,6-dione (75 mg, 0.26 mmol) and TEA (52 mg, 0.52 mmol) in DCM (5 mL) was added 4-methylbenzenesulfonyl chloride (59 mg, 0.31 mmol) at room temperature, then it was stirred at room temperature for 2 h. The mixture was concentrated and purified by reverse phase chromatography to give the desired product (54 mg, 46% yield) as a light yellow solid. MS (ESI) m/z: 453.1 [M+H].sup.+.

Step 2. Synthesis of 3-(6-(3-(4-(6-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)prop-1-yn-1-yl)-1-oxoisindolin-2-yl)piperidine-2,6-dione

(1056) ##STR00623##

(1057) A mixture of 3-(2-(2,6-dioxopiperidin-3-yl)-3-oxoisindolin-5-yl)prop-2-yn-1-yl 4-methylbenzenesulfonate (40 mg, 0.08 mmol), (R)-6-(2-(3-fluorophenyl)pyrrolidin-1-yl)-3-(6-(piperazin-1-yl)pyridin-2-yl)imidazo[1,2-b]pyridazine (37 mg, 0.08 mmol), K.sub.2CO.sub.3 (22 mg, 0.16 mmol) and NaI (1.5 mg, 0.01 mmol) in CH.sub.3CN (3 mL) were stirred at 80° C. for 5 h. LCMS showed the reaction was completed. The mixture was concentrated and purified by reverse phase chromatography to give the desired product (29 mg, 39% yield) as a light yellow solid. MS (ESI) m/z: 724.3 [M+H].sup.+.

Example 238:3-(5-(3-(4-(6-(6-((R)-2-(3-Fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)prop-1-yn-1-yl)-1-oxoisindolin-2-yl)piperidine-2,6-dione (TR-188)

(1058) ##STR00624##

(1059) Step 1. Synthesis of 3-(2-(2,6-dioxopiperidin-3-yl)-1-oxoisindolin-5-yl)prop-2-yn-1-yl 4-methylbenzenesulfonate

(1060) ##STR00625##

(1061) To a solution 3-(5-(3-hydroxyprop-1-yn-1-yl)-1-oxoisindolin-2-yl)piperidine-2,6-dione (75 mg, 0.26 mmol) and TEA (52 mg, 0.52 mmol) in DCM (5 mL) was added 4-methylbenzenesulfonyl chloride (59 mg, 0.31 mmol) at room temperature, then it was stirred at room temperature for 2 h. The mixture was concentrated and purified by reverse phase chromatography to give the desired product (58 mg, 47% yield) as a light yellow solid. MS (ESI) m/z: 453.1 [M+H].sup.+.

Step 2. Synthesis of 3-(5-(3-(4-(6-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)prop-1-yn-1-yl)-1-oxoisindolin-2-yl)piperidine-2,6-dione (TR-188)

(1062) ##STR00626##

(1063) A mixture of 3-(2-(2,6-dioxopiperidin-3-yl)-1-oxoisindolin-5-yl)prop-2-yn-1-yl 4-methylbenzenesulfonate (40 mg, 0.08 mmol), (R)-6-(2-(3-fluorophenyl)pyrrolidin-1-yl)-3-(6-(piperazin-1-yl)pyridin-2-yl)imidazo[1,2-b]pyridazine (37 mg, 0.08 mmol), K.sub.2CO.sub.3 (22 mg, 0.16 mmol) and NaI (1.5 mg, 0.01 mmol) in CH.sub.3CN (3 mL) were stirred at 80° C. for 5 h. LCMS showed the reaction was completed. The mixture was concentrated and purified by reverse phase chromatography to give the desired product (21 mg, 36% yield) as a light yellow solid. MS (ESI) m/z: 724.3 [M+H].sup.+.

Example 239:3-(6-((2-(4-(6-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)-2-oxoethyl)amino)-1-oxoisindolin-2-yl)piperidine-2,6-dione (TR-189)

(1064) ##STR00627##

Step 1. Synthesis of (2-(2,6-dioxopiperidin-3-yl)-3-oxoisindolin-5-yl)glycine

(1065) ##STR00628##

(1066) To a mixture of To the mixture of 3-(6-amino-1-oxoisindolin-2-yl)piperidine-2,6-dione (130 mg, 501.43 umol) and oxaldehydic acid (55.69 mg, 752.14 umol) in MeOH (5 mL) and THF (1 mL) was added Pd/C (10%, 10 mg) at room temperature, the resulting mixture was stirred at 45° C. under H.sub.2 for 16 h and monitored by LCMS. The reaction mixture was filtrated and concentrated to give crude product which was slurried (EtOAc/MeOH=5/1, 5 mL), filtrated to give desired product (150 mg, 94% yield) as a brown solid. MS (ESI) m/z: 318.3 [M+H].sup.+.

Step 2. Synthesis of 3-(6-((2-(4-(6-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)-2-oxoethyl)amino)-1-oxoisindolin-2-yl)piperidine-2,6-dione (TR-189)

(1067) ##STR00629##

(1068) To a solution of (R)-6-(2-(3-fluorophenyl)pyrrolidin-1-yl)-3-(6-(piperazin-1-yl)pyridin-2-yl)imidazo[1,2-b]pyridazine (4.4 mg, 9.92 umol) and (2-(2,6-dioxopiperidin-3-yl)-3-oxoisindolin-5-yl)glycine (3.15 mg, 9.92 umol) in DMSO (2 mL) were added HATU (4.52 mg, 11.90 umol) and DIEA (2.56 mg, 19.84 umol) at RT. The reaction mixture was stirred at room temperature for 16 h, then it was quenched with H.sub.2O (10 mL) and extracted with DCM (10 mL*3). The combined organic layers were washed with aqueous NaCl (10 mL*2), dried over Na.sub.2SO.sub.4, concentrated and purified by Prep-TLC (DCM/MeOH=10/1) to give desired product (6 mg, 81% yield) as a yellow solid. MS (ESI) m/z: 743.8 [M+H].sup.+.

Example 240:3-(5-((2-(4-(6-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)-2-oxoethyl)amino)-1-oxoisindolin-2-yl)piperidine-2,6-dione (TR-190)

(1069) ##STR00630##

Step 1. Synthesis of (2-(2,6-dioxopiperidin-3-yl)-1-oxoisindolin-5-yl)glycine

(1070) ##STR00631##

(1071) To a solution of 3-(5-amino-1-oxoisindolin-2-yl)piperidine-2,6-dione (130 mg, 501.43 umol) and oxaldehydic acid (55.69 mg, 752.14 umol) in MeOH (5 mL) and THF (1 mL) was added Pd/C (6.09 mg, 50.14 umol) at rt. the resulting mixture was stirred at 45° C. under H.sub.2 for 16 h and monitored by LCMS. The reaction mixture was filtrated and concentrated to give crude product which was slurried (EtOAc/MeOH=5/1, 5 mL), filtrated to desired product (130 mg, 81.712% yield) as a brown solid. MS (ESI) m/z: 318.3 [M+H].sup.+.

Step 2. Synthesis of 3-(5-((2-(4-(6-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)-2-oxoethyl)amino)-1-oxoisindolin-2-yl)piperidine-2,6-dione (TR-190)

(1072) ##STR00632##

(1073) To a solution of (R)-6-(2-(3-fluorophenyl)pyrrolidin-1-yl)-3-(6-(piperazin-1-yl)pyridin-2-yl)imidazo[1,2-b]pyridazine (4.4 mg, 9.92 umol) and (2-(2,6-dioxopiperidin-3-yl)-1-oxoisindolin-5-yl)glycine (3.15 mg, 9.92 umol) in DMSO (2 mL) was added HATU (4.52 mg, 11.90 umol) and DIEA (2.56 mg, 19.84 umol) at RT. The reaction mixture was stirred at room temperature for 16 h, then it was quenched with H.sub.2O (10 mL) and extracted with DCM (10 mL*3). The combined organic layers were washed with aqueous NaCl (10 mL*2), dried over Na.sub.2SO.sub.4, concentrated and purified by Prep-TLC (DCM/MeOH=10/1) to give desired product (5.8 mg, 79% yield) as a yellow solid. MS (ESI) m/z: 743.9 [M+H].sup.+.

Example 241:3-(5-((2-(4-(6-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)ethyl)amino)-1-oxoisindolin-2-yl)piperidine-2,6-dione (TR-191)

(1074) ##STR00633##

Step 1. Synthesis of (R)-3-(6-(4-(2,2-dimethoxyethyl)piperazin-1-yl)pyridin-2-yl)-6-(2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazine

(1075) ##STR00634##

(1076) To a suspension of (R)-6-(2-(3-fluorophenyl)pyrrolidin-1-yl)-3-(6-(piperazin-1-yl)pyridin-2-yl)imidazo[1,2-b]pyridazine (200 mg, 450.94 umol) in CH.sub.3CN (5 mL) were added 2-bromo-

1,1-dimethoxy-ethane (114.32 mg, 676.41 μmol) and K.sub.2CO.sub.3 (124.46 mg, 901.88 μmol) at room temperature under the N₂, the resulting mixture was warmed to 75° C. and stirred for 16 h. Then it was cooled to room temperature and concentrated, the residue was purified by silica gel chromatography (DCM/MeOH=20/1-10/1) to give desired product (200 mg, 83% yield) as a yellow solid. MS (ESI) m/z: 532.7 [M+H].sup.+.

Step 2. Synthesis of (R)-2-(4-(6-(6-(2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)acetaldehyde

(1077) ##STR00635##

(1078) To a solution of (R)-3-(6-(4-(2,2-dimethoxyethyl)piperazin-1-yl)pyridin-2-yl)-6-(2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazine (250 mg, 470.26 μmol) in dioxane (6 mL) was added conc. HCl (3 mL) at RT, the reaction mixture was warmed to 50° C. and stirred for 16 h. Then it was concentrated to give crude product (220 mg, 96% yield) as a yellow solid which was used directly in the next step without further purification. MS (ESI) m/z: 504.5 [M+18+H].sup.+.

Step 3. Synthesis of 3-(5-((2-(4-(6-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)ethyl)amino)-1-oxoisindolin-2-yl)piperidine-2,6-dione (TR-191)

(1079) ##STR00636##

(1080) To a solution of (R)-2-(4-(6-(6-(2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)acetaldehyde (48 mg, 98.86 μmol) and 3-(6-amino-1-oxoisindolin-2-yl)piperidine-2,6-dione (38.44 mg, 148.28 μmol) in MeOH (1 mL), DCE (2 mL) and AcOH (1 mL) was added NaBH₃CN (12.46 mg, 197.71 μmol) at RT. The reaction mixture was stirred at room temperature for 16 h, then it was purified by prep-HPLC to give 10 mg crude product which was further purified by prep-TLC (DCM/MeOH=10/1) to desired product (1.4 mg, 2% yield) as a yellow solid. MS (ESI) m/z: 729.7 [M+H].sup.+.

Example 242: Synthesis of 2-(2,6-Dioxopiperidin-3-yl)-5-((2-((1-(6-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)azetidin-3-yl)amino)ethyl)amino) isoindoline-1,3-dione (TR-192)

(1081) ##STR00637##

Step 1. Synthesis of tert-butyl (R)-(1-(6-(6-(2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)azetidin-3-yl)carbamate

(1082) ##STR00638##

(1083) To a solution of (R)-6-(2-(3-fluorophenyl)pyrrolidin-1-yl)-3-(6-fluoropyridin-2-yl)imidazo[1,2-b]pyridazine (50 mg, 132.6 μmol) in DMSO (3 mL) were added KF (23 mg, 397.8 μmol) and tert-butyl azetidin-3-ylcarbamate (68.2 mg, 397.8 μmol). The resulting mixture was stirred at 130° C. for 1 hr. After the Imide was totally consumed, the reaction was poured into water (20 mL) and extracted with ethyl acetate (3×10 mL). The combined organic layers were washed with saturated brine (20 mL), dried over anhydrous sodium sulfate, filtered and evaporated under reduced pressure. The resulting residue was purified by reverse-phase chromatography to afford the desired product tert-butyl (R)-(1-(6-(6-(2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)azetidin-3-yl)carbamate (21 mg, 30% yield) as a light yellow solid. MS (ESI) m/z: 530.3 [M+H].sup.+.

Step 2. Synthesis of (R)-1-(6-(6-(2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)azetidin-3-amine

(1084) ##STR00639##

(1085) To a solution of tert-butyl (R)-(1-(6-(6-(2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)azetidin-3-yl)carbamate (100 mg, 188.9 μmol) in DCM (2 mL) was added TFA (2 mL). The resulting mixture was stirred at 25° C. for 5 hr. After the starting material was totally consumed, the reaction was evaporated under reduced pressure. The resulting residue was purified by reverse-phase chromatography to yield the desired product (R)-1-(6-(6-(2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)azetidin-3-amine (73 mg,

91% yield) as a light yellow solid. MS (ESI) m/z: 430.2 [M+H].sup.+.

Step 3. Synthesis of tert-butyl (R)-(2-((1-(6-(6-(2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)azetidin-3-yl)amino)ethyl)carbamate

(1086) ##STR00640##

(1087) To a solution of (R)-1-(6-(6-(2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)azetidin-3-amin (100 mg, 232.9 umol) in DMAC (2 mL) were added Na.sub.2CO.sub.3 (74 mg, 698.8 umol), NaI (34.7 mg, 232.9 umol) and tert-butyl (2-bromoethyl)carbamate (155 mg, 698.8 umol). The resulting mixture was stirred at 60° C. for 15 hr. After the starting material was totally consumed, the reaction was evaporated under reduced pressure. The resulting residue was purified by reverse-phase chromatography to yield the desired product tert-butyl (R)-(2-((1-(6-(6-(2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)azetidin-3-yl)amino)ethyl)carbamate (39 mg, 30% yield) as a light yellow solid. MS (ESI) m/z: 573.3 [M+H].sup.+.

Step 4. Synthesis of (R)—NI-(1-(6-(6-(2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)azetidin-3-yl)ethane-1,2-diamine

(1088) ##STR00641##

(1089) To a solution of tert-butyl (R)-(2-((1-(6-(6-(2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)azetidin-3-yl)amino)ethyl)carbamate (39 mg, 69.6 umol) in DCM (2 mL) was added TFA (2 mL). The resulting mixture was stirred at 25° C. for 5 hr. After the starting material was totally consumed, the reaction was evaporated under reduced pressure. The resulting residue was purified by reverse-phase chromatography to yield the desired product (R)-N1-(1-(6-(6-(2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)azetidin-3-yl)ethane-1,2-diamine (25 mg, 76% yield) as a light yellow solid. MS (ESI) m/z: 473.2 [M+H].sup.+.

Step 5. Synthesis of 2~ (2,6-dioxopiperidin-3-yl)-5-((2-((1-(6-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)azetidin-3-yl)amino)ethyl)amino) isoindoline-1,3-dione (TR-192)

(1090) ##STR00642##

(1091) To a solution of (R)—NI-(1-(6-(6-(2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)azetidin-3-yl)ethane-1,2-diamine (20 mg, 42.3 umol) in DMSO (3 mL) were added KF (7.4 mg, 127.1 umol) and 2-(2,6-dioxopiperidin-3-yl)-5-fluoroisoindoline-1,3-dione (35 mg, 127.1 umol). The resulting mixture was stirred at 130° C. for 1 hr. After the Primary amine was totally consumed, the reaction was poured into water (20 mL) and extracted with ethyl acetate (3×10 mL). The combined organic layers were washed with saturated brine (20 mL), dried over anhydrous sodium sulfate, filtered and evaporated under reduced pressure. The resulting residue was purified by reverse-phase chromatography to afford the desired product 2-(2,6-dioxopiperidin-3-yl)-5-((2-((1-(6-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)azetidin-3-yl)amino)ethyl)amino) isoindoline-1,3-dione (9.2 mg, 30% yield) as a light yellow solid. MS (ESI) m/z: 729.3 [M+H].sup.+.

Example 243: Synthesis of 2-(2,6-dioxopiperidin-3-yl)-5-((2-(1-(6-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)azetidin-3-yl)ethyl)amino) isoindoline-1,3-dione (TR-193)

(1092) ##STR00643##

(1093) TR-193 was synthesized following the standard procedure for preparing TR-175. MS (ESI) m/z: 714.3 [M+H].sup.+.

Example 244: 2-(2,6-Dioxopiperidin-3-yl)-5-(3-(4-(6-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)-3-oxopropyl)isoindoline-1,3-dione (TR-194)

(1094) ##STR00644##

Step 1. Synthesis of (E)-3-(2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisoindolin-5-yl)acrylic acid

(1095) ##STR00645##

(1096) A mixture of 5-bromo-2-(2,6-dioxopiperidin-3-yl)isoindoline-1,3-dione (200 mg, 0.60 mmol), acrylic acid (65 mg, 0.90 mmol), Pd(PPh₃)₄ (69 mg, 0.06 mmol) and TEA (182 mg, 1.8 mmol) in DMSO (15 mL) was stirred at 90° C. for 16 h. The reaction mixture was concentrated and the residue was purified by reverse phase chromatography to give the desired product (82 mg, 42% yield) as a pale brown solid. MS (ESI) m/z: 327.1 [M-H]⁺.

Step 2. Synthesis of 3-(2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-5-yl)propanoic acid

(1097) ##STR00646##

(1098) A mixture of (E)-3-(2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-5-yl)acrylic acid (50 mg, 0.15 mmol) and Pd/C (10 mg) in methanol (5 mL) was stirred under H₂ (1 atm, balloon) at room temperature for 3 h. The reaction mixture was purified by reverse phase chromatography to give the desired product (31 mg, 62% yield) as a light brown solid. MS (ESI) m/z: 329.1 [M-H]⁺.

Step 3. Synthesis of 2-(2,6-dioxopiperidin-3-yl)-5-(3-(4-(6-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)-3-oxopropyl)isoindoline-1,3-dione (TR-194)

(1099) ##STR00647##

(1100) A mixture of 3-(2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-5-yl)propanoic acid (20 mg, 0.06 mmol), (R)-6-(2-(3-fluorophenyl)pyrrolidin-1-yl)-3-(6-(piperazin-1-yl)pyridin-2-yl)imidazo[1,2-b]pyridazine (27 mg, 0.06 mmol), EDCI (17 mg, 0.09 mmol), NMM (30 mg, 0.30 mmol) and HOAT (12 mg, 0.09 mmol) in DMSO (1.5 mL) were stirred at room temperature for 16 h. LCMS showed the reaction was completed. The mixture was concentrated and purified by reverse phase chromatography to give the desired product (16 mg, 36% yield) as a light yellow solid. MS (ESI) m/z: 756.3 [M+H]⁺.

Example 245: 2-(2,6-Dioxopiperidin-3-yl)-5-((E)-3-(4-(6-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)-3-oxoprop-1-en-1-yl)isoindoline-1,3-dione (TR-195)

(1101) ##STR00648##

(1102) A mixture of (E)-3-(2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-5-yl)acrylic acid (20 mg, 0.06 mmol), (R)-6-(2-(3-fluorophenyl)pyrrolidin-1-yl)-3-(6-(piperazin-1-yl)pyridin-2-yl)imidazo[1,2-b]pyridazine (27 mg, 0.06 mmol), EDCI (17 mg, 0.09 mmol), NMM (30 mg, 0.30 mmol) and HOAT (12 mg, 0.09 mmol) in DMSO (1.5 mL) were stirred at room temperature for 16 h. LCMS showed the reaction was completed. The mixture was concentrated and purified by reverse phase chromatography to give the desired product (21 mg, 46% yield) as a light yellow solid. MS (ESI) m/z: 754.3 [M+H]⁺.

Example 246: 3-(6-((2-(4-(6-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)ethyl)amino)-1-oxoisindolin-2-yl)piperidine-2,6-dione (TR-196)

(1103) ##STR00649##

(1104) To a solution of (R)-2-(4-(6-(6-(2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)acetaldehyde (5 mg, 10.30 μmol) and 3-(6-amino-1-oxoisindolin-2-yl)piperidine-2,6-dione (3 mg, 11.57 μmol) in DMF (1 mL) were added TMSCl (3.34 mg, 30.89 μmol) and NaBH₄ (782.61 μg, 20.59 μmol) at 0° C. The reaction mixture was stirred at 0° C. for another 1.5h, then it was warmed to room temperature and stirred for 16 h. The reaction mixture was purified by prep-HPLC to give 5 mg of crude product which was further purified by prep-TLC (DMC/MeOH=10/1) to give desired product (2.15 mg, 29% yield) as a white solid. MS (ESI) m/z: 729.8 [M+H]⁺.

Example 247: 2-(2,6-Dioxopiperidin-3-yl)-5-((4-(6-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)methyl)isoindoline-1,3-dione (TR-197)

(1105) ##STR00650##

Step 1. Synthesis of 2-(2,6-dioxopiperidin-3-yl)-5-methyl]isoindoline-1,3-dione

(1106) ##STR00651##

(1107) The mixture of isobenzofuran-1,3-dione (15 g, 101.27 mmol), 3-aminopiperidine-2,6-dione (19.46 g, 151.91 mmol) and NaOAc (101.27 mmol) were stirred in AcOH (300 mL) at 100° C. for 16 h. LCMS showed the reaction was completed. The solvent was removed under vacuum to give the crude product which was used for next step without further purification (25 g, 90% yield). MS (ESI) m/z: 273.4 [M+H].sup.+.

Step 2. Synthesis of 5-(bromomethyl)-2-(2,6-dioxopiperidin-3-yl)isoindoline-1,3-dione

(1108) ##STR00652##

(1109) The mixture of 2-(2,6-dioxopiperidin-3-yl)-5-methyl]isoindoline-1,3-dione (1 g, 3.67 mmol), NBS (719.10 mg, 4.04 mmol) and BPO (45.41 mg, 734.60 umol) in CH.sub.3CN (50 mL) were stirred at 90° C. for 4 h. The solvent was removed under vacuum to give the crude product which was purified by silica gel chromatography (petroleum ether/EtOAc=5/1) to give the desired product as an off-white solid (1 g, 58% yield). MS (ESI) m/z: 353.3 [M+H].sup.+.

Step 3. Synthesis of 2-(2,6-dioxopiperidin-3-yl)-5-((4-(6-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)methyl)isoindoline-1,3-dione (TR-197)

(1110) ##STR00653##

(1111) To a solution of (R)-6-(2-(3-fluorophenyl)pyrrolidin-1-yl)-3-(6-(piperazin-1-yl)pyridin-2-yl)imidazo[1,2-b]pyridazine (0.020 g, 45.09 umol) in DCM (5 mL) was added 5-(bromomethyl)-2-(2,6-dioxopiperidin-3-yl)isoindoline-1,3-dione (47.50 mg, 135.28 umol), then the reaction was stirred for 16 h. LCMS showed the reaction was completed. The mixture was purified by Prep-TLC (DCM/MeOH=100/2) to give the desired product as an off-white solid (1.9 mg, 6% yield). MS (ESI) m/z: 714.7 [M+H].sup.+.

Example 248: 2-(2,6-Dioxopiperidin-3-yl)-5-(3-((4-(6-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)methyl)azetidin-1-yl)isoindoline-1,3-dione (TR-198)

(1112) ##STR00654##

Step 1. Synthesis of 2-(2,6-dioxopiperidin-3-yl)-5-(3-(hydroxymethyl)azetidin-1-yl)isoindoline-1,3-dione

(1113) ##STR00655##

(1114) The mixture of 2-(2,6-dioxopiperidin-3-yl)-5-fluoroisoindoline-1,3-dione (2 g, 7.24 mmol), azetidin-3-ylmethanol (693.88 mg, 7.96 mmol) and TEA (2.20 g, 21.72 mmol) were stirred in DMF (10 mL) at 85° C. for 1 h. LCMS showed the reaction was completed. The mixture was extracted with EtOAc, washed with brine, dried over Na.sub.2SO.sub.4, concentrated to give the crude product which was purified by reverse phase chromatography to give the desired product as a yellow solid (800 mg, 32% yield). MS (ESI) m/z: 344.2 [M+H].sup.+.

Step 2. Synthesis of (1-(2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisoindolin-5-yl)azetidin-3-yl)methyl 4-methylbenzenesulfonate

(1115) ##STR00656##

(1116) To a solution of 2-(2,6-dioxopiperidin-3-yl)-5-(3-(hydroxymethyl)azetidin-1-yl)isoindoline-1,3-dione (0.034 g, 99.03 umol) and DMAP (24.20 mg, 198.06 umol) in pyridine (2 mL) was added TsCl (13.97 mg, 198.06 umol), then the reaction was stirred for 4 h. The mixture was purified by silica gel chromatography (petroleum ether/EtOAc=1/1) to give the product as a yellow solid which was further purified by reverse phase chromatography to give the purified product as a yellow solid (25 mg, 51% yield). MS (ESI) m/z: 498.4 [M+H].sup.+.

Step 3. Synthesis of 2-(2,6-Dioxopiperidin-3-yl)-5-(3-((4-(6-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)methyl)azetidin-1-yl)isoindoline-1,3-dione (TR-198)

(1117) ##STR00657##

(1118) The mixture of (R)-6-(2-(3-fluorophenyl)pyrrolidin-1-yl)-3-(6-(piperazin-1-yl)pyridin-2-yl)imidazo[1,2-b]pyridazine (0.015 g, 33.82 μ mol), (1-(2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-5-yl)azetidin-3-yl)methyl 4-methylbenzenesulfonate (21.00 mg, 42.21 μ mol), K.sub.2CO.sub.3 (14.00 mg, 101.46 μ mol) and Nal (10.14 mg, 67.64 μ mol) in CH.sub.3CN (3 mL) were stirred at 82° C. for 16 h. LCMS showed the reaction was completed. The mixture was purified by reverse phase chromatography to give the desired product as a yellow solid (12 mg, 46% yield). MS (ESI) m/z: 769.8 [M+H].sup.+.

Example 249: 2-(2,6-Dioxopiperidin-3-yl)-5-((3-(2-(4-(6-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)ethoxy)propyl)amino) isoindoline-1,3-dione (TR-199)

(1119) ##STR00658##

Step 1. Synthesis of 2-(2-cyanoethoxy)ethyl 4-methylbenzenesulfonate

(1120) ##STR00659##

(1121) To a solution 3-(2-hydroxyethoxy)propanenitrile (100 mg, 868.58 μ mol) in DCM (5 mL) were added C.sub.4-methylbenzenesulfonyl chloride (250 mg, 1.31 mmol) and DMAP (21.22 mg, 173.72 μ mol) DIPEA (336.77 mg, 2.61 mmol). The resulting solution was stirred at 10° C. overnight. The resulting residue was purified by reverse-phase chromatography to yield the desired product as a white solid (178 mg, 76% yield). MS (ESI) m/z: 270.8 [M+H].sup.+.

Step 2. Synthesis of (R)-3-(2-(4-(6-(6-(2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)ethoxy)propanenitrile

(1122) ##STR00660##

(1123) A solution (R)-6-(2-(3-fluorophenyl)pyrrolidin-1-yl)-3-(6-(piperazin-1-yl)pyridin-2-yl)imidazo[1,2-b]pyridazine (108 mg, 243.51 μ mol) in ACN (5 mL) were added C.sub.2-(2-cyanoethoxy)ethyl 4-methylbenzenesulfonate (178 mg, 660.93 μ mol) and Nal (11.9 mg, 79.39 μ mol) K2CO.sub.3 (165 mg, 1.20 mmol). The resulting solution was stirred at 80° C. overnight. The resulting residue was purified by reverse-phase chromatography to yield the desired product as a white solid (100 mg, 76% yield). MS (ESI) m/z: 541.5 [M+H].sup.+.

Step 3. Synthesis of (R)-3-(2-(4-(6-(6-(2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)ethoxy)propan-1-amine

(1124) ##STR00661##

(1125) A solution (R)-3-(2-(4-(6-(6-(2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)ethoxy)propanenitrile (100 mg, 184.97 μ mol) in MeOH (5 mL) were added NaBH4 (130 mg, 184.97 μ mol) CoCl (50 mg, 184.97 μ mol). The resulting solution was stirred at 0° C. overnight. The resulting residue was quenched by ammonia, concentrated, diluted with 10 ml 1% HCl and extracted with DCM/MeOH=10/1 (10 mL). The organic phase was concentrated and the residue was purified by reverse-phase chromatography to yield the desired product as a white solid (20 mg, 20% yield). MS (ESI) m/z: 545.6 [M+H].sup.+.

Step 4. Synthesis of 2-(2,6-dioxopiperidin-3-yl)-5-((3-(2-(4-(6-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)ethoxy)propyl)amino) isoindoline-1,3-dione (TR-199)

(1126) ##STR00662##

(1127) A solution (R)-3-(2-(4-(6-(6-(2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)ethoxy)propan-1-amine (20 mg, 36.72 μ mol) in DMSO (1 mL) were added KF (2.13 mg, 36.72 μ mol) and 2-(2,6-dioxo-3-piperidyl)-5-fluoro-isoindoline-1,3-dione (10 mg, 36.20 μ mol). The resulting solution was stirred at 105° C. for 8 h. The resulting residue was purified by reverse-phase chromatography to yield the desired product as a white solid (5 mg, 17% yield). MS (ESI) m/z: 801.8 [M+H].sup.+.

Example 250: 3-(5-(2-(4-(6-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)ethoxy)-1-oxoisindolin-2-yl)piperidine-2,6-dione (TR-200)

(1128) ##STR00663##

(1129) TR-200 was synthesized following the standard procedure for preparing TR-201. MS (ESI) m/z: 730.7 [M+H].sup.+.

Example 251: 3-(6-(2-(4-(6-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)ethoxy)-1-oxoisindolin-2-yl)piperidine-2,6-dione (TR-201)

(1130) ##STR00664##

Step 1. Synthesis of (R)-2-(4-(6-(6-(2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)ethan-1-ol

(1131) ##STR00665##

(1132) A mixture of (R)-6-(2-(3-fluorophenyl)pyrrolidin-1-yl)-3-(6-(piperazin-1-yl)pyridin-2-yl)imidazo[1,2-b]pyridazine (500 mg, 1.12 mmol), 2-bromoethan-1-ol (211.6 mg, 1.69 mmol), K.sub.2CO.sub.3 (463 mg, 3.36 mmol) and NaI (169.3 mg, 1.12 mmol) in DMF (20 mL) were stirred at 100° C. for 16 h. The mixture was filtered and purified by reverse phase chromatography to give the desired product (159 mg, 29% yield). MS (ESI) m/z: 488.3 [M+H].sup.+.

Step 2. Synthesis of (R)-2-(4-(6-(6-(2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)ethyl 4-methylbenzenesulfonate

(1133) ##STR00666##

(1134) To a solution (R)-2~ (4-(6-(6-(2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)ethan-1-ol (159 mg, 0.327 mmol) and TEA (99 mg, 0.98 mmol) in DCM (15 mL) was added 4-methylbenzenesulfonyl chloride (93 mg, 0.49 mmol) at room temperature, then it was stirred at room temperature for 4 h. The mixture was concentrated and purified by reverse phase chromatography to give the desired product (121 mg, 58% yield) as white solid. MS (ESI) m/z: 642.1 [M+H].sup.+.

Step 3. Synthesis of 3-(6-(2-(4-(6-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)ethoxy)-1-oxoisindolin-2-yl)piperidine-2,6-dione

(1135) ##STR00667##

(1136) A mixture of (R)-2-(4-(6-(6-(2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)ethyl 4-methylbenzenesulfonate (100 mg, 0.15 mmol), 3-(6-hydroxy-1-oxoisindolin-2-yl)piperidine-2,6-dione (60.8 mg, 0.23 mmol), K.sub.2CO3 (61.6 mg, 0.45 mmol) and NaI (22.4 mg, 0.15 mmol) in CH.sub.3CN (3 mL) were stirred at 80° C. for 5 h. LCMS showed the reaction was completed. The mixture was concentrated and purified by reverse phase chromatography to give the desired product (32.8 mg, 30% yield) as a white solid. MS (ESI) m/z: 730.7 [M+H].sup.+.

Example 252: 5-((2-(4-(6-(6-((R)-2-(3-Fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)-2-oxoethyl)amino)-2-(1-methyl-2,6-dioxopiperidin-3-yl)isoindoline-1,3-dione (TR-123 neg)

(1137) ##STR00668##

Step 1. Synthesis of tert-butyl (2-(1-methyl-2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-5-yl)glycinate

(1138) ##STR00669##

(1139) To a solution of 5-fluoro-2-(1-methyl-2,6-dioxopiperidin-3-yl)isoindoline-1,3-dione (100 mg, 0.34 mmol) in DMSO (3 mL) were added KF (39 mg, 0.68 mmol) and tert-butyl glycinate (89 mg, 0.68 mmol). The resulting mixture was stirred at 130° C. for 5 h. The reaction was cooled to room temperature before H.sub.2O (50 mL) was added. The mixture was extracted with EtOAc (10 mL×3). The combined organic layers were concentrated and the resulting residue was purified by reverse phase chromatography to give the desired product (82 mg, 61% yield) as pale brown solid. MS (ESI) m/z 402.2 [M+H].sup.+.

Step 2. Synthesis of (2-(1-methyl-2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-5-yl)glycine

(1140) ##STR00670##

(1141) A mixture of tert-butyl (2-(1-methyl-2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-5-yl)glycinate (80 mg, 0.20 mmol) in HCOOH (88%, 5 mL) was stirred at room temperature for 16 h.

The mixture was concentrated and the resulting residue was purified by reverse phase chromatography to give the desired product (54 mg, 78% yield) as pale brown solid. MS (ESI) m/z 346.1 [M+H].sup.+.

Step 3. Synthesis of 5-((2-(4-(6-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)-2-oxoethyl)amino)-2-(1-methyl-2,6-dioxopiperidin-3-yl)isoindoline-1,3-dione

(1142) ##STR00671##

(1143) TR-123 neg was synthesized following the standard procedure for preparing TR-053 (18 mg, yield 51%). MS (ESI) m/z: 771.3 [M+H].sup.+.

Example 253: 3-(6-(3-(4-(6-(6-((R)-2-(3-Fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)-3-oxoprop-1-yn-1-yl)-1-oxoisindolin-2-yl)piperidine-2,6-dione (TR-202)

(1144) ##STR00672##

Step 1. Synthesis of 3-(2-(2,6-dioxopiperidin-3-yl)-3-oxoisindolin-5-yl)propionic acid

(1145) ##STR00673##

(1146) A mixture of 3-(6-(3-hydroxyprop-1-yn-1-yl)-1-oxoisindolin-2-yl)piperidine-2,6-dione (100 mg, 0.34 mmol), TEMPO (2,2,6,6-tetramethyl-1-piperidinyloxy free radical) (5.2 mg, 0.034 mmol), acetonitrile (3 mL) and sodium phosphate buffer (1.3 mL, 0.67 M, pH=6.7) was stirred at 35° C. A solution of sodium chlorite was prepared by dissolving 80% NaClO.sub.2 (9.14 g, 80.0 mmol) in water (40 mL) and a solution of dilute NaOCl was prepared by diluting household bleach (5.25% NaOCl, 1.06 mL, ca. 2.0 mol %) with water (19 mL). Then NaClO.sub.2 solution (0.4 mL) was added followed by the dilute solution of NaOCl solution (0.2 mL). The resulting mixture was stirred at 35° C. until the reaction was completed. The reaction mixture was purified by reverse-phase chromatography to give the desired product (72 mg, 69% yield) as a light yellow solid. MS (ESI) m/z: 313.1 [M+H].sup.+.

Step 2. Synthesis of 3-(6-(3-(4-(6-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)-3-oxoprop-1-yn-1-yl)-1-oxoisindolin-2-yl)piperidine-2,6-dione

(1147) ##STR00674##

(1148) To a solution of 3-(2-(2,6-dioxopiperidin-3-yl)-3-oxoisindolin-5-yl)propionic acid (10 mg, 32.1 umol) in DMSO (1.5 mL) were added HOAt (6.5 mg, 48.1 umol), EDCl (9.2 mg, 48.1 umol), NMM (32.4 mg, 321.0 umol) and (R)-6-(2-(3-fluorophenyl)pyrrolidin-1-yl)-3-(6-(piperazin-1-yl)pyridin-2-yl)imidazo[1,2-b]pyridazine (14.2 mg, 32.1 umol) sequentially. After the resulting solution was stirred at 25° C. for 16 h, the reaction was poured into water (50 mL) and extracted with ethyl acetate (3×20 mL). The combined organic layers were washed with saturated brine (50 mL), dried over anhydrous sodium sulfate, filtered and concentrated under reduced pressure. The resulting residue was purified by reverse-phase chromatography to give the desired product 3-(6-(3-(4-(6-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)-3-oxoprop-1-yn-1-yl)-1-oxoisindolin-2-yl)piperidine-2,6-dione (11.3 mg, 48% yield) as a light yellow solid. MS (ESI) m/z: 738.3 [M+H].sup.+.

Example 254: 3-(5-(3-(4-(6-(6-((R)-2-(3-Fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)-3-oxoprop-1-yn-1-yl)-1-oxoisindolin-2-yl)piperidine-2,6-dione (TR-203)

(1149) ##STR00675##

Step 1. Synthesis of 3-(2-(2,6-dioxopiperidin-3-yl)-1-oxoisindolin-5-yl)propionic acid

(1150) ##STR00676##

(1151) A mixture of 3-(5-(3-hydroxyprop-1-yn-1-yl)-1-oxoisindolin-2-yl)piperidine-2,6-dione (100 mg, 0.34 mmol), TEMPO (2,2,6,6-tetramethyl-1-piperidinyloxy free radical) (5.2 mg, 0.034 mmol), acetonitrile (3 mL) and sodium phosphate buffer (1.3 mL, 0.67 M, pH=6.7) was stirred at 35° C. A solution of sodium chlorite is prepared by dissolving 80% NaClO.sub.2 (9.14 g, 80.0

mmol) in water (40 mL) and a solution of dilute NaOCl is prepared by diluting household bleach (5.25% NaOCl, 1.06 mL, ca. 2.0 mol %) with water (19 mL). Then NaClO.sub.2 solution (0.4 mL) was added followed by the dilute NaOCl solution (0.2 mL). The resulting mixture was stirred at 35° C. until the reaction was completed. The reaction mixture was purified by reverse-phase chromatography to give the desired product (65 mg, 63% yield) as a light yellow solid. MS (ESI) m/z: 313.1 [M+H].sup.+.

Step 2. Synthesis of 3-(5-(3-(4-(6-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)-3-oxoprop-1-yn-1-yl)-1-oxoisindolin-2-yl)piperidine-2,6-dione

(1152) ##STR00677##

(1153) TR-203 was synthesized following the standard procedure for preparing TR-202(9.8 mg, 42% yield) as a light yellow solid. MS (ESI) m/z: 738.3 [M+H].sup.+.

Example 255:2-(2,6-Dioxopiperidin-3-yl)-5-(2-(4-(6-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)ethyl)isoindoline-1,3-dione (TR-204)

(1154) ##STR00678##

Step 1. Synthesis of dimethyl 4-bromophthalate

(1155) ##STR00679##

(1156) To a solution of 5-bromoisobenzofuran-1,3-dione (4.6 g, 20.26 mmol) in MeOH (50 mL) was added H.sub.2SO.sub.4 (992.90 mg, 10.13 mmol) at room temperature. After the reaction mixture was stirred at 80° C. for 16 h, the crude product was dissolved in DCM (50 mL), washed with aq. NaHCO.sub.3 (100 mL) and brine (100 mL). The organic phase was dried over Na.sub.2SO.sub.4, filtered, and concentrated. The resulting residue was purified by silica gel chromatography (petroleum ether/EtOAc=10:1) to give dimethyl 4-bromophthalate (5.0 g, 90% yield) as colorless oil. MS (ESI) m/z: 375.0 [M+H].sup.+.

Step 2. Synthesis of dimethyl 4-(2-(benzyloxy)ethyl)phthalate

(1157) ##STR00680##

(1158) To a solution of 2-benzyloxyethyl (trifluoro)boronpotassium hydride (800 mg, 3.30 mmol) and dimethyl 4-bromophthalate (1.08 g, 3.97 mmol) in toluene (20 mL) and H.sub.2O (10 mL) were added bis(di-tert-butyl (4-dimethylaminophenyl)phosphine)dichloropalladium (II)(233.97 mg, 330.46 umol) and Cs.sub.2CO.sub.3 (2.15 g, 6.61 mmol) at room temperature under N.sub.2. After the reaction mixture was stirred at 100° C. for 16 h, the solvent was removed under reduced pressure. The resulting crude product was purified by silica gel chromatography (petroleum ether/EtOAc=10:1) to give dimethyl 4-(2-(benzyloxy)ethyl)phthalate (960 mg, 88% yield) as colorless oil. MS (ESI) m/z: 329.6 [M+H].sup.+.

Step 3. Synthesis of 4-(2-(benzyloxy)ethyl)phthalic acid

(1159) ##STR00681##

(1160) To a solution of dimethyl 4-(2-(benzyloxy)ethyl)phthalate (1.0 g, 3.05 mmol) in MeOH (10 mL), THF (10 mL) and H.sub.2O (10 mL) was added NaOH (730.91 mg, 18.27 mmol) at room temperature. After the reaction mixture was stirred at 80° C. for 3 h, organic solvents were removed under reduced pressure. The resulting aqueous mixture was acidified with conc. HCl to pH=2, before being extracted with DCM (3×20 mL). The organic layers were combined, dried over Na.sub.2SO.sub.4, filtered and concentrated to give crude 4-(2-benzyloxyethyl)phthalic acid (900 mg, 98% yield) as a yellow solid. This product was used in the next step directly without further purification. MS (ESI) m/z: 301.3 [M+H].sup.+.

Step 4. Synthesis of 5-(2-(benzyloxy)ethyl)-2-(2,6-dioxopiperidin-3-yl)isoindoline-1,3-dione

(1161) ##STR00682##

(1162) A solution of 4-(2-benzyloxyethyl)phthalic acid (800 mg, 2.66 mmol) and 3-aminopiperidine-2,6-dione (341.3 mg, 2.66 mmol) in pyridine (40 mL) was stirred at 120° C. for 16 h. Then the reaction mixture was concentrated and purified by silica gel chromatography (DCM/MeOH=20:1) to give 5-(2-(benzyloxy)ethyl)-2-(2,6-dioxopiperidin-3-yl)isoindoline-1,3-

dione (0.98 g, 94% yield) as a yellow solid. MS (ESI) m/z: 393.3 [M+H].sup.+.

Step 5. Synthesis of 2-(2,6-dioxopiperidin-3-yl)-5-(2-hydroxyethyl)isoindoline-1,3-dione (1163) ##STR00683##

(1164) To a solution of 5-(2-(benzyloxy)ethyl)-2-(2,6-dioxopiperidin-3-yl)isoindoline-1,3-dione (700 mg, 1.78 mmol) in THF (25 mL) were added Pd(OH) 2(50 mg) and Pd/C (50 mg) at room temperature. Then the reaction mixture was stirred at 50° C. for 16 h under hydrogen atmosphere pressure. After cooled down to room temperature, the mixture was filtered and concentrated to give crude 2-(2,6-dioxopiperidin-3-yl)-5-(2-hydroxyethyl)isoindoline-1,3-dione (500 mg, 93% yield) as a white solid. This product was used in the next step directly without further purification. MS (ESI) m/z: 303.2 [M+H].sup.+.

Step 6. Synthesis of 2-(2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-5-yl)ethyl 4-methylbenzenesulfonate (1165) ##STR00684##

(1166) To a solution of 2-(2,6-dioxopiperidin-3-yl)-5-(2-hydroxyethyl)isoindoline-1,3-dione (25 mg, 82.70 umol) in TEA (5 mL) was added 4-methylbenzenesulfonyl chloride (15.77 mg, 82.70 umol) at 0° C. After the reaction mixture was stirred at room temperature for 2 h, the mixture was concentrated and purified by prep-TLC (DCM/MeOH=20:1) to give crude 2-(2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-5-yl)ethyl 4-methylbenzenesulfonate (20 mg, 53% yield). This product was used in the next step directly without further purification. MS (ESI) m/z: 457.3 [M+H].sup.+.

Step 7. Synthesis of 2-(2,6-dioxopiperidin-3-yl)-5-(2-(4-(6-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)ethyl)isoindoline-1,3-dione (1167) ##STR00685##

(1168) To a solution (R)-6-(2-(3-fluorophenyl)pyrrolidin-1-yl)-3-(6-(piperazin-1-yl)pyridin-2-yl)imidazo[1,2-b]pyridazine (14.57 mg, 32.86 umol) and 2-(2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-5-yl)ethyl 4-methylbenzenesulfonate (15 mg, 32.86 umol) in CH.sub.3CN (4 mL) were added NaI (9.85 mg, 65.72 umol) and K.sub.2CO.sub.3 (18.14 mg, 131.44 umol) at room temperature under N.sub.2. After the reaction mixture was stirred at 80° C. for 16 h, the solvent was removed and the resulting residue was purified by prep-TLC (DCM/MeOH=10:1) to give 2-(2,6-dioxopiperidin-3-yl)-5-(2-(4-(6-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)ethyl)isoindoline-1,3-dione (14 mg, 59% yield) as a light yellow solid. MS (ESI) m/z: 729.6 [M+H].sup.+.

Example 256:3-(5-(3-((4-(6-(6-((R)-2-(3-Fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)methyl)azetidin-1-yl)-1-oxoisindolin-2-yl)piperidine-2,6-dione (TR-205)

(1169) ##STR00686## ##STR00687##

Step 1. Synthesis of tert-butyl 3-((benzyloxy)methyl)azetidine-1-carboxylate (1170) ##STR00688##

(1171) To a solution of tert-butyl 3-(hydroxymethyl)azetidine-1-carboxylate (2 g, 10.0 mmol) and NaH (720 mg, 30.0 mmol) in THF (50 mL) was added BnBr (2.74 g, 15.0 mmol) in portions at 0° C. under N.sub.2. The resulting mixture was stirred at room temperature for 5 h, before the reaction was quenched with H.sub.2O (100 mL) at 0° C. and extracted with DCM (100 mL×2). The organic layers were combined and dried over Na.sub.2SO.sub.4, filtered, and concentrated. The resulting residue was purified by silica gel flash chromatography (petroleum/ethyl acetate=2:1) to give tert-butyl 3-((benzyloxy)methyl)azetidine-1-carboxylate (1.5 g, 51% yield) as a light yellow oil. MS (ESI) m/z: 278.3 [M+H].sup.+.

Step 2. Synthesis of 3-((benzyloxy)methyl)azetidine (1172) ##STR00689##

(1173) To a solution of tert-butyl 3-((benzyloxy)methyl)azetidine-1-carboxylate (1.5 g, 5.4 mmol) in DCM (15 mL) was added TFA (3 mL) at 0° C. After the reaction mixture was stirred at room

temperature for 1 h, the solvent was removed under vacuum. The resulting residue was washed with diethyl ether to give 3-((benzyloxy)methyl)azetidine (1.0 g, 76% yield) as a yellow oil. MS (ESI) m/z: 178.2 [M+H].sup.+.

Step 3. Synthesis of 5-(3-((benzyloxy)methyl)azetidin-1-yl)isobenzofuran-1 (3H)-one (1174) ##STR00690##

(1175) To a solution of 3-((benzyloxy)methyl)azetidine (1.0 g, 5.6 mmol) and 5-bromoisobenzofuran-1 (3H)-one (1.19 g, 5.6 mmol) in DMSO (15 mL) were added Cs.sub.2CO.sub.3 (5.4 g, 16.8 mmol) and Ruphos-Pd-G3 (470 mg, 0.54 mmol) at room temperature under N.sub.2. After the reaction mixture was stirred at 80° C. for 12 h, the resulting black mixture was diluted with ethyl acetate (20 mL), washed with saturated aqueous brine (2×20 mL). The combined organic layers were dried over sodium sulfate, filtered, and concentrated. The resulting residue was purified by silica gel flash chromatography (petroleum/ethyl acetate=1:1) to give 5-(3-((benzyloxy)methyl)azetidin-1-yl)isobenzofuran-1 (3H)-one (470 mg, 27% yield) as a brown solid. MS (ESI) m/z: 310.1 [M+H].sup.+.

Step 4. Synthesis of 4-(3-((benzyloxy)methyl)azetidin-1-yl)-2-(hydroxymethyl)benzoic acid (1176) ##STR00691##

(1177) To a solution of 5-(3-((benzyloxy)methyl)azetidin-1-yl)isobenzofuran-1 (470 mg, 1.5 mmol) in MeOH (15 mL) was added NaOH (181 mg, 4.5 mmol) at room temperature. Then the mixture was stirred at 60° C. for 2 h. After cooled down to room temperature, the solution was acidified to pH=2-3 with 1 N HCl and filtered. The filter cake was washed with EtOAc/petroleum ether (50 mL, 1:10) to give 4-(3-((benzyloxy)methyl)azetidin-1-yl)-2-(hydroxymethyl)benzoic acid (350 mg, 74% yield) as a yellow solid. MS (ESI) m/z: 328.2 [M+H].sup.+.

Step 5. Synthesis of methyl 4-(3-((benzyloxy)methyl)azetidin-1-yl)-2-(hydroxymethyl)benzoate (1178) ##STR00692##

(1179) To a solution of 4-(3-((benzyloxy)methyl)azetidin-1-yl)-2-(hydroxymethyl)benzoic acid (350 mg, 1.06 mmol) in EtOAc/MeOH (15 mL) was added TMSCHN.sub.2 (1 mL) at 0° C. After the completion of addition, the reaction mixture was stirred at room temperature for 1 h, then concentrated to give crude methyl 4-(3-((benzyloxy)methyl)azetidin-1-yl)-2-(hydroxymethyl)benzoate (220 mg, 62% yield) as a yellow solid. This product was used in the next step directly without further purification. MS (ESI) m/z: 342.2 [M+H].sup.+.

Step 6. Synthesis of methyl 4-(3-((benzyloxy)methyl)azetidin-1-yl)-2-(((methylsulfonyl)oxy)methyl)benzoate (1180) ##STR00693##

(1181) To a solution of methyl 4-(3-((benzyloxy)methyl)azetidin-1-yl)-2-(hydroxymethyl)benzoate (220 mg, 0.64 mmol) in DCM (10 mL) were added MsCl (0.5 mL) and DIEA (248 mg, 1.92 mmol) at 0° C. Then the reaction was stirred at room temperature for 1 h, quenched the reaction with water (30 mL). The resulting solution was extracted with DCM (20 mL×2) and the organic layer was concentrated to give methyl 4-(3-((benzyloxy)methyl)azetidin-1-yl)-2-(((methylsulfonyl)oxy)methyl)benzoate (150 mg, 55% yield) as a yellow solid. This product was used in the next step directly without further purification. MS (ESI) m/z: 342.2 [M+H].sup.+.

Step 7. Synthesis of 3-(5-(3-((benzyloxy)methyl)azetidin-1-yl)-1-oxoisindolin-2-yl)piperidine-2,6-dione (1182) ##STR00694##

(1183) To a solution of methyl 4-(3-((benzyloxy)methyl)azetidin-1-yl)-2-(((methylsulfonyl)oxy)methyl)benzoate (150 mg, 0.35 mmol) in acetonitrile (5 mL) were added 3-aminopiperidine-2,6-dione (44 mg, 0.35 mmol) and potassium carbonate (150 mg, 1.05 mmol) at room temperature. The mixture was stirred at room temperature for 2 h. The solvent was removed under vacuum and the residue was purified by silica gel flash chromatography (MeOH/DCM=1:10) to give 3-(S-(3-((benzyloxy)methyl)azetidin-1-yl)-1-oxoisindolin-2-yl)piperidine-2,6-dione (40 mg, 26% yield) as a white solid. MS (ESI) m/z: 420.8 [M+H].sup.+.

Step 8. Synthesis of 3-(5-(3-(hydroxymethyl)azetidin-1-yl)-1-oxoisindolin-2-yl)piperidine-2,6-dione

(1184) ##STR00695##

(1185) To a solution of 3-(5-(3-((benzyloxy)methyl)azetidin-1-yl)-1-oxoisindolin-2-yl)piperidine-2,6-dione (40 mg, 0.09 mmol) in DCM (5 mL) was added MsOH (1 mL) at 0° C. The reaction was stirred at room temperature for 2 h, then the solvent was removed under vacuum. The residue was purified by silica gel flash chromatography (MeOH/H.sub.2O=1:1) to give 3-(5-(3-(hydroxymethyl)azetidin-1-yl)-1-oxoisindolin-2-yl)piperidine-2,6-dione (26 mg, 83% yield) as a yellow solid. MS (ESI) m/z: 330.1 [M+H].sup.+.

Step 9. Synthesis of (1-(2-(2,6-dioxopiperidin-3-yl)-1-oxoisindolin-5-yl)azetidin-3-yl)methyl 4-methylbenzenesulfonate

(1186) ##STR00696##

(1187) To a solution of 3-(5-(3-(hydroxymethyl)azetidin-1-yl)-1-oxoisindolin-2-yl)piperidine-2,6-dione (26 mg, 0.08 mmol) in dichloromethane (2 mL) were added TsCl (30 mg, 0.16 mmol) and DMAP (20 mg, 0.08 mmol) at room temperature. The mixture was stirred at room temperature for 3 h. The reaction was concentrated and purified by silica gel flash chromatography (MeOH/DCM=1:10) to give (1-(2-(2,6-dioxopiperidin-3-yl)-1-oxoisindolin-5-yl)azetidin-3-yl)methyl 4-methylbenzenesulfonate (13 mg, 34% yield) as a yellow solid. MS (ESI) m/z: 484.2 [M+H].sup.+.

Step 10. Synthesis of 3-(5-(3-((4-(6-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)methyl)azetidin-1-yl)-1-oxoisindolin-2-yl)piperidine-2,6-dione

(1188) ##STR00697##

(1189) To a solution of (1-(2-(2,6-dioxopiperidin-3-yl)-1-oxoisindolin-5-yl)azetidin-3-yl)methyl 4-methylbenzenesulfonate (13 mg, 0.03 mmol) and (S)-6-(2-(3-fluorophenyl)pyrrolidin-1-yl)-3-(6-(piperazin-1-yl)pyridin-2-yl)imidazo[1,2-b]pyridazine (12 mg, 0.03 mmol) in acetonitrile (1 mL) were added potassium carbonate (11 mg, 0.09 mmol) and sodium iodide (4 mg, 0.03 mmol) at room temperature. The reaction mixture was stirred at 80° C. for 2 h, then concentrated and purified by silica gel flash chromatography (MeOH/DCM=1:10) to give 3-(5-(3-((4-(6-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)methyl)azetidin-1-yl)-1-oxoisindolin-2-yl)piperidine-2,6-dione (3.5 mg, 18% yield) as a white solid. MS (ESI) m/z: 755.4 [M+H].sup.+.

Example 257: 3-(5-((1-(6-(6-((R)-2-(3-Fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)azetidin-3-yl)amino)-1-oxoisindolin-2-yl)piperidine-2,6-dione (TR-206)

(1190) ##STR00698##

Step 1. Synthesis of tert-butyl 3-((2-(2,6-dioxopiperidin-3-yl)-1-oxoisindolin-5-yl)amino)azetidine-1-carboxylate

(1191) ##STR00699##

(1192) To a solution of 3-(5-amino-1-oxoisindolin-2-yl)piperidine-2,6-dione (200 mg, 0.77 mmol) and AcOH (2 mL) in DCM (6 mL) was added 1M borane tetrahydrofuran complex solution (1.54 mL, 1.54 mmol) at 0° C., then it was stirred at 0° C. for 2 h. The mixture was concentrated and purified by reverse-phase chromatography to give the desired product (61 mg, 19% yield) as a light yellow solid. MS (ESI) m/z: 415.2 [M+H].sup.+.

Step 2. Synthesis of 3-(5-(azetidin-3-ylamino)-1-oxoisindolin-2-yl)piperidine-2,6-dione

(1193) ##STR00700##

(1194) To a solution of tert-butyl 3-((2-(2,6-dioxopiperidin-3-yl)-1-oxoisindolin-5-yl)amino)azetidine-1-carboxylate (30 mg, 0.072 mmol) in DCM (5 mL) was added TFA (4 mL) at room temperature, then it was stirred at room temperature for 2 h. The mixture was concentrated and purified by reverse-phase chromatography to give the desired product (18 mg, 79% yield) as a light yellow solid. MS (ESI) m/z: 315.1 [M+H].sup.+.

Step 3. Synthesis of 3-(5-((1-(6-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)azetidin-3-yl)amino)-1-oxoisindolin-2-yl)piperidine-2,6-dione (1195) ##STR00701##

(1196) To a solution of 3-(5-(azetidin-3-ylamino)-1-oxoisindolin-2-yl)piperidine-2,6-dione (18 mg, 0.057 mmol) and (R)-6-(2-(3-fluorophenyl)pyrrolidin-1-yl)-3-(6-fluoropyridin-2-yl)imidazo[1,2-b]pyridazine (21 mg, 0.057 mmol) in DMSO (5 mL) was added KF (3.3 mg, 0.057 mmol) at room temperature, then it was stirred at 130° C. for 16 h. The reaction was cooled to room temperature before H.sub.2O (50 mL) was added. The mixture was extracted with EtOAc (10 mL×3). The combined organic layers were concentrated and purified by reverse-phase chromatography to give the desired product 3-(5-((1-(6-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)azetidin-3-yl)amino)-1-oxoisindolin-2-yl)piperidine-2,6-dione (5.5 mg, 14% yield) as a light yellow solid. MS (ESI) m/z: 672.3 [M+H].sup.+.

Example 258: 3-(5-(4-(6-(6-((R)-2-(3-Fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)-1-oxoisindolin-2-yl)piperidine-2,6-dione (TR-207)

(1197) ##STR00702##

(1198) TR-207 was synthesized following the standard procedure for preparing TR-206 (4.8 mg, 15% yield) as a light yellow solid. MS (ESI) m/z: 686.3 [M+H].sup.+.

Example 259: 3-(5-(4-(6-(6-((R)-2-(3-Fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)-1-oxoisindolin-2-yl)piperidine-2,6-dione (TR-208)

(1199) ##STR00703##

Step 1. Synthesis of tert-butyl 4-((2-(2,6-dioxopiperidin-3-yl)-1-oxoisindolin-5-yl)ethynyl)piperidine-1-carboxylate

(1200) ##STR00704##

(1201) A mixture of 3-(5-bromo-1-oxoisindolin-2-yl)piperidine-2,6-dione (100 mg, 0.31 mmol), tert-butyl 4-ethynylpiperidine-1-carboxylate (71 mg, 0.34 mmol), Pd(dppf) Cl.sub.2 (22 mg, 0.03 mmol), Cul (5.7 mg, 0.03 mmol) and DIPEA (79 mg, 0.62 mmol) in DMSO (10 mL) was stirred at 80° C. for 16 h. The reaction mixture was concentrated and the residue was purified by reverse-phase chromatography to give the desired product (55 mg, 39% yield) as a pale brown solid. MS (ESI) m/z: 452.2 [M+H].sup.+.

Step 2. Synthesis of 3-(1-oxo-5-(piperidin-4-ylethynyl)isindolin-2-yl)piperidine-2,6-dione

(1202) ##STR00705##

(1203) To a solution of tert-butyl 4-((2-(2,6-dioxopiperidin-3-yl)-1-oxoisindolin-5-yl)ethynyl)piperidine-1-carboxylate (55 mg, 0.12 mmol) in DCM (5 mL) was added TFA (4 mL) at room temperature. After it was stirred at room temperature for 2 h, the mixture was concentrated and purified by reverse-phase chromatography to give the desired product (36 mg, 85% yield) as a light yellow solid. MS (ESI) m/z: 352.2 [M+H].sup.+.

Step 3. Synthesis of 3-(5-((1-(6-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperidin-4-yl)ethynyl)-1-oxoisindolin-2-yl)piperidine-2,6-dione

(1204) ##STR00706##

(1205) TR-208 was synthesized following the standard procedure for preparing TR-206 (6.2 mg, 15% yield) as a light yellow solid. MS (ESI) m/z: 709.3 [M+H].sup.+.

Example 260: N-(2-(2,6-Dioxopiperidin-3-yl)-1,3-dioxoisindolin-5-yl)-1-(6-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)azetidine-3-carboxamide (TR-209)

(1206) ##STR00707##

(1207) Step 1. Synthesis of (R)-1-(6-(2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)azetidine-3-carboxylic acid

(1208) ##STR00708##

(1209) To a solution of azetidine-3-carboxylic acid (16 mg, 0.16 mmol) and (R)-6-(2-(3-fluorophenyl)pyrrolidin-1-yl)-3-(6-fluoropyridin-2-yl)imidazo[1,2-b]pyridazine (50 mg, 0.13

mmol) in DMSO (8 mL) was added KF (7.5 mg, 0.13 mmol) at room temperature. After it was stirred at 130° C. for 16 h, the reaction was cooled to room temperature before H.sub.2O (50 mL) was added. The mixture was extracted with EtOAc (3×10 mL). The combined organic layers were concentrated and purified by reverse-phase chromatography to give the desired product (22 mg, 37% yield) as a light yellow solid. MS (ESI) m/z: 459.2 [M+H].sup.+.

Step 2. Synthesis of N-(2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-5-yl)-1-(6-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl) azetidine-3-carboxamide (1210) ##STR00709##

(1211) TR-209 was synthesized following the standard procedure for preparing TR-202 (9.8 mg, 21% yield) as a light yellow solid. MS (ESI) m/z: 714.3 [M+H].sup.+.

Example 261: 3-(5-(((1-(6-(6-((R)-2-(3-Fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)azetidin-3-yl)methyl)amino)-1-oxoisindolin-2-yl)piperidine-2,6-dione (TR-210) (1212) ##STR00710##

Step 1. Synthesis of tert-butyl 3-(((2-(2,6-dioxopiperidin-3-yl)-1-oxoisindolin-5-yl)amino)methyl)azetidine-1-carboxylate

(1213) ##STR00711##

(1214) To a solution of 3-(5-amino-1-oxoisindolin-2-yl)piperidine-2,6-dione (100 mg, 0.38 mmol) and AcOH (2 mL) in DCM (6 mL) was added 1M borane tetrahydrofuran complex solution (0.76 mL, 0.76 mmol) at 0° C. After it was stirred at 0° C. for 2 h, the mixture was concentrated and purified by reverse-phase chromatography to give the desired product (42 mg, 26% yield) as a light yellow solid. MS (ESI) m/z: 429.2 [M+H].sup.+.

Step 2. Synthesis of 3-(5-((azetidin-3-ylmethyl)amino)-1-oxoisindolin-2-yl)piperidine-2,6-dione (1215) ##STR00712##

(1216) To a solution of tert-butyl 3-(((2-(2,6-dioxopiperidin-3-yl)-1-oxoisindolin-5-yl)amino)methyl)azetidine-1-carboxylate (42 mg, 0.10 mmol) in DCM (5 mL) was added TFA (4 mL) at room temperature. After it was stirred at room temperature for 2 h, the mixture was concentrated and purified by reverse-phase chromatography to give the desired product (25 mg, 78% yield) as a light yellow solid. MS (ESI) m/z: 329.2 [M+H].sup.+.

Step 3. Synthesis of 3-(5-(((1-(6-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)azetidin-3-yl)methyl)amino)-1-oxoisindolin-2-yl)piperidine-2,6-dione

(1217) ##STR00713##

(1218) TR-210 was synthesized following the standard procedure for preparing TR-208 (6.7 mg, 15% yield) as a light yellow solid. MS (ESI) m/z: 686.3 [M+H].sup.+.

Example 262: 2-(2,6-Dioxopiperidin-3-yl)-5-(3-(4-(6-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazine-1-carbonyl)azetidin-1-yl)isoindoline-1,3-dione (TR-211)

(1219) ##STR00714##

Step 1. Synthesis of 1-(2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-5-yl)azetidine-3-carboxylic acid

(1220) ##STR00715##

(1221) To a solution of 2-(2,6-dioxopiperidin-3-yl)-5-fluoroisoindoline-1,3-dione (276 mg, 1.0 mmol) and azetidine-3-carboxylic acid (171 mg, 1.5 mmol) in DMSO (3 mL) was added KF (232 mg, 4.0 mmol). The resulting solution was stirred at 120° C. for 5 h, before the reaction was quenched with water (10 mL), and extracted with ethyl acetate (2×20 mL). The organic layers were combined, dried over Na.sub.2SO.sub.4, and concentrated. The resulting residue was purified by reverse-phase chromatography to give 1-(2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-5-yl)azetidine-3-carboxylic acid (175 mg, 49% yield) as a yellow solid. MS (BSI) m/z: 358.1 [M+H].sup.+.

Step 2. Synthesis of 2-(2,6-dioxopiperidin-3-yl)-5-(3-(4-(6-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-

yl)imidazo[1,2-b]pyridazin-3-yl)piperidin-2-yl)piperazine-1-carbonyl)azetidin-1-yl)isoindoline-1,3-dione

(1222) ##STR00716##

(1223) To a solution of 1-(2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-5-yl)azetidine-3-carboxylic acid (175 mg, 0.49 mmol) and 6-[(2R)-2-(3-fluorophenyl)pyrrolidin-1-yl]-3-(6-piperazin-1-yl-2-pyridyl)imidazo[1,2-b]pyridazine (217 mg, 0.49 mmol) in N,N-dimethylformamide (10 mL) were added HATU (371 mg, 0.98 mmol) and DIEA (189 mg, 1.46 mmol) at room temperature. The reaction mixture was stirred at room temperature for 2 h, before it was quenched with water (30 mL). The resulting mixture was extracted with ethyl acetate (3×20 mL). And the combined organic layers were dried over sodium sulfate, filtered, and concentrated. The resultant residue was purified by silica gel chromatography (DCM/MeOH=10:1) to give 2-(2,6-dioxopiperidin-3-yl)-5-(3-(4-(6-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazine-1-carbonyl)azetidin-1-yl)isoindoline-1,3-dione (128 mg, 33% yield) as a yellow solid. MS (ESI) m/z: 783.8 [M+H].sup.+.

Example 263: 3-(5-((1-(6-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperidin-4-yl)amino)-1-oxoisindolin-2-yl)piperidine-2,6-dione (TR-212)

(1224) ##STR00717##

Step 1. Synthesis of tert-butyl 4-((2-(2,6-dioxopiperidin-3-yl)-1-oxoisindolin-5-yl)amino)piperidine-1-carboxylate

(1225) ##STR00718##

(1226) To a solution of 3-(5-amino-1-oxoisindolin-2-yl)piperidine-2,6-dione (130 mg, 0.5 mmol) in dichloromethane (10 mL) and AcOH (2.0 mL) were added tert-butyl 4-oxopiperidine-1-carboxylate (119 mg, 1.0 mmol) and BH.sub.3/THF (0.5 mL) at room temperature. The mixture was stirred at room temperature for 16 h, then quenched with H.sub.2O (20 mL) and extracted with dichloromethane (3×20 mL). The organic layer was dried over sodium sulfate, filtered and concentrated to afford crude tert-butyl 4-((2-(2,6-dioxopiperidin-3-yl)-1-oxoisindolin-5-yl)amino)piperidine-1-carboxylate (160 mg, 72% yield) as a green solid. This product was used in the next step directly without further purification. MS (ESI) m/z: 443.2 [M+H].sup.+.

Step 2. Synthesis of 3-(1-oxo-5-(piperidin-4-ylamino) isoindolin-2-yl)piperidine-2,6-dione

(1227) ##STR00719##

(1228) To a solution of tert-butyl 4-((2-(2,6-dioxopiperidin-3-yl)-1-oxoisindolin-5-yl)amino)piperidine-1-carboxylate (160 mg, 0.36 mmol) in DCM (5 mL) was added TFA (1 mL) at 0° C. The reaction mixture was stirred at room temperature for 16 h, before the solvent was removed under vacuum. The resulting residue was washed with diethyl ether to give 3-(1-oxo-5-(piperidin-4-ylamino) isoindolin-2-yl)piperidine-2,6-dione (100 mg, 80% yield) as a yellow solid. This product was used in the next step directly without further purification. MS (ESI) m/z: 343.2 [M+H].sup.+.

Step 3. Synthesis of 3-(5-((1-(6-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperidin-4-yl)amino)-1-oxoisindolin-2-yl)piperidine-2,6-dione

(1229) ##STR00720##

(1230) To a solution of 3-(1-oxo-5-(piperidin-4-ylamino) isoindolin-2-yl)piperidine-2,6-dione (100 mg, 0.29 mmol) and (R)-6-(2-(3-fluorophenyl)pyrrolidin-1-yl)-3-(6-fluoropyridin-2-yl)imidazo[1,2-b]pyridazine (110 mg, 0.29 mmol) in DMSO (5 mL) was added KF (67 mg, 1.16 mmol). The resulting solution was stirred at 120° C. for 5 h, before the reaction was quenched with water (10 mL) and extracted with ethyl acetate (3×20 mL). The combined organic layers were dried over sodium sulfate, and concentrated. The resulting residue was purified by silica gel chromatography column (DCM/MeOH=12:1) to give 3-(5-((1-(6-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperidin-4-yl)amino)-1-oxoisindolin-2-yl)piperidine-2,6-dione (23 mg, 11% yield) as a yellow solid. MS (ESI) m/z: 700.8 [M+H].sup.+.

Example 264: 2-((2-(2,6-Dioxopiperidin-3-yl)-1,3-dioxoisindolin-5-yl)amino)-N-(1-(6-(6-((R)-2-

(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)azetidin-3-yl)acetamide (TR-213)

(1231) ##STR00721##

Step 1. Synthesis of tert-butyl (R)-(1-(6-(6-(2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)azetidin-3-yl)carbamate

(1232) ##STR00722##

(1233) To a solution of (R)-6-(2-(3-fluorophenyl)pyrrolidin-1-yl)-3-(6-fluoropyridin-2-yl)imidazo[1,2-b]pyridazine (189 mg, 0.5 mmol) and tert-butyl azetidin-3-ylcarbamate (86 mg, 0.5 mmol) in DMSO (5 mL) was added KF (116 mg, 2 mmol). The resulting solution was stirred at 120° C. for 5 h, before it was quenched with water (10 mL) and extracted with ethyl acetate (3×20 mL). The combined organic layers were dried over sodium sulfate, and concentrated. The resulting residue was purified by silica gel chromatography column (DCM/MeOH=10:1) to give tert-butyl (R)-(1-(6-(6-(2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)azetidin-3-yl)carbamate (181 mg, 68% yield) as a yellow solid. MS (ESI) m/z: 530.2 [M+H].sup.+.

Step 2. Synthesis of 3-(1-oxo-5-(piperidin-4-ylamino) isoindolin-2-yl)piperidine-2,6-dione

(1234) ##STR00723##

(1235) To a solution of tert-butyl (R)-(1-(6-(6-(2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)azetidin-3-yl)carbamate (181 mg, 0.34 mmol) in DCM (5 mL) was added TFA (1 mL) at 0° C. The reaction was stirred at room temperature for 1 h, before solvent was removed under vacuum. The resulting residue was washed with diethyl ether to give 3-(1-oxo-5-(piperidin-4-ylamino) isoindolin-2-yl)piperidine-2,6-dione (130 mg, 89% yield) as a yellow solid. This product was used in the next step directly without further purification. MS (ESI) m/z: 430.2 [M+H].sup.+.

Step 3. Synthesis of 2-((2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisoindolin-5-yl)amino)-N-(1-(6-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)azetidin-3-yl)acetamide

(1236) ##STR00724##

(1237) To a solution of 1-(2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisoindolin-5-yl)azetidine-3-carboxylic acid (120 mg, 0.27 mmol) and (2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisoindolin-5-yl)glycine (89 mg, 0.27 mmol) in N,N-dimethylformamide (10 mL) were added HATU (159 mg, 0.54 mmol) and DIEA (108 mg, 0.81 mmol) at 0° C. The mixture was stirred at room temperature for 2 h, before the reaction was quenched with water (20 mL) and extracted with ethyl acetate (3×20 mL). The combined organic layers were dried over sodium sulfate, and concentrated. The resulting residue was purified by silica gel chromatography (DCM/MeOH=10:1) to give 2-((2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisoindolin-5-yl)amino)-N-(1-(6-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)azetidin-3-yl)acetamide (97 mg, 46% yield) as a yellow solid. MS (ESI) m/z: 744.0 [M+H].sup.+.

Example 265: 2-(2,6-Dioxo-3-piperidyl)-5-(2-(4-(6-(6-((2R)-2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)-2-pyridyl)piperazin-1-yl)ethyl-((4-methoxyphenyl)methyl)amino) isoindoline-1,3-dione (TR-214)

(1238) ##STR00725##

Step 1. Synthesis of 2-((4-methoxyphenyl)methylamino)ethanol

(1239) ##STR00726##

(1240) 1-(Chloromethyl)-4-methoxy-benzene (8 g, 51.08 mmol) was added to 2-aminoethanol (31.20 g, 510.83 mmol) at 0° C. After the resulting reaction mixture was stirred at room temperature for 2 h, the reaction was quenched with water (100 mL), and extracted with DCM (2×100 mL). The combined organic layers were dried over sodium sulfate, and concentrated. The resulting residue was purified by silica gel chromatography (DCM/MeOH=20:1 to 10:1) to give 2-((4-methoxyphenyl)methylamino)ethanol (8 g, 86% yield). MS (ESI) m/z: 182.4 [M+H].sup.+.

Step 2. Synthesis of 2-((2,6-dioxo-3-piperidyl)-5-(2-hydroxyethyl)-((4-methoxyphenyl)methyl)amino) isoindoline-1,3-dione

(1241) ##STR00727##

(1242) To a solution of 2-((4-methoxyphenyl)methylamino)ethanol (1.09 g, 6.00 mmol) and 2-(2,6-dioxo-3-piperidyl)-5-fluoro-isoindoline-1,3-dione (1.38 g, 5.00 mmol) in DMSO (20 mL) was added DIPEA (2.58 g, 19.98 mmol) at room temperature. The reaction mixture was stirred at 120° C. for 16 h, before the mixture was quenched with water (100 mL), and extracted with DCM (50 mL×2). The organic layer was washed with brine (50 mL×2), dried over Na.sub.2SO.sub.4, filtered and concentrated. The resulting residue was slurried in (petroleum ether/EtOAc=5:1, 50 mL), filtered to give 2-(2,6-dioxo-3-piperidyl)-5-(2-hydroxyethyl)-((4-methoxyphenyl)methyl)amino) isoindoline-1,3-dione (1.5 g, 69% yield) as a yellow solid. MS (ESI) m/z: 438.4 [M+H].sup.+.

Step 3. Synthesis of 2-((2-(2,6-dioxo-3-piperidyl)-1,3-dioxo-isoindolin-5-yl)-((4-methoxyphenyl)methyl)amino)ethyl 4-methylbenzenesulfonate

(1243) ##STR00728##

(1244) To a solution of 2-(2,6-dioxo-3-piperidyl)-5-(2-hydroxyethyl)-((4-methoxyphenyl)methyl)amino) isoindoline-1,3-dione (437 mg, 998.98 umol) in TEA (5 mL) and DCM (5 mL) were added DMAP (122.05 mg, 998.98 umol) and 4-methylbenzenesulfonyl chloride (285.68 mg, 1.50 mmol) at room temperature. After the reaction mixture was stirred at room temperature for 2 h, the reaction was concentrated and purified by silica gel chromatography (DCM/MeOH=50:1) to give 2-((2-(2,6-dioxo-3-piperidyl)-1,3-dioxo-isoindolin-5-yl)-((4-methoxyphenyl)methyl)amino)ethyl 4-methylbenzenesulfonate (580 mg, 98% yield). MS (ESI) m/z: 592.4 [M+H].sup.+.

Step 4. Synthesis of 2-(2,6-dioxo-3-piperidyl)-5-(2-(4-(6-(6-((2R)-2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)-2-pyridyl)piperazin-1-yl)ethyl)-((4-methoxyphenyl)methyl)amino) isoindoline-1,3-dione

(1245) ##STR00729##

(1246) To a solution of 2-((2-(2,6-dioxo-3-piperidyl)-1,3-dioxo-isoindolin-5-yl)-((4-methoxyphenyl)methyl)amino)ethyl 4-methylbenzenesulfonate (480.22 mg, 811.69 umol) and 6-((2R)-2-(3-fluorophenyl)pyrrolidin-1-yl)-3-(6-piperazin-1-yl-2-pyridyl)imidazo[1,2-b]pyridazine (300 mg, 676.41 umol) in CH.sub.3CN (20 mL) were added K.sub.2CO.sub.3 (560.07 mg, 4.06 mmol) and NaI (202.77 mg, 1.35 mmol) at room temperature. After the mixture was stirred at 98° C. for 16 h, the reaction was concentrated and purified by silica gel chromatography (DCM/MeOH=50:1 to 20:1) to give 2-(2,6-dioxo-3-piperidyl)-5-(2-(4-(6-(6-((2R)-2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)-2-pyridyl)piperazin-1-yl)ethyl)-((4-methoxyphenyl)methyl)amino) isoindoline-1,3-dione (380 mg, 65% yield) as a yellow solid. MS (ESI) m/z: 864.0 [M+H].sup.+.

Example 266: N-(2-(2,6-Dioxopiperidin-3-yl)-1,3-dioxoisoindolin-5-yl)-1-(6-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperidine-4-carboxamide (TR-215)

(1247) ##STR00730##

Step 1. Synthesis of (R)-1-(6-(6-(2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperidine-4-carboxylic acid

(1248) ##STR00731##

(1249) To a solution of piperidine-4-carboxylic acid (38 mg, 0.30 mmol) and (R)-6-(2-(3-fluorophenyl)pyrrolidin-1-yl)-3-(6-fluoropyridin-2-yl)imidazo[1,2-b]pyridazine (100 mg, 0.26 mmol) in DMSO (8 mL) was added KF (15 mg, 0.26 mmol) at room temperature. After it was stirred at 130° C. for 16 h, the reaction was cooled to room temperature and quenched by H.sub.2O (50 mL). The mixture was extracted with EtOAc (3×10 mL). The combined organic layers were concentrated and purified by reverse-phase chromatography to give the desired product (65 mg, 52% yield) as a light yellow solid. MS (ESI) m/z: 487.2 [M+H].sup.+.

Step 2. Synthesis of (R)-1-(6-(6-(2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperidine-4-carbonyl chloride
(1250) ##STR00732##

(1251) To a mixture of (R)-1-(6-(6-(2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperidine-4-carboxylic acid (65 mg, 0.13 mmol) and DMF (3 mg, 0.04 mmol) in DCM (10 mL) was added oxalyl chloride (330 mg, 2.6 mmol) at room temperature. After it was stirred at reflux for 2 h, the reaction was concentrated and used directly in the next step without further purification (60 mg, 92% yield) as a light yellow solid.

Step 3. Synthesis of N-(2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-5-yl)-1-(6-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperidine-4-carboxamide
(1252) ##STR00733##

(1253) To a mixture of 5-amino-2-(2,6-dioxopiperidin-3-yl)isoindoline-1,3-dione (35 mg, 0.13 mmol) in THF (5 mL) was added (R)-1-(6-(6-(2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperidine-4-carbonyl chloride (60 mg, 0.12 mmol) at room temperature. After it was stirred at room temperature for 1 h, the mixture was concentrated and purified by reverse-phase chromatography to give the desired product N-(2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-5-yl)-1-(6-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperidine-4-carboxamide (12 mg, 13% yield) as a light yellow solid. MS (ESI) m/z: 742.3 [M+H].sup.+.

Example 267: 3-[4-[2-[4-[6-[6-[(2R)-2-(3-Fluorophenyl)pyrrolidin-1-yl]imidazo[1,2-b]pyridazin-3-yl]-2-pyridyl]piperazin-1-yl]ethylamino]-3-methyl-2-oxo-benzimidazol-1-yl]piperidine-2,6-dione (TR-216)

(1254) ##STR00734##

Step 1. Synthesis of 2-(methylamino)-3-nitrobenzoic acid

(1255) ##STR00735##

(1256) To a solution of 2-fluoro-3-nitrobenzoic acid (20 g, 0.108 mol) and methanamine hydrochloride (8.7 g, 0.129 mol) in EtOH (200 mL) was added DIEA (70 g, 0.54 mol) at room temperature. After the reaction mixture was stirred at 80° C. overnight, the reaction was cooled down to room temperature. After concentration, the resulting residue was diluted with water (100 mL) and adjusted the pH to 3 with conc. HCl, and extracted with EtOAc (2×150 mL). The combined organic layers were washed with brine twice, dried with sodium sulfate, filtered and concentrated to give crude 2-(methylamino)-3-nitrobenzoic acid (21 g, 100% yield) as a yellow solid. This product was used in the next step directly without further purification. MS (ESI) m/z: 197.3 [M+H].sup.+.

Step 2. Synthesis of 1-methyl-7-nitro-1,3-dihydro-2H-benzo[d]imidazol-2-one

(1257) ##STR00736##

(1258) To a solution of 2-(methylamino)-3-nitrobenzoic acid (24 g, 0.122 mol) in t-BuOH (300 mL) were added DIPEA (31 g, 0.244 mol) and DPPA (37 g, 0.134 mol) at room temperature. After the reaction mixture was stirred at 90° C. for 16 h, the reaction solution was concentrated. The resulting residue was triturated with water (500 mL), filtered and washed with EtOAc. The solid was collected and dried to give 1-methyl-7-nitro-1,3-dihydro-2H-benzo[d]imidazol-2-one (22 g, 93% yield) as a yellow solid. MS (ESI) m/z: 194. 1 [M+H].sup.+.

Step 3. Synthesis of 3-(3-methyl-5-nitro-2-oxo-benzimidazol-1-yl)piperidine-2,6-dione

(1259) ##STR00737##

(1260) To a suspension of NaH (48 mg, 1.20 mmol) in DMF (5 mL) was added 3-methyl-5-nitro-1H-benzimidazol-2-one (193 mg, 999.18 μmol) at 0° C. The reaction mixture was stirred at 0° C. for 0.5 h, before a solution of 3-bromopiperidine-2,6-dione (383.70 mg, 2.00 mmol) in DMF (5 mL) was added dropwise. After the completion of addition, the reaction mixture was stirred at 80° C. for 2 h. After concentration, the reaction was purified by prep-HPLC to give 3-(3-methyl-5-nitro-2-oxo-benzimidazol-1-yl)piperidine-2,6-dione (80 mg, 26% yield) as a black solid. MS (ESI)

m/z: 305.3 [M+H].sup.+.

Step 4. Synthesis of 3-(4-amino-3-methyl-2-oxo-benzimidazol-1-yl)piperidine-2,6-dione (1261) ##STR00738##

(1262) To a solution of 3-(3-methyl-4-nitro-2-oxo-benzimidazol-1-yl)piperidine-2,6-dione (80 mg, 262.93 μmol) in MeOH (10 mL) and THF (10 mL) was added Pd/C (20 mg) at room temperature. After the reaction mixture was stirred at room temperature for 1 h under hydrogen atmosphere, the reaction was filtered and concentrated to give 3-(4-amino-3-methyl-2-oxo-benzimidazol-1-yl)piperidine-2,6-dione (70 mg, 97% yield) as a brown solid. This product was used in the next step directly without further purification. MS (ESI) m/z: 275.3 [M+H].sup.+.

Step 5. Synthesis of 3-[4-[2-[4-[6-[6-[(2R)-2-(3-fluorophenyl)pyrrolidin-1-yl]imidazo[1,2-b]pyridazin-3-yl]-2-pyridyl]piperazin-1-ylethylamino]-3-methyl-2-oxo-benzimidazol-1-yl]piperidine-2,6-dione (1263) ##STR00739##

(1264) To a solution of 2-[4-[6-[6-[(2R)-2-(3-fluorophenyl)pyrrolidin-1-yl]imidazo[1,2-b]pyridazin-3-yl].sup.-. 2-pyridyl]piperazin-1-yl]acetaldehyde (61.96 mg, 127.61 μmol) and 3-(4-amino-3-methyl-2-oxo-benzimidazol-1-yl)piperidine-2,6-dione (35 mg, 127.61 μmol) in DMF (10 mL) were added TMSCl (41.35 mg, 382.83 μmol) and NaBH.sub.4 (9.70 mg, 255.22 μmol) at 0° C. After the reaction mixture was stirred at 0° C. for 4 h, the reaction was warmed to room temperature and stirred for 16 h. After concentration, the reaction was purified by prep-HPLC to give the product with some impurities. The crude material was further purified by prep-TLC (DCM/MeOH=15:1) to give 3-[4-[2-[4-[6-[6-[(2R)-2-(3-fluorophenyl)pyrrolidin-1-yl]imidazo[1,2-b]pyridazin-3-yl]-2-pyridyl]piperazin-1-yl]ethylamino]-3-methyl-2-oxo-benzimidazol-1-yl]piperidine-2,6-dione (14 mg, 15% yield) as a white solid. MS (ESI) m/z: 744.9 [M+H].sup.+.

Example 268: 3-[4-[3-[4-[6-[6-[(2R)-2-(3-Fluorophenyl)pyrrolidin-1-yl]imidazo[1,2-b]pyridazin-3-yl]-2-pyridyl]piperazin-1-yl]propylamino]-3-methyl-2-oxo-benzimidazol-1-yl]piperidine-2,6-dione (TR-217)

(1265) ##STR00740##

(1266) To a solution of 3-[4-[6-[6-[(2R)-2-(3-fluorophenyl)pyrrolidin-1-yl]imidazo[1,2-b]pyridazin-3-yl].sup.-. 2-pyridyl]piperazin-1-yl]propanal (54.64 mg, 109.38 μmol) and 3-(4-amino-3-methyl-2-oxo-benzimidazol-1-yl)piperidine-2,6-dione (30 mg, 109.38 μmol) in DMF (5 mL) were added TMSCl (41.35 mg, 328.14 μmol) and NaBH.sub.4 (8.31 mg, 218.76 μmol) at 0° C. After the reaction mixture was stirred at 0° C. for 4 h, the reaction was warmed to room temperature and stirred for 16 h. After concentration, the reaction was purified by prep-HPLC to give the crude product, which was further purified a by prep-TLC (DCM/MeOH=15:1) to give 3-[4-[3-[4-[6-[6-[(2R)-2-(3-fluorophenyl)pyrrolidin-1-yl]imidazo[1,2-b]pyridazin-3-yl]-2-pyridyl]piperazin-1-yl]propylamino]-3-methyl-2-oxo-benzimidazol-1-yl]piperidine-2,6-dione (10 mg, 12% yield) as a white solid. MS (ESI) m/z: 759.0 [M+H].sup.+.

Example 269: 3-(5-(((1~ (6-(6-((R)-2-(3-Fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperidin-4-yl)methyl)amino)-1-oxoisindolin-2-yl)piperidine-2,6-dione (TR-218)

(1267) ##STR00741##

Step 1. Synthesis of tert-butyl 4-(((2-(2,6-dioxopiperidin-3-yl)-1-oxoisindolin-5-yl)amino)methyl)piperidine-1-carboxylate

(1268) ##STR00742##

(1269) To a solution of 3-(5-amino-1-oxoisindolin-2-yl)piperidine-2,6-dione (130 mg, 0.5 mmol) in dichloromethane (10 mL) and AcOH (2.0 mL) were added tert-butyl 4-formylpiperidine-1-carboxylate (130 mg, 1.0 mmol) and BH.sub.3/THF (1 M, 0.5 mL). After the mixture was stirred at room temperature for 16 h, the reaction was quenched with water (20 mL) and extracted with ethyl acetate (3×20 mL). The combined organic layers were dried over sodium sulfate, filtered and concentrated to give tert-butyl 4-(((2-(2,6-dioxopiperidin-3-yl)-1-oxoisindolin-5-yl)amino)methyl)piperidine-1-carboxylate (230 mg, crude) as a green solid. This crude product was

used in the next step directly without further purification. MS (ESI) m/z: 457.3 [M+H].sup.+.
Step 2. Synthesis of 3-(1-oxo-5-((piperidin-4-ylmethyl)amino) isoindolin-2-yl)piperidine-2,6-dione (1270) ##STR00743##

(1271) To a solution of tert-butyl 4-(((2-(2,6-dioxopiperidin-3-yl)-1-oxoisoindolin-5-yl)amino)methyl)piperidine-1-carboxylate (230 mg, crude) in DCM (10 mL) was added TFA (2 mL) at 0° C. After the reaction mixture was stirred at room temperature for 16 h, the solvent was removed under vacuum. The resulting residue was washed with diethyl ether to give 3-(1-oxo-5-((piperidin-4-ylmethyl)amino) isoindolin-2-yl)piperidine-2,6-dione (200 mg, crude) as a yellow solid. This product was used in the next step directly without further purification. MS (ESI) m/z: 357.3 [M+H].sup.+.

Step 3. Synthesis of 3-(5-(((1-(6-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperidin-4-yl)methyl)amino)-1-oxoisoindolin-2-yl)piperidine-2,6-dione

(1272) ##STR00744##

(1273) To a solution of 3-(1-oxo-5-((piperidin-4-ylmethyl)amino) isoindolin-2-yl)piperidine-2,6-dione (200 mg, crude) and (R)-6-(2-(3-fluorophenyl)pyrrolidin-1-yl)-3-(6-fluoropyridin-2-yl)imidazo[1,2-b]pyridazine (88 mg, 0.23 mmol) in DMSO (5 mL) was added KF (270 mg, 0.92 mmol). The resulting solution was stirred at 120° C. for 5 h, before the reaction was quenched with water (10 mL) and extracted with ethyl acetate (3×10 mL). The combined organic layers were dried over sodium sulfate, and concentrated. The resulting residue was purified by silica gel column (DCM/MeOH=10:1) to give 3-(5-(((1-(6-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperidin-4-yl)methyl)amino)-1-oxoisoindolin-2-yl)piperidine-2,6-dione (14 mg, 8% yield) as a yellow solid. MS (ESI) m/z: 714.8 [M+H].sup.+.

Example 270:3-(5-(((1-(6-(6-((R)-2-(3-Fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)azetidin-3-yl)ethynyl)-1-oxoisoindolin-2-yl)piperidine-2,6-dione (TR-219)

(1274) ##STR00745##

Step 1. Synthesis of tert-butyl 3-((2-(2,6-dioxopiperidin-3-yl)-1-oxoisoindolin-5-yl)ethynyl)azetidine-1-carboxylate

(1275) ##STR00746##

(1276) To a solution of 3-(5-bromo-1-oxoisoindolin-2-yl)piperidine-2,6-dione (136 mg, 0.42 mmol) in DMF (5 mL) were added TEA (170 mg, 4.2 mmol), Cul (8 mg, 0.04 mmol), Pd(dppf) Cl.sub.2 (31 mg, 0.04 mmol), and tert-butyl 3-ethynylazetidine-1-carboxylate (100 mg, 0.42 mmol). The reaction mixture was stirred at 80° C. for 12 h. Then the resulting black mixture was diluted with ethyl acetate (20 mL), washed with brine (3×20 mL). The organic layer was dried over sodium sulfate, filtered, and concentrated. The resulting residue was purified by silica gel flash chromatography (petroleum/ethyl acetate=3:7) to give tert-butyl 3-((2-(2,6-dioxopiperidin-3-yl)-1-oxoisoindolin-5-yl)ethynyl)azetidine-1-carboxylate (130 mg, 73% yield) as a brown solid. MS (ESI) m/z: 424.2 [M+H].sup.+.

Step 2. Synthesis of 3-(5-(azetidin-3-ylethynyl)-1-oxoisoindolin-2-yl)piperidine-2,6-dione

(1277) ##STR00747##

(1278) To a solution of tert-butyl 3-((2-(2,6-dioxopiperidin-3-yl)-1-oxoisoindolin-5-yl)ethynyl)azetidine-1-carboxylate (130 mg, 0.30 mmol) in DCM (5 mL) was added TFA (1 mL) at 0° C. After the reaction was stirred at room temperature for 1 h, the solvent was removed under vacuum. The residue was washed with diethyl ether to give crude 3-(5-(azetidin-3-ylethynyl)-1-oxoisoindolin-2-yl)piperidine-2,6-dione (86 mg, 89% yield) as a yellow solid. This product was used in the next step directly without further purification. MS (ESI) m/z: 324.2 [M+H].sup.+.

Step 3. Synthesis of 3-(5-(((1-(6-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)azetidin-3-yl)ethynyl)-1-oxoisoindolin-2-yl)piperidine-2,6-dione

(1279) ##STR00748##

(1280) To a solution of 3-(5-(azetidin-3-ylethynyl)-1-oxoisoindolin-2-yl)piperidine-2,6-dione (86

mg, 0.26 mmol) and (R)-6-(2-(3-fluorophenyl)pyrrolidin-1-yl)-3-(6-fluoropyridin-2-yl)imidazo[1,2-b]pyridazine (60 mg, 0.26 mmol) in DMSO (5 mL) was added KF (77 mg, 1.04 mmol). The resulting solution was stirred at 120° C. for 5 h, before the reaction was quenched with water (20 mL) and extracted with ethyl acetate (3×20 mL). The combined organic layers were dried over sodium sulfate, filtered and concentrated. The resulting residue was purified by silica gel column (DCM/MeOH=10:1) to give 3-(5-((1-(6-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)azetid-3-yl)ethynyl)-1-oxoisindolin-2-yl)piperidine-2,6-dione (47 mg, 26% yield) as a yellow solid. MS (ESI) m/z: 681.2 [M+H].sup.+.

Example 271: 3-(5-((2-(4-(6-(6-((R)-2-(3-Fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)ethyl)amino)-3-methyl-2-oxo-2,3-dihydro-1H-benzo[d]imidazol-1-yl)piperidine-2,6-dione (TR-220)

(1281) ##STR00749##

Step 1. Synthesis of 2-(methylamino)-4-nitro-benzoic acid

(1282) ##STR00750##

(1283) To a solution of 2-fluoro-4-nitro-benzoic acid (37 g, 199.88 mmol) and methanamine hydrochloride (12.42 g, 399.76 mmol) in NMP (300 mL) was added DIPEA (103.33 g, 799.53 mmol) at room temperature. The reaction mixture was stirred at 120° C. for 48 h. After the reaction was cooled down to room temperature, the reaction was quenched with water (100 mL), and the pH was adjusted to 3 with conc. HCl. The resulting solid was filtered and dried to give crude 2-(methylamino)-4-nitro-benzoic acid (38 g, 97% yield) as a yellow solid. This product was used in the next step directly without further purification. MS (ESI) m/z: 197.3 [M+H].sup.+.

Step 2. Synthesis of 3-methyl-5-nitro-1H-benzimidazol-2-one

(1284) ##STR00751##

(1285) To a solution of 2-(methylamino)-4-nitro-benzoic acid (38 g, 193.72 mmol) in t-BuOH (30 mL) were added DIPEA (75.11 g, 581.16 mmol) and DPPA (63.93 g, 232.46 mmol) at room temperature. Then the reaction mixture was stirred at 90° C. for 16 h. After the reaction was cooled down to room temperature, the mixture was filtered and resulting solid was dried to give crude 3-methyl-5-nitro-1H-benzimidazol-2-one (33 g, 88% yield) as a yellow solid. This product was used in the next step directly without further purification. MS (ESI) m/z: 194.1 [M+H].sup.+.

Step 3. Synthesis of 3-(5-amino-3-methyl-2-oxo-2,3-dihydro-1H-benzo[d]imidazol-1-yl)piperidine-2,6-dione

(1286) ##STR00752##

(1287) To a suspension of NaH (48 mg, 1.20 mmol) in DMF (5 mL) was added 1-methyl-6-nitro-1,3-dihydro-2H-benzo[d]imidazol-2-one (193 mg, 999.18 umol) at 0° C. The reaction mixture was stirred at 0° C. for 0.5 h, before a solution of 3-bromopiperidine-2,6-dione (383.70 mg, 2.00 mmol) in DMF (5 mL) was added dropwise. After the completion of addition, the reaction mixture was stirred at 80° C. for 2 h. After the reaction is cooled down to room temperature, the reaction was concentrated and the resulting residue was purified by prep-HPLC to give 3-(5-amino-3-methyl-2-oxo-2,3-dihydro-1H-benzo[d]imidazol-1-yl)piperidine-2,6-dione (80 mg, 26% yield) as a black solid. MS (ESI) m/z: 305.3 [M+H].sup.+.

Step 4. Synthesis of 3-(5-amino-3-methyl-2-oxo-2,3-dihydro-1H-benzo[d]imidazol-1-yl)piperidine-2,6-dione

(1288) ##STR00753##

(1289) To a solution of 3-(5-amino-3-methyl-2-oxo-2,3-dihydro-1H-benzo[d]imidazol-1-yl)piperidine-2,6-dione (80 mg, 262.93 umol) in MeOH (10 mL) and THF (10 mL) was added Pd/C (20 mg) at room temperature. After the reaction mixture was stirred at room temperature for 1 h under hydrogen atmosphere, the reaction was filtered and the filtrate was concentrated to give crude 3-(5-amino-3-methyl-2-oxo-2,3-dihydro-1H-benzo[d]imidazol-1-yl)piperidine-2,6-dione (70 mg, 97% yield) as a brown solid. This product was used in the next step directly without further purification. MS (ESI) m/z: 275.3 [M+H].sup.+.

Step 5. Synthesis of 3-(5-((2-(4-(6-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)ethyl)amino)-3-methyl-2-oxo-2,3-dihydro-1H-benzo[d]imidazol-1-yl)piperidine-2,6-dione

(1290) ##STR00754##

(1291) To a solution of (R)-2-(4-(6-(6-(2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)acetaldehyde (57 mg, 0.12 mmol) in DMF (3 ml) were added 3-(5-amino-3-methyl-2-oxo-2,3-dihydro-1H-benzo[d]imidazol-1-yl)piperidine-2,6-dione (31 mg, 0.11 mmol), TMSCl (39 mg, 0.36 mmol) and NaBH.sub.4 (9 mg, 0.24 mmol). The reaction mixture was stirred at 0° C. for 48 h. After the reaction was concentrated, the resulting residue was purified by prep-HPLC to give 3-(5-((2-(4-(6-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)ethyl)amino)-3-methyl-2-oxo-2,3-dihydro-1H-benzo[d]imidazol-1-yl)piperidine-2,6-dione (7 mg, 8% yield) as a yellow solid. MS (ESI) m/z: 744.8 [M+H].sup.+.

Example 272: 3-(5-((3-(4-(6-(6-((R)-2-(3-Fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)propyl)amino)-3-methyl-2-oxo-2,3-dihydro-1H-benzo[d]imidazol-1-yl)piperidine-2,6-dione (TR-221)

(1292) ##STR00755##

(1293) To a solution of (R)-3-(4-(6-(6-(2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)propanal (132.92 mg, 266.06 umol) and 3-(5-amino-3-methyl-2-oxo-2,3-dihydro-1H-benzo[d]imidazol-1-yl)piperidine-2,6-dione (82 mg, 298.97 umol) in DMF (10 mL) were added TMSCl (86.20 mg, 798.17 umol) and NaBH.sub.4 (20.22 mg, 532.11 umol) at 0° C. After the reaction mixture was stirred at 0° C. for 1.5 h, the reaction was warmed to room temperature and stirred for 16 h. After concentration, the mixture was purified by prep-HPLC to give a crude product, which was further purified by prep-TLC (DMC/MeOH=10:1) to give 3-(5-((3-(4-(6-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)propyl)amino)-3-methyl-2-oxo-2,3-dihydro-1H-benzo[d]imidazol-1-yl)piperidine-2,6-dione (17 mg, 8% yield) as a white solid. MS (ESI) m/z: 759.0 [M+H].sup.+.

Example 273: 2-(2,6-Dioxopiperidin-3-yl)-5-(3-((4-(3-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)-1H-pyrrol-1-yl)piperidin-1-yl)methyl)azetidin-1-yl)isoindoline-1,3-dione (TR-222)

(1294) ##STR00756## ##STR00757##

Step 1. Synthesis of tert-butyl (R)-3-(6-(2-(3~fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)-2,5-dihydro-1H-pyrrole-1-carboxylate

(1295) ##STR00758##

(1296) To a solution of (R)-3-bromo-6-(2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazine (360 mg, 996.65 umol) and tert-butyl 3-(4,4,5,5-tetramethyl-1,3,2-dioxaborolan-2-yl)-2,5-dihydropyrrole-1-carboxylate (294.19 mg, 996.65 umol) in dioxane (10 mL) and H.sub.2O (5 mL) were added K.sub.2CO.sub.3 (275.07 mg, 1.99 mmol) and Pd(dppf) Cl.sub.2 (72.85 mg, 99.66 umol) at room temperature under N.sub.2. After the reaction mixture was stirred at 100° C. for 4 h, the reaction was concentrated and purified by silica gel chromatography (DCM/MeOH=30:1) to give tert-butyl (R)-3-(6-(2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2~b]pyridazin-3-yl)-2,5-dihydro-1H-pyrrole-1-carboxylate (440 mg, 98% yield) as a black solid. MS (ESI) m/z: 451.0 [M+H].sup.+.

Step 2. Synthesis of (R)-3-(2,5-dihydro-1H-pyrrol-3-yl)-6-(2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazine hydrochloride

(1297) ##STR00759##

(1298) To a solution of tert-butyl (R)-3-(6-(2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)-2,5-dihydro-1H-pyrrole-1-carboxylate (450 mg, 1.00 mmol) in DCM (5 mL) was added HCl/dioxane (4 mL) at room temperature. The reaction mixture was stirred at room temperature for 2 h, before it was concentrated to give crude (R)-3-(2,5-dihydro-1H-pyrrol-3-yl)-6-

(2~ (3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazine hydrochloride (350 mg, 100% yield) as solid (HCl salt). This product was used in the next step directly without further purification. MS (ESI) m/z: 350.6 [M+H].sup.+.

Step 3. Synthesis of benzyl (R)-4-(3-(6-(2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)-2,5-dihydro-1H-pyrrol-1-yl)piperidine-1-carboxylate
(1299) ##STR00760##

(1300) To a solution of 3-(2,5-dihydro-1H-pyrrol-3-yl)-6-[(2R)-2-(3-fluorophenyl)pyrrolidin-1-yl]imidazo[1,2-b]pyridazine (600 mg, 1.72 mmol) and benzyl 4-methylsulfonyloxypiperidine-1-carboxylate (2.15 g, 6.87 mmol) in CH₂Cl₂ (20 mL) and DMF (20 mL) were added NaI (514.79 mg, 3.43 mmol) and K₂CO₃ (1.42 g, 10.30 mmol). The reaction mixture was stirred at 100° C. for 16 h, before it was concentrated and purified by silica gel chromatography (DCM/MeOH=20:1) to give benzyl (R)-4-(3-(6-(2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)-2,5-dihydro-1H-pyrrol-1-yl)piperidine-1-carboxylate (230 mg, 24% yield). MS (ESI) m/z: 567.9 [M+H].sup.+.

Step 4. Synthesis of (R)-6-(2-(3-fluorophenyl)pyrrolidin-1-yl)-3-(1-(piperidin-4-yl)-2,5-dihydro-1H-pyrrol-3-yl)imidazo[1,2-b]pyridazine

(1301) ##STR00761##

(1302) To a solution of benzyl (R)-4-(3-(6-(2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)-2,5-dihydro-1H-pyrrol-1-yl)piperidine-1-carboxylate (200 mg, 352.94 μmol) in AcOH (10 mL) was added HBr (5 mL) at room temperature. The reaction mixture was stirred at 50° C. for 16 h, before the reaction was concentrated. The resulting residue was diluted with water and pH value was adjusted to 9 with aq. NaHCO₃. The resulting aqueous mixture was extracted with DCM (2×30 mL). The combined organic layers were washed with brine (2×20 mL), dried over Na₂SO₄, filtered, and concentrated. The resulting residue was purified by prep-TLC (DCM/MeOH=10:1) to give (R)-6-(2-(3-fluorophenyl)pyrrolidin-1-yl)-3-(1-(piperidin-4-yl)-2,5-dihydro-1H-pyrrol-3-yl)imidazo[1,2-b]pyridazine (120 mg, 79% yield) as a white solid. MS (ESI) m/z: 433.6 [M+H].sup.+.

Step 5. Synthesis of 2-(2,6-dioxopiperidin-3-yl)-5-(3-((4-(3-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)-1H-pyrrol-1-yl)piperidin-1-yl)methyl)azetidin-1-yl)isoindoline-1,3-dione

(1303) ##STR00762##

(1304) To a solution of (R)-6-(2-(3-fluorophenyl)pyrrolidin-1-yl)-3-(1-(piperidin-4-yl)-2,5-dihydro-1H-pyrrol-3-yl)imidazo[1,2-b]pyridazine (60 mg, 138.72 μmol) and (1-(2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisoindolin-5-yl)azetidin-3-yl)methyl 4-methylbenzenesulfonate (69.01 mg, 138.72 μmol) in CH₂Cl₂ (10 mL) were added NaI (41.58 mg, 277.43 μmol) and K₂CO₃ (114.86 mg, 832.30 μmol) at room temperature. After the mixture was stirred at 95° C. for 16 h, the reaction was concentrated and purified by prep-TLC (DCM/MeOH=10:1) to give crude product, then purified again by prep-TLC (DCM/MeOH=20:1) to give 2-(2,6-dioxopiperidin-3-yl)-5-(3-((4-(3-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)-1H-pyrrol-1-yl)piperidin-1-yl)methyl)azetidin-1-yl)isoindoline-1,3-dione (17 mg, 16% yield) as a yellow solid. MS (ESI) m/z: 757.0 [M+H].sup.+.

Example 274:3—((S)-5-(3-(3-(4-(6-(6-((R)-2-(3-Fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)prop-1-yn-1-yl)phenyl)-2-oxooxazolidin-3-yl)piperidine-2,6-dione (TR-223)

(1305) ##STR00763##

Step 1. Synthesis of 3-(3-((5S)-3-(2,6-dioxopiperidin-3-yl)-2-oxooxazolidin-5-yl)phenyl)prop-2-yn-1-yl 4-methylbenzenesulfonate

(1306) ##STR00764##

(1307) To a solution of 3-((S)-5-(3-(3-hydroxyprop-1-yn-1-yl)phenyl)-2-oxooxazolidin-3-yl)piperidine-2,6-dione (30 mg, 0.1 mmol) in dichloromethane (2 mL) were added TsCl (36 mg,

0.18 mmol) and DMAP (21.9 mg, 0.2 mmol). After the mixture was stirred at room temperature for 4 h, the reaction was quenched with water (20 mL) and extracted with ethyl acetate (3×20 mL). The combined organic layers were dried over sodium sulfate, filtered, and concentrated. The resulting residue was purified by silica gel chromatography (MeOH/DCM=1:10) to give 3-(3-((5S)-3-(2,6-dioxopiperidin-3-yl)-2-oxooxazolidin-5-yl)phenyl)prop-2-yn-1-yl 4-methylbenzenesulfonate (22 mg, 70% yield) as a yellow solid. MS (ESI) m/z: 348.3 [M+H].sup.+.

Step 2. Synthesis of 3-((S)-5-(3-(3-(4-(6-(6-((S)-2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)prop-1-yn-1-yl)phenyl)-2-oxooxazolidin-3-yl)piperidine-2,6-dione

(1308) ##STR00765##

(1309) To a solution of 3-(3-((5S)-3-(2,6-dioxopiperidin-3-yl)-2-oxooxazolidin-5-yl)phenyl)prop-2-yn-1-yl 4-methylbenzenesulfonate (22 mg, 0.06 mmol) and (R)-6-(2-(3-fluorophenyl)pyrrolidin-1-yl)-3-(6-(piperazin-1-yl)pyridin-2-yl)imidazo[1,2-b]pyridazine (26 mg, 0.06 mmol) in acetonitrile (2 mL) were added potassium carbonate (13 mg, 0.12 mmol) and sodium iodide (8 mg, 0.2 mmol). The reaction mixture was stirred at 80° C. for 2 h, before the reaction was concentrated and purified by silica gel chromatography (DCM/MeOH=10:1) to give 3-((S)-5-(3-(3-(4-(6-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)prop-1-yn-1-yl)phenyl)-2-oxooxazolidin-3-yl)piperidine-2,6-dione (25 mg, 38% yield) as a yellow solid. MS (ESI) m/z: 754.8 [M+H].sup.+.

Example 275: 3~(S)-5-(4-(3-(4-(6-(6-((R)-2-(3-Fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)prop-1-yn-1-yl)phenyl)-2-oxooxazolidin-3-yl)piperidine-2,6-dione (TR-224)

(1310) ##STR00766##

Step 1. Synthesis of 3-(4-((5S)-3-(2,6-dioxopiperidin-3-yl)-2-oxooxazolidin-5-yl)phenyl)prop-2-yn-1-yl 4-methylbenzenesulfonate

(1311) ##STR00767##

(1312) To a solution of 3-((S)-5-(4-(3-hydroxyprop-1-yn-1-yl)phenyl)-2-oxooxazolidin-3-yl)piperidine-2,6-dione (40 mg, 0.12 mmol) in dichloromethane (2 mL) were added TsCl (68 mg, 0.35 mmol) and DMAP (76 mg, 0.6 mmol) at room temperature. The reaction mixture was stirred at room temperature for 4 h, before the reaction was quenched with water (20 mL) and extracted with ethyl acetate (3×20 mL). The combined organic layers were dried over sodium sulfate, filtered, and concentrated. The resulting residue was purified by silica gel chromatography (MeOH/DCM=1:10) to give 3-(4-((5S)-3-(2,6-dioxopiperidin-3-yl)-2-oxooxazolidin-5-yl)phenyl)prop-2-yn-1-yl 4-methylbenzenesulfonate (30 mg, 71% yield) as a yellow solid. MS (ESI) m/z: 483.3 [M+H].sup.+.

Step 2. Synthesis of 3-((S)-5-(4-(3-(4-(6-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)prop-1-yn-1-yl)phenyl)-2-oxooxazolidin-3-yl)piperidine-2,6

(1313) ##STR00768##

(1314) To a solution of 3-(4-((5S)-3-(2,6-dioxopiperidin-3-yl)-2-oxooxazolidin-5-yl)phenyl)prop-2-yn-1-yl 4-methylbenzenesulfonate (30 mg, 0.08 mmol) and (R)-6-(2-(3-fluorophenyl)pyrrolidin-1-yl)-3-(6-(piperazin-1-yl)pyridin-2-yl)imidazo[1,2-b]pyridazine (27 mg, 0.08 mmol) in acetonitrile (3 mL) were added potassium carbonate (17 mg, 0.16 mmol) and sodium iodide (9 mg, 0.08 mmol). The reaction mixture was stirred at 80° C. for 2 h, before it was concentrated and purified by silica gel chromatography (DCM/MeOH=10:1) to give 3-((S)-5-(4-(3-(4-(6-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)prop-1-yn-1-yl)phenyl)-2-oxooxazolidin-3-yl)piperidine-2,6-dione (17 mg, 36% yield) as a yellow solid. MS (ESI) m/z: 754.8 [M+H].sup.+.

Example 276: 2-(2,6-Dioxopiperidin-3-yl)-5-(3-((4-(6-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)methyl)pyrrolidin-1-yl)isoindoline-

1,3-dione (TR-225)

(1315) ##STR00769##

Step 1. Synthesis of 2-(2,6-dioxopiperidin-3-yl)-5-(3-(hydroxymethyl)pyrrolidin-1-yl)isoindoline-1,3-dione

(1316) ##STR00770##

(1317) A solution of 2-(2,6-dioxopiperidin-3-yl)-5-fluoroisoindoline-1,3-dione (100 mg, 0.362 mmol), pyrrolidin-3-ylmethanol (55 mg, 0.543 mmol) and DIEA (234 mg, 1.81 mmol) in NMP (5 mL) was stirred at 120° C. for half hour. After the reaction was cooled down to room temperature, the resulting solution was diluted with EtOAc (100 mL), washed with brine twice, dried over sodium sulfate and concentrated to give the crude 2-(2,6-dioxopiperidin-3-yl)-5-(3-(hydroxymethyl)pyrrolidin-1-yl)isoindoline-1,3-dione (140 mg, 108% yield) as a colorless oil. This product was used in the next step directly without further purification. MS (ESI) m/z: 358.5 [M+H].sup.+.

Step 2. Synthesis of 1-(2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-5-yl)pyrrolidine-3-carbaldehyde

(1318) ##STR00771##

(1319) A mixture of 2-(2,6-dioxopiperidin-3-yl)-5-(3-(hydroxymethyl)pyrrolidin-1-yl)isoindoline-1,3-dione (140 mg, 0.39 mmol) and Dess-Martin oxidant (826 mg, 1.95 mmol) in DCM (5 mL) and DMF (2 mL) was stirred at 40° C. for 20 mins. After the reaction was cooled down to room temperature, the reaction mixture was diluted with brine (100 mL) and EtOAc (100 mL). The organic phase was washed with brine twice, dried over sodium sulfate, and concentrated. The resulting residue was purified by silica gel chromatography (DCM/MeOH=15:1) to afford 1-(2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-5-yl)pyrrolidine-3-carbaldehyde (80 mg, 58% yield) as a yellow solid. MS (ESI) m/z: 356.7 [M+H].sup.+.

Step 3. Synthesis of 2-(2,6-dioxopiperidin-3-yl)-5-(3-((4-(6-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)methyl)pyrrolidin-1-yl)isoindoline-1,3-dione

(1320) ##STR00772##

(1321) To a solution of 1-(2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-5-yl)pyrrolidine-3-carbaldehyde (80 mg, 0.225 mmol) and (R)-6-(2-(3-fluorophenyl)pyrrolidin-1-yl)-3-(6-(piperazin-1-yl)pyridin-2-yl)imidazo[1,2-b]pyridazine hydrochloride (107 mg, 0.225 mmol) in DCE (3 mL) and THF (3 mL) were added DIEA (30 mg, 0.225 mmol) and titanium tetrakisopropanolate (639 mg, 2.25 mmol). After the reaction mixture was stirred at 40° C. for 0.5 h, sodium cyanoborohydride (140 mg, 2.25 mmol) was added. The resulting reaction mixture was stirred at room temperature for 1 h, before the reaction mixture was diluted with DCM/MeOH (10:1, 150 mL) and saturated NaHCO₃ (20 mL). After being stirred for 5 mins, the mixture was filtered through celite. The filtrate was washed with brine twice, dried over sodium sulfate, filtered and concentrated. The resulting residue was purified by prep-HPLC to give 2-(2,6-dioxopiperidin-3-yl)-5-((4-(6-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)methyl)pyrrolidin-1-yl)isoindoline-1,3-dione (9.6 mg, 6% yield) as a yellow solid. MS (ESI) m/z: 784.0 [M+H].sup.+.

Example 277: 2-(2,6-Dioxopiperidin-3-yl)-5-(4-((4-(6-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)methyl)piperidin-1-yl)isoindoline-1,3-dione (TR-226)

(1322) ##STR00773##

Step 1. Synthesis of 2-(2,6-dioxopiperidin-3-yl)-5-(4-(hydroxymethyl)piperidin-1-yl)isoindoline-1,3-dione

(1323) ##STR00774##

(1324) A solution of 2-(2,6-dioxopiperidin-3-yl)-5-fluoroisoindoline-1,3-dione (130 mg, 0.471 mmol), piperidin-4-ylmethanol (81 mg, 0.706 mmol) and DIEA (303 mg, 2.355 mmol) in NMP (5

ml) was stirred at 120° C. for 0.5 h. After the reaction was cooled down to room temperature, the resulting solution was diluted with EtOAc (100 mL), washed with brine twice, dried over sodium sulfate, filtered and concentrated to give the crude product 2-(2,6-dioxopiperidin-3-yl)-5-(4-(hydroxymethyl)piperidin-1-yl)isoindoline-1,3-dione (200 mg, 114% yield) as a colorless oil. This product was used in the next step directly without further purification. MS (ESI) m/z: 372.5 [M+H].sup.+.

Step 2. Synthesis of 1-(2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-5-yl)pyrrolidine-3-carbaldehyde

(1325) ##STR00775##

(1326) A mixture of 2-(2,6-dioxopiperidin-3-yl)-5-(4-(hydroxymethyl)piperidin-1-yl)isoindoline-1,3-dione (200 mg, 0.539 mmol) and Dess-Martin oxidant (1.14 mg, 2.69 mmol) in DCM (5 ml) and DMF (2 ml) was stirred at 40° C. for 20 mins. After the reaction was cooled down to room temperature, the reaction mixture was diluted with brine (100 mL) and EtOAc (100 mL). The organic phase was washed with brine twice, dried, filtered and concentrated. The resulting residue was purified by silica gel chromatography (DCM/MeOH=15:1) to afford 1-(2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-5-yl)pyrrolidine-3-carbaldehyde (50 mg, 25% yield) as a yellow solid. MS (ESI) m/z: 370.5 [M+H].sup.+.

Step 3. Synthesis of 2-(2,6-dioxopiperidin-3-yl)-5-(4-((4-(6-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)methyl)piperidin-1-yl)isoindoline-1,3-dione

(1327) ##STR00776##

(1328) To a solution of 1-(2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-5-yl)piperidine-4-carbaldehyde (50 mg, 0.135 mmol) and (R)-6-(2-(3-fluorophenyl)pyrrolidin-1-yl)-3-(6-(piperazin-1-yl)pyridin-2-yl)imidazo[1,2-b]pyridazine hydrochloride (51 mg, 0.135 mmol) in DCE (6 mL) were added DIEA (18 mg, 0.135 mmol) and titanium tetrakisopropanolate (383 mg, 1.35 mmol). The reaction mixture was stirred at 40° C. for 0.5 h, before sodium cyanoborohydride (140 mg, 2.25 mmol) and MeOH (3 ml) was added. After the reaction mixture was stirred at room temperature for 1 h, the reaction mixture was diluted with DCM/MeOH (10:1, 150 mL) and saturated aqueous NaHCO₃ (20 ml). After being stirred for 5 mins, the mixture was filtered through celite. And the filtrate was washed with brine twice. The organic layer was dried over sodium sulfate, filtered and concentrated. The resulting residue was purified by prep-HPLC to give 2-(2,6-dioxopiperidin-3-yl)-5-(4-((4-(6-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)methyl)piperidin-1-yl)isoindoline-1,3-dione (8.6 mg, 8% yield) as a yellow solid. MS (ESI) m/z: 798.0 [M+H].sup.+.

Example 278: 3-(3-(2-(4-(6-(6-((R)-2-(3-Fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)ethyl)-2-oxo-2,3-dihydro-1H-benzo[d]imidazol-1-yl)piperidine-2,6-dione (TR-227)

(1329) ##STR00777## ##STR00778##

Step 1. Synthesis of 2-((2-nitrophenyl)amino)ethan-1-ol

(1330) ##STR00779##

(1331) To a solution of 1-fluoro-2-nitrobenzene (10 g, 70.9 mmol) in EtOH (80 mL) were added K₂CO₃ (11.7 g, 85.04 mmol) and 2-aminoethan-1-ol (13.0 g, 212.6 mmol). The reaction was stirred at reflux for 12 h. After the reaction was cooled to room temperature, water and ethyl acetate were added. The organic phase was separated, dried over anhydrous sodium sulfate, filtered and concentrated. The resulting crude product was triturated with EtOAc/petroleum ether (20 mL, 10:1) to give 2-((2-nitrophenyl)amino)ethan-1-ol (11 g, 85% yield) as a yellow solid. MS (ESI) m/z: 183.4 [M+H].sup.+.

Step 2. Synthesis of 2-((2-aminophenyl)amino)ethan-1-ol

(1332) ##STR00780##

(1333) To a solution of 2-((2-nitrophenyl)amino)ethan-1-ol (7.0 g, 38.5 mmol) in EtOH (30 mL)

was added 10% Pd/C (0.5 g). The reaction mixture was stirred at room temperature overnight under hydrogen atmosphere. After filtration through a celite cup, the filtrate was concentrated to give crude 2-((2-aminophenyl)amino)ethan-1-ol (5.9 g, 99% yield). This product was used in the next step directly without further purification. MS (ESI) m/z: 153.3 [M+H].sup.+.

Step 3. Synthesis of N'-(2-((tert-butyldimethylsilyl)oxy)ethyl)benzene-1,2-diamine (1334) ##STR00781##

(1335) To a solution of 2-((2-aminophenyl)amino)ethan-1-ol (1.0 g, 6.58 mmol) in DCM (20 mL) were added imidazole (671 mg, 9.87 mmol) and tert-butyldimethylsilyl chloride (1.19 g, 7.89 mmol). After the mixture was stirred at room temperature for 15 h, the reaction was quenched with water and extracted with ethyl acetate (50 mL). The organic layer was washed with water twice, saturated brine, dried over sodium sulfate, filtered and concentrated. The resulting residue was triturated with petroleum ether/EtOAc (20 mL/2 mL) to give N-(2-((tert-butyldimethylsilyl)oxy)ethyl)benzene-1,2-diamine (1.1 g, 63% yield) as a pale solid. MS (ESI) m/z: 268.0 [M+H].sup.+.

Step 4. Synthesis of 1-(2-((tert-butyldimethylsilyl)oxy)ethyl)-1,3-dihydro-2H-benzo[d]imidazol-2-one (1336) ##STR00782##

(1337) To a solution of N-(2-((tert-butyldimethylsilyl)oxy)ethyl)benzene-1,2-diamine (1.7 g, 6.39 mmol), in THF (25 mL) was added N, N'-carbonyldiimidazole (2.07 g, 12.78 mmol) under N.sub.2. The reaction mixture was stirred at room temperature for 4 h, before the solvent was removed under vacuum. The resulting residue was recrystallized from methanol and n-hexane to give 1-(2-((tert-butyldimethylsilyl)oxy)ethyl)-1,3-dihydro-2H-benzo[d]imidazol-2-one (1.4 g, 75% yield) as a white solid. MS (ESI) m/z: 293.3 [M+H].sup.+.

Step 5. Synthesis of 3-(3-(2-((tert-butyldimethylsilyl)oxy)ethyl)-2-oxo-2,3-dihydro-1H-benzo[d]imidazol-1-yl)piperidine-2,6-dione (1338) ##STR00783##

(1339) To a stirred solution of 1-(2-((tert-butyldimethylsilyl)oxy)ethyl)-1,3-dihydro-2H-benzo[d]imidazol-2-one (458 mg, 1.57 mmol) in DMF (2 mL) was added NaH (69.08 mg, 60% w/w dispersed into mineral oil, 1.73 mmol) at 0° C. under N.sub.2. After the reaction mixture was stirred for 20 min at 0° C., a solution of 3-bromopiperidine-2, 6-dione (150 mg, 0.79 mmol) in DMF (0.5 mL) was added dropwise at 0° C. After the resulting mixture was stirred for additional 3 h at room temperature, the reaction was quenched with AcOH (0.5 mL) and concentrated under reduced pressure. The residue was purified by silica gel chromatography column (petroleum ether/EtOAc=4:1) to afford 3-(3-(2-((tert-butyldimethylsilyl)oxy)ethyl)-2-oxo-2,3-dihydro-1H-benzo[d]imidazol-1-yl)piperidine-2,6-dione (132 mg, 21% yield) as a white solid. MS (ESI) m/z: 404.9 [M+H].sup.+.

Step 6. Synthesis of 3-(3-(2-hydroxyethyl)-2-oxo-2,3-dihydro-1H-benzo[d]imidazol-1-yl)piperidine-2,6-dione (1340) ##STR00784##

(1341) To a solution of 3-(3-(2-((tert-butyldimethylsilyl)oxy)ethyl)-2-oxo-2,3-dihydro-1H-benzo[d]imidazol-1-yl)piperidine-2,6-dione (130 mg, 0.32 mmol) in tetrahydrofuran (10 mL) was added tetra-n-butyl ammonium fluoride (168.71 mg, 0.65 mmol). The mixture was stirred at room temperature for 1 b. After removal of the solvent, the resulting residue was purified by silica gel chromatography column (hexane/EtOAc=1:1) to afford 3-(3-(2-hydroxyethyl)-2-oxo-2,3-dihydro-1H-benzo[d]imidazol-1-yl)piperidine-2,6-dione (83 mg, 90% yield) as a white solid. MS (ESI) m/z: 290.3 [M+H].sup.+.

Step 7. Synthesis of 2-(3-(2,6-dioxopiperidin-3-yl)-2-oxo-2,3-dihydro-1H-benzo[d]imidazol-1-yl)ethyl 4-methylbenzenesulfonate (1342) ##STR00785##

(1343) To a solution of 3-(3-(2-hydroxyethyl)-2-oxo-2,3-dihydro-1H-benzo[d]imidazol-1-

yl)piperidine-2,6-dione (83 mg, 0.29 mmol) in dry CH₂Cl₂ (5 mL) were added p-toluenesulfonyl chloride (82.1 mg, 0.43 mmol), 4-dimethylaminopyridine (12 mg, 0.1 mmol) and Et₃N (279 mL, 2 mmol) at 0° C. After the reaction mixture was stirred at room temperature for 24 h. the reaction was quenched with water, and extracted with DCM. The organic layer was washed with brine and dried over MgSO₄, filtered and concentrated. The resulting residue was purified by prep-TLC (petroleum ether/EtOAc=2:1) to afford 2-(3-(2,6-dioxopiperidin-3-yl)-2-oxo-2,3-dihydro-1H-benzo[d]imidazol-1-yl)ethyl 4-methylbenzenesulfonate (60 mg, 47% yield) as a white solid. MS (ESI) m/z: 444.5 [M+H]⁺.

Step 8. Synthesis of 3-(3-(2-(4-(6-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)ethyl)-2-oxo-2,3-dihydro-1H-benzo[d]imidazol-1-yl)piperidine-2,6-dione

(1344) ##STR00786##

(1345) To a solution of 2-(3-(2,6-dioxopiperidin-3-yl)-2-oxo-2,3-dihydro-1H-benzo[d]imidazol-1-yl)ethyl 4-methylbenzenesulfonate (60 mg, 0.14 mmol) and (R)-6-(2-(3-fluorophenyl)pyrrolidin-1-yl)-3-(6-(piperazin-1-yl)pyridin-2-yl)imidazo[1,2-b]pyridazine (67 mg, 0.14 mmol) in DMF (2 mL) was added DIEA (89.6 mg, 0.70 mmol). The reaction mixture was stirred at 60° C. for 12 h. After the reaction was cooled down to room temperature, the solution was quenched with water and extracted with EtOAc. The organic layer was washed with brine and dried over MgSO₄, filtered and concentrated. The resulting residue was purified by prep-TLC to give 3-(3-(2-(4-(6-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)ethyl)-2-oxo-2,3-dihydro-1H-benzo[d]imidazol-1-yl)piperidine-2,6-dione (17 mg, 17% yield) as a white solid. MS (ESI) m/z: 715.9 [M+H]⁺.

Example 279: 3-(3-(3-(4-(6-(6-((R)-2-(3-Fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)propyl)-2-oxo-2,3-dihydro-1H-benzo[d]imidazol-1-yl)piperidine-2,6-dione (TR-228)

(1346) ##STR00787## ##STR00788##

Step 1. Synthesis of 3-((2-nitrophenyl)amino)propan-1-ol

(1347) ##STR00789##

(1348) To a solution of 1-fluoro-2-nitrobenzene (10 g, 70.87 mmol) in EtOH (80 mL) were added K₂CO₃ (11.7 g, 85.04 mmol) and 3-aminopropan-1-ol (15.9 g, 212.6 mmol). After the reaction was stirred at reflux for 12 h, the reaction was cooled to room temperature, quenched with water, and diluted with ethyl acetate. The organic phase was separated, dried over anhydrous sodium sulfate, filtered and concentrated. The resulting crude product was triturated with EtOAc/petroleum ether (20 mL, 10:1) to give 3-((2-nitrophenyl)amino)propan-1-ol (11.1 g, 80% yield) as a yellow solid. MS (ESI) m/z: 197.3 [M+H]⁺.

Step 2. Synthesis of 3-((2-aminophenyl)amino)propan-1-ol

(1349) ##STR00790##

(1350) To a solution of 3-((2-nitrophenyl)amino)propan-1-ol (7.0 g, 38.5 mmol) in EtOH (30 mL) was added 10% Pd/C (1.0g). The reaction mixture was stirred at room temperature overnight under hydrogen atmosphere. After the catalyst was filtered off through a celite cup, the filtrate was concentrated to give crude 3-((2-aminophenyl)amino)propan-1-ol (8.82 g, 99% yield). This product was used in the next step directly without further purification. MS (ESI) m/z: 168.3 [M+H]⁺.

Step 3. Synthesis of N-(3-((tert-butyldimethylsilyl)oxy)propyl)benzene-1,2-diamine

(1351) ##STR00791##

(1352) To a solution of 3-((2-aminophenyl)amino)propan-1-ol (3.0 g, 18.07 mmol) in DCM (25 mL) were added imidazole (1.85 g, 27.1 mmol) and tert-butyldimethylsilyl chloride (3.27 g, 21.68 mmol). The mixture was stirred at room temperature for 15 h, before the reaction was quenched with water and extracted with ethyl acetate (50 mL). The organic layer was washed with water twice and saturated aqueous brine, dried over sodium sulfate, filtered and concentrated. The residue

was triturated with petroleum ether/EtOAc (40 mL/5 mL) to give N1-(3-((tert-butyltrimethylsilyl)oxy)propyl)benzene-1,2-diamine (5.6 g, 74% yield) as a pale solid. MS (ESI) m/z: 282.0 [M+H].sup.+.

Step 4, Synthesis of 1-(3-((tert-butyltrimethylsilyl)oxy)propyl)-1,3-dihydro-2H-benzo[d]imidazol-2-one

(1353) ##STR00792##

(1354) To a solution of N-(3-((tert-butyltrimethylsilyl)oxy)propyl)benzene-1,2-diamine (5.6 g, 18.4 mmol), in THF (50 mL) was added N,N'-carbonyldiimidazole (5.97 g, 36.86 mmol) under N.sub.2. The reaction mixture was stirred at room temperature for 4 h, at which time the solvent was removed under vacuum. The resulting residue was recrystallized from methanol and n-hexane to give 1-(3-((tert-butyltrimethylsilyl)oxy)propyl)-1,3-dihydro-2H-benzo[d]imidazol-2-one (5 g, 88% yield) as a white solid. MS (ESI) m/z: 308.0 [M+H].sup.+.

Step 5. Synthesis of 3-(3-(3-((tert-butyltrimethylsilyl)oxy)propyl)-2-oxo-2,3-dihydro-1H-benzo[d]imidazol-1-yl)piperidine-2,6-dione

(1355) ##STR00793##

(1356) To a stirred solution of 1-(3-((tert-butyltrimethylsilyl)oxy)propyl)-1,3-dihydro-2H-benzo[d]imidazol-2-one (465 mg, 1.57 mmol) in DMF (2 mL) was added NaH (69.08 mg, 60% w/w dispersed into mineral oil, 1.73 mmol) at 0° C. under N.sub.2. After the reaction mixture was stirred for 20 min at 0° C., a solution of 3-bromopiperidine-2, 6-dione (150 mg, 0.79 mmol) in DMF (0.5 mL) was added dropwise at 0° C. The resulting mixture was stirred for additional 3 h at room temperature, before the reaction was quenched with AcOH (0.5 mL) and concentrated under reduced pressure. The resulting residue was purified by silica gel chromatography column (petroleum ether/EtOAc=4:1) to afford 3-(3-(3-((tert-butyltrimethylsilyl)oxy)propyl)-2-oxo-2,3-dihydro-1H-benzo[d]imidazol-1-yl)piperidine-2,6-dione (150.5 mg, 23% yield) as a white solid. MS (ESI) m/z: 418.9 [M+H].sup.+.

Step 6. Synthesis of 3-(3-(3-hydroxypropyl)-2-oxo-2,3-dihydro-1H-benzo[d]imidazol-1-yl)piperidine-2,6-dione

(1357) ##STR00794##

(1358) To a solution of 3-(3-(3-((tert-butyltrimethylsilyl)oxy)propyl)-2-oxo-2,3-dihydro-1H-benzo[d]imidazol-1-yl)piperidine-2,6-dione (300 mg, 0.74 mmol) in tetrahydrofuran (10 mL) was added tetra-n-butyl ammonium fluoride (384.5 mg, 1.47 mmol). The mixture was stirred at room temperature for 1 h. After removal of the solvent, the resulting residue was purified by silica gel chromatography column (hexane/EtOAc=1:1) to afford 3-(3-(3-hydroxypropyl)-2-oxo-2,3-dihydro-1H-benzo[d]imidazol-1-yl)piperidine-2,6-dione (83 mg, 90% yield) as a white solid. MS (ESI) m/z: 304.4 [M+H].sup.+.

Step 7. Synthesis of 3-(3-(2,6-dioxopiperidin-3-yl)-2-oxo-2,3-dihydro-1H-benzo[d]imidazol-1-yl)propyl 4-methylbenzenesulfonate

(1359) ##STR00795##

(1360) To a solution of 3-(3-(3-hydroxypropyl)-2-oxo-2,3-dihydro-1H-benzo[d]imidazol-1-yl)piperidine-2,6-dione (220 mg, 0.66 mmol) in dry CH.sub.2Cl.sub.2(5 mL) were added p-toluenesulfonyl chloride (188 mg, 0.99 mmol), 4-dimethylaminopyridine (12 mg, 0.1 mmol) and Et₃N (279 mL, 2 mmol) at 0° C. After the reaction mixture was stirred at room temperature for 24 h, the reaction was quenched with water, and extracted with DCM. The organic layer was washed with brine and dried over MgSO.sub.4, filtered and concentrated. The resulting residue was purified by prep-TLC (petroleum ether/EtOAc=2:1) to afford 3-(3-(2,6-dioxopiperidin-3-yl)-2-oxo-2,3-dihydro-1H-benzo[d]imidazol-1-yl)propyl 4-methylbenzenesulfonate (150 mg, 50% yield) as a white solid. MS (ESI) m/z: 458.5 [M+H].sup.+.

Step 8. Synthesis of 3-(3-(3-(4-(6-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)propyl)-2-oxo-2,3-dihydro-1H-benzo[d]imidazol-1-yl)piperidine-2,6-dione

(1361) ##STR00796##

(1362) To a solution of 3-(3-(2,6-dioxopiperidin-3-yl)-2-oxo-2,3-dihydro-1H-benzo[d]imidazol-1-yl)propyl 4-methylbenzenesulfonate (70 mg, 0.16 mmol) and (R)-6-(2-(3-fluorophenyl)pyrrolidin-1-yl)-3-(6-(piperazin-1-yl)pyridin-2-yl)imidazo[1,2-b]pyridazine (76.6 mg, 0.16 mmol) in DMF (2 mL) was added DIEA (89.6 mg, 0.70 mmol). The reaction mixture was stirred at 60° C. for 12 h. After the reaction was cooled down to room temperature, the reaction was quenched with water and extracted with EtOAc. The organic layer was washed with brine, dried over MgSO₄, filtered and concentrated. The resulting residue was purified by prep-TLC to give 3-(3-(3-(4-(6-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)propyl)-2-oxo-2,3-dihydro-1H-benzo[d]imidazol-1-yl)piperidine-2,6-dione (22 mg, 17% yield) as a white solid. MS (ESI) m/z: 729.9[M+H].sup.+.

Example 280:3—((S)-5-(4-((4-(6-(6-((R)-2-(3-Fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)methyl)phenyl)-2-oxooxazolidin-3-yl)piperidine-2,6-dione (TR-229)

(1363) ##STR00797##

(1364) To a solution of (R)-6-(2-(3-fluorophenyl)pyrrolidin-1-yl)-3-(6-fluoropyridin-2-yl)imidazo[1,2-b]pyridazine hydrochloride (40 mg, 0.041 mmol) and 3-((S)-2-oxo-5-(4-(piperazin-1-ylmethyl)phenyl)oxazolidin-3-yl)piperidine-2,6-dione (15.2 mg, 0.041 mmol) in DMSO (4 mL) was added KF (24 mg, 0.41 mmol). The reaction mixture was stirred at 130° C. overnight. After the reaction was cooled down to room temperature, the reaction mixture was quenched with saturated aqueous NaHCO₃ (100 mL) and extracted with EtOAc (100 mL). The organic layer was washed with brine, dried over sodium sulfate, filtered and concentrated. The resulting residue was purified by prep-TLC (DCM/MeOH=10:1) to give 3-((S)-5-(4-((4-(6-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)methyl)phenyl)-2-oxooxazolidin-3-yl)piperidine-2,6-dione (4 mg, 13% yield) as a white solid. MS (ESI) m/z: 731.0 [M+H].sup.+.

Example 281:2-(2,6-Dioxopiperidin-3-yl)-5-(3-((4-(5-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)-3,6-dihydropyridin-1 (2H)-yl)piperidin-1-yl)methyl)azetidin-1-yl)isoindoline-1,3-dione (TR-230)

(1365) ##STR00798##

Step 1. Synthesis of tert-butyl (R)-5-(6-(2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)-3,6-dihydropyridine-1 (2H)-carboxylate

(1366) ##STR00799##

(1367) To a solution of (R)-3-bromo-6-(2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazine (1.5 g, 4.1 mmol) and tert-butyl 5-(4,4,5,5-tetramethyl-1,3,2-dioxaborolan-2-yl)-3,6-dihydropyridine-1 (2H)-carboxylate (1.44 g, 4.5 mmol) in 1,4-dioxane (20 mL) and H₂O (5 mL) were added potassium carbonate (1.14 g, 8.2 mmol) and Pd(dppf) Cl₂ (306 mg, 0.41 mmol) at room temperature. The reaction mixture was stirred at 80° C. for 4 h. The resulting black mixture was diluted with ethyl acetate (50 mL), washed with saturated aqueous brine (3×50 mL). The combined organic layers were dried over sodium sulfate, filtered, and concentrated. The resulting residue was purified by silica gel flash chromatography (petroleum ether/ethyl acetate=1:4) to give tert-butyl (R)-5-(6-(2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)-3,6-dihydropyridine-1 (2H)-carboxylate (1.5 g, 78% yield) as a yellow oil. MS (ESI) m/z: 464.2 [M+H].sup.+.

Step 2. Synthesis of (R)-6-(2-(3-fluorophenyl)pyrrolidin-1-yl)-3-(1,2,5,6-tetrahydropyridin-3-yl)imidazo[1,2-b]pyridazine

(1368) ##STR00800##

(1369) To a solution of tert-butyl (R)-5-(6-(2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl).sub.0-3,6-dihydropyridine-1 (2H)-carboxylate (1.5 g, 3.2 mmol) in DCM (15 mL) was added TFA (3 mL) at 0° C. The reaction mixture was stirred at room temperature for 3 h,

before the solvent was removed under vacuum. The resulting residue was washed with diethyl ether to give (R)-6-(2-(3-fluorophenyl)pyrrolidin-1-yl)-3-(1,2,5,6-tetrahydropyridin-3-yl)imidazo[1,2-b]pyridazine (1.0g, 88% yield) as a yellow solid. This product was used in the next step directly without further purification. MS (ESI) m/z: 364.2 [M+H].sup.+.

Step 3. Synthesis of tert-butyl (R)-4-(5-(6-(2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)-3,6-dihydropyridin-1 (2H)-yl)piperidine-1-carboxylate

(1370) ##STR00801##

(1371) To a solution of (R)-6-(2-(3-fluorophenyl)pyrrolidin-1-yl)-3-(1,2,5,6-tetrahydropyridin-3-yl)imidazo[1,2-b]pyridazine (350 mg, 0.96 mmol) and tert-butyl 4-(tosyloxyl)piperidine-1-carboxylate (1024 mg, 2.88 mmol) in acetonitrile (10 mL) were added potassium carbonate (398 mg, 3.56 mmol) and sodium iodide (144 mg, 0.96 mmol) at room temperature. The reaction mixture was stirred at 100° C. for 2 h, before the reaction was concentrated. The resulting residue was purified by silica gel flash chromatography (MeOH/DCM=1:10) to give tert-butyl (R)-4-(5-(6-(2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)-3,6-dihydropyridin-1 (2H)-yl)piperidine-1-carboxylate (260 mg, 49% yield) as a yellow solid. MS (ESI) m/z: 547.8 [M+H].sup.+.

Step 4. Synthesis of (R)-6-(2-(3-fluorophenyl)pyrrolidin-1-yl)-3-(1-(piperidin-4-yl)-1,2,5,6-tetrahydropyridin-3-yl)imidazo[1,2-b]pyridazine

(1372) ##STR00802##

(1373) To a solution of tert-butyl (R)-4-(5-(6-(2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl).sub.0-3,6-dihydropyridin-1 (2H)-yl)piperidine-1-carboxylate (260 mg, 0.47 mmol) in DCM (5 mL) was added TFA (1 mL) at 0° C. The reaction mixture was stirred at room temperature for 1 h, before the solvent was removed under vacuum. The resulting residue was washed with diethyl ether to give (R)-6-(2-(3-fluorophenyl)pyrrolidin-1-yl)-3-(1-(piperidin-4-yl)-1,2,5,6-tetrahydropyridin-3-yl)imidazo[1,2-b]pyridazine (160 mg, 75% yield) as a yellow solid. This product was used in the next step directly without further purification. MS (ESI) m/z: 447.8 [M+H].sup.+.

Step 5. Synthesis of 2-(2,6-dioxopiperidin-3-yl)-5-(3-((4-(5-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)-3,6-dihydropyridin-1 (2H)-yl)piperidin-1-yl)methyl)azetidin-1-yl)isoindoline-1,3-dione

(1374) ##STR00803##

(1375) To a solution of (R)-6-(2-(3-fluorophenyl)pyrrolidin-1-yl)-3-(1-(piperidin-4-yl)-1,2,5,6-tetrahydropyridin-3-yl)imidazo[1,2-b]pyridazine (92 mg, 0.2 mmol) and (1-(2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisoindolin-5-yl)azetidin-3-yl)methyl 4-methylbenzenesulfonate (105 mg, 0.24 mmol) in acetonitrile (5 mL) were added potassium carbonate (75 mg, 0.6 mmol) and sodium iodide (26 mg, 0.2 mmol). After the reaction mixture was stirred at 80° C. for 2 h, the reaction was concentrated and the resulting residue was purified by silica gel chromatography (DCM/MeOH=10:1) to give 2-(2,6-dioxopiperidin-3-yl)-5-(3-((4-(5-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl).sub.0-3,6-dihydropyridin-1 (2H)-yl)piperidin-1-yl)methyl)azetidin-1-yl)isoindoline-1,3-dione (3.7 mg, 1% yield) as a yellow solid. MS (ESI) m/z: 772. 1 [M+H].sup.+.

Example 282: N-(5-(3,5-Difluorobenzyl)-1H-indazol-3-yl)-4-(4-((1-(2~ (2,6-dioxopiperidin-3-yl)-1,3-dioxoisoindolin-5-yl)azetidin-3-yl)methyl)piperazin-1-yl)-2-((tetrahydro-2H-pyran-4-yl)amino)benzamide (TR-231)

(1376) ##STR00804##

(1377) To a solution of N-(5-(3,5-difluorobenzyl)-1H-indazol-3-yl)-4-(piperazin-1-yl)-2-((tetrahydro-2H-pyran-4-yl)amino)benzamide (15 mg, 27.44 umol) and (1-(2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisoindolin-5-yl)azetidin-3-yl)methyl 4-methylbenzenesulfonate (16.38 mg, 32.93 umol) in MeCN (5 mL) was added DIPEA (21.28 mg, 164.65 umol). After the reaction mixture was stirred at 80° C. for 16 h, the solvent was removed under vacuum. The resulting residue was

purified by prep-TLC to give N-(5-(3,5-difluorobenzyl)-1H-indazol-3-yl)-4-(4-((1-(2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-5-yl) azetidin-3-yl)methyl)piperazin-1-yl)-2-((tetrahydro-2H-pyran-4-yl)amino)benzamide (9.5 mg, 39% yield) as a yellow solid. MS (ESI) m/z: 873.1 [M+H].sup.+.

Example 283: N-(5-(3,5-Difluorobenzyl)-1H-indazol-3-yl)-4-(4-(2-((2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-5-yl)amino)ethyl)piperazin-1-yl)-2-((tetrahydro-2H-pyran-4-yl)amino)benzamide (TR-232)

(1378) ##STR00805##

Step 1. Synthesis of N-(5-(3,5-difluorobenzyl)-1H-indazol-3-yl)-4-(4-(2-((2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-5-yl)(4-methoxybenzyl)amino)ethyl)piperazin-1-yl)-2-((tetrahydro-2H-pyran-4-yl)amino)benzamide

(1379) ##STR00806##

(1380) To a solution of N-(5-(3,5-difluorobenzyl)-1H-indazol-3-yl)-4-(piperazin-1-yl)-2-((tetrahydro-2H-pyran-4-yl)amino)benzamide (80 mg, 146.36 μ mol) and 2-((2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-5-yl)(4-methoxybenzyl)amino)ethyl 4-methylbenzenesulfonate (103.91 mg, 175.63 μ mol) in MeCN (5 mL) was added DIPEA (113.49 mg, 878.14 μ mol). After the reaction mixture was stirred at 90° C. for 24 h, the solvent was removed under vacuum. The resulting residue was purified by prep-TLC to give N-(5-(3,5-difluorobenzyl)-1H-indazol-3-yl)-4-(4-(2-((2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-5-yl)(4-methoxybenzyl)amino)ethyl)piperazin-1-yl)-2-((tetrahydro-2H-pyran-4-yl)amino)benzamide (80 mg, 57% yield) as a yellow solid. MS (ESI) m/z: 967.2 [M+H].sup.+.

Step 2. Synthesis of N-(5-(3,5-difluorobenzyl)-1H-indazol-3-yl)-4-(4-(2-((2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-5-yl)amino)ethyl)piperazin-1-yl)-2-((tetrahydro-2H-pyran-4-yl)amino)benzamide

(1381) ##STR00807##

(1382) To a solution of N-(5-(3,5-difluorobenzyl)-1H-indazol-3-yl)-4-(4-(2-((2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-5-yl)(4-methoxybenzyl)amino)ethyl)piperazin-1-yl)-2-((tetrahydro-2H-pyran-4-yl)amino)benzamide (30 mg, 31.05 μ mol) in DCM (2 mL) was added TFA (1 mL). After the reaction mixture was stirred at room temperature for 1 h, the reaction was concentrated. The resulting residue was purified by prep-HPLC to give N-(5-(3,5-difluorobenzyl)-1H-indazol-3-yl)-4-(4-(2-((2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-5-yl)amino)ethyl)piperazin-1-yl)-2-((tetrahydro-2H-pyran-4-yl)amino)benzamide (5 mg, 17% yield) as a yellow solid. MS (ESI) m/z: 847.1 [M+H].sup.+.

Example 284: N-(2-(2,6-Dioxopiperidin-3-yl)-1,3-dioxoisindolin-5-yl)-2-(4-(6-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)acetamide (TR-233)

(1383) ##STR00808##

Step 1. Synthesis of 2-chloro-N-(2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-5-yl)acetamide

(1384) ##STR00809##

(1385) A mixture of 5-amino-2-(2,6-dioxopiperidin-3-yl)isoindoline-1,3-dione (40 mg, 0.146 mmol) and 2-chloroacetyl chloride (82 mg, 0.732 mmol) in THF (5 mL) was stirred at 70° C. for 1 h. After the reaction was completed, the reaction mixture was concentrated. The resulting residue was triturated with petroleum ether, filtered and dried to give 2-chloro-N-(2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-5-yl)acetamide (55 mg, 107% yield) as a white solid. This product was used in the next step directly without further purification. MS (ESI) m/z: 350.4 [M+H].sup.+.

Step 2. Synthesis of N-(2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-5-yl)-2-(4-(6-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)acetamide

(1386) ##STR00810##

(1387) A solution of 2-chloro-N-(2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-5-yl)acetamide (55 mg, 0.157 mmol), (R)-6-(2-(3-fluorophenyl)pyrrolidin-1-yl)-3-(6-(piperazin-1-yl)pyridin-2-

yl)imidazo[1,2-b]pyridazine hydrochloride (75 mg, 0.157 mmol) and DIEA (101 mg, 0.785 mmol) in DMF (5 mL) was stirred at 80° C. for 1 h. After the reaction was completed, the reaction solution was diluted with EtOAc (200 mL), washed with brine twice, dried over sodium sulfate, filtered and concentrated. The resulting residue was purified by prep-TLC (DCM/MeOH=10:1) to give N-(2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-5-yl)-2-(4-(6-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)acetamide (15.5 mg, 13% yield) as a white solid. MS (ESI) m/z: 757.9 [M+H].sup.+.

Example 285: 2-(2,6-Dioxopiperidin-3-yl)-5-(3-((4-(6-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-a]pyridin-3-yl)pyridin-2-yl)piperazin-1-yl)methyl)azetidin-1-yl)isoindoline-1,3-dione (TR-234)

(1388) ##STR00811##

(1389) To a solution of (R)-6-(2-(3-fluorophenyl)pyrrolidin-1-yl)-3-(6-(piperazin-1-yl)pyridin-2-yl)imidazo[1,2-a]pyridine (22 mg, 4.98 μ mol) and (1-(2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-5-yl)azetidin-3-yl)methyl 4-methylbenzenesulfonate (24.8 mg, 5.0 μ mol) in CH₂Cl₂ (4 mL) were added NaI (15 mg, 10.0 μ mol) and K₂CO₃ (28 mg, 20.00 μ mol) at room temperature under N₂. The reaction mixture was stirred at 98° C. for 16 h, before the solvent was removed under vacuum. The resulting residue was purified by prep-TLC (DCM/MeOH=20:1) to give 2-(2,6-dioxopiperidin-3-yl)-5-(3-((4-(6-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-a]pyridin-3-yl)pyridin-2-yl)piperazin-1-yl)methyl)azetidin-1-yl)isoindoline-1,3-dione (19 mg, 50% yield). MS (ESI) m/z: 769.0 [M+H].sup.+.

Example 286: 3-((S)-5-(3-((4-(6-(6-((R)-2-(3-Fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)methyl)phenyl)-2-oxooxazolidin-3-yl)piperidine-2,6-dione (TR-235)

(1390) ##STR00812##

(1391) A mixture of (R)-6-(2-(3-fluorophenyl)pyrrolidin-1-yl)-3-(6-fluoropyridin-2-yl)imidazo[1,2-b]pyridazine (80 mg, 0.21 mmol), 2,2,2-trifluoroacetaldehyde compound with 3-((S)-2-oxo-5-(3-(piperazin-1-yl)methyl)phenyl)oxazolidin-3-yl)piperidine-2,6-dione (103.23 mg, 0.21 mmol) and KF (122.01 mg, 2.1 mmol) in DMSO (1 mL) was stirred at 130° C. for 3 h. After the reaction was cooled down to room temperature, the reaction was quenched with H₂O (10 mL), extracted with EtOAc. The organic layer was washed with brine, dried over MgSO₄, filtered and concentrated. The residue was purified by prep-TLC (DCM/CH₂OH=10:1) to give 3-((S)-5-((4-(6-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)methyl)phenyl)-2-oxooxazolidin-3-yl)piperidine-2,6-dione (21 mg, 14% yield) as a white solid. MS (ESI) m/z: 730.9 [M+H].sup.+.

Example 287: 3-(4-((2-(4-(6-(6-((R)-2-(3-Fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)ethyl)amino)-2-oxoindolin-1-yl)piperidine-2,6-dione (TR-236)

(1392) ##STR00813##

Step 1. Synthesis of 4-nitroindolin-2-one

(1393) ##STR00814##

(1394) To a solution of 4-nitro-1H-indole (5.0 g, 30.9 mmol) in CHCl₃ (200 mL) was added NCS (N-chlorosuccinimide) (2.67 g, 20 mmol) at room temperature. The reaction mixture was stirred at 80° C. for 48 h. The solvent was evaporated under reduced pressure to give a yellow solid. The solid was dissolved in AcOH (80 mL) and 85% H₃PO₄ (30 mL). The resulting mixture was stirred at 100° C. for 3 h. After the reaction was cooled down to room temperature, the solution was acidified with 10 N NaOH to pH=6 and extracted with ethyl acetate (3×100 mL). The combined organic layers were washed with saturated brine (100 mL), dried over anhydrous sodium sulfate, filtered and evaporated under reduced pressure. The resulting residue was purified by silica gel column chromatography (EtOAc/petroleum ether=1:1 to 1:0) to afford the 4-nitroindolin-2-one (3.5 g, 64% yield) as a yellow solid. MS (ESI) m/z: 179 [M+H].sup.+.

Step 2. Synthesis of 3-(4-

nitro-2-oxoindolin-1-yl)piperidine-2,6-dione

(1395) ##STR00815##

(1396) To a solution of 4-nitroindolin-2-one (1.0 g, 5.6 mmol) in DMF (40 mL) was added NaH (60% in mineral oil, 247 mg, 6.2 mmol) at 0° C. The reaction mixture was warmed to room temperature for 0.5 h, before 3-bromopiperidine-2,6-dione (2.14 g, 11.2 mmol) was added. The resulting mixture was stirred at 80° C. for 1 h, before the reaction was cooled down to room temperature. The reaction was quenched with water (50 mL) and extracted with ethyl acetate (3×20 mL). The combined organic layers were washed with saturated brine (20 mL), dried over anhydrous sodium sulfate, filtered and evaporated under reduced pressure. The resulting residue was purified by silica gel column chromatography (DCM/MeOH=30:1 to 15:1) to give a mixture of 3-(4-nitro-2-oxoindolin-3-yl)piperidine-2,6-dione, 3-(4-nitro-2-oxoindolin-1-yl)piperidine-2,6-dione and 3-((4-nitro-1H-indol-2-yl)oxyl)piperidine-2,6-dione (400 mg, ratio=1:1:1) as a brown solid. These products were used in the next step directly without further purification. MS (ESI) m/z: 290 [M+H].sup.+.

Step 3. Synthesis of 3-(4-nitro-2-oxoindolin-1-yl)piperidine-2,6-dione

(1397) ##STR00816##

(1398) A mixture of 3-(4-nitro-2-oxoindolin-3-yl)piperidine-2,6-dione, 3-(4-nitro-2-oxoindolin-1-yl)piperidine-2,6-dione and 3-((4-nitro-1H-indol-2-yl)oxyl)piperidine-2,6-dione (400 mg, 1:1:1) and 10% Pd/C (80 mg) in THF (10 mL) was stirred under H.sub.2(1 atm) at room temperature for 5 h. After the reaction was filtered through celite, the filtrate was removed under reduced pressure and the residue was purified by prep-TLC (DCM/MeOH=10:1) to give 3-(4-nitro-2-oxoindolin-1-yl)piperidine-2,6-dione (30 mg, 2% yield over 2 steps) as a light yellow solid. MS (ESI) m/z: 260 [M+H].sup.+.

Step 4. Synthesis of 3-(4-((2-(4-(6-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)ethyl)amino)-2-oxoindolin-1-yl)piperidine-2,6-dione

(1399) ##STR00817##

(1400) To a solution of (R)-2-(4-(6-(6-(2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)acetaldehyde (57 mg, 0.12 mmol) and 3-(4-amino-2-oxoindolin-1-yl)piperidine-2,6-dione (30 mg, 0.12 mmol) in DMF (1 mL) were added TMSCl (39 mg, 0.36 mmol) and NaBH.sub.4 (9.1 mg, 0.24 mmol) at 0° C. The reaction mixture was stirred at 0° C. for 1.5 h, before it was warmed to room temperature and stirred for 16 h. The reaction mixture was purified by prep-HPLC to give the crude product, which was further purified by prep-TLC (DMC/MeOH=15:1) to give 3-(4-((2-(4-(6-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)ethyl)amino)-2-oxoindolin-1-yl)piperidine-2,6-dione (7.7 mg, 9% yield) as a white solid. MS (ESI) m/z: 729.9 [M+H].sup.+.

Example 288: 2~ (2,6-Dioxopiperidin-3-yl)-5-((3~ (4-(6-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)azetidin-1-yl)methyl)isoindoline-1,3-dione (TR-237)

(1401) ##STR00818##

Step 1. Synthesis of 5-(bromomethyl)-2-(2,6-dioxopiperidin-3-yl)isoindoline-1,3-dione

(1402) ##STR00819##

(1403) To a solution of 2-(2,6-dioxopiperidin-3-yl)-5-methyl]isoindoline-1,3-dione (1 g, 3.67 mmol) in CCL (50 mL) were added NBS (719.10 mg, 4.04 mmol) and AIBN (azobisisobutyronitrile)(67.32 mg, 0.41 mmol). After the reaction mixture was stirred at 90° C. for 16 h, the solvent was removed under vacuum. The resulting residue was purified by prep-HPLC to give 5-(bromomethyl)-2-(2,6-dioxopiperidin-3-yl)isoindoline-1,3-dione (500 mg, 40% yield) as an off-white solid. MS (ESI) m/z: 353.3 [M+H].sup.+.

Step 2. Synthesis of (R)-tert-butyl 3-(4-(6-(6-(2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)azetidine-1-carboxylate

(1404) ##STR00820##

(1405) To a solution of (R)-6-(2-(3-fluorophenyl)pyrrolidin-1-yl)-3-(6-(piperazin-1-yl)pyridin-2-yl)imidazo[1,2-b]pyridazine (300 mg, 0.68 mmol) and tert-butyl 3-oxoazetidine-1-carboxylate (139 mg, 0.81 mmol) in THF (10 mL) was added titanium tetrakisopropanolate (384 mg, 1.35 mmol) at 0° C. Then the mixture was stirred at 60° C. for 4 h. After the reaction was cooled down to room temperature, NaBH.sub.3CN (85 mg, 1.35 mmol) was added. After the reaction mixture was stirred at room temperature for another 16 h, the reaction solution was poured into water (20 mL) and extracted with ethyl acetate (3×10 mL). The combined organic layers were washed with saturated brine (20 mL), dried over anhydrous sodium sulfate, filtered and evaporated under reduced pressure. The resulting residue was purified by reverse-phase chromatography to afford the desired product (R)-tert-butyl 3-(4-(6-(6-(2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)azetidine-1-carboxylate (350 mg, 86% yield) as a yellow resin. MS (ESI) m/z: 599.8 [M+H].sup.+.

Step 3. Synthesis of (R)-3-(6-(4-(azetidin-3-yl)piperazin-1-yl)pyridin-2-yl)-6-(2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazine

(1406) ##STR00821##

(1407) To a solution of (R)-tert-butyl 3-(4-(6-(6-(2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)azetidine-1-carboxylate (350 mg, 0.59 mmol) in DCM (5 mL) was added TFA (2 mL). The reaction mixture was stirred at 25° C. for 5 h. After the starting material was totally consumed, the reaction was evaporated under reduced pressure to give crude (R)-3-(6-(4-(azetidin-3-yl)piperazin-1-yl)pyridin-2-yl)-6-(2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazine (500 mg, 100% yield) as TFA salt. This product was used directly in the next step without further purification. MS (ESI) m/z: 499.8 [M+H].sup.+.

Step 4. Synthesis of 2-(2,6-dioxopiperidin-3-yl)-5-((3-(4-(6-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)azetidin-1-yl)methyl)isoindoline-1,3-dione

(1408) ##STR00822##

(1409) To a solution of crude (R)-3-(6-(4-(azetidin-3-yl)piperazin-1-yl)pyridin-2-yl)-6-(2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazine (50 mg, 0.06 mmol) in NMP (1 mL) was added 5-(bromomethyl)-2-(2,6-dioxopiperidin-3-yl)isoindoline-1,3-dione (35 mg, 0.1 mmol) and DIPEA (0.5 mL) at room temperature. The reaction mixture was heated to 120° C. under microwave for 1 h. After the reaction was cooled down to room temperature, the reaction solution was poured into water (10 mL) and extracted with ethyl acetate (3×10 mL). The combined organic layers were washed with saturated brine (20 mL), dried over anhydrous sodium sulfate, filtered and evaporated under reduced pressure. The residue was purified by prep-TLC (DCM/MeOH=15:1) to give 2-(2,6-dioxopiperidin-3-yl)-5-((3-(4-(6-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)azetidin-1-yl)methyl)isoindoline-1,3-dione (13 mg, 28% yield) as a yellow solid. MS (ESI) m/z: 769.9 [M+H].sup.+.

Example 289:3-(4-(3-(4-(6-(6-((R)-2-(3-Fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)prop-1-yn-1-yl)-3-methyl-2-oxo-2,3-dihydro-1H-benzo[d]imidazol-1-yl)piperidine-2,6-dione (TR-238)

(1410) ##STR00823##

Step 1. Synthesis of 2-bromo-N-methyl-6-nitroaniline

(1411) ##STR00824##

(1412) To a solution of 1-bromo-2-fluoro-3-nitrobenzene (15.5 g, 70.5 mmol) in EtOH (80 mL) was added DIEA (45 g, 353 mmol) and methanamine hydrochloride (14.3 g, 211.5 mmol). After the solution was stirred at 85° C. for 12 h, the reaction was cooled to room temperature, concentrated and diluted with water and ethyl acetate. The organic phase was separated, dried over anhydrous sodium sulfate, filtered and concentrated to give crude 2-bromo-N-methyl-6-nitroaniline (15 g, 92% yield) as a yellow solid which was used directly in the next step without further purification. MS (ESI) m/z: 231. 1 [M+H].sup.+.

Step 2.: Synthesis of 6-bromo-N-methylbenzene-1,2-diamine

(1413) ##STR00825##

(1414) A mixture of 2-bromo-N-methyl-6-nitroaniline (14.0 g, 60.6 mmol) in THF (40 mL) and AcOH (80 mL) was stirred at 85° C. for 2 h in the presence of Fe (17 g, 303 mmol). The reaction mixture was filtered and the filtrate was concentrated. The residue was purified by silica gel chromatography (petroleum ether/EtOAc=10:1) to give 6-bromo-N-methylbenzene-1,2-diamine (8.2 g, 68% yield). MS (ESI) m/z: 201.2 [M+H].sup.+.

Step 3. Synthesis of 7-bromo-1-methyl-1,3-dihydro-2H-benzo[d]imidazol-2-one

(1415) ##STR00826##

(1416) To a solution of 6-bromo-N-methylbenzene-1,2-diamine (8.9 g, 44.5 mmol) in THF (100 mL) was added N,N'-carbonyldiimidazole (29 g, 178.2 mmol) at room temperature under N.sub.2. After the reaction mixture was stirred at room temperature overnight, the solvent was removed. The residue was dissolved in DCM/MeOH (10:1) and washed with water. The organic layer was dried over Na.sub.2SO.sub.4, filtered and concentrated. The residue was purified by silica gel chromatography (DCM/MeOH=60:1) to give 7-bromo-1-methyl-1,3-dihydro-2H-benzo[d]imidazol-2-one (4.4 g, 44% yield) as a white solid. MS (ESI) m/z: 227.2 [M+H].sup.+.

Step 4. Synthesis of 3-(4-bromo-3-methyl-2-oxo-2,3-dihydro-1H-benzo[d]imidazol-1-yl)piperidine-2,6-dione

(1417) ##STR00827##

(1418) To a solution of 7-bromo-1-methyl-1,3-dihydro-2H-benzo[d]imidazol-2-one (2.5 g, 11.0 mmol) in DMF (9 mL) was added NaH (529 mg, 60% w/w dispersed into mineral oil, 13.21 mmol) at 0° C. under nitrogen atmosphere. The reaction mixture was stirred at 0° C. for 20 min, before a solution of 3-bromopiperidine-2, 6-dione (1.47 g, 7.71 mmol) in DMF (6 mL) was added dropwise at 0° C. The resulting mixture was stirred at room temperature overnight, before the reaction was quenched with water and extracted with ethyl acetate (3 x). The organic layers were combined, dried, filtered, and concentrated. The resulting residue was purified by silica gel chromatography (petroleum ether/EtOAc=2:1 to DCM/MeOH=60:1) to give 3-(4-bromo-3-methyl-2-oxo-2,3-dihydro-1H-benzo[d]imidazol-1-yl)piperidine-2,6-dione as a white solid (247 mg, 7% yield). MS (ESI) m/z: 338 [M+H].sup.+.

Step 5. Synthesis of 3-(4-(3-hydroxyprop-1-yn-1-yl)-3-methyl-2-oxo-2,3-dihydro-1H-benzo[d]imidazol-1-yl)piperidine-2,6-dione

(1419) ##STR00828##

(1420) To a solution of 3-(4-bromo-3-methyl-2-oxo-2,3-dihydro-1H-benzo[d]imidazol-1-yl)piperidine-2,6-dione (130 mg, 0.39 mmol) in DMF (3.5 mL) were added prop-2-yn-1-ol (43 mg, 0.77 mmol), CuI (15 mg, 0.079 mmol), Pd(dppf) Cl.sub.2(29 mg, 0.04 mmol) and TEA (394 mg, 3.9 mmol). The mixture was stirred at 80° C. overnight, before the reaction was quenched with water and extracted with ethyl acetate (3 x). The organic layers were combined, dried, filtered, and concentrated. The resulting residue was purified by reverse phase chromatography to give 3-(4-(3-hydroxyprop-1-yn-1-yl)-3-methyl-2-oxo-2,3-dihydro-1H-benzo[d]imidazol-1-yl)piperidine-2,6-dione (94 mg, 78% yield) as a white solid. MS (ESI) m/z: 314.4 [M+H].sup.+.

Step 6. Synthesis of 3-(4-(3-chloroprop-1-yn-1-yl)-3-methyl-2-oxo-2,3-dihydro-1H-benzo[d]imidazol-1-yl)piperidine-2,6-dione

(1421) ##STR00829##

(1422) To a solution of 3-(4-(3-hydroxyprop-1-yn-1-yl)-3-methyl-2-oxo-2,3-dihydro-1H-benzo[d]imidazol-1-yl)piperidine-2,6-dione (30 mg, 0.096 mmol) in dry CH.sub.2Cl.sub.2(2.5 mL) and anhydrous THF (1 mL) were added TsCl (27.6 mg, 0.14 mmol) and DMAP (35 mg, 0.29 mmol). Then the reaction mixture was stirred at room temperature for 12 h, before the reaction was quenched with water and extracted with ethyl acetate (3 x). The organic layers were combined, dried, filtered, and concentrated. The resulting residue was purified by prep-TLC (DCM:/MeOH=25:1) to give 3-(4-(3-chloroprop-1-yn-1-yl)-3-methyl-2-oxo-2,3-dihydro-1H-

benzo[d]imidazol-1-yl)piperidine-2,6-dione (12 mg, 38% yield) as a white solid. MS (ESI) m/z: 332.4 [M+H].sup.+.

Step 7. Synthesis of 3-(4-(3-(4-(6-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)prop-1-yn-1-yl)-3-methyl-2-oxo-2,3-dihydro-1H-benzo[d]imidazol-1-yl)piperidine-2,6-dione

(1423) ##STR00830##

(1424) A mixture of 3-(4-(3-chloroprop-1-yn-1-yl)-3-methyl-2-oxo-2,3-dihydro-1H-benzo[d]imidazol-1-yl)piperidine-2,6-dione (12 mg, 0.036 mmol), K.sub.2CO.sub.3 (15 mg, 0.108 mmol), NaI (8 mg, 0.054 mmol) and (R)-6-(2-(3-fluorophenyl)pyrrolidin-1-yl)-3-(6-(piperazin-1-yl)pyridin-2-yl)imidazo[1,2-b]pyridazine hydrochloride (21 mg, 0.044 mmol) in CH.sub.3CN (0.8 ml) was stirred at 90° C. for 4 h, before the reaction was quenched with water and extracted with ethyl acetate (3 x). The organic layers were combined, dried, filtered, and concentrated. The resulting residue was purified by prep-TLC (DCM/MeOH=12:1) to give 3-(4-(3-(4-(6-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)prop-1-yn-1-yl)-3-methyl-2-oxo-2,3-dihydro-1H-benzo[d]imidazol-1-yl)piperidine-2,6-dione (8.6 mg, 32% yield) as a white solid. MS (ESI) m/z: 740.0 [M+H].sup.+.

Example 290:3-(4-(4-(4-(6-(6-((R)-2-(3-Fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)but-1-yn-1-yl)-3-methyl-2-oxo-2,3-dihydro-1H-benzo[d]imidazol-1-yl)piperidine-2,6-dione (TR-239)

(1425) ##STR00831##

(1426) TR-239 was synthesized according to the standard procedure for preparing TR-238 (5.8 mg, 16% yield). MS (ESI) m/z: 754.0 [M+H].sup.+.

Example 291:3-(4-(3-(4-(6-(6-((R)-2-(3-Fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)propyl)-3-methyl-2-oxo-2,3-dihydro-1H-benzo[d]imidazol-1-yl)piperidine-2,6-dione (TR-240)

(1427) ##STR00832##

Step 1. Synthesis of 3-(4-(3-hydroxypropyl)-3-methyl-2-oxo-2,3-dihydro-1H-benzo[d]imidazol-1-yl)piperidine-2,6-dione

(1428) ##STR00833##

(1429) A solution of 3-(4-(3-hydroxyprop-1-yn-1-yl)-3-methyl-2-oxo-2,3-dihydro-1H-benzo[d]imidazol-1-yl)piperidine-2,6-dione (25 mg, 0.08 mmol) in THF (3 mL) was stirred in the presence of 10% Pd/C (15 mg) under hydrogen (1 atm) for 4 h, before the reaction mixture was filtered. The filtrate was concentrated to give crude 3-(4-(3-hydroxypropyl)-3-methyl-2-oxo-2,3-dihydro-1H-benzo[d]imidazol-1-yl)piperidine-2,6-dione (30 mg, 99% yield) which was used directly in the next step without further purification. MS (ESI) m/z: 318.4 [M+H].sup.+.

Step 2. Synthesis of 3-(1-(2,6-dioxopiperidin-3-yl)-3-methyl-2-oxo-2,3-dihydro-1H-benzo[d]imidazol-4-yl)propyl 4-methylbenzenesulfonate

(1430) ##STR00834##

(1431) To a solution of 3-(4-(3-hydroxypropyl)-3-methyl-2-oxo-2,3-dihydro-1H-benzo[d]imidazol-1-yl)piperidine-2,6-dione (30 mg, 0.095 mmol) in dry CH.sub.2Cl.sub.2 (2 mL) and anhydrous THF (1 mL) were added TsCl (36 mg, 0.19 mmol) and DMAP (46 mg, 0.38 mmol). The reaction mixture was stirred at room temperature for 12 h, before the reaction was concentrated. The resulting residue was purified by prep-TLC (DCM/MeOH=25:1) to give 3-(1-(2,6-dioxopiperidin-3-yl)-3-methyl-2-oxo-2,3-dihydro-1H-benzo[d]imidazol-4-yl)propyl 4-methylbenzenesulfonate (6 mg, 13% yield) as a white solid. MS (ESI) m/z: 472.5 [M+H].sup.+.

Step 3. Synthesis of 3-(4-(3-(4-(6-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)propyl)-3-methyl-2-oxo-2,3-dihydro-1H-benzo[d]imidazol-1-yl)piperidine-2,6-dione

(1432) ##STR00835##

(1433) A mixture of 3-(1-(2,6-dioxopiperidin-3-yl)-3-methyl-2-oxo-2,3-dihydro-1H-

benzo[d]imidazol-4-yl)propyl 4-methylbenzenesulfonate (6 mg, 0.013 mmol), K.sub.2CO.sub.3 (5.3 mg, 0.038 mmol), NaI (3 mg, 0.02 mmol) and (R)-6-(2-(3-fluorophenyl)pyrrolidin-1-yl)-3-(6-(piperazin-1-yl)pyridin-2-yl)imidazo[1,2-b]pyridazine hydrochloride (7.5 mg, 0.016 mmol) in CH.sub.3CN (0.8 mL) was stirred at 90° C. for 4 h, before the reaction was quenched with water and extracted with ethyl acetate (3 x). The organic layers were combined, dried, filtered, and concentrated. The resulting residue was purified by prep-TLC (DCM/MeOH=10:1) to give 3-(4-(3-(4-(6-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)propyl)-3-methyl-2-oxo-2,3-dihydro-1H-benzo[d]imidazol-1-yl)piperidine-2,6-dione e (4.0 mg, 42% yield) as a white solid. MS (ESI) m/z: 744.0 [M+H].sup.+.

Example 292: 3-(4-(4-(4-(6-(6-((R)-2-(3-Fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)butyl)-3-methyl-2-oxo-2,3-dihydro-1H-benzo[d]imidazol-1-yl)piperidine-2,6-dione (TR-241)

(1434) ##STR00836##

(1435) TR-241 was synthesized according to the standard procedure for preparing TR-240 (12 mg, 45% yield). MS (ESI) m/z: 758.1 [M+H].sup.+.

Example 293: 3-(3-(2-(4-(6-(6-((R).SUB.2-(3-Fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-a]pyridin-3-yl)pyridin-2-yl)piperazin-1-yl)ethyl)-2-oxo-2,3-dihydro-1H-benzo[d]imidazol-1-yl)piperidine-2,6-dione (TR-242)

(1436) ##STR00837##

(1437) To a solution of 2-(3-(2,6-dioxopiperidin-3-yl)-2-oxo-2,3-dihydro-1H-benzo[d]imidazol-1-yl)ethyl 4-methylbenzenesulfonate (44.3 mg, 0.10 mmol) and (R)-6-(2-(3-fluorophenyl)pyrrolidin-1-yl)-3-(6-(piperazin-1-yl)pyridin-2-yl)imidazo[1,2-a]pyridine (44.2 mg, 0.10 mmol) in CH.sub.3CN (1mL) were added sodium iodide (22.5 mg, 0.15 mmol) and K.sub.2CO.sub.3 (41.4 mg, 0.3 mmol) at room temperature. The reaction mixture was stirred at 90° C. for 12 h. After cooled down to room temperature, the reaction was quenched with water and extracted with ethyl acetate (3×10 mL). The combined organic layers were washed with saturated brine (10 mL), dried over anhydrous sodium sulfate, filtered and evaporated under reduced pressure. The residue was purified by reverse-phase chromatography to afford 3-(3-(2-(4-(6-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-a]pyridin-3-yl)pyridin-2-yl)piperazin-1-yl)ethyl)-2-oxo-2,3-dihydro-1H-benzo[d]imidazol-1-yl)piperidine-2,6-dione (35 mg, 49% yield) as a white solid. MS (ESI) m/z: 714.9 [M+H].sup.+.

Example 294: 3-(3-(3-(4-(6-(6-((R)-2-(3-Fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-a]pyridin-3-yl)pyridin-2-yl)piperazin-1-yl)propyl)-2-oxo-2,3-dihydro-1H-benzo[d]imidazol-1-yl)piperidine-2,6-dione (TR-243)

(1438) ##STR00838##

(1439) To a solution of 3-(3-(2,6-dioxopiperidin-3-yl)-2-oxo-2,3-dihydro-1H-benzo[d]imidazol-1-yl)propyl 4-methylbenzenesulfonate (45.3 mg, 0.10 mmol) and (R)-6-(2-(3-fluorophenyl)pyrrolidin-1-yl)-3-(6-(piperazin-1-yl)pyridin-2-yl)imidazo[1,2-a]pyridine (44.2 mg, 0.10 mmol) in CH.sub.3CN (1 mL) were added sodium iodide (22.5 mg, 0.15 mmol) and K.sub.2CO.sub.3 (41.4 mg, 0.3 mmol) at room temperature. The reaction mixture was stirred at 90° C. for 12 h. After cooled down to room temperature, the reaction was quenched with water and extracted with ethyl acetate (3×10 mL). The combined organic layers were washed with saturated brine (10 mL), dried over anhydrous sodium sulfate, filtered and evaporated under reduced pressure. The residue was purified by reverse-phase chromatography to afford 3-(3-(3-(4-(6-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-a]pyridin-3-yl)pyridin-2-yl)piperazin-1-yl)propyl)-2-oxo-2,3-dihydro-1H-benzo[d]imidazol-1-yl)piperidine-2,6-dione (22 mg, 30% yield) as a white solid. MS (ESI) m/z: 729.0 [M+H].sup.+.

Example 295: 3-(4~(2-(4-(6-(6-((R)-2-(3-Fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)ethoxy)-3-methyl-2-oxo-2,3-dihydro-1H-benzo[d]imidazol-1-yl)piperidine-2,6-dione (TR-244)

(1440) ##STR00839## ##STR00840##

Step 1. Synthesis of (R)-2-(4-(6-(6-(2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)ethan-1-ol

(1441) ##STR00841##

(1442) To a solution of (R)-6-(2-(3-fluorophenyl)pyrrolidin-1-yl)-3-(6-(piperazin-1-yl)pyridin-2-yl)imidazo[1,2-b]pyridazine (200 mg, 0.42 mmol) and 2-bromoethanol (157.45 mg, 1.26 mmol) in DMSO (3 mL) were added K.sub.2CO.sub.3 (173.89 mg, 1.26 mmol) and NaI (63.0 mg, 0.42 mmol) at room temperature. The reaction mixture was stirred at 80° C. overnight. After cooled down to room temperature, the reaction was quenched with water (15 mL) and extracted with ethyl acetate (3×10 mL). The combined organic layers were washed with saturated brine (10 mL), dried over anhydrous sodium sulfate, filtered and evaporated under reduced pressure. The residue was purified by silica gel chromatography to afford (R)-2-(4-(6-(6-(2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)ethan-1-ol (130 mg, 64% yield) as a brown oil. MS (ESI) m/z: 488.7 [M+H].sup.+.

Step 2. Synthesis of (R)-2-(4-(6-(6-(2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)ethyl 4-methylbenzenesulfonate

(1443) ##STR00842##

(1444) To a solution of (R)-2-(4-(6-(6-(2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)ethan-1-ol (130 mg, 0.27 mmol) in dry CH.sub.2Cl.sub.2 (5 mL) were added p-toluensulfonyl chloride (80.46 mg, 0.53 mmol), 4-dimethylaminopyridine (12 mg, 0.1 mmol) and Et.sub.3N (136.35 mg, 1.35 mmol) at 0° C. Then the reaction mixture was stirred at room temperature for 24 h. The reaction was quenched with water (5 mL) and extracted with CH.sub.2Cl.sub.2 (3×5 mL). The combined organic layers were washed with brine and dried over MgSO.sub.4, filtered and concentrated under reduced pressure.

(1445) The residue was purified by prep-TLC to give (R)-2-(4-(6-(6-(2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)ethyl 4-methylbenzenesulfonate (80 mg, 46% yield) as a white solid. MS (ESI) m/z: 642.3 [M+H].sup.+.

Step 3. Synthesis of 2-(methylamino)-3-nitrophenol

(1446) ##STR00843##

(1447) To a solution of 2-fluoro-3-nitrophenol (5 g, 31.85 mmol) in EtOH (20 mL) were added DIEA (26 mL, 159.25 mmol) and methanamine hydrochloride (6.45 g, 95.55 mmol). Then the reaction mixture was stirred at 90° C. for 12 h. After cooled down to room temperature, the reaction was quenched with water (30 mL) and extracted with ethyl acetate (3×50 mL). The combined organic layers were washed with brine, dried over MgSO.sub.4, filtered and concentrated under reduced pressure. The resulting crude product was triturated with petroleum ether/EtOAc (10:1, 20 mL) to give 2-(methylamino)-3-nitrophenol (3.5 g, 65% yield) as a yellow solid. MS (ESI) m/z: 169.1 [M+H].sup.+.

Step 4. Synthesis of 2-((tert-butyldimethylsilyl)oxy)-N-methyl-6-nitroaniline

(1448) ##STR00844##

(1449) To a solution of 2-(methylamino)-3-nitrophenol (3.6 g, 21.43 mmol) and DMAP (200 mg, 1.6 mmol) in DCM (30 mL) were added triethylamine (21.6 g, 214.3 mmol) and tert-butyldimethylsilyl chloride (6.4 g, 42.86 mmol). The reaction mixture was stirred at room temperature for 14 h, before the reaction was quenched with water (15 mL) and extracted with DCM (3×30 mL). The combined organic layers were washed with brine, dried over MgSO.sub.4, filtered and concentrated under reduced pressure. The resulting residue was triturated with petroleum ether/EtOAc (10:1, 22 mL) to give 2-((tert-butyldimethylsilyl)oxy)-N-methyl-6-nitroaniline (3.4 g, 56% yield) as a yellow solid. MS (ESI) m/z: 283.9 [M+H].sup.+.

Step 5. Synthesis of 6-((tert-butyldimethylsilyl)oxy)-N-methylbenzene-1,2-diamine

(1450) ##STR00845##

(1451) To a solution of 2-((tert-butyldimethylsilyl)oxy)-N-methyl-6-nitroaniline (3.4 g, 12.06

mmol) in EtOH (30 mL) was added 10% Pd/C (0.5 g). Then the mixture was stirred under hydrogen atmosphere (1 atm) overnight. After the catalyst was filtered through celite, the filtrate was evaporated to dryness to give crude 6-((tert-butyldimethylsilyl)oxy)-N1-methylbenzene-1,2-diamine (3.0 g, 99% yield). This product was used in the next step directly without further purification. MS (ESI) m/z: 254.0 [M+H].sup.+.

Step 6. Synthesis of 7-((tert-butyldimethylsilyl)oxy)-1-methyl-1,3-dihydro-2H-benzo[d]imidazol-2-one

(1452) ##STR00846##

(1453) To a solution of 6-((tert-butyldimethylsilyl)oxy)-N'-methylbenzene-1,2-diamine (3.0 g, 11.91 mmol) in THE (30 mL) was added N,N'-carbonyldiimidazole (5.79 g, 35.73 mmol) at room temperature under N.sub.2. The reaction mixture was stirred at room temperature for 4 h, before the solvent was removed under reduced pressure. The residue was recrystallized with methanol and n-hexane to give 7-((tert-butyldimethylsilyl)oxy)-1-methyl-1,3-dihydro-2H-benzo[d]imidazol-2-one (2.8 g, 84% yield) as a white solid. MS (ESI) m/z: 279.6 [M+H].sup.+.

Step 7. Synthesis of 3-(4-((tert-butyldimethylsilyl)oxy)-3-methyl-2-oxo-2,3-dihydro-1H-benzo[d]imidazol-1-yl)piperidine-2,6-dione

(1454) ##STR00847##

(1455) To a stirred solution of 7-((tert-butyldimethylsilyl)oxy)-1-methyl-1,3-dihydro-2H-benzo[d]imidazol-2-one (600 mg, 2.16 mmol) in DMF (5 mL) was added NaH (104 mg, 60% w/w dispersed into mineral oil, 2.60 mmol) at 0° C. under nitrogen atmosphere. After the reaction mixture was stirred at 0° C. for 20 min, a solution of 3-bromopiperidine-2, 6-dione (291.84 mg, 1.52 mmol) in DMF (3 mL) was added dropwise at 0° C. The resulting mixture was stirred at room temperature for additional 3 h, before the reaction was quenched with AcOH (0.5 mL) and concentrated under reduced pressure to give crude 3-(4-((tert-butyldimethylsilyl)oxy)-3-methyl-2-oxo-2,3-dihydro-1H-benzo[d]imidazol-1-yl)piperidine-2,6-dione. This product was used in the next step directly without further purification.

Step 8. Synthesis of 3-(4-hydroxy-3-methyl-2-oxo-2,3-dihydro-1H-benzo[d]imidazol-1-yl)piperidine-2,6-dione

(1456) ##STR00848##

(1457) To a stirred solution of 3-(4-((tert-butyldimethylsilyl)oxy)-3-methyl-2-oxo-2,3-dihydro-1H-benzo[d]imidazol-1-yl)piperidine-2,6-dione (crude) in DCM (10 mL) was added TFA (2 mL). The reaction mixture was stirred at rt for 1 h, before the mixture was concentrated. The residue was purified by silica gel chromatography to give 3-(4-hydroxy-3-methyl-2-oxo-2,3-dihydro-1H-benzo[d]imidazol-1-yl)piperidine-2,6-dione (70 mg, 12% yield over two steps). MS (ESI) m/z: 276.3 [M+H].sup.+.

Step 9. Synthesis of 3-(4-(2-(4-(6-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)ethoxy)-3-methyl-2-oxo-2,3-dihydro-1H-benzo[d]imidazol-1-yl)piperidine-2,6-dione

(1458) ##STR00849##

(1459) To a solution of 3-(4-hydroxy-3-methyl-2-oxo-2,3-dihydro-1H-benzo[d]imidazol-1-yl)piperidine-2,6-dione (3.05 mg, 0.012 mmol) and (R)-2-(4-(6-(6-(2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)ethyl 4-methylbenzenesulfonate (5.0 mg, 0.01 mmol) in DMSO (0.5 mL) were added K.sub.2CO.sub.3 (5.52 mg, 0.04 mmol) and NaI (2.25 mg, 0.015 mmol) at room temperature. The reaction mixture was stirred at 70° C. for 2 h. After the reaction was cooled down to room temperature, the reaction was quenched with water (10 mL) and extracted with ethyl acetate (3×10 mL). The combined organic layers were washed with saturated brine (10 mL), dried over anhydrous sodium sulfate, filtered and evaporated under reduced pressure. The resulting residue was purified by prep-HPLC to afford 3-(4-(2-(4-(6-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)ethoxy)-3-methyl-2-oxo-2,3-dihydro-1H-benzo[d]imidazol-1-yl)piperidine-2,6-dione (1.2 mg,

16% yield) as a white solid. MS (ESI) m/z: 745.9 [M+H].sup.+.

Example 296: 3-(4-(3-(4-(6-(6-((R)-2-(3-Fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)propoxy)-3-methyl-2-oxo-2,3-dihydro-1H-benzo[d]imidazol-1-yl)piperidine-2,6-dione (TR-245)

(1460) ##STR00850##

(1461) TR-245 was synthesized according to the standard procedures for preparing TR-244 (1.5 mg, 17% yield). MS (ESI) m/z: 759.9 [M+H].sup.+.

Example 297: 3-(5-(2-(4-(6-(6-((R)-2-(3-Fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)ethoxy)-3-methyl-2-oxo-2,3-dihydro-1H-benzo[d]imidazol-1-yl)piperidine-2,6-dione (TR-246)

(1462) ##STR00851##

(1463) TR-246 was synthesized according to the standard procedures for preparing TR-244 (5.2 mg, 19% yield). MS (ESI) m/z: 745.8 [M+H].sup.+.

Example 298: 3-(7-((2-(4-(6-(6-((R)-2-(3-Fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)ethyl)amino)-2-oxobenzo[d]oxazol-3 (2H)-yl)piperidine-2,6-dione (TR-247)

(1464) ##STR00852##

Step 1. Synthesis of 7-nitrobenzo[d]oxazol-2(3H)-one

(1465) ##STR00853##

(1466) A mixture of 2-hydroxy-3-nitro-benzoic acid (500 mg, 2.73 mmol), DPPA (1.50 g, 5.46 mmol), TEA (828.89 mg, 8.19 mmol) in toluene (5 mL) was stirred at 110° C. for 20 h under N.sub.2. The reaction was purified by silica gel chromatography (DCM/MeOH=60:1) to give 7-nitrobenzo[d]oxazol-2(3H)-one (206 mg, 1.14 mmol, 42% yield) as a yellow solid. MS (ESI) m/z: 179.2 [M-H].sup.-.

Step 2. Synthesis of 3-(7-nitro-2-oxobenzo[d]oxazol-3 (2H)-yl)piperidine-2,6-dione

(1467) ##STR00854##

(1468) To a 50 mL of three-necked round-bottomed flask was added 7-nitrobenzo[d]oxazol-2(3H)-one (170 mg, 943.83 umol), 3-bromopiperidine-2,6-dione (163.10 mg, 849.45 umol), dry K2CO.sub.3 (117.22 mg, 849.45 umol), and DMF (3 mL) under N.sub.2. After the mixture was stirred at 25° C. for 12 h, the reaction mixture was purified by silica gel column chromatography (petroleum ether/EtOAc=2:1 to 1:1) to give 3-(7-nitro-2-oxobenzo[d]oxazol-3 (2/1)-yl)piperidine-2,6-dione (50 mg, 18% yield) as a yellow solid. MS (ESI) m/z: 290.3 [M-H].sup.-.

Step 3. Synthesis of 3-(7-amino-2-oxobenzo[d]oxazol-3 (2H)-yl)piperidine-2,6-dione

(1469) ##STR00855##

(1470) To a solution of 3-(7-nitro-2-oxobenzo[d]oxazol-3 (2H)-yl)piperidine-2,6-dione (43 mg, 147.66 umol) in THF (0.7 mL) was added Pd/C (50.59 mg, 208.27 umol, 50% purity) under H.sub.2. The mixture was stirred at 25° C. for 3 h. LCMS showed the reaction was complete. Pd/C was filtered over Celite. After solvents were removed under vacuum, the resulting residue was purified by silica gel column chromatography (petroleum ether/EtOAc=2:1 to 1:1) to give 3-(7-amino-2-oxobenzo[d]oxazol-3 (2H)-yl)piperidine-2,6-dione (38 mg, 99% yield) as a white solid. MS (ESI) m/z: 262.2 [M+H].sup.+.

Step 4. Synthesis of 3-[7-[2-[4-[6-[6-[(2R)-2-(3-fluorophenyl)pyrrolidin-1-yl]imidazo[1,2-b]pyridazin-3-yl]-2-pyridyl]piperazin-1-ylethylamino]-2-oxo-1,3-benzoxazol-3-yl]piperidine-2,6-dione

(1471) ##STR00856##

(1472) To a solution of 2-[4-[6-[6-[(2R)-2-(3-fluorophenyl)pyrrolidin-1-yl]imidazo[1,2-b]pyridazin-3-yl].sup.-. 2-pyridyl]piperazin-1-yl]acetaldehyde (18.6 mg, 38.31 umol) and 3-(7-amino-2-oxo-1,3-benzoxazol-3-yl)piperidine-2,6-dione (10.01 mg, 38.31 umol) in DMF (2 mL) were added TMSCl (12.41 mg, 114.92 umol) and NaBH.sub.4 (2.91 mg, 76.61 umol) at 0° C. The reaction mixture was stirred at 0° C. for 2 h, before being stirred at room temperature for 16 h. The

reaction mixture was concentrated and purified by prep-TLC (DCM/MeOH=15/1) to give 3-[7-[2-[4-[6-[6-[(2R)-2-(3-fluorophenyl)pyrrolidin-1-yl]imidazo[1,2-b]pyridazin-3-yl]-2-pyridyl]piperazin-1-ylethylamino]-2-oxo-1,3-benzoxazol-3-yl]piperidine-2,6-dione (9 mg, 32% yield). MS (ESI) m/z: 731.8 [M+H].sup.+.

Example 299: 2-(2,6-Dioxopiperidin-3-yl)-5-((3-((4-(6-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)methyl)azetidin-1-yl)methyl)isoindoline-1,3-dione (TR-248)

(1473) ##STR00857##

Step 1: Synthesis of tert-butyl (R)-3-((4-(6-(6-(2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)methyl)azetidine-1-carboxylate

(1474) ##STR00858##

(1475) To a solution of 6-[(2R)-2-(3-fluorophenyl)pyrrolidin-1-yl]-3-(6-piperazin-1-yl-2-pyridyl)imidazo[1,2-b]pyridazine (400 mg, 901.88 umol) and tert-butyl 3-formylazetidine-1-carboxylate (167.05 mg, 901.88 umol) in THF (10 mL) was added Ti (OiPr).sub.4 (512.27 mg, 1.80 mmol) at room temperature. The resulting mixture was stirred at 60° C. for 3 b, before being cooled to room temperature. NaBH.sub.3 CN (335.50 mg, 3.61 mmol) was added in portions. The resulting mixture was stirred at rt for 16 h, before being quenched with water (20 mL), and extracted with DCM (3×20 mL). The DCM layers were combined, dried over Na.sub.2SO.sub.4, and concentrated under vacuum. The resulting residue was purified by reverse phase chromatography to give tert-butyl 3-[[4-[6-[6-[(2R)-2-(3-fluorophenyl)pyrrolidin-1-yl]imidazo[1,2-b]pyridazin-3-yl]-2-pyridyl]piperazin-1-yl]methyl]azetidine-1-carboxylate (350 mg, 63% yield). MS (ESI) m/z: 613.7 [M+H].sup.+.

Step 2: Synthesis of (R)-3-(6-(4-(azetidin-3-ylmethyl)piperazin-1-yl)pyridin-2-yl)-6-(2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazine

(1476) ##STR00859##

(1477) To a solution of tert-butyl 3-[[4-[6-[6-[(2R)-2-(3-fluorophenyl)pyrrolidin-1-yl]imidazo[1,2-b]pyridazin-3-yl]-2-pyridyl]piperazin-1-yl]methyl]azetidine-1-carboxylate (100 mg, 163.20 umol) in DCM (5 mL) was added TFA (2 mL) at rt. The reaction mixture was stirred for 1 h, before being concentrated to give 3-[6-[4-(azetidin-3-ylmethyl)piperazin-1-yl]-2-pyridyl]-6-[(2R)-2-(3-fluorophenyl)pyrrolidin-1-yl]imidazo[1,2-b]pyridazine (100 mg, 100% yield), which was used directly in the next step without further purification. MS (ESI) m/z: 513.7 [M+H].sup.+.

Step 3: Synthesis of 2-(2,6-dioxopiperidin-3-yl)-5-((3-((4-(6-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)methyl)azetidin-1-yl)methyl)isoindoline-1,3-dione

(1478) ##STR00860##

(1479) To a solution of 5-(bromomethyl)-2-(2,6-dioxo-3-piperidyl)isoindoline-1,3-dione (35 mg, 99.67 umol) and 3-[6-[4-(azetidin-3-ylmethyl)piperazin-1-yl]-2-pyridyl]-6-[(2R)-2-(3-fluorophenyl)pyrrolidin-1-yl]imidazo[1,2-b]pyridazine (51.09 mg, 99.67 umol) in NMP (5 mL) was added DIPEA (128.82 mg, 996.72 umol) at rt. The reaction mixture was stirred at 120° C. for 0.5 h, before being purified by prep-HPLC to give a crude product, which was further purified by prep-TLC (DCM/MeOH=15/1) to give 2-(2,6-dioxo-3-piperidyl)-5-[[3-[4-[6-[6-[(2R)-2-(3-fluorophenyl)pyrrolidin-1-yl]imidazo[1,2-b]pyridazin-3-yl]-2-pyridyl]piperazin-1-yl]methyl]azetidin-1-yl]methyl]isoindoline-1,3-dione (4 mg, 5% yield). MS (ESI) m/z: 782.9 [M+H].sup.+.

Example 300: N-(5-(3,5-Difluorobenzyl)-1H-indazol-3-yl)-4-(4-((1-((2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisoindolin-5-yl)methyl)azetidin-3-yl)methyl)piperazin-1-yl)-2-((tetrahydro-2H-pyran-4-yl)amino)benzamide (TR-249)

(1480) ##STR00861##

Step 1. Synthesis of tert-butyl 3-((4-(4-((5-(3,5-difluorobenzyl)-1H-indazol-3-yl)carbamoyl)-3-((tetrahydro-2H-pyran-4-yl)amino)phenyl)piperazin-1-yl)methyl)azetidine-1-carboxylate

(1481) ##STR00862##

(1482) To a solution of N-(5-(3,5-difluorobenzyl)-1H-indazol-3-yl)-4-(piperazin-1-yl)-2-((tetrahydro-2H-pyran-4-yl)amino)benzamide (100 mg, 0.18 mmol) and tert-butyl 3-formylazetidine-1-carboxylate (67 mg, 0.36 mmol) in DCM (10 mL)/AcOH (1 mL) was added BH.sub.3/THF (0.5 mL, 1M in THF, 0.54 mmol) in portions at 0° C. under N.sub.2. The resulting mixture was stirred at room temperature for 2 h. The reaction was quenched with H.sub.2O (100 mL) at 0° C. and extracted with DCM (2×100 mL). The organic layers were combined, dried over Na.sub.2SO.sub.4, filtered, and concentrated. The resulting residue was purified by silica gel flash chromatography to give tert-butyl 3-((4-(4-((5-(3,5-difluorobenzyl)-1H-indazol-3-yl)carbamoyl)-3-((tetrahydro-2H-pyran-4-yl)amino)phenyl)piperazin-1-yl)methyl)azetidine-1-carboxylate (85 mg, 65% yield) as a yellow oil. MS (ESI) m/z: 716.7 [M+H].sup.+.

Step 2. Synthesis of 4-(4-(azetidin-3-ylmethyl)piperazin-1-yl)-N-(5-(3,5-difluorobenzyl)-1H-indazol-3-yl)-2-((tetrahydro-2H-pyran-4-yl)amino)benzamide

(1483) ##STR00863##

(1484) To a solution of tert-butyl 3-((4-(4-((5-(3,5-difluorobenzyl)-1H-indazol-3-yl)carbamoyl)-3-((tetrahydro-2H-pyran-4-yl)amino)phenyl)piperazin-1-yl)methyl)azetidine-1-carboxylate (85 mg, 0.12 mmol) in DCM (2 mL) was added TFA (1 mL) at 0° C. After the reaction mixture was stirred at room temperature for 1 h, the solvent was removed under vacuum. The resulting residue was purified by reverse-phase chromatography to give the desired product (43 mg, 59% yield) as a white solid. MS (ESI) m/z: 616.7 [M+H].sup.+.

Step 3. Synthesis of N-(5-(3,5-difluorobenzyl)-1H-indazol-3-yl)-4-(4-((1-((2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-5-yl)methyl)azetidin-3-yl)methyl)piperazin-1-yl)-2-((tetrahydro-2H-pyran-4-yl)amino)benzamide

(1485) ##STR00864##

(1486) To a solution of 4-(4-(azetidin-3-ylmethyl)piperazin-1-yl)-N-(5-(3,5-difluorobenzyl)-1H-indazol-3-yl)-2-((tetrahydro-2H-pyran-4-yl)amino)benzamide (43 mg, 0.07 mmol) and 5-(bromomethyl)-2-(2,6-dioxopiperidin-3-yl)isoindoline-1,3-dione (30 mg, 0.08 mmol) in dimethylacetamide (DMAc) (5 mL) was added DIPEA (30 mg, 0.21 mmol) at room temperature under N.sub.2. After the reaction mixture was stirred at 50° C. for 1 h, the resulting residue was purified by reverse-phase chromatography, followed by prep-TLC to give the desired product (15 mg, 26% yield) as a white solid. MS (ESI) m/z: 886.9 [M+H].sup.+.

Example 301: 3-(5-(3-((4-(6-(6-((R)-2-(3-Fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)methyl)azetidin-1-yl)-3-methyl-2-oxo-2,3-dihydro-1H-benzo[d]imidazol-1-yl)piperidine-2,6-dione (TR-250)

(1487) ##STR00865##

Step 1. Synthesis of 5-chloro-N-methyl-2-nitroaniline

(1488) ##STR00866##

(1489) A solution of 4-chloro-2-fluoro-1-nitrobenzene (3 g, 17.14 mmol), methylamine hydrochloride (1.26 g, 18.85 mmol) and DIPEA (6.63 g, 51.46 mmol) in DMSO (40 mL) was stirred at room temperature overnight. The reaction solution was quenched with water (200 mL), extracted with EtOAc (2×100 mL), washed with brine, dried over Na.sub.2SO.sub.4, filtered and concentrated to give 5-chloro-N-methyl-2-nitroaniline (3.2 g, 100% yield) as a yellow solid. MS (ESI) m/z: 187.2 [M+H].sup.+.

Step 2. Synthesis of 5-(3-((benzyloxy)methyl)azetidin-1-yl)-N-methyl-2-nitroaniline

(1490) ##STR00867##

(1491) A solution of 5-chloro-N-methyl-2-nitroaniline (3.2 g, 17.2 mmol), 3-((benzyloxy)methyl)azetidine (3.34 g, 18.92 mmol) and K.sub.2CO.sub.3 (11.8 g, 86 mmol) in DMSO (60 mL) was stirred at 100° C. overnight. The reaction mixture was diluted with water (300 mL), extracted with EtOAc (2×100 mL), washed with brine, dried over Na.sub.2SO.sub.4, filtered and concentrated. The resulting residue was purified by silica gel flash chromatography (petroleum

ether/EtOAc=1:1) to give 5-(3-((benzyloxy)methyl)azetidin-1-yl)-N-methyl-2-nitroaniline (3.2 g, 57% yield) as a yellow solid. MS (ESI) m/z: 328.7 [M+H].sup.+.

Step 3. Synthesis of 6-(3-((benzyloxy)methyl)azetidin-1-yl)-1-methyl-1,3-dihydro-2H-benzo[d]imidazol-2-one

(1492) ##STR00868##

(1493) A mixture of 5-(3-((benzyloxy)methyl)azetidin-1-yl)-N-methyl-2-nitroaniline (2 g, 6.11 mmol), Raney Nickel (0.1 g) in THF (200 mL) was stirred at room temperature under H.sub.2 atmosphere (1 atm) overnight. The supernatant liquid of the reaction mixture was poured into a 250 mL glass bottle. CDI (2.9 g, 18.33 mmol) was added portionwise. The reaction mixture was stirred for 0.5 h at room temperature under N.sub.2 atmosphere. The reaction solution was poured into water (300 mL), extracted with EtOAc (2×200 mL), washed with brine, dried over Na.sub.2SO.sub.4, filtered and concentrated. The residue was triturated with EtOAc (100 mL) to give 6-(3-((benzyloxy)methyl)azetidin-1-yl)-1-methyl-1,3-dihydro-2H-benzo[d]imidazol-2-one (1.4 g, 70% yield) as a white solid. MS (ESI) m/z: 324.4 [M+H].sup.+.

Step 4. Synthesis of 3-(5-(3-((benzyloxy)methyl)azetidin-1-yl)-3-methyl-2-oxo-2,3-dihydro-1H-benzo[d]imidazol-1-yl)piperidine-2,6-dione

(1494) ##STR00869##

(1495) To a solution of 6-(3-((benzyloxy)methyl)azetidin-1-yl)-1-methyl-1,3-dihydro-2H-benzo[d]imidazol-2-one (1.4 g, 4.33 mmol) in anhydrous DMF (30 mL) was added NaH (0.208 g, 5.20 mmol) portionwise in ice bath under N.sub.2 atmosphere. After the reaction was stirred for 0.5 h, 3-bromopiperidine-2,6-dione (575 mg, 3.03 mmol) in DMF (2 mL) was added dropwise at the same temperature. After the reaction mixture was stirred at room temperature overnight, the reaction was quenched with acetic acid (1 mL) in ice bath, diluted with water (200 mL), extracted with EA (2×80 mL), washed with brine, dried over Na.sub.2SO.sub.4, filtered and concentrated. The resulting residue was purified by silica gel flash chromatography (petroleum ether/EtOAc=2:1 to pure EtOAc) to give the 3-(5-(3-((benzyloxy)methyl)azetidin-1-yl)-3-methyl-2-oxo-2,3-dihydro-1H-benzo[d]imidazol-1-yl)piperidine-2,6-dione (400 mg, 30% yield) as a white solid. MS (ESI) m/z: 435.9 [M+H].sup.+.

Step 5. Synthesis of 3-(5-(3-(hydroxymethyl)azetidin-1-yl)-3-methyl-2-oxo-2,3-dihydro-1H-benzo[d]imidazol-1-yl)piperidine-2,6-dione

(1496) ##STR00870##

(1497) A solution of 3-(5-(3-((benzyloxy)methyl)azetidin-1-yl)-3-methyl-2-oxo-2,3-dihydro-1H-benzo[d]imidazol-1-yl)piperidine-2,6-dione (400 mg, 0.921 mmol) in MsOH (1 mL) and DCM (2 mL) was stirred at room temperature for 3 h. The reaction solution was diluted with ice water (100 mL), neutralized with saturated aqueous NaHCO.sub.3 solution, extracted with DCM/MeOH (10:1, 3×50 mL). The combined organic layers were dried over Na.sub.2SO.sub.4, filtered and concentrated to give 3-(5-(3-(hydroxymethyl)azetidin-1-yl)-3-methyl-2-oxo-2,3-dihydro-1H-benzo[d]imidazol-1-yl)piperidine-2,6-dione (210 mg, 66% yield) as a white solid. MS (ESI) m/z: 345.5 [M+H].sup.+.

Step 6. Synthesis of (1-(1-(2,6-dioxopiperidin-3-yl)-3-methyl-2-oxo-2,3-dihydro-1H-benzo[d]imidazol-5-yl)azetidin-3-yl)methylmethanesulfonate

(1498) ##STR00871##

(1499) To a solution of 3-(5-(3-(hydroxymethyl)azetidin-1-yl)-3-methyl-2-oxo-2,3-dihydro-1H-benzo[d]imidazol-1-yl)piperidine-2,6-dione (100 mg, 0.29 mmol) and DIPEA (187 mg, 1.45 mmol) in DCM (5 mL) was added Ms2O (100 mg, 0.58 mmol) in DCM (1 mL) dropwise at 0° C. After the reaction was stirred at room temperature for 2 h, the reaction solution was diluted with water (200 mL), extracted with DCM (2×100 mL), washed with brine, dried over Na.sub.2SO.sub.4, filtered and concentrated to give (1-(1-(2,6-dioxopiperidin-3-yl)-3-methyl-2-oxo-2,3-dihydro-1H-benzo[d]imidazol-5-yl)azetidin-3-yl)methylmethanesulfonate (58 mg, 47% yield) as a light yellow solid. MS (ESI) m/z: 423.5 [M+H].sup.+.

Step 7. Synthesis of 3-(5-(3-((4-(6-(6-((R)-2-(3-

fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)methyl)azetidin-1-yl)-3-methyl-2-oxo-2,3-dihydro-1H-benzo[d]imidazol-1-yl)piperidine-2,6-dione

(1500) ##STR00872##

(1501) A solution of (1-(1-(2,6-dioxopiperidin-3-yl)-3-methyl-2-oxo-2,3-dihydro-1H-benzo[d]imidazol-5-yl)azetidin-3-yl)methyl methanesulfonate (58 mg, 0.137 mmol), (R)-6-(2-(3-fluorophenyl)pyrrolidin-1-yl).sub.0-3-(6-(piperazin-1-yl)pyridin-2-yl)imidazo[1,2-b]pyridazine (72 mg, 0.164 mmol), KI (22 mg, 0.137 mmol), and DIEA (88 mg, 0.685 mmol) in NMP (1 mL) was stirred at 130° C. for 0.5 h under microwave irradiation. The reaction solution was diluted with water (100 mL), extracted with EtOAc (2×50 mL), washed with brine, dried over Na.sub.2SO.sub.4, filtered and concentrated. The residue was purified with prep-TLC (DCM:MeOH=7:1) to give 3-(5-(3-((4-(6-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)methyl)azetidin-1-yl)-3-methyl-2-oxo-2,3-dihydro-1H-benzo[d]imidazol-1-yl)piperidine-2,6-dione (4 mg, 4% yield) as a white solid. MS (ESI) m/z: 770.8 [M+H].sup.+.

Example 302: N-(5-(3,5-Difluorobenzyl)-1H-indazol-3-yl)-4-(4-((1-((2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-5-yl)methyl)azetidin-3-yl)methyl)piperazin-1-yl)-2-((2-fluornethylaminolhenzamide (TR_251)

(1502) ##STR00873##

(1503) TR-251 was synthesized following the standard procedure for preparing TR-249 (5.5 mg, 13% yield by 3 steps). MS (ESI) m/z: 848.3 [M+H].sup.+.

Example 303: N—(S-(3,5-Difluorobenzyl)-1H-indazol-3-yl)-4-(4-(1-((2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-5-yl)methyl)azetidin-3-yl)piperazin-1-yl)-2-((2-fluoroethyl)amino)benzamide (TR-252)

(1504) ##STR00874##

(1505) TR-252 was synthesized following the standard procedure for preparing TR-249 (7.1 mg, 11% yield by 3 steps). MS (ESI) m/z: 834.3 [M+H].sup.+.

Example 304: N-(5-(3,5-Difluorobenzyl)-1H-indazol-3-yl)-4-(4-(3-(2-(2,6-dioxopiperidin-3-yl)-1-oxoisindolin-5-yl)prop-2-yn-1-yl)piperazin-1-yl)-2-((tetrahydro-2H-pyran-4-yl)amino)benzamide (TR-253)

(1506) ##STR00875##

Step 1. Synthesis of 3-(5-(3-hydroxyprop-1-yn-1-yl)-1-oxoisindolin-2-yl)piperidine-2,6-dione

(1507) ##STR00876##

(1508) To a solution of 3-(5-bromo-1-oxoisindolin-2-yl)piperidine-2,6-dione (500 mg, 1.5 mmol) and prop-2-yn-1-ol (126 mg, 2.25 mmol) in DMSO (15 mL) were added DIEA (975 mg, 7.5 mmol), Cul (28 mg, 0.15 mmol) and Pd(dppf) Cl.sub.2(110 mg, 0.15 mmol) at room temperature under N.sub.2. The reaction mixture was stirred at 100° C. for 12 h. The resulting residue was purified by reverse-phase chromatography to give the desired product (430 mg, 96% yield) as a brown solid. MS (ESI) m/z: 299.4 [M+H].sup.+.

Step 2. Synthesis of 3-(2-(2,6-dioxopiperidin-3-yl)-1-oxoisindolin-5-yl)prop-2-yn-1-yl 4-methylbenzenesulfonate

(1509) ##STR00877##

(1510) To a solution of 3-(5-(3-hydroxyprop-1-yn-1-yl)-1-oxoisindolin-2-yl)piperidine-2,6-dione (100 mg, 0.33 mmol) in DCM (5 mL) were added TsCl (95 mg, 0.49 mmol) and KOH (37 mg, 0.66 mmol) at 0° C. After the reaction was stirred at room temperature for 1 h, the reaction was quenched with water (30 mL). The resulting mixture was extracted with DCM (2×20 mL). And the organic layers were combined, dried over Na.sub.2SO.sub.4, filtered and concentrated. The resulting residue was purified by prep-TLC to give 3-(2-(2,6-dioxopiperidin-3-yl)-1-oxoisindolin-5-yl)prop-2-yn-1-yl 4-methylbenzenesulfonate (30 mg, 20% yield) as a white solid. MS (ESI) m/z: 453.2 [M+H].sup.+.

Step 3. Synthesis of N—(S-(3,5-difluorobenzyl)-1H-indazol-3-yl)-4-(4-(3-(2-(2,6-dioxopiperidin-3-yl)-1-oxoisindolin-5-yl)prop-2-yn-1-yl)piperazin-1-yl)-2-((tetrahydro-2H-pyran-4-yl)amino)benzamide

(1511) ##STR00878##

(1512) To a solution of N—(S-(3,5-difluorobenzyl)-1H-indazol-3-yl)-4-(piperazin-1-yl)-2-((tetrahydro-2H-pyran-4-yl)amino)benzamide (15 mg, 0.03 mmol) and 3-(2-(2,6-dioxopiperidin-3-yl)-1-oxoisindolin-5-yl)prop-2-yn-1-yl 4-methylbenzenesulfonate (15 mg, 0.03 mmol) in MeCN (2 mL) were added NaI (9 mg, 0.06 mmol) and DIPEA (20 mg, 0.15 mmol) at room temperature. The reaction mixture was stirred at 85° C. for 12 h, before being purified by prep-TLC to give N-(5-(3,5-difluorobenzyl)-1H-indazol-3-yl)-4-(4-(3-(2-(2,6-dioxopiperidin-3-yl)-1-oxoisindolin-5-yl)prop-2-yn-1-yl)piperazin-1-yl)-2-((tetrahydro-2H-pyran-4-yl)amino)benzamide (10 mg, 42% yield) as a white solid. MS (ESI) m/z: 827.9 [M+H].sup.+.

Example 305: N-(5-(3,5-Difluorobenzyl)-1H-indazol-3-yl)-4-(4-(1-((2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-5-yl)methyl)azetidin-3-yl)piperazin-1-yl)-2-((tetrahydro-2H-pyran-4-yl)amino)benzamide (TR-254)

(1513) ##STR00879##

(1514) TR-254 was synthesized following the standard procedure for preparing TR-249 (12 mg, 9% yield by 3 steps). MS (ESI) m/z: 872.8 [M+H].sup.+.

Example 306: N—(S-(3,5-Difluorobenzyl)-1H-indazol-3-yl)-4-(4-(3-(2-(2,6-dioxopiperidin-3-yl)-1-oxoisindolin-5-yl)prop-2-yn-1-yl)piperazin-1-yl)-2-((2-fluoroethyl)amino)benzamide (TR-255)

(1515) ##STR00880##

(1516) TR-255 was synthesized following the standard procedure for preparing TR-253 (1.5 mg, 12% yield). MS (ESI) m/z: 789.9 [M+H].sup.+.

Example 307: N—(S-(3,5-Difluorobenzyl)-1H-indazol-3-yl)-4-(4-(3-(2-(2,6-dioxopiperidin-3-yl)-1-oxoisindolin-5-yl)prop-2-yn-1-yl)piperazin-1-yl)benzamide (TR-256)

(1517) ##STR00881##

(1518) TR-256 was synthesized following the standard procedure for preparing TR-253 (2.1 mg, 12% yield). MS (ESI) m/z: 728.7 [M+H].sup.+.

Example 308: 3-(5-(3-(4-(6-(6-((R)-2-(3-Fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)propoxy)-3-methyl-2-oxo-2,3-dihydro-1H-benzo[d]imidazol-1-yl)piperidine-2,6-dione (TR-257)

(1519) ##STR00882##

(1520) A mixture of 3-(5-hydroxy-3-methyl-2-oxo-2,3-dihydro-1H-benzo[d]imidazol-1-yl)piperidine-2,6-dione (10.0 mg, 0.036 mmol), (R)-3-(4-(6-(6-(2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)propyl 4-methylbenzenesulfonate (15.0 mg, 0.030 mmol), sodium iodide (6.75 mg, 0.045 mmol) and K₂CO₃ (16.56 mg, 0.120 mmol) in DMSO (1 mL) was stirred at 70° C. for 12 h. The reaction mixture was diluted with water (10 mL) and extracted with EtOAc (10 mL). The organic phase was dried over anhydrous sodium sulfate, filtered and concentrated. The resulting residue was purified by prep-TLC to give the desired product (5.1 mg, 23% yield) as a white solid. MS (ESI) m/z: 760.0 [M+H].sup.+.

Example 309: 3-(4-(3-((4-(6-(6-((R)-2-(3-Fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)methyl)azetidin-1-yl)-3-methyl-2-oxo-2,3-dihydro-1H-benzo[d]imidazol-1-yl)piperidine-2,6-dione (TR-258)

(1521) ##STR00883##

(1522) TR-258 was synthesized following the standard procedure for preparing TR-250 (9 mg, 2.3% yield). MS (ESI) m/z: 770.9 [M+H].sup.+.

Example 310: 3-(5-(4-(4-(6-(6-((R)-2-(3-Fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)butyl)-3-methyl-2-oxo-2,3-dihydro-1H-benzo[d]imidazol-1-yl)piperidine-2,6-dione (TR-259)

(1523) ##STR00884##

Step 1. Synthesis of 5-bromo-N-methyl-2-nitroaniline

(1524) ##STR00885##

(1525) A mixture of 4-bromo-2-fluoro-1-nitrobenzene (10 g, 45.87 mmol), and K_2CO_3 (12.66 g, 91.74 mmol), methanamine hydrochloride (3.68 g, 55 mmol) in DMSO (100 mL) was stirred at 100° C., overnight. The mixture was poured into water (400 mL), extracted with EtOAc (2×200 mL), washed with brine, dried over Na_2SO_4 , filtered and concentrated to give the 5-bromo-N-methyl-2-nitroaniline (10.4 g, 99% yield) as a yellow solid. MS (ESI) m/z : 231.2 $[\text{M}+\text{H}]^+$.

Step 2. Synthesis of 5-bromo-NI-methylbenzene-1,2-diamine

(1526) ##STR00886##

(1527) To a solution of 5-bromo-N-methyl-2-nitroaniline (10.4 g, 45.2 mmol) in AcOH (40 mL)/EtOH (40 mL)/water (20 mL) was added Fe (12.65 g, 226 mmol) portionwise at room temperature. The reaction was stirred at 50° C. for 2 h, before the reaction mixture was filtered over Celite and washed with DCM/MeOH (10:1). The filtrate was washed with brine, dried over Na_2SO_4 , filtered and concentrated to give the 5-bromo-NI-methylbenzene-1,2-diamine (9.6 g, 100% yield) as a black solid. MS (ESI) m/z : 201.2 $[\text{M}+\text{H}]^+$.

Step 3. Synthesis of 6-bromo-1-methyl-1,3-dihydro-2H-benzo[d]imidazol-2-one

(1528) ##STR00887##

(1529) To a solution of 5-bromo-NI-methylbenzene-1,2-diamine (9.5 g, 47.5 mmol) in DMF (100 mL) was added CDI (10 g, 61.75 mmol) portionwise at room temperature. After the reaction was stirred at 60° C. for 1 h, the solution was poured into water (400 mL), extracted with EtOAc (2×200 mL), washed with brine, dried over Na_2SO_4 , filtered and concentrated. The resulting residue was triturated with EtOAc (100 mL) to give 6-bromo-1-methyl-1,3-dihydro-2H-benzo[d]imidazol-2-one (6.2 g, 58% yield) as a gray solid. MS (ESI) m/z : 227.2 $[\text{M}+\text{H}]^+$.

Step 4. Synthesis of 3-(5-bromo-3-methyl-2-oxo-2,3-dihydro-1H-benzo[d]imidazol-1-yl)piperidine-2,6-dione

(1530) ##STR00888##

(1531) To a solution of 6-bromo-1-methyl-1,3-dihydro-2H-benzo[d]imidazol-2-one (2 g, 8.85 mmol) in anhydrous DMF (30 mL) was added NaH (0.424 g, 10.62 mmol) portionwise at 0° C. under N_2 atmosphere. After the reaction was stirred for 0.5 h, a solution of 3-bromopiperidine-2,6-dione (845 mg, 4.42 mmol) in DMF (2 mL) was added dropwise at the same temperature. After the reaction mixture was stirred at room temperature overnight, the reaction was quenched with AcOH (1.2 mL) in ice bath, diluted with water (200 mL), extracted with EtOAc (2×80 mL), washed with brine, dried over Na_2SO_4 , filtered, and concentrated. The resulting residue was purified by silica gel column (petroleum ether/EtOAc=2:1 to 1:1) to give 3-(5-bromo-3-methyl-2-oxo-2,3-dihydro-1H-benzo[d]imidazol-1-yl)piperidine-2,6-dione (300 mg, 20% yield) as a white solid. MS (ESI) m/z : 340.3 $[\text{M}+\text{H}]^+$.

Step 5. Synthesis of 3-(5-bromo-3-methyl-2-oxo-2,3-dihydro-1H-benzo[d]imidazol-1-yl)piperidine-2,6-dione

(1532) ##STR00889##

(1533) A mixture of 3-(5-bromo-3-methyl-2-oxo-2,3-dihydro-1H-benzo[d]imidazol-1-yl)piperidine-2,6-dione (250 mg, 0.74 mmol), but-3-yn-1-ol (260 mg, 3.7 mmol), Pd(dppf) Cl_2 (54 mg, 0.074 mmol), CuI (28 mg, 0.148 mmol) and TEA (373.7 mg, 3.7 mmol) in DMSO (5 mL) was stirred at 80° C. overnight under N_2 atmosphere. After the mixture was filtered over Celite and washed with DCM, the filtrate was diluted with water (200 mL), extracted with EtOAc (2×100 mL), washed with brine, dried over Na_2SO_4 , filtered and concentrated. The resulting residue was purified by silica gel flash chromatography (DCM: MeOH=15:1) to give 3-(5-(4-hydroxybut-1-yn-1-yl)-3-methyl-2-oxo-2,3-dihydro-1H-benzo[d]imidazol-1-yl)piperidine-2,6-dione (160 mg, 66% yield) as a red brown solid. MS (ESI) m/z : 328.2 $[\text{M}+\text{H}]^+$.

Step 6. Synthesis of 3-(5-(4-hydroxybutyl)-3-methyl-2-oxo-2,3-dihydro-1H-benzo[d]imidazol-1-yl)piperidine-2,6-dione
(1534) ##STR00890##

(1535) A mixture of 3-(5-(4-hydroxybut-1-yn-1-yl)-3-methyl-2-oxo-2,3-dihydro-1H-benzo[d]imidazol-1-yl)piperidine-2,6-dione (40 mg, 0.122 mmol), Pd/C (20 mg) in MeOH (20 mL) was stirred at room temperature for 2 h under H.sub.2 atmosphere. The mixture was filtered with Celite. The filtrate was concentrated to give 3-(5-(4-hydroxybutyl)-3-methyl-2-oxo-2,3-dihydro-1H-benzo[d]imidazol-1-yl)piperidine-2,6-dione (40 mg, 99% yield) as a gray solid. MS (ESI) m/z: 332.3 [M+H].sup.+.

Step 7. Synthesis of 4-(1-(2,6-dioxopiperidin-3-yl)-3-methyl-2-oxo-2,3-dihydro-1H-benzo[d]imidazol-5-yl)butyl methanesulfonate

(1536) ##STR00891##

(1537) To a solution of 3-(5-(4-hydroxybutyl)-3-methyl-2-oxo-2,3-dihydro-1H-benzo[d]imidazol-1-yl)piperidine-2,6-dione (40 mg, 0.122 mmol) and DIEA (80 mg, 0.61 mmol) in DCM (20 mL) was added MsCl (0.05 mL, 0.61 mmol) in DCM (2 mL) at 0° C. After the reaction was stirred for 0.5 h, the reaction solution was quenched with water (100 mL), extracted with DCM (2×50 mL), washed with brine, dried over Na.sub.2SO.sub.4, filtered and concentrated. The resulting residue was purified by prep-TLC (DCM/MeOH=10:1) to give 4-(1-(2,6-dioxopiperidin-3-yl)-3-methyl-2-oxo-2,3-dihydro-1H-benzo[d]imidazol-5-yl)butyl methanesulfonate (17 mg, 34% yield) as a gray solid. MS (ESI) m/z: 410.4 [M+H].sup.+.

Step 8. Synthesis of 3-(5-(4-(4-(6-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)butyl)-3-methyl-2-oxo-2,3-dihydro-1H-benzo[d]imidazol-1-yl)piperidine-2,6-dione

(1538) ##STR00892##

(1539) A mixture of 4-(1-(2,6-dioxopiperidin-3-yl)-3-methyl-2-oxo-2,3-dihydro-1H-benzo[d]imidazol-5-yl)butyl methanesulfonate (17 mg, 0.041 mmol), (R)-6-(2-(3-fluorophenyl)pyrrolidin-1-yl)-3-(6-(piperazin-1-yl)pyridin-2-yl)imidazo[1,2-b]pyridazine (21 mg, 0.049 mmol), NaI (12.3 mg, 0.082 mmol) and DIEA (26 mg, 0.205 mmol) in CH.sub.3CN (5 mL) was stirred at 80° C. overnight. The mixture was diluted with water (50 mL), extracted with EtOAc (2×20 mL), washed with brine, dried over Na.sub.2SO.sub.4, filtered and concentrated. The resulting residue was purified by prep-TLC (DCM/MeOH=7:1) to give 3-(5-(4-(4-(6-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)butyl)-3-methyl-2-oxo-2,3-dihydro-1H-benzo[d]imidazol-1-yl)piperidine-2,6-dione (6 mg, 19% yield) as a white solid. MS (ESI) m/z: 757.9 [M+H].sup.+.

Example 311: 3-(5-(4-(4-(6-(6-((R)-2-(3-Fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)but-1-yn-1-yl)-3-methyl-2-oxo-2,3-dihydro-1H-benzo[d]imidazol-1-yl)piperidine-2,6-dione (TR-260)

(1540) ##STR00893##

(1541) TR-260 was synthesized following the standard procedure for preparing TR-259 (3.5 mg, 13% yield). MS (ESI) m/z: 753.8 [M+H].sup.+.

Example 312: N-(5-(3,5-Difluorobenzyl)-1H-indazol-3-yl)-4-(4-(1-((2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-5-yl)methyl)azetidin-3-yl)piperazin-1-yl)benzamide (TR-261)

(1542) ##STR00894##

(1543) TR-261 was synthesized following the standard procedure for preparing TR-249 (6.3 mg, 11% yield by 3 steps). MS (ESI) m/z: 773.3 [M+H].sup.+.

Example 313: N-(5-(3,5-Difluorobenzyl)-1H-indazol-3-yl)-4-(4-((1-((2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-5-yl)methyl)azetidin-3-yl)methyl)piperazin-1-yl)benzamide (TR-262)

(1544) ##STR00895##

(1545) TR-262 was synthesized following the standard procedure for preparing TR-249 (5.7 mg, 15% yield by 3 steps). MS (ESI) m/z: 787.3 [M+H].sup.+.

Example 314: N-(5-(3,5-Difluorobenzyl)-1H-indazol-3-yl)-4-(4-(2-((2-(2,6-dioxopiperidin-3-

yl)-1,3-dioxoisindolin-5-yl)amino)ethyl)piperazin-1-yl)benzamide (TR-263)

(1546) ##STR00896##

Step 1. Synthesis of N-(5-(3,5-difluorobenzyl)-1H-indazol-3-yl)-4-(4-(2-((2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-5-yl)(4-methoxybenzyl)amino)ethyl)piperazin-1-yl)benzamide

(1547) ##STR00897##

(1548) To a solution of N-(5-(3,5-difluorobenzyl)-1H-indazol-3-yl)-4-(piperazin-1-yl)benzamide (20.1 mg, 0.04 mmol) and 2-((2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-5-yl)(4-methoxybenzyl)amino)ethyl 4-methylbenzenesulfonate (40.6 mg, 0.07 mmol) in acetonitrile (5 mL) were added potassium carbonate (18 mg, 0.13 mmol) and sodium iodide (7 mg, 0.04 mmol) at room temperature. The reaction mixture was stirred at 80° C. for 48 h, before being concentrated and purified by silica gel column chromatography (MeOH/DCM=1:10) to give the desired product (9.3 mg, 23% yield) as a yellow solid. MS (ESI) m/z: 867.3 [M+H].sup.+.

Step 2. Synthesis of N-(5-(3,5-difluorobenzyl)-1H-indazol-3-yl)-4-(4-(2-((2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-5-yl)amino)ethyl)piperazin-1-yl)benzamide

(1549) ##STR00898##

(1550) To a solution of N-(5-(3,5-difluorobenzyl)-1H-indazol-3-yl)-4-(4-(2-((2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-5-yl)(4-methoxybenzyl)amino)ethyl)piperazin-1-yl)benzamide (9.3 mg, 0.01 mmol) in DCM (1 mL) was added TFA (11 mg, 0.09 mmol) at room temperature. The reaction mixture was stirred at room temperature for 1 h, before being concentrated and purified by reverse-phase chromatography to give N-(5-(3,5-difluorobenzyl)-1H-indazol-3-yl)-4-(4-(2-((2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-5-yl)amino)ethyl)piperazin-1-yl)benzamide (3.5 mg, 46% yield) as a white solid. MS (ESI) m/z: 747.3 [M+H].sup.+.

Example 315: N-(5-(3,5-Difluorobenzyl)-1H-indazol-3-yl)-4-(4-((2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-5-yl)glycyl)piperazin-1-yl)benzamide (TR-264)

(1551) ##STR00899##

(1552) To a solution of N-(5-(3,5-difluorobenzyl)-1H-indazol-3-yl)-4-(piperazin-1-yl)benzamide (10.7 mg, 0.02 mmol) in DMSO (1 mL) were added HOAT (4.5 mg, 0.03 mmol), EDCI (6.4 mg, 0.03 mmol), NMM (11.3 mg, 0.11 mmol) and (2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-5-yl)glycine (11.1 mg, 0.03 mmol) sequentially. After the resulting solution was stirred at room temperature for 16 h, the reaction was poured into water (20 mL) and extracted with ethyl acetate (3×20 mL). The combined organic layers were washed with saturated brine (30 mL), dried over anhydrous sodium sulfate, filtered and evaporated under reduced pressure. The resulting residue was purified by reverse-phase chromatography to give N-(5-(3,5-difluorobenzyl)-1H-indazol-3-yl)-4-(4-((2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-5-yl)glycyl)piperazin-1-yl)benzamide (6.3 mg, 38% yield) as a white solid. MS (ESI) m/z: 761.3 [M+H].sup.+.

Example 316: N-(5-(3,5-Difluorobenzyl)-1H-indazol-3-yl)-4-(4-((1-(2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-5-yl)azetidin-3-yl)methyl)piperazin-1-yl)benzamide (TR-265)

(1553) ##STR00900##

(1554) TR-265 was synthesized following the standard procedure for preparing TR-231 (3.5 mg, 20% yield). MS (ESI) m/z: 773.3 [M+H].sup.+.

Example 317: N-(5-(3,5-Difluorobenzyl)-1H-indazol-3-yl)-4-(4-((2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-5-yl)glycyl)piperazin-1-yl)-2-((2-fluoroethyl)amino)benzamide (TR-266)

(1555) ##STR00901##

(1556) TR-266 was synthesized following the standard procedure for preparing TR-264 (5 mg, 20% yield). MS (ESI) m/z: 822.7 [M+H].sup.+.

Example 318: N—(S-(3,5-Difluorobenzyl)-1H-indazol-3-yl)-4-(4-((1-(2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-5-yl)azetidin-3-yl)methyl)piperazin-1-yl)-2-((2-fluoroethyl)amino)benzamide (TR-267)

(1557) ##STR00902##

(1558) TR-267 was synthesized following the standard procedure for preparing TR-231 (15 mg, 31% yield). MS (ESI) m/z: 834.8 [M+H].sup.+.

Example 319: N-(5-(3,5-Difluorobenzyl)-1H-indazol-3-yl)-4-(4-(2-((2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-5-yl)amino)ethyl)piperazin-1-yl)-2-((2-fluoroethyl)amino)benzamide (TR-268)

(1559) ##STR00903##

(1560) TR-268 was synthesized following the standard procedure for preparing TR-263 (12 mg, 14% yield by 2 steps). MS (ESI) m/z: 808.8 [M+H].sup.+.

Example 320: N-(5-(3,5-Difluorobenzyl)-1H-indazol-3-yl)-4-(4-(2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-5-yl)piperazin-1-yl)-2-((tetrahydro-2H-pyran-4-yl)amino)benzamide (TR-269)

(1561) ##STR00904##

(1562) To a solution of N-(5-(3,5-difluorobenzyl)-1H-indazol-3-yl)-4-(piperazin-1-yl)-2-((tetrahydro-2H-pyran-4-yl)amino)benzamide (40 mg, 0.07 mmol) and 2-(2,6-dioxopiperidin-3-yl)-5-fluoroisindoline-1,3-dione (24 mg, 0.08 mmol) in DMSO (5 mL) was added DIPEA (45 mg, 0.35 mmol) at room temperature under N.sub.2. After the reaction mixture was stirred at 100° C. for 1 h, the reaction was purified by reverse-phase chromatography, followed by prep-TLC to give N-(5-(3,5-difluorobenzyl)-1H-indazol-3-yl)-4-(4-(2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-5-yl)piperazin-1-yl)-2-((tetrahydro-2H-pyran-4-yl)amino)benzamide (20 mg, 35% yield) as a yellow solid. MS (ESI) m/z: 803.8 [M+H].sup.+.

Example 321: N-(5-(3,5-Difluorobenzyl)-1H-indazol-3-yl)-4-(4-((1-(2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-5-yl)pyrrolidin-3-yl)methyl)piperazin-1-yl)-2-((tetrahydro-2H-pyran-4-yl)amino)benzamide (TR-270)

(1563) ##STR00905##

(1564) TR-270 was synthesized following the standard procedure for preparing TR-231 (13 mg, 9% yield by 3 steps). MS (ESI) m/z: 886.9 [M+H].sup.+.

Example 322: N-(5-(3,5-Difluorobenzyl)-1H-indazol-3-yl)-4-(4-((1-(2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-5-yl)piperidin-4-yl)methyl)piperazin-1-yl)-2-((tetrahydro-2H-pyran-4-yl)amino)benzamide (TR-271)

(1565) ##STR00906##

(1566) TR-271 was synthesized following the standard procedure for preparing TR-231 (15 mg, 15% yield by 3 steps). MS (ESI) m/z: 900.9 [M+H].sup.+.

Example 323: N-(5-(3,5-Difluorobenzyl)-1H-indazol-3-yl)-4-(4-((1-(2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-5-yl)piperidin-3-yl)methyl)piperazin-1-yl)-2-(tetrahydro-2H-pyran-4-yl)amino)benzamide (TR-272)

(1567) ##STR00907##

(1568) TR-272 was synthesized following the standard procedure for preparing TR-231 (13 mg, 6% yield by 3 steps), MS (ESI) m/z: 900.9 [M+H].sup.+.

Example 324: N-(5-(3,5-Difluorobenzyl)-1H-indazol-3-yl)-4-(4-(2-((2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-5-yl)amino)-2-oxoethyl)piperazin-1-yl)-2-((tetrahydro-2H-pyran-4-yl)amino)benzamide (TR-273)

(1569) ##STR00908##

Step 1. Synthesis of 2-chloro-N-(2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-5-yl)acetamide

(1570) ##STR00909##

(1571) To a solution of 5-amino-2-(2,6-dioxopiperidin-3-yl)isindoline-1,3-dione (60 mg, 0.22 mmol) in THF (5 mL) was added 2-chloroacetyl chloride (30 mg, 0.26 mmol) at room temperature. After the reaction was refluxed for 1 h, the reaction mixture was concentrated. The resulting residue was purified by reverse-phase chromatography to give the desired product (65 mg, 85% yield) as a pale yellow solid. MS (ESI) m/z: 350.0 [M+H].sup.+.

Step 2. Synthesis of N-(5-(3,5-difluorobenzyl)-1H-indazol-3-yl)-4-(4-(2-((2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-5-yl)amino)-2-oxoethyl)piperazin-1-yl)-2-((tetrahydro-2H-pyran-4-

yl)amino)benzamide
(1572) ##STR00910##

(1573) To a solution of N-(5-(3,5-difluorobenzyl)-1H-indazol-3-yl)-4-(piperazin-1-yl)-2-((tetrahydro-2H-pyran-4-yl)amino)benzamide (10 mg, 0.018 mmol) in MeCN (3 mL) were added K.sub.2CO.sub.3 (7.5 mg, 0.054 mmol), NaI (2.7 mg, 0.018 mmol) and 2-chloro-N-(2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-5-yl)acetamide (6.3 mg, 0.018 mmol) at room temperature. After the reaction was stirred at 80° C. for 2 h, the mixture was concentrated and purified by reverse-phase chromatography to give the desired product N-(5-(3,5-difluorobenzyl)-1H-indazol-3-yl)-4-(4-(2-((2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-5-yl)amino)-2-oxoethyl)piperazin-1-yl)-2-((tetrahydro-2H-pyran-4-yl)amino)benzamide (5.6 mg, 36% yield) as a light yellow solid. MS (ESI) m/z: 860.3 [M+H].sup.+.

Example 325: N-(5-(3,5-Difluorobenzyl)-1H-indazol-3-yl)-4-(4-((2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-5-yl)methyl)piperazin-1-yl)-2-((tetrahydro-2H-pyran-4-yl)amino)benzamide (TR-274)

(1574) ##STR00911##

(1575) To a solution of N-(5-(3,5-difluorobenzyl)-1H-indazol-3-yl)-4-(piperazin-1-yl)-2-((tetrahydro-2H-pyran-4-yl)amino)benzamide (10 mg, 0.018 mmol) and 5-(bromomethyl)-2-(2,6-dioxopiperidin-3-yl)isoindoline-1,3-dione (6.4 mg, 0.018 mmol) in DMAc (2 mL) was added DIPEA (7.7 mg, 0.060 mmol) at room temperature. The reaction was stirred at 60° C. for 1 h. The reaction was cooled to room temperature before being quenched with H.sub.2O (30 mL). The mixture was extracted with EtOAc (3×10 mL). The combined organic layers were dried over Na.sub.2SO.sub.4, concentrated and purified by reverse-phase chromatography to give the desired product N-(5-(3,5-difluorobenzyl)-1H-indazol-3-yl)-4-(4-((2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-5-yl)methyl)piperazin-1-yl)-2-((tetrahydro-2H-pyran-4-yl)amino)benzamide (6.1 mg, 36% yield) as a light yellow solid. MS (ESI) m/z: 817.3 [M+H].sup.+.

Example 326: N-(5-(3,5-Difluorobenzyl)-1H-indazol-3-yl)-4-(4-(1-(2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-5-yl)azetidin-3-yl)piperazin-1-yl)-2-((tetrahydro-2H-pyran-4-yl)amino)benzamide (TR-275)

(1576) ##STR00912##

Step 1. Synthesis of tert-butyl 3-(4-(4-((5-(3,5-difluorobenzyl)-1H-indazol-3-yl)carbamoyl)-3-((tetrahydro-2H-pyran-4-yl)amino)phenyl)piperazin-1-yl)azetidine-1-carboxylate

(1577) ##STR00913##

(1578) A mixture of tert-butyl 3-oxoazetidine-1-carboxylate (14.2 mg, 0.08 mmol), NaBH.sub.3CN (5.2 mg, 0.08 mmol) and N-(5-(3,5-difluorobenzyl)-1H-indazol-3-yl)-4-(piperazin-1-yl)-2-((tetrahydro-2H-pyran-4-yl)amino)benzamide (30.3 mg, 0.05 mmol) in MeOH (5 mL) was added AcOH (3.8 mg, 0.06 mmol) at room temperature. After the reaction was stirred overnight, the mixture was poured into water (30 mL) and extracted with ethyl acetate (3×10 mL). The combined organic layers were washed with saturated brine (30 mL), dried over anhydrous sodium sulfate, filtered and evaporated under reduced pressure. The resulting residue was concentrated and purified by reverse-phase chromatography to give the desired product (18.0 mg, 46% yield) as a white solid. MS (ESI) m/z: 702.4 [M+H].sup.+.

Step 2. Synthesis of 4-(4-(azetidin-3-yl)piperazin-1-yl)-N-(5-(3,5-difluorobenzyl)-1H-indazol-3-yl)-2-((tetrahydro-2H-pyran-4-yl)amino)benzamide

(1579) ##STR00914##

(1580) To a solution of tert-butyl 3-(4-(4-((5-(3,5-difluorobenzyl)-1H-indazol-3-yl)carbamoyl)-3-((tetrahydro-2H-pyran-4-yl)amino)phenyl)piperazin-1-yl)azetidine-1-carboxylate (18.0 mg, 0.02 mmol) in DCM (1 mL) was added TFA (0.5 mL) at room temperature. The reaction mixture was stirred at room temperature for 1 h, before being concentrated and purified by reverse-phase chromatography to give the desired product (12.8 mg, 83% yield) as a pale white solid. MS (ESI) m/z: 602.3 [M+H].sup.+.

Step 3. Synthesis of N-(5-(3,5-difluorobenzyl)-1H-indazol-3-yl)-4-(4-(1-(2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-5-yl)azetidin-3-yl)piperazin-1-yl)-2-((tetrahydro-2H-pyran-4-yl)amino)benzamide

(1581) ##STR00915##

(1582) To a solution of 4-(4-(azetidin-3-yl)piperazin-1-yl)-N-(5-(3,5-difluorobenzyl)-1H-indazol-3-yl)-2-((tetrahydro-2H-pyran-4-yl)amino)benzamide (10.2 mg, 0.02 mmol) in DMSO (2 mL) were added 2-(2,6-dioxopiperidin-3-yl)-5-fluoroisindoline-1,3-dione (7 mg, 0.03 mmol) and DIPEA (6.6 mg, 0.05 mmol) at room temperature. The reaction mixture was stirred at 80° C. overnight, before being purified by reverse-phase chromatography to give N-(5-(3,5-difluorobenzyl)-1H-indazol-3-yl)-4-(4-(1-(2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-5-yl)azetidin-3-yl)piperazin-1-yl)-2-((tetrahydro-2H-pyran-4-yl)amino)benzamide (5.7 mg, 39% yield) as a white solid. MS (ESI) m/z: 858.4 [M+H].sup.+.

Example 327: N-(5-(3,5-Difluorobenzyl)-1H-indazol-3-yl)-4-(4-(3-(2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-5-yl)propyl)piperazin-1-yl)-2-((tetrahydro-2H-pyran-4-yl)amino)benzamide (TR-276)

(1583) ##STR00916##

(1584) To a mixture of 3-(2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-5-yl)propyl 4-methylbenzenesulfonate (10.3 mg, 0.02 mmol), potassium carbonate (6.0 mg, 0.04 mmol) and sodium iodide (2.6 mg, 0.02 mmol) in acetonitrile (2 mL) was added N-(5-(3,5-difluorobenzyl)-1H-indazol-3-yl)-4-(piperazin-1-yl)-2-((tetrahydro-2H-pyran-4-yl)amino)benzamide (17.9 mg, 0.03 mmol) at room temperature. After the reaction was stirred at 80° C. for 16 h, the mixture was concentrated and purified by reverse-phase chromatography to give the desired product N-(5-(3,5-difluorobenzyl)-1H-indazol-3-yl)-4-(4-(3-(2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-5-yl)propyl)piperazin-1-yl)-2-((tetrahydro-2H-pyran-4-yl)amino)benzamide (7.2 mg, 39% yield) as a light yellow solid. MS (ESI) m/z: 844.4 [M+H].sup.+.

Example 328: N-(5-(3,5-Difluorobenzyl)-1H-indazol-3-yl)-4-(4-(2-(2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-5-yl)ethyl)piperazin-1-yl)-2-((tetrahydro-2H-pyran-4-yl)amino)benzamide (TR-277)

(1585) ##STR00917##

(1586) TR-277 was synthesized following the standard procedure for preparing TR-276 (9.5 mg, 22% yield by 2 steps). MS (ESI) m/z: 831.3 [M+H].sup.+.

Example 329: N-(5-(3,5-Difluorobenzyl)-1H-indazol-3-yl)-4-(4-(1-(2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-5-yl)piperidin-3-yl)piperazin-1-yl)-2-((tetrahydro-2H-pyran-4-yl)amino)benzamide (TR-278)

(1587) ##STR00918##

(1588) TR-278 was synthesized following the standard procedure for preparing TR-275 (9.4 mg, 22% yield by 3 steps). MS (ESI) m/z: 886.4 [M+H].sup.+.

Example 330: N-(5-(3,5-Difluorobenzyl)-1H-indazol-3-yl)-4-(4-(1-((2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-5-yl)methyl)piperidin-3-yl)piperazin-1-yl)-2-((tetrahydro-2H-pyran-4-yl)amino)benzamide (TR-279)

(1589) ##STR00919##

(1590) TR-279 was synthesized following the standard procedure for preparing TR-274 (6.8 mg, 47% yield). MS (ESI) m/z: 900.4 [M+H].sup.+.

Example 331: N-(5-(3,5-Difluorobenzyl)-1H-indazol-3-yl)-4-(4-(1-(2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-5-yl)piperidin-4-yl)piperazin-1-yl)-2-((tetrahydro-2H-pyran-4-yl)amino)benzamide (TR-280)

(1591) ##STR00920##

(1592) TR-280 was synthesized following the standard procedure for preparing TR-275 (10.8 mg, 26% yield by 3 steps). MS (ESI) m/z: 886.4 [M+H].sup.+.

Example 332: N-(5-(3,5-Difluorobenzyl)-1H-indazol-3-yl)-4-(4-(1-((2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-5-yl)methyl)piperidin-4-yl)piperazin-1-yl)-2-((tetrahydro-2H-pyran-4-yl)amino)benzamide (TR-281)

(1593) ##STR00921##

(1594) TR-281 was synthesized following the standard procedure for preparing TR-274 (7.5 mg, 52% yield). MS (ESI) m/z: 900.4 [M+H].sup.+.

Example 333: N-(5-(3,5-Difluorobenzyl)-1H-indazol-3-yl)-4-(4-((4-(2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-5-yl)morpholin-2-yl)methyl)piperazin-1-yl)-2-((tetrahydro-2H-pyran-4-yl)amino)benzamide (TR-282)

(1595) ##STR00922##

(1596) TR-282 was synthesized following the standard procedure for preparing TR-231 (10 mg, 24% yield). MS (ESI) m/z: 902.9 [M+H].sup.+.

Example 334: N-(5-(3,5-Difluorobenzyl)-1H-indazol-3-yl)-4-(4-(3-(2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-5-yl)prop-2-yn-1-yl)piperazin-1-yl)-2-((tetrahydro-2H-pyran-4-yl)amino)benzamide (TR-283)

(1597) ##STR00923##

Step 1. Synthesis of 3-(2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-5-yl)prop-2-yn-1-yl methanesulfonate

(1598) ##STR00924##

(1599) To a solution of 2-(2,6-dioxopiperidin-3-yl)-5-(3-hydroxyprop-1-yn-1-yl)isoindoline-1,3-dione (50 mg, 0.16 mmol) in DCM (3 mL) were added MsCl (0.1 mL) and TEA (50 mg, 0.48 mmol) at 0° C. The reaction mixture was stirred at room temperature for 1 h, before being quenched with water (10 mL) and extracted with DCM (10 mL×2). The combined organic layers were dried over anhydrous sodium sulfate, filtered and concentrated to give 3-(2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-5-yl)prop-2-yn-1-yl methanesulfonate (35 mg, 56% yield) as a yellow solid. This crude product was used in the next step directly without further purification. MS (ESI) m/z: 391.3 [M+H].sup.+.

Step 2. Synthesis of N-(5-(3,5-difluorobenzyl)-1H-indazol-3-yl)-4-(4-(3-(2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-5-yl)prop-2-yn-1-yl)piperazin-1-yl)-2-((tetrahydro-2H-pyran-4-yl)amino)benzamide

(1600) ##STR00925##

(1601) To a solution of N—(S-(3,5-difluorobenzyl)-1H-indazol-3-yl)-4-(piperazin-1-yl)-2-((tetrahydro-2H-pyran-4-yl)amino)benzamide (20 mg, 0.04 mmol) and 3-(2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-5-yl)prop-2-yn-1-yl methanesulfonate (18 mg, 0.05 mmol) in MeCN (5 mL) were added NaI (12 mg, 0.08 mmol) and DIPEA (30 mg, 0.20 mmol) at room temperature. After the reaction mixture was stirred at 85° C. for 12 h, the mixture was purified by prep-TLC to give N-(5-(3,5-difluorobenzyl)-1H-indazol-3-yl)-4-(4-(3-(2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-5-yl)prop-2-yn-1-yl)piperazin-1-yl)-2-((tetrahydro-2H-pyran-4-yl)amino)benzamide (11 mg, 33% yield) as a white solid. MS (ESI) m/z: 841.8 [M+H].sup.+.

Example 335: N—(S-(3,5-Difluorobenzyl)-1H-indazol-3-yl)-4-(4-(1-(2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-5-yl)pyrrolidin-3-yl)piperazin-1-yl)-2-((tetrahydro-2H-pyran-4-yl)amino)benzamide (TR-284)

(1602) ##STR00926##

(1603) TR-284 was synthesized following the standard procedure for preparing TR-275 (15 mg, 31% yield by 3 steps). MS (ESI) m/z: 872.8 [M+H].sup.+.

Example 336: N-(5-(3,5-Difluorobenzyl)-1H-indazol-3-yl)-4-(4-(1-((2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-5-yl)methyl)pyrrolidin-3-yl)piperazin-1-yl)-2-((tetrahydro-2H-pyran-4-yl)amino)benzamide (TR-285)

(1604) ##STR00927##

(1605) TR-285 was synthesized following the standard procedure for preparing TR-274 (15 mg,

57% yield). MS (ESI) m/z: 886.8 [M+H].sup.+.

Example 337: 2-(2,6-Dioxopiperidin-3-yl)-5-(2-((4-(6-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)methyl)morpholino) isoindoline-1,3-dione (TR-286)

(1606) ##STR00928##

Step 1. Synthesis of 2-(2,6-dioxopiperidin-3-yl)-5-(2-(hydroxymethyl)morpholino) isoindoline-1,3-dione

(1607) ##STR00929##

(1608) To a solution of 2-(2,6-dioxopiperidin-3-yl)-5-fluoroisoindoline-1,3-dione (500 mg, 1.8 mmol) and DIEA (700 mg, 5.4 mmol) in DMSO (5 mL) was added morpholin-2-ylmethanol (320 mg, 2.7 mmol). After the resulting mixture was stirred at 120° C. for 2 b, the mixture was purified by reverse phase column chromatography to give 2-(2,6-dioxopiperidin-3-yl)-5-(2-(hydroxymethyl) morpholino) isoindoline-1,3-dione (547 mg, 81% yield) as a yellow oil. MS (ESI) m/z: 374.3 [M+H].sup.+.

Step 2. Synthesis of (4-(2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisoindolin-5-yl)morpholin-2-yl)methyl 4-methylbenzenesulfonate

(1609) ##STR00930##

(1610) To a solution of 2-(2,6-dioxopiperidin-3-yl)-5-(2-(hydroxymethyl)morpholino) isoindoline-1,3-dione (170 mg, 0.46 mmol), TEA (232 mg, 2.3 mmol), DMAP (5.6 mg, 0.046 mmol) in DCM (5 mL) and DMSO (5 mL) was added TsCl (350 mg, 1.8 mmol) at 0° C. The reaction mixture was stirred at room temperature for 1 h, before DCM was removed under vacuum. The resulting residue was purified by reverse phase column chromatography to give (4-(2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisoindolin-5-yl)morpholin-2-yl)methyl 4-methylbenzenesulfonate (173 mg, 72% yield) as a yellow solid. MS (ESI) m/z: 528.4 [M+H].sup.+.

Step 3. Synthesis of 2-(2,6-dioxopiperidin-3-yl)-5-(2-((4-(6-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)methyl)morpholino) isoindoline-1,3-dione

(1611) ##STR00931##

(1612) A mixture of (4-(2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisoindolin-5-yl)morpholin-2-yl)methyl 4-methylbenzenesulfonate (48 mg, 0.09 mmol), (S)-6-(2-(3-fluorophenyl)pyrrolidin-1-yl)-3-(6-(piperazin-1-yl)pyridin-2-yl)imidazo[1,2-pyridazine (40 mg, 0.09 mmol), potassium carbonate (37 mg, 0.27 mmol) and sodium iodide (20 mg, 0.14 mmol) in acetonitrile (1.5 mL) was stirred at 90° C. overnight. The mixture was purified by reverse phase column chromatography to give 2-(2,6-dioxopiperidin-3-yl)-5-(2-((4-(6-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)methyl)morpholino) isoindoline-1,3-dione (547 mg, 81% yield) as a yellow solid. MS (ESI) m/z: 799.9 [M+H].sup.+.

Example 338: 2-(2,6-Dioxopiperidin-3-yl)-5-(3-((1-(6-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperidin-4-yl)oxy]azetidin-1-yl]isoindoline-1,3-dione (TR-287)

(1613) ##STR00932##

Step 1: Synthesis of tert-butyl 3-(pyridin-4-yloxy) azetidine-1-carboxylate

(1614) ##STR00933##

(1615) To a solution of pyridin-4-ol (0.95 g, 9.99 mmol) and tert-butyl 3-methylsulfonyloxazetidine-1-carboxylate (2.51 g, 9.99 mmol) in DMF (100 mL) was added K.sub.2CO.sub.3 (4.14 g, 29.97 mmol) at room temperature. After the resulting mixture was stirred at 100° C. for 16 h, the reaction mixture was quenched with water, extracted with EtOAc (3×10 mL). The combined EtOAc layers were washed with brine, dried over Na.sub.2SO.sub.4, and concentrated. The resulting residue was purified by silica gel chromatography (petroleum ether/EtOAc=1/1) to give tert-butyl 3-(4-pyridyloxy) azetidine-1-carboxylate (2.0 g, 80% yield) as white solid. MS (ESI) m/z: 251.4 [M+H].sup.+.

Step 2: Synthesis of tert-butyl 3-(piperidin-4-yloxy) azetidine-1-carboxylate 4-

methylbenzenesulfonic acid salt

(1616) ##STR00934##

(1617) To a solution of tert-butyl 3-(4-pyridyloxy) azetidine-1-carboxylate (1.5 g, 5.99 mmol) in EtOH (20 mL) were added TsOH (1.14 g, 5.99 mmol) and PtO.sub.2 (700 mg) at room temperature under H.sub.2. After the reaction mixture was stirred at 50° C. for 16 h, the reaction was filtered and concentrated to give tert-butyl 3-(4-piperidyloxy) azetidine-1-carboxylate (2.7g, 100% yield) as a TsOH salt. MS (ESI) m/z: 157.0 [M-Boc+H].sup.+.

Step 3: Synthesis of benzyl 4-((1-(tert-butoxycarbonyl)azetidin-3-yl)oxyl)piperidine-1-carboxylate

(1618) ##STR00935##

(1619) To a solution of tert-butyl 3-(4-piperidyloxy) azetidine-1-carboxylate (2.5 g, 5.83 mmol) in DCM (20 mL) was added TEA (1.77 g, 17.50 mmol) and Cbz-Cl (991.74 mg, 5.83 mmol) at 0° C. After the reaction mixture was stirred at room temperature for 4 h, the reaction was concentrated. The resulting residue was purified by silica gel chromatography (petroleum ether/EtOAc=5/1 to 2/1) to give benzyl 4-(1-tert-butoxycarbonylazetidin-3-yl)oxypiperidine-1-carboxylate (2.2 g, 97% yield) as colorless oil. MS (ESI) m/z: 391.5 [M+H].sup.+.

Step 4: Synthesis of tert-butyl 3-(piperidin-4-yloxy) azetidine-1-carboxylate

(1620) ##STR00936##

(1621) To a solution of benzyl 4-(1-tert-butoxy carbonylazetidin-3-yl)oxypiperidine-1-carboxylate (800 mg, 2.05 mmol) in THF (20 mL) was added Pd/C (100 mg, 823.38 umol) at room temperature under the H.sub.2. After the reaction mixture was stirred for 16 h, the reaction was filtered over Celite. The filtrate was concentrated to give tert-butyl 3-(4-piperidyloxy) azetidine-1-carboxylate (520 mg, 99% yield) as colorless oil. MS (ESI) m/z: 257.5 [M+H].sup.+.

Step 5: Synthesis of tert-butyl (R)-3-((1-(6-(6-(2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperidin-4-yl)oxy) azetidine-1-carboxylate

(1622) ##STR00937##

(1623) To a solution of 6-[(2R)-2-(3-fluorophenyl)pyrrolidin-1-yl]-3-(6-fluoro-2-pyridyl)imidazo[1,2-b]pyridazine (377 mg, 998.97 umol) and tert-butyl 3-(4-piperidyloxy) azetidine-1-carboxylate (256.08 mg, 998.97 umol) in DMSO (20 mL) was added DIPEA (774.64 mg, 5.99 mmol). After the reaction mixture was stirred at 120° C. for 48 h, the reaction was purified by reverse phase chromatography to give tert-butyl 3-[1-[6-[6-[(2R)-2-(3-fluorophenyl)pyrrolidin-1-yl]imidazo[1,2-b]pyridazin-3-yl]-2-pyridyl]-4-piperidyl]oxy]azetidine-1-carboxylate (400 mg, 65% yield). MS (ESI) m/z: 614.8 [M+H].sup.+.

Step 6: Synthesis of (R)-3-(6-(4-(azetidin-3-yloxy)piperidin-1-yl)pyridin-2-yl)-6-(2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazine

(1624) ##STR00938##

(1625) To a solution of tert-butyl 3-[[1-[6-[6-[(2R)-2-(3-fluorophenyl)pyrrolidin-1-yl]imidazo[1,2-b]pyridazin-3-yl]-2-pyridyl]-4-piperidyl]oxy]azetidine-1-carboxylate (400 mg, 651.76 umol) in DCM (10 mL) was added TFA (74.31 mg, 651.76 umol). After the reaction mixture was stirred at room temperature for 2 h, the reaction was concentrated to give 3-[6-[4-(azetidin-3-yloxy)-1-piperidyl]-2-pyridyl]-6-[(2R)-2-(3-fluorophenyl)pyrrolidin-1-yl]imidazo[1,2-b]pyridazine (410 mg, 100% yield) which was directly used in the next step. MS (ESI) m/z: 514.8 [M+H].sup.+.

Step 7: Synthesis of 2-(2,6-dioxo-3-piperidyl)-5-[3-[1-[6-[6-[(2R)-2-(3-fluorophenyl)pyrrolidin-1-yl]imidazo[1,2-b]pyridazin-3-yl]-2-pyridyl]-4-piperidyl]oxy]azetidin-1-yl]isoindoline-1,3-dione

(1626) ##STR00939##

(1627) To a solution of 3-[6-[4-(azetidin-3-yloxy)-1-piperidyl]-2-pyridyl]-6-[(2R)-2-(3-fluorophenyl)pyrrolidin-1-yl]imidazo[1,2-b]pyridazine (200 mg, 318.66 umol) and 2-(2,6-dioxo-3-piperidyl)-5-fluoro-isoindoline-1,3-dione (88.02 mg, 318.66 umol) in DMSO (5 mL) was added DIPEA (411.83 mg, 3.19 mmol). After the mixture was stirred at 120° C. for 16 b, the reaction mixture was purified by reverse phase chromatography to give a crude product, which was further purified by prep-TLC (DCM/MeOH=10/1) to give 2-(2,6-dioxo-3-piperidyl)-5-[3-[1-[6-[6-[(2R)-2-

(3-fluorophenyl)pyrrolidin-1-yl]imidazo[1,2-b]pyridazin-3-yl]-2-pyridyl]-4-piperidyl]oxy]azetidin-1-yl]isoindoline-1,3-dione (150 mg, 61% yield) as yellow solid. MS (ESI) m/z: 770.8 [M+H].sup.+.

Example 339: 2-(2,6-Dioxopiperidin-3-yl)-5-(3-((1-(6-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperidin-4-yl)amino) azetidin-1-yl)isoindoline-1,3-dione (TR-288)

(1628) ##STR00940##

Step 1. Synthesis of (R)-1-(6-(6-(2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperidin-4-ol

(1629) ##STR00941##

(1630) A mixture of (R)-6-(2-(3-fluorophenyl)pyrrolidin-1-yl)-3-(6-fluoropyridin-2-yl)imidazo[1,2-b]pyridazine (500 mg, 1.33 mmol), piperidin-4-ol (362 mg, 0.106 mmol) and DIEA (0.2 mL) in NMP (0.5 mL) was heated at 140° C. for 48 h. After the reaction was diluted with water (2 mL), the solution was extracted with EtOAc (3×5 mL). The combined organic layers were washed with brine and dried over MgSO₄, filtered and concentrated. The resulting residue was purified by silica gel chromatography to afford (R)-1-(6-(6-(2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperidin-4-ol (500 mg, 82% yield) as a white solid. MS (ESI) m/z: 459.8 [M+H].sup.+.

Step 2. Synthesis of (R)-1-(6-(6-(2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperidin-4-one

(1631) ##STR00942##

(1632) Oxalyl chloride (55.46 mg, 0.437 mmol) was dissolved in DCM (5 mL) and cooled to -78° C. DMSO (68.64 mg, 0.88 mmol) in DCM (1 mL) was added to the solution dropwise via an addition funnel. Upon completion of addition, the reaction was stirred for 10 min, before a solution of (R)-1-(6-(6-(2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperidin-4-ol (100 mg, 0.22 mmol) in DMSO (0.2 mL) and DCM (0.5 mL) was added dropwise via addition funnel. The reaction was stirred for 15 min, before being quenched by slow addition of TEA (177.76 mg, 1.76 mmol). The reaction was allowed to warm to room temperature. The reaction was stirred until alcohol was consumed as indicated by TLC. The solvent was removed under reduced pressure and the residue was taken up in EtOAc. The suspension was washed twice with H₂O, followed by brine. The organic layer was dried with Na₂SO₄, filtered and concentrated. The resulting residue was purified by prep-TLC to get title compound (33 mg, 33% yield) as a white solid. MS (ESI) m/z: 457.5 [M+H].sup.+.

Step 3. Synthesis of tert-butyl (1-(2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-5-yl)azetidin-3-yl) carbamate

(1633) ##STR00943##

(1634) To a solution of 2-(2,6-dioxo-3-piperidyl)-5-fluoro-isoindoline-1,3-dione (200 mg, 1.81 mmol) in DMSO (3 mL) were added DIEA (278.4 mg, 2.175 mmol) and tert-butyl azetidin-3-ylcarbamate (124.8 mg, 0.725 mmol). After the reaction was irradiated at 110° C. by microwave for 30 min, the resulting mixture was purified by silica gel column chromatography to get tert-butyl (1-(2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-5-yl)azetidin-3-yl)carbamate (279 mg, 90% yield) as a yellow solid. MS (ESI) m/z: 429.4 [M+H].sup.+.

Step 4. Synthesis of 5-(3-aminoazetidin-1-yl)-2-(2,6-dioxopiperidin-3-yl)isoindoline-1,3-dione

(1635) ##STR00944##

(1636) To a stirred solution of tert-butyl (1-(2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-5-yl)azetidin-3-yl)carbamate (650 mg, 1.52 mmol) in DCM (5 mL) was added TFA (2 mL). After the reaction was stirred at rt for 3 h, the mixture was concentrated to provide 5-(3-aminoazetidin-1-yl)-2-(2,6-dioxopiperidin-3-yl)isoindoline-1,3-dione (890 mg, 99% yield) as a white solid. MS (ESI) m/z: 329.3 [M+H].sup.+.

Step 5. Synthesis of 2-(2,6-dioxopiperidin-3-yl)-5-(3-((1-(6-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-

yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperidin-4-yl)amino) azetidin-1-yl)isoindoline-1,3-dione

(1637) ##STR00945##

(1638) To a solution of 5~ (3-aminoazetidin-1-yl)-2-(2,6-dioxopiperidin-3-yl)isoindoline-1,3-dione (35 mg, 0.08 mmol) and (R)-1-(6-(6-(2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperidin-4-one (25.25 mg, 0.08 mmol) in MeOH (3 mL) was added NaBH.sub.3CN (15.08 mg, 0.24 mmol). After the reaction was stirred at rt for 6 h, the mixture was concentrated. The resulting residue was purified by prep-TLC to provide 2-(2,6-dioxopiperidin-3-yl)-5-(3-((1-(6-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperidin-4-yl)amino) azetidin-1-yl)isoindoline-1,3-dione (20 mg, 33% yield) as a white solid. MS (ESI) m/z: 769.8 [M+H].sup.+.

Example 340: N-(5-(3,5-Difluorobenzyl)-1H-indazol-3-yl)-4-(4-(4-((2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-5-yl)amino)cyclohexyl)piperazin-1-yl)-2-((tetrahydro-2H-pyran-4-yl)amino)benzamide (TR-289)

(1639) ##STR00946##

Step 1. Synthesis of 2-(2,6-dioxopiperidin-3-yl)-5-((4 oxocyclohexyl)amino) isoindoline-1,3-dione (1640) ##STR00947##

(1641) A mixture of 4-aminocyclohexan-1-one (407 mg, 3.6 mmol), 2-(2,6-dioxopiperidin-3-yl)-5-fluoroisoindoline-1,3-dione (100 mg, 0.36 mmol) and DIPEA (232 mg, 1.8 mmol) in DMSO (8 mL) was stirred at 110° C. under microwave irradiation for 2 h. The reaction was cooled to room temperature before H.sub.2O (100 mL) was added. The mixture was extracted with EtOAc (3×40 mL). The combined organic layers were dried over Na.sub.2SO.sub.4, filtered and concentrated. The resulting residue was purified by reverse-phase chromatography to give the desired product (38 mg, 28% yield) as a light yellow solid. MS (ESI) m/z: 370.1 [M+H].sup.+.

Step 2. Synthesis of N-(5-(3,5-difluorobenzyl)-1H-indazol-3-yl)-4-(4-(4-((2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-5-yl)amino)cyclohexyl)piperazin-1-yl)-2-((tetrahydro-2H-pyran-4-yl)amino)benzamide

(1642) ##STR00948##

(1643) To a solution of N—(S-(3,5-difluorobenzyl)-1H-indazol-3-yl)-4-(piperazin-1-yl)-2-((tetrahydro-2H-pyran-4-yl)amino)benzamide (22 mg, 0.041 mmol) and 2-(2,6-dioxopiperidin-3-yl)-5-((4-oxocyclohexyl)amino) isoindoline-1,3-dione (15 mg, 0.041 mmol) in MeOH (5 mL) was added NaBH.sub.3CN (3.6 mg, 0.062 mmol) and AcOH (0.5 mg, 0.0082 mmol) at room temperature. After the reaction was stirred at room temperature for 2 h, the mixture was concentrated. The resulting residue was purified by reverse-phase chromatography to give the desired product N-(5-(3,5-difluorobenzyl)-1H-indazol-3-yl)-4-(4-(4-((2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-5-yl)amino)cyclohexyl)piperazin-1-yl)-2-((tetrahydro-2H-pyran-4-yl)amino)benzamide (12.7 mg, 35% yield) as a light yellow solid. MS (ESI) m/z: 900.4 [M+H].sup.+.

Example 341: N-(5-(3,5-Difluorobenzyl)-1H-indazol-3-yl)-4-(4-(6-(2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-S-yl)-6-azaspiro[3.4]octan-2-yl)piperazin-1-yl)-2-((tetrahydro-2H-pyran-4-yl)amino)benzamide (TR-290)

(1644) ##STR00949##

(1645) TR-290 was synthesized following the standard procedure for preparing TR-275 (8.4 mg, 19% yield by 3 steps). MS (ESI) m/z: 912.4 [M+H].sup.+.

Example 342: N-(5-(3,5~Difluorobenzyl)-1H-indazol-3-yl)-4-(4-(2-(2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-5-yl)-2-azaspiro[3.4]octan-6-yl)piperazin-1-yl)-2-((tetrahydro-2H-pyran~4-yl)amino)benzamide (TR-291)

(1646) ##STR00950##

(1647) TR-291 was synthesized following the standard procedure for preparing TR-275 (7.4 mg, 13% yield by 3 steps). MS (ESI) m/z: 912.4 [M+H].sup.+.

Example 343:2-(2,6-Dioxopiperidin-3-yl)-5-(1~(2-(4-(6~(6-((R)-2~(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)ethyl)azetidin-3-yl)isoindoline-1,3-dione (TR-292)

(1648) ##STR00951## ##STR00952##

Step 1: Synthesis of dimethyl 4-bromophthalate

(1649) ##STR00953##

(1650) To a solution of 5-bromoisobenzofuran-1,3-dione (22.7 g, 99.99 mmol) in MeOH (200 mL) was added H.sub.2SO.sub.4 (9.80 g, 99.99 mmol). After the reaction mixture was stirred at 80° C. for 16 h, the reaction was concentrated. The residue was dissolved in DCM (50 mL), washed with aq. NaHCO.sub.3 and brine, dried over Na.sub.2SO.sub.4 and concentrated. The resulting residue was purified by silica gel chromatography (petroleum ether/EtOAc=10/1) to give dimethyl 4-bromophthalate (27 g, 99% yield) as colorless oil. MS (ESI) m/z: 274.9 [M+H].sup.+.

Step 2: Synthesis of dimethyl 4-(4,4,5,5-tetramethyl-1,3,2-dioxaborolan-2-yl)phthalate

(1651) ##STR00954##

(1652) To a solution of dimethyl 4-bromobenzene-1,2-dicarboxylate (27.3 g, 99.97 mmol) and 4,4,5,5-tetramethyl-2-(4,4,5,5-tetramethyl-1,3,2-dioxaborolan-2-yl)-1,3,2-dioxaborolane (30.46 g, 119.96 mmol) in dioxane (300 mL) were added CH.sub.3CO.sub.2K (19.59 g, 199.94 mmol) and Pd(dppf) Cl.sub.2 (3.65 g, 5.00 mmol). The reaction mixture was stirred at 90° C. for 16 h, before it was concentrated. The residue was purified by silica gel chromatography (petroleum ether/EtOAc=20/1 to 10/1) to give dimethyl 4-(4,4,5,5-tetramethyl-1,3,2-dioxaborolan-2-yl)benzene-1,2-dicarboxylate (27 g, 84% yield) as oil. MS (ESI) m/z: 321.4 [M+H].sup.+.

Step 3: Synthesis of (3,4-bis(methoxycarbonyl)phenyl)boronic acid

(1653) ##STR00955##

(1654) To a solution of dimethyl 4-(4,4,5,5-tetramethyl-1,3,2-dioxaborolan-2-yl)benzene-1,2-dicarboxylate (30 g, 93.71 mmol) in acetone (400 mL) and water (200 mL) were added NaIO.sub.4 (20.05 g, 93.71 mmol) and CH.sub.3CO.sub.2NH.sub.4 (72.15 g, 937.08 mmol). After the reaction mixture was stirred at room temperature for 16 h, the reaction was extracted with EtOAc (3×100 mL). The combined organic layers were washed with brine (2×100 mL), dried over Na.sub.2SO.sub.4, filtered and concentrated. The resulting residue was purified by silica gel chromatography (petroleum ether/EtOAc=10/1) to provide [3,4-bis(methoxycarbonyl)phenyl]boronic acid (14 g, 63% yield) as white solid. MS (ESI) m/z: 238.9 [M+H].sup.+.

Step 4: Synthesis of dimethyl 4-(1-tert-butoxycarbonylazetidin-3-yl)benzene-1,2-dicarboxylate

(1655) ##STR00956##

(1656) A mixture of 4-methoxybenzenesulfonylhydrazide (2.02 g, 9.99 mmol) and tert-butyl 3-oxoazetidine-1-carboxylate (1.71 g, 9.99 mmol) in dioxane (100 mL) was stirred at 80° C. for 4 h. After the reaction was cooled to room temperature, [3,4-bis(methoxycarbonyl)phenyl]boronic acid (3.00 g, 12.61 mmol) was added. The resulting reaction mixture was stirred at 100° C. for 16 h, before it was concentrated. The residue was purified by silica gel chromatography (petroleum ether/EtOAc=3/1) to give dimethyl 4-(1-tert-butoxycarbonylazetidin-3-yl)benzene-1,2-dicarboxylate (1.1 g, 32% yield). MS (ESI) m/z: 699.7 [2M+H].sup.+.

Step 5: Synthesis of 4-(1-(tert-butoxycarbonyl)azetidin-3-yl)phthalic acid

(1657) ##STR00957##

(1658) To a solution of dimethyl 4-(1-tert-butoxycarbonylazetidin-3-yl)benzene-1,2-dicarboxylate (1.1 g, 3.15 mmol) in MeOH (40 mL) and water (40 mL) was added NaOH (629.69 mg, 15.74 mmol). The reaction mixture was stirred at 80° C. for 2 h. After being cooled to room temperature, the pH of the reaction mixture was adjusted to 4 by HCl (IN). The reaction mixture was extracted with DCM (2×100 mL), dried over Na.sub.2SO.sub.4 and concentrated to give 4-(1-tert-butoxycarbonylazetidin-3-yl)phthalic acid (1.0 g, 99% yield) as white solid. MS (ESI) m/z: 266.0 [M-56+H].sup.+.

Step 6: Synthesis of tert-butyl 3-(2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-5-yl)azetidine-1-carboxylate

(1659) ##STR00958##

(1660) A mixture of 4-(1-tert-butoxycarbonylazetidin-3-yl)phthalic acid (1.0 g, 3.11 mmol) and 3-aminopiperidine-2,6-dione (768.34 mg, 4.67 mmol, HCl salt) in pyridine (50 mL) was stirred at 120° C. 16 h. Then it was concentrated and the residue was purified by silica gel chromatography (DCM/MeOH=20/1) to give tert-butyl 3-[2-(2,6-dioxo-3-piperidyl)-1,3-dioxo-isindolin-5-yl]azetidine-1-carboxylate (850 mg, 66% yield). MS (ESI) m/z: 827.8 [2M+H].sup.+.

Step 7: Synthesis of 5-(azetidin-3-yl)-2-(2,6-dioxopiperidin-3-yl)isoindoline-1,3-dione

(1661) ##STR00959##

(1662) To a solution of tert-butyl 3-[2-(2,6-dioxo-3-piperidyl)-1,3-dioxo-isindolin-5-yl]azetidine-1-carboxylate (850 mg, 2.06 mmol) in DCM (10 mL) was added TFA (5 mL). After the mixture was stirred at room temperature for 1 h, the reaction was concentrated to give 5-(azetidin-3-yl)-2-(2,6-dioxo-3-piperidyl)isoindoline-1,3-dione (880 mg, 100% yield) as TFA salt. MS (ESI) m/z: 314.2 [M+H].sup.+.

Step 8: Synthesis of 2-(2,6-dioxo-3-piperidyl)-5-[1-[2-[4-[6-[6-[(2R)-2-(3-fluorophenyl)pyrrolidin-1-yl]imidazo[1,2-b]pyridazin-3-yl]-2-pyridyl]piperazin-1-yl]ethyl]azetidin-3-yl]isoindoline-1,3-dione

(1663) ##STR00960##

(1664) To a solution of 2-[4-[6-[6-[(2R)-2-(3-fluorophenyl)pyrrolidin-1-yl]imidazo[1,2-b]pyridazin-3-yl].sup.-. 2-pyridyl]piperazin-1-yl]acetaldehyde (50 mg, 102.97 umol) and 5-(azetidin-3-yl)-2-(2,6-dioxo-3-piperidyl)isoindoline-1,3-dione (32.26 mg, 75.50 umol) in MeOH (5 mL) was added NaBH₄CN (51.08 mg, 823.80 umol) at 0° C. The reaction mixture was stirred at 0° C. for 4 h, and stirred at room temperature for 16 h, before the reaction mixture was concentrated. The resulting residue was purified by prep-TLC (DCM/MeOH=15/1) to give a crude product, which was further purified by reverse phase chromatography to give 2-(2,6-dioxo-3-piperidyl)-5-[1-[2-[4-[6-[6-[(2R)-2-(3-fluorophenyl)pyrrolidin-1-yl]imidazo[1,2-b]pyridazin-3-yl]-2-pyridyl]piperazin-1-ylethyl]azetidin-3-yl]isoindoline-1,3-dione (5 mg, 6% yield). MS (ESI) m/z: 783.0 [M+H].sup.+.

Example 344:2-(2,6-Dioxopiperidin-3-yl)-5-(1-((1-(6-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperidin-4-yl)methyl)azetidin-3-yl)isoindoline-1,3-dione (TR-293)

(1665) ##STR00961##

(1666) TR-293 was synthesized following the standard procedure for preparing TR-286 (8.6 mg, 20% yield). MS (ESI) m/z: 768.7 [M+H].sup.+.

Example 345:2-(2,6-Dioxopiperidin-3-yl)-4-((3-((4-(6-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)methyl)azetidin-1-yl)methyl)isoindoline-1,3-dione (TR-294)

(1667) ##STR00962##

Step 1. Synthesis of tert-butyl 3-(((methylsulfonyl)oxy)methyl)azetidine-1-carboxylate

(1668) ##STR00963##

(1669) To a solution of tert-butyl 3-(hydroxymethyl)azetidine-1-carboxylate (100 mg, 0.53 mmol), and TEA (161 mg, 1.59 mmol) in DCM (2 mL) was added MsCl (121 mg, 1.07 mmol) at 0° C. After the reaction mixture was stirred at 0° C. for 1 h, the mixture was purified by silica gel chromatography (petroleum ether/EtOAc=4:1 to 1:1) to give tert-butyl 3-(((methylsulfonyl)oxy)methyl)azetidine-1-carboxylate (121 mg, 85% yield) as a colorless solid which was used immediately in the next step.

Step 2. Synthesis of tert-butyl (R)-3-((4-(6-(6-(2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)methyl)azetidine-1-carboxylate

(1670) ##STR00964##

(1671) A mixture of (R)-6-(2-(3-fluorophenyl)pyrrolidin-1-yl)-3-(6-(piperazin-1-yl)pyridin-2-yl)imidazo[1,2-b]pyridazine (202 mg, 0.46 mmol), tert-butyl 3-(((methylsulfonyl)oxy)methyl)azetidine-1-carboxylate (121 mg, 0.46 mmol), potassium carbonate (186 mg, 1.38 mmol) and sodium iodide (103 mg, 0.69 mmol) in acetonitrile (5 mL) was stirred at 90° C. overnight under N.sub.2. The mixture was purified by reverse phase column chromatography to give tert-butyl (R)-3-((4-(6-(6-(2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)methyl)azetidine-1-carboxylate (225 mg, 80% yield) as a brown solid. MS (ESI) m/z: 613.7 [M+H].sup.+.

Step 3. Synthesis of (R)-3-(6-(4-(azetidin-3-ylmethyl)piperazin-1-yl)pyridin-2-yl)-6-(2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazine

(1672) ##STR00965##

(1673) A mixture of tert-butyl (R)-3-((4-(6-(6-(2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)methyl)azetidine-1-carboxylate (225 mg, 0.37 mmol) and TFA (0.5 mL) in DCM (1 mL) was stirred at room temperature for 1 h. The reaction was concentrated to give (R)-3-(6-(4-(azetidin-3-ylmethyl)piperazin-1-yl)pyridin-2-yl)-6-(2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazine (188 mg, 100% yield) as a colorless oil. MS (ESI) m/z: 513.5 [M+H].sup.+.

Step 4. Synthesis of 2-(2,6-dioxopiperidin-3-yl)-4-((3-((4-(6-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)methyl)azetidin-1-yl)methyl)isoindoline-1,3-dione

(1674) ##STR00966##

(1675) A mixture of (R)-3-(6-(4-(azetidin-3-ylmethyl)piperazin-1-yl)pyridin-2-yl)-6-(2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazine (45 mg, 0.07 mmol), 4-(bromomethyl)-2-(2,6-dioxopiperidin-3-yl)isoindoline-1,3-dione (25 mg, 0.07 mmol) and potassium carbonate (29 mg, 0.21 mmol) in acetonitrile (1.5 mL) was stirred at room temperature for 2 h. The reaction mixture was purified by reverse phase column chromatography to give 2-(2,6-dioxopiperidin-3-yl)-4-((3-((4-(6-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)methyl)azetidin-1-yl)methyl)isoindoline-1,3-dione (11.7 mg, 21% yield) as a brown solid. MS (ESI) m/z: 783.8 [M+H].sup.+.

Example 346: 3-(5-(3-((4-(6-(6-((R)-2-(3-Fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)methyl)azetidin-1-yl)-2-methyl-4-oxoquinazolin-3 (4H)-yl)piperidine-2,6-dione (TR-295)

(1676) ##STR00967##

(1677) A mixture of (R)-3-(6-(4-(azetidin-3-ylmethyl)piperazin-1-yl)pyridin-2-yl)-6-(2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazine (10.0 mg, 0.02 mmol), 3-(5-fluoro-2-methyl-4-oxoquinazolin-3 (4H)-yl)piperidine-2,6-dione (8.7 mg, 0.03 mmol) and DIEA (13.0 mg, 0.10 mmol) in DMAc (2 mL) was stirred at 90° C. for 16 h. The reaction mixture was purified by reverse-phase chromatography to afford 3-(5-(3-((4-(6-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)methyl)azetidin-1-yl)-2-methyl-4-oxoquinazolin-3 (4H)-yl)piperidine-2,6-dione (7.7 mg, 44% yield) as a light yellow solid. MS (ESI) m/z=782.7 [M+H].sup.+.

Example 347: 3-(5-(3-((4-(6-(6-((R)-2-(3-Fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)methyl)azetidin-1-yl)-4-oxobenzo[d][1,2,3]triazin-3 (4H)-yl)piperidine-2,6-dione (TR-296)

(1678) ##STR00968##

(1679) TR-296 was synthesized following the standard procedure for preparing TR-295 (8.2 mg, 48% yield). MS (ESI) m/z: 769.7 [M+H].sup.+.

Example 348: 3-(5-(3-(2-(4-(6-(6-((R)-2-(3-Fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)methyl)azetidin-1-yl)-4-oxobenzo[d][1,2,3]triazin-3 (4H)-yl)piperidine-2,6-dione (TR-297)

(1680) ##STR00969##

(1681) TR-297 was synthesized following the standard procedure for preparing TR-286 (5 mg, 28% yield). MS (ESI) m/z: 783.7 [M+H].sup.+.

Example 349: N—(S-(3,5-Difluorobenzyl)-1H-indazol-3-yl)-4-(4-(3-(2-(2,6-dioxopiperidin-3-yl)-3-oxoisindolin-5-yl)prop-2-yn-1-yl)piperazin-1-yl)-2-((tetrahydro-2/-pyran-4-yl)amino)benzamide (TR-298)

(1682) ##STR00970##

(1683) To a solution of N-(5-(3,5-difluorobenzyl)-1H-indazol-3-yl)-4-(piperazin-1-yl)-2-((tetrahydro-2H-pyran-4-yl)amino)benzamide (20 mg, 0.04 mmol) and 3-(2-(2,6-dioxopiperidin-3-yl)-3-oxoisindolin-5-yl)prop-2-yn-1-yl 4-methylbenzenesulfonate (18 mg, 0.05 mmol) in MeCN (5 mL) were added NaI (12 mg, 0.08 mmol) and DIPEA (30 mg, 0.20 mmol) at room temperature. After the reaction mixture was stirred at 85° C. for 12 h, the mixture was purified by prep-TLC to give N-(5-(3,5-difluorobenzyl)-1H-indazol-3-yl)-4-(4-(3-(2-(2,6-dioxopiperidin-3-yl)-3-oxoisindolin-5-yl)prop-2-yn-1-yl)piperazin-1-yl)-2-((tetrahydro-2H-pyran-4-yl)amino)benzamide (1.5 mg, 6% yield) as a white solid. MS (ESI) m/z: 827.9 [M+H].sup.+.

Example 350: N-(5-(3,5-Difluorobenzyl)-1H-indazol-3-yl)-4-(4-(3-((2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-5-yl)amino)cyclopentyl)piperazin-1-yl)-2-((tetrahydro-2/-pyran-4-yl)amino)benzamide (TR-299)

(1684) ##STR00971##

(1685) TR-299 was synthesized following the standard procedure for preparing TR-275 (9 mg, 35% yield by 3 steps). This product contains two unseparated isomers. MS (ESI) m/z: 912.4 [M+H].sup.+.

Example 351: N—(S-(3,5-Difluorobenzyl)-1H-indazol-3-yl)-4-(4-((1~ (2~ (2,6-dioxopiperidin-3-yl)-1-oxoisindolin-5-yl)azetidin-3-yl)methyl)piperazin-1-yl)-2-((tetrahydro-2H-pyran-4-yl)amino)benzamide (TR-300)

(1686) ##STR00972##

(1687) To a solution of N-(5-(3,5-difluorobenzyl)-1H-indazol-3-yl)-4-(piperazin-1-yl)-2-((tetrahydro-2H-pyran-4-yl)amino)benzamide (15 mg, 0.03 mmol) and (1-(2-(2,6-dioxopiperidin-3-yl)-1-oxoisindolin-5-yl)azetidin-3-yl)methyl 4-methylbenzenesulfonate (15 mg, 0.03 mmol) in MeCN (5 mL) were added DIPEA (20 mg, 0.15 mmol) and NaI (10 mg, 0.06 mmol) at room temperature. The reaction mixture was stirred at 80° C. for 12 h, before the reaction was concentrated. The resulting residue was purified by prep-TLC to give N—(S-(3,5-difluorobenzyl)-1H-indazol-3-yl)-4-(4-((1-(2-(2,6-dioxopiperidin-3-yl)-1-oxoisindolin-5-yl)azetidin-3-yl)methyl)piperazin-1-yl)-2-((tetrahydro-2H-pyran-4-yl)amino)benzamide (3 mg, 12% yield) as a white solid. MS (ESI) m/z: 858.7 [M+H].sup.+.

Example 352: N—(S-(3,5-Difluorobenzyl)-1H-indazol-3-yl)-4-(4-(1-(1-(2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-5-yl)azetidin-3-yl)ethyl)piperazin-1-yl)-2-((tetrahydro-2H-pyran-4-yl)amino)benzamide (TR-301)

(1688) ##STR00973##

(1689) TR-301 was synthesized following the standard procedure for preparing TR-275 (13 mg, 10% yield by 3 steps). MS (ESI) m/z: 886.8 [M+H].sup.+.

Example 353: 3-(5~ (1-(1-(6~ (6-((R)-2-(3-Fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperidin-4-yl)azetidin-3-yl)-1-oxoisindolin-2-yl)piperidine-2,6-dione (TR-302)

(1690) ##STR00974##

(1691) TR-302 was synthesized following the standard procedure for preparing TR-292 (37 mg, 80% yield). MS (ESI) m/z: 740.7 [M+H].sup.+.

Example 354: N-(5-(3,5-Difluorobenzyl)-1H-indazol-3-yl)-4-(4-(3-(1-(2,6-dioxopiperidin-3-yl)-3-methyl-2-oxo-2,3~dihydro-1/7-benzo[d]imidazol-4-yl)prop-2-yn-1-yl)piperazin-1-yl)-2-((tetrahydro-2/-pyran-4-yl)amino)benzamide (TR-303)

(1692) ##STR00975##

(1693) A mixture of 3-(4-(3-chloroprop-1-yn-1-yl)-3-methyl-2-oxo-2,3-dihydro-1H-benzo[d]imidazol-1-yl)piperidine-2,6-dione (8 mg, 0.026 mmol), N-(5-(3,5-difluorobenzyl)-1H-indazol-3-yl)-4-(piperazin-1-yl)-2-((tetrahydro-2H-pyran-4-yl)amino)benzamide (15 mg, 0.028 mmol), DIEA (10 mg, 0.078 mmol) and sodium iodide (6 mg, 0.039 mmol) in acetonitrile (1.5 mL) was stirred at 80° C. overnight under N.sub.2. The mixture was purified by reverse phase column chromatography to give N-(5-(3,5-difluorobenzyl)-1H-indazol-3-yl)-4-(4-(3~ (1-(2,6-dioxopiperidin-3-yl)-3-methyl-2-oxo-2,3-dihydro-1H-benzo[d]imidazol-4-yl)prop-2-yn-1-yl)piperazin-1-yl)-2-((tetrahydro-2H-pyran-4-yl)amino)benzamide (5.9 mg, 27% yield) as a white solid. MS (ESI) m/z: 842.7 [M+H].sup.+.

Example 355: N-(5-(3,5-Difluorobenzyl)-1H-indazol-3-yl)-4-(4-(4-(1-(2,6-dioxopiperidin-3-yl)-3-methyl-2-oxo-2,3-dihydro-1H-benzo[d]imidazol-4-yl)butyl)piperazin-1-yl)-2-((tetrahydro-2H-pyran-4-yl)amino)benzamide (TR-304)

(1694) ##STR00976##

(1695) TR-304 was synthesized following the standard procedure for preparing TR-303 (39.9 mg, 68% yield). MS (ESI) m/z: 861.0 [M+H].sup.+.

Example 356: N-(5-(3,5~Difluorobenzyl)-1H-indazol-3-yl)-4-(4-(3-(1-(2,6-dioxopiperidin-3-yl)-3-methyl-2-oxo-2,3-dihydro-1H-benzo[d]imidazol-4-yl)propyl)piperazin-1-yl)-2-((tetrahydro-2H-pyran-4-yl)amino)benzamide (TR-305)

(1696) ##STR00977##

(1697) TR-305 was synthesized following the standard procedure for preparing TR-303 (9.3 mg, 36% yield). MS (ESI) m/z: 846.9 [M+H].sup.+.

Example 357: N-(5-(3,5-Difluorobenzyl)-1H-indazol-3-yl)-4-(4-(4-(1-(2,6-dioxopiperidin-3-yl)-3-methyl-2-oxo-2,3-dihydro-1H-benzo[d]imidazol-4-yl)but-3-yn-1-yl)piperazin-1-yl)-2-((tetrahydro-2H-pyran-4-yl)amino)benzamide (TR-306)

(1698) ##STR00978##

(1699) TR-306 was synthesized following the standard procedure for preparing TR-303 (5.6 mg, 15% yield). MS (ESI) m/z: 856.7 [M+H].sup.+.

Example 358: 3-(5~ (3-(2-(4-(6-(6-((R)-2-(3-Fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)ethyl)azetidin-1-yl)-2-methyl-4-oxoquinazolin-3 (4H)-yl)piperidine-2,6-dione (TR-307)

(1700) ##STR00979##

(1701) TR-307 was synthesized following the standard procedure for preparing TR-295 (2.4 mg, 4% yield). MS (ESI) m/z: 796.9 [M+H].sup.+.

Example 359: 2~ (2,6-Dioxopiperidin-3-yl)-5-(1-(1-(6-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperidin-4-yl]azetidin-3-yl)isoindoline-1,3-dione (TR-308)

(1702) ##STR00980##

Step 1. Synthesis of (R)-8-(6-(6-(2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)-1,4-dioxo-8-azaspiro[4.5]decane

(1703) ##STR00981##

(1704) A mixture of (R)-6-(2-(3-fluorophenyl)pyrrolidin-1-yl)-3-(6-fluoropyridin-2-yl)imidazo[1,2-b]pyridazine (500 mg, 1.35 mmol), 1,4-dioxo-8-azaspiro[4.5]decane (379.8 mg, 2.65 mmol) and DIEA (1 ml) in NMP (2.5 mL) was heated at 140° C. for 48 h. After addition of water, the reaction mixture was extracted with EtOAc. The organic layer was washed with brine and dried over MgSO.sub.4, filtered and concentrated. The resulting residue was purified by silica gel chromatography to afford (R)-8-(6-(6-(2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)-1,4-dioxo-8-azaspiro[4.5]decane (410 mg, 61% yield) as a white solid. MS (ESI) m/z: 501.7 [M+H].sup.+.

Step 2. Synthesis of (R)-1-(6-(6-(2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperidin-4-one

(1705) ##STR00982##

(1706) To a solution of (R)-8-(6-(6-(2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)-1,4-dioxo-8-azaspiro[4.5]decane (400 mg, 0.79 mmol) in 1,4-dioxane (4 mL) were added H₂O (0.5 mL) and H₂SO₄ (con. 1.0 mL) dropwise. After the mixture was stirred at rt for 2 h, the reaction was quenched with NaHCO₃ aqueous solution. The mixture was extracted with EtOAc. The organic layer was washed with brine and dried over MgSO₄, filtered and concentrated. The residue was purified by silica gel chromatography to afford (R)-1-(6-(6-(2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperidin-4-one (340 mg, 94% yield) as a white solid. MS (ESI) m/z: 457.6 [M+H]⁺.

Step 3. Synthesis of 2-(2,6-dioxopiperidin-3-yl)-5-(1-(1-(6-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperidin-4-yl)azetidin-3-yl)isoindoline-1,3-dione (1707) ##STR00983##

(1708) To a solution of 5-(azetidin-3-yl)-2-(2,6-dioxopiperidin-3-yl)isoindoline-1,3-dione (30 mg, 0.096 mmol) and (R)-1-(6-(6-(2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperidin-4-one (43.7 mg, 0.096 mmol) in CH₃OH (3 mL) was added NaBH₃CN (11.94 mg, 0.192 mmol). After the reaction was stirred at rt for 6 h, the mixture was concentrated. The resulting residue was purified by prep-TLC to provide 2-(2,6-dioxopiperidin-3-yl)-5-(1-(1-(6-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperidin-4-yl)azetidin-3-yl)isoindoline-1,3-dione (20 mg, 28% yield) as a white solid. MS (ESI) m/z: 754.6 [M+H]⁺.

Example 360: 3-(6-(1-(1-(6-(6-((R)-2-(3-Fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperidin-4-yl)azetidin-3-yl)-1-oxoisoindolin-2-yl)piperidine-2,6-dione (TR-309) (1709) ##STR00984##

(1710) TR-309 was synthesized following the standard procedure for preparing TR-292 (50 mg, 52% yield). MS (ESI) m/z: 740.7 [M+H]⁺.

Example 361: N—(S-(3,5-Difluorobenzyl)-1H-indazol-3-yl)-4-(4-(2-(3-(2,6-dioxopiperidin-3-yl)-2-oxo-2,3-dihydro-1H-benzo[d]imidazol-1-yl)ethyl)piperazin-1-yl)-2-((tetrahydro-2H-pyran-4-yl)amino)benzamide (TR-310)

(1711) ##STR00985##

(1712) TR-310 was synthesized following the standard procedure for preparing TR-303 (28 mg, 30% yield). MS (ESI) m/z: 818.7 [M+H]⁺.

Example 362: N—(S-(3,5-Difluorobenzyl)-1H-indazol-3-yl)-4-(4-(3-(3-(2,6-dioxopiperidin-3-yl)-2-oxo-2,3-dihydro-1H-benzo[d]imidazol-1-yl)propyl)piperazin-1-yl)-2-((tetrahydro-2H-pyran-4-yl)amino)benzamide (TR-311)

(1713) ##STR00986##

(1714) TR-311 was synthesized following the standard procedure for preparing TR-303 (22.4 mg, 30% yield). MS (ESI) m/z: 832.9 [M+H]⁺.

Example 363: N-(5-(3,5-Difluorobenzyl)-1H-indazol-3-yl)-4-(4-(3-((1-(2,6-dioxopiperidin-3-yl)-3-methyl-2-oxo-2,3-dihydro-1H-benzo[d]imidazol-4-yl)oxy)propyl)piperazin-1-yl)-2-((tetrahydro-2H-pyran-4-yl)amino)benzamide (TR-312)

(1715) ##STR00987##

(1716) A mixture of 3-(4-hydroxy-3-methyl-2-oxo-2,3-dihydro-1H-benzo[d]imidazol-1-yl)piperidine-2,6-dione (5.0 mg, 0.018 mmol), 4-(4-(3-chloropropyl)piperazin-1-yl)-N-(5-(3,5-difluorobenzyl)-1H-indazol-3-yl)-2-((tetrahydro-2H-pyran-4-yl)amino)benzamide (11.3 mg, 0.018 mmol), sodium iodide (4.05 mg, 0.027 mmol) and K₂CO₃ (9.94 mg, 0.072 mmol) in DMSO (1 mL) was heated at 60° C. for 4 h. Then the mixture was partitioned between water (10 mL) and EtOAc (10 mL). The organic phase was dried over anhydrous sodium sulfate, and concentrated. The residue was purified by prep-TLC to give desired product (5.3 mg, 34% yield) as a white solid. MS (ESI) m/z: 862.9 [M+H]⁺.

Example 364: N-(5-(3,5-Difluorobenzyl)-1H-indazol-3-yl)-4-(4-(3-((1-(2,6-dioxopiperidin-3-yl)-3-

methyl-2-oxo-2,3-dihydro-1H-benzo[d]imidazol-5-yl)oxy)propyl)piperazin-1-yl)-2-((tetrahydro-2H-pyran-4-yl)amino)benzamide (TR-313)

(1717) ##STR00988##

(1718) TR-313 was synthesized following the standard procedure for preparing TR-312(10 mg, 41% yield). MS (ESI) m/z: 862.9 [M+H].sup.+.

Example 365: N-(5-(3,5-Difluorobenzyl)-1H-indazol-3-yl)-4-(4-(2-((1-(2,6-dioxopiperidin-3-yl)-3-methyl-2-oxo-2,3-dihydro-1H-benzo[d]imidazol-5-yl)oxy)ethyl)piperazin-1-yl)-2-((tetrahydro-2H-pyran-4-yl)amino)benzamide (TR-314)

(1719) ##STR00989##

(1720) TR-314 was synthesized following the standard procedure for preparing TR-312(8.2 mg, 33% yield). MS (ESI) m/z: 848.8 [M+H].sup.+.

Example 366: N-(5-(3,5-Difluorobenzyl)-1H-indazol-3-yl)-4-(4-(3-((1-(2,6-dioxopiperidin-3-yl)-3-methyl-2-oxo-2,3-dihydro-1H-benzo[d]imidazol-4-yl)amino)propyl)piperazin-1-yl)-2-((tetrahydro-2H-pyran-4-yl)amino)benzamide (TR-315)

(1721) ##STR00990##

Step 1. Synthesis of 3-(4-((3-chloropropyl)amino)-3-methyl-2-oxo-2,3-dihydro-1H-benzo[d]imidazol-1-yl)piperidine-2,6-dione

(1722) ##STR00991##

(1723) A mixture of 3-(4-amino-3-methyl-2-oxo-2,3-dihydro-1H-benzo[d]imidazol-1-yl)piperidine-2,6-dione (80.0 mg, 0.29 mmol), 1-6chloro-3-iodopropane (77.1 mg, 0.38 mmol) and DIEA (112.5 mg, 0.87 mmol) in DMSO (2 mL) was stirred at 40° C. overnight. The resulting mixture was partitioned between water (5 mL) and EtOAc (5 mL). The organic phase was dried over anhydrous sodium sulfate, filtered and concentrated. The resulting residue was purified by silica gel column chromatography (DCM/MeOH=10:1) to afford 3-(4-((3-chloropropyl)amino)-3-methyl-2-oxo-2,3-dihydro-1H-benzo[d]imidazol-1-yl)piperidine-2,6-dione (32.2 mg, 30% yield) as a white solid. MS (ESI) m/z: 351.2 [M+H].sup.+.

Step 2. Synthesis of N-(5-(3,5-difluorobenzyl)-1H-indazol-3-yl)-4-(4-(3-((1-(2,6-dioxopiperidin-3-yl)-3-methyl-2-oxo-2,3-dihydro-1H-benzo[d]imidazol-4-yl)amino)propyl)piperazin-1-yl)-2-((tetrahydro-2H-pyran-4-yl)amino)benzamide

(1724) ##STR00992##

(1725) A mixture of 3-(4-((3-chloropropyl)amino)-3-methyl-2-oxo-2,3-dihydro-1H-benzo[d]imidazol-1-yl)piperidine-2,6-dione (32.2 mg, 0.09 mmol), N-(5-(3,5-difluorobenzyl)-1H-indazol-3-yl)-4-(piperazin-1-yl)-2~((tetrahydro-2H-pyran-4-yl)amino)benzamide (49.8 mg, 0.09 mmol), K.sub.2CO.sub.3 (37.9 mg, 0.27 mmol) and NaI (13.6 mg, 0.09 mmol) in DMSO (1 mL) was stirred at 80° C. for 1 h. The resulting mixture was partitioned between water (5 mL) and EtOAc (5 mL). The organic phase was dried over anhydrous sodium sulfate, filtered and concentrated. The resulting residue was purified by prep-TLC (DCM/MeOH=12:1) to afford N-(5-(3,5-difluorobenzyl)-1H-indazol-3-yl)-4-(4-(3-((1-(2,6-dioxopiperidin-3-yl)-3-methyl-2-oxo-2,3-dihydro-1H-benzo[d]imidazol-4-yl)amino)propyl)piperazin-1-yl)-2-((tetrahydro-2H-pyran-4-yl)amino)benzamide as a white solid (5.4 mg, 7% yield). MS (ESI) m/z: 861.7 [M+H].sup.+.

Example 367: N-(5-(3,5-Difluorobenzyl)-1H-indazol-3-yl)-4-(4-(3-((1-(2,6-dioxopiperidin-3-yl)-3-methyl-2-oxo-2,3-dihydro-1H-benzo[d]imidazol-5-yl)amino)propyl)piperazin-1-yl)-2-((tetrahydro-2H-pyran-4-yl)amino)benzamide (TR-316)

(1726) ##STR00993##

(1727) TR-316 was synthesized following the standard procedure for preparing TR-315 (4.7 mg, 10% yield). MS (ESI) m/z: 861.7 [M+H].sup.+.

Example 368: 3-(6-(3-((4-(6-(6-((R)-2-(3-Fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)methyl)azetidin-1-yl)-1-oxoisindolin-2-yl)piperidine-2,6-dione (TR-317)

(1728) ##STR00994## ##STR00995##

Step 1 and step 2. Synthesis of (2-(2,6-dioxopiperidin-3-yl)-3-oxoisindolin-5-yl)boronic acid (1729) ##STR00996##

(1730) To a solution of 3-(6-bromo-1-oxo-isindolin-2-yl)piperidine-2,6-dione (3.00 g, 9.28 mmol) and 4,4,5,5-tetramethyl-2-(4,4,5,5-tetramethyl-1,3,2-dioxaborolan-2-yl)-1,3,2-dioxaborolane (2.83 g, 11.14 mmol) in DMF (80 mL) was added Pd(dppf) Cl.sub.2 (678.65 mg, 928.39 μ mol) and CH.sub.3CO.sub.2K (2.73 g, 27.85 mmol). The reaction mixture was stirred at 100° C. for 16 h, before it was quenched with water (100 mL). After addition of NaIO.sub.4 (5.93 g, 27.85 mmol), the pH of the reaction was adjusted to 2 by IN HCL. The mixture was purified by prep-HPLC to give [2-(2,6-dioxo-3-piperidyl)-3-oxo-isindolin-5-yl]boronic acid (2.2 g, 82% yield) as a white solid. MS (ESI) m/z: 288.9 [M+H].sup.+.

Step 3. Synthesis of 3-(6-(3-((benzyloxy)methyl)azetidin-1-yl)-1-oxoisindolin-2-yl)piperidine-2,6-dione

(1731) ##STR00997##

(1732) To a solution of [2-(2,6-dioxo-3-piperidyl)-3-oxo-isindolin-5-yl]boronic acid (949 mg, 3.29 mmol) and 3-(benzyloxymethyl)azetidine (700.69 mg, 3.95 mmol) in DMF (50 mL) were added Cu (OAc).sub.2 (1.19 g, 6.59 mmol), TEA (2.00 g, 19.77 mmol) and 4A molecular sieves (2 g). The reaction mixture was stirred at 40° C. for 16 h, before it was purified by prep-HPLC to give a crude product, which was further purified by prep-TLC (DCM/MeOH=20:1) to give 3-[6-[3-(benzyloxymethyl)azetidin-1-yl]-1-oxo-isindolin-2-yl]piperidine-2,6-dione (400 mg, 29% yield). MS (ESI) m/z: 420.5 [M+H].sup.+.

Step 4. Synthesis of 3-(6-(3-(hydroxymethyl)azetidin-1-yl)-1-oxoisindolin-2-yl)piperidine-2,6-dione

(1733) ##STR00998##

(1734) A mixture of 3-(6-(3-((benzyloxy)methyl)azetidin-1-yl)-1-oxoisindolin-2-yl)piperidine-2,6-dione (200 mg, 0.48 mmol) and MsOH (2.0 mL) in DCM (4.0 mL) was stirred at room temperature for 2 h. The reaction was concentrated and the resulting residue was purified by reverse-phase chromatography (CH.sub.3CN/H.sub.2O=20:1) to give 3-(6-(3-(hydroxymethyl)azetidin-1-yl)-1-oxoisindolin-2-yl)piperidine-2,6-dione (127 mg, 81% yield) as a white solid. MS (ESI) m/z: 330.3 [M+H].sup.+.

Step 5. Synthesis of (1-(2-(2,6-dioxopiperidin-3-yl)-3-oxoisindolin-5-yl)azetidin-3-yl)methyl 4-methylbenzenesulfonate

(1735) ##STR00999##

(1736) To a solution of 3-(6-(3-(hydroxymethyl)azetidin-1-yl)-1-oxoisindolin-2-yl)piperidine-2,6-dione (60 mg, 0.18 mmol), TEA (2 mL), and DMAP (44 mg, 0.36 mmol) in DCM (2 mL) was added TsCl (69 mg, 0.36 mmol) at 0° C. The reaction mixture was stirred at room temperature overnight, before DCM was removed under vacuum. The resulting residue was purified by prep-TLC (DCM/MeOH=16:1) to give (1-(2-(2,6-dioxopiperidin-3-yl)-3-oxoisindolin-5-yl)azetidin-3-yl)methyl 4-methylbenzenesulfonate (69 mg, 86% yield) as a white solid. MS (ESI) m/z: 484.6 [M+H].sup.+.

Step 6. Synthesis of 3-(6-(3-((4-(6-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)methyl)azetidin-1-yl)-1-oxoisindolin-2-yl)piperidine-2,6-dione

(1737) ##STR01000##

(1738) A mixture of (1-(2-(2,6-dioxopiperidin-3-yl)-3-oxoisindolin-5-yl)azetidin-3-yl)methyl 4-methylbenzenesulfonate (20 mg, 0.04 mmol), (R)-6-(2-(3-fluorophenyl)pyrrolidin-1-yl)-3-(6-(piperazin-1-yl)pyridin-2-yl)imidazo[1,2-b]pyridazine (20 mg, 0.045 mmol), K.sub.2CO.sub.3 (17 mg, 0.12 mmol) and sodium iodide (9 mg, 0.06 mmol) in acetonitrile (2.0 mL) was stirred at 80° C. overnight under N.sub.2. Then the mixture was purified by reverse-phase chromatography (CH.sub.3CN/H.sub.2O=20:1), followed by prep-TLC (DCM/MeOH=8:1) to give 3-(6-(3-((4-(6-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-

b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)methyl)azetidin-1-yl)-1-oxoisindolin-2-yl)piperidine-2,6-dione (19 mg, 79% yield) as a white solid. MS (ESI) m/z: 755.7 [M+H].sup.+.

Example 369: N-(5-(3,5-Difluorobenzyl)-1H-indazol-3-yl)-4-(4-((1~ (2-(2,6-dioxopiperidin-3-yl)-3-oxoisindolin-5-yl)azetidin-3-yl)methyl)piperazin-1-yl)-2-((tetrahydro-2H-pyran-4-yl)amino)benzamide (TR-318)

(1739) ##STR01001##

(1740) TR-318 was synthesized following the standard procedure for preparing TR-317 (11 mg, 31% yield). MS (ESI) m/z: 858.9 [M+H].sup.+.

Example 370: N-(5-(3,5-Difluorobenzyl)-1H-indazol-3-yl)-4-(4-(4-(1-(2,6-dioxopiperidin-3-yl)-3-methyl-2-oxo-2,3-dihydro-1H-benzo[d]imidazol-5-yl)but-3-yn-1-yl)piperazin-1~yl)-2-((tetrahydro-2H-pyran-4-yl)amino)benzamide (TR-319)

(1741) ##STR01002##

(1742) TR-319 was synthesized following the standard procedure for preparing TR-303 (10 mg, 12% yield). MS (ESI) m/z: 856.6 [M+H].sup.+.

Example 371: N-(5-(3,5-Difluorobenzyl)-1H-indazol-3-yl)-4-(4-(4-(1-(2,6-dioxopiperidin-3-yl)-3-methyl-2-oxo-2,3-dihydro-1H-benzo[d]imidazol-5-yl)butyl)piperazin-1-yl)-2-((tetrahydro-2/-pyran-4-yl)amino)benzamide (TR-320)

(1743) ##STR01003##

(1744) TR-320 was synthesized following the standard procedure for preparing TR-303 (15 mg, 29% yield). MS (ESI) m/z: 860.8 [M+H].sup.+.

Example 372: N-(5-(3,5-Difluorobenzyl)-1H-indazol-3-yl)-4-(4-((1-(1-(2,6-dioxopiperidin-3-yl)-3-methyl-2-oxo-2,3~dihydro-1H-benzo[d]imidazol-5-yl)azetidin-3-yl)methyl)piperazin-1-yl)-2-((tetrahydro-2/-pyran-4-yl)amino)benzamide (TR-321)

(1745) ##STR01004##

(1746) TR-321 was synthesized following the standard procedure for preparing TR-303 (8 mg, 20% yield). MS (ESI) m/z: 873.7 [M+H].sup.+.

Example 373: N-(5-(3,5~Difluorobenzyl)-1H-indazol-3-yl)-4-(4-(2-((1-(2,6-dioxopiperidin-3-yl)-3-methyl-2-oxo-2,3-dihydro-1H-benzo[d]imidazol-4-yl)oxy)ethyl)piperazin-1-yl)-2-((tetrahydro-2H-pyran~4-yl)amino)benzamide (TR-322)

(1747) ##STR01005##

(1748) TR-322 was synthesized following the standard procedure for preparing TR-312(15 mg, 43% yield). MS (ESI) m/z: 848.7 [M+H].sup.+.

Example 374: N-(5-(3,5-Difluorobenzyl)-1H-indazol-3-yl)-4-(4-((2-(2,6-dioxopiperidin-3-yl)-1-oxoisindolin-5-yl)ethynyl)piperidin-1-yl)-2-((tetrahydro-2H-pyran-4-yl)amino)benzamide (TR-323)

(1749) ##STR01006## ##STR01007##

Step 1. Synthesis of tert-butyl 4-fluoro-2-nitrobenzoate

(1750) ##STR01008##

(1751) A solution of 4-fluoro-2-nitro-benzoic acid (10 g, 54.1 mmol), di-tert-butyl-dicarbonate (23.5 g, 108.2 mmol) and 4-dimethylaminopyridine (1.6 g, 13.5 mmol) in tert-butanol (100 mL) and dichloromethane (100 mL) was stirred at room temperature for about 20 h. The reaction mixture was then diluted with ethyl acetate (500 mL), washed with 1N HCl (500 mL), water (500 mL) and brine (500 mL). After drying over sodium sulfate, the organic layer was filtered and concentrated to afford tert-butyl 4-fluoro-2-nitrobenzoate (13 g crude, quantity) as yellow oil, which was used in the next step without further purification. MS (ESI) m/z=264.1 [M+Na].sup.+.

Step 2. Synthesis of tert-butyl 4-(4-ethynylpiperidin-1-yl)-2-nitrobenzoate

(1752) ##STR01009##

(1753) To a solution of tert-butyl 4-fluoro-2-nitrobenzoate (1 g, 4. 1 mmol) in DMF (20 mL) was added 4-ethynylpiperidine (675 mg, 6.2 mmol) and TEA (1.2 g, 12.3 mmol). The mixture was stirred at 70° C. for 16 h, before being poured into water and extracted with ethyl acetate (3×100

mL). The combined organic layers were washed with water and brine, dried over Na.sub.2SO.sub.4, filtered and concentrated to give tert-butyl 4-(4-ethynylpiperidin-1-yl)-2-nitrobenzoate (600 mg, 46% yield) as a yellow oil, which was used in the next step without further purification. MS (ESI) m/z=331.4 [M+H].sup.+.

Step 3. Synthesis of tert-butyl 2-amino-4-(4-ethynylpiperidin-1-yl)benzoate

(1754) ##STR01010##

(1755) To a solution of tert-butyl 4-(4-ethynylpiperidin-1-yl)-2-nitrobenzoate (650 mg, 1.9 mmol) in EtOH (10 mL) and AcOH (1 mL) was added Fe (1.1 g, 19 mmol). The reaction mixture was stirred at 90° C. for 16 h. The mixture was concentrated, diluted with water, and extracted with ethyl acetate. The organic layer was washed with NaHCO.sub.3 aqueous solution and brine, dried over Na.sub.2SO.sub.4, filtered and concentrated. The resulting residue was purified by flash chromatography to give tert-butyl 2-amino-4-(4-ethynylpiperidin-1-yl)benzoate (550 mg, 96% yield) as a yellow oil. MS (ESI) m/z=301.5 [M+H].sup.+.

Step 4. Synthesis of tert-butyl 4-(4-ethynylpiperidin-1-yl)-2-((tetrahydro-2H-pyran-4-yl)amino)benzoate

(1756) ##STR01011##

(1757) To a solution of tert-butyl 2-amino-4-(4-ethynylpiperidin-1-yl)benzoate (100 mg, 0.18 mmol) and tert-butyl 3-formylazetidine-1-carboxylate (605 mg, 2.01 mmol) in DCM (10 mL) and AcOH (1 mL) was added BH.sub.3 (2.1 mL, 2.01 mmol, 1M in THF) in portions at 0° C. under N.sub.2. The resulting mixture was stirred at room temperature for 2 h, before being quenched with H.sub.2O (100 mL) at 0° C. and extracted with DCM (2×100 mL). The organic layers were combined, dried over Na.sub.2SO.sub.4, filtered, and concentrated. The resulting residue was purified by silica gel flash chromatography to give tert-butyl 4-(4-ethynylpiperidin-1-yl)-2-((tetrahydro-2H-pyran-4-yl)amino)benzoate (520 mg, 65% yield) as a yellow oil. MS (ESI) m/z: 385.7 [M+H].sup.+.

Step 5. Synthesis of tert-butyl 4-(4-ethynylpiperidin-1-yl)-2-(2,2,2-trifluoro-N-(tetrahydro-2H-pyran-4-yl)acetamido)benzoate

(1758) ##STR01012##

(1759) To a solution of tert-butyl 4-(4-ethynylpiperidin-1-yl)-2-((tetrahydro-2H-pyran-4-yl)amino)benzoate (470 mg, 1.2 mmol) in DCM (10 mL) were added TEA (607 mg, 6.0 mmol) and TFAA (308 mg, 1.4 mmol) at 0° C. After the mixture was stirred at room temperature for 1 h, the reaction was washed with water and brine, dried over Na.sub.2SO.sub.4, filtered and concentrated. The resulting residue was purified by flash chromatography to give tert-butyl 4-(4-ethynylpiperidin-1-yl)-2-(2,2,2-trifluoro-N-(tetrahydro-2H-pyran-4-yl)acetamido)benzoate (300 mg, 52% yield) as a yellow oil. MS (ESI) m/z=481.5 [M+H].sup.+.

Step 6. Synthesis of 4-(4-ethynylpiperidin-1-yl)-2-(2,2,2-trifluoro-N-(tetrahydro-2H-pyran-4-yl)acetamido)benzoic acid

(1760) ##STR01013##

(1761) To a solution of tert-butyl 4-(4-ethynylpiperidin-1-yl)-2-(2,2,2-trifluoro-N-(tetrahydro-2H-pyran-4-yl)acetamido)benzoate (300 mg, 0.62 mmol) in DCM (2 mL) was added TFA (1 mL) at 0° C. After the reaction mixture was stirred at room temperature for 16 h, the reaction was concentrated under vacuum. The resulting residue was purified by reverse-phase chromatography to give t 4-(4-ethynylpiperidin-1-yl)-2-(2,2,2-trifluoro-N-(tetrahydro-2H-pyran-4-yl)acetamido)benzoic acid (245 mg, 93% yield) as a white solid. MS (ESI) m/z: 425.5 [M+H].sup.+.

Step 7. Synthesis of N-(5-(3,5-difluorobenzyl)-1H-indazol-3-yl)-4-(4-ethynylpiperidin-1-yl)-2-(2,2,2-trifluoro-N-(tetrahydro-2H-pyran-4-yl)acetamido)benzamide

(1762) ##STR01014##

(1763) To a suspension of 4-(4-ethynylpiperidin-1-yl)-2-(2,2,2-trifluoro-N-(tetrahydro-2H-pyran-4-yl)acetamido)benzoic acid (100 mg, 0.23 mmol) in dry dichloromethane (5 mL) were added

catalytic amount of DMF, and oxalyl chloride (45 mg, 0.35 mmol) at 0° C. The mixture was stirred for about 1.5 h. The reaction was concentrated, and the residue was azeotroped with dry dichloromethane twice. The resulting residue was diluted with THF (50 mL). And the resulting suspension was added slowly and gradually to a solution of 5-(3,5-difluorobenzyl)-1H-indazol-3-ylamine (72 mg, 0.27 mmol) and pyridine (181 mg, 2.3 mmol) in dry THF (10 mL) at -35° C. After the mixture was stirred at -35° C. for 1 h, the reaction was washed with 1N HCl solution and brine, dried over Na.sub.2SO.sub.4, filtered and concentrated to give N-(5-(3,5-difluorobenzyl)-1H-indazol-3-yl)-4-(4-ethynylpiperidin-1-yl)-2-(2,2,2-trifluoro-N-(tetrahydro-2H-pyran-4-yl)acetamido)benzamide (180 mg, crude) as a gray solid, which was used in the next step without further purification. MS (ESI) m/z=666.8 [M+H].sup.+.

Step 8. Synthesis of N—(S-(3,5-difluorobenzyl)-1H-indazol-3-yl)-4-(4-ethynylpiperidin-1-yl)-2-((tetrahydro-2H-pyran-4-yl)amino)benzamide
(1764) ##STR01015##

(1765) To a solution of N-(5-(3,5-difluorobenzyl)-1H-indazol-3-yl)-4-(4-ethynylpiperidin-1-yl)-2-(2,2,2-trifluoro-N-(tetrahydro-2H-pyran-4-yl)acetamido)benzamide (180 mg, 0.27 mmol) in MeOH (5 mL) was added K.sub.2CO.sub.3 (75 mg, 0.54 mmol) at 0° C. The reaction was stirred at 50° C. for 1 h, before being concentrated. The resulting residue was purified by prep-TLC to give N-(5-(3,5-difluorobenzyl)-1H-indazol-3-yl)-4-(4-ethynylpiperidin-1-yl)-2-((tetrahydro-2H-pyran-4-yl)amino)benzamide (105 mg, 80% yield by two steps) as a gray solid. MS (ESI) m/z: 570.5 [M+H].sup.+.

Step 9. Synthesis of N—(S-(3,5-difluorobenzyl)-1H-indazol-3-yl)-4-(4-((2-(2,6-dioxopiperidin-3-yl)-1-oxoisindolin-5-yl)ethynyl)piperidin-1-yl)-2-((tetrahydro-2H-pyran-4-yl)amino)benzamide
(1766) ##STR01016##

(1767) To a solution of N-(5-(3,5-difluorobenzyl)-1H-indazol-3-yl)-4-(4-ethynylpiperidin-1-yl)-2-((tetrahydro-2H-pyran-4-yl)amino)benzamide (50 mg, 0.08 mmol) and 3-(5-bromo-1-oxoisindolin-2-yl)piperidine-2,6-dione (34 mg, 0.11 mmol) in DMSO (15 mL) were added TEA (40 mg, 0.4 mmol), PPh.sub.3 (24 mg, 0.08 mmol), CuI (7.6 mg, 0.05 mmol) and Pd(PPh.sub.3).sub.2Cl.sub.2 (28 mg, 0.05 mmol) at room temperature under N.sub.2. The reaction mixture was irradiated at 100° C. for 1 h under microwave. The resulting residue was purified by reverse-phase chromatography, followed by prep-TLC to give N-(5-(3,5-difluorobenzyl)-1H-indazol-3-yl)-4-(4-((2-(2,6-dioxopiperidin-3-yl)-1-oxoisindolin-5-yl)ethynyl)piperidin-1-yl)-2-((tetrahydro-2H-pyran-4-yl)amino)benzamide (13 mg, 20% yield) as a white solid. MS (ESI) m/z: 812.7 [M+H].sup.+.

Example 375: N-(5-(3,5-Difluorobenzyl)-1H-indazol-3-yl)-4-(4-(2-((1-(2,6-dioxopiperidin-3-yl)-3-methyl-2-oxo-2,3-dihydro-1H-benzo[d]imidazol-5-yl)amino)ethyl)piperazin-1-yl)-2-((tetrahydro-2H-pyran-4-yl)amino)benzamide (TR-324)

(1768) ##STR01017##

(1769) TR-324 was synthesized following the standard procedure for preparing TR-315 (3.2 mg, 8% yield). MS (ESI) m/z: 847.4 [M+H].sup.+.

Example 376: N-(5-(3,5-Difluorobenzyl)-1H-indazol-3-yl)-4-(4-(2-((1-(2,6-dioxopiperidin-3-yl)-3-methyl-2-oxo-2,3-dihydro-1H-benzo[d]imidazol-4-yl)amino)ethyl)piperazin-1-yl)-2-((tetrahydro-2H-pyran-4-yl)amino)benzamide (TR-325)

(1770) ##STR01018##

(1771) TR-325 was synthesized following the standard procedure for preparing TR-315 (2.2 mg, 6% yield). MS (ESI) m/z: 847.4 [M+H].sup.+.

Example 377: N-(5-(3,5-Difluorobenzyl)-1H-indazol-3-yl)-4-(4-((1~(3-(2,6-dioxopiperidin-3-yl)-4-oxo-3,4-dihydrobenzo[d][1,2,3]triazin-5-yl)azetidin-3-yl)methyl)piperazin-1-yl)-2-((tetrahydro-2H-pyran-4-yl)amino)benzamide (TR-326)

(1772) ##STR01019##

(1773) TR-326 was synthesized following the standard procedure for preparing TR-297 (3.2 mg,

18% yield). MS (ESI) m/z: 872.7 [M+H].sup.+.

Example 378: N-(5-(3,5-Difluorobenzyl)-1H-indazol-3-yl)-4-(4-(2-(1-(3-(2,6-dioxopiperidin-3-yl)-4-oxo-3,4-dihydrobenzo[d][1,2,3]triazin-5-yl)azetidin-3-yl)ethyl)piperazin-1-yl)-2-((tetrahydro-2/-pyran-4-yl)amino)benzamide (TR-327)

(1774) ##STR01020##

(1775) TR-327 was synthesized following the standard procedure for preparing TR-295 (2.3 mg, 8% yield). MS (ESI) m/z: 886.8 [M+H].sup.+.

Example 379: N-(5-(3,5-Difluorobenzyl)-1H-indazol-3-yl)-4-(4-((3~ (2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-5-yl)azetidin-1-yl)methyl)piperidin-1-yl)-2-((tetrahydro-2/7-pyran-4-yl)amino)benzamide (TR-328)

(1776) ##STR01021## ##STR01022##

Step 1. Synthesis of tert-butyl 4-((benzyloxy)methyl)piperidine-1-carboxylate

(1777) ##STR01023##

(1778) To a solution of tert-butyl 4-(hydroxymethyl)piperidine-1-carboxylate (4.8 g, 0.022 mol) in DMF (60 mL) was added sodium hydride (60% dispersion in mineral oil, 1.32 g, 0.033 mol) slowly at 0° C. After the reaction was stirred at 0° C. for 2 h, a solution of benzyl bromide (4.42 g, 0.026 mol) in DMF (50 mL) was added slowly over 15 minutes. After the resulting reaction was stirred at rt for 2 h, the reaction mixture was poured into water (500 mL), and extracted with EtOAc (3×200 mL). The combined organic layers were derived over Na.sub.2SO.sub.4, filtered and concentrated. The resulting residue was purified by silica gel column chromatography to give the desired product (6.4 g, 95% yield) as light yellow solid. MS (ESI) m/z: 306.2 [M+H].sup.+.

Step 2. Synthesis of 4-((benzyloxy)methyl)piperidine

(1779) ##STR01024##

(1780) To a solution of tert-butyl 4-((benzyloxy)methyl)piperidine-1-carboxylate (6.4 g, 0.021 mol) in DCM (30 mL) was added TFA (20 mL) at room temperature. After the reaction was stirred at room temperature for 2 h, the mixture was concentrated. The resulting residue was purified by reverse-phase chromatography to give the desired product (4.1 g, 95% yield) as light yellow solid. MS (ESI) m/z: 206.1 [M+H].sup.+.

Step 3. Synthesis of tert-butyl 4-(4-((benzyloxy)methyl)piperidin-1-yl)-2-nitrobenzoate

(1781) ##STR01025##

(1782) To a solution of 4-((benzyloxy)methyl)piperidine (3.0 g, 0.015 mol) and tert-butyl 4-fluoro-2-nitrobenzoate (3.5 g, 0.015 mol) in DMF (50 mL) was added DIPEA (5.8 g, 0.045 mol) at room temperature. The reaction was stirred at 70° C. for 3 h. The reaction was cooled to room temperature before H.sub.2O (500 mL) was added. The mixture was extracted with EtOAc (3×200 mL). The combined organic layers were concentrated and purified by silica gel column chromatography to give the desired product (4.8 g, 75% yield) as light yellow solid. MS (ESI) m/z: 427.2 [M+H].sup.+.

Step 4. Synthesis of tert-butyl 2-amino-4-(4-((benzyloxy)methyl)piperidin-1-yl)benzoate

(1783) ##STR01026##

(1784) To a solution of tert-butyl 4~ (4-((benzyloxy)methyl)piperidin-1-yl)-2-nitrobenzoate (1.2 g, 2.82 mmol) in EtOH (15 mL) was added palladium on carbon (10%)(50% wet with water, 150 mg) at room temperature. After the reaction was stirred at room temperature for 16 h, the mixture was concentrated. The resulting residue was purified by silica gel column chromatography to give the desired product (0.99 g, 89% yield) as light yellow solid. MS (ESI) m/z: 397.2 [M+H].sup.+.

Step 5. Synthesis of tert-butyl 4-(4-((benzyloxy)methyl)piperidin-1-yl)-2-((tetrahydro-2H-pyran-4-yl)amino)benzoate

(1785) ##STR01027##

(1786) To a solution of tert-butyl 2-amino-4-(4-((benzyloxy)methyl)piperidin-1-yl)benzoate (900 mg, 2.27 mmol), tetrahydro-4H-pyran-4-one (454 mg, 4.54 mmol) and AcOH (5 mL) in DCM (15 mL) was added 1M borane tetrahydrofuran complex solution (9.08 mL, 9.08 mmol) at 0° C. After

the reaction was stirred at rt for 16 h, the mixture was concentrated. The resulting residue was purified by silica gel column chromatography to give the desired product (1.04 g, 95% yield) as light yellow solid. MS (ESI) m/z : 481.3 $[M+H].sup.+$.

Step 6. Synthesis of tert-butyl 4-(4-((benzyloxy)methyl)piperidin-1-yl)-2-(2,2,2-trifluoro-N-(tetrahydro-2H-pyran-4-yl)acetamido)benzoate

(1787) ##STR01028##

(1788) To a solution of tert-butyl 4-(4-((benzyloxy)methyl)piperidin-1-yl)-2-((tetrahydro-2H-pyran-4-yl)amino)benzoate (1.04 g, 2.16 mmol) and TEA (545 mg, 5.40 mmol) in DCM (15 mL) was added TFAA (680 mg, 3.24 mmol) at 0° C. After the reaction was stirred at room temperature for 2 h, the mixture was concentrated. The resulting residue was purified by silica gel column chromatography to give the desired product (1.18 g, 95% yield) as light yellow solid. MS (ESI) m/z : 577.3 $[M+H].sup.+$.

Step 7. Synthesis of 4-(4-((benzyloxy)methyl)piperidin-1-yl)-2-(2,2,2-trifluoro-N-(tetrahydro-2H-pyran-4-yl)acetamido)benzoic acid

(1789) ##STR01029##

(1790) To a solution of tert-butyl 4-(4-((benzyloxy)methyl)piperidin-1-yl)-2-(2,2,2-trifluoro-N-(tetrahydro-2H-pyran-4-yl)acetamido)benzoate (1.18 g, 2.04 mmol) in DCM (30 mL) was added TFA (20 mL) at room temperature. After the reaction was stirred at room temperature for 2 h, the mixture was concentrated. The resulting residue was purified by reverse-phase chromatography to give the desired product (980 mg, 92% yield) as light yellow solid. MS (ESI) m/z : 521.2 $[M+H].sup.+$.

Step 8. Synthesis of 4-(4-((benzyloxy)methyl)piperidin-1-yl)-2-(2,2,2-trifluoro-N-(tetrahydro-2H-pyran-4-yl)acetamido)benzoyl chloride

(1791) ##STR01030##

(1792) To a solution of 4-(4-((benzyloxy)methyl)piperidin-1-yl)-2-(2,2,2-trifluoro-N-(tetrahydro-2H-pyran-4-yl)acetamido)benzoic acid (0.5 g, 0.96 mmol) and DMF (20 mg, 0.27 mmol) in DCM (15 mL) was added oxalyl chloride (15 mL) at room temperature. After the reaction was stirred at 45° C. for 2 h, the mixture was concentrated to give the desired product (510 mg, 98% yield) as light yellow solid, which was used directly in the next step without further purification.

Step 9. Synthesis of 4-(4-((benzyloxy)methyl)piperidin-1-yl)-N-(5-(3,5-difluorobenzyl)-1H-indazol-3-yl)-2-(2,2,2-trifluoro-N-(tetrahydro-2H-pyran-4-yl)acetamido)benzamide

(1793) ##STR01031##

(1794) To a solution of 5-(3,5-difluorobenzyl)-1H-indazol-3-amine (260 mg, 1.00 mmol) and pyridine (237 mg, 3.00 mmol) in THF (10 mL) was added a solution of 4-(4-((benzyloxy)methyl)piperidin-1-yl)-2-(2,2,2-trifluoro-N-(tetrahydro-2H-pyran-4-yl)acetamido)benzoyl chloride (510 mg, 0.95 mmol) in THF (5 mL) over 15 minutes at -60° C. After the reaction was stirred at -40° C. for 2 h, the reaction mixture was poured into water (100 mL), and extracted with EtOAc (3×30 mL). The combined organic layers were dried over Na.sub.2SO.sub.4, filtered and concentrated. The resulting residue was purified by silica gel column chromatography to give the desired product (540 mg, 77% yield) as light yellow solid. MS (ESI) m/z : 762.3 $[M+H].sup.+$.

Step 10. Synthesis of N-(5-(3,5-difluorobenzyl)-1H-indazol-3-yl)-4-(4-(hydroxymethyl)piperidin-1-yl)-2-(2,2,2-trifluoro-N-(tetrahydro-2H-pyran-4-yl)acetamido)benzamide

(1795) ##STR01032##

(1796) To a solution of 4-(4-((benzyloxy)methyl)piperidin-1-yl)-N-(5-(3,5-difluorobenzyl)-1H-indazol-3-yl)-2-(2,2,2-trifluoro-N-(tetrahydro-2H-pyran-4-yl)acetamido)benzamide (500 mg, 0.66 mmol) in DCM (10 mL) was added methanesulfonic acid (315 mg, 3.28 mmol) at 0° C. After the reaction was stirred at room temperature for 2 h, the mixture was concentrated. The resulting residue was purified by reverse-phase chromatography to give the desired product (382 mg, 86% yield) as light yellow solid. MS (ESI) m/z : 672.3 $[M+H].sup.+$.

Step 11. Synthesis of 4-(4-(bromomethyl)piperidin-1-yl)-N-(5-(3,5-difluorobenzyl)-1H-indazol-3-yl)-2-((tetrahydro-2H-pyran-4-yl)amino)benzamide

(1797) ##STR01033##

(1798) A solution of N-(5-(3,5-difluorobenzyl)-1H-indazol-3-yl)-4-(4-(hydroxymethyl)piperidin-1-yl)-2-(2,2,2-trifluoro-N-(tetrahydro-2H-pyran-4-yl)acetamido)benzamide (100 mg, 0.15 mmol) in hydrobromic acid (48% aqueous solution, 5 mL) was stirred at 120° C. for 8 h. After the mixture was concentrated, the resulting residue was purified by reverse-phase chromatography to give the desired product (36 mg, 38% yield) as light yellow solid. MS (ESI) m/z: 638.2 [M+H].sup.+.

Step 12. Synthesis of N-(5-(3,5-difluorobenzyl)-1H-indazol-3-yl)-4-(4-((3-(2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-5-yl)azetidin-1-yl)methyl)piperidin-1-yl)-2-((tetrahydro-2H-pyran-4-yl)amino)benzamide

(1799) ##STR01034##

(1800) To a solution of 4-(4-(bromomethyl)piperidin-1-yl)-N-(5-(3,5-difluorobenzyl)-1H-indazol-3-yl)-2-((tetrahydro-2H-pyran-4-yl)amino)benzamide (10 mg, 0.016 mmol) in CH₃CN (2 mL) were added K₂CO₃ (6.6 mg, 0.048 mmol), NaI (2.7 mg, 0.018 mmol) and 5-(azetidin-3-yl)-2-(2,6-dioxopiperidin-3-yl)isoindoline-1,3-dione (5.6 mg, 0.018 mmol) at room temperature. After the reaction was stirred at 80° C. for 5 h, the mixture was concentrated. The resulting mixture was purified by reverse-phase chromatography to give the desired product N-(5-(3,5-difluorobenzyl)-1H-indazol-3-yl)-4-(4-((3-(2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-5-yl)azetidin-1-yl)methyl)piperidin-1-yl)-2-((tetrahydro-2H-pyran-4-yl)amino)benzamide (3.6 mg, 26% yield) as light yellow solid. MS (ESI) m/z: 871.4 [M+H].sup.+.

Example 380: N-(5-(3,5-Difluorobenzyl)-1H-indazol-3-yl)-4-(4-((3-(2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-5-yl)-2,6-diazaspiro[3.3]heptan-2-yl)methyl)piperidin-1-yl)-2-((tetrahydro-2H-pyran-4-yl)amino)benzamide (TR-329)

(1801) ##STR01035##

(1802) TR-329 was synthesized following the standard procedure for preparing TR-328 (2.3 mg, 16% yield). MS (ESI) m/z: 912.4 [M+H].sup.+.

Example 381: N-(5-(3,5-Difluorobenzyl)-1H-indazol-3-yl)-4-(4-((3-(2-(2,6-dioxopiperidin-3-yl)-1-oxoisindolin-5-yl)azetidin-1-yl)methyl)piperidin-1-yl)-2-((tetrahydro-2H-pyran-4-yl)amino)benzamide (TR-330)

(1803) ##STR01036##

(1804) TR-330 was synthesized following the standard procedure for preparing TR-328 (3.1 mg, 22% yield). MS (ESI) m/z: 857.4 [M+H].sup.+.

Example 382: 3-(5-(3-(4-(6-(6-((R)-2-(3-Fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)azetidin-1-yl)-1-oxoisindolin-2-yl)piperidine-2,6-dione (TR-331)

(1805) ##STR01037##

Step 1 to step 4. Please see the standard procedures for preparing TR-317

Step 5. Synthesis of 3-(1-oxo-5-(3-oxoazetidin-1-yl)isoindolin-2-yl)piperidine-2,6-dione

(1806) ##STR01038##

(1807) A solution of pyridine sulfur trioxide complex (420 mg, 2.6 mmol) in DMSO (2 mL) was added dropwise to a solution of 3-(5-(3-hydroxyazetidin-1-yl)-1-oxoisindolin-2-yl)piperidine-2,6-dione (104 mg, 0.33 mmol) and TEA (333 mg, 3.3 mmol) in DMSO (2 mL) at room temperature. The reaction mixture was stirred at room temperature for 40 min, before the mixture was purified by reverse-phase chromatography (CH₃CN/H₂O=20:1) to give 3-(1-oxo-5-(3-oxoazetidin-1-yl)isoindolin-2-yl)piperidine-2,6-dione (87 mg, 84% yield) as white solid. MS (ESI) m/z: 314.0 [M+H].sup.+.

Step 6. Synthesis of 3-(5-(3-(4-(6-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)azetidin-1-yl)-1-oxoisindolin-2-yl)piperidine-2,6-dione

(1808) ##STR01039##

(1809) A mixture of 3-(1-oxo-5-(3-oxoazetidin-1-yl)isoindolin-2-yl)piperidine-2,6-dione (24 mg, 0.076 mmol), (R)-6-(2-(3-fluorophenyl)pyrrolidin-1-yl)-3-(6-(piperazin-1-yl)pyridin-2-yl)imidazo[1,2-b]pyridazine (67 mg, 0.15 mmol) and a drop of acetic acid in THF (3.0 mL) was refluxed for 3 h under Ar atmosphere. After the reaction was cooled to room temperature, NaBH.sub.3CN (24 mg, 0.38 mmol) was added. The reaction mixture was stirred at room temperature overnight, before being purified by reverse-phase chromatography (CH.sub.3CN/H.sub.2O=20:1), followed by prep-TLC (DCM/MeOH=12:1) to give 3-(5-(3-(4-(6-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)azetidin-1-yl)-1-oxoisindolin-2-yl)piperidine-2,6-dione (11.6 mg, 21% yield) as a white solid. MS (ESI) m/z: 741.6 [M+H].sup.+.

Example 383: N-(5-(3,5-Difluorobenzyl)-1H-indazol-3-yl)-4-(4-(1-(2-(2,6-dioxopiperidin-3-yl)-1-oxoisindolin-5-yl)azetidin-3-yl)piperazin-1-yl)-2-((tetrahydro-2H-pyran-4-yl)amino)benzamide (TR-332)

(1810) ##STR01040##

(1811) TR-332 was synthesized following the standard procedure for preparing TR-331 (9.3 mg, 13% yield). MS (ESI) m/z: 844.8 [M+H].sup.+.

Example 384: N-(5-(3,5-Difluorobenzyl)-1H-indazol-3-yl)-4-(4-((1-(3-(2,6-dioxopiperidin-3-yl)-2-methyl-4-oxo-3,4-dihydroquinazolin-5-yl)azetidin-3-yl)methyl)piperazin-1-yl)-2-((tetrahydro-2H-pyran-4-yl)amino)benzamide (TR-333)

(1812) ##STR01041##

(1813) TR-333 was synthesized following the standard procedure for preparing TR-295 (2.2 mg, 16% yield). MS (ESI) m/z: 885.7 [M+H].sup.+.

Example 385: N-(5-(3,5-Difluorobenzyl)-1H-indazol-3-yl)-4-(4-(2-(1-(3-(2,6-dioxopiperidin-3-yl)-2-methyl-4-oxo-3,4-dihydroquinazolin-5-yl)azetidin-3-yl)ethyl)piperazin-1-yl)-2-((tetrahydro-2H-pyran-4-yl)amino)benzamide (TR-334)

(1814) ##STR01042##

(1815) TR-334 was synthesized following the standard procedure for preparing TR-295 (2.9 mg, 10% yield). MS (ESI) m/z: 899.9 [M+H].sup.+.

Example 386: 3-(6-(3-(4-(6-(6-((R)-2-(3-Fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)azetidin-1-yl)-1-oxoisindolin-2-yl)piperidine-2,6-dione (TR-335)

(1816) ##STR01043##

(1817) TR-335 was synthesized following the standard procedure for preparing TR-331 (4.3 mg, 7% yield). MS (ESI) m/z: 741.4 [M+H].sup.+.

Example 387: N-(5-(3,5-Difluorobenzyl)-1H-indazol-3-yl)-4-(4-(1-(2-(2,6-dioxopiperidin-3-yl)-3-oxoisindolin-5-yl)azetidin-3-yl)piperazin-1-yl)-2-((tetrahydro-2H-pyran-4-yl)amino)benzamide (TR-336)

(1818) ##STR01044##

(1819) TR-336 was synthesized following the standard procedure for preparing TR-331 (12 mg, 20% yield). MS (ESI) m/z: 844.9 [M+H].sup.+.

Example 388: N-(5-(3,5-Difluorobenzyl)-1H-indazol-3-yl)-4-(4-((1-(2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-5-yl)azetidin-3-yl)amino) piperidin-1-yl)-2-((tetrahydro-2H-pyran-4-yl)amino)benzamide (TR-337)

(1820) ##STR01045## ##STR01046##

Step 1 to step 7. Please see the standard procedures for preparing TR-328

Step 8. Synthesis of 4-(4-(benzyloxyl)piperidin-1-yl)-N-(5-(3,5-difluorobenzyl)-1H-indazol-3-yl)-2-((tetrahydro-2H-pyran-4-yl)amino)benzamide

(1821) ##STR01047##

(1822) To a solution of 4-(4-(benzyloxyl)piperidin-1-yl)-N-(5-(3,5-difluorobenzyl)-1H-indazol-3-yl)-2-(2,2,2-trifluoro-N-(tetrahydro-2H-pyran-4-yl)acetamido)benzamide (352 mg, 0.47 mmol) in MeOH (8 mL) was added K.sub.2CO.sub.3 (195.08 mg, 1.42 mmol) at room temperature. The

reaction mixture was stirred at rt for 16 h. The solvent was removed under reduced pressure to give a residue, which was diluted with ethyl acetate and washed with H₂O. The organic layer was separated, dried over sodium sulfate, filtered and concentrated under reduced pressure. The resulting residue was purified by silica gel column (DCM/CH₂OH=20/1) to give 4-(4-(benzyloxyl)piperidin-1-yl)-N-(5-(3,5-difluorobenzyl)-1H-indazol-3-yl)-2-((tetrahydro-2H-pyran-4-yl)amino)benzamide (300 mg, 98% yield) as a white solid. MS (ESI) m/z 652.6 [M+H]⁺.
Step 9. Synthesis of N-(5-(3,5-difluorobenzyl)-1H-indazol-3-yl)-4-(4-hydroxypiperidin-1-yl)-2-((tetrahydro-2H-pyran-4-yl)amino)benzamide

(1823) ##STR01048##

(1824) To a solution of 4-(4-(benzyloxyl)piperidin-1-yl)-N-(5-(3,5-difluorobenzyl)-1H-indazol-3-yl)-2-((tetrahydro-2H-pyran-4-yl)amino)benzamide (1.6 g, 2.46 mmol) in DCM (3 mL) was added MsOH (3 mL). After the reaction was stirred at rt for 2 h, the resulting mixture was concentrated under reduced pressure. The resulting residue was purified by silica gel column chromatography to give N-(5-(3,5-difluorobenzyl)-1H-indazol-3-yl)-4-(4-hydroxypiperidin-1-yl)-2-((tetrahydro-2H-pyran-4-yl)amino)benzamide (1.2 g, 87% yield) as a white solid. MS (ESI) m/z=562.7

[M+H]⁺. Step 10. Synthesis of N-(5-(3,5-difluorobenzyl)-1H-indazol-3-yl)-4-(4-oxopiperidin-1-yl)-2-((tetrahydro-2H-pyran-4-yl)amino)benzamide

(1825) ##STR01049##

(1826) To a solution of N-(5-(3,5-difluorobenzyl)-1H-indazol-3-yl)-4-(4-hydroxypiperidin-1-yl)-2-((tetrahydro-2H-pyran-4-yl)amino)benzamide (135 mg, 0.24 mmol) in dimethylsulfoxide (1 mL) were added triethylamine (0.5 mL) and sulphur trioxide pyridine complex (307.3 mg, 1.93 mmol) at 5° C. After the mixture was stirred at rt for 2 h, the reaction mixture was diluted with ethyl acetate (15 mL) and water (10 mL). The aqueous layer was separated and extracted with ethyl acetate (2×10 mL). The combined organic layers were washed with water (15 mL) and brine (15 mL), dried over sodium sulfate, filtered and concentrated under vacuum. The resulting residue was purified by prep-TLC (DCM/CH₂OH=10/1) to give N-(5-(3,5-difluorobenzyl)-1H-indazol-3-yl)-4-(4-oxopiperidin-1-yl)-2-((tetrahydro-2H-pyran-4-yl)amino)benzamide (65 mg, 49% yield) as a yellow solid. MS (ESI) m/z=562.7 [M+H]⁺.

Step 11. Synthesis of N-(5-(3,5-difluorobenzyl)-1H-indazol-3-yl)-4-(4-((1-(2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-5-yl)azetidin-3-yl)amino)piperidin-1-yl)-2-((tetrahydro-2H-pyran-4-yl)amino)benzamide

(1827) ##STR01050##

(1828) A mixture of N-(5-(3,5-difluorobenzyl)-1H-indazol-3-yl)-4-(4-oxopiperidin-1-yl)-2-((tetrahydro-2H-pyran-4-yl)amino)benzamide (10 mg, 0.018 mmol), 5-(3-aminoazetidin-1-yl)-2-(2,6-dioxopiperidin-3-yl)isoindoline-1,3-dione (5.9 mg, 0.018 mmol) and NaBH₄.3CN (3.39 mg, 0.054 mmol) in CH₂OH (0.5 mL) was stirred at rt for 2 h. The resulting mixture was concentrated and the residue was purified by prep-TLC to give N-(5-(3,5-difluorobenzyl)-1H-indazol-3-yl)-4-(4-((1-(2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-5-yl)azetidin-3-yl)amino)piperidin-1-yl)-2-((tetrahydro-2H-pyran-4-yl)amino)benzamide (7.0 mg, 45% yield) as yellow solid. MS (ESI) m/z=872.8 [M+H]⁺.

Example 389: N-(5-(3,5-Difluorobenzyl)-1H-indazol-3-yl)-4-(4-(3-(2-(2,6-dioxopiperidin-3-yl)-1-oxoisindolin-5-yl)azetidin-1-yl)piperidin-1-yl)-2-((tetrahydro-2H-pyran-4-yl)amino)benzamide (TR-338)

(1829) ##STR01051##

(1830) TR-338 was synthesized following the standard procedure for preparing TR-337 (8.5 mg, 56% yield). MS (ESI) m/z: 843.8 [M+H]⁺.

Example 390: N-(5-(3,5-Difluorobenzyl)-1H-indazol-3-yl)-4-(4-(3-(2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-5-yl)azetidin-1-yl)piperidin-1-yl)-2-((tetrahydro-2H-pyran-4-yl)amino)benzamide (TR-339)

(1831) ##STR01052##

(1832) TR-339 was synthesized following the standard procedure for preparing TR-337 (8.9 mg, 58% yield). MS (ESI) m/z: 857.8 [M+H].sup.+.

Example 391: N-(5-(3,5-Difluorobenzyl)-1H-indazol-3-yl)-4-(4-(6-(2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-5-yl)-2,6-diazaspiro[3.3]heptan-2-yl)piperidin-1-yl)-2-((tetrahydro-2/-pyran-4-yl)amino)benzamide (TR-340)

(1833) ##STR01053##

(1834) TR-340 was synthesized following the standard procedure for preparing TR-337 (5.6 mg, 34% yield). MS (ESI) m/z: 898.4 [M+H].sup.+.

Example 392: (S)—N-(5-(3,5-difluorobenzyl)-1H-indazol-3-yl)-4-(4-((1-(2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-5-yl)azetidin-3-yl)methyl)piperazin-1-yl)-2-((tetrahydro-2/-pyran-4-yl)amino)benzamide (TR-341)

(1835) The enantiomers of TR-231 were separated using a Daicel IG (10 µm, 250×20 mm) chiral HPLC column. The mobile phase was MeOH: DCM=48:52 with 0.2% 7N ammonia in methanol as additive. And the flow rate was 40 mL/min. The(S)-form of TR-231 was assigned to TR-341.

Example 393: (R)—N-(5-(3,5-difluorobenzyl)-1H-indazol-3-yl)-4-(4-((1-(2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-5-yl)azetidin-3-yl)methyl)piperazin-1-yl)-2-((tetrahydro-2H-pyran-4-yl)amino)benzamide (TR-342)

(1836) The enantiomers of TR-231 were separated using a Daicel IG (10 µm, 250×20 mm) chiral HPLC column. The mobile phase was MeOH: DCM=48:52 with 0.2% 7N ammonia in methanol as additive. And the flow rate was 40 mL/min. The (R)-form of TR-231 was assigned to TR-342.

Example 394: (S)—N-(5-(3,5-difluorobenzyl)-1H-indazol-3-yl)-4-(4-(1-(2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-5-yl)azetidin-3-yl)piperazin-1-yl)-2-((tetrahydro-2/-pyran-4-yl)amino)benzamide (TR-343)

(1837) The enantiomers of TR-275 were separated using a Daicel IG (10 µm, 250×20 mm) chiral HPLC column. The mobile phase was MeOH: DCM=60:40. And the flow rate was 40 mL/min. The(S)-form of TR-275 was assigned to TR-343.

Example 395: (R)—N-(5-(3,5-difluorobenzyl)-1H-indazol-3-yl)-4-(4-(1-(2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-5-yl)azetidin-3-yl)piperazin-1-yl)-2-((tetrahydro-2H-pyran-4-yl)amino)benzamide (TR-344)

(1838) The enantiomers of TR-275 were separated using a Daicel IG (10 µm, 250×20 mm) chiral HPLC column. The mobile phase was MeOH: DCM=60:40. And the flow rate was 40 mL/min. The (R)-form of TR-231 was assigned to TR-344.

Example 396: N-(5-((R)-2-(2,5-Difluorophenyl)pyrrolidin-1-yl)pyrazolo[1,5-a]pyrimidin-3-yl)-6-((1-((2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-5-yl)glycyl)piperidin-4-yl)amino)picolinamide (CPD-460)

(1839) ##STR01054##

Step 1. Synthesis of methyl 6-((1-(tert-butoxycarbonyl)piperidin-4-yl)amino)picolinate

(1840) ##STR01055##

(1841) A mixture of methyl 6-fluoropyridine-2-carboxylate (100 mg, 0.645 mmol), tert-butyl 4-aminopiperidine-1-carboxylate (193.6 mg, 0.97 mmol) and N-ethyl-N-isopropylpropane-2-amine (0.42 mL) in 2.0 mL of 1-methylpyrrolidin-2-one was stirred at 100° C. overnight. The mixture was extracted with ethyl acetate, the organic phase were concentrated and the residue was purified by flash chromatography (C.sub.18) to get desired compound (130 mg, yield: 60%) as white solid. MS (ESI) m/z=336.5 [M+H].sup.+.

Step 2. Synthesis of 6-((1-(tert-butoxycarbonyl)piperidin-4-yl)amino)picolinic acid

(1842) ##STR01056##

(1843) To a solution of methyl 6-((1-(tert-butoxycarbonyl)piperidin-4-yl)amino)picolinate (130 mg, 0.389 mmol) in CH.sub.3OH (2 mL) was added 2 N NaOH (1 mL). After the mixture was stirred at rt for 3 h, the mixture was concentrated and the residue was purified by column chromatography on C.sub.18 column to afford the title compound (90 mg, yield: 72%) as a white solid. MS (ESI)

m/z=322.3 [M+H].sup.+.

Step 3. Synthesis of tert-butyl (R)-4-((6-((5-(2-(2,5-difluorophenyl)pyrrolidin-1-yl)pyrazolo[1,5-a]pyrimidin-3-yl)carbamoyl)pyridin-2-yl)amino) piperidine-1-carboxylate

(1844) ###STR01057##

(1845) A mixture of 6-((1-(tert-butoxycarbonyl)piperidin-4-yl)amino)picolinic acid (60 mg, 0.187 mmol), (R)-5-(2-(2,5-difluorophenyl)pyrrolidin-1-yl)pyrazolo[1,5-a]pyrimidin-3-amine (58.88 mg, 0.187 mmol), HOAT (37.87 mg, 0.281 mmol), EDCI (55.11 mg, 0.281 mmol) and 4-Methylmorpholine (56.76 mg, 0.562 mmol) in DMSO (1 mL) was stirred at room temperature for 12 h. Then the mixture was purified by flash chromatography on silica C.sub.18 to afford the title compound (70 mg, yield: 61%) as white solid. MS (ESI) m/z=619.8 [M+H].sup.+.

Step 4. Synthesis of (R)—N-(5-(2-(2,5-difluorophenyl)pyrrolidin-1-yl)pyrazolo[1,5-a]pyrimidin-3-yl)-6-(piperidin-4-ylamino)picolinamide

(1846) ###STR01058##

(1847) To a solution of tert-butyl (R)-4-((6-((5-(2-(2,5-difluorophenyl)pyrrolidin-1-yl)pyrazolo[1,5-a]pyrimidin-3-yl)carbamoyl)pyridin-2-yl)amino) piperidine-1-carboxylate (70 mg, 0.113 mmol) in DCM (2 mL) was added TFA (1 mL). After it was stirred at room temperature for 2 h, the resulting mixture was concentrated in vacuo to get the crude product (73 mg, yield: 99.9%) as a yellow solid, which was used directly in the next step. MS (ESI) m/z=519.6 [M+H].sup.+.

Step 5. Synthesis of N-(5-((R)-2-(2,5-difluorophenyl)pyrrolidin-1-yl)pyrazolo[1,5-a]pyrimidin-3-yl)-6-((1-((2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-5-yl)glycyl)piperidin-4-yl)amino)picolinamide

(1848) A mixture of (R)—N-(5-(2-(2,5-difluorophenyl)pyrrolidin-1-yl)pyrazolo[1,5-a]pyrimidin-3-yl)-6-(piperidin-4-ylamino)picolinamide (8 mg, 0.015 mmol), (2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-5-yl)glycine (5.11 mg, 0.015 mmol), HOAT (3.03 mg, 0.0225 mmol), EDCI (4.32 mg, 0.0225 mmol) and 4-Methylmorpholine (7.58 mg, 0.075 mmol) in DMSO (1 mL) was stirred at room temperature for 12 h. Then the mixture was purified by reverse phase chromatography to afford the title compound (9.5 mg, yield: 76%) as yellow solid. MS (ESI) m/z=832.7 [M+H].sup.+.

Example 397: N-(5-((R)-2-(2,5-Difluorophenyl)pyrrolidin-1-yl)pyrazolo[1,5-a]pyrimidin-3-yl)-6-((1-(4-(((S)-1-((2S,4R)-4-hydroxy-2-((4-(4-methylthiazol-5-yl)benzyl)carbamoyl)pyrrolidin-1-yl)-3,3-dimethyl-1-oxobutan-2-yl)amino)-4-oxobutanoyl)piperidin-4-yl)amino)picolinamide (CPD-461)

(1849) ###STR01059##

(1850) CPD-461 was synthesized following the same procedure as CPD-460 (10.2 mg, yield: 65%). MS (ESI) m/z=1032. 1 [M+H].sup.+.

Example 398: N-(5-((R)-2-(2,5-Difluorophenyl)pyrrolidin-1-yl)pyrazolo[1,5-a]pyrimidin-3-yl)-6-((1~(6-(((S)-1-((2S,4R)-4-hydroxy-2-((4-(4-methylthiazol-5-yl)benzyl)carbamoyl)pyrrolidin-1-yl)-3,3-dimethyl-1-oxobutan-2-yl)amino)-6-oxohexanoyl)piperidin-4-yl)amino)picolinamide (CPD-462)

(1851) ###STR01060##

(1852) CPD-462 was synthesized following the same procedure as CPD-460 (11.2 mg, yield: 70%). MS (ESI) m/z=1060.2 [M+H].sup.+.

Example 399: N-(5-((R)-2-(2,5-Difluorophenyl)pyrrolidin-1-yl)pyrazolo[1,5-a]pyrimidin-3-yl)-6-((1-(8-(((S)-1-((2S,4R)-4-hydroxy-2-((4-(4-methylthiazol-5-yl)benzyl)carbamoyl)pyrrolidin-1-yl)-3,3-dimethyl-1-oxobutan-2-yl)amino)-8-oxooctanoyl)piperidin-4-yl)amino)picolinamide (CPD-463)

(1853) ###STR01061##

(1854) CPD-463 was synthesized following the same procedure as CPD-460. (11.4 mg, yield: 70%). MS (ESI) m/z=1088.1 [M+H].sup.+.

Example 400: N-(5-((R)-2-(2,5-Difluorophenyl)pyrrolidin-1-yl)pyrazolo[1,5-a]pyrimidin-3-yl)-6-((1-(10-(((S)-1-((2S,4R)-4-hydroxy-2-((4-(4-methylthiazol-5-yl)benzyl)carbamoyl)pyrrolidin-1-

yl)-3,3-dimethyl-1-oxobutan-2-yl)amino)-10-oxodecanoyl)piperidin-4-yl)amino)picolinamide (CPD-464)

(1855) ##STR01062##

(1856) CPD-464 was synthesized following the same procedure as CPD-460 (10.6 mg, yield: 63%). MS (ESI) m/z=1116.3 [M+H].sup.+.

Example 401: N-(5-((R)-2-(2,5-Difluorophenyl)pyrrolidin-1-yl)pyrazolo[1,5-a]pyrimidin-3-yl)-6-((1~(3-(3-(((S)-1-((2S,4R)-4-hydroxy-2-((4-(4-methylthiazol-5-yl)benzyl)carbamoyl)pyrrolidin-1-yl)-3,3-dimethyl-1-oxobutan-2-yl)amino)-3-oxopropoxy)propanoyl)piperidin-4-yl)amino)picolinamide (CPD-465)

(1857) ##STR01063##

(1858) CPD-465 was synthesized following the same procedure as CPD-460 (10.5 mg, yield: 65%). MS (ESI) m/z=1076.1 [M+H].sup.+.

Example 402: N-(5-((R)-2-(2,5-Difluorophenyl)pyrrolidin-1-yl)pyrazolo[1,5-a]pyrimidin-3-yl)-6-((1-(3-(2-(3-(((S)-1-((2,4R)-4-hydroxy-2-((4-(4-methylthiazol-5-yl)benzyl)carbamoyl)pyrrolidin-1-yl)-3,3-dimethyl-1-oxobutan-2-yl)amino)-3-oxopropoxy)ethoxy)propanoyl)piperidin-4-yl)amino)picolinamide (CPD-466)

(1859) ##STR01064##

(1860) CPD-466 was synthesized following the same procedure as CPD-460 (9.8 mg, yield: 58%). MS (ESI) m/z=1120.2 [M+H].sup.+.

Example 403: N-(5-((R)-2-(2,5~Difluorophenyl)pyrrolidin-1-yl)pyrazolo[1,5-a]pyrimidin-3-yl)-6-((1-((S)-15-((2S,4R)-4-hydroxy-2-((4-(4-methylthiazol-5-yl)benzyl)carbamoyl)pyrrolidine-1-carbonyl)-16, 16-dimethyl-13-oxo-4,7, 10-trioxa-14-azaheptadecanoyl)piperidin-4-yl)amino)picolinamide (CPD-467)

(1861) ##STR01065##

(1862) CPD-467 was synthesized following the same procedure as CPD-460 (11.2 mg, yield: 64%). MS (ESI) m/z=1164.2 [M+H].sup.+.

Example 404: N-(5-((R)-2-(2,5-Difluorophenyl)pyrrolidin-1-yl)pyrazolo[1,5-a]pyrimidin-3-yl)-6-((1-((S)-18-((2S,4R)-4-hydroxy-2-((4-(4-methylthiazol-5-yl)benzyl)carbamoyl)pyrrolidine-1-carbonyl)-19,19-dimethyl-16-oxo-4, 7, 10, 13-tetraoxa-17-azaicosanoyl)piperidin-4-yl)amino)picolinamide (CPD-468)

(1863) ##STR01066##

(1864) CPD-468 was synthesized following the same procedure as CPD-460 (11.6 mg, yield: 64%). MS (ESI) m/z=1208.3 [M+H].sup.+.

Example 405: N-(5-((R)-2-(2,5-Difluorophenyl)pyrrolidin-1-yl)pyrazolo[1,5-a]pyrimidin-3-yl)-6-((1-((S)-21-((2S,4R)-4-hydroxy-2-((4-(4-methylthiazol-5-yl)benzyl)carbamoyl)pyrrolidine-1-carbonyl)-22,22-dimethyl-19-oxo-4,7,10, 13, 16-pentaoxa-20-azatricosanoyl)piperidin-4-yl)amino)picolinamide (CPD-469)

(1865) ##STR01067##

(1866) CPD-469 was synthesized following the same procedure as CPD-460 (10.0 mg, yield: 53%). MS (ESI) m/z=1252.3 [M+H].sup.+.

Example 406: N-(5-((R)-2-(2,5-Difluorophenyl)pyrrolidin-1-yl)pyrazolo[1,5-a]pyrimidin-3-yl)-6-((1-(4-((2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-5-yl)amino)butanoyl)piperidin-4-yl)amino)picolinamide (CPD-470)

(1867) ##STR01068##

(1868) CPD-470 was synthesized following the same procedure as CPD-460 (9.2 mg, yield: 71%). MS (ESI) m/z=860.8 [M+H].sup.+.

Example 407: N-(5-((R)-2-(2,5-Difluorophenyl)pyrrolidin-1-yl)pyrazolo[1,5-a]pyrimidin-3-yl)-6-((1-(6-((2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-5-yl)amino)hexanoyl)piperidin-4-yl)amino)picolinamide (CPD-471)

(1869) ##STR01069##

(1870) CPD-471 was synthesized following the same procedure as CPD-460. (8.7 mg, yield: 65%). MS (ESI) m/z=888.9 [M+H].sup.+.

Example 408: N-(5-((R)-2-(2,5-Difluorophenyl)pyrrolidin-1-yl)pyrazolo[1,5-a]pyrimidin-3-yl)-6-((1-(8-((2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-5-yl)amino) octanoyl)piperidin-4-yl)amino)picolinamide (CPD-472)

(1871) ##STR01070##

(1872) CPD-473 was synthesized following the same procedure as CPD-460 (10.0 mg, yield: 73%). MS (ESI) m/z=916.9 [M+H].sup.+.

Example 409: N-(5-((R)-2-(2,5-Difluorophenyl)pyrrolidin-1-yl)pyrazolo[1,5-a]pyrimidin-3-yl)-6-((1~ (3-(2-((2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-5-yl)amino)ethoxy)propanoyl)piperidin-4-yl)amino)picolinamide (CPD-473)

(1873) ##STR01071##

(1874) CPD-473 was synthesized following the same procedure as CPD-460 (10.2 mg, yield: 76%). MS (ESI) m/z=890.8 [M+H].sup.+.

Example 410: N-(5-((R)-2-(2,5-Difluorophenyl)pyrrolidin-1-yl)pyrazolo[1,5-a]pyrimidin-3-yl)-6-((1-(3-(2-(2-((2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-5-yl)amino)ethoxy)ethoxy)propanoyl)piperidin-4-yl)amino)picolinamide (CPD-474)

(1875) ##STR01072##

(1876) CPD-474 was synthesized following the same procedure as CPD-460 (10.8 mg, yield: 77%). MS (ESI) m/z=934.8 [M+H].sup.+.

Example 411: N—(S—((R)-2-(2,5-Difluorophenyl)pyrrolidin-1-yl)pyrazolo[1,5-a]pyrimidin-3-yl)-6-((1-(3-(2-(2-(2-((2~ (2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-5-yl)amino)ethoxy)ethoxy)ethoxy)propanoyl)piperidin-4-yl)amino)picolinamide (CPD-475)

(1877) ##STR01073##

(1878) CPD-475 was synthesized following the same procedure as CPD-460 (10.1 mg, yield: 75%). MS (ESI) m/z=978.9 [M+H].sup.+.

Example 412: N-(5-((R)-2-(2,5-Difluorophenyl)pyrrolidin-1-yl)pyrazolo[1,5-a]pyrimidin-3-yl)-6-((1~ (1-((2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-5-yl)amino)-3,6,9,12-tetraoxapentadecan-15-oyl)piperidin-4-yl)amino)picolinamide (CPD-476)

(1879) ##STR01074##

(1880) CPD-476 was synthesized following the same procedure as CPD-460 (11.3 mg, yield: 74%). MS (ESI) m/z=1022.9 [M+H].sup.+.

Example 413: N-(5-((R)-2-(2,5-Difluorophenyl)pyrrolidin-1-yl)pyrazolo[1,5-a]pyrimidin-3-yl)-6-((1-(1-((2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-5-yl)amino)-3,6,9,12, 15-pentaoxaoctadecan-18-oyl)piperidin-4-yl)amino)picolinamide (CPD-477)

(1881) ##STR01075##

(1882) CPD-477 was synthesized following the same procedure as CPD-460 (11.6 mg, yield: 72%), MS (ESI) m/z=1067.0 [M+H].sup.+.

Example 414: N-(2-(4-Carbamoylpiperidin-1-yl)-4-((4-((2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-4-yl)glycyl)piperazin-1-yl)methyl)phenyl)-2-morpholinoxazole-4-carboxamide (CPD-478)

(1883) ##STR01076##

Step 1. Synthesis of ter-butyl 4-(3-fluoro-4-nitrobenzyl)piperazine-1-carboxylate

(1884) ##STR01077##

(1885) To a solution of 3-fluoro-4-nitrobenzaldehyde (1 g, 5.9 mmol) and tert-butyl piperazine-1-carboxylate (1.6 g, 8.8 mmol) in DCM (20 mL) was added NaBH (OAc) 3 (2.5 g, 11.8 mmol) at 0° C. After it was stirred at room temperature for 16 h, the mixture was concentrated and purified by reverse-phase chromatography to give the desired product (1.5 g, yield: 75%) as white solid. MS (ESI) m/z=340.3 [M+H].sup.+.

Step 2. Synthesis of tert-butyl 4-(3-(4-carbamoylpiperidin-1-yl)-4-nitrobenzyl)piperazine-1-

carboxylate

(1886) ##STR01078##

(1887) To a solution of tert-butyl 4-(3-fluoro-4-nitrobenzyl)piperazine-1-carboxylate (1 g, 2.9 mmol) and piperidine-4-carboxamide (1 g, 4.4 mmol) in CH₂Cl₂ (10 mL) was added DIPEA (760 mg, 5.9 mmol) at rt. The reaction mixture was stirred at 80° C. for 2 h. The mixture was concentrated and purified by reverse-phase chromatography to afford the title product (1.2 g, yield: 91%) as white solid. MS (ESI) m/z=448.3 [M+H]⁺.

Step 3. Synthesis of tert-butyl 4-(4-amino-3-(4-carbamoylpiperidin-1-yl)benzyl)piperazine-1-carboxylate

(1888) ##STR01079##

(1889) To a solution of tert-butyl 4-(4-amino-3-(4-carbamoylpiperidin-1-yl)benzyl)piperazine-1-carboxylate (1.2 g, 2.7 mmol) in DCM (10 mL) was added Pd/C (60 mg, 5% weight). After the mixture was stirred at room temperature for 16 h under hydrogen atmosphere. The mixture was filtered and concentrated. The residue was purified by reverse-phase chromatography to afford the title compound (700 mg, yield: 62%) as light yellow solid. MS (ESI) m/z=418.3 [M+H]⁺.

Step 4. Synthesis of tert-butyl 4-(3-(4-carbamoylpiperidin-1-yl)-4-(2-morpholinooxazole-4-carboxamido)benzyl)piperazine-1-carboxylate

(1890) ##STR01080##

(1891) To a solution of tert-butyl 4-(4-amino-3-(4-carbamoylpiperidin-1-yl)benzyl)piperazine-1-carboxylate (600 mg, 1.4 mmol) and 2-morpholinooxazole-4-carboxylic acid (277 mg, 1.4 mmol) in DMSO (10 mL) were added HOAt (293 mg, 2.1 mmol), EDCI (414 mg, 2.1 mmol), NMM (725 mg, 7.2 mmol). After the reaction was stirred at room temperature for 16 h, the mixture was purified by reverse-phase chromatography to give the desired product (390 mg, yield: 45%) as white solid. MS (ESI) m/z=598.4 [M+H]⁺.

Step 5. Synthesis of N-(2-(4-carbamoylpiperidin-1-yl)-4-(piperazin-1-ylmethyl)phenyl)-2-morpholinooxazole-4-carboxamide

(1892) ##STR01081##

(1893) To a solution of tert-butyl 4-(3-(4-carbamoylpiperidin-1-yl)-4-(2-morpholinooxazole-4-carboxamido)benzyl)piperazine-1-carboxylate (390 mg, 0.65 mmol) in DCM (10 mL) was added TFA (5 mL). After the reaction was stirred at room temperature for 2 h, the mixture was concentrated and purified by reverse-phase chromatography to afford the title compound (320 mg, yield: 98%) as white solid. MS (ESI) m/z=498.5 [M+H]⁺.

Step 6. Synthesis of N-(2-(4-carbamoylpiperidin-1-yl)-4-((4-((2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-4-yl)glycyl)piperazin-1-yl)methyl)phenyl)-2-morpholinooxazole-4-carboxamide

(1894) ##STR01082##

(1895) To a solution of N-(2-(4-carbamoyl-1-piperidyl)-4-(piperazin-1-ylmethyl)phenyl)-2-morpholino-oxazole-4-carboxamide (3.00 mg, 6.03 μmol) in DMSO (0.5 mL) were added HOAt (1.23 mg, 9.04 μmol), EDCI (1.74 mg, 9.04 μmol) and NMM (3.05 mg, 30.15 μmol). The mixture was stirred at room temperature for 2 min, before 2-((2-(2,6-dioxo-3-piperidyl)-1,3-dioxoisindolin-4-yl)amino)acetic acid was added (2.00 mg, 6.03 μmol) to above mixture. After the reaction was stirred at 25° C. for 12 h, the mixture was purified by reverse-phase chromatography to give N-(2-(4-carbamoylpiperidin-1-yl)-4-((4-((2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-4-yl)glycyl)piperazin-1-yl)methyl)phenyl)-2-morpholinooxazole-4-carboxamide (2.80 mg, yield: 53%) as light yellow solid. MS (ESI) m/z=811.4 [M+H]⁺.

Example 415: N-(2-(4-carbamoylpiperidin-1-yl)-4-((4-(3-((2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-4-yl)amino)propanoyl)piperazin-1-yl)methyl)phenyl)-2-morpholinooxazole-4-carboxamide (CPD-479)

(1896) ##STR01083##

(1897) CPD-479 was synthesized following the standard procedure for preparing CPD-478 (2.8 mg, 56% yield) as a light yellow solid. MS (ESI) m/z=825.4 [M+H]⁺.

Example 416: N-(2-(4-carbamoylpiperidin-1-yl)-4-((4-(4-((2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-4-yl)amino)butanoyl)piperazin-1-yl)methyl)phenyl)-2-morpholinooxazole-4-carboxamide (CPD-480)

(1898) ##STR01084##

(1899) CPD-480 was synthesized following the standard procedure for preparing CPD-478 (3.2 mg, yield: 63%) as light yellow solid. MS (ESI) m/z=839.4 [M+H].sup.+.

Example 417: N-(2-(4-carbamoylpiperidin-1-yl)-4-((4-(5-((2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-4-yl)amino)pentanoyl)piperazin-1-yl)methyl)phenyl)-2-morpholinooxazole-4-carboxamide (CPD-481)

(1900) ##STR01085##

(1901) CPD-481 was synthesized following the standard procedure for preparing CPD-478 (3.1 mg, yield: 60%) as light yellow solid. MS (ESI) m/z=853.5 [M+H].sup.+.

Example 418: N-(2-(4-carbamoylpiperidin-1-yl)-4-((4-(6-((2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-4-yl)amino)hexanoyl)piperazin-1-yl)methyl)phenyl)-2-morpholinooxazole-4-carboxamide (CPD-482)

(1902) ##STR01086##

(1903) CPD-482 was synthesized following the standard procedure for preparing CPD-478 (2.9 mg, yield: 55%) as light yellow solid. MS (ESI) m/z=867.5 [M+H].sup.+.

Example 419: N-(2-(4-carbamoylpiperidin-1-yl)-4-((4-(7-((2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-4-yl)amino) heptanoyl)piperazin-1-yl)methyl)phenyl)-2-morpholinooxazole-4-carboxamide (CPD-483)

(1904) ##STR01087##

(1905) CPD-483 was synthesized following the standard procedure for preparing CPD-478 (3.2 mg, yield: 60%) as light yellow solid. MS (ESI) m/z=881.5 [M+H].sup.+.

Example 420: N-(2-(4-carbamoylpiperidin-1-yl)-4-((4-(8-((2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-4-yl)amino) octanoyl)piperazin-1-yl)methyl)phenyl)-2-morpholinooxazole-4-carboxamide (CPD-484)

(1906) ##STR01088##

(1907) CPD-484 was synthesized following the standard procedure for preparing CPD-478 (3.0 mg, yield: 54%) as light yellow solid. MS (ESI) m/z=895.5 [M+H].sup.+.

Example 421: N-(2-(4-carbamoylpiperidin-1-yl)-4-((4-(3-(2-((2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-4-yl)amino)ethoxy)propanoyl)piperazin-1-yl)methyl)phenyl)-2-morpholinooxazole-4-carboxamide (CPD-485)

(1908) ##STR01089##

(1909) CPD-485 was synthesized following the standard procedure for preparing CPD-478 (2.8 mg, yield: 52%) as light yellow solid. MS (ESI) m/z=869.5 [M+H].sup.+.

Example 422: N-(2-(4-carbamoylpiperidin-1-yl)-4-((4-(3-(2-(2-((2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-4-yl)amino)ethoxy)ethoxy)propanoyl)piperazin-1-yl)methyl)phenyl)-2-morpholinooxazole-4-carboxamide (CPD-486)

(1910) ##STR01090##

(1911) CPD-486 was synthesized following the standard procedure for preparing CPD-478 (2.9 mg, yield: 52%) as light yellow solid. MS (ESI) m/z=913.5 [M+H].sup.+.

Example 423: N-(2-(4-carbamoylpiperidin-1-yl)-4-((4-(3-(2-(2-(2-((2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-4-yl)amino)ethoxy)ethoxy)ethoxy)propanoyl)piperazin-1-yl)methyl)phenyl)-2-morpholinooxazole-4-carboxamide (CPD-487)

(1912) ##STR01091##

(1913) CPD-487 was synthesized following the standard procedure for preparing CPD-478 (3.3 mg, yield: 55%) as light yellow solid. MS (ESI) m/z=957.5 [M+H].sup.+.

Example 424: N-(2-(4-carbamoylpiperidin-1-yl)-4-((4-(1-((2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-4-yl)amino)-3,6,9,12-tetraoxapentadecan-15-oyl)piperazin-1-yl)methyl)phenyl)-2-

morpholinooxazole-4-carboxamide (CPD-488)

(1914) ##STR01092##

(1915) CPD-488 was synthesized following the standard procedure for preparing CPD-478 (3.5 mg, yield: 56%) as light yellow solid. MS (ESI) m/z=1001.6 [M+H].sup.+.

Example 425: N-(2-(4-carbamoylpiperidin-1-yl)-4-((4-(1-((2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-4-yl)amino)-3,6,9,12, 15-pentaoxaoctadecan-18-oyl)piperazin-1-yl)methyl)phenyl)-2-morpholinooxazole-4-carboxamide (CPD-489)

(1916) ##STR01093##

(1917) CPD-489 was synthesized following the standard procedure for preparing CPD-478 (3.4 mg, yield: 55%) as light yellow solid. MS (ESI) m/z=1045.6 [M+H].sup.+.

Example 426: N-(2-(4-carbamoylpiperidin-1-yl)-4-((4-(4-(((S)-1-((2S,4R)-4-hydroxy-2-((4-(4-methylthiazol-5-yl)benzyl)carbamoyl)pyrrolidin-1-yl)-3,3-dimethyl-1-oxobutan-2-yl)amino)-4-oxobutanoyl)piperazin-1-yl)methyl)phenyl)-2-morpholinooxazole-4-carboxamide (CPD-490)

(1918) ##STR01094##

(1919) CPD-490 was synthesized following the standard procedure for preparing CPD-478 (10.6 mg, yield: 70%) as light yellow solid. MS (ESI) m/z=1010.5 [M+H].sup.+.

Example 427: N-(2-(4-carbamoylpiperidin-1-yl)-4-((4-(6-(((S)-1-((2S,4R)-4-hydroxy-2-((4-(4-methylthiazol-5-yl)benzyl)carbamoyl)pyrrolidin-1-yl)-3,3-dimethyl-1-oxobutan-2-yl)amino)-6-oxohexanoyl)piperazin-1-yl)methyl)phenyl)-2-morpholinooxazole-4-carboxamide (CPD-491)

(1920) ##STR01095##

(1921) CPD-491 was synthesized following the standard procedure for preparing CPD-478 (9.8 mg, yield: 67%) as light yellow solid. MS (ESI) m/z=1038.5 [M+H].sup.+.

Example 428: N-(2-(4-carbamoylpiperidin-1-yl)-4-((4-(8-(((S)-1-((2S,4R)-4-hydroxy-2-((4-(4-methylthiazol-5-yl)benzyl)carbamoyl)pyrrolidin-1-yl)-3,3-dimethyl-1-oxobutan-2-yl)amino)-8-oxooctanoyl)piperazin-1-yl)methyl)phenyl)-2-morpholinooxazole-4-carboxamide (CPD-492)

(1922) ##STR01096##

(1923) CPD-492 was synthesized following the standard procedure for preparing CPD-478 (9.3 mg, yield: 69%) as light yellow solid. MS (ESI) m/z=1066.5 [M+H].sup.+.

Example 429: N-(2-(4-carbamoylpiperidin-1-yl)-4-((4-(10-(((S)-1-((2S,4R)-4-hydroxy-2-((4-(4-methylthiazol-5-yl)benzyl)carbamoyl)pyrrolidin-1-yl)-3,3-dimethyl-1-oxobutan-2-yl)amino)-10-oxodecanoyl)piperazin-1-yl)methyl)phenyl)-2-morpholinooxazole-4-carboxamide (CPD-493)

(1924) ##STR01097##

(1925) CPD-493 was synthesized following the standard procedure for preparing CPD-478 (10.8 mg, yield: 73%) as light yellow solid. MS (ESI) m/z=1094.6 [M+H].sup.+.

Example 430: N-(2-(4-carbamoylpiperidin-1-yl)-4-((4-(10-(((S)-1-((2S,4R)-4-hydroxy-2-((4-(4-methylthiazol-5-yl)benzyl)carbamoyl)pyrrolidin-1-yl)-3,3-dimethyl-1-oxobutan-2-yl)amino)-10-oxodecanoyl)piperazin-1-yl)methyl)phenyl)-2-morpholinooxazole-4-carboxamide (CPD-494)

(1926) ##STR01098##

(1927) CPD-494 was synthesized following the standard procedure for preparing CPD-478 (11.2 mg, yield: 75%) as light yellow solid. MS (ESI) m/z=1054.5 [M+H].sup.+.

Example 431: N-(2-(4-carbamoylpiperidin-1-yl)-4-((4-(3-(2-(3-(((S)-1-((2S,4R)-4-hydroxy-2-((4-(4-methylthiazol-5-yl)benzyl)carbamoyl)pyrrolidin-1-yl)-3,3-dimethyl-1-oxobutan-2-yl)amino)-3-oxopropoxy)ethoxy)propanoyl)piperazin-1-yl)methyl)phenyl)-2-morpholinooxazole-4-carboxamide (CPD-495)

(1928) ##STR01099##

(1929) CPD-495 was synthesized following the standard procedure for preparing CPD-478 (10.5 mg, yield: 73%) as light yellow solid. MS (ESI) m/z=1098.5 [M+H].sup.+.

Example 432: N-(2-(4-carbamoylpiperidin-1-yl)-4-((4-((S)-15-((2S,4R)-4-hydroxy-2-((4-(4-methylthiazol-5-yl)benzyl)carbamoyl)pyrrolidine-1-carbonyl)-16,16-dimethyl-13-oxo-4,7,10-trioxa-14-azaheptadecanoyl)piperazin-1-yl)methyl)phenyl)-2-morpholinooxazole-4-carboxamide

(CPD-496)

(1930) ##STR01100##

(1931) CPD-496 was synthesized following the standard procedure for preparing CPD-478 (12.5 mg, yield: 75%) as light yellow solid. MS (ESI) $m/z=1142.6$ $[M+H].sup.+.$

Example 433: N-(2-(4-carbamoylpiperidin-1-yl)-4-((4-((S)-18-((2S,4R)-4-hydroxy-2-((4-(4-methylthiazol-S-yl)benzyl)carbamoyl)pyrrolidine-1-carbonyl)-19,19-dimethyl-16-oxo-4,7,10,13-tetraoxa-17-azaicosanoyl)piperazin-1-yl)methyl)phenyl)-2-morpholinooxazole-4-carboxamide (CPD-497)

(1932) ##STR01101##

(1933) CPD-497 was synthesized following the standard procedure for preparing CPD-478 (10.7 mg, yield: 72%) as light yellow solid. MS (ESI) $m/z=1186.6$ $[M+H].sup.+.$

Example 434: 2-(4-(2-Amino-3-(3-methoxy-4-((4-methoxybenzyl)oxy)benzyl)-3/-imidazo[4,5-b]pyridin-6-yl)-1H-pyrazol-1-yl)-N-(2-((2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-4-yl)amino)ethyl)acetamide (CPD-498)

(1934) ##STR01102##

Step 1. Synthesis of 3-methoxy-4-((4-methoxybenzyl)oxy)benzonitrile

(1935) ##STR01103##

(1936) A solution of 4-hydroxy-3-methoxybenzonitrile (10.0 g, 67.1 mmol), Cs.sub.2CO.sub.3 (27.3 g, 83.8 mmol) and p-methoxybenzyl chloride (13.1 g, 83.8 mmol) in MeCN (200 mL) was stirred at 90° C. for 1 h. Then the mixture was filtered and the filtrate cake was washed with ethyl acetate. The combined organic solvents were concentrated under reduced pressure. The residue was washed with MTBE and dried to give the crude title compound (16.0 g, yield: 89%) as yellow solid. MS (ESI) $m/z=270.1$ $[M+H].sup.+.$

Step 2. Synthesis of (3-methoxy-4-((4-methoxybenzyl)oxyl)phenyl)methanamine

(1937) ##STR01104##

(1938) To a stirred solution of 3-methoxy-4-((4-methoxybenzyl)oxy)benzonitrile (10.0 g, 37.2 mmol) in tetrahydrofuran (60 mL) was added lithium aluminum hydride (2.12 g, 56.0 mmol). The resulting mixture was stirred at room temperature for 1 h. After the reaction was cooled to 0° C., the reaction was quenched with water (2.8 mL). The mixture was treated with 1 N sodium hydroxide solution (2.8 mL) and additional water (8.5 mL). After stirring for 15 min, the mixture was filtered through celite, dried over magnesium sulfate, filtered, and concentrated to give (3-methoxy-4-((4-methoxybenzyl)oxyl)phenyl)methanamine (8.0 g, 79% yield) as an off white solid. MS (ESI) $m/z=274.1$ $[M+H].sup.+.$

Step 3. Synthesis of 5-bromo-N-(3-methoxy-4-((4-methoxybenzyl)oxy)benzyl)-3-nitropyridin-2-amine

(1939) ##STR01105##

(1940) A solution of (3-methoxy-4-((4-methoxybenzyl)oxyl)phenyl)methanamine (8.0 g, 29.3 mmol), 5-bromo-2-chloro-3-nitropyridine (6.61 g, 27.9 mmol) and DIPEA (4.5 g, 34.8 mmol) in MeCN (150 mL) was heated at 90° C. for 72 h. The mixture was concentrated under reduced pressure. The residue was diluted with water, and extracted with DCM (2×200 mL). The combined organic layers were dried over anhydrous sodium sulfate and purified by silica gel flash chromatography (DCM/MeOH=20:1) to give 5-bromo-N-(3-methoxy-4-((4-methoxybenzyl)oxy)benzyl)-3-nitropyridin-2-amine (10.0 g, yield: 75%) as yellow solid. MS (ESI) $m/z=474.1$ $[M+H].sup.+.$

Step 4. Synthesis of 5-bromo-N2-(3-methoxy-4-((4-methoxybenzyl)oxy)benzyl)pyridine-2,3-diamine

(1941) ##STR01106##

(1942) To a stirred solution of 5-bromo-N-(3-methoxy-4-((4-methoxybenzyl)oxyl)benzyl)-3-nitropyridin-2-amine (10.0 g, 21.14 mmol) in tetrahydrofuran (150 mL), ethanol (150 mL), and water (150 mL) was added sodium hydrosulfite (22.1 g, 126.8 mmol). The resulting mixture was

heated to reflux for 4 h. After it was cooled to room temperature, the mixture was diluted with water and extracted with DCM (2×200 mL). The combined organic layers were dried over anhydrous sodium sulfate and purified by silica gel flash chromatography (DCM/MeOH=20:1) to give 5-bromo-N-(3-methoxy-4-((4-methoxybenzyl)oxy)benzyl)pyridine-2,3-diamine (4.5 g, yield: 48%) as off white solid. MS (ESI) m/z=444.1 [M+H].sup.+.

Step 5. Synthesis of 6-bromo-3-(3-methoxy-4-((4-methoxybenzyl)oxy)benzyl)-3H-imidazo[4,5-b]pyridin-2-amine

(1943) ##STR01107##

(1944) To a stirred solution of 5-bromo-N-(3-methoxy-4-((4-methoxybenzyloxy)benzyl)pyridine-2,3-diamine (4.5 g, 10.16 mmol) in dichloromethane (100 mL) and methanol (100 mL) was added cyanogen bromide (3.1 mL, 5.0 M in acetonitrile, 15.5 mmol). The resulting solution was stirred at room temperature for 24 h, before a second aliquot of cyanogen bromide solution was added (3.1 mL). After 48 h, a third aliquot of cyanogen bromide solution (3.1 mL) was added and stirred for another 24 h. Then the reaction mixture was concentrated and the residue was dissolved in dichloromethane. The solution was washed with 1 N sodium hydroxide solution, dried over anhydrous magnesium sulfate, filtered, concentrated and purified by silica gel flash chromatography (DCM/MeOH=20:1) to give 6-bromo-3-(3-methoxy-4-((4-methoxybenzyl)oxy)benzyl)-3H-imidazo[4,5-b]pyridin-2-amine (1.5 g, yield: 31%) as off white solid. ¹H NMR (400 MHz, DMSO-d₆) δ 7.92(d, J=1.6 Hz, 1H), 7.59 (d, J=1.2 Hz, 1H), 7.32 (d, J=8.4 Hz, 2H), 7.12(s, 2H), 7.05 (s, 1H), 6.94-6.91 (m, 3H), 6.68 (d, J=8.4 Hz, 1H), 5.17 (s, 2H), 4.92(s, 2H), 3.74 (s, 3H), 3.70 (s, 3H). MS (ESI) m/z: 469.1 [M+H].sup.+.

Step 6. Synthesis of tert-butyl 2-(4-(2-amino-3-(3-methoxy-4-((4-methoxybenzyl)oxy)benzyl)-3H-imidazo[4,5-b]pyridin-6-yl)-1H-pyrazol-1-yl)acetate

(1945) ##STR01108##

(1946) To a solution of 6-bromo-3-(3-methoxy-4-((4-methoxybenzyl)oxy)benzyl)-3H-imidazo[4,5-b]pyridin-2-amine (200 mg, 0.42 mmol) and tert-butyl 2-(4-(4,4,5,5-tetramethyl-1,3,2-dioxaborolan-2-yl)-1H-pyrazol-1-yl)acetate (194 mg, 0.63 mmol) in dioxane (10 mL) and H₂O (2 mL) were added Na₂CO₃ (133 mg, 1.26 mmol) and Pd(dppf) Cl₂ (20 mg, 0.02 mmol) at room temperature under N₂. After the reaction was stirred at 100° C. for 16 h, the reaction mixture was purified by reverse-phase chromatography to give tert-butyl 2-(4-(2-amino-3-(3-methoxy-4-((4-methoxybenzyl)oxy)benzyl)-3H-imidazo[4,5-b]pyridin-6-yl)-1H-pyrazol-1-yl)acetate (180 mg, yield: 75%) as white solid. MS (ESI) m/z=571.8 [M+H].sup.+.

Step 7. Synthesis of 2-(4-(2-amino-3-(3-methoxy-4-((4-methoxybenzyl)oxy)benzyl)-3H-imidazo[4,5-b]pyridin-6-yl)-1H-pyrazol-1-yl)acetic acid

(1947) ##STR01109##

(1948) To a solution of tert-butyl 2-(4-(2-amino-3-(3-methoxy-4-((4-methoxybenzyl)oxy)benzyl)-3H-imidazo[4,5-b]pyridin-6-yl)-1H-pyrazol-1-yl)acetate (180 mg, 0.32 mmol) in THF (10 mL) and H₂O (3 mL) was added NaOH (38 mg, 0.94 mmol). After the reaction was stirred at 100° C. for 16 h, the reaction mixture was purified by reverse-phase chromatography to give 2-(4-(2-amino-3-(3-methoxy-4-((4-methoxybenzyl)oxy)benzyl)-3H-imidazo[4,5-b]pyridin-6-yl)-1H-pyrazol-1-yl)acetic acid (120 mg, yield: 73%) as white solid. MS (ESI) m/z=515.4 [M+H].sup.+.

Step 8. Synthesis of 2-(4-(2-amino-3-(3-methoxy-4-((4-methoxybenzyl)oxy)benzyl)-3H-imidazo[4,5-b]pyridin-6-yl)-1H-pyrazol-1-yl)-N-(2-((2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-4-yl)amino)ethyl)acetamide

(1949) ##STR01110##

(1950) To a solution of 2-(4-(2-amino-3-(3-methoxy-4-((4-methoxybenzyl)oxy)benzyl)-3H-imidazo[4,5-b]pyridin-6-yl)-1H-pyrazol-1-yl)acetic acid (5 mg, 0.009 mmol), 4-((2-aminoethyl)amino)-2-(2,6-dioxopiperidin-3-yl)isindoline-1,3-dione (4.6 mg, 0.015 mmol) in DMSO (2 mL) were added EDCI (3.4 mg, 0.018 mmol), HOAT (2.5 mg, 0.018 mmol) and NMM

(4.5 mg, 0.045 mmol). After the reaction was stirred at rt for 16 h, the reaction mixture was purified by reverse-phase chromatography to give 2-(4-(2-amino-3-(3-methoxy-4-((4-methoxybenzyl)oxy)benzyl)-3H-imidazo[4,5-b]pyridin-6-yl)-1H-pyrazol-1-yl)-N-(2-((2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-4-yl)amino)ethyl)acetamide (1.1 mg, yield: 15%) as yellow solid. MS (ESI) m/z =813.7 [M+H].sup.+.

Example 435: 2-(4-(2-Amino-3-(3-methoxy-4-((4-methoxybenzyl)oxy)benzyl)-3H-imidazo[4,5-b]pyridin-6-yl)-1H-pyrazol-1-yl)-N-(4-((2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-4-yl)amino)butyl)acetamide (CPD-499)

(1951) ##STR01111##

(1952) CPD-499 was synthesized following the standard procedure for preparing CPD-498 (1.2 mg, yield: 16%) as light yellow solid. MS (ESI) m/z =841.4 [M+H].sup.+.

Example 436: 2-(4-(2-Amino-3-(3-methoxy-4-((4-methoxybenzyl)oxy)benzyl)-3H-imidazo[4,5-b]pyridin-6-yl)-1H-pyrazol-1-yl)-N-(6-((2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-4-yl)amino)hexyl)acetamide (CPD-500)

(1953) ##STR01112##

(1954) CPD-500 was synthesized following the standard procedure for preparing CPD-498 (1.2 mg, yield: 16%) as light yellow solid. MS (ESI) m/z =869.4 [M+H].sup.+.

Example 437: 2-(4-(2-Amino-3-(3-methoxy-4-((4-methoxybenzyl)oxy)benzyl)-3H-imidazo[4,5-b]pyridin-6-yl)-1H-pyrazol-1-yl)-N-(8-((2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-4-yl)amino)octyl)acetamide (CPD-501)

(1955) ##STR01113##

(1956) CPD-501 was synthesized following the standard procedure for preparing CPD-498 (1.4 mg, yield: 17%) as light yellow solid. MS (ESI) m/z =897.4 [M+H].sup.+.

Example 438: 2-(4-(2-Amino-3-(3-methoxy-4-((4-methoxybenzyl)oxy)benzyl)-3H-imidazo[4,5-b]pyridin-6-yl)-1H-pyrazol-1-yl)-N-(2-(2-((2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-4-yl)amino)ethoxy)ethyl)acetamide (CPD-502)

(1957) ##STR01114##

(1958) CPD-502 was synthesized following the standard procedure for preparing CPD-498 (1.5 mg, yield: 19%) as light yellow solid. MS (ESI) m/z =857.4 [M+H].sup.+.

Example 439: 2-(4-(2-Amino-3-(3-methoxy-4-((4-methoxybenzyl)oxy)benzyl)-3H-imidazo[4,5-b]pyridin-6-yl)-1H-pyrazol-1-yl)-N-(2-(2-(2-((2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-4-yl)amino)ethoxy)ethoxy)ethyl)acetamide (CPD-503)

(1959) ##STR01115##

(1960) CPD-503 was synthesized following the standard procedure for preparing CPD-498 (1.6 mg, yield: 20%) as light yellow solid. MS (ESI) m/z =901.4 [M+H].sup.+.

Example 440: 2-(4-(2-Amino-3-(3-methoxy-4-((4-methoxybenzyl)oxy)benzyl)-3H-imidazo[4,5-b]pyridin-6-yl)-1H-pyrazol-1-yl)-N-(2-(2-(2-(2-((2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-4-yl)amino)ethoxy)ethoxy)ethoxy)ethyl)acetamide (CPD-504)

(1961) ##STR01116##

(1962) CPD-504 was synthesized following the standard procedure for preparing CPD-498 (1.4 mg, yield: 16%) as light yellow solid. MS (ESI) m/z =945.5 [M+H].sup.+.

Example 441: 2-(4-(2-Amino-3-(3-methoxy-4-(4-methoxybenzyl)oxy)benzyl)-3H-imidazo[4,5-b]pyridin-6-yl)-1H-pyrazol-1-yl)-N-(14-((2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-4-yl)amino)-3,6,9,12-tetraoxatetradecyl)acetamide (CPD-505)

(1963) ##STR01117##

(1964) CPD-505 was synthesized following the standard procedure for preparing CPD-498 (1.4 mg, yield: 16%) as light yellow solid. MS (ESI) m/z : 989.5 [M+H].sup.+.

Example 442: 2-(4-(2-Amino-3-(3-methoxy-4-((4-methoxybenzyl)oxy)benzyl)-3H-imidazo[4,5-b]pyridin-6-yl)-1H-pyrazol-1-yl)-N-(17-((2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-4-yl)amino)-3,6,9,12, 15-pentaoxaheptadecyl)acetamide (CPD-506)

(1965) ##STR01118##

(1966) CPD-506 was synthesized following the standard procedure for preparing CPD-498 (1.4 mg, yield: 15%) as light yellow solid. MS (ESI) m/z: 1033.5 [M+H].sup.+.

Example 443: (2S,4R)-1-((S)-2-(2-(2-(4-(2-amino-3-(3-methoxy-4-((4-methoxybenzyl)oxy)benzyl)-3/7-imidazo[4,5-b]pyridin-6-yl)-1H-pyrazol-1-yl)acetamido)acetamido)-3,3-dimethylbutanoyl)-4-hydroxy-N-(4-(4-methylthiazol-5-yl)benzyl)pyrrolidine-2-carboxamide (CPD-507)

(1967) ##STR01119##

(1968) CPD-507 was synthesized following the standard procedure for preparing CPD-498 (1.1 mg, yield: 13%) as light yellow solid. MS (ESI) m/z=984.5 [M+H].sup.+.

Example 444: (2S,4R)-1-((S)-2-(4-(2-(4-(2-amino-3-(3-methoxy-4-((4-methoxybenzyl)oxy)benzyl)-3H-imidazo[4,5-b]pyridin-6-yl)-1H-pyrazol-1-yl)acetamido)butanamido)-3,3-dimethylbutanoyl)-4-hydroxy-N-(4-(4-methylthiazol-5-yl)benzyl)pyrrolidine-2-carboxamide (CPD-508)

(1969) ##STR01120##

(1970) CPD-508 was synthesized following the standard procedure for preparing CPD-498 (1.7 mg, yield: 19%) as light yellow solid. MS (ESI) m/z=1012.5 [M+H].sup.+.

Example 445: (2S,4R)-1-((S)-2-(6-(2-(4-(2-amino-3-(3-methoxy-4-((4-methoxybenzyl)oxy)benzyl)-3H-imidazo[4,5-b]pyridin-6-yl)-1H-pyrazol-1-yl)acetamido)hexanamido)-3,3-dimethylbutanoyl)-4-hydroxy-N-(4-(4-methylthiazol-5-yl)benzyl)pyrrolidine-2-carboxamide (CPD-509)

(1971) ##STR01121##

(1972) CPD-509 was synthesized following the standard procedure for preparing CPD-498 (1.5 mg, yield: 16%) as light yellow solid. MS (ESI) m/z=1040.5 [M+H].sup.+.

Example 446: (2S,4R)-1-((S)-2-(8-(2-(4-(2-amino-3-(3-methoxy-4-((4-methoxybenzyl)oxy)benzyl)-3H-imidazo[4,5-b]pyridin-6-yl)-1H-pyrazol-1-yl)acetamido)octanamido)-3,3-dimethylbutanoyl)-4-hydroxy-N-(4-(4-methylthiazol-5-yl)benzyl)pyrrolidine-2-carboxamide (CPD-510)

(1973) ##STR01122##

(1974) CPD-510 was synthesized following the standard procedure for preparing CPD-498 (1.5 mg, yield: 16%) as light yellow solid. MS (ESI) m/z=1068.6 [M+H].sup.+.

Example 447: (2S,4R)-1-((S)-2-(10-(2-(4-(2-amino-3-(3-methoxy-4-((4-methoxybenzyl)oxy)benzyl)-3H-imidazo[4,5-b]pyridin-6-yl)-1H-pyrazol-1-yl)acetamido)decanamido)-3,3-dimethylbutanoyl)-4-hydroxy-N-(4-(4-methylthiazol-5-yl)benzyl)pyrrolidine-2-carboxamide (CPD-511)

(1975) ##STR01123##

(1976) CPD-511 was synthesized following the standard procedure for preparing CPD-498 (1.5 mg, yield: 15%) as light yellow solid. MS (ESI) m/z=1096.6 [M+H].sup.+.

Example 448: (2S,4R)-1-((S)-2-(3-(2-(2-(4-(2-amino-3-(3-methoxy-4-((4-methoxybenzyl)oxy)benzyl)-3H-imidazo[4,5-b]pyridin-6-yl)-1H-pyrazol-1-yl)acetamido)ethoxy)propanamido)-3,3-dimethylbutanoyl)-4-hydroxy-N-(4-(4-methylthiazol-5-yl)benzyl)pyrrolidine-2-carboxamide (CPD-512)

(1977) ##STR01124##

(1978) CPD-512 was synthesized following the standard procedure for preparing CPD-498 (1.2 mg, yield: 13%) as light yellow solid. MS (ESI) m/z=1042.5 [M+H].sup.+.

Example 449: (2S,4R)-1-((S)-1-(4-(2-amino-3-(3-methoxy-4-((4-methoxybenzyl)oxy)benzyl)-3H-imidazo[4,5-b]pyridin-6-yl)-1H-pyrazol-1-yl)-14-(tert-butyl)-2,12-dioxo-6,9-dioxo-3,13-diazapentadecan-15-oyl)-4-hydroxy-N-(4-(4-methylthiazol-5-yl)benzyl)pyrrolidine-2-carboxamide (CPD-513)

(1979) ##STR01125##

(1980) CPD-513 was synthesized following the standard procedure for preparing CPD-498 (1.7 mg, yield: 17%) as light yellow solid. MS (ESI) m/z=1086.5 [M+H].sup.+.

Example 450: (2S,4R)-1-((S)-1-(4-(2-amino-3-(3-methoxy-4-((4-methoxybenzyl)oxy)benzyl)-3H-imidazo[4,5-b]pyridin-6-yl)-1H-pyrazol-1-yl)-17-(tert-butyl)-2,15-dioxo-6,9,12-trioxa-3,16-diazaoctadecan-18-oyl)-4-hydroxy-N-(4-(4-methylthiazol-5-yl)benzyl)pyrrolidine-2-carboxamide (CPD-514)

(1981) ##STR01126##

(1982) CPD-514 was synthesized following the standard procedure for preparing CPD-498 (1.5 mg, yield: 15%) as light yellow solid. MS (ESI) m/z=1130.6 [M+H].sup.+.

Example 451: (2S,4R)-1-((S)-1-(4-(2-amino-3-(3-methoxy-4-((4-methoxybenzyl)oxy)benzyl)-3H-imidazo[4,5-b]pyridin-6-yl)-1H-pyrazol-1-yl)-20-(tert-butyl)-2,18-dioxo-6,9,12, 15-tetraoxa-3,19-diazahenicosan-21-oyl)-4-hydroxy-N-(4-(4-methylthiazol-5-yl)benzyl)pyrrolidine-2-carboxamide (CPD-515)

(1983) ##STR01127##

(1984) CPD-515 was synthesized following the standard procedure for preparing CPD-498 (1.8 mg, yield: 17%) as light yellow solid. MS (ESI) m/z=1174.6 [M+H].sup.+.

Example 452: (2S,4R)-1-((S)-1-(4-(2-amino-3-(3-methoxy-4-((4-methoxybenzyl)oxy)benzyl)-3H-imidazo[4,5-b]pyridin-6-yl)-1H-pyrazol-1-yl)-23-(tert-butyl)-2,21-dioxo-6,9,12,15, 18-pentaoxa-3,22-diazatetracosan-24-oyl)-4-hydroxy-N-(4-(4-methylthiazol-5-yl)benzyl)pyrrolidine-2-carboxamide (CPD-516)

(1985) ##STR01128##

(1986) CPD-516 was synthesized following the standard procedure for preparing CPD-498 (1.8 mg, yield: 17%) as light yellow solid. MS (ESI) m/z=1218.6 [M+H].sup.+.

(1987) Certain compounds disclosed herein have the structures shown in Table 1.

(1988) TABLE-US-00001 TABLE 1 Cpd. Code Structure/Name CPD-001 (TR-001)

 embedded image CPD-002 (TR-002)  embedded image CPD-003 (TR-003)
 embedded image CPD-004 (TR-004)  embedded image CPD-005 (TR-005)  embedded image
CPD-006 (TR-006)  embedded image CPD-007 (TR-007)  embedded image CPD-008 (TR-008)
 embedded image CPD-009 (TR-009)  embedded image CPD-010 (TR-010)  embedded image
CPD-011 (TR-011)  embedded image CPD-012 (TR-012)  embedded image CPD-013 (TR-
013)  embedded image CPD-014 (TR-014)  embedded image CPD-015 (TR-015)
 embedded image CPD-016 (TR-016)  embedded image CPD-017 (TR-017)  embedded image
CPD-018 (TR-018)  embedded image CPD-019 (TR-019)  embedded image CPD-020 (TR-020)
 embedded image CPD-021 (TR-021)  embedded image CPD-022 (TR-022) 0
 embedded image CPD-023 (TR-023)  embedded image CPD-024 (TR-024)  embedded image
CPD-025 (TR-025)  embedded image CPD-026 (TR-026)  embedded image CPD-027 (TR-027)
 embedded image CPD-028 (TR-028)  embedded image CPD-029 (TR-029)  embedded image
CPD-030 (TR-030)  embedded image CPD-031 (TR-031)  embedded image CPD-032 (TR-032)
0  embedded image CPD-033 (TR-033)  embedded image CPD-034 (TR-034)
 embedded image CPD-035 (TR-035)  embedded image CPD-036 (TR-036)  embedded image
CPD-037 (TR-037)  embedded image CPD-038 (TR-038)  embedded image CPD-039 (TR-039)
 embedded image CPD-040 (TR-040)  embedded image CPD-041 (TR-041)  embedded image
CPD-042 (TR-042) 0  embedded image CPD-043 (TR-043)  embedded image CPD-044 (TR-
044)  embedded image CPD-045 (TR-045)  embedded image CPD-046 (TR-046)
 embedded image CPD-047 (TR-047)  embedded image CPD-048 (TR-048)  embedded image
CPD-049 (TR-049)  embedded image CPD-050 (TR-050)  embedded image CPD-051 (TR-051)
 embedded image CPD-052 (TR-052) 0  embedded image CPD-053 (TR-053)
 embedded image CPD-054 (TR-054)  embedded image CPD-055 (TR-055)  embedded image
CPD-056 (TR-056)  embedded image CPD-057 (TR-057)  embedded image CPD-058 (TR-058)
 embedded image CPD-059 (TR-059)  embedded image CPD-060 (TR-060)  embedded image

CPD-061 (TR-061)  embedded image CPD-062 (TR-062) 0  embedded image CPD-063 (TR-063)  embedded image CPD-064 (TR-064)  embedded image CPD-065 (TR-065)  embedded image CPD-066  embedded image CPD-067  embedded image CPD-068  embedded image CPD-069  embedded image CPD-070  embedded image CPD-071  embedded image CPD-072 00  embedded image CPD-073 01  embedded image CPD-074 02  embedded image CPD-075 03  embedded image CPD-076 04  embedded image CPD-077 05  embedded image CPD-078 06  embedded image CPD-079 07  embedded image CPD-080 (TR-190) 08  embedded image CPD-081 (TR-107) 09  embedded image CPD-082 0  embedded image CPD-083  embedded image CPD-084  embedded image CPD-085 (TR-106)  embedded image CPD-086 (TR-105)  embedded image CPD-087 (TR-108)  embedded image CPD-088 (TR-104)  embedded image CPD-089  embedded image CPD-090 (TR-111)  embedded image CPD-091 (TR-110)  embedded image CPD-092 (TR-102) 0  embedded image CPD-093 (TR-059)  embedded image CPD-094  embedded image CPD-095  embedded image CPD-096  embedded image CPD-097  embedded image CPD-098  embedded image CPD-099  embedded image CPD-100  embedded image CPD-101  embedded image CPD-102 0  embedded image CPD-103  embedded image CPD-104  embedded image CPD-105  embedded image CPD-106 (TR-155)  embedded image CPD-107 (TR-152)  embedded image CPD-108  embedded image CPD-109  embedded image CPD-110  embedded image CPD-111 (TR-151)  embedded image CPD-112 (TR-153) 0  embedded image CPD-113 (TR-150)  embedded image CPD-114 (TR-149)  embedded image CPD-115 (TR-147)  embedded image CPD-116 (TR-148)  embedded image CPD-117 (TR-154)  embedded image CPD-118  embedded image CPD-119 (TR-161)  embedded image CPD-120  embedded image CPD-121 (TR-163)  embedded image CPD-122 (TR-160) 0  embedded image CPD-123 (TR-166)  embedded image CPD-124 (TR-144)  embedded image CPD-125 (TR-145)  embedded image CPD-126 (TR-169)  embedded image CPD-127 (TR-142)  embedded image CPD-128 (TR-138)  embedded image CPD-129 (TR-139)  embedded image CPD-130 (TR-136)  embedded image CPD-131 (TR-165)  embedded image CPD-132 (TR-167) 0  embedded image CPD-133 (TR-156)  embedded image CPD-134 (TR-164)  embedded image CPD-135 (TR-158)  embedded image CPD-136 (TR-157)  embedded image CPD-137 (TR-159)  embedded image CPD-138 (TR-143)  embedded image CPD-139 (TR-141)  embedded image CPD-140 (TR-146)  embedded image CPD-141 (TR-140)  embedded image CPD-142 (TR-137) 0  embedded image CPD-143 (TR-135)  embedded image CPD-144 (TR-134)  embedded image CPD-145 (TR-130)  embedded image CPD-146 (TR-128)  embedded image CPD-147 (TR-120)  embedded image CPD-148 (TR-119)  embedded image CPD-149 (TR-116)  embedded image CPD-150 (TR-113)  embedded image CPD-151 (TR-114)  embedded image CPD-152 (TR-122) 0  embedded image CPD-153 (TR-117)  embedded image CPD-154 (TR-129)  embedded image CPD-155 (TR-127)  embedded image CPD-156 (TR-124)  embedded image CPD-157 (TR-126)  embedded image CPD-158 (TR-168)  embedded image CPD-159 (TR-115)  embedded image CPD-160 (TR-123)  embedded image CPD-161 (TR-121)  embedded image CPD-162 (TR-118) 0  embedded image CPD-163 (TR-131)  embedded image CPD-164 (TR-132)  embedded image CPD-165 (TR-133)  embedded image CPD-166 (TR-125)  embedded image CPD-167 (TR-103)  embedded image CPD-168 (TR-162)  embedded image CPD-169 (TR-123- neg)  embedded image CPD-170 (TR-170)  embedded image CPD-171 (TR-171)  embedded image CPD-172 (TR-172) 00  embedded image CPD-173 (TR-173) 01  embedded image CPD-174 (TR-174) 02  embedded image CPD-175 (TR-175) 03  embedded image CPD-176 (TR-176) 04  embedded image CPD-177 (TR-177) 05  embedded image CPD-178 (TR-178) 06  embedded image CPD-179 (TR-179) 07  embedded image CPD-180 (TR-180) 08  embedded image CPD-181 (TR-181) 09  embedded image CPD-182 (TR-182) 0  embedded image CPD-183 (TR-183)  embedded image CPD-184 (TR-184)  embedded image

CPD-185 (TR-185)  CPD-186 (TR-186)  CPD-187 (TR-187) 
 CPD-188 (TR-188)  CPD-189 (TR-189)  CPD-190 (TR-190)  CPD-191 (TR-191)  CPD-192 (TR-192) 
0  CPD-193 (TR-193)  CPD-194 (TR-194) 
 CPD-195 (TR-195)  CPD-196 (TR-196)  CPD-197 (TR-197)  CPD-198 (TR-198)  CPD-199 (TR-199) 
 CPD-200 (TR-200)  CPD-201 (TR-201)  CPD-202 (TR-202) 0  CPD-203 (TR-203)  CPD-204 (TR-204) 
 CPD-205 (TR-205)  CPD-206 (TR-206)  CPD-207 (TR-207)  CPD-208 (TR-208)  CPD-209 (TR-209)  CPD-210 (TR-210)  CPD-211 (TR-211) 
 CPD-212 (TR-212) 0  CPD-213 (TR-213)  CPD-214 (TR-214)  CPD-215 (TR-215)  CPD-216 (TR-216)  CPD-217 (TR-217)  CPD-218 (TR-218) 
 CPD-219 (TR-219)  CPD-220 (TR-220)  CPD-221 (TR-221)  CPD-222 (TR-222) 0  CPD-223 (TR-223)  CPD-224 (TR-224)  CPD-225 (TR-225) 
 CPD-226 (TR-226)  CPD-227 (TR-227)  CPD-228 (TR-228)  CPD-229 (TR-229)  CPD-230 (TR-230) 
 CPD-231 (TR-231)  CPD-232 (TR-232) 0  CPD-233 (TR-233)  CPD-234 (TR-234)  CPD-235 (TR-235) 
 CPD-236 (TR-236)  CPD-237 (TR-237)  CPD-238 (TR-238)  CPD-239 (TR-239)  CPD-240 (TR-240)  CPD-241 (TR-241)  CPD-242 (TR-242) 
0  CPD-243 (TR-243)  CPD-244 (TR-244)  CPD-245 (TR-245)  CPD-246 (TR-246)  CPD-247 (TR-247) 
 CPD-248 (TR-248)  CPD-249 (TR-249)  CPD-250 (TR-250)  CPD-251 (TR-251)  CPD-252 (TR-252) 0  CPD-253 (TR-253)  CPD-254 (TR-254) 
 CPD-255 (TR-255)  CPD-256 (TR-256)  CPD-257 (TR-257)  CPD-258 (TR-258)  CPD-259 (TR-259)  CPD-260 (TR-260)  CPD-261 (TR-261) 
 CPD-262 (TR-262) 0  CPD-263 (TR-263)  CPD-264 (TR-264)  CPD-265 (TR-265)  CPD-266 (TR-266) 
 CPD-267 (TR-267)  CPD-268 (TR-268)  CPD-269 (TR-269)  CPD-270 (TR-270)  CPD-271 (TR-271)  CPD-272 (TR-272) 00  CPD-273 (TR-273) 01  CPD-274 (TR-274) 02  CPD-275 (TR-275) 03 
 CPD-276 (TR-276) 04  CPD-277 (TR-277) 05  CPD-278 (TR-278) 06  CPD-279 (TR-279) 07 
 CPD-280 (TR-280) 08  CPD-281 (TR-281) 09  CPD-282 (TR-282) 0  CPD-283 (TR-283) 
 CPD-284 (TR-284)  CPD-285 (TR-285)  CPD-286 (TR-286)  CPD-287 (TR-287)  CPD-288 (TR-288)  CPD-289 (TR-289)  CPD-290 (TR-290) 
 CPD-291 (TR-291)  CPD-292 (TR-292) 0  CPD-293 (TR-293)  CPD-294 (TR-294)  CPD-295 (TR-295)  CPD-296 (TR-296)  CPD-297 (TR-297) 
 CPD-298 (TR-298)  CPD-299 (TR-299)  CPD-300 (TR-300)  CPD-301 (TR-301)  CPD-302 (TR-302) 0

 CPD-303 (TR-303)  CPD-304 (TR-304) 
CPD-305 (TR-305)  CPD-306 (TR-306)  CPD-307 (TR-307)
 CPD-308 (TR-308)  CPD-309 (TR-309) 
CPD-310 (TR-310)  CPD-311 (TR-311)  CPD-312 (TR-312)
0  CPD-313 (TR-313)  CPD-314 (TR-314)
 CPD-315 (TR-315)  CPD-316 (TR-316) 
CPD-317 (TR-317)  CPD-318 (TR-318)  CPD-319 (TR-319)
 CPD-320 (TR-320)  CPD-321 (TR-321) 
CPD-322 (TR-322) 0  CPD-323 (TR-323)  CPD-324 (TR-
324)  CPD-325 (TR-325)  CPD-326 (TR-326)
 CPD-327 (TR-327)  CPD-328 (TR-328) 
CPD-329 (TR-329)  CPD-330 (TR-330)  CPD-331 (TR-331)
 CPD-332 (TR-332) 0  CPD-333 (TR-333)
 CPD-334 (TR-334)  CPD-335 (TR-335) 
CPD-336 (TR-336)  CPD-337 (TR-337)  CPD-338 (TR-338)
 CPD-339 (TR-339)  CPD-340 (TR-340) 
CPD-341 (TR-341)  CPD-342 (TR-342) 0  CPD-343 (TR-
343)  CPD-344 (TR-344)  CPD-345 
CPD-346  CPD-347  CPD-348  CPD-349
 CPD-350  CPD-351  CPD-352 0
 CPD-353  CPD-354  CPD-355
 CPD-356  CPD-357  CPD-358
 CPD-359  CPD-360  CPD-361
 CPD-362 0  CPD-363  CPD-364
 CPD-365  CPD-366  CPD-367
 CPD-368  CPD-369  CPD-370
 CPD-371  CPD-372 00  CPD-373 01
 CPD-374 02  CPD-375 03  CPD-376 04
 CPD-377 05  CPD-378 06  CPD-379 07
 CPD-380 08  CPD-381 09  CPD-382 0
 CPD-383  CPD-384  CPD-385
 CPD-386  CPD-387  CPD-388
 CPD-389  CPD-390  CPD-391
 CPD-392 0  CPD-393  CPD-394
 CPD-395  CPD-396  CPD-397
 CPD-398  CPD-399  CPD-400
 CPD-401  CPD-402 0  CPD-403
 CPD-404  CPD-405  CPD-406
 CPD-407  CPD-408  CPD-409
 CPD-410  CPD-411  CPD-412 0
 CPD-413  CPD-414  CPD-415
 CPD-416  CPD-417  CPD-418
 CPD-419  CPD-420  CPD-421
 CPD-422 0  CPD-423  CPD-424
 CPD-425  CPD-426  CPD-427
 CPD-428  CPD-429  CPD-430
 CPD-431  CPD-432 0  CPD-433
 CPD-434  CPD-435  CPD-436
 CPD-437  CPD-438  CPD-439
 CPD-440  CPD-441  CPD-442 0

 embedded image CPD-443  embedded image CPD-444  embedded image CPD-445
 embedded image CPD-446  embedded image CPD-447  embedded image CPD-448
 embedded image CPD-449  embedded image CPD-450  embedded image CPD-451
 embedded image CPD-452 0  embedded image CPD-453  embedded image CPD-454
 embedded image CPD-455  embedded image CPD-456  embedded image CPD-457
 embedded image CPD-458  embedded image CPD-459  embedded image CPD-460
 embedded image CPD-461  embedded image CPD-462 0  embedded image CPD-463
 embedded image CPD-464  embedded image CPD-465  embedded image CPD-466
 embedded image CPD-467  embedded image CPD-468  embedded image CPD-469
 embedded image CPD-470  embedded image CPD-471  embedded image CPD-472 00
 embedded image CPD-473 01  embedded image CPD-474 02  embedded image CPD-475 03
 embedded image CPD-476 04  embedded image CPD-477 05  embedded image CPD-478 06
 embedded image CPD-479 07  embedded image CPD-480 08  embedded image CPD-481 09
 embedded image CPD-482 0  embedded image CPD-483  embedded image CPD-484
 embedded image CPD-485  embedded image CPD-486  embedded image CPD-487
 embedded image CPD-488  embedded image CPD-489  embedded image CPD-490
 embedded image CPD-491  embedded image CPD-492 0  embedded image CPD-493
 embedded image CPD-494  embedded image CPD-495  embedded image CPD-496
 embedded image CPD-497  embedded image CPD-498  embedded image CPD-499
 embedded image CPD-500  embedded image CPD-501  embedded image CPD-502 0
 embedded image CPD-503  embedded image CPD-504  embedded image CPD-505
 embedded image CPD-506  embedded image CPD-507  embedded image CPD-508
 embedded image CPD-509  embedded image CPD-510  embedded image CPD-511
 embedded image CPD-512 0  embedded image CPD-513  embedded image CPD-514
 embedded image CPD-515  embedded image CPD-516  embedded image Cpd. Code
Chemical Name CPD-001 (2S,4R)-1-((S)-2-(6-(4-(6-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)-6-oxohexanamido)-3,3-dimethylbutanoyl)-4-hydroxy-N-(4-(4-methylthiazol- (TR-001) 5-yl)benzyl)pyrrolidine-2-carboxamide CPD-002 (2S,4R)-1-((S)-2-(2-(2-(2-(4-(6-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)-2-oxoethoxy)ethoxy)acetamido)-3,3-dimethylbutanoyl)- (TR-002) 4-hydroxy-N-(4-(4-methylthiazol-5-yl)benzyl)pyrrolidine-2-carboxamide CPD-003 (2S,4R)-1-((S)-2-(tert-butyl)-20-(4-(6-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)-4,20-dioxo-6,9,12,15,18-pentaoxa-3-azaicosanoyl)-4- (TR-003) (hydroxy-N-(4-(4-methylthiazol-5-yl)benzyl)pyrrolidine-2-carboxamide CPD-004 (2S,4R)-1-((S)-2-(7-(4-(6-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)-7-oxoheptanamido)-3,3-dimethylbutanoyl)-4-hydroxy-N-(4-(4- (TR-004) methylthiazol-5-yl)benzyl)pyrrolidine-2-carboxamide CPD-005 (2S,4R)-1-((S)-2-(tert-butyl)-22-(4-(6-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)-4,22-dioxo-7,10,13,16,19-pentaoxa-3-azadocosanoyl)-4- (TR-005) hydroxy-N-(4-(4-methylthiazol-5-yl)benzyl)pyrrolidine-2-carboxamide CPD-006 (2S,4R)-1-((S)-2-(4-(4-(6-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)-4-oxobutanamido)-3,3-dimethylbutanoyl)-4-hydroxy-N-(4-(4- (TR-006) methylthiazol-5-yl)benzyl)pyrrolidine-2-carboxamide CPD-007 (2S,4R)-1-((S)-2-(3-(3-(4-(6-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)-3-oxopropoxy)propanamido)-3,3-dimethylbutanoyl)-4-hydroxy-N-(4-(4- (TR-007) methylthiazol-5-yl)benzyl)pyrrolidine-2-carboxamide CPD-008 (2S,4R)-1-((S)-2-(2-(2-(4-(6-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)-2-oxoethoxy)acetamido)-3,3-dimethylbutanoyl)-4-hydroxy-N-(4-(4- (TR-008) methylthiazol-5-yl)benzyl)pyrrolidine-2-carboxamide CPD-009 2-(2,6-dioxopiperidin-3-yl)-4-((7-(4-(6-(6-((R)-2-(3-

fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)-7-oxoheptyl)amino)isoindoline-1,3-dione (TR-009) CPD-010 2-(2,6-dioxopiperidin-3-yl)-4-((2-(2-(2-(3-(4-(6-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)-3-oxopropoxy)ethoxy)ethoxy)ethyl)amino) (TR-010) isoindoline-1,3-dione CPD-011 (2S,4R)-1-((S)-2-(tert-butyl)-14-(4-(6-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)-4,14-dioxo-6,9,12-trioxa-3-azatetradecanoyl)-4- (TR-011) hydroxy-N-(4-(4-methylthiazol-5-yl)benzyl)pyrrolidine-2-carboxamide CPD-012 (2S,4R)-1-((S)-2-(5-(4-(6-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)-5-oxopentanamido)-3,3-dimethylbutanoyl)-4-hydroxy-N-(4- (TR-012) (4-methylthiazol-5-yl)benzyl)pyrrolidine-2-carboxamide CPD-013 2-(2,6-dioxopiperidin-3-yl)-4-((4-(4-(6-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)-4-oxobutyl)amino)isoindoline-1,3-dione (TR-013) CPD-014 2-(2,6-dioxopiperidin-3-yl)-4-((5-(4-(6-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)-5-oxopentyl)amino)isoindoline-1,3-dione (TR-014) CPD-015 2-(2,6-dioxopiperidin-3-yl)-4-((6-(4-(6-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)-6-oxohexyl)amino)isoindoline-1,3-dione (TR-015) CPD-016 (2S,4R)-1-((S)-2-(tert-butyl)-16-(4-(6-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)-4,16-dioxo-7,10,13-trioxa-3-azahexadecanoyl)-4-hydroxy-N- (TR-016) (4-(4-methylthiazol-5-yl)benzyl)pyrrolidine-2-carboxamide CPD-017 (2S,4R)-1-((S)-2-(10-(4-(6-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)-10-oxodecanamido)-3,3-dimethylbutanoyl)-4-hydroxy-N-(4-(4- (TR-017) methylthiazol-5-yl)benzyl)pyrrolidine-2-carboxamide CPD-018 (2S,4R)-1-((S)-2-(9-(4-(6-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)-9-oxononanamido)-3,3-dimethylbutanoyl)-4-hydroxy-N-(4-(4- (TR-018) methylthiazol-5-yl)benzyl)pyrrolidine-2-carboxamide CPD-019 (2S,4R)-1-((S)-2-(tert-butyl)-19-(4-(6-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)-4,19-dioxo-7,10,13,16-tetraoxa-3-azanonadecanoyl)-4-hydroxy- (TR-019) N-(4-(4-methylthiazol-5-yl)benzyl)pyrrolidine-2-carboxamide CPD-020 (2S,4R)-1-((S)-2-(3-(2-(3-(4-(6-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)-3-oxopropoxy)ethoxy)propanamido)-3,3-dimethylbutanoyl)-4- (TR-020) hydroxy-N-(4-(4-methylthiazol-5-yl)benzyl)pyrrolidine-2-carboxamide CPD-021 2-(2,6-dioxopiperidin-3-yl)-4-((15-(4-(6-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)-15-oxo-3,6,9,12-tetraoxapentadecyl)amino)isoindoline-1,3- (TR-021) dione CPD-022 (2S,4R)-1-((S)-2-(11-(4-(6-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)-11-oxoundecanamido)-3,3-dimethylbutanoyl)-4-hydroxy-N-(4-(4- (TR-022) methylthiazol-5-yl)benzyl)pyrrolidine-2-carboxamide CPD-023 2-(2,6-dioxopiperidin-3-yl)-4-((2-(6-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)-2-oxoethyl)amino)isoindoline-1,3-dione (TR-023) CPD-024 2-(2,6-dioxopiperidin-3-yl)-4-((8-(4-(6-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)-8-oxooctyl)amino)isoindoline-1,3-dione (TR-024) CPD-025 2-(2,6-dioxopiperidin-3-yl)-4-((18-(4-(6-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)-18-oxo-3,6,9,12,15-pentaoxaoctadecyl)amino) (TR-025) isoindoline-1,3-dione CPD-026 2-(2,6-dioxopiperidin-3-yl)-4-((3-(4-(6-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)-3-oxopropyl)amino)isoindoline-1,3-dione (TR-026) CPD-027 2-(2,6-dioxopiperidin-3-yl)-4-((2-(4-(6-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)-3-oxopropoxy)ethyl)amino)isoindoline-1,3-dione (TR-027) CPD-028 2-(2,6-dioxopiperidin-3-yl)-4-((2-(2-(3-(4-(6-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)-3-

oxopropoxy)ethyl)amino)isoidole-1,3-dione (TR-028) CPD-029 (2S,4R)-1-((S)-2-(8-(4-(6-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)-8-oxooctanamido)-3,3-dimethylbutanoyl)-4-hydroxy-N-(4-(4- (TR-029) methylthiazol-5-yl)benzyl)pyrrolidine-2-carboxamide CPD-030 N-(5-(3,5-difluorobenzyl)-1H-indazol-3-yl)-4-(4-(3-(2-((2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-4-yl)amino)ethoxy)propanoyl)piperazin-1-yl)-2-((tetrahydro-2H-pyran-4-yl)amino)benzamide (TR-030) CPD-031 N-(5-(3,5-difluorobenzyl)-1H-indazol-3-yl)-4-(4-(1-((2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-4-yl)amino)-3,6,9,12-tetraoxapentadecan-15-oyl)piperazin-2-yl)-1-((tetrahydro-2H-pyran-4- (TR-031) yl)amino)benzamide CPD-032 N-(5-(3,5-difluorobenzyl)-1H-indazol-3-yl)-4-(4-(3-(2-(2-((2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-4-yl)amino)ethoxy)ethoxy)propanoyl)piperazin-1-yl)-2-((tetrahydro-2H-pyran-4- (TR-033) yl)amino)benzamide CPD-033 (2S,4R)-1-((S)-2-(8-(4-(4-((5-(3,5-difluorobenzyl)-1H-indazol-3-yl)carbamoyl)-3-((tetrahydro-2H-pyran-4-yl)amino)phenyl)piperazin-1-yl)-8-oxooctanamido)-3,3-dimethylbutanoyl)-4-hydroxy-N- (TR-033) (4-(4-methylthiazol-5-yl)benzyl)pyrrolidine-2-carboxamide CPD-034 (2S,4R)-1-((S)-2-(10-(4-(4-((5-(3,5-difluorobenzyl)-1H-indazol-3-yl)carbamoyl)-3-((tetrahydro-2H-pyran-4-yl)amino)phenyl)piperazin-1-yl)-10-oxodecanamido)-3,3-dimethylbutanoyl)-4- (TR-034) hydroxy-N-(4-(4-methylthiazol-5-yl)benzyl)pyrrolidine-2-carboxamide CPD-035 (2S,4R)-1-((S)-2-((11-(4-(4-((5-(3,5-difluorobenzyl)-1H-indazol-3-yl)carbamoyl)-3-((tetrahydro-2H-pyran-4-yl)amino)phenyl)piperazin-1-yl)-11-oxoundecanamido)-3,3-dimethylbutanoyl)-4- (TR-035) hydroxy-N-(4-(4-methylthiazol-5-yl)benzyl)pyrrolidine-2-carboxamide CPD-036 (2S,4R)-1-((S)-2-(5-(4-(4-((5-(3,5-difluorobenzyl)-1H-indazol-3-yl)carbamoyl)-3-((tetrahydro-2H-pyran-4-yl)amino)phenyl)piperazin-1-yl)-5-oxopentanamido)-3,3-dimethylbutanoyl)-4- (TR-036) hydroxy-N-(4-(4-methylthiazol-5-yl)benzyl)pyrrolidine-2-carboxamide CPD-037 (2S,4R)-1-((S)-2-(9-(4-(4-((5-(3,5-difluorobenzyl)-1H-indazol-3-yl)carbamoyl)-3-((tetrahydro-2H-pyran-4-yl)amino)phenyl)piperazin-1-yl)-9-oxononanamido)-3,3-dimethylbutanoyl)-4- (TR-037) hydroxy-N-(4-(4-methylthiazol-5-yl)benzyl)pyrrolidine-2-carboxamide CPD-038 (2S,4R)-1-((S)-2-(3-(2-(3-(4-(4-((5-(3,5-difluorobenzyl)-1H-indazol-3-yl)carbamoyl)-3-((tetrahydro-2H-pyran-4-yl)amino)phenyl)piperazin-1-yl)-3-oxopropoxy)ethoxy)propanamido)-3,3- (TR-038) dimethylbutanoyl)-4-hydroxy-N-(4-(4-methylthiazol-5-yl)benzyl)pyrrolidine-2-carboxamide CPD-039 (2S,4R)-1-((S)-2-(tert-butyl)-16-(4-(4-((5-(3,5-difluorobenzyl)-1H-indazol-3-yl)carbamoyl)-3-((tetrahydro-2H-pyran-4-yl)amino)phenyl)piperazin-1-yl)-4,16-dioxo-7,10,13-trioxa-3-azaheptadecanoyl)-4- (TR-039) hydroxy-N-(4-(4-methylthiazol-5-yl)benzyl)pyrrolidine-2-carboxamide CPD-040 (2S,4R)-1-((S)-2-(tert-butyl)-20-(4-(4-((5-(3,5-difluorobenzyl)-1H-indazol-3-yl)carbamoyl)-3-((tetrahydro-2H-pyran-4-yl)amino)phenyl)piperazin-1-yl)-4,20-dioxo-6,9,12,15,18-pentaoxa-3- (TR-040) azaicosanoyl)-4-hydroxy-N-(4-(4-methylthiazol-5-yl)benzyl)pyrrolidine-2-carboxamide CPD-041 (2S,4R)-1-((S)-2-(tert-butyl)-19-(4-(4-((5-(3,5-difluorobenzyl)-1H-indazol-3-yl)carbamoyl)-3-((tetrahydro-2H-pyran-4-yl)amino)phenyl)piperazin-1-yl)-4,19-dioxo-7,10,13,16-tetraoxa-3- (TR-041) azanonadecanoyl)-4-hydroxy-N-(4-(4-methylthiazol-5-yl)benzyl)pyrrolidine-2-carboxamide CPD-042 (2S,4R)-1-((S)-2-(2-(2-(2-(4-(4-((5-(3,5-difluorobenzyl)-1H-indazol-3-yl)carbamoyl)-3-((tetrahydro-2H-pyran-4-yl)amino)phenyl)piperazin-1-yl)-2-oxoethoxy)ethoxy)acetamido)-3,3- (TR-042) dimethylbutanoyl)-4-hydroxy-N-(4-(4-methylthiazol-5-yl)benzyl)pyrrolidine-2-carboxamide CPD-043 (2S,4R)-1-((S)-2-(tert-butyl)-22-(4-(4-((5-(3,5-difluorobenzyl)-1H-indazol-3-yl)carbamoyl)-3-((tetrahydro-2H-pyran-4-yl)amino)phenyl)piperazin-1-yl)-4,22-dioxo-7,10,13,16,19-pentaoxa-3- (TR-043) azadocosanoyl)-4-hydroxy-N-(4-(4-methylthiazol-5-yl)benzyl)pyrrolidine-2-carboxamide CPD-044 (2S,4R)-1-((S)-2-(6-(4-(4-((5-(3,5-difluorobenzyl)-1H-indazol-3-yl)carbamoyl)-3-((tetrahydro-2H-pyran-4-yl)amino)phenyl)piperazin-1-yl)-6-oxohexanamido)-3,3-dimethylbutanoyl)-4-hydroxy-N- (TR-044) (4-(4-methylthiazol-5-yl)benzyl)pyrrolidine-2-carboxamide CPD-045 (2S,4R)-1-((S)-2-(2-(2-(4-(4-((5-(3,5-difluorobenzyl)-1H-indazol-3-yl)carbamoyl)-3-((tetrahydro-2H-pyran-4-yl)amino)phenyl)piperazin-1-yl)-2-

oxoethoxy)acetamido)-3,3-dimethylbutanoyl)-4- (TR-045) hydroxy-N-(4-(4-methylthiazol-5-yl)benzyl)pyrrolidine-2-carboxamide CPD-046 (2S,4R)-1-((S)-2-(4-(4-(4-((5-(3,5-difluorobenzyl)-1H-indazol-3-yl)carbamoyl)-3-((tetrahydro-2H-pyran-4-yl)amino)phenyl)piperazin-1-yl)-4-oxobutanamido)-3,3-dimethylbutanoyl)-4- (TR-046) hydroxy-N-(4-(4-methylthiazol-5-yl)benzyl)pyrrolidine-2-carboxamide CPD-047 (2S,4R)-1-((S)-2-(7-(4-(4-((5-(3,5-difluorobenzyl)-1H-indazol-3-yl)carbamoyl)-3-((tetrahydro-2H-pyran-4-yl)amino)phenyl)piperazin-1-yl)-7-oxoheptanamido)-3,3-dimethylbutanoyl)-4- (TR-047) hydroxy-N-(4-(4-methylthiazol-5-yl)benzyl)pyrrolidine-2-carboxamide CPD-048 (2S,4R)-1-((S)-2-(3-(3-(4-(4-((5-(3,5-difluorobenzyl)-1H-indazol-3-yl)carbamoyl)-3-((tetrahydro-2H-pyran-4-yl)amino)phenyl)piperazin-1-yl)-3-oxopropoxy)propanamido)-3,3-dimethylbutanoyl)- (TR-048) 4-hydroxy-N-(4-(4-methylthiazol-5-yl)benzyl)pyrrolidine-2-carboxamide CPD-049 N-(5-(3,5-difluorobenzyl)-1H-indazol-3-yl)-4-(4-(7-((2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-4-yl)amino)heptanoyl)piperazin-1-yl)-2-((tetrahydro-2H-pyran-4-yl)amino)benzamide (TR-049) CPD-050 N-(5-(3,5-difluorobenzyl)-1H-indazol-3-yl)-4-(4-(5-((2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-4-yl)amino)pentanoyl)piperazin-1-yl)-2-((tetrahydro-2H-pyran-4-yl)amino)benzamide (TR-050) CPD-051 N-(5-(3,5-difluorobenzyl)-1H-indazol-3-yl)-4-(4-(3-((2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-4-yl)amino)propanoyl)piperazin-1-yl)-2-((tetrahydro-2H-pyran-4-yl)amino)benzamide (TR-051) CPD-052 N-(5-(3,5-difluorobenzyl)-1H-indazol-3-yl)-4-(4-(8-((2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-4-yl)amino)octanoyl)piperazin-1-yl)-2-((tetrahydro-2H-pyran-4-yl)amino)benzamide (TR-052) CPD-053 N-(5-(3,5-difluorobenzyl)-1H-indazol-3-yl)-4-(4-(4-(4-((2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-4-yl)amino)butanoyl)piperazin-1-yl)-2-((tetrahydro-2H-pyran-4-yl)amino)benzamide (TR-053) CPD-054 N-(5-(3,5-difluorobenzyl)-1H-indazol-3-yl)-4-(4-(1-((2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-4-yl)amino)-3,6,9,12,15-pentaoxaoctadecan-18-oyl)piperazin-1-yl)-2-((tetrahydro-2H-pyran-4-yl)amino)benzamide (TR-054) CPD-055 N-(5-(3,5-difluorobenzyl)-1H-indazol-3-yl)-4-(4-(2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-4-yl)glycyl)piperazin-1-yl)-2-((tetrahydro-2H-pyran-4-yl)amino)benzamide (TR-055) CPD-056 N-(5-(3,5-difluorobenzyl)-1H-indazol-3-yl)-4-(4-(6-((2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-4-yl)amino)hexanoyl)piperazin-1-yl)-2-((tetrahydro-2H-pyran-4-yl)amino)benzamide (TR-056) CPD-057 N-(5-(3,5-difluorobenzyl)-1H-indazol-3-yl)-4-(4-(3-(2-(2-(2-((2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-4-yl)amino)ethoxy)ethoxy)ethoxy)propanoyl)piperazin-1-yl)-2-((tetrahydro-2H-pyran-4-yl)amino)benzamide (TR-057) CPD-058 (2S,4R)-1-((S)-2-(tert-butyl)-14-(4-(4-((5-(3,5-difluorobenzyl)-1H-indazol-3-yl)carbamoyl)-3-((tetrahydro-2H-pyran-4-yl)amino)phenyl)piperazin-1-yl)-4,14-dioxo-6,9,12-trioxa-3-azatetradecanoyl)-4- (TR-058) hydroxy-N-(4-(4-methylthiazol-5-yl)benzyl)pyrrolidine-2-carboxamide CPD-059 N-(2-(2-(2-(2-((2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-4-yl)amino)ethoxy)ethoxy)ethoxy)ethyl)-2-(4-(6-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)acetamide (TR-059) CPD-060 N-(2-((2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-4-yl)amino)ethyl)-2-(4-(6-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)acetamide (TR-060) CPD-061 (2S,4R)-1-((S)-2-(8-(2-(4-(4-((5-(3,5-difluorobenzyl)-1H-indazol-3-yl)carbamoyl)-3-((tetrahydro-2H-pyran-4-yl)amino)phenyl)piperazin-1-yl)acetamido)octanamido)-3,3-dimethylbutanoyl)-4- (TR-061) hydroxy-N-(4-(4-methylthiazol-5-yl)benzyl)pyrrolidine-2-carboxamide CPD-062 N-(5-(3,5-difluorobenzyl)-1H-indazol-3-yl)-4-(4-(2-((2-(2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-4-yl)amino)ethyl)amino)-2-oxoethyl)piperazin-1-yl)-2-((tetrahydro-2H-pyran-4-yl)amino)benzamide (TR-062) CPD-063 (2S,4R)-1-((S)-2-(2-(2-(4-(4-((5-(3,5-difluorobenzyl)-1H-indazol-3-yl)carbamoyl)-3-((tetrahydro-2H-pyran-4-yl)amino)phenyl)piperazin-1-yl)acetamido)acetamido)-3,3-dimethylbutanoyl)-4-hydroxy- (TR-063) N-(4-(4-methylthiazol-5-yl)benzyl)pyrrolidine-2-carboxamide CPD-064 N-(5-(3,5-difluorobenzyl)-1H-indazol-3-yl)-4-(4-

(14-((2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-4-yl)amino)-2-oxo-6,9,12-trioxa-3-azatetradecyl)piperazin-1-yl)-2-((tetrahydro-2H-pyran-4- (TR-064) yl)amino)benzamide CPD-065 N-(5-(3,5-difluorobenzyl)-1H-indazol-3-yl)-4-(4-(20-((2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-4-yl)amino)-2-oxo-6,9,12,15,18-pentaoxa-3-azaicosyl)piperazin-1-yl)-2-((tetrahydro-2H-pyran-4- (TR-065) yl)amino)benzamide CPD-066 (2S,4R)-1-((S)-2-(9-(2-(4-(6-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)acetamido)nonanamido)-3,3-dimethylbutanoyl)-4-hydroxy-N-(4-(4- (TR-066) methylthiazol-5-yl)benzyl)pyrrolidine-2-carboxamide CPD-067 (2S,4R)-1-((S)-2-(tert-butyl)-17-(4-(6-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)-4,16-dioxo-6,9,12-trioxa-3,15-diazaheptadecanoyl)-4-hydroxy- (TR-067) N-(4-(4-methylthiazol-5-yl)benzyl)pyrrolidine-2-carboxamide CPD-068 (2S,4R)-1-((S)-2-(tert-butyl)-14-(4-(6-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)-4,13-dioxo-6,9-dioxa-3,12-diazatetradecanoyl)-4-hydroxy- (TR-068) N-(4-(4-methylthiazol-5-yl)benzyl)pyrrolidine-2-carboxamide CPD-069 (2S,4R)-1-((S)-2-(3-(2-(2-(4-(6-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)acetamido)ethoxy)propanamido)-3,3-dimethylbutanoyl)-4- (TR-069) hydroxy-N-(4-(4-methylthiazol-5-yl)benzyl)pyrrolidine-2-carboxamide CPD-070 (2S,4R)-1-((S)-2-(6-(2-(4-(6-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)acetamido)hexanamido)-3,3-dimethylbutanoyl)-4-hydroxy-N-(4-(4- methylthiazol-5-yl)benzyl)pyrrolidine-2-carboxamide CPD-071 (2S,4R)-1-((S)-2-(4-(2-(4-(6-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)acetamido)butanamido)-3,3-dimethylbutanoyl)-4-hydroxy-N-(4-(4- methylthiazol-5-yl)benzyl)pyrrolidine-2-carboxamide CPD-072 (2S,4R)-1-((S)-2-(5-(2-(4-(6-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)acetamido)pentanamido)-3,3-dimethylbutanoyl)-4-hydroxy-N-(4-(4- methylthiazol-5-yl)benzyl)pyrrolidine-2-carboxamide CPD-073 (2S,4R)-1-((S)-2-(10-(2-(4-(6-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)acetamido)decanamido)-3,3-dimethylbutanoyl)-4-hydroxy-N-(4-(4- methylthiazol-5-yl)benzyl)pyrrolidine-2-carboxamide CPD-074 (2S,4R)-1-((S)-2-(7-(2-(4-(6-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)acetamido)heptanamido)-3,3-dimethylbutanoyl)-4-hydroxy-N-(4-(4- methylthiazol-5-yl)benzyl)pyrrolidine-2-carboxamide CPD-075 (2S,4R)-1-((S)-2-(2-(2-(4-(6-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)acetamido)acetamido)-3,3-dimethylbutanoyl)-4-hydroxy-N-(4-(4- methylthiazol-5-yl)benzyl)pyrrolidine-2-carboxamide CPD-076 (2S,4R)-1-((S)-2-(2-(2-(2-(4-(6-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)acetamido)ethoxy)acetamido)-3,3-dimethylbutanoyl)-4-hydroxy- N-(4-(4-methylthiazol-5-yl)benzyl)pyrrolidine-2-carboxamide CPD-077 (2S,4R)-1-((S)-20-(tert-butyl)-1-(4-(6-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)-2,18-dioxo-6,9,12,15-tetraoxa-3,19-diazahenicosan-21-oyl)-4- hydroxy-N-(4-(4-methylthiazol-5-yl)benzyl)pyrrolidine-2-carboxamide CPD-078 (2S,4R)-1-((S)-17-(tert-butyl)-1-(4-(6-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)-2,15-dioxo-6,9,12-trioxa-3,16-diazaoctadecan-18-oyl)-4- hydroxy-N-(4-(4-methylthiazol-5-yl)benzyl)pyrrolidine-2-carboxamide CPD-079 (2S,4R)-1-((S)-23-(tert-butyl)-1-(4-(6-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)-2,21-dioxo-6,9,12,15,18-pentaoxa-3,22-diazatetracosan-24-oyl)- 4-hydroxy-N-(4-(4-methylthiazol-5-yl)benzyl)pyrrolidine-2-carboxamide CPD-080 N-(3-((2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-4-yl)amino)propyl)-2-(4-(6-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)acetamide (TR-109) CPD-081 N-(4-((2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-4-yl)amino)butyl)-2-(4-(6-(6-((R)-2-(3-

fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)acetamide (TR-107) CPD-082 (2S,4R)-1-((S)-2-(3-(2-(4-(6-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)acetamido)propanamido)-3,3-dimethylbutanoyl)-4-hydroxy-N-(4-(4-methylthiazol-5-yl)benzyl)pyrrolidine-2-carboxamide CPD-083 (2S,4R)-1-((S)-2-((11-(2-(4-(6-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)acetamido)undecanamido)-3,3-dimethylbutanoyl)-4-hydroxy-N-(4-(4-methylthiazol-5-yl)benzyl)pyrrolidine-2-carboxamide CPD-084 (2S,4R)-1-((S)-14-(tert-butyl)-1-(4-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)-2,12-dioxo-6,9-dioxo-3,13-diazapentadecan-15-oyl)-4-hydroxy-N-(4-(4-methylthiazol-5-yl)benzyl)pyrrolidine-2-carboxamide CPD-085 N-(2-(2-((2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-4-yl)amino)ethoxy)ethyl)-2-(4-(6-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)acetamide (TR-106) CPD-086 N-(2-(2-(2-((2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-4-yl)amino)ethoxy)ethoxy)ethyl)-2-(4-(6-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl) (TR-105) acetamide CPD-087 N-(14-((2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-4-yl)amino)-3,6,9,12-tetraoxatetradecyl)-2-(4-(6-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl) (TR-108) yl)acetamide CPD-088 N-(17-((2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-4-yl)amino)-3,6,9,12,15-pentaoxaheptadecyl)-2-(4-(6-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl) (TR-104) piperazin-1-yl)acetamide CPD-089 (2S,4R)-1-((S)-2-(8-(2-(4-(6-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)acetamido)octanamido)-3,3-dimethylbutanoyl)-4-hydroxy-N-(4-(4-methylthiazol-5-yl)benzyl)pyrrolidine-2-carboxamide CPD-090 N-(6-((2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-4-yl)amino)hexyl)-2-(4-(6-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)acetamide (TR-111) CPD-091 N-(7-((2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-4-yl)amino)heptyl)-2-(4-(6-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)acetamide (TR-110) CPD-092 N-(8-((2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-4-yl)amino)octyl)-2-(4-(6-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)acetamide (TR-102) CPD-093 N-(2-(2-(2-(2-((2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-4-yl)amino)ethoxy)ethoxy)ethoxy)ethyl)-2-(4-(6-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2- (TR-059) yl)piperazin-1-yl)acetamide CPD-094 (2S,4R)-1-((S)-2-(9-(2-(4-(4-((5-(3,5-difluorobenzyl)-1H-indazol-3-yl)carbamoyl)-3-((tetrahydro-2H-pyran-4-yl)amino)phenyl)piperazin-1-yl)acetamido)nonanamido)-3,3-dimethylbutanoyl)-4-hydroxy-N-(4-(4-methylthiazol-5-yl)benzyl)pyrrolidine-2-carboxamide CPD-095 (2S,4R)-1-((S)-2-(tert-butyl)-17-(4-(4-((5-(3,5-difluorobenzyl)-1H-indazol-3-yl)carbamoyl)-3-((tetrahydro-2H-pyran-4-yl)amino)phenyl)piperazin-1-yl)-4,16-dioxo-6,9,12-trioxa-3,15-diazaheptadecanoyl)-4-hydroxy-N-(4-(4-methylthiazol-5-yl)benzyl)pyrrolidine-2-carboxamide CPD-096 (2S,4R)-1-((S)-2-(3-(2-(2-(4-(4-((5-(3,5-difluorobenzyl)-1H-indazol-3-yl)carbamoyl)-3-((tetrahydro-2H-pyran-4-yl)amino)phenyl)piperazin-1-yl)acetamido)ethoxy)propanamido)-3,3-dimethylbutanoyl)-4-hydroxy-N-(4-(4-methylthiazol-5-yl)benzyl)pyrrolidine-2-carboxamide CPD-097 (2S,4R)-1-((S)-2-(4-(2-(4-(4-((5-(3,5-difluorobenzyl)-1H-indazol-3-yl)carbamoyl)-3-((tetrahydro-2H-pyran-4-yl)amino)phenyl)piperazin-1-yl)acetamido)butanamido)-3,3-dimethylbutanoyl)-4-hydroxy-N-(4-(4-methylthiazol-5-yl)benzyl)pyrrolidine-2-carboxamide CPD-098 (2S,4R)-1-((S)-2-(3-(2-(4-(4-((5-(3,5-difluorobenzyl)-1H-indazol-3-yl)carbamoyl)-3-((tetrahydro-2H-pyran-4-yl)amino)phenyl)piperazin-1-yl)acetamido)propanamido)-3,3-dimethylbutanoyl)-4-hydroxy-N-(4-(4-methylthiazol-5-yl)benzyl)pyrrolidine-2-carboxamide CPD-099 (2S,4R)-1-((S)-2-(5-(2-(4-(4-((5-(3,5-difluorobenzyl)-1H-indazol-3-yl)carbamoyl)-3-((tetrahydro-2H-pyran-4-yl)amino)phenyl)piperazin-1-yl)acetamido)pentanamido)-3,3-dimethylbutanoyl)-4-hydroxy-N-(4-

(4-methylthiazol-5-yl)benzyl)pyrrolidine-2-carboxamide CPD-100 (2S,4R)-1-((S)-2-(10-(2-(4-(4-((5-(3,5-difluorobenzyl)-1H-indazol-3-yl)carbamoyl)-3-((tetrahydro-2H-pyran-4-yl)amino)phenyl)piperazin-1-yl)acetamido)decanamido)-3,3-dimethylbutanoyl)-4-hydroxy-N-(4-(4-methylthiazol-5-yl)benzyl)pyrrolidine-2-carboxamide CPD-101 (2S,4R)-1-((S)-2-(2-(2-(2-(4-(4-((5-(3,5-difluorobenzyl)-1H-indazol-3-yl)carbamoyl)-3-((tetrahydro-2H-pyran-4-yl)amino)phenyl)piperazin-1-yl)acetamido)ethoxy)acetamido)-3,3-dimethylbutanoyl)-4-hydroxy-N-(4-(4-methylthiazol-5-yl)benzyl)pyrrolidine-2-carboxamide CPD-102 (2S,4R)-1-((S)-2-(11-(2-(4-(4-((5-(3,5-difluorobenzyl)-1H-indazol-3-yl)carbamoyl)-3-((tetrahydro-2H-pyran-4-yl)amino)phenyl)piperazin-1-yl)acetamido)undecanamido)-3,3-dimethylbutanoyl)-4-hydroxy-N-(4-(4-methylthiazol-5-yl)benzyl)pyrrolidine-2-carboxamide CPD-103 (2S,4R)-1-((S)-2-(7-(2-(4-(4-((5-(3,5-difluorobenzyl)-1H-indazol-3-yl)carbamoyl)-3-((tetrahydro-2H-pyran-4-yl)amino)phenyl)piperazin-1-yl)acetamido)heptanamido)-3,3-dimethylbutanoyl)-4-hydroxy-N-(4-(4-methylthiazol-5-yl)benzyl)pyrrolidine-2-carboxamide CPD-104 (2S,4R)-1-((S)-2-(tert-butyl)-14-(4-(4-((5-(3,5-difluorobenzyl)-1H-indazol-3-yl)carbamoyl)-3-((tetrahydro-2H-pyran-4-yl)amino)phenyl)piperazin-1-yl)-4,13-dioxo-6,9-dioxo-3,12-diazatetradecanoyl)-4-hydroxy-N-(4-(4-methylthiazol-5-yl)benzyl)pyrrolidine-2-carboxamide CPD-105 (2S,4R)-1-((S)-2-(6-(2-(4-(4-((5-(3,5-difluorobenzyl)-1H-indazol-3-yl)carbamoyl)-3-((tetrahydro-2H-pyran-4-yl)amino)phenyl)piperazin-1-yl)acetamido)hexanamido)-3,3-dimethylbutanoyl)-4-hydroxy-N-(4-(4-methylthiazol-5-yl)benzyl)pyrrolidine-2-carboxamide CPD-106 N-(5-(3,5-difluorobenzyl)-1H-indazol-3-yl)-4-(4-(2-((4-((2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-4-yl)amino)butyl)amino)-2-oxoethyl)piperazin-1-yl)-2-((tetrahydro-2H-pyran-4-yl) (TR-155) amino)benzamide CPD-107 N-(5-(3,5-difluorobenzyl)-1H-indazol-3-yl)-4-(4-(2-((3-(2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-4-yl)amino)propyl)amino)-2-oxoethyl)piperazin-1-yl)-2-((tetrahydro-2H-pyran-4-yl)amino) (TR-152) benzamide CPD-108 (2S,4R)-1-((S)-23-(tert-butyl)-1-(4-(4-((5-(3,5-difluorobenzyl)-1H-indazol-3-yl)carbamoyl)-3-((tetrahydro-2H-pyran-4-yl)amino)phenyl)piperazin-1-yl)-2,21-dioxo-6,9,12,15,18-pentaoxa-3,22-diazatetracosan-24-oyl)-4-hydroxy-N-(4-(4-methylthiazol-5-yl)benzyl)pyrrolidine-2-carboxamide CPD-109 (2S,4R)-1-((S)-17-(tert-butyl)-1-(4-(4-((5-(3,5-difluorobenzyl)-1H-indazol-3-yl)carbamoyl)-3-((tetrahydro-2H-pyran-4-yl)amino)phenyl)piperazin-1-yl)-2,5-dioxo-6,9,12-trioxa-3,16-diazaoctadecan-18-oyl)-4-hydroxy-N-(4-(4-methylthiazol-5-yl)benzyl)pyrrolidine-2-carboxamide CPD-110 (2S,4R)-1-((S)-20-(tert-butyl)-1-(4-(4-((5-(3,5-difluorobenzyl)-1H-indazol-3-yl)carbamoyl)-3-((tetrahydro-2H-pyran-4-yl)amino)phenyl)piperazin-1-yl)-2,18-dioxo-6,9,12,15-tetraoxa-3,19-diazahenicosan-21-oyl)-4-hydroxy-N-(4-(4-methylthiazol-5-yl)benzyl)pyrrolidine-2-carboxamide CPD-111 N-(5-(3,5-difluorobenzyl)-1H-indazol-3-yl)-4-(4-(2-((2-(2-((2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-4-yl)amino)ethoxy)ethyl)amino)-2-oxoethyl)piperazin-1-yl)-2-((tetrahydro-2H- (TR-151) pyran-4-yl)amino)benzamide CPD-112 N-(5-(3,5-difluorobenzyl)-1H-indazol-3-yl)-4-(4-(2-((2-(2-((2-((2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-4-yl)amino)ethoxy)ethoxy)ethyl)amino)-2-oxoethyl)piperazin-1-yl)-2-((tetrahydro-2H- (TR-153) pyran-4-yl)amino)benzamide CPD-113 N-(5-(3,5-difluorobenzyl)-1H-indazol-3-yl)-4-(4-(17-((2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-4-yl)amino)-2-oxo-6,9,12,15-tetraoxa-3-azaheptadecyl)piperazin-1-yl)-2-((tetrahydro-2H- (TR-150) pyran-4-yl)amino)benzamide CPD-114 N-(5-(3,5-difluorobenzyl)-1H-indazol-3-yl)-4-(4-(2-((5-((2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-4-yl)amino)pentyl)amino)-2-oxoethyl)piperazin-1-yl)-2-((tetrahydro-2H-pyran-4-yl)amino) (TR-149) benzamide CPD-115 N-(5-(3,5-difluorobenzyl)-1H-indazol-3-yl)-4-(4-(2-((6-((2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-4-yl)amino)hexyl)amino)-2-oxoethyl)piperazin-1-yl)-2-((tetrahydro-2H- (TR-147) pyran-4-yl)amino)benzamide CPD-116 N-(5-(3,5-difluorobenzyl)-1H-indazol-3-yl)-4-(4-(2-((8-((2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-4-yl)amino)octyl)amino)-2-oxoethyl)piperazin-1-yl)-2-((tetrahydro-2H-pyran-4- (TR-148) yl)amino)benzamide CPD-117 N-(5-(3,5-difluorobenzyl)-1H-indazol-3-yl)-4-(4-(2-((7-((2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-4-

yl)amino)heptyl)amino)-2-oxoethyl)piperazin-1-yl)-2-((tetrahydro-2H-pyran-4-yl)amino) (TR-154) benzamide CPD-118 (2S,4R)-1-((S)-14-(tert-butyl)-1-(4-(4-((5-(3,5-difluorobenzyl)-1H-indazol-3-yl)carbamoyl)-3-((tetrahydro-2H-pyran-4-yl)amino)phenyl)piperazin-1-yl)-2,12-dioxo-6,9-dioxo-3,13-diazapentadecan-15-oyl)-4-hydroxy-N-(4-(4-methylthiazol-5-yl)benzyl)pyrrolidine-2-carboxamide CPD-119 N-(5-(3,5-difluorobenzyl)-1H-indazol-3-yl)-4-(4-(2-((2-(2-((2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-5-yl)amino)ethoxy)ethyl)amino)-2-oxoethyl)piperazin-1-yl)-2-((tetrahydro-2H-pyran-4-yl) (TR-161) amino)benzamide CPD-120 N-(5-(3,5-difluorobenzyl)-1H-indazol-3-yl)-4-(4-(2-(((2-(2-((2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-5-yl)amino)ethoxy)ethoxy)methyl)amino)-2-oxoethyl)piperazin-1-yl)-2-((tetrahydro-2H-pyran-4-yl)amino)benzamide CPD-121 N-(5-(3,5-difluorobenzyl)-1H-indazol-3-yl)-4-(4-(14-((2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-5-yl)amino)-2-oxo-6,9,12-trioxa-3-azatetradecyl)piperazin-1-yl)-2-((tetrahydro-2H-pyran-4- (TR-163) yl)amino)benzamide CPD-122 N-(5-(3,5-difluorobenzyl)-1H-indazol-3-yl)-4-(4-(17-((2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-5-yl)amino)-2-oxo-6,9,12,15-tetraoxa-3-azaheptadecyl)piperazin-1-yl)-2-((tetrahydro-2H-pyran-4- (TR-160) yl)amino)benzamide CPD-123 N-(5-(3,5-difluorobenzyl)-1H-indazol-3-yl)-4-(4-(20-((2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-5-yl)amino)-2-oxo-6,9,12,15,18-pentaoxa-3-azaicosyl)piperazin-1-yl)-2-((tetrahydro-2H-pyran-4- (TR-166) yl)amino)benzamide CPD-124 N-(5-(3,5-difluorobenzyl)-1H-indazol-3-yl)-4-(4-((2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-5-yl)glycyl)piperazin-1-yl)-2-((tetrahydro-2H-pyran-4-yl)amino)benzamide (TR-144) CPD-125 N-(5-(3,5-difluorobenzyl)-1H-indazol-3-yl)-4-(4-(3-((2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-5-yl)amino)propanoyl)piperazin-1-yl)-2-((tetrahydro-2H-pyran-4-yl)amino)benzamide (TR-145) CPD-126 N-(5-(3,5-difluorobenzyl)-1H-indazol-3-yl)-4-(4-(4-((2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-5-yl)amino)butanoyl)piperazin-1-yl)-2-((tetrahydro-2H-pyran-4-yl)amino)benzamide (TR-169) CPD-127 N-(5-(3,5-difluorobenzyl)-1H-indazol-3-yl)-4-(4-(5-((2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-5-yl)amino)pentanoyl)piperazin-1-yl)-2-((tetrahydro-2H-pyran-4-yl)amino)benzamide (TR-142) CPD-128 N-(5-(3,5-difluorobenzyl)-1H-indazol-3-yl)-4-(4-(6-((2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-5-yl)amino)hexanoyl)piperazin-1-yl)-2-((tetrahydro-2H-pyran-4-yl)amino)benzamide (TR-138) CPD-129 N-(5-(3,5-difluorobenzyl)-1H-indazol-3-yl)-4-(4-(7-((2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-5-yl)amino)heptanoyl)piperazin-1-yl)-2-((tetrahydro-2H-pyran-4-yl)amino)benzamide (TR-139) CPD-130 N-(5-(3,5-difluorobenzyl)-1H-indazol-3-yl)-4-(4-(8-((2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-5-yl)amino)octanoyl)piperazin-1-yl)-2-((tetrahydro-2H-pyran-4-yl)amino)benzamide (TR-136) CPD-131 N-(5-(3,5-difluorobenzyl)-1H-indazol-3-yl)-4-(4-(2-((2-((2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-5-yl)amino)ethyl)amino)-2-oxoethyl)piperazin-1-yl)-2-((tetrahydro-2H-pyran-4-yl)amino) (TR-165) benzamide CPD-132 N-(5-(3,5-difluorobenzyl)-1H-indazol-3-yl)-4-(4-(2-((3-((2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-5-yl)amino)propyl)amino)-2-oxoethyl)piperazin-1-yl)-2-((tetrahydro-2H-pyran-4-yl)amino) (TR-167) benzamide CPD-133 N-(5-(3,5-difluorobenzyl)-1H-indazol-3-yl)-4-(4-(2-((4-((2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-5-yl)amino)butyl)amino)-2-oxoethyl)piperazin-1-yl)-2-((tetrahydro-2H-pyran-4-yl)amino) (TR-156) benzamide CPD-134 N-(5-(3,5-difluorobenzyl)-1H-indazol-3-yl)-4-(4-(2-((5-((2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-5-yl)amino)pentyl)amino)-2-oxoethyl)piperazin-1-yl)-2-((tetrahydro-2H-pyran-4-yl)amino) (TR-164) benzamide CPD-135 N-(5-(3,5-difluorobenzyl)-1H-indazol-3-yl)-4-(4-(2-((6-((2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-5-yl)amino)hexyl)amino)-2-oxoethyl)piperazin-1-yl)-2-((tetrahydro-2H-pyran-4-yl)amino) (TR-158) benzamide CPD-136 N-(5-(3,5-difluorobenzyl)-1H-indazol-3-yl)-4-(4-(2-((7-((2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-5-yl)amino)heptyl)amino)-2-oxoethyl)piperazin-1-yl)-2-((tetrahydro-2H-pyran-4-yl)amino) (TR-157) benzamide CPD-137 N-(5-(3,5-difluorobenzyl)-1H-indazol-3-yl)-4-(4-(2-((8-((2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-5-yl)amino)octyl)amino)-2-oxoethyl)piperazin-1-yl)-2-((tetrahydro-2H-pyran-4-yl)amino) (TR-159)

benzamide CPD-138 N-(5-(3,5-difluorobenzyl)-1H-indazol-3-yl)-4-(4-(3-(2-((2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-5-yl)amino)ethoxy)propanoyl)piperazin-1-yl)-2-((tetrahydro-2H-pyran-4-yl)amino)benzamide (TR-143) CPD-139 N-(5-(3,5-difluorobenzyl)-1H-indazol-3-yl)-4-(4-(3-(2-(2-((2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-5-yl)amino)ethoxy)ethoxy)propanoyl)piperazin-1-yl)-2-((tetrahydro-2H-pyran-4-yl)amino) (TR-141) benzamide CPD-140 N-(5-(3,5-difluorobenzyl)-1H-indazol-3-yl)-4-(4-(3-(2-(2-(2-((2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-5-yl)amino)ethoxy)ethoxy)ethoxy)propanoyl)piperazin-1-yl)-2-((tetrahydro-2H-pyran-4- (TR-146) yl)amino)benzamide CPD-141 N-(5-(3,5-difluorobenzyl)-1H-indazol-3-yl)-4-(4-(1-((2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-5-yl)amino)-3,6,9,12-tetraoxapentadecan-15-oyl)piperazin-1-yl)-2-((tetrahydro-2H-pyran-4- (TR-140) yl)amino)benzamide CPD-142 N-(5-(3,5-difluorobenzyl)-1H-indazol-3-yl)-4-(4-(1-((2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-5-yl)amino)-3,6,9,12,15-pentaoxaoctadecan-18-oyl)piperazin-1-yl)-2-((tetrahydro-2H-pyran-4- (TR-137) yl)amino)benzamide CPD-143 N-(2-(2-((2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-5-yl)amino)ethoxy)ethyl)-2-(4-(6-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)acetamide (TR-135) CPD-144 N-(2-(2-(2-((2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-5-yl)amino)ethoxy)ethoxy)ethyl)-2-(4-(6-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1- (TR-134) yl)acetamide CPD-145 N-(2-((2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-5-yl)amino)ethyl)-2-(4-(6-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)acetamide (TR-130) CPD-146 N-(6-((2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-5-yl)amino)hexyl)-2-(4-(6-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)acetamide (TR-128) CPD-147 2-(2,6-dioxopiperidin-3-yl)-5-((15-(4-(6-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)-15-oxo-3,6,9,12-tetraoxapentadecyl)amino)isoindoline-1,3- (TR-120) dione CPD-148 2-(2,6-dioxopiperidin-3-yl)-5-((2-(3-(4-(6-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)-3-oxopropoxy)ethyl)amino)isoindoline-1,3-dione (TR-119) CPD-149 2-(2,6-dioxopiperidin-3-yl)-5-((18-(4-(6-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)-18-oxo-3,6,9,12,15-pentaoxaoctadecyl)amino)isoindoline-1,3- (TR-116) dione CPD-150 2-(2,6-dioxopiperidin-3-yl)-5-((2-(2-(3-(4-(6-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)-3-oxopropoxy)ethoxy)ethyl)amino)isoindoline-1,3-dione (TR-113) CPD-151 2-(2,6-dioxopiperidin-3-yl)-5-((2-(2-(2-(3-(4-(6-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)-3-oxopropoxy)ethoxy)ethoxy)ethyl)amino) (TR-114) isoindoline-1,3-dione CPD-152 2-(2,6-dioxopiperidin-3-yl)-5-((6-(4-(6-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)-6-oxohexyl)amino)isoindoline-1,3-dione (TR-122) CPD-153 2-(2,6-dioxopiperidin-3-yl)-5-((7-(4-(6-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)-7-oxoheptyl)amino)isoindoline-1,3-dione (TR-117) CPD-154 N-(2-(2-(2-(2-((2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-5-yl)amino)ethoxy)ethoxy)ethoxy)ethyl)-2-(4-(6-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2- (TR-129) yl)piperazin-1-yl)acetamide CPD-155 N-(3-((2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-5-yl)amino)propyl)-2-(4-(6-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)acetamide (TR-127) CPD-156 N-(4-((2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-5-yl)amino)butyl)-2-(4-(6-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)acetamide (TR-124) CPD-157 N-(7-((2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-5-yl)amino)heptyl)-2-(4-(6-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)acetamide (TR-126) CPD-158 2-(2,6-dioxopiperidin-3-yl)-5-((4-(4-(6-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-

1-yl)-4-oxobutyl)amino)isoindoline-1,3-dione (TR-168) CPD-159 2-(2,6-dioxopiperidin-3-yl)-5-((5-(4-(6-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)-5-oxopentyl)amino)isoindoline-1,3-dione (TR-115) CPD-160 2-(2,6-dioxopiperidin-3-yl)-5-((2-(4-(6-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)-2-oxoethyl)amino)isoindoline-1,3-dione (TR-123) CPD-161 2-(2,6-dioxopiperidin-3-yl)-5-((3-(4-(6-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)-3-oxopropyl)amino)isoindoline-1,3-dione (TR-121) CPD-162 2-(2,6-dioxopiperidin-3-yl)-5-((8-(4-(6-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)-8-oxooctyl)amino)isoindoline-1,3-dione (TR-118) CPD-163 N-(5-((2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisoindolin-5-yl)amino)pentyl)-2-(4-(6-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)acetamide (TR-131) CPD-164 N-(17-((2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisoindolin-5-yl)amino)-3,6,9,12,15-pentaoxaheptadecyl)-2-(4-(6-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl) (TR-132) piperazin-1-yl)acetamide CPD-165 N-(8-((2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisoindolin-5-yl)amino)octyl)-2-(4-(6-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)acetamide (TR-133) CPD-166 N-(14-((2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisoindolin-5-yl)amino)-3,6,9,12-tetraoxatetradecyl)-2-(4-(6-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin- (TR-125) 1-yl)acetamide CPD-167 N-(5-((2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisoindolin-4-yl)amino)pentyl)-2-(4-(6-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)acetamide (TR-103) CPD-168 N-(5-(3,5-Difluorobenzyl)-1H-indazol-3-yl)-4-(4-(2-((2-(2-(2-((2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisoindolin-5-yl)amino)ethoxy)ethoxy)ethyl)amino)-2-oxoethyl)piperazin-1-yl)-2-((tetrahydro-2H- (TR-162) pyran-4-yl)amino)benzamide CPD-169 5-((2-(4-(6-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)-2-oxoethyl)amino)-2-(1-methyl-2,6-dioxopiperidin-3-yl)isoindoline-1,3-dione (TR-123- neg) CPD-170 2-(2,6-dioxopiperidin-3-yl)-5-(4-(6-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)isoindoline-1,3-dione (TR-170) CPD-171 2-(2,6-dioxopiperidin-3-yl)-5-((3-(4-(6-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)propyl)amino)isoindoline-1,3-dione (TR-171) CPD-172 2-(2,6-dioxopiperidin-3-yl)-5-(3-(4-(6-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)propyl)isoindoline-1,3-dione (TR-172) CPD-173 2-(2,6-dioxopiperidin-3-yl)-5-(2-(4-(6-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)ethoxy)isoindoline-1,3-dione (TR-173) CPD-174 2-(2,6-dioxopiperidin-3-yl)-5-((1-(6-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperidin-4-yl)amino)isoindoline-1,3-dione (TR-174) CPD-175 2-(2,6-dioxopiperidin-3-yl)-5-((1-(6-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)azetidin-3-yl)amino)isoindoline-1,3-dione (TR-175) CPD-176 2-(2,6-dioxopiperidin-3-yl)-5-((2-(2-(4-(6-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)ethoxy)ethyl)amino)isoindoline-1,3-dione (TR-176) CPD-177 2-(2,6-dioxopiperidin-3-yl)-5-((2-(2-(2-(4-(6-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)ethoxy)ethoxy)ethyl)amino)isoindoline-1,3-dione (TR-177) CPD-178 2-(2,6-dioxopiperidin-3-yl)-5-(((1-(6-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperidin-4-yl)methyl)amino)isoindoline-1,3-dione (TR-178) CPD-179 2-(2,6-dioxopiperidin-3-yl)-5-(((1-(6-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)azetidin-3-yl)methyl)amino)isoindoline-1,3-dione (TR-179) CPD-180 2-(2,6-dioxopiperidin-3-yl)-5-((2-(1-(6-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperidine-4-yl)ethyl)amino)isoindoline-1,3-dione (TR-180) CPD-181 2-(2,6-dioxopiperidin-3-yl)-5-((2-(4-(6-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-

b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)ethyl)amino)isoindoline-1,3-dione (TR-181) CPD-182 2-(2,6-Dioxopiperidin-3-yl)-5-(4-(4-(6-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)piperidin-1-yl)isoindoline-1,3-dione (TR-182) CPD-183 2-(2,6-Dioxopiperidin-3-yl)-5-(3-(4-(6-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)prop-1-yn-1-yl)isoindoline-1,3-dione (TR-183) CPD-184 2-(2,6-Dioxopiperidin-3-yl)-5-(3-(4-(6-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)azetidin-1-yl)isoindoline-1,3-dione (TR-184) CPD-185 3-(6-(3-(4-(6-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)propyl-1-oxoisoindolin-2-yl)piperidine-2,6-dione (TR-185) CPD-186 3-(5-(3-(4-(6-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)propyl)-1-oxoisoindolin-2-yl)piperidine-2,6-dione (TR-186) CPD-187 2-(2,6-Dioxopiperidin-3-yl)-5-(3-(4-(6-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)prop-1-yn-1-yl)isoindoline-1,3-dione (TR-187) CPD-188 3-(5-(3-(4-(6-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)prop-1-yn-1-yl)-1-oxoisoindolin-2-yl)piperidine-2,6-dione (TR-188) CPD-189 3-(6-((2-(4-(6-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)-2-oxoethyl)amino)-1-oxoisoindolin-2-yl)piperidine-2,6-dione (TR-189) CPD-190 3-(5-((2-(4-(6-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)-2-oxoethyl)amino)-1-oxoisoindolin-2-yl)piperidine-2,6-dione (TR-190) CPD-191 3-(5-((2-(4-(6-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)ethyl)amino)-1-oxoisoindolin-2-yl)piperidine-2,6-dione (TR-191) CPD-192 2-(2,6-Dioxopiperidin-3-yl)-5-((2-((1-(6-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)azetidin-3-yl)amino)ethyl)amino)isoindoline-1,3-dione (TR-192) CPD-193 2-(2,6-Dioxopiperidin-3-yl)-5-((2-(1-(6-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)azetidin-3-yl)ethyl)amino)isoindoline-1,3-dione (TR-193) CPD-194 2-(2,6-Dioxopiperidin-3-yl)-5-(3-(4-(6-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)-3-oxopropyl)isoindoline-1,3-dione (TR-194) CPD-195 2-(2,6-Dioxopiperidin-3-yl)-5-((E)-3-(4-(6-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)-3-oxoprop-1-en-1-yl)isoindoline-1,3-dione (TR-195) CPD-196 3-(6-((2-(4-(6-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)ethyl)amino)-1-oxoisoindolin-2-yl)piperidine-2,6-dione (TR-196) CPD-197 2-(2,6-Dioxopiperidin-3-yl)-5-((4-(6-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)methyl)isoindoline-1,3-dione (TR-197) CPD-198 2-(2,6-Dioxopiperidin-3-yl)-5-(3-((4-(6-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)methyl)azetidin-1-yl)isoindoline-1,3-dione (TR-198) CPD-199 2-(2,6-Dioxopiperidin-3-yl)-5-((3-(2-(4-(6-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)ethoxy)propyl)amino)isoindoline-1,3-dione (TR-199) CPD-200 3-(5-(2-(4-(6-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)ethoxy)-1-oxoisoindolin-2-yl)piperidine-2,6-dione (TR-200) CPD-201 3-(6-(2-(4-(6-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)ethoxy)-1-oxoisoindolin-2-yl)piperidine-2,6-dione (TR-201) CPD-202 3-(6-(3-(4-(6-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)-3-oxoprop-1-yn-1-yl)-1-oxoisoindolin-2-yl)piperidine-2,6-dione (TR-202) CPD-203 3-(5-(3-(4-(6-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)-3-oxoprop-1-yn-1-yl)-1-oxoisoindolin-2-yl)piperidine-2,6-dione (TR-203) CPD-204 2-(2,6-dioxopiperidin-3-yl)-5-(2-(4-(6-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)ethyl)isoindoline-1,3-dione (TR-204) CPD-205 3-(5-(3-((4-(6-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-

yl)methyl)azetidin-1-yl)-1-oxoisindolin-2-yl)piperidine-2,6-dione (TR-205) CPD-206 3-(5-((1-(6-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)azetidin-3-yl)amino)-1-oxoisindolin-2-yl)piperidine-2,6-dione (TR-206) CPD-207 3-(5-(4-(6-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)-1-oxoisindolin-2-yl)piperidine-2,6-dione (TR-207) CPD-208 3-(5-((1-(6-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperidin-4-yl)ethynyl)-1-oxoisindolin-2-yl)piperidine-2,6-dione (TR-208) CPD-209 N-(2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-5-yl)-1-(6-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)azetidine-3-carboxamide (TR-209) CPD-210 3-(5-(((1-(6-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)azetidin-3-yl)methyl)amino)-1-oxoisindolin-2-yl)piperidine-2,6-dione (TR-210) CPD-211 2-(2,6-dioxopiperidin-3-yl)-5-(3-(4-(6-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazine-1-carbonyl)azetidin-1-yl)isoindoline-1,3-dione (TR-211) CPD-212 3-(5-((1-(6-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperidin-4-yl)amino)-1-oxoisindolin-2-yl)piperidine-2,6-dione (TR-212) CPD-213 2-((2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-5-yl)amino)-N-(1-(6-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)azetidin-3-yl)acetamide (TR-213) CPD-214 2-(2,6-dioxo-3-piperidyl)-5-[2-[4-[6-[6-[(2R)-2-(3-fluorophenyl)pyrrolidin-1-yl]imidazo[1,2-b]pyridazin-3-yl]-2-pyridyl]piperazin-1-yl]ethyl-[(4-methoxyphenyl)methyl]amino]isoindoline-1,3-dione (TR-214) CPD-215 N-(2-(2,6-Dioxopiperidin-3-yl)-1,3-dioxoisindolin-5-yl)-1-(6-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperidine-4-carboxamide (TR-215) CPD-216 3-[4-[2-[4-[6-[6-[(2R)-2-(3-fluorophenyl)pyrrolidin-1-yl]imidazo[1,2-b]pyridazin-3-yl]-2-pyridyl]piperazin-1-yl]ethylamino]-3-methyl-2-oxo-benzimidazol-1-yl]piperidine-2,6-dione (TR-216) CPD-217 3-[4-[3-[4-[6-[6-[(2R)-2-(3-fluorophenyl)pyrrolidin-1-yl]imidazo[1,2-b]pyridazin-3-yl]-2-pyridyl]piperazin-1-yl]propylamino]-3-methyl-2-oxo-benzimidazol-1-yl]piperidine-2,6-dione (TR-217) CPD-218 3-(5-(((1-(6-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperidin-4-yl)methyl)amino)-1-oxoisindolin-2-yl)piperidine-2,6-dione (TR-218) CPD-219 3-(5-((1-(6-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)azetidin-3-yl)ethynyl)-1-oxoisindolin-2-yl)piperidine-2,6-dione (TR-219) CPD-220 3-(5-((2-(4-(6-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)ethyl)amino)-3-methyl-2-oxo-2,3-dihydro-1H-benzo (TR-220) [d]imidazol-1-yl)piperidine-2,6-dione CPD-221 3-[5-[3-[4-[6-[6-[(2R)-2-(3-fluorophenyl)pyrrolidin-1-yl]imidazo[1,2-b]pyridazin-3-yl]-2-pyridyl]piperazin-1-yl]propylamino]-3-methyl-2-oxo-benzimidazol-1-yl]piperidine-2,6-dione (TR-221) CPD-222 2-(2,6-dioxo-3-piperidyl)-5-[3-[[4-[3-[6-[(2R)-2-(3-fluorophenyl)pyrrolidin-1-yl]imidazo[1,2-b]pyridazin-3-yl]pyrrol-1-yl]-1-piperidyl]methyl]azetidin-1-yl]isoindoline-1,3-dione (TR-222) CPD-223 3-((S)-5-(3-(3-(4-(6-(6-((S)-2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)prop-1-yn-1-yl)phenyl)-2-oxooxazolidin-3-yl)piperidine-2,6-dione (TR-223) CPD-224 3-((S)-5-(4-(3-(4-(6-(6-((S)-2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)prop-1-yn-1-yl)phenyl)-2-oxooxazolidin-3-yl)piperidine-2,6-dione (TR-224) CPD-225 2-(2,6-dioxopiperidin-3-yl)-5-(3-((4-(6-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)methyl)pyrrolidin-1-yl)isoindoline-1,3-dione (TR-225) CPD-226 2-(2,6-dioxopiperidin-3-yl)-5-(4-((4-(6-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)methyl)piperidin-1-yl)isoindoline-1,3-dione (TR-226) CPD-227 3-(3-(2-(4-(6-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)ethyl)-2-oxo-2,3-dihydro-1H-benzo[d]imidazol-1-yl)piperidine-2,6-dione (TR-227) CPD-228 3-(3-(3-(4-(6-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)propyl-2-oxo-2,3-dihydro-1H-benzo[d]imidazol-1-yl)piperidine-2,6-dione (TR-

228) CPD-229 3-((S)-5-(4-((4-(6-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)methyl)phenyl)-2-oxooxazolidin-3-yl)piperidine-2,6-dione (TR-229) CPD-230 2-(2,6-dioxopiperidin-3-yl)-5-(3-((4-(5-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)-3,6-dihydropyridin-1(2H)-yl)piperidin-1-yl)methyl)azetidin-1-yl)isoindoline-1,3-dione (TR-230) CPD-231 N-[5-[(3,5-difluorophenyl)methyl]-1H-indazol-3-yl]-4-[4-[[1-[2-(2,6-dioxo-3-piperidyl)-1,3-dioxo-isoindolin-5-yl]azetidin-3-yl]methyl]piperazin-1-yl]-2-(tetrahydropyran-4-ylamino)benzamide (TR-231) CPD-232 N-[5-[(3,5-difluorophenyl)methyl]-1H-indazol-3-yl]-4-[4-[2-[[2-(2,6-dioxo-3-piperidyl)-1,3-isoindolin-5-yl]amino]ethyl]piperazin-1-yl]-2-(tetrahydropyran-4-ylamino)benzamide (TR-232) CPD-233 N-(2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisoindolin-5-yl)-2-(4-(6-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)acetamide (TR-233) CPD-234 2-(2,6-dioxopiperidin-3-yl)-5-(3-((4-(6-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-a]pyridin-3-yl)pyridin-2-yl)piperazin-1-yl)methyl)azetidin-1-yl)isoindoline-1,3-dione (TR-234) CPD-235 3-((S)-5-(3-((4-(6-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)methyl)phenyl)-2-oxooxazolidin-3-yl)piperidine-2,6-dione (TR-235) CPD-236 3-(4-((2-(4-(6-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)ethyl)amino)-2-oxoindolin-1-yl)piperidine-2,6-dione (TR-236) CPD-237 2-(2,6-dioxopiperidin-3-yl)-5-((3-(4-(6-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)azetidin-1-yl)methyl)isoindoline-1,3-dione (TR-237) CPD-238 3-(4-(3-(4-(6-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)prop-1-yn-1-yl)-3-methyl-2-oxo-2,3-dihydro-1H-benzo[d]imidazol-1-yl)piperidine-2,6-dione CPD-239 3-(4-(4-(4-(6-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)but-1-yn-1-yl)-3-methyl-2-oxo-2,3-dihydro-1H-benzo[d]imidazol-1-yl) (TR-239) piperidine-2,6-dione CPD-240 3-(4-(3-(4-(6-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)propyl)-3-methyl-2-oxo-2,3-dihydro-1H-benzo[d]imidazol-1-yl)piperidine-2,6-dione (TR-240) CPD-241 3-(4-(4-(4-(6-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)butyl)-3-methyl-2-oxo-2,3-dihydro-1H-benzo[d]imidazol-1-yl)piperidine-2,6-dione (TR-241) CPD-242 3-(3-(2-(4-(6-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-a]pyridin-3-yl)pyridin-2-yl)piperazin-1-yl)ethyl)-2-oxo-2,3-dihydro-1H-benzo[d]imidazol-1-yl)piperidine-2,6-dione (TR-242) CPD-243 3-(3-(3-(4-(6-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-a]pyridin-3-yl)pyridin-2-yl)piperazin-1-yl)propyl)-2-oxo-2,3-dihydro-1H-benzo[d]imidazol-1-yl)piperidine-2,6-dione (TR-243) CPD-244 3-(4-(2-(4-(6-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)ethoxy)-3-methyl-2-oxo-2,3-dihydro-1H-benzo[d]imidazol-1-yl)piperidine-2,6-dione (TR-244) CPD-245 3-(4-(3-(4-(6-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)propoxy)-3-methyl-2-oxo-2,3-dihydro-1H-benzo[d]imidazol-1-yl)piperidine-2,6-dione (TR-245) CPD-246 3-(5-(2-(4-(6-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)ethoxy)-3-methyl-2-oxo-2,3-dihydro-1H-benzo[d]imidazol-1-yl)piperidine-2,6-dione (TR-246) CPD-247 3-(7-((2-(4-(6-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)ethyl)amino)-2-oxobenzo[d]oxazol-3(2H)-yl)piperidine-2,6-dione (TR-247) CPD-248 2-(2,6-dioxopiperidin-3-yl)-5-((3-((4-(6-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)methyl)azetidin-1-yl)methyl)isoindoline-1,3-dione (TR-248) CPD-249 N-(5-(3,5-difluorobenzyl)-1H-indazol-3-yl)-4-(4-((1-((2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisoindolin-5-yl)methyl)azetidin-3-yl)methyl)piperazin-1-yl)-2-((tetrahydro-2H-pyran-4-yl)amino)benzamide (TR-249) CPD-250 3-(5-(3-((4-(6-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)methyl)azetidin-1-yl)-3-methyl-2-oxo-2,3-dihydro-1H-benzo[d]imidazol-1-yl)piperidine- (TR-250) 2,6-dione CPD-251 N-(5-(3,5-

di fluorobenzyl)-1H-indazol-3-yl)-4-(4-((1-((2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-5-yl)methyl)azetidin-3-yl)methyl)piperazin-1-yl)-2-((2-fluoroethyl)amino)benzamide (TR-251) CPD-252 N-(5-(3,5-difluorobenzyl)-1H-indazol-3-yl)-4-(4-(1-((2-(2,6-dioxopiperidin-yl)-1,3-dioxoisindolin-5-yl)methyl)azetidin-3-yl)piperazin-1-yl)-2-((2-fluoroethyl)amino)benzamide (TR-252) CPD-253 N-(5-(3,5-difluorobenzyl)-1H-indazol-3-yl)-4-(4-(3-(2-(2,6-dioxopiperidin-3-yl)-1-oxoisindolin-5-yl)prop-2-yn-1-yl)piperazin-1-yl)-2-((tetrahydro-2H-pyran-4-yl)amino)benzamide (TR-253) CPD-254 N-(5-(3,5-difluorobenzyl)-1H-indazol-3-yl)-4-(4-(1-((2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-5-yl)methyl)azetidin-3-yl)piperazin-1-yl)-2-((tetrahydro-2H-pyran-4-yl)amino)benzamide (TR-254) CPD-255 N-(5-(3,5-difluorobenzyl)-1H-indazol-3-yl)-4-(4-(3-(2-(2,6-dioxopiperidin-3-yl)-1-oxoisindolin-5-yl)prop-2-yn-1-yl)piperazin-1-yl)-2-((2-fluoroethyl)amino)benzamide (TR-255) CPD-256 N-(5-(3,5-difluorobenzyl)-1H-indazol-3-yl)-4-(4-(3-(2-(2,6-dioxopiperidin-3-yl)-1-oxoisindolin-yl)prop-2-yn-1-yl)piperazin-1-yl)benzamide (TR-256) CPD-257 3-(5-(3-(4-(6-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)propoxy)-3-methyl-2-oxo-2,3-dihydro-1H-benzo[d]imidazol-1-yl)piperidine-2,6-dione (TR-257) CPD-258 3-(4-(3-((4-(6-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)methyl)azetidin-1-yl)-3-methyl-2-oxo-2,3-dihydro-1H-benzo[d]imidazol-1-yl)piperidine- (TR-258) 2,6-dione CPD-259 3-(5-(4-(4-(6-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)butyl)-3-methyl-2-oxo-2,3-dihydro-1H-benzo[d]imidazol-1-yl)piperidine-2,6-dione (TR-259) CPD-260 3-(5-(4-(4-(6-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)but-1-yn-1-yl)-3-methyl-2-oxo-2,3-dihydro-1H-benzo[d]imidazol-1-yl) (TR-260) piperidine-2,6-dione CPD-261 N-(5-(3,5-difluorobenzyl)-1H-indazol-3-yl)-4-(4-(1-((2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-5-yl)methyl)azetidin-3-yl)piperazin-1-yl)benzamide (TR-261) CPD-262 N-(5-(3,5-difluorobenzyl)-1H-indazol-3-yl)-4-(4-((1-((2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-5-yl)methyl)azetidin-3-yl)methyl)piperazin-1-yl)benzamide (TR-262) CPD-263 N-(5-(3,5-difluorobenzyl)-1H-indazol-3-yl)-4-(4-(2-((2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-5-yl)amino)ethyl)piperazin-1-yl)benzamide (TR-263) CPD-264 N-(5-(3,5-difluorobenzyl)-1H-indazol-3-yl)-4-(4-((2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-5-yl)glycyl)piperazin-1-yl)benzamide (TR-264) CPD-265 N-(5-(3,5-difluorobenzyl)-1H-indazol-3-yl)-4-(4-((1-(2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-5-yl)azetidin-3-yl)methyl)piperazin-1-yl)benzamide (TR-265) CPD-266 N-(5-(3,5-difluorobenzyl)-1H-indazol-3-yl)-4-(4-((2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-5-yl)glycyl)piperazin-1-yl)-2-((2-fluoroethyl)amino)benzamide (TR-266) CPD-267 N-(5-(3,5-difluorobenzyl)-1H-indazol-3-yl)-4-(4-((1-(2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-5-yl)azetidin-3-yl)methyl)piperazin-1-yl)-2-((2-fluoroethyl)amino)benzamide (TR-267) CPD-268 N-(5-(3,5-difluorobenzyl)-1H-indazol-3-yl)-4-(4-(2-((2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-5-yl)amino)ethyl)piperazin-1-yl)-2-((2-fluoroethyl)amino)benzamide (TR-268) CPD-269 N-(5-(3,5-difluorobenzyl)-1H-indazol-3-yl)-4-(4-(2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-5-yl)piperazin-1-yl)-2-((tetrahydro-2H-pyran-4-yl)amino)benzamide (TR-269) CPD-270 N-(5-(3,5-difluorobenzyl)-1H-indazol-3-yl)-4-(4-((1-(2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-5-yl)pyrrolidin-3-yl)methyl)piperazin-1-yl)-2-((tetrahydro-2H-pyran-4-yl)amino)benzamide (TR-270) CPD-271 N-(5-(3,5-difluorobenzyl)-1H-indazol-3-yl)-4-(4-((1-(2-(2,6-difluoropiperidin-3-yl)-1,3-dioxoisindolin-5-yl)piperidin-4-yl)methyl)piperazin-1-yl)-2-((tetrahydro-2H-pyran-4-yl)amino)benzamide (TR-271) CPD-272 N-(5-(3,5-difluorobenzyl)-1H-indazol-3-yl)-3-(4-((1-(2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-5-yl)piperidin-3-yl)methyl)piperazin-1-yl)-2-((tetrahydro-2H-pyran-4-yl)amino)benzamide (TR-272) CPD-273 N-(5-(3,5-difluorobenzyl)-1H-indazol-3-yl)-4-(4-(2-((2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-5-yl)amino)-2-oxoethyl)piperazin-1-yl)-2-((tetrahydro-2H-pyran-4-yl)amino)benzamide (TR-273) CPD-274 N-(5-(3,5-difluorobenzyl)-1H-indazol-3-yl)-4-(4-((2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-5-yl)methyl)piperazin-1-yl)-2-((tetrahydro-2H-pyran-4-

yl)amino)benzamide (TR-274) CPD-275 N-(5-(3,5-difluorobenzyl)-1H-indazol-3-yl)-4-(4-(1-(2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-5-yl)azetidin-3-yl)piperazin-1-yl)-2-((tetrahydro-2H-pyran-4-yl)amino)benzamide (TR-275) CPD-276 N-(5-(3,5-difluorobenzyl)-1H-indazol-3-yl)-4-(4-(3-(2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-5-yl)propyl)piperazin-1-yl)-2-((tetrahydro-2H-pyran-4-yl)amino)benzamide (TR-276) CPD-277 N-(5-(3,5-difluorobenzyl)-1H-indazol-3-yl)-4-(4-(2-(2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-5-yl)ethyl)piperazin-1-yl)-2-((tetrahydro-2H-pyran-4-yl)amino)benzamide (TR-277) CPD-278 N-(5-(3,5-difluorobenzyl)-1H-indazol-3-yl)-4-(4-(1-(2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-5-yl)piperidin-3-yl)piperazin-1-yl)-2-((tetrahydro-2H-pyran-4-yl)amino)benzamide (TR-278) CPD-279 N-(5-(3,5-difluorobenzyl)-1H-indazol-3-yl)-4-(4-(1-((2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-5-yl)methyl)piperidin-3-yl)piperazin-1-yl)-2-((tetrahydro-2H-pyran-4-yl)amino)benzamide (TR-279) CPD-280 N-(5-(3,5-difluorobenzyl)-1H-indazol-3-yl)-4-(4-(1-(2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-5-yl)piperidin-4-yl)piperazin-1-yl)-2-((tetrahydro-2H-pyran-4-yl)amino)benzamide (TR-280) CPD-281 N-(5-(3,5-difluorobenzyl)-1H-indazol-3-yl)-4-(4-(1-((2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-5-yl)methyl)piperidin-4-yl)piperazin-1-yl)-2-((tetrahydro-2H-pyran-4-yl)amino)benzamide (TR-281) CPD-282 N-(5-(3,5-difluorobenzyl)-1H-indazol-3-yl)-4-(4-((4-(2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-5-yl)morpholin-2-yl)methyl)piperazin-1-yl)-2-((tetrahydro-2H-pyran-4-yl)amino)benzamide (TR-282) CPD-283 N-(5-(3,5-difluorobenzyl)-1H-indazol-3-yl)-4-(4-(3-(2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-5-yl)prop-2-yn-1-yl)piperazin-1-yl)-2-((tetrahydro-2H-pyran-4-yl)amino)benzamide (TR-283) CPD-284 N-(5-(3,5-difluorobenzyl)-1H-indazol-3-yl)-4-(4-(1-(2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-5-yl)pyrrolidin-3-yl)piperazin-1-yl)-2-((tetrahydro-2H-pyran-4-yl)amino)benzamide (TR-284) CPD-285 N-(5-(3,5-difluorobenzyl)-1H-indazol-3-yl)-4-(4-(1-((2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-5-yl)methyl)pyrrolidin-3-yl)piperazin-1-yl)-2-((tetrahydro-2H-pyran-4-yl)amino)benzamide (TR-285) CPD-286 2-(2,6-dioxopiperidin-3-yl)-5-(2-((4-(6-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)methyl)morpholino)isoindoline-1,3-dione (TR-286) CPD-287 2-(2,6-dioxopiperidin-3-yl)-5-(3-((1-(6-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperidin-4-yl)oxy)azetidin-1-yl)isoindoline-1,3-dione (TR-287) CPD-288 2-(2,6-dioxopiperidin-3-yl)-5-(3-((1-(6-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperidin-4-yl)amino)azetidin-1-yl)isoindoline-1,3-dione (TR-288) CPD-289 N-(5-(3,5-difluorobenzyl)-1H-indazol-3-yl)-4-(4-(4-((2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-5-yl)amino)cyclohexyl)piperazin-1-yl)-2-((tetrahydro-2H-pyran-4-yl)amino)benzamide (TR-289) CPD-290 N-(5-(3,5-difluorobenzyl)-1H-indazol-3-yl)-4-(4-(6-(2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-5-yl)-6-azaspiro[3.4]octan-2-yl)piperazin-1-yl)-2-((tetrahydro-2H-pyran-4-yl)amino)benzamide (TR-290) CPD-291 N-(5-(3,5-difluorobenzyl)-1H-indazol-3-yl)-4-(4-(2-(2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-5-yl)-2-azaspiro[3.4]octan-6-yl)piperazin-1-yl)-2-((tetrahydro-2H-pyran-4-yl)amino)benzamide (TR-291) CPD-292 2-(2,6-dioxopiperidin-3-yl)-5-(1-(2-(4-(6-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)ethyl)azetidin-3-yl)isoindoline-1,3-dione (TR-292) CPD-293 2-(2,6-dioxopiperidin-3-yl)-5-(1-((1-(6-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperidin-4-yl)methyl)azetidin-3-yl)isoindoline-1,3-dione (TR-293) CPD-294 2-(2,6-dioxopiperidin-3-yl)-4-((3-((4-(6-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)methyl)azetidin-1-yl)methyl)isoindoline-1,3-dione (TR-294) CPD-295 3-(5-(3-((4-(6-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)methyl)azetidin-1-yl)-2-methyl-4-oxoquinazolin-3(4H)-yl)piperidine-2,6-dione (TR-295) CPD-296 3-(5-(3-((4-(6-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)methyl)azetidin-1-yl)-4-oxobenzo[d][1,2,3]triazin-3(4H)-yl)piperidine-2,6-dione (TR-296) CPD-297 3-(5-(3-(2-(4-(6-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-

b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)ethyl)azetidin-1-yl)-4-oxobenzo[d][1,2,3]triazin-3(4H)-yl)piperidine-2,6-dione (TR-297) CPD-298 N-(5-(3,5-difluorobenzyl)-1H-indazol-3-yl)-4-(4-(3-(2-(2,6-dioxopiperidin-3-yl)-3-oxoisindolin-5-yl)prop-2-yn-1-yl)piperazin-1-yl)-2-((tetrahydro-2H-pyran-4-yl)amino)benzamide (TR-298) CPD-299 N-(5-(3,5-difluorobenzyl)-1H-indazol-3-yl)-4-(4-(3-((2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-5-yl)amino)cyclopentyl)piperazin-1-yl)-2-((tetrahydro-2H-pyran-4-yl)amino)benzamide (TR-299) CPD-300 N-(5-(3,5-difluorobenzyl)-1H-indazol-3-yl)-4-(4-((1-(2-(2,6-dioxopiperidin-3-yl)-1-oxoisindolin-5-yl)azetidin-3-yl)methyl)piperazin-1-yl)-2-((tetrahydro-2H-pyran-4-yl)amino)benzamide (TR-300) CPD-301 N-(5-(3,5-difluorobenzyl)-1H-indazol-3-yl)-4-(4-(1-(1-(2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-5-yl)azetidin-3-yl)ethyl)piperazin-1-yl)-2-((tetrahydro-2H-pyran-4-yl)amino)benzamide (TR-301) CPD-302 3-(5-(1-(1-(6-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperidin-4-yl)azetidin-3-yl)-1-oxoisindolin-2-yl)piperidine-2,6-dione (TR-300) CPD-303 N-(5-(3,5-difluorobenzyl)-1H-indazol-3-yl)-4-(4-(3-(1-(2,6-dioxopiperidin-3-yl)-3-methyl-2-oxo-2,3-dihydro-1H-benzo[d]imidazol-4-yl)prop-2-yn-1-yl)piperazin-1-yl)-2-((tetrahydro-2H-pyran-4-yl)amino)benzamide (TR-303) CPD-304 N-(5-(3,5-difluorobenzyl)-1H-indazol-3-yl)-4-(4-(4-(1-(2,6-dioxopiperidin-3-yl)-3-methyl-2-oxo-2,3-dihydro-1H-benzo[d]imidazol-4-yl)butyl)piperazin-1-yl)-2-((tetrahydro-2H-pyran-4-yl)amino)benzamide (TR-304) CPD-305 N-(5-(3,5-difluorobenzyl)-1H-indazol-3-yl)-4-(4-(3-(1-(2,6-dioxopiperidin-3-yl)-3-methyl-2-oxo-2,3-dihydro-1H-benzo[d]imidazol-4-yl)propyl)piperazin-1-yl)-2-((tetrahydro-2H-pyran-4-yl)amino)benzamide (TR-305) CPD-306 N-(5-(3,5-difluorobenzyl)-1H-indazol-3-yl)-4-(4-(4-(1-(2,6-dioxopiperidin-3-yl)-3-methyl-2-oxo-2,3-dihydro-1H-benzo[d]imidazol-4-yl)but-3-yn-1-yl)piperazin-1-yl)-2-((tetrahydro-2H-pyran-4-yl)amino)benzamide (TR-306) CPD-307 3-(5-(3-(2-(4-(6-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)ethyl)azetidin-1-yl)-2-methyl-4-oxoquinazolin-3(4H)-yl)piperidine-2,6-dione (TR-307) CPD-308 2-(2,6-dioxopiperidin-3-yl)-5-(1-(1-(6-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperidin-4-yl)azetidin-3-yl)isoindoline-1,3-dione (TR-308) CPD-309 3-(6-(1-(1-(6-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperidin-4-yl)azetidin-3-yl)-1-oxoisindolin-2-yl)piperidine-2,6-dione (TR-309) CPD-310 N-(5-(3,5-difluorobenzyl)-1H-indazol-3-yl)-4-(4-(2-(3-(2,6-dioxopiperidin-3-yl)-2-oxo-2,3-dihydro-1H-benzo[d]imidazol-1-yl)ethyl)piperazin-1-yl)-2-((tetrahydro-2H-pyran-4-yl)amino)benzamide (TR-310) CPD-311 N-(5-(3,5-difluorobenzyl)-1H-indazol-3-yl)-4-(4-(3-(3-(2,6-dioxopiperidin-3-yl)-2-oxo-2,3-dihydro-1H-benzo[d]imidazol-1-yl)propyl)piperazin-1-yl)-2-((tetrahydro-2H-pyran-4-yl)amino)benzamide (TR-311) CPD-312 N-(5-(3,5-difluorobenzyl)-1H-indazol-3-yl)-4-(4-(3-((1-(2,6-dioxopiperidin-3-yl)-3-methyl-2-oxo-2,3-dihydro-1H-benzo[d]imidazol-4-yl)oxy)propyl)piperazin-1-yl)-2-((tetrahydro-2H-pyran-4-yl)amino)benzamide (TR-312) CPD-313 N-(5-(3,5-difluorobenzyl)-1H-indazol-3-yl)-4-(4-(3-((1-(2,6-dioxopiperidin-3-yl)-3-methyl-2-oxo-2,3-dihydro-1H-benzo[d]imidazol-5-yl)oxy)propyl)piperazin-1-yl)-2-((tetrahydro-2H-pyran-4-yl)amino)benzamide (TR-313) CPD-314 N-(5-(3,5-difluorobenzyl)-1H-indazol-3-yl)-4-(4-(2-((1-(2,6-dioxopiperidin-3-yl)-3-methyl-2-oxo-2,3-dihydro-1H-benzo[d]imidazol-5-yl)oxy)ethyl)piperazin-1-yl)-2-((tetrahydro-2H-pyran-4-yl)amino)benzamide (TR-314) CPD-315 N-(5-(3,5-difluorobenzyl)-1H-indazol-3-yl)-4-(4-(3-((1-(2,6-dioxopiperidin-3-yl)-3-methyl-2-oxo-2,3-dihydro-1H-benzo[d]imidazol-4-yl)amino)propyl)piperazin-1-yl)-2-((tetrahydro-2H-pyran-4-yl)amino)benzamide (TR-315) CPD-316 N-(5-(3,5-difluorobenzyl)-1H-indazol-3-yl)-4-(4-(3-((1-(2,6-dioxopiperidin-3-yl)-3-methyl-2-oxo-2,3-dihydro-1H-benzo[d]imidazol-5-yl)amino)propyl)piperazin-1-yl)-2-((tetrahydro-2H-pyran-4-yl)amino)benzamide (TR-316) CPD-317 3-(6-(3-((4-(6-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)methyl)azetidin-1-yl)-1-oxoisindolin-2-yl)piperidine-2,6-dione (TR-317) CPD-318 N-(5-(3,5-difluorobenzyl)-1H-indazol-3-yl)-4-(4-((1-(2-(2,6-dioxopiperidin-3-yl)-2-oxoisindolin-5-

yl)azetidin-3-yl)methyl)piperazin-1-yl)-2-((tetrahydro-2H-pyran-4-yl)amino)benzamide (TR-318) CPD-319 N-(5-(3,5-difluorobenzyl)-1H-indazol-3-yl)-4-(4-(4-(1-(2,6-dioxopiperidin-3-yl)-3-methyl-2-oxo-2,3-dihydro-1H-benzo[d]imidazol-5-yl)but-3-yn-1-yl)piperazin-1-yl)-2-((tetrahydro-2H-pyran-4-yl)amino)benzamide CPD-320 N-(5-(3,5-difluorobenzyl)-1H-indazol-3-yl)-4-(4-(4-(1-(2,6-dioxopiperidin-3-yl)-3-methyl-2-oxo-2,3-dihydro-1H-benzo[d]imidazol-5-yl)butyl)piperazin-1-yl)-2-((tetrahydro-2H-pyran-4-yl)amino) (TR-320) benzamide CPD-321 N-(5-(3,5-difluorobenzyl)-1H-indazol-3-yl)-4-(4-((1-(1-(2,6-dioxopiperidin-3-yl)-3-methyl-2-oxo-2,3-dihydro-1H-benzo[d]imidazol-5-yl)azetidin-3-yl)methyl)piperazin-1-yl)-2-((tetrahydro-2H-pyran-4-yl)amino)benzamide CPD-322 N-(5-(3,5-difluorobenzyl)-1H-indazol-3-yl)-4-(4-(2-((1-(2,6-dioxopiperidin-3-yl)-3-methyl-2-oxo-2,3-dihydro-1H-benzo[d]imidazol-4-yl)oxy)ethyl)piperazin-1-yl)-2-((tetrahydro-2H-pyran-4-yl)amino) (TR-322) benzamide CPD-323 N-(5-(3,5-difluorobenzyl)-1H-indazol-3-yl)-4-(4-((2-(2,6-dioxopiperidin-3-yl)-1-oxoisindolin-5-yl)ethynyl)piperidin-1-yl)-2-((tetrahydro-2H-pyran-4-yl)amino)benzamide (TR-323) CPD-324 N-(5-(3,5-difluorobenzyl)-1H-indazol-3-yl)-4-(4-(2-((1-(2,6-dioxopiperidin-3-yl)-3-methyl-2-oxo-2,3-dihydro-1H-benzo[d]imidazol-5-yl)amino)ethyl)piperazin-1-yl)-2-((tetrahydro-2H-pyran-4-yl)amino)benzamide CPD-325 N-(5-(3,5-difluorobenzyl)-1H-indazol-3-yl)-4-(4-(2-((1-(2,6-dioxopiperidin-3-yl)-3-methyl-2-oxo-2,3-dihydro-1H-benzo[d]imidazol-4-yl)amino)ethyl)piperazin-1-yl)-2-((tetrahydro-2H-pyran-4-yl)amino)benzamide CPD-326 N-(5-(3,5-difluorobenzyl)-1H-indazol-3-yl)-4-(4-((1-(3-(2,6-dioxopiperidin-3-yl)-4-oxo-3,4-dihydrobenzo[d][1,2,3]triazin-5-yl)azetidin-3-yl)methyl)piperazin-1-yl)-2-((tetrahydro-2H-pyran-4-yl)amino) (TR-326) benzamide CPD-327 N-(5-(3,5-difluorobenzyl)-1H-indazol-3-yl)-4-(4-(2-(1-(3-(2,6-dioxopiperidin-3-yl)-4-oxo-3,4-dihydrobenzo[d][1,2,3]triazin-5-yl)azetidin-3-yl)ethyl)piperazin-1-yl)-2-((tetrahydro-2H-pyran-4-yl)amino) (TR-327) benzamide CPD-328 N-(5-(3,5-difluorobenzyl)-1H-indazol-3-yl)-4-(4-((3-(2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-5-yl)azetidin-1-yl)methyl)piperidin-1-yl)-2-((tetrahydro-2H-pyran-4-yl)amino)benzamide (TR-328) CPD-329 N-(5-(3,5-difluorobenzyl)-1H-indazol-3-yl)-4-(4-((6-(2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-5-yl)-2,6-diazaspiro[3.3]heptan-2-yl)methyl)piperidin-1-yl)-2-((tetrahydro-2H-pyran-4-yl)amino) (TR-329) benzamide CPD-330 N-(5-(3,5-difluorobenzyl)-1H-indazol-3-yl)-4-(4-((3-(2-(2,6-dioxopiperidin-3-yl)-1-oxoisindolin-5-yl)azetidin-1-yl)methyl)piperidin-1-yl)-2-((tetrahydro-2H-pyran-4-yl)amino)benzamide (TR-330) CPD-331 3-(5-(3-(4-(6-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)azetidin-1-yl)-1-oxoisindolin-2-yl)piperidine-2,6-dione (TR-331) CPD-332 N-(5-(3,5-difluorobenzyl)-1H-indazol-3-yl)-4-(4-(1-(2-(2,6-dioxopiperidin-3-yl)-1-oxoisindolin-5-yl)azetidin-3-yl)piperazin-1-yl)-2-((tetrahydro-2H-pyran-4-yl)amino)benzamide (TR-332) CPD-333 N-(5-(3,5-difluorobenzyl)-1H-indazol-3-yl)-4-(4-((1-(3-(2,6-dihydroquinazolin-5-yl)azetidin-3-yl)methyl)piperazin-1-yl)-2-((tetrahydro-2H-pyran-4-yl)amino)benzamide (TR-333) CPD-334 N-(5-(3,5-difluorobenzyl)-1H-indazol-3-yl)-4-(4-(2-(1-(3-(2,6-dioxopiperidin-3-yl)-2-methyl-4-oxo-3,4-dihydroquinazolin-5-yl)azetidin-3-yl)ethyl)piperazin-1-yl)-2-((tetrahydro-2H-pyran-4-yl)amino) (TR-334) benzamide CPD-335 3-(6-(3-(4-(6-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)azetidin-1-yl)-1-oxoisindolin-2-yl)piperidine-2,6-dione (TR-335) CPD-336 N-(5-(3,5-difluorobenzyl)-1H-indazol-3-yl)-4-(4-(1-(2-(2,6-dioxopiperidin-3-yl)-3-oxoisindolin-5-yl)azetidin-3-yl)piperazin-1-yl)-2-((tetrahydro-2H-pyran-4-yl)amino)benzamide (TR-336) CPD-337 N-(5-(3,5-difluorobenzyl)-1H-indazol-3-yl)-4-(4-((1-(2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-5-yl)azetidin-3-yl)amino)piperidin-1-yl)-2-((tetrahydro-2H-pyran-4-yl)amino)benzamide (TR-337) CPD-338 N-(5-(3,5-difluorobenzyl)-1H-indazol-3-yl)-4-(4-(3-(2-(2,6-dioxopiperidin-3-yl)-1-oxoisindolin-5-yl)azetidin-1-yl)piperidin-1-yl)-2-((tetrahydro-2H-pyran-4-yl)amino)benzamide (TR-338) CPD-339 N-(5-(3,5-difluorobenzyl)-1H-indazol-3-yl)-4-(4-(3-(2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-5-yl)azetidin-1-yl)piperidin-1-yl)-2-((tetrahydro-2H-pyran-4-yl)amino)benzamide (TR-339) CPD-340 N-(5-(3,5-difluorobenzyl)-1H-indazol-3-yl)-4-(4-(6-(2-(2,6-dioxopiperidin-3-yl)-1,3-

dioxoisindolin-5-yl)-2,6-diazaspiro[3.3]heptan-2-yl)piperidin-1-yl)-2-((tetrahydro-2H-pyran-4-yl)amino)benzamide (TR-340) CPD-341 (S)-N-(5-(3,5-difluorobenzyl)-1H-indazol-3-yl)-4-(4-((1-(2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-5-yl)azetidin-3-yl)methyl)piperazin-1-yl)-2-((tetrahydro-2H-pyran-4-yl)amino)benzamide (TR-341) CPD-342 (R)-N-(5-(3,5-difluorobenzyl)-1H-indazol-3-yl)-4-(4-((1-(2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-5-yl)azetidin-3-yl)methyl)piperazin-1-yl)-2-((tetrahydro-2H-pyran-4-yl)amino)benzamide (TR-342) CPD-343 (S)-N-(5-(3,5-difluorobenzyl)-1H-indazol-3-yl)-4-(4-(1-(2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-5-yl)azetidin-3-yl)piperazin-1-yl)-2-((tetrahydro-2H-pyran-4-yl)amino)benzamide (TR-343) CPD-344 (R)-N-(5-(3,5-difluorobenzyl)-1H-indazol-3-yl)-4-(4-(1-(2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-5-yl)azetidin-3-yl)piperazin-1-yl)-2-((tetrahydro-2H-pyran-4-yl)amino)benzamide (TR-344) CPD-345 N-(5-(3,5-difluorobenzyl)-1H-indazol-3-yl)-4-(4-(1-((2-(2,6-dioxopiperidin-3-yl)-3-oxoisindolin-5-yl)methyl)piperidin-3-yl)piperazin-1-yl)-2-((tetrahydro-2H-pyran-4-yl)amino)benzamide CPD-346 N-(5-(3,5-difluorobenzyl)-1H-indazol-3-yl)-4-(4-(1-((2-(2,6-dioxopiperidin-3-yl)-1-oxoisindolin-5-yl)methyl)piperidin-3-yl)piperazin-1-yl)-2-((tetrahydro-2H-pyran-4-yl)amino)benzamide CPD-347 N-(5-(3,5-difluorobenzyl)-1H-indazol-3-yl)-4-(4-(1-(2-(2,6-dioxopiperidin-3-yl)-1-oxoisindolin-5-yl)piperidin-4-yl)piperazin-1-yl)-2-((tetrahydro-2H-pyran-4-yl)amino)benzamide CPD-348 N-(5-(3,5-difluorobenzyl)-1H-indazol-3-yl)-4-(4-(1-(2-(2,6-dioxopiperidin-3-yl)-3-oxoisindolin-5-yl)piperidin-4-yl)piperazin-1-yl)-2-((tetrahydro-2H-pyran-4-yl)amino)benzamide CPD-349 N-(5-(3,5-difluorobenzyl)-1H-indazol-3-yl)-4-(4-(3-(2-(2,6-dioxopiperidin-3-yl)-3-oxoisindolin-5-yl)azetidin-1-yl)piperidin-1-yl)-2-((tetrahydro-2H-pyran-4-yl)amino)benzamide CPD-350 N-(5-(3,5-difluorobenzyl)-1H-indazol-3-yl)-4-(1-((1-(2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-5-yl)azetidin-3-yl)methyl)piperidin-4-yl)-2-((tetrahydro-2H-pyran-4-yl)amino)benzamide CPD-351 N-(5-(3,5-difluorobenzyl)-1H-indazol-3-yl)-4-(4-((1-(2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-5-yl)azetidin-3-yl)methyl)piperidin-1-yl)-2-((tetrahydro-2H-pyran-4-yl)amino)benzamide CPD-352 N-(5-(3,5-difluorobenzyl)-1H-indazol-3-yl)-4-(4-((1-(2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-5-yl)azetidin-3-yl)oxy)piperidin-1-yl)-2-((tetrahydro-2H-pyran-4-yl)amino)benzamide CPD-353 N-(5-(3,5-difluorobenzyl)-1H-indazol-3-yl)-4-(4-(3-((2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-5-yl)oxy)cyclopentyl)piperazin-1-yl)-2-((tetrahydro-2H-pyran-4-yl)amino)benzamide CPD-354 N-(5-(3,5-difluorobenzyl)-1H-indazol-3-yl)-4-(4-(1-(1-(2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-5-yl)azetidin-3-yl)cyclopropyl)piperazin-1-yl)-2-((tetrahydro-2H-pyran-4-yl)amino)benzamide CPD-355 N-(5-(3,5-difluorobenzyl)-1H-indazol-3-yl)-4-(4-(4-((2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-5-yl)oxy)cyclohexyl)piperazin-1-yl)-2-((tetrahydro-2H-pyran-4-yl)amino)benzamide CPD-356 N-(5-(3,5-difluorobenzyl)-1H-indazol-3-yl)-4-(4-(6-(2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-5-yl)-2,6-diazaspiro[3.4]octan-2-yl)piperidin-1-yl)-2-((tetrahydro-2H-pyran-4-yl)amino)benzamide CPD-357 N-(5-(3,5-difluorobenzyl)-1H-indazol-3-yl)-4-(4-(2-(2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-5-yl)-2,6-diazaspiro[3.4]octan-6-yl)piperidin-1-yl)-2-((tetrahydro-2H-pyran-4-yl)amino)benzamide CPD-358 N-(5-(3,5-difluorobenzyl)-1H-indazol-3-yl)-4-(4-(3-(2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-5-yl)cyclobutyl)piperazin-1-yl)-2-((tetrahydro-2H-pyran-4-yl)amino)benzamide CPD-359 N-(5-(3,5-difluorobenzyl)-1H-indazol-3-yl)-4-(4-((3-(2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-5-yl)cyclobutyl)methyl)piperazin-1-yl)-2-((tetrahydro-2H-pyran-4-yl)amino)benzamide CPD-360 N-(5-(3,5-difluorobenzyl)-1H-indazol-3-yl)-4-(4-(3-(2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-5-yl)cyclobutane-1-carbonyl)piperazin-1-yl)-2-((tetrahydro-2H-pyran-4-yl)amino)benzamide CPD-361 N-(5-(3,5-difluorobenzyl)-1H-indazol-3-yl)-4-(4-((1-(2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-5-yl)-2-oxoazetidin-3-yl)methyl)piperazin-1-yl)-2-((tetrahydro-2H-pyran-4-yl)amino)benzamide CPD-362 N-(5-(3,5-difluorobenzyl)-1H-indazol-3-yl)-4-(4-((4-(2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-5-yl)azetidin-2-yl)methyl)piperazin-1-yl)-2-((tetrahydro-2H-pyran-4-yl)amino)benzamide CPD-363 N-(5-(3,5-difluorobenzyl)-1H-indazol-3-yl)-4-(4-((4-(2-(2,6-dioxopiperidin-3-yl)-1,3-

[illegible]

dioxopiperidin-3-yl)-1-methyl-1H-indazol-7-yl)oxy)ethyl)piperazin-1-yl)-2-((tetrahydro-2H-pyran-4-yl)amino)benzamide CPD-389 N-(5-(3,5-difluorobenzyl)-1H-indazol-3-yl)-4-(4-(2-((3-(2,6-dioxopiperidin-3-yl)-1-methyl-1H-indazol-7-yl)amino)ethyl)piperazin-1-yl)-2-((tetrahydro-2H-pyran-4-yl)amino)benzamide CPD-390 N-(5-(3,5-difluorobenzyl)-1H-indazol-3-yl)-4-(4-(3-((3-(2,6-dioxopiperidin-3-yl)-1-methyl-1H-indazol-7-yl)oxy)propyl)piperazin-1-yl)-2-((tetrahydro-2H-pyran-4-yl)amino)benzamide CPD-391 N-(5-(3,5-difluorobenzyl)-1H-indazol-3-yl)-4-(4-(3-((3-(2,6-dioxopiperidin-3-yl)-1-methyl-1H-indazol-7-yl)amino)propyl)piperazin-1-yl)-2-((tetrahydro-2H-pyran-4-yl)amino)benzamide CPD-392 N-(5-(3,5-difluorobenzyl)-1H-indazol-3-yl)-4-(4-(2-((1-(2,6-dioxopiperidin-3-yl)-1H-benzo[d]imidazol-4-yl)oxy)ethyl)piperazin-1-yl)-2-((tetrahydro-2H-pyran-4-yl)amino)benzamide CPD-393 N-(5-(3,5-difluorobenzyl)-1H-indazol-3-yl)-4-(4-(2-((1-(2,6-dioxopiperidin-3-yl)-1H-benzo[d]imidazol-4-yl)amino)ethyl)piperazin-1-yl)-2-((tetrahydro-2H-pyran-4-yl)amino)benzamide CPD-394 N-(5-(3,5-difluorobenzyl)-1H-indazol-3-yl)-4-(4-(3-((1-(2,6-dioxopiperidin-3-yl)-1H-benzo[d]imidazol-4-yl)oxy)propyl)piperazin-1-yl)-2-((tetrahydro-2H-pyran-4-yl)amino)benzamide CPD-395 N-(5-(3,5-difluorobenzyl)-1H-indazol-3-yl)-4-(4-(3-((1-(2,6-dioxopiperidin-3-yl)-1H-benzo[d]imidazol-4-yl)amino)propyl)piperazin-1-yl)-2-((tetrahydro-2H-pyran-4-yl)amino)benzamide CPD-396 N-(5-(3,5-difluorobenzyl)-1H-indazol-3-yl)-4-(4-(2-((1-(2,6-dioxopiperidin-3-yl)-1H-indol-4-yl)oxy)ethyl)piperazin-1-yl)-2-((tetrahydro-2H-pyran-4-yl)amino)benzamide CPD-397 N-(5-(3,5-difluorobenzyl)-1H-indazol-3-yl)-4-(4-(2-((1-(2,6-dioxopiperidin-3-yl)-1H-indol-4-yl)amino)ethyl)piperazin-1-yl)-2-((tetrahydro-2H-pyran-4-yl)amino)benzamide CPD-398 N-(5-(3,5-difluorobenzyl)-1H-indazol-3-yl)-4-(4-(3-((1-(2,6-dioxopiperidin-3-yl)-1H-indol-4-yl)oxy)propyl)piperazin-1-yl)-2-((tetrahydro-2H-pyran-4-yl)amino)benzamide CPD-399 N-(5-(3,5-difluorobenzyl)-1H-indazol-3-yl)-4-(4-(3-((1-(2,6-dioxopiperidin-3-yl)-1H-indol-4-yl)amino)propyl)piperazin-1-yl)-2-((tetrahydro-2H-pyran-4-yl)amino)benzamide CPD-400 N-(5-(3,5-difluorobenzyl)-1H-indazol-3-yl)-4-(4-(2-((3-(2,6-dioxopiperidin-3-yl)-1-methyl-1H-indazol-6-yl)oxy)ethyl)piperazin-1-yl)-2-((tetrahydro-2H-pyran-4-yl)amino)benzamide CPD-401 N-(5-(3,5-difluorobenzyl)-1H-indazol-3-yl)-4-(4-(2-((3-(2,6-dioxopiperidin-3-yl)-1-methyl-1H-indazol-6-yl)amino)ethyl)piperazin-1-yl)-2-((tetrahydro-2H-pyran-4-yl)amino)benzamide CPD-402 N-(5-(3,5-difluorobenzyl)-1H-indazol-3-yl)-4-(4-(3-((3-(2,6-dioxopiperidin-3-yl)-1-methyl-1H-indazol-6-yl)oxy)propyl)piperazin-1-yl)-2-((tetrahydro-2H-pyran-4-yl)amino)benzamide CPD-403 N-(5-(3,5-difluorobenzyl)-1H-indazol-3-yl)-4-(4-(3-((3-(2,6-dioxopiperidin-3-yl)-1-methyl-1H-indazol-6-yl)amino)propyl)piperazin-1-yl)-2-((tetrahydro-2H-pyran-4-yl)amino)benzamide CPD-404 N-(5-(3,5-difluorobenzyl)-1H-indazol-3-yl)-4-(4-(2-((1-(2,6-dioxopiperidin-3-yl)-1H-benzo[d]imidazol-5-yl)oxy)ethyl)piperazin-1-yl)-2-((tetrahydro-2H-pyran-4-yl)amino)benzamide CPD-405 N-(5-(3,5-difluorobenzyl)-1H-indazol-3-yl)-4-(4-(2-((1-(2,6-dioxopiperidin-3-yl)-1H-benzo[d]imidazol-5-yl)amino)ethyl)piperazin-1-yl)-2-((tetrahydro-2H-pyran-4-yl)amino)benzamide CPD-406 N-(5-(3,5-difluorobenzyl)-1H-indazol-3-yl)-4-(4-(3-((1-(2,6-dioxopiperidin-3-yl)-1H-benzo[d]imidazol-5-yl)oxy)propyl)piperazin-1-yl)-2-((tetrahydro-2H-pyran-4-yl)amino)benzamide CPD-407 N-(5-(3,5-difluorobenzyl)-1H-indazol-3-yl)-4-(4-(3-((1-(2,6-dioxopiperidin-3-yl)-1H-benzo[d]imidazol-5-yl)amino)propyl)piperazin-1-yl)-2-((tetrahydro-2H-pyran-4-yl)amino)benzamide CPD-408 N-(5-(3,5-difluorobenzyl)-1H-indazol-3-yl)-4-(4-(2-((1-(2,6-dioxopiperidin-3-yl)-1H-indol-5-yl)oxy)ethyl)piperazin-1-yl)-2-((tetrahydro-2H-pyran-4-yl)amino)benzamide CPD-409 N-(5-(3,5-difluorobenzyl)-1H-indazol-3-yl)-4-(4-(2-((1-(2,6-dioxopiperidin-3-yl)-1H-indol-5-yl)amino)ethyl)piperazin-1-yl)-2-((tetrahydro-2H-pyran-4-yl)amino)benzamide CPD-410 N-(5-(3,5-difluorobenzyl)-1H-indazol-3-yl)-4-(4-(3-((1-(2,6-dioxopiperidin-3-yl)-1H-indol-5-yl)oxy)propyl)piperazin-1-yl)-2-((tetrahydro-2H-pyran-4-yl)amino)benzamide CPD-411 N-(5-(3,5-difluorobenzyl)-1H-indazol-3-yl)-4-(4-(3-((1-(2,6-dioxopiperidin-3-yl)-1H-indol-5-yl)amino)propyl)piperazin-1-yl)-2-((tetrahydro-2H-pyran-4-yl)amino)benzamide CPD-412 3-(7-(3-(4-(6-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)prop-1-yn-1-yl)-1-methyl-1H-indazol-3-yl)piperidine-2,6-dione CPD-413 3-(7-(3-(4-(6-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-

yl)imidazo[1,2-b]pyridazin-3-yl)piperidin-2-yl)piperazin-1-yl)propyl)-1-methyl-1H-indazol-3-
yl)piperidine-2,6-dione CPD-414 3-(7-(2-(4-(6-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-
yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)ethoxy)-1-methyl-1H-indazol-3-
yl)piperidine-2,6-dione CPD-415 3-(7-((2-(4-(6-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-
yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)ethylamino)-1-methyl-1H-indazol-3-
yl)piperidine-2,6-dione CPD-416 3-(7-(4-(4-(6-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-
yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)but-1-yn-1-yl)-1-methyl-1H-indazol-
3-yl)piperidine-2,6-dione CPD-417 3-(7-(4-(4-(6-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-
yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)butyl-1-methyl-1H-indazol-3-
yl)piperidine-2,6-dione CPD-418 3-(7-(3-(4-(6-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-
yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)propoxy)-1-methyl-1H-indazol-3-
yl)piperidine-2,6-dione CPD-419 3-(7-((3-(4-(6-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-
yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)propylamino)-1-methyl-1H-indazol-
3-yl)piperidine-2,6-dione CPD-420 3-(4-(3-(4-(6-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-
yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)prop-1-yn-1-yl)-1H-benzo[d]imidazol-
1-yl)piperidine-2,6-dione CPD-421 3-(4-(3-(4-(6-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-
yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)propyl)-1H-benzo[d]imidazol-1-
yl)piperidine-2,6-dione CPD-422 3-(4-(2-(4-(6-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-
yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)ethoxy)-1H-benzo[d]imidazol-1-
yl)piperidine-2,6-dione CPD-423 3-(4-((2-(4-(6-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-
yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)ethylamino)-1H-benzo[d]imidazol-1-
yl)piperidine-2,6-dione CPD-424 3-(4-(4-(4-(6-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-
yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)but-1-yn-1-yl)-1H-benzo[d]imidazol-
1-yl)piperidine-2,6-dione CPD-425 3-(4-(4-(4-(6-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-
yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)butyl)-1H-benzo[d]imidazol-1-
yl)piperidine-2,6-dione CPD-426 3-(4-(3-(4-(6-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-
yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)propoxy)-1H-benzo[d]imidazol-1-
yl)piperidine-2,6-dione CPD-427 3-(4-((3-(4-(6-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-
yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)propylamino)-1H-benzo[d]imidazol-
1-yl)piperidine-2,6-dione CPD-428 3-(4-(3-(4-(6-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-
yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)prop-1-yn-1-yl)-1H-indol-1-
yl)piperidine-2,6-dione CPD-429 3-(4-(3-(4-(6-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-
yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)propyl)-1H-indol-1-yl)piperidine-2,6-
dione CPD-430 3-(4-(2-(4-(6-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-
yl)pyridin-2-yl)piperazin-1-yl)ethoxy)-1H-indol-1-yl)piperidine-2,6-dione CPD-431 3-(4-((2-(4-(6-
(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-
yl)ethylamino)-1H-indol-1-yl)piperidine-2,6-dione CPD-432 3-(4-(4-(4-(6-(6-((R)-2-(3-
fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)but-1-yn-1-
yl)-1H-indol-1-yl)piperidine-2,6-dione CPD-433 3-(4-(4-(4-(6-(6-((R)-2-(3-
fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)butyl)-1H-
indol-1-yl)piperidine-2,6-dione CPD-434 3-(4-(3-(4-(6-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-
yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)propoxy)-1H-indol-1-yl)piperidine-
2,6-dione CPD-435 3-(4-((3-(4-(6-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-
b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)propylamino)-1H-indol-1-yl)piperidine-2,6-dione
CPD-436 3-(6-(3-(4-(6-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-
yl)pyridin-2-yl)piperazin-1-yl)prop-1-yn-1-yl)-1-methyl-1H-indazol-3-yl)piperidine-2,6-dione
CPD-437 3-(6-(3-(4-(6-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-
yl)pyridin-2-yl)piperazin-1-yl)propyl-1-methyl-1H-indazol-3-yl)piperidine-2,6-dione CPD-438 3-
(6-(2-(4-(6-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-
yl)piperazin-1-yl)ethoxy)-1-methyl-1H-indazol-3-yl)piperidine-2,6-dione CPD-439 3-(6-((2-(4-(6-

(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)ethyl)amino)-1-methyl-1H-indazol-3-yl)piperidine-2,6-dione CPD-440 3-(6-(4-(4-(6-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)but-1-yn-1-yl)-1-methyl-1H-indazol-3-yl)piperidine-2,6-dione CPD-441 3-(6-(4-(4-(6-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)butyl)-1-methyl-1H-indazol-3-yl)piperidine-2,6-dione CPD-442 3-(6-(3-(4-(6-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)propoxy)-1-methyl-1H-indazol-3-yl)piperidine-2,6-dione CPD-443 3-(6-((3-(4-(6-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)propyl)amino)-1-methyl-1H-indazol-3-yl)piperidine-2,6-dione CPD-444 3-(5-(3-(4-(6-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)prop-1-yn-1-yl)-1H-benzo[d]imidazol-1-yl)piperidine-2,6-dione CPD-445 3-(5-(3-(4-(6-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)propyl)-1H-benzo[d]imidazol-1-yl)piperidine-2,6-dione CPD-446 3-(5-(2-(4-(6-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)ethoxy)-1H-benzo[d]imidazol-1-yl)piperidine-2,6-dione CPD-447 3-(5-((2-(4-(6-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)ethyl)amino)-1H-benzo[d]imidazol-1-yl)piperidine-2,6-dione CPD-448 3-(5-(4-(4-(6-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)but-1-yn-1-yl)-1H-benzo[d]imidazol-1-yl)piperidine-2,6-dione CPD-449 3-(5-(4-(4-(6-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)butyl)-1H-benzo[d]imidazol-1-yl)piperidine-2,6-dione CPD-450 3-(5-(3-(4-(6-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)propoxy)-1H-benzo[d]imidazol-1-yl)piperidine-2,6-dione CPD-451 3-(5-((3-(4-(6-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)propyl)amino)-1H-benzo[d]imidazol-1-yl)piperidine-2,6-dione CPD-452 3-(5-(3-(4-(6-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)prop-1-yn-1-yl)-1H-indol-1-yl)piperidine-2,6-dione CPD-453 3-(5-(3-(4-(6-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)propyl)-1H-indol-1-yl)piperidine-2,6-dione CPD-454 3-(5-(2-(4-(6-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)ethoxy)-1H-indol-1-yl)piperidine-2,6-dione CPD-455 3-(5-((2-(4-(6-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)ethyl)amino)-1H-indol-1-yl)piperidine-2,6-dione CPD-456 3-(5-(4-(4-(6-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)but-1-yn-1-yl)-1H-indol-1-yl)piperidine-2,6-dione CPD-457 3-(5-(4-(4-(6-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)butyl)-1H-indol-1-yl)piperidine-2,6-dione CPD-458 3-(5-(3-(4-(6-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)propoxy)-1H-indol-1-yl)piperidine-2,6-dione CPD-459 3-(5-((3-(4-(6-(6-((R)-2-(3-fluorophenyl)pyrrolidin-1-yl)imidazo[1,2-b]pyridazin-3-yl)pyridin-2-yl)piperazin-1-yl)propyl)amino)-1H-indol-1-yl)piperidine-2,6-dione CPD-460 N-(5-((R)-2-(2,5-difluorophenyl)pyrrolidin-1-yl)pyrazolo[1,5-a]pyrimidin-3-yl)-6-((1-((2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-5-yl)glycyl)piperidin-4-yl)amino)picolinamide CPD-461 N-(5-((R)-2-(2,5-difluorophenyl)pyrrolidin-1-yl)pyrazolo[1,2-a]pyrimidin-3-yl)-6-((1-(4-(((S)-1-((2S,4R)-4-hydroxy-2-((4-(4-methylthiazol-5-yl)benzyl)carbamoyl)pyrrolidin-1-yl)-3,3-dimethyl-1-oxobutan-2-yl)amino)-4-oxobutanoyl)piperidin-4-yl)amino)picolinamide CPD-462 N-(5-((R)-2-(2,5-difluorophenyl)pyrrolidin-1-yl)pyrazolo[1,5-a]pyrimidin-3-yl)-6-((1-(6-(((S)-1-((2S,4R)-4-hydroxy-2-((4-(4-methylthiazol-5-yl)benzyl)carbamoyl)pyrrolidin-1-yl)-3,3-dimethyl-1-oxobutan-2-yl)amino)-6-oxohexanoyl)piperidin-4-yl)amino)picolinamide CPD-463 N-(5-((R)-2-(2,5-difluorophenyl)pyrrolidin-1-yl)pyrazolo[1,5-a]pyrimidin-3-yl)-6-((1-(8-(((S)-1-((2S,4R)-4-

hydroxy-2-((4-(4-methylthiazol-5-yl)benzyl)carbamoyl)pyrrolidin-1-yl)-3,3-dimethyl 1-oxobutan-2-yl)amino)-8-oxooctanoyl)piperidin-4-yl)amino)picolinamide CPD-464 N-(5-((R)-2-(2,5-difluorophenyl)pyrrolidin-1-yl)pyrazolo[1,5-a]pyrimidin-3-yl)-6-((1-(10-(((S)-1-((2S,4R)-4-hydroxy-2-((4-(4-methylthiazol-5-yl)benzyl)carbamoyl)pyrrolidin-1-yl)-3,3-dimethyl-1-oxobutan-2-yl)amino)-10-oxodecanoyl)piperidin-4-yl)amino)picolinamide CPD-465 N-(5-((R)-2-(2,5-difluorophenyl)pyrrolidin-1-yl)pyrazolo[1,5-a]pyrimidin-3-yl)-6-((1-(3-(3-(((S)-1-((2S,4R)-4-hydroxy-2-((4-(4-methylthiazol-5-yl)benzyl)carbamoyl)pyrrolidin-1-yl)-3,3-dimethyl-1-oxobutan-2-yl)amino)-3-oxopropoxy)propanoyl)piperidin-4-yl)amino)picolinamide CPD-466 N-(5-((R)-2-(2,5-difluorophenyl)pyrrolidin-1-yl)pyrazolo[1,5-a]pyrimidin-3-yl)-6-((1-(3-(2-(3-(((S)-1-((2S,4R)-4-hydroxy-2-((4-(4-methylthiazol-5-yl)benzyl)carbamoyl)pyrrolidin-1-yl)-3,3-dimethyl-1-oxobutan-2-yl)amino)-3-oxopropoxy)ethoxy)propanoyl)piperidin-4-yl)amino)picolinamide CPD-467 N-(5-((R)-2-(2,5-difluorophenyl)pyrrolidin-1-yl)pyrazolo[1,5-a]pyrimidin-3-yl)-6-((1-((S)-15-((2S,4R)-4-hydroxy-2-((4-(4-methylthiazol-5-yl)benzyl)carbamoyl)pyrrolidine-1-carbonyl)-16,16-dimethyl-13-oxo-4,7,10-trioxa-14-azaheptadecanoyl)piperidin-4-yl)amino)picolinamide CPD-468 N-(5-((R)-2-(2,5-difluorophenyl)pyrrolidin-1-yl)pyrazolo[1,5-a]pyrimidin-3-yl)-6-((1-((S)-18-((2S,4R)-4-hydroxy-2-((4-(4-methylthiazol-5-yl)benzyl)carbamoyl)pyrrolidine-1-carbonyl)-19,19-dimethyl-16-oxo-4,7,10,13-tetraoxa-17-azaicosanoyl)piperidin-4-yl)amino)picolinamide CPD-469 N-(5-((R)-2-(2,5-difluorophenyl)pyrrolidin-1-yl)pyrazolo[1,5-a]pyrimidin-3-yl)-6-((1-((S)-21-((2S,4R)-4-hydroxy-2-((4-(4-methylthiazol-5-yl)benzyl)carbamoyl)pyrrolidine-1-carbonyl)-22,22-dimethyl-19-oxo-4,7,10,13,16-pentaoxa-20-azatricosanoyl)piperidin-4-yl)amino)picolinamide CPD-470 N-(5-((R)-2-(2,5-difluorophenyl)pyrrolidin-1-yl)pyrazolo[1,5-a]pyrimidin-3-yl)-6-((1-(4-((2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-5-yl)amino)butanoyl)piperidin-4-yl)amino)picolinamide CPD-471 N-(5-((R)-2-(2,5-difluorophenyl)pyrrolidin-1-yl)pyrazolo[1,5-a]pyrimidin-3-yl)-6-((1-(6-((2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-5-yl)amino)hexanoyl)piperidin-4-yl)amino)picolinamide CPD-472 N-(5-((R)-2-(2,5-difluorophenyl)pyrrolidin-1-yl)pyrazolo[1,5-a]pyrimidin-3-yl)-6-((1-(8-((2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-5-yl)amino)octanoyl)piperidin-4-yl)amino)picolinamide CPD-473 N-(5-((R)-2-(2,5-difluorophenyl)pyrrolidin-1-yl)pyrazolo[1,5-a]pyrimidin-3-yl)-6-((1-(3-(2-((2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-5-yl)amino)ethoxy)propanoyl)piperidin-4-yl)amino)picolinamide CPD-474 N-(5-((R)-2-(2,5-difluorophenyl)pyrrolidin-1-yl)pyrazolo[1,5-a]pyrimidin-3-yl)-6-((1-(3-(2-(2-((2-(2,6-difluoropiperidin-3-yl)-1,3-dioxoisindolin-5-yl)amino)ethoxy)ethoxy)propanoyl)piperidin-4-yl)amino)picolinamide CPD-475 N-(5-((R)-2-(2,5-difluorophenyl)pyrrolidin-1-yl)pyrazolo[1,5-a]pyrimidin-3-yl)-6-((3-(2-(2-(2-((2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-5-yl)amino)ethoxy)ethoxy)ethoxy)propanoyl)piperidin-4-yl)amino)picolinamide CPD-476 N-(5-((R)-2-(2,5-difluorophenyl)pyrrolidin-1-yl)pyrazolo[1,5-a]pyrimidin-3-yl)-6-((1-(1-((2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-5-yl)amino)-3,6,9,12-tetraoxapentadecan-15-oyl)piperidin-4-yl)amino)picolinamide CPD-477 N-(5-((R)-2-(2,5-difluorophenyl)pyrrolidin-1-yl)pyrazolo[1,5-a]pyrimidin-3-yl)-6-((1-(1-((2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-5-yl)amino)-3,6,9,12,15-pentaoxaoctadecan-18-oyl)piperidin-4-yl)amino)picolinamide CPD-478 N-(2-(4-carbamoylpiperidin-1-yl)-4-((4-((2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-4-yl)glycyl)piperazin-1-yl)methyl)phenyl)-2-morpholinoxazole-4-carboxamide CPD-479 N-(2-(4-carbamoylpiperidin-1-yl)-4-((4-(3-((2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-4-yl)amino)propanoyl)piperazin-1-yl)methyl)phenyl)-2-morpholinoxazole-4-carboxamide CPD-480 N-(2-(4-carbamoylpiperidin-1-yl)-4-((4-(4-((2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-4-yl)amino)butanoyl)piperazin-1-yl)methyl)phenyl)-2-morpholinoxazole-4-carboxamide CPD-481 N-(2-(4-carbamoylpiperidin-1-yl)-4-((4-(5-((2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-4-yl)amino)pentanoyl)piperazin-1-yl)methyl)phenyl)-2-morpholinoxazole-4-carboxamide CPD-482 N-(2-(4-carbamoylpiperidin-1-yl)-4-((4-(6-((2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-4-yl)amino)hexanoyl)piperazin-1-yl)methyl)phenyl)-2-morpholinoxazole-4-carboxamide CPD-483 N-(2-(4-carbamoylpiperidin-1-yl)-4-((4-(7-((2-(2,6-

dioxopiperidin-3-yl)-1,3-dioxoisindolin-4-yl)heptanoyl)piperazin-1-yl)methyl)phenyl)-2-morpholinoxazole-4-carboxamide CPD-484 N-(2-(4-carbamoylpiperidin-1-yl)-4-((4-(8-((2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-4-yl)amino)octanoyl)piperazin-1-yl)methyl)phenyl)-2-morpholinoxazole-4-carboxamide CPD-485 N-(2-(4-carbamoylpiperidin-1-yl)-4-((4-(3-(2-((2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-4-yl)amino)ethoxy)propanoyl)piperazin-1-yl)methyl)phenyl)-2-morpholinoxazole-4-carboxamide CPD-486 N-(2-(4-carbamoylpiperidin-1-yl)-4-((4-(3-(2-(2-((2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-4-yl)amino)ethoxy)ethoxy)ethoxy)propanoyl)piperazin-1-yl)methyl)phenyl)-2-morpholinoxazole-4-carboxamide CPD-487 N-(2-(4-carbamoylpiperidin-1-yl)-4-((4-(3-(2-(2-(2-((2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-4-yl)amino)ethoxy)ethoxy)ethoxy)ethoxy)propanoyl)piperazin-1-yl)methyl)phenyl)-2-morpholinoxazole-4-carboxamide CPD-488 N-(2-(4-carbamoylpiperidin-1-yl)-4-((4-(1-((2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-4-yl)amino)-3,6,9,12-tetraoxapentadecan-15-oyl)piperazin-1-yl)methyl)phenyl)-2-morpholinoxazole-4-carboxamide CPD-489 N-(2-(4-carbamoylpiperidin-1-yl)-4-((4-(1-((2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-4-yl)amino)-3,6,9,12,15-pentaoxaoctadecan-18-oyl)piperazin-1-yl)methyl)phenyl)-2-morpholinoxazole-4-carboxamide CPD-490 N-(2-(4-carbamoylpiperidin-1-yl)-4-((4-(4-(((S)-1-((2S,4R)-4-hydroxy-2-((4-(4-methylthiazol-5-yl)benzyl)carbamoyl)pyrrolidin-1-yl)-3,3-dimethyl-1-oxobutan-2-yl)amino)-4-oxobutanoyl)piperazin-1-yl)methyl)phenyl)-2-morpholinoxazole-4-carboxamide CPD-491 N-(2-(4-carbamoylpiperidin-1-yl)-4-((4-(6-(((S)-1-((2S,4R)-4-hydroxy-2-((4-(4-methylthiazol-5-yl)benzyl)carbamoyl)pyrrolidin-1-yl)-3,3-dimethyl-1-oxobutan-2-yl)amino)-6-oxohexanoyl)piperazin-1-yl)methyl)phenyl)-2-morpholinoxazole-4-carboxamide CPD-492 N-(2-(4-carbamoylpiperidin-1-yl)-4-((4-(8-(((S)-1-((2S,4R)-4-hydroxy-2-((4-(4-methylthiazol-5-yl)benzyl)carbamoyl)pyrrolidin-1-yl)-3,3-dimethyl-1-oxobutan-2-yl)amino)-8-oxooctanoyl)piperazin-1-yl)methyl)phenyl)-2-morpholinoxazole-4-carboxamide CPD-493 N-(2-(4-carbamoylpiperidin-1-yl)-4-((4-(10-(((S)-1-((2S,4R)-4-hydroxy-2-((4-(4-methylthiazol-5-yl)benzyl)carbamoyl)pyrrolidin-1-yl)-3,3-dimethyl-1-oxobutan-2-yl)amino)-10-oxodecanoyl)piperazin-1-yl)methyl)phenyl)-2-morpholinoxazole-4-carboxamide CPD-494 N-(2-(4-carbamoylpiperidin-1-yl)-4-((4-(3-(3-(((S)-1-((2S,4R)-4-hydroxy-2-((4-(4-methylthiazol-5-yl)benzyl)carbamoyl)pyrrolidin-1-yl)-3,3-dimethyl-1-oxobutan-2-yl)amino)-3-oxopropyl)ethoxy)propanoyl)piperidin-1-yl)methyl)phenyl)-2-morpholinoxazole-4-carboxamide CPD-495 N-(2-(4-carbamoylpiperidin-1-yl)-4-((4-(3-(2-(3-(((S)-1-((2S,4R)-4-hydroxy-2-((4-(4-methylthiazol-5-yl)benzyl)carbamoyl)pyrrolidin-1-yl)-3,3-dimethyl-1-oxobutan-2-yl)amino)-3-oxopropoxy)ethoxy)propanoyl)piperazin-1-yl)methyl)phenyl)-2-morpholinoxazole-4-carboxamide CPD-496 N-(2-(4-carbamoylpiperidin-1-yl)-4-((4-((S)-15-((2S,4R)-4-hydroxy-2-((4-(4-methylthiazol-5-yl)benzyl)carbamoyl)pyrrolidine-1-carbonyl)-16,16-dimethyl-13-oxo-4,7,10-trioxa-14-azaheptadecanoyl)piperazin-1-yl)methyl)phenyl)-2-morpholinoxazole-4-carboxamide CPD-497 N-(2-(4-carbamoylpiperidin-1-yl)-4-((4-((S)-18-((2S,4R)-4-hydroxy-2-((4-(4-methylthiazol-5-yl)benzyl)carbamoyl)pyrrolidine-1-carbonyl)-19,19-dimethyl-16-oxo-4,7,10,13-tetraoxa-17-azaicosanoyl)piperazin-1-yl)methyl)phenyl)-2-morpholinoxazole-4-carboxamide CPD-498 2-(4-(2-amino-3-(3-methoxy-4-((4-methoxybenzyl)oxy)benzyl)-3H-imidazo[4,5-b]pyridin-6-yl)-1H-pyrazol-1-yl)-N-(2-((2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-4-yl)amino)ethyl)acetamide CPD-499 2-(4-(2-amino-3-(3-methoxy-4-((4-methoxybenzyl)oxy)benzyl)-3H-imidazo[4,5-b]pyridin-6-yl)-1H-pyrazol-1-yl)-N-(4-((2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-4-yl)amino)butyl)acetamide CPD-500 2-(4-(2-amino-3-(3-methoxy-4-((4-methoxybenzyl)oxy)benzyl)-3H-imidazo[4,5-b]pyridin-6-yl)-1H-pyrazol-1-yl)-N-(6-((2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-4-yl)amino)hexyl)acetamide CPD-501 2-(4-(2-amino-3-(3-methoxy-4-((4-methoxybenzyl)oxy)benzyl)-3H-imidazo[4,5-b]pyridin-6-yl)-1H-pyrazol-1-yl)-N-(8-((2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-4-yl)amino)octyl)acetamide CPD-502 2-(4-(2-amino-3-(3-methoxy-4-((4-methoxybenzyl)oxy)benzyl)-3H-imidazo[4,5-b]pyridin-6-yl)-1H-pyrazol-1-yl)-N-(2-(2-((2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-4-yl)amino)ethoxy)ethyl)

acetamide CPD-503 2-(4-(2-amino-3-(3-methoxy-4-((4-methoxybenzyl)oxy)benzyl)-3H-imidazo[4,5-b]pyridin-6-yl)-1H-pyrazol-1-yl)-N-(2-(2-(2-((2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-4-yl)amino)ethoxy)ethoxy) ethyl)acetamide CPD-504 2-(4-(2-amino-3-(3-methoxy-4-((4-methoxybenzyl)oxy)benzyl)-3H-imidazo[4,5-b]pyridin-6-yl)-1H-pyrazol-1-yl)-N-(2-(2-(2-(2-((2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-4-yl)amino)ethoxy)ethoxy)ethoxy)ethyl)acetamide CPD-505 2-(4-(2-amino-3-(3-methoxy-4-((4-methoxybenzyl)oxy)benzyl)-3H-imidazo[4,5-b]pyridin-6-yl)-1H-pyrazol-1-yl)-N-(14-((2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-4-yl)amino)-3,6,9,12-tetraoxatetradecyl)acetamide CPD-506 2-(4-(2-amino-3-(3-methoxy-4-((4-methoxybenzyl)oxy)benzyl)-3H-imidazo[4,5-b]pyridin-6-yl)-1H-pyrazol-1-yl)-N-(17-((2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-4-yl)amino)-3,6,9,12,15-pentaoxaheptadecyl)acetamide CPD-507 (2S,4R)-1-((S)-2-(2-(2-(4-(2-amino-3-(3-methoxy-4-((4-methoxybenzyl)oxy)benzyl)-3H-imidazo[4,5-b]pyridin-6-yl)-1H-pyrazol-1-yl)acetamido)-3,3-dimethylbutanoyl)-4-hydroxy-N-(4-(4-methylthiazol-5-yl)benzyl)pyrrolidine-2-carboxamide CPD-508 (2S,4R)-1-((S)-2-(4-(2-(4-(2-amino-3-(3-methoxy-4-((4-methoxybenzyl)oxy)benzyl)-3H-imidazo[4,5-b]pyridin-6-yl)-1H-pyrazol-1-yl)acetamido)butanamido)-3,3-dimethylbutanoyl)-4-hydroxy-N-(4-(4-methylthiazol-5-yl)benzyl)pyrrolidine-2-carboxamide CPD-509 (2S,4R)-1-((S)-2-(6-(2-(4-(2-amino-3-(3-methoxy-4-((4-methoxybenzyl)oxy)benzyl)-3H-imidazo[4,5-b]pyridin-6-yl)-1H-pyrazol-1-yl)acetamido)hexanamido)-3,3-dimethylbutanoyl)-4-hydroxy-N-(4-(4-methylthiazol-5-yl)benzyl)pyrrolidine-2-carboxamide CPD-510 (2S,4R)-1-((S)-2-(8-(2-(4-(2-amino-3-(3-methoxy-4-((4-methoxybenzyl)oxy)benzyl)-3H-imidazo[4,5-b]pyridin-6-yl)-1H-pyrazol-1-yl)acetamido)octanamido)-3,3-dimethylbutanoyl)-4-hydroxy-N-(4-(4-methylthiazol-5-yl)benzyl)pyrrolidine-2-carboxamide CPD-511 (2S,4R)-1-((S)-2-(10-(2-(4-(2-amino-3-(3-methoxy-4-((4-methoxybenzyl)oxy)benzyl)-3H-imidazo[4,5-b]pyridin-6-yl)-1H-pyrazol-1-yl)acetamido)decanamido)-3,3-dimethylbutanoyl)-4-hydroxy-N-(4-(4-methylthiazol-5-yl)benzyl)pyrrolidine-2-carboxamide CPD-512 (2S,4R)-1-((S)-2-(3-(2-(2-(4-(2-amino-3-(3-methoxy-4-((4-methoxybenzyl)oxy)benzyl)-3H-imidazo[4,5-b]pyridin-6-yl)-1H-pyrazol-1-yl)acetamido)ethoxy)propanamido)-3,3-dimethylbutanoyl)-4-hydroxy-N-(4-(4-methylthiazol-5-yl)benzyl)pyrrolidine-2-carboxamide CPD-513 (2S,4R)-1-((S)-1-(4-(2-amino-3-(3-methoxy-4-((4-methoxybenzyl)oxy)benzyl)-3H-imidazo[4,5-b]pyridin-6-yl)-1H-pyrazol-1-yl)-14-(tert-butyl)-2,12-dioxo-6,9-dioxo-3,13-diazapentadecan-15-oyl)-4-hydroxy-N-(4-(4-methylthiazol-5-yl)benzyl)pyrrolidine-2-carboxamide CPD-514 (2S,4R)-1-((S)-1-(4-(2-amino-3-(3-methoxy-4-((4-methoxybenzyl)oxy)benzyl)-3H-imidazo[4,5-b]pyridin-6-yl)-1H-pyrazol-1-yl)-17-(tert-butyl)-2,15-dioxo-6,9,12-trioxa-3,16-diazaoctadecan-18-oyl)-4-hydroxy-N-(4-(4-methylthiazol-5-yl)benzyl)pyrrolidine-2-carboxamide CPD-515 (2S,4R)-1-((S)-1-(4-(2-amino-3-(3-methoxy-4-((4-methoxybenzyl)oxy)benzyl)-3H-imidazo[4,5-b]pyridin-6-yl)-1H-pyrazol-1-yl)-20-(tert-butyl)-2,18-dioxo-6,9,12,15-tetraoxa-3,19-diazahenicosan-21-oyl)-4-hydroxy-N-(4-(4-methylthiazol-5-yl)benzyl)pyrrolidine-2-carboxamide CPD-516 (2S,4R)-1-((S)-1-(4-(2-amino-3-(3-methoxy-4-((4-methoxybenzyl)oxy)benzyl)-3H-imidazo[4,5-b]pyridin-6-yl)-1H-pyrazol-1-yl)-23-(tert-butyl)-2,21-dioxo-6,9,12,15,18-pentaoxa-3,22-diazatetracosan-24-oyl)-4-hydroxy-N-(4-(4-methylthiazol-5-yl)benzyl)pyrrolidine-2-carboxamide

(1989) As used herein, in case of discrepancy between the structure and chemical name provided for a particular compound, the structure shall control.

Example 453. Assessing the effect of selected compounds on reducing TPM3-TRKA fusion protein levels in KM12 colon cancer cells (FIG. 1A-1C)

(1990) KM12 cells were treated with DMSO or indicated compounds at 100 nM for 16 hours. The Western blot results showed that multiple compounds significantly reduced TPM3-TRKA protein levels.

Example 454. Bivalent compounds time-dependently reduced the TPM3-TRKA fusion protein levels in KM12 colon cancer cells (FIG. 2)

(1991) KM12 cells were treated with CPD-027, CPD-053, and CPD-060 at 100 nM for indicated incubation time. The Western blot results showed that CPD-027, CPD-053, and CPD-060 rapidly reduced TPM3-TRKA protein levels.

Example 455. Bivalent compounds reduce TPM3-TRKA protein levels in KM12 subcutaneous xenograft tumors (FIG. 3)

(1992) Athymic nude mice bearing KM12 subcutaneous xenograft tumors at the right flank were intraperitoneally treated with 10, 20, or 50 mg/kg CPD-053, CPD-027, or CPD-060, as indicated in FIG. 3. Four hours after drug administration, animals were sacrificed for immunoblotting of TPM3-TPKA in homogenized xenograft tumor chunks. The label “a” or “b” represents two different chunks of the same xenograft tumor.

Example 456. Bivalent compounds suppressed viability of KM12 colorectal cancer cells (FIG. 4A-4B)

(1993) KM12 cells seeded in 96-well plates were treated with 500 nM entrectinib or bivalent compounds, i.e. CPD-010, CPD-053, and CPD-057 following a 12-point 2-fold serial dilution for 3 days. CPD-010, CPD-053, and CPD-057 inhibited the viability of KM12 cells (FIG. 4A). A summary of IC₅₀ values of selected compounds is presented in Table 2.

(1994) KM12 or H358 cells were treated with 1000 nM CPD-053 following an 8-point 3-fold serial dilution for 3 days. CPD-053 significantly inhibited the viability of KM12 cells but not H358 cells (FIG. 4B).

Example 457. Assessing the effect of selected compounds on reducing TPM3-TRKA fusion protein levels in KM12 colon cancer cells (FIG. 5A-SC)

(1995) KM12 cells were treated with DMSO or indicated compounds at 0.1 nM, 1 nM or 10 nM for 16 hours. Immunoblotting data showed that all the selected compounds at 1 or 10 nM significantly reduced TPM3-TRKA protein levels, while at 0.1 nM the extents of degradation varied.

Example 458. Bivalent compounds concentration-dependently reduced the TPM3-TRKA fusion protein levels in KM12 colon cancer cells (FIG. 6A-6D)

(1996) KM12 cells were treated with DMSO or indicated compounds at various concentrations for 16 hours. Immunoblotting results showed that TPM3-TRKA was significantly degraded in a concentration-dependent manner.

Example 459. Bivalent compound concentration-dependently reduced the overexpressed TPM3-TRKA, AGBLA-TRKB and ETV6-TRKC fusion protein levels (FIG. 7)

(1997) KM12 cells were infected with lentivirus that directed expression of TPM3-TRKA, AGBL4-TRKB or ETV6-TRKC fusion. Cells were treated with DMSO or TR-123 at a range of dose for 18 hours.

Example 460. Bivalent compound-mediated degradation of TRKA is dependent on the interaction with cereblon (FIG. 5A-8B)

(1998) KM12 or HEL cells were treated with a dose range of compound TR-123 or TR-123-negative. The latter lost binding to cereblon (CRBN) due to a chemical modification. Data showed that TR-123 reduced TRKA protein levels in a concentration-dependent manner while TR-123-negative had none or less effects on TRKA protein levels.

Example 461. Bivalent compound reduced TRKA degradation is dependent on the ubiquitin-proteasome system (FIG. 9A-9B)

(1999) KM12 or HEL cells were treated with a single dose of compound TR-123 or combination with MG-132, Bortezomib, MLN4924 or Pomalidomide. Data showed that TR-123 reduced TRKA degradation is compromised by proteasome inhibitors, MG-132 or Bortezomib, cullin E₃ ligase inhibitor, MLN4924 or CRBN ligand, Pomalidomide.

Example 462. Bivalent compounds reduced TPM3-TRKA protein levels in KM12 subcutaneous xenograft tumors (FIG. 10A-10B)

(2000) Athymic nude mice bearing KM12 subcutaneous xenograft tumors at the right flank were

intraperitoneally or orally treated with bivalent compounds at indicated dose. Eight hours after drug administration, animals were sacrificed for immunoblotting of TPM3-TPKA in homogenized xenograft tumor samples.

Example 463. The pharmacokinetics of TR-123 (FIG. 11)

(2001) A single 20 mg/kg intraperitoneal injection of compound TR-123 was evaluated. Plasma concentrations of TR-123 reported at each of the 6 time points (30 min, 1 h, 2 h, 4 h, 8 h, and 12 h post compound administration) are the average values from 3 test animals. Data showed a good plasma exposure of TR-123 over 12 hours.

Example 464. Bivalent compound showed in vivo anti-tumor activity (FIG. 12A-12B)

(2002) Athymic nude mice bearing KM12 subcutaneous xenograft tumors at the right flank were intraperitoneally treated with vehicle or 25 mg/kg CPD-060 once per day (qd), twice per day (bid), or once every two days (q2d). Tumor volumes were assessed every two days and expressed as $\text{length} \times \text{width} \times \text{width} / 2$. Body weights of experimental mice were also measured to assess the toxicity of compounds.

Example 465. Bivalent compounds concentration-dependently reduced the TPM3-TRKA fusion protein levels in KM12 colon cancer cells (FIG. 13A-13E)

(2003) KM12 cells were treated with DMSO or indicated compounds at various concentrations for 4 hours. Immunoblotting results showed that TPM3-TRKA was significantly degraded in a concentration-dependent manner.

Example 466. The pharmacokinetics of TR-198 in mice (FIG. 14)

(2004) A single 2 mg/kg intravenous injection and 20 mg/kg oral gavage of TR-198 were evaluated in mice. Plasma concentrations of TR-198 reported at each of the 6 time points (5 min, 30 min, 2 h, 4 h, 8 h, and 12 h post compound administration via IV, 30 min, 1 h, 2 h, 4 h, 8 h, and 12 h post compound administration via p.o.) are the average values from 3 test animals. Data showed around 16% oral bioavailability of TR-198 over 12 hours.

Example 467. Bivalent compound showed in vivo anti-tumor activity (FIG. 15A-15B)

(2005) Athymic nude mice bearing KM12 subcutaneous xenograft tumors at the right flank were treated with vehicle, 10-20 mg/kg TR-181 or TR-198 once per day (qd), or twice per day (bid) via oral gavage. Tumor volumes were assessed every two days and expressed as $\text{length} \times \text{width} \times \text{width} / 2$. Body weights of experimental mice were also measured to assess the toxicity of compounds. Data showed significant in vivo anti-tumor activity of TR-181 and TR-198.

Example 468. Bivalent compound exhibited analgesic activity in monoiodoacetate-induced osteoarthritis models of rats and guinea pigs (FIG. 16A-16B)

(2006) Osteoarthritis was induced in the right knee of adult male animals using monoiodoacetate injection. One week later, animals were treated with vehicle (Veh), 20 mg/kg TR-181, or 100 mg/kg ibuprofen (Ibu) as the positive control. Weight distribution was determined as the percentage of weight born on the injured limb using an incapacitance meter. Response in the ibuprofen group was determined 2 hours after drug administration. The other two groups were measured 6 hours following treatment.

Example 469. Bivalent compounds decreased TPM3-TRKA protein levels in KM12 cells in a concentration-dependent manner (FIG. 17A-21C)

(2007) KM12 cells (a cancer model comprising a colon cancer cell line) were treated with DMSO or indicated compounds at various concentrations for 4 hours. The western blot results showed that multiple compounds significantly reduced TPM3-TRKA protein levels. Some compounds were more effective than others, as shown in the data.

Example 470. Bivalent compounds downregulated NPM-ALK protein levels in SU-DHL-1 cells following exposure to indicated compounds for 8 hours (FIG. 22)

(2008) SU-DHL-1 cells (a cancer model comprising an anaplastic large cell lymphoma cell line) were treated with DMSO or indicated compounds at various concentrations for 4 hours. The western blot results showed that multiple compounds significantly reduced NPM-ALK protein

levels.

Example 471. Plasma exposure and degradation of target protein in xenograft tumors following single dose of bivalent compounds (FIG. 23A-23B)

(2009) Plasma concentrations of TR-231 and TR-275 were determined following a single dose of 10 mg/kg orally or 1 mg/kg intravenously in male ICR mice (FIG. 23A). Plasma concentrations of TR-231 or TR-275 are examined at indicated different time points post compound administration. The reported values are the average values from 3 test animals. Data showed around 48% oral bioavailability for TR-231 and 27% oral bioavailability for TR-275.

(2010) Athymic nude mice bearing KM12 subcutaneous xenograft tumors at the right flank were orally treated with TR-231 and TR-275 at 10, 20 or 40 mg/kg doses. Animals were sacrificed for immunoblotting of TPM3-TPKA in homogenized xenograft tumor samples after 6, 16, and 24 hours, the results of which are shown in FIG. 23B.

Example 472. Bivalent compounds reduced KM12 xenograft tumors growth (FIG. 24A-24B)

(2011) Mice bearing KM12 xenograft tumors were treated with vehicle, TR-231 or TR-275 orally at 40 mg/kg, twice per day for 14 days. Tumor sizes were measured every two days and presented as $\text{volume} = (\text{length} \times \text{width} \times \text{width}) / 2$. These and other data included herein indicate that the compounds described herein may be useful for treating a cancer such as colon cancer. For example, a compound described herein may be useful for reducing tumor volume or limiting tumor growth.

Example 473. Heterobifunctional compounds reduced the TPM3-TRKA fusion protein levels in KM12 colon cancer cells (FIG. 25A-25C)

(2012) KM12 cells were treated with heterobifunctional compounds or their corresponding control compounds at 10 nM or 100 nM for 16 hours. The Western blot results showed that multiple heterobifunctional compounds significantly reduced TPM3-TRKA protein levels even at 10 nM.

Example 474. Heterobifunctional compounds reduced the wildtype TRKA protein levels in HEL erythroleukemia cells (FIG. 26)

(2013) HEL cells were treated with heterobifunctional compounds at 10 nM or 100 nM for 16 hours. The Western blot results showed that some heterobifunctional compounds significantly reduced TRKA protein levels.

Example 475. Heterobifunctional compounds suppressed viability of KM12 colorectal cancer cells (FIG. 27A-27C)

(2014) KM12 cells seeded in 96-well plates were treated with heterobifunctional compounds following a 11-point 3-fold serial dilution for 3 days. Selected compounds inhibited the viability of KM12 cells (FIG. 27A-27C). A summary of IC₅₀ values of selected compounds is presented in Table 3.

Example 476. Heterobifunctional compounds reduced TPM3-TRKA fusion protein levels in a cell line (FIG. 28A-28B)

(2015) KM12 cells were treated with heterobifunctional compounds or their corresponding control compounds at 0.4-30 nM for 8-16 hours. The Western blot results showed that multiple heterobifunctional compounds significantly reduced TPM3-TRKA protein levels even at 0.4 nM. TR-342 appeared more effective at degrading TRKA at lower concentrations than TR-342. TR-343 appeared more effective at degrading TRKA at lower concentrations than TR-344.

Example 477. Clinical Trials

(2016) The compounds described herein will be tested in human clinical studies to determine their efficacy for degrading TRK proteins (e.g. TRKA and/or TRKC) in a human. Compounds will be selected based on TRK binding data, TRK kinase activity inhibition data, TRK protein degradation data, cellular anti-proliferation data, and/or other experiments described herein. For example, heterobifunctional compounds that bind TRKA or degrade TRKA in vitro or in vivo will be tested for similar effects in humans. Likewise, heterobifunctional compounds that inhibit cells in vivo or in vitro will be tested in humans with similar cancers. In some embodiments, the compounds described herein are effective against cancer in humans.

Materials and Methods

(2017) General Chemistry Methods:

(2018) All chemicals and reagents were purchased from commercial suppliers and used without further purification. LCMS spectra for all compounds were acquired using a Shimadzu LC-MS 2020 system comprising a pump (LC-20AD) with degasser (DGU-20A_{sup}.3), an autosampler (SIL-20AHT), a column oven (CTO-20A)(set at 40° C., unless otherwise indicated), a photo-diode array (PDA)(SPD-M20A) detector, an evaporative light-scattering (ELSD)(Alltech 3300ELSD) detector. Chromatography was performed on a Shimadzu SunFire C_{sub}.18 (5 µm 50 * 4.6 mm) with water containing 0.1% formic acid as solvent A and acetonitrile containing 0.1% formic acid as solvent B at a flow rate of 2.0 ml/min. Flow from the column was split to a MS spectrometer. The MS detector was configured with an electrospray ionization source. Nitrogen was used as the nebulizer gas. Data acquisition was performed with a Labsolution data system. Proton Nuclear Magnetic Resonance (_{sup}.1H-NMR) spectra were recorded on a Bruker Avance III 400 spectrometer. Chemical shifts are expressed in parts per million (ppm) and reported as δ value (chemical shift δ). Coupling constants are reported in units of hertz (J value, Hz; Integration and splitting patterns: where s=singlet, d=double, t=triplet, q=quartet, brs=broad singlet, m=multiple). Preparative HPLC was performed on Agilent Prep 1260 series with UV detector set to 254 nm or 220 nm. Samples were injected onto a Phenomenex Luna 75×30 mm, 5 µm, C_{sub}.18 column at room temperature. The flow rate was 40 ml/min. A linear gradient was used with 10% (or 50%) of MeOH (A) in H₂O (with 0.1% TFA)(B) to 100% of MeOH (A). All compounds showed >90% purity using the LCMS methods described above.

(2019) Plasmids

(2020) The coding sequences of human TPM3-TRKA, AGBL4-TRKB and ETV6-TRKC fusion genes were constructed into pLVX-EF1 α -mCherry-C_{sub}.1 lentiviral vector.

(2021) Cell Culture

(2022) KM12, H358, HEL, SU-DHL-1 and other cells were cultured at 37° C. with 5% CO_{sub}.2 in RPMI 1640 Medium supplemented with 10% fetal bovine serum. Cells were authenticated using the short tandem repeat (STR) assays. *Mycoplasma* test results were negative. Stable cell lines were established by lentivirus transduction, selected and maintained in medium containing 1 µg/mL puromycin (Beyotime Biotechnology).

(2023) Antibodies and Reagents

(2024) Rabbit anti-TRK antibody (929918), phosphor-ERK antibody (4370S) and Vinculin antibody (18799S) were purchased from Cell Signaling Technology. HRP-conjugated anti-GAPDH antibody, anti-B-actin antibody and anti- α -tubulin antibodies were purchased from GNI. Media and other cell culture reagents were purchased from Thermo Fisher. The CellTiter-Glo Assay kit was purchased from Promega.

(2025) Immunoblotting

(2026) Cultured cells were washed with cold PBS once and lysed in cold RIPA buffer supplemented with protease inhibitors and phosphatase inhibitors (Beyotime Biotechnology). The solutions were then incubated at 4° C. for 30 minutes with gentle agitation to fully lyse cells. For tumor tissues, tumors were cut into small pieces, milled with cold RIPA buffer using a mechanical homogenizer and lysed for 1 hour at 4° C. Cell lysates were centrifuged at 13,000 rpm for 10 minutes at 4° C. and pellets were discarded. Total protein concentrations in the lysates were determined by BCA assays (Beyotime Biotechnology). Cell lysates were mixed with Laemmli loading buffer to 1× and heated at 99° C. for 5 min. Proteins were resolved on SDS-PAGE and visualized by chemiluminescence. Images were taken by a ChemiDoc MP Imaging system (Bio-Rad). Protein bands were quantitated using the software provided by Bio-Rad.

(2027) Cell Viability Assays

(2028) Cells were seeded at a density of 5000 cells per well in 96-well assay plates and treated with test compounds following a 12-point 2-fold serial dilution or a 8-point 3-fold dilution, or 11-point

3-fold serial dilution. Three days later, cell viability was determined using the CellTiter-Glo assay kit (Promega) according to the manufacturer's instructions. The dose-response curves were determined and IC_{sub}.50 values were calculated using the GraphPad Prism software following a nonlinear regression (least squares fit) method.

(2029) The activities of selected compounds on KM12 cell viability (IC_{sub}.50 values) and TMP3-TRKA degradation (percentage degradation at 10 nM) in KM12 cells are listed in Table 2 and Table 3

(2030) TABLE-US-00002 TABLE 2 Degradation Cpd. ID IC_{sub}.50 (nM) (at 10 nM) CPD-001 >1000 – CPD-002 >1000 – CPD-003 >1000 – CPD-004 176 – CPD-005 >1000 – CPD-006 500 – CPD-007 >1000 – CPD-008 >1000 – CPD-009 35 ++ CPD-010 30 ++ CPD-011 >1000 – CPD-012 >1000 – CPD-013 10.2 +++ CPD-014 7.9 +++ CPD-015 14.9 +++ CPD-016 150 – CPD-017 145 + CPD-018 258 – CPD-019 500 – CPD-020 200 – CPD-021 27.4 +++ CPD-022 100 ++ CPD-023 14.9 +++ CPD-024 10 +++ CPD-025 16.7 +++ CPD-026 82.3 ++ CPD-027 4.9 +++ CPD-028 4.3 +++ CPD-029 74 ++ CPD-030 20.5 +++ CPD-031 68.5 +++ CPD-032 5.4 +++ CPD-033 100 +++ CPD-034 300 + CPD-035 300 + CPD-036 300 – CPD-037 150 ++ CPD-038 359 – CPD-039 222 + CPD-040 600 – CPD-041 400 + CPD-042 >1000 – CPD-043 >1000 – CPD-044 73 ++ CPD-045 >1000 ++ CPD-046 700 ++ CPD-047 94 +++ CPD-048 >1000 + CPD-049 73 +++ CPD-050 28 +++ CPD-051 11 +++ CPD-052 60 +++ CPD-053 1.5 +++ CPD-054 13 +++ CPD-055 13 +++ CPD-056 12 +++ CPD-057 6.4 +++ CPD-058 >1000 + CPD-059 6 +++ CPD-060 6 +++ CPD-061 >1000 + CPD-062 29 +++ CPD-063 >1000 – CPD-064 14.9 +++ CPD-065 72.6 +++ TR-102 44.4 N/A TR-103 13.5 N/A TR-104 8.1 +++ TR-105 10.1 +++ TR-106 3.8 ++ TR-107 8.4 +++ TR-108 4.0 +++ TR-109 3.6 +++ TR-110 35.9 N/A TR-111 19.7 N/A TR-113 1.3 +++ TR-114 16.9 N/A TR-115 0.9 +++ TR-116 2.0 +++ TR-117 4.3 +++ TR-118 6.3 +++ TR-119 1.6 +++ TR-120 10.0 +++ TR-121 2.1 +++ TR-122 2.7 +++ TR-123 1.7 +++ TR-124 2.8 +++ TR-125 2.8 +++ TR-126 12.5 N/A TR-127 2.7 +++ TR-128 3.3 +++ TR-129 4.4 +++ TR-130 0.9 +++ TR-131 2.7 +++ TR-132 4.4 +++ TR-133 22.8 N/A TR-134 5.6 +++ TR-135 0.9 +++ TR-136 23.6 N/A TR-137 2.5 +++ TR-138 31.1 N/A TR-139 27.1 N/A TR-140 1.6 +++ TR-141 2.6 +++ TR-142 3.4 +++ TR-143 5.6 +++ TR-144 2.2 +++ TR-145 3.1 +++ TR-146 4.2 +++ TR-147 6.4 +++ TR-148 23.2 N/A TR-149 4.6 +++ TR-150 16.6 +++ TR-151 3.1 +++ TR-152 3.1 +++ TR-153 6.3 +++ TR-154 20.8 N/A TR-155 2.9 +++ TR-156 3.3 +++ TR-157 8.3 +++ TR-158 6.4 +++ TR-159 30.4 N/A TR-160 4.8 +++ TR-161 4.7 +++ TR-162 4.1 +++ TR-163 6.3 +++ TR-164 7.8 +++ TR-165 3.5 +++ TR-166 4.4 +++ TR-167 3.3 +++ TR-168 2.9 +++ TR-169 1.6 +++ TR-170 15.1 N/A TR-171 5.2 +++ TR-172 2 +++ TR-173 8.9 +++ TR-174 18.7 N/A TR-175 40.8 N/A TR-176 18 +++ TR-177 1.9 +++ TR-178 67.7 N/A TR-179 23.4 N/A TR-180 33.8 N/A TR-181 2.9 +++ TR-182 16 +++ TR-183 129 N/A TR-184 9.3 +++ TR-185 17.1 N/A TR-186 5.8 +++ TR-187 18.6 N/A TR-188 16.8 +++ TR-189 2.5 +++ TR-190 8.7 +++ TR-191 5.4 +++ TR-192 103.0 N/A TR-193 1103.0 N/A TR-194 7.4 ++ TR-195 23.6 N/A TR-196 3.1 ++ TR-197 67.1 N/A TR-198 5.5 +++ TR-199 39.8 N/A TR-200 100.6 N/A TR-201 >500 N/A TR-202 4.3 ++ TR-203 3.7 ++ TR-204 10.6 +++ TR-205 122 N/A TR-206 15.8 N/A TR-207 31.3 N/A TR-208 7 ++ TR-209 >500 N/A TR-210 14 ++ TR-211 2 +++ TR-212 15.8 N/A TR-213 22.7 N/A TR-214 2.7 + TR-215 >500 – TR-216 3.6 +++ TR-217 4 +++ TR-218 15.3 ++ TR-219 44.9 ++ TR-220 5.5 +++ TR-221 7.8 +++ TR-222 >500 – TR-223 11.3 ++ TR-224 8.7 +++ TR-225 8.4 +++ TR-226 13.8 +++ TR-227 23.6 ++ TR-228 56.3 ++ TR-229 50.8 ++ TR-230 >500 – TR-231 3 +++ TR-232 2.6 +++ TR-233 5 +++ TR-234 284 – TR-235 3.3 ++ TR-236 99.25 + TR-237 137.8 +++ TR-238 16.22 +++ TR-239 45.94 +++ TR-240 16.05 +++ TR-241 6.1 +++ TR-242 >500 – TR-243 >500 – TR-244 297.9 + TR-245 >500 + TR-246 100.7 +++ TR-247 8.2 +++ TR-248 27.4 +++ TR-249 15.1 +++ TR-250 16.1 +++ TR-251 46.5 +++ TR-252 47.6 +++ TR-253 6.9 +++ TR-254 20.3 +++ TR-255 14.4 +++ TR-256 46.9 ++ TR-257 >500 – TR-258 10.4 +++ TR-259 5.3 +++ TR-260 26.4 +++ TR-261 365.6 ++ TR-262 81 +++ TR-263 19.2 ++ TR-264 9.9 +++ TR-265 13.3 +++ TR-266 7.2 +++ TR-267 11.7 +++ TR-268 6.2 +++ TR-269 44.6 + TR-270 10.3 +++ TR-271 22 +++ TR-272 20.1 ++ TR-273 41.1 +++ TR-274

439.8 – TR-275 8.9 +++ TR-276 9.9 +++ TR-277 >500 – TR-278 34.7 ++ TR-279 19 +++ TR-280 4.3 +++ TR-281 17.4 +++ TR-282 6.3 +++ TR-283 54 +++ TR-284 5 +++ TR-285 9.7 +++ TR-286 7.8 +++ TR-287 19.3 +++ TR-288 9.2 +++ TR-289 3.8 +++ TR-290 11.2 +++ TR-291 446 – TR-292 8.2 +++ TR-293 9.5 +++ TR-294 12.5 +++ TR-295 30 ++ TR-296 22 ++ TR-297 39 ++ TR-298 30 +++ TR-299 23 +++ TR-300 53 ++ TR-301 6 +++ TR-302 1.7 +++ TR-303 8.4 +++ TR-304 1.5 +++ TR-305 2.5 +++ TR-306 4 +++ TR-307 12.7 – TR-308 11.8 +++ TR-309 4.3 ++ TR-310 12.5 – TR-311 14.8 – TR-312 107 – TR-313 13.8 – TR-314 213 ++ TR-315 4.5 +++ TR-316 2.3 +++ TR-317 5 +++ TR-318 2.4 +++ TR-319 5.7 +++ TR-320 1.4 +++ TR-321 2.3 +++ TR-322 168 + TR-323 31 +++ TR-324 1.3 +++ TR-325 13 +++ TR-326 14 ++ TR-327 13.3 ++ TR-328 35.9 +++ TR-329 >500 – TR-330 28.5 +++ TR-331 1.6 +++ TR-332 1.7 +++ TR-333 10.5 ++ TR-334 24.6 ++ TR-335 8.3 ++ TR-336 3.2 +++ TR-337 5.5 +++ TR-338 1.3 +++ TR-339 8.1 +++ TR-340 7.5 +++ TR-341 3 +++ TR-342 3 +++ TR-343 9 +++ TR-344 9 +++

(2031) The IC₅₀ value of each compound was determined as described in FIG. 4A and calculated using the GraphPad Prism 5.0 software. The percentage of degradation of each compound was indicated in the following categories: N/A: data not available; -: no degradation; +: degradation less than 50%; ++; degradation equal or greater than 50% but less than 80%; +++: degradation equal or greater than 80%

(2032) TABLE-US-00003 TABLE 3 Degradation Cpd. ID IC₅₀ (nM) (at 10 nM) CPD-460 32 – CPD-461 >1000 N/A CPD-462 >1000 N/A CPD-463 443 N/A CPD-464 87 – CPD-465 2018 N/A CPD-466 253 N/A CPD-467 343 N/A CPD-468 >1000 N/A CPD-469 >1000 N/A CPD-470 21 +++ CPD-471 19 +++ CPD-472 37 +++ CPD-473 24 +++ CPD-474 18 +++ CPD-475 17 +++ CPD-476 22.4 +++ CPD-477 51.7 ++ CPD-478 28.0 +++ CPD-479 132.0 ++ CPD-480 22.0 +++ CPD-481 17.0 +++ CPD-482 4.7 +++ CPD-483 5.5 +++ CPD-484 42.0 +++ CPD-485 94.0 ++ CPD-486 230.0 N/A CPD-487 121.0 ++ CPD-488 147.0 N/A CPD-489 141.0 N/A CPD-490 2796.0 N/A CPD-491 3701.0 N/A CPD-492 178.0 N/A CPD-493 46.0 ++ CPD-494 >10000 N/A CPD-495 >10000 N/A CPD-496 8764.0 N/A CPD-497 >10000 N/A CPD-498 36.5 ++ CPD-499 11.3 +++ CPD-500 13.2 +++ CPD-501 13.2 +++ CPD-502 682.4 – CPD-503 503.5 – CPD-504 333.9 – CPD-505 195.2 – CPD-506 204.1 N/A CPD-507 1520 N/A CPD-508 2095 N/A CPD-509 53.1 – CPD-510 58.8 + CPD-511 85.8 + CPD-512 307.2 N/A CPD-513 260.8 N/A CPD-514 174 N/A CPD-515 343.2 N/A CPD-516 473.2

(2033) The IC₅₀ value of each compound was determined as described in FIG. 27 and calculated using the GraphPad Prism 5.0 software. The percentage of degradation of each compound was indicated in the following categories: N/A: data not available; -: no degradation; +: degradation less than 50%; ++: degradation equal or greater than 50% but less than 80%; +++: degradation equal or greater than 80%.

(2034) Tumor Xenograft Studies

(2035) All animal experiments were performed under protocols approved by the Institutional Animal Care and Use Committee (IACUC) of Cullgen. Athymic nude mice (male, 5-weeks old) received 5 million KM12 cells subcutaneously inoculated at the right flank site. For pharmacodynamic studies, animals were treated when tumors were approximately 500 mm³ in size. For tumor growth assays, animals were treated when tumors were larger than 100 mm³. Tumor volumes were calculated as (length×width×width)/2. Tumor-bearing mice were treated intraperitoneally or orally with vehicle or bivalent compounds as indicated in figures. For tumor growth assays, tumor sizes and body weight of mice were measured every 2 days. For pharmacodynamic studies, the levels of TPM3-TPKA in xenograft tumors was determined following a single dose of drug administration. At indicated hours after drug administration, selected mice were sacrificed, tumors were resected. Small chunks of tumors were homogenized for immunoblotting of TPM3-TPKA and other proteins as indicated.

(2036) Mouse Pharmacokinetic (PK) Studies

(2037) Standard PK studies were conducted using male Swiss Albino mice by Sai Life Sciences or

using male ICR mice by Cullgen. Plasma exposures of TR-123 (10 mg/kg, intraperitoneal injection), TR-198 (20 mg/kg, oral gavage, or 2 mg/kg, intravenous injection), TR-231 and TR-275 (10 mg/kg, oral gavage, or 1 mg/kg, intravenous injection) were determined. Plasma concentrations were reported at indicated time points as the average values derived from three experimental animals. Error bars represent standard deviations.

(2038) Monoiodoacetate (MIA)-Induced Pain Models

(2039) Adult male Sprague Dawley rats or guinea pigs were used in these experiments. The baseline of hind limb weight distribution was determined prior to MIA injection using an incapacitance tester (Yuyan Instruments, Shanghai, China). The weight distribution was expressed in the following formula:

$$\text{Weight on the right paw(g)} \div \text{weight on(left+right)paws} \times 100$$

(2040) The next day, 2 mg MIA in 50 μ l PBS solution was injected into the right knee of experimental rats. For guinea pigs, 3 mg MIA was used. One week following MIA injection, hind limb weight distribution was measured to validate the symptoms. Prior to treatment, animals were randomized into 2 or 3 groups (n=7), each group was treated with vehicle, 100 mg/kg ibuprofen (i.m), or 20 mg/kg TR-181 (i.p.). Response to the compounds was subsequently determined as the changes in weight distribution between two hind limbs. Aguilar, A., Lu, J., Liu, L., Du, D.,

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OTHER EMBODIMENTS

(2041) It is to be understood that while the invention has been described in conjunction with the detailed description thereof, the foregoing description is intended to illustrate and not limit the scope of the invention, which is defined by the scope of the appended claims. Other aspects, advantages, and modifications are within the scope of the following claims.

Claims

1. A bivalent compound, selected from the group consisting of: (S)-N-(5-(3,5-Difluorobenzyl)-1H-indazol-3-yl)-4-(4-((1-(2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-5-yl)azetidin-3-yl)methyl)piperazin-1-yl)-2-((tetrahydro-2H-pyran-4-yl)amino)benzamide (TR-341); and (R)-N-(5-(3,5-Difluorobenzyl)-1H-indazol-3-yl)-4-(4-((1-(2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-5-yl)azetidin-3-yl)methyl)piperazin-1-yl)-2-((tetrahydro-2H-pyran-4-yl)amino)benzamide (TR-342); or a pharmaceutically acceptable salt thereof.
2. The bivalent compound of claim 1, which is (S)-N-(5-(3,5-Difluorobenzyl)-1H-indazol-3-yl)-4-(4-((1-(2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-5-yl)azetidin-3-yl)methyl)piperazin-1-yl)-2-((tetrahydro-2H-pyran-4-yl)amino)benzamide (TR-341), or a pharmaceutically acceptable salt thereof.
3. The bivalent compound of claim 1, which is (R)-N-(5-(3,5-Difluorobenzyl)-1H-indazol-3-yl)-4-(4-((1-(2-(2,6-dioxopiperidin-3-yl)-1,3-dioxoisindolin-5-yl)azetidin-3-yl)methyl)piperazin-1-yl)-2-((tetrahydro-2H-pyran-4-yl)amino)benzamide (TR-342), or a pharmaceutically acceptable salt thereof.
4. A pharmaceutical composition comprising a bivalent compound of claim 1, or a pharmaceutically acceptable salt thereof, and a pharmaceutically acceptable carrier or diluent.
5. A method of treating a tropomyosin receptor kinase (TRK)-mediated disease, comprising administering to a subject with a TRK-mediated disease a bivalent compound of claim 1, or a pharmaceutically acceptable salt thereof.
6. The method of claim 5, wherein the TRK-mediated disease results from TRK expression, mutation, splicing, or fusion.
7. The method of claim 5, wherein the TRK-mediated disease is cancer.
8. The method of claim 7, wherein the cancer is a relapsed cancer.
9. The method of claim 7, wherein the cancer is refractory to one or more previous treatments.
10. The method of claim 7, further comprising administering to the subject an additional therapeutic regimen for treating cancer.
11. The method of claim 10, wherein the additional therapeutic regimen is selected from the group consisting of surgery, chemotherapy, radiation therapy, hormone therapy, and immunotherapy.
12. The method of claim 5, wherein the TRK-mediated disease is acute pain or chronic pain.

